EDUCORN'S

MATHEMATICS

PRACTICE PROBLEMS

MATHEMATICAL CONCEPTS FOR HIGHER CLASS

VOLUME-I

MATHEMATICS IS TRUE FRIEND....

Complex Numbers

Topic 1

Algebraic Operations and Equality of Complex Numbers

Q 1. If $(x + iy)^{1/3} = 2 + 3i$, then $3x + 2y$ is equal to

[Kerala CEE]

(a) -20

(b) -60

(c) -120

(d) 60

(e) 156

Q 2. The value of $\frac{\cos 30^\circ + i \sin 30^\circ}{\cos 60^\circ - i \sin 60^\circ}$ is equal to

[VTTEEE]

(a) i

(b) –i

(c) $\frac{1+\sqrt{3}i}{2}$

(d) $\frac{1-\sqrt{3}i}{2}$

Q 3. If $i=\sqrt{-1}$ and n is a positive integer, then $i^n + i^{n+1} + i^{n+2} + i^{n+3}$ is equal to

[WB JEE]

(a) 1

(b) i

(c) i^n

(d) 0

Q 4. If $2\alpha = -1 - i\sqrt{3}$ and $2\beta = -1 + i\sqrt{3}$, then $5\alpha^4 + 5\beta^4 + 7\alpha^{-1}\beta^{-1}$ is equal to

[Kerala CEE]

(a) -1

(b) -2

(c) 0

(d) 1

(e) 2

Q 5. Let $a_n = i^{(n+1)^2}$, where $i = \sqrt{-1}$ and n = 1,2,3,....

Then, the value of $a_1 + a_3 + a_5 + \ldots + a_{25}$ is

[Kerala CEE]

(a) 13

(b) 13 + i

(c) 13 − i

(d) 12

(e) $12 - i$

Q 6. If $a = \cos\theta + i \sin\theta$, then $\frac{1+a}{1-a}$ is equal to

[OJEE]

(a) $i\cot\frac{\theta}{2}$

(b) $i\tan\frac{\theta}{2}$

(c) $i\cos\frac{\theta}{2}$

(d) $i\operatorname{cosec}\frac{\theta}{2}$

Q 7. The complex number $\frac{(-\sqrt{3}+3i)(1-i)}{(3+\sqrt{3}i)(i)(\sqrt{3}+\sqrt{3}i)}$ when represented in the Argand diagram is

(a) in the second quadrant

(b) in the first quadrant

[Jamia Millia Islamia]

(c) on the y−axis (imaginary axis)

(d) on the x−axis (real axis)

Q 8. In which quadrant of the complex plane, the point $\frac{1+2i}{1-i}$ lies ?

[UP SEE]

(a) Fourth

(b) First

(c) Second

(d) Third

Q 9. Let $z = \frac{11-3i}{1+i}$. If α is a real number such that $z - i\alpha$ is real, then the value of α is

[Kerala CEE]

(a) 4

(b) −4

(c) 7

(d) −7

(e) 3

Q 10. The value of sum $\sum_{n=1}^{13}(i^n + i^{n+1})$, where $i = \sqrt{-1}$,

equals

[AMU, OJEE]

(a) i

(b) $i - 1$

(c) $-i$

(d) 0

Q 11. The real part of $(1 - \cos\theta + 2i \sin\theta)^{-1}$ is

[WBJEE]

(a) $\frac{1}{3+5\cos\theta}$

(b) $\frac{1}{5-3\cos\theta}$

(c) $\frac{1}{3-5\cos\theta}$

(d) $\frac{1}{5-3\cos\theta}$

Q 12. The number of non−zero integral solutions of the equation $|1 - i|^x = 2^x$ is

[WB JEE]

(a) infinite

(b) 1

(c) 2

(d) None of these

Q 13. The imaginary part of $\frac{(1+i)^2}{i(2i-1)}$ is

[VITEEE]

(a) 4/5

(b) 0

(c) $2/5$

(d) $-(4/5)$

Q 14. If $z = x - iy$ and $z^{1/3} = p + iq$, then

$\left(\frac{x}{p}+\frac{y}{q}\right)\Big/(p^2+q^2)$ is equal to

[Jamia Millia Islamia, MP PET]

(a) 1

(b) -1

(c) 2

(d) -2

Q 15. If $\frac{(1+i)^2}{2-i}$ = x + iy, then x + y is equa

[J&K CET]

(a) $-\frac{2}{5}$

(b) $\frac{6}{5}$

(c) $\frac{2}{5}$

(d) $-\frac{6}{5}$

Q 16. If z is a complex number such that Re (z) = Im (z), then

[Kerala CEE]

(a) $\text{Re}(z^2) = 0$

(b) $\text{Im}(z^2) = 0$

(c) $\text{Re}\ (z^2) = \text{Im}\ (z^2)$

(d) $\text{Re}\ (z^2) = -\ \text{Im}\ (z^2)$

(e) $z^2 = 0$

Q 17. The smallest positive integer n for which $(1 + i)^{2n} = (1 - i)^{2n}$ is

[AMU]

(a) 1

(b) 2

(c) 3

(d) 4

Q 18. The smallest positive integer n for which

$\left(\frac{1+i}{1-i}\right)^n = 1$, is

[WBJEE]

(a) n = 8

(b) n = 12

(c) n = 16

(d) None of these

Q 19. If $z = x + iy$, $z^{1/3} = a - ib$ and $\frac{x}{a} - \frac{y}{b} = k(a^2 - b^2)$, then value of k equals

[DCE]

(a) 2

(b) 4

(c) 6

(d) 1

Q 20. The values of x and y satisfying the equation $\frac{(1+i)x-2i}{3+i}+\frac{(2-3i)y+i}{3-i}=i$ are

[BCECE]

(a) x = −1, y = 3

(b) x = 3, y = −1

(c) x = 0, y = 1

(d) x = l, y = 0.

Q 21. The additive inverse of $1 - i$ is

[UP SEE]

(a) $0 + 0i$

(b) $-1 + i$

(c) $-1 + i$

(d) None of these

Q 22. The value of $\sum_{n=0}^{\infty}\left(\frac{2i}{3}\right)^n$ is

[EAMCET]

(a) $\frac{9+6i}{13}$

(b) $\frac{9-6i}{13}$

(c) $9 + 6i$

(d) $9 - 6i$

Q 23. $\left(\frac{1}{1-2i}+\frac{3}{1+i}\right)\left(\frac{3+4i}{2-4i}\right)$ is equal to

[Jamia Millia Islamia]

(a) $\frac{1}{2}+\frac{9}{2}i$

(b) $\frac{1}{2}-\frac{9}{2}i$

(c) $\frac{1}{4}-\frac{9}{4}i$

(d) $\frac{1}{4}+\frac{9}{4}i$

Q 24. $\frac{3+2i\sin\theta}{1-2i\sin\theta}$ will be purely imaginary, if θ is equal to

[Jamia Millia Islamia]

(a) $2n\pi\pm\frac{\pi}{3}$

(b) $n\pi+\frac{\pi}{3}$

(c) $n\pi\pm\frac{\pi}{3}$

(d) None of these

Q 25. If $\left(\frac{1+i}{1-i}\right)^{x}=1$, then

[AIEEE]

(a) x = 4n, where n is any positive integer (b)

x=2n, where n is any positive integer

(c) x = 4n +1, where n is any positive integer

(d) x = 2n +1, where n is any positive integer

Q 26. If $x = 3 + i$, then $x^3 - 3x^2 - 8x + 15$ is equal to

[UPSEE]

(a) 45

(b) -15

(c) 10

(d) 6

Topic 2

Conjugate, Modulus and Argument

Q 1. The number of complex numbers z such that $|z - 1| = |z + 1| = |z - i|$ equals

[AIEEE]

(a) 0

(b) 1

(c) 2

(d) ∞

Q 2. If $z = \frac{4}{1-i}$, then $\bar{z}$ is (where $\bar{z}$ is complex conjugate of z).

[WBJEE]

(a) $2(1 + i)$

(b) $(1 + i)$

(c) $\frac{2}{1-i}$

(d) $\frac{4}{1+i}$

Q 3. If $-\pi < \arg(z) < -\frac{\pi}{2}$ then $\arg(\bar{z}) - \arg(-\bar{z})$ is

[WB JEE]

(a) π

(b) $-\pi$

(c) $\pi/2$

(d) $-\pi/2$

Q 4. The modulus of the complex number z such that $|z + 3 - i| = 1$ and $\arg(z) = \pi$ is equal to

[Kerala CEE]

(a) 1

(b) 2

(c) 9

(d) 4

(e) 3

Q 5. If $z_1, z_2, \ldots, z_n$ are complex numbers such

that $|z_1| = |z_2| = \ldots = |z_n| = 1$, then $|z_1 + z_2 + \ldots + z_n|$ is equal to

[Kerala CEE]

(a) $|z_1 z_2 z_3 \ldots z_n|$

(b) $|z_1| + |z_2| + \ldots + |z_n|$

(c) $\left|\frac{1}{z_1} + \frac{1}{z_2} + \ldots + \frac{1}{z_n}\right|$

(d) n

(e) $\sqrt{n}$

Q 6. If $z_1 = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$ and $z_2 = \sqrt{3}\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$,

then $|z_1 z_2|$ is

[Kerala CEE]

(a) 6

(b) $\sqrt{2}$

(c) $\sqrt{6}$

(d) $\sqrt{3}$

(e) $\sqrt{2} + \sqrt{3}$

Q 7. If $z = r(\cos\theta + i \sin\theta)$, then the value of

$\frac{z}{\bar{z}}+\frac{\bar{z}}{z}$ is

[Kerala CEE]

(a) $\cos 2\theta$

(b) $2 \cos 2\theta$

(c) $2 \cos \theta$

(d) $2 \sin \theta$

(e) $2 \sin 2\theta$

Q 8. If then the maximum value of |z| is equal to

$\left|z-\frac{4}{z}\right|=2$.

[AIEEE]

(a) $\sqrt{3}+1$

(b) $\sqrt{5}+1$

(c) 2

(d) $2+\sqrt{2}$

Q 9. The number of solutions of the equation

$z^2 + \bar{z} = 0$ is

[DCE, MP PET]

(a) 2

(b) 4

(c) 6

(d) 8

Q 10. If $z = \sqrt{3} + i$, then the argument of $z^2 e^{z-i}$ is

equal to

[Kerala CEE]

(a) $\frac{\pi}{2}$

(b) $\frac{\pi}{6}$

(c) $e^{\pi/6}$

(d) $e^{\pi/3}$

(e) $\pi/3$

Q 11. Let z and w be two complex numbers such

that $|z| \leq 1$, $|w| \leq 1$ and $|z + iw| = |z - \overline{iw}| = 2$.

Then, z is equal to

[AMU]

(a) 1 or i

(b) i or −i

(c) 1 or -1

(d) i or -1

Q 12. The locus of z satisfying the inequality

$\left|\frac{z+2i}{2z+i}\right|<1$, where z = x + iy, is

[EAMCET]

(a) $x^2 + y^2 < 1$

(b) $x^2 - y^2 < 1$

(c) $x^2 + y^2 > 1$

(d) $2x^2 + 3y^2 < 1$

Q 13. The solution of equation |z| − z = 1 + 2i is

[OJEE]

(a) $\frac{3}{2}+2i$

(b) $\frac{3}{2}-2i$

(c) $3 - 2i$

(d) None of these

Q 14. For any complex number z, the minimum value of $|z| + |z - 1|$ is

[WBJEE]

(a) 0

(b) 1

(c) 2

(d) -1

Q 15. The modulus of $\frac{1-i}{3+i}+\frac{4i}{5}$ is

[WBJEE]

(a) $\sqrt{5}$ unit

(b) $\frac{\sqrt{11}}{5}$ unit

(c) $\frac{\sqrt{5}}{5}$ unit

(d) $\frac{\sqrt{12}}{5}$ unit

Q 16. If $(3+i)(z+\bar{z})-(2+i)(z-\bar{z})+14i=0$, then $z\bar{z}$ is equal to

[J&K CET]

(a) 5

(b) 8

(c) 10

(d) 40

Q 17. The conjugate of a complex number is $\frac{1}{i-1}$.

Then, that complex number is

[AIEEE]

(a) $\frac{1}{i-1}$

(b) $-\frac{1}{i-1}$

(c) $\frac{1}{i+1}$

(d) $-\frac{1}{i+1}$

Q 18. The amplitude of $\sin\frac{\pi}{5}+i\left(1-\cos\frac{\pi}{5}\right)$ is

[DCE]

(a) $\frac{2\pi}{5}$

(b) $\frac{\pi}{15}$

(c) $\frac{\pi}{10}$

(d) $\frac{\pi}{5}$

Q 19. $\left|\frac{1}{2}(z_1+z_2)+\sqrt{z_1z_2}\right|+\left|\frac{1}{2}(z_1+z_2)-\sqrt{z_1z_2}\right|$ is equal to

[DCE]

(a) $|z_1 + z_2|$

(b) $|z_1 - z_2|$

(c) $|z_1| + |z_2|$

(d) $|z_1| - |z_2|$

Q 20. If z is a complex number such that $z = -z$,

then

[KCET]

(a) z is purely real

(b) z is purely imaginary

(c) z is any complex number

(d) real part of z is the same as its imaginary part

Q 21. If $(\sqrt{5}+\sqrt{3}i)^{33} = 2^{49}z$, then modulus of the complex number z is equal to

[Kerala CEE]

(a) 1

(b) $\sqrt{2}$

(c) $2\sqrt{2}$

(d) 4

(e) 8

Q 22. If $\frac{5z_2}{11z_1}$ is purely imaginary, then the value of

$\left|\frac{2z_1+3z_2}{2z_1-3z_2}\right|$ is

[Kerala CEE]

(a) 37/33

(b) 2

(c) 1

(d) 3

(e) $33/37$

Q $23.$ If m_1 ,m_2,m_3 and m_4 respectively denote the moduli of the complex numbers $1 + 4i, 3 + i, 1 - i$ and $2 - 3i$, then the correct one, among the following is

[EAMCET]

(a) $m_1 < m_2 < m_3 < m_4$

(b) $m_4 < m_3 < m_2 < m_1$

(c) $m_3 < m_2 < m_4 < m_1$

(d) $m_3 < m_1 < m_2 < m_4$

Q 24. The points in the set $\left\{ z \in C : \arg\left(\frac{z-2}{z-6i} \right) = \frac{\pi}{2} \right\}$

(where C denotes the set of all complex numbers)

lie on the curve which is a

[EAMCET]

(a) circle

(b) pair of lines

(c) parabola

(d) hyperbola

Q 25. The principal amplitude of $(\sin 40° + i \cos 40°)^5$ is

[WBJEE]

(a) $70°$

(b) -110°

(c) 110°

(d) -70°

Q 26. A and B are two points on the Argand plane such that the segment AB is bisected at the point $(0, 0)$. If the point A, which is in the third quadrant has principal amplitude θ, then the principal amplitude of the point B is

[WBJEE]

(a) $-\theta$

(b) $\pi - \theta$

(c) $\theta - \pi$

(d) $\pi + \theta$

Q 27. For two complex numbers z_1, z_2 the relation $|z_1 + z_2| = |z_1| + |z_2|$ holds, if

[WBJEE]

(a) arg (z_1) = arg (z_2)

(b) arg (z_1) + arg $(z_2) = \frac{\pi}{2}$

(c) $z_1 z_2 = 1$

(d) $|z_1| = |z_2|$

Q 28. If $\left|\frac{z-25}{z-1}\right| = 5,$ find the value of |z|

[VTTEEE]

(a) 3

(b) 4

(c) 5

(d) 6

Q 29. Argument of the complex number $\left(\frac{-1-3i}{2+i}\right)$ is

[VITEEE]

(a) 45°

(b) 135°

(c) 225°

(d) 240°

Q 30. The modulus and amplitude of $(1+i\sqrt{3})^8$ are

respectively

[Jamia Millia Islamia]

(a) 256 and $\frac{\pi}{3}$

(b) 256 and $\frac{2\pi}{3}$

(c) 2 and $\frac{2\pi}{3}$

(d) 256 and $\frac{8\pi}{3}$

Q 31. The conjugate of the complex number

$\frac{(1+i)^2}{1-i}$ is

[Manipal]

(a) $1 - i$

(b) $1 + i$

(c) $-1 + i$

(d) $-1 - i$

Q 32. The magnitude and amplitude of

$\frac{(1+i\sqrt{3})(2+2i)}{(\sqrt{3}-i)}$ are respectively

[Guj.CET]

(a) $2, \frac{3\pi}{4}$

(b) $2\sqrt{2}, \frac{3\pi}{4}$

(c) $2\sqrt{2}, \frac{\pi}{4}$

(d) $2\sqrt{2}, \frac{\pi}{2}$

Q 33. The argument of the complex number

$\frac{13-5i}{4-9i}$ is **[UP SEE]**

(a) $\pi/3$

(b) $\pi/4$

(c) $\pi/5$

(d) $\pi/6$

Q 34. If arg(z) = θ, then arg($\bar{z}$) is equal to

[UP SEE]

(a) $\theta - \pi$

(b) $\pi - \theta$

(c) θ

(d) $-\theta$

Q 35. If $z = \frac{1 - i\sqrt{3}}{2}$, then the correct matching of

List I from List II is

[EAMCET]

List I List II

(i)	$a\bar{a}$	(A)	$-\frac{\pi}{3}$
(ii)	$a-\bar{a}$	(B)	$-i\sqrt{3}$
(iii)	$a-\bar{a}$	(C)	$2i/\sqrt{3}$
(iv)	$\mathrm{Im}\left(\frac{4}{3a}\right)$	(D)	1
		(E)	$\frac{\pi}{3}$
		(F)	$\frac{2}{\sqrt{3}}$

correct match is

(i) (ii) (iii) (iv)

(a) D E C B

(b) D A B F

(c) F E B C

(d) D A B C

Q 36. If $\frac{z-1}{z+1}$ is purely imaginary number $(z \neq -1)$, then | z | is equal to

[OJEE]

(a) 1

(b) 2

(c) 3

(d) 4

Q 37. For any two complex numbers z_1 and z_2 and any real numbers a and b, $|(az_1 - bz_2)|^2 + |(bz_1 + az_2)|^2$ is equal to

[WBJEE]

(a) $(a^2+b^2)(|z_1| + |z_2|)$

(b) $(a^2+b^2)(|z_1|^2+|z_2|^2)$

(c) $(a^2 + b^2(|z_1|^2 - |z_2|^2)$

(d) None of the above

Q 38. The complex numbers sin x + i cos 2x

and cos x − i sin 2x are conjugate to each other fo

[WB JEE]

(a) $x = n\pi$

(b) $x = \left(n + \frac{1}{2}\right)\pi$

(c) $x = 0$

(d) No value of x

Q 39. Let z, w be complex numbers such that $\bar{z} + i\bar{w} = 0$ and arg (zw) = π. Then, arg (z) equals

[Jamia Millia Islamia]

(a) $\frac{\pi}{4}$

(b) $\frac{\pi}{2}$

(c) $\frac{3\pi}{4}$

(d) $\frac{5\pi}{4}$

Q 40. If $|z + 4| \leq 3$, then the greatest and the least value of $| z + l|$ are respectively

[RPET]

(a) $6,-6$

(b) $6,0$

(c) $7,2$

(d) $0,-1$

Q 41. If $w = \alpha + i\beta$, where $\beta \neq 0$ and $z \neq 1$, satisfies the condition that $\left(\frac{w - \bar{w}z}{1 - z}\right)$ is purely real, then the set of values of z is

[IIT JEE]

(a) $|z| = 1, z \neq 2$

(b) $|z| = 1$ and $z \neq 1$

(c) $z = \bar{z}$

(d) None of these

Q 42. The amplitude of $\frac{1+i\sqrt{3}}{\sqrt{3}+i}$ is

[BITSAT]

(a) $\frac{\pi}{3}$

(b) $\frac{\pi}{4}$

(c) $\frac{2\pi}{3}$

(d) $\frac{\pi}{6}$

Q 43. The modulus and amplitude of $\frac{1+2i}{1-(1-i)^2}$ are

(a) $\sqrt{2}$ and $\frac{\pi}{6}$

(b) 1 and 0

(c) 1 and $\frac{\pi}{3}$

(d) 1 and $\frac{\pi}{4}$

[BITSAT]

Q 44. If z = x + iy is a variable complex number such that arg $\frac{z-1}{z+1}=\frac{\pi}{4}$, then

[UP SEE]

(a) $x^2 - y^2 - 2x = l$

(b) $x^2 + y^2 - 2x = l$

(c) $x^2 + y^2 - 2y = 1$

(d) $x^2 + y^2 + 2x = l$

Q 45. If α and β are different complex numbers with $|\beta| = 1$, then $\left|\frac{\beta - \alpha}{1 - \bar{\alpha}\beta}\right|$ is

[Kerala CEE]

(a) 0

(b) $3/2$

(c) $1/2$

(d) 1

(e) 2

Q 46. If z_1, z_2 are any two complex numbers,

then

[WB JEE]

(a) $|z_1 + z_2| \geq |z_1| + |z_2|$

(b) $|z_1 + z_2| > |z_1| + |z_2|$

(c) $|z_1 + z_2| \leq |z_1| + |z_2|$

(d) $|z_1 + z_2 = |z_1| + |z_2|$

Q 47. Let z_1 and z_2 be complex numbers, then

$|z_1 + z_2|^2 + |z_1 - z_2|^2$ is equal to

[MPPET]

(a) $|-z_1|^2 + |z_2|^2$

(b) $2(|z_1|^2 + |z_2|^2)$

(c) $2(z^2 + z_2^2)$

(d) $4z_1z_2$

Q 48. The argument of $\frac{1+i\sqrt{3}}{1-i\sqrt{3}}$ is

[J&KCET]

(a) $2\pi/3$

(b) $\pi/3$

(c) $-\pi/3$

(d) $-2\pi/3$

Q 49. The number of solutions for the equations

$|z - 1| = |z - 2| = |z - i|$ is

[BITSAT, OJEE]

(a) one solution

(b) 3 solutions

(c) 2 solutions

(d) no solution

Q 50. If $(3 + i)z = (3 - i)\bar{z}$ then the complex number z is

[AMU]

(a) $a(3 - i)$, $a \in R$

(b) $\frac{a}{(3+i)}$, $a \in R$

(c) $a(3 + i)$, $a \in R$

(d) $a(-3 + i)$, $a \in R$

Q 51. If α_1, α_2, α_3 respectively denote the moduli of the complex numbers $-i, \frac{1}{3}(1+i)$ and $-1 + i$, then their increasing order is

[EAMCET, RPET]

(a) α_1, α_2, α_3

(b) α_3, α_2, α_1

(c) α_2, α_1, α_3

(d) α_3, α_1, α_2

Q 52. If $z_1 = 1 + 2i$ and $z_2 = 3 + 5i$, then Re $[\bar{z}_2 z_1 / z_2]$ is equal to

[J&KCET]

(a) $-31/17$

(b) $17/22$

(c) $-17/31$

(d) $22/17$

Q 53. The argument of $(1-i\sqrt{3})/(1+i\sqrt{3})$ is

[UP SEE]

(a) $60°$

(b) $120°$

(c) $210°$

(d) $240°$

Q 54. The real part of $\left[1+\cos\left(\frac{\pi}{5}\right)+i\sin\left(\frac{\pi}{5}\right)\right]$ is

[UP SEE]

(a) 1

(b) $\frac{1}{2}$

(c) $\frac{1}{2}\cos\left(\frac{\pi}{10}\right)$

(d) $\frac{1}{2}\cos\left(\frac{\pi}{5}\right)$

(e) $\frac{1}{2}\sec\left(\frac{\pi}{10}\right)$

Q 55. If $\frac{2z_1}{3z_2}$ is a purely imaginary, then $\left|\frac{z_1 - z_2}{z_1 + z_2}\right|$ Is

[Kerala CEE]

(a) $\frac{2}{3}$

(b) $\frac{3}{2}$

(c) $\frac{4}{9}$

(d) 1

(e) $\frac{9}{4}$

Q 56. If z_1, z_2 are two complex numbers satisfying $\left|\frac{z_1 - 3z_2}{3 - z_1\overline{z}_2}\right| = 1, |z_1| \neq 3$, then $|z_2|$ is equal to

[EAMCET]

(a) 1

(b) 2

(c) 3

(d) 4

Q 57. Let z_1 be a complex number with $|z_1| = 1$ and z_2 be any complex number, then $\left|\frac{z_1 - z_2}{1 - z_1 z_2}\right|$ is equal to

(a) 0

(b) 1

(c) -1

(d) 2

[OJEE]

Q 58. The complex numbers z having positive argument and satisfying $12 - 3i < |z|$, is

[J&K CET]

(a) $\frac{12}{5}+\frac{16}{5}i$

(b) $\frac{4}{5}+\frac{6}{5}i$

(c) $\frac{6}{5}-\frac{5}{2}i$

(d) None of these

Q 59. If |z| = I and $w=\frac{z-1}{z+1}$(where $z \neq -1$), then

Re (w) is

[IIT JEE]

(a) 0

(b) $\frac{1}{|z+1|^2}$

(c) $\left|\frac{1}{z+1}\right| \cdot \frac{1}{|z+1|^2}$

(d) $\frac{\sqrt{2}}{|z+1|^2}$

Q 60. If z and w are two non–zero complex numbers such that | zw | = 1 and arg (z) − arg (w) = $\frac{\pi}{2}$, then $\bar{z}$w is equal to

[AIEEE]

(a) 1

(b) −1

(c) i

(d) −i

Q 61. If z = x + iy and $\left|\frac{1-iz}{z-i}\right|=1$, the locus of z is

[UP SEE]

(a) x−axis

(b) y−axis

(c) circle with unity radius

(d) None of the above

Q 62. The complex number z = x + iy, which satisfy the equation $\left|\frac{z-5i}{z+5i}\right|=1$ lies on

[AMU]

(a) real axis

(b) the line y = 5

(c) a circle passing through the origin

(d) None of the above

Q 63. If the amplitude of $z - 2 - 3i$ is $\frac{\pi}{4}$, then the locus of $z = x + iy$, is

[EAMCET]

(a) $x + y - 1 = 0$

(b) $x - y - 1 = 0$

(c) $x + y + 1 = 0$

(d) $x - y + 1 = 0$

Q 64. A real value of x will satisfy the equation

$\left(\frac{3-4ix}{3+4ix}\right) = \alpha - i\beta$($\alpha$, β are real), if

[OJEE]

(a) $\alpha^2 - \beta^2 = -1$

(b) $\alpha^2 - \beta^2 = 1$

(c) $\alpha^2 + \beta^2 = 1$

(d) $\alpha^2 - \beta^2 = 2$

Q 65. The conjugate of complex number $\frac{2-3i}{4-i}$ is

[MPPET]

(a) $\frac{3i}{4}$

(b) $\frac{11+10i}{17}$

(c) $\frac{11-10i}{17}$

(d) $\frac{2+3i}{4i}$

Topic 3

De–Moivre's Theorem, Square Root and Roots of Unity

Q 1. Let z = cos θ + i sin θ. Then, the value of

$\sum_{m=1}^{15} \mathrm{Im}(z^{2n-1})$ at $\theta = 2°$ is

[IIT JEE]

(a) $\frac{1}{\sin 2^o}$

(b) $\frac{1}{3\sin 2^o}$

(c) $\frac{1}{2\sin 2^o}$

(d) $\frac{1}{4\sin 2^{o}}$

Q 2. If a = cos θ + i sin θ, then $\frac{1+a}{1-a}$ is equal to

[BITSAT]

(a) $\cot\frac{\theta}{2}$

(b) cot θ

(c) $i\cot\frac{\theta}{2}$

(d) $i\tan\frac{\theta}{2}$

Q 3. One of the values of is $\left(\frac{1+i}{\sqrt{2}}\right)^{2/3}$

(a) $\sqrt{3}+i$

(b) –i

(c) i

(d) $-\sqrt{3}+i$

Q 4. The smallest positive integral value of n such that $\left[\frac{1+\sin\frac{\pi}{8}+i\cos\frac{\pi}{8}}{1+\sin\frac{\pi}{8}-i\cos\frac{\pi}{8}}\right]^n$ is purely imaginary, is equal to

[KCET]

(a) 4

(b) 3

(c) 2

(d) 8

Q 5. If n is an integer which leaves remainder one when divided by three, then $(1+\sqrt{3}i)^n + (1-\sqrt{3}i)^n$ equals

[EAMCET]

(a) -2^{n+1}

(b) 2^{n+1}

(c) $-(-2)^n$

(d) -2^n

Q 6. If $x + iy = \sqrt{\frac{a+ib}{c+id}}$, then $x^2 + y^2$ is equal to

[OJEE]

(a) $\frac{a^2-b^2}{c^2+d^2}$

(b) $\frac{a^2+b^2}{c^2+d^2}$

(c) $\frac{a^2+b^2}{c^2-d^2}$

(d) None of these

Q 7. If n is a positive integer, then

$(1+i\sqrt{3})^n+(1-i\sqrt{3})^n$ is equal to

[OJEE]

(a) $2^{n-1}\cos\frac{n\pi}{3}$

(b) $2^n\cos\frac{n\pi}{3}$

(c) $2^{n+1}\cos\frac{n\pi}{3}$

(d) None of these

Q 8. If $\frac{3}{2+\cos\theta+i\sin\theta} = a + ib$, then $[(a-2)^2+b^2]$

is equal to

[BCECE]

(a) 0

(b) 1

(c) -1

(d) 2

Q 9. If the expression $\frac{\left[\sin\left(\frac{x}{2}\right)+\cos\left(\frac{x}{2}\right)-i\tan(x)\right]}{\left[1+2i\sin\left(\frac{x}{2}\right)\right]}$ is

real, then the set of all possible value of x is

[Jamia Millia Islamia]

(a) $n\pi + \alpha$

(b) $2n\pi$

(c) $\frac{n\pi}{2}+\alpha$

(d) None of these

Q 10. If $z^2 + z + 1 = 0$, where z is a complex number, then the value of

$$\left(z+\frac{1}{z}\right)^2+\left(z^2+\frac{1}{z^2}\right)^2+\left(z^3+\frac{1}{z^3}\right)^2+.....+\left(z^6+\frac{1}{z^6}\right)^2 \text{ is}$$

[Jamia Millia Islamia]

(a) 6

(b) 12

(c) 18

(d) 24

Q 11. If $(x + iy) = \sqrt{\frac{1+2i}{3+4i}}$, then $(x^2 + y^2)^2$ is equal to

[MPPET]

(a) 5

(b) 1/5

(c) 2/5

(d) 5/2

Q 12. If $\left(\frac{1+\cos\phi+i\sin\phi}{1+\cos\phi-i\sin\phi}\right)^n$ = u + iv, where u and v all real numbers, then u is

[RPET]

(a) n cos ϕ

(b) cos nϕ

(c) $\cos\left(\frac{n\phi}{2}\right)$

(d) $\sin\left(\frac{n\phi}{2}\right)$

Q 13. If square root of $-7+24i$ is x + iy, then x is

[RPET]

(a) ± 1

(b) ± 2

(c) ± 3

(d) ± 4

Q 14. The value of $\frac{a+b\omega+c\omega^2}{b+c\omega+a\omega^2}+\frac{a+b\omega+c\omega^2}{c+a\omega+b\omega^2}$ will be

[DCE]

(a) 1

(b) -1

(c) 2

(d) -2

Q 15. If is ω a complex cube root of unity, then

$\sin\left\{(\omega^{10}+\omega^{23})\pi-\frac{\pi}{4}\right\}$ is equal to

[BITSAT, EAMCET]

(a) $\frac{1}{\sqrt{2}}$

(b) $\frac{1}{2}$

(c) 1

(d) $\frac{\sqrt{3}}{2}$

Q 16. If α is a complex number satisfying the equation $\alpha^2 + \alpha + 1 = 0$, then α^{31} is equal to

[KCET]

(a) α

(b) α^2

(c) 1

(d) i

Q 17. If $\sqrt{x+iy} = \pm(a+ib)$, then $\sqrt{-x-iy}$ is equal to

[Kerala CEE]

(a) $\pm$(b + ia)

(b) $\pm$(a – ib)

(c) (ai + b)

(d) $\pm$(b – ia) (e) None of these

Q 18. The value of $\left(-\frac{1}{2}+\frac{\sqrt{3}i}{2}\right)^{1000}$ is

[OJEE]

(a) ω^3

(b) ω^2

(c) $\omega^3 - \omega$

(d) ω

Q 19. If ω is a cube root of unity, then the value of $(1 - \omega + \omega^2)^5 + (1 + \omega - \omega^2)^5$ is

[OJEE,WBJEE]

(a) 30

(b) 32

(c) 2

(d) None of these

Q 20. If $2x = -1 + \sqrt{3}i$, then the value of $(1 - x^2 + x)^6 - (1 - x + x^2)^6$ is equal to

[Jamia Millia Islamia]

(a) 32

(b) − 64

(c) 64

(d) 0

Q 21. If ω is an imaginary cube root of unity and $x = a + b$, $y = a\omega + b\omega^2$, $z = a\omega^2 + b\omega$, then $x^2 + y^2 + z^2$ is equal to

[MPPET]

(a) $6ab$

(b) $3ab$

(c) $6a^2b^2$

(d) $3a^2b^2$

Q 22. The value of

$$\frac{\left[(\cos 20^\circ + i\sin 20^\circ)(\cos 75^\circ + i\sin 75^\circ)(\cos 10^\circ + i\sin 10^\circ)\right]}{\sin 15^\circ - i\cos 15^\circ}$$ is

[MPPET]

(a) 0

(b) -1

(c) i

(d) 1

Q 23. The square roots of $-7-24\sqrt{-1}$ are

[J&KCET]

(a) $\pm(4+3\sqrt{-1})$

(b) $\pm(3+4\sqrt{-1})$

(c) $\pm(3-4\sqrt{-1})$

(d) $\pm(4-3\sqrt{-1})$

Q 24. If $(\cos\theta + i\sin\theta)(\cos 2\theta + i \sin 2\theta)\ldots$

$(\cos n\theta + i \sin n\theta) = 1$, then the value of θ is

[BITSAT]

(a) $\frac{2m\pi}{n(n+1)}$

(b) $4m\pi$

(c) $\frac{4m\pi}{n(n+1)}$

(d) $\frac{m\pi}{n(n+1)}$

Q 25. If $1, \omega, \omega^2$ are the cube roots of unity, then $(1 + \omega)(1 + \omega^2)(1 + \omega^4)(1 + \omega^8)$ is equal to

[KCET]

(a) 1

(b) 0

(c) ω^2

(d) ω

Q 26. If α, β, γ are the cube roots of a negative number p, then for any three real numbers x, y, z the value of $\frac{x\alpha + y\beta + z\gamma}{x\beta + y\gamma + z\alpha}$ is

[Kerala CEE]

(a) $\frac{1 - i\sqrt{3}}{2}$

(b) $\frac{-1 - i\sqrt{3}}{2}$

(c) (x + y + z)i

(d) π

(d) $\left(\frac{x+y+z}{2}\right)\pi$

Q 27. If ω is a complex cube root of unity, then the value of $\sin\left\{(\omega^{10}+\omega^{23})\pi-\frac{\pi}{6}\right\}$ is

[Kerala CEE]

(a) $\frac{1}{\sqrt{2}}$

(b) $\frac{\sqrt{3}}{2}$

(c) $-\frac{1}{\sqrt{2}}$

(d) $-\frac{\sqrt{3}}{2}$

(e) $\frac{1}{2}$

Q 28. If $x + x^2 = \sqrt{3}x$, then $\sum_{n=1}^{24}\left(x^n - \frac{1}{x^n}\right)^2$ is equal to

[Kerala CEE]

(a) 0

(b) 48

(c) −24

(d) 24

(e)−48

Q 29. A value of n such that $\left(\frac{\sqrt{3}}{2}+\frac{i}{2}\right)^n = 1$ is

[EAMCET]

(a) 12

(b) 3

(c) 2

(d) 1

Q 30. The value of the expression

$$2\left(1+\frac{1}{\omega}\right)\left(1+\frac{1}{\omega^2}\right)+3\left(2+\frac{1}{\omega}\right)\left(2+\frac{1}{\omega^2}\right)+.....+(n+1)\left(n+\frac{1}{\omega}\right)\left(n+\frac{1}{\omega}\right) \text{ is}$$

[OJEE]

(a) $\left[\frac{n(n+1)}{2}\right]^2$

(b) $\left[\frac{n(n+1)}{2}\right]^2 - n$

(c) $\left[\frac{n(n+1)}{2}\right]^2 + n$

(d) None of these

Q 31. If $a = \cos\alpha + i\sin\alpha$, $b = \cos\beta + i\sin\beta$, $c = \cos\gamma + i\sin\gamma$ and $b/c + c/a + a/b = 1$, then $\cos(\beta - \gamma) + \cos(\gamma - \alpha) + \cos(\alpha - \beta)$ is equal to

[OJEE]

(a) 0

(b) 1

(c) -1

(d) None of these

Q 32. The value of

$\left(\cos\frac{\pi}{2}+i\sin\frac{\pi}{2}\right)\left(\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}\right)\left(\cos\frac{\pi}{8}+i\sin\frac{\pi}{8}\right).....\infty$ is

[OJEE]

(a) 1

(b) 0

(c) −1

(d) None of these

Q 33. If $\omega(\neq 1)$ be a cube root of unity and $(1 + \omega^2)^n = (1 + \omega^4)^n$, then the least positive value of n is

(a) 2

(b) 3

(c) 5

(d) 6

[WB JEE]

Q 34. If 1, a_1, a_2,,.....a_{n-1} are the n roots of unity, then the value of $(1 - a_1)(1 - a_2)(1 - a_3)...(1 - a_{n-1})$ is equal to

(a) $\sqrt{3}$

(b) $\frac{1}{2}$

(c) n

(d) 0

[BCECE]

Q 35. The product of all values of $(\cos \alpha + i \sin \alpha)^{3/5}$ is

[VITEEE]

(a) 1

(b) $\cos \alpha + i \sin \alpha$

(c) $\cos 3\alpha + i \sin 3\alpha$

(d) $\cos 5\alpha + i \sin 5\alpha$

Q 36. If $1, \omega, \omega^2$ are the cube roots of unity, then $(1 - \omega + \omega^2)(1 - \omega^2 + \omega^4)(1 - \omega^4 + \omega^8)(1 - \omega^8 + \omega^{16})$ ... upto $2n$ factors is

[J&KCET]

(a) $2n$

(b) 2^{2n}

(c) 1

(d) -2^{2n}

Q 37. One root of $(\mathrm{I})^{1/3}$ is

[DCE]

(a) $\frac{\sqrt{3}\mathrm{i}}{2}$

(b) $\frac{1+\sqrt{3}\mathrm{i}}{2}$

(c) $\frac{1-\sqrt{3}\mathrm{i}}{4}$

(d) $\frac{-1-\sqrt{3}\mathrm{i}}{2}$

Q 38. The value of $\sum_{k=1}^{10}\left(\sin\frac{2k\pi}{11}+\mathrm{i}\cos\frac{2k\pi}{11}\right)$ is

[AIEEE]

(a) 1

(b) -1

(c) $-i$

(d) i

Q 39. Let $a = e^{i\frac{2\pi}{3}}$. Then, the equation whose roots are $a + a^{-2}$ and $a^2 + a^{-4}$ is

[Kerala CEE]

(a) $x^2 - 2x + 4 + 0$

(b) $x^2 - x + 1 = 0$

(c) $x^2 + x + 4 = 0$

(d) $x^2 + 2x - 4 = 0$

(e) $x^2 + 2x + 4 = 0$

Q 40. If $i = \sqrt{-1}$, then $4 + 5\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right) + 3\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)^{365}$ is equal to

[Kerala CEE]

(a) $1 - i\sqrt{3}$

(b) $-1+i\sqrt{3}$

(c) $i\sqrt{3}$

(d) $-i\sqrt{3}$

(e) $1+i\sqrt{3}$

Q 41. If $\omega=\frac{-1+\sqrt{3}i}{2}$, then $(3 + \omega + 3\omega^2)^4$ is

[AMU]

(a) 16

(b) −16

(c) 16ω

(d) $16\omega^2$

Q 42. If ω is an imaginary cube root of unity, then $(1 + \omega - \omega^2)^7$ equals

[OJEE]

(a) $128\ \omega$

(b) -128ω

(c) $128\omega^2$

(d) $-128\omega^2$

Q 43. $\left(\frac{-1+\sqrt{-3}}{2}\right)^{100}+\left(\frac{-1-\sqrt{-3}}{2}\right)^{100}$ is equal to

[WBJEE]

(a) 2

(b) zero

(c) -1

(d) 1

Q 44. The value of $1+\sum_{k=0}^{14}\left\{\cos\frac{(2k+1)}{15}\pi+i\sin\frac{(2k+1)}{15}\pi\right\}$ is

[BCECE]

(a) 0

(b) -1

(c) 1

(d) i

Q 45. If $iz^4 + 1 = 0$, then z can take the value

[MP PET]

(a) $\frac{1+i}{\sqrt{2}}$

(b) $\cos\frac{\pi}{8}+i\sin\frac{\pi}{8}$

(c) $\frac{1}{4i}$

(d) i

Q 46. The minimum value of $|a + b\omega + c\omega^2|$, where a, b and c are all not equal integers and $\omega(\neq 1)$ is a cube root of unity, is

[IITJEE]

(a) $\sqrt{3}$

(b) $1/2$

(c) 1

(d) 0

Q 47. Value of $\sum_{k=1}^{6}\left[\sin\left(\frac{2k\pi}{7}\right)-i\cos\left(\frac{2k\pi}{7}\right)\right]$ is equal to

[BITSAT, OJEE]

(a) -1

(b) 1

(c) 0

(d) None of these

Q 48. If $z = \frac{7-i}{3-4i}$, then z^{14} is equal to

[Kerala CEE]

(a) 2^7

(b) $2^7 i$

(c) $2^{14} i$

(d) $-2^7 i$

(e) -2^{14}

Q 49. If $1, \omega, \omega^2$ are the cube roots of unity, then $\omega^2(1 + \omega^3) - (1 + \omega^2)\omega$ is equal to

[OJEE]

(a) 1

(b) -1

(c) i

(d) 0

Q 50. Let $x = \alpha + \beta$, $y = \alpha\omega + \beta\omega^2$, $z = \alpha\omega^2 + \beta\omega$, ω is an imaginary cube root of unity. The value of xyz is

[OJEE, RPET]

(a) $\alpha^2 + \beta^2$

(b) $\alpha^2 - \beta^2$

(c) $\alpha^3 + \beta^3$

(d) $\alpha^3 - \beta^3$

Q 51. If $1, \omega, \omega^2$ are the cube roots of unity, then $(3 + \omega^2 + \omega^4)^6$ is equal to

[BCECE]

(a) 64

(b) 729

(c) 2

(d) 0

Q 52. If is ω a complex cube root of unity, then $\frac{a+b\omega+c\omega^2}{c+a\omega+b\omega^2}+\frac{c+a\omega+b\omega^2}{a+b\omega+c\omega^2}+\frac{b+c\omega+a\omega^2}{b+c\omega^4+a\omega^5}$ is equal to

[MPPET]

(a) 1

(b) ω

(c) ω^2

(d) 0

Q 53. If α, β and γ are angles such that $\tan \alpha + \tan \beta + \tan \gamma = \tan \alpha \tan \beta \tan \gamma$ and $x = \cos \alpha + i \sin \alpha$, $y = \cos \beta + i \sin \beta$ and $z = \cos \gamma + i \sin \gamma$, then xyz is equal to

[MP PET]

(a) 1, but not -1

(b) -1, but not 1

(c) $+1$ or -1

(d) 0

Q 54. Which of the following is a fourth root of

$\frac{1}{2}+i\frac{\sqrt{3}}{2}$?

[Guj.CET]

(a) $\text{cis}\frac{\pi}{12}$

(b) $\text{cis}\frac{\pi}{2}$

(c) $\text{cis}\frac{\pi}{6}$

(d) $\text{cis}\frac{\pi}{3}$

Q 55. The product of cube roots of -1 is equal to

[DCE]

(a) -1

(b) 0

(c) -2

(d) 4

Q 56. The value of $(2-\omega)(2-\omega^2)(2-\omega^{10})(2-\omega^{11})$, where ω is the complex cube root of unity, is

[Kerala CEE]

(a) 49

(b) 50

(c) 48

(d) 47

(e) 64

Q 57. If $z_r = \cos\left(\frac{\pi}{2^r}\right) + i\sin\left(\frac{\pi}{2^r}\right)$, then $z_1 \cdot z_2 \cdot z_3 \ldots\ldots$ upto ∞ equals

[Kerala CEE, EAMCET]

(a) -3

(b) -2

(c) 1

(d) 0

(e) -1

Q 58. The value of the expression $(2 - \omega)(2 - \omega^2) + 2(3 - \omega)(3 - \omega^2) + \ldots\ldots\ldots + (n - 1)(n - \omega)(n - \omega^2)$ where ω is an imaginary cube root of unit is

[OJEE]

(a) $(n-1)n(n^2+3n+4)/4$

(b) $(n-1)n(n^2+3n+4)/2$

(c) $(n+1)n(n^2+3n+4)/2$

(d) None of the above

Q 59. If co is an imaginary cube root of unity, n is a positive integer but not a multiple of 3, then the value of $1+\omega^n+\omega^{2n}$ is

[MPPET]

(a) 3

(b) $\omega + 2$

(c) 0

(d) $\omega^2 + 1$

Q 60. If ω is a complex cube root of unity, then the value of $\omega^{99} + \omega^{100} + \omega^{101}$ is

[Guj.CET]

(a) 1

(b) -1

(c) 3

(d) 0

Q 61. If $x+\frac{1}{x}=2\sin\alpha, y+\frac{1}{y}=2\cos\beta$, then $x^3y^3+\frac{1}{x^3y^3}$ is

[Kerala CEE]

(a) $2\cos 3(\beta-\alpha)$

(b) $2\cos 3(\beta+\alpha)$

(c) $2\sin 3(\beta-\alpha)$

(d) $2 \sin 3(\beta + \alpha)$

(e) $\sin 3 (\beta - \alpha)$

Q 62. $\frac{(\cos\theta + i\sin\theta)^4}{(\sin\theta + i\cos\theta)^5}$ is equal to

[Kerala CEE]

(a) $\cos \theta - i \sin \theta$

(b) $\sin \theta - i \cos \theta$

(c) $\cos 9\theta - i \sin 9\theta$

(c) $\sin 9\theta - i \cos 9\theta$

(e) $\cos \theta + i \sin \theta$

Q 63. If ω is a complex cube root of unity, then

$225 + (3\omega + 8\omega^2)^2 + (3\omega^2 + 8\omega)^2$ is equal to

[EAMCET]

(a) 72

(b) 192

(c) 200

(d) 248

Q 64. If n is a positive integer, then $(1 + i)^n + (1 - i)^n$ is equal to

[OJEE]

(a) $(\sqrt{2})^{n-2} \cos\left(\frac{n\pi}{4}\right)$

(b) $(\sqrt{2})^{n-2} \sin\left(\frac{n\pi}{4}\right)$

(c) $(\sqrt{2})^{n+2} \cos\left(\frac{n\pi}{4}\right)$

(d) $(\sqrt{2})^{n+2} \sin\left(\frac{n\pi}{4}\right)$

Topic 4 Applications in Geometry

Q 1. Let z_1 and z_2 be two distinct complex numbers and let $z = (1 - t) z_1 + tz_2$ for some real number t with $0 < t < 1$. If arg (w) denotes the principal argument of a non−zero complex number w, then

(a) $|z - z_1| + |z - z_2| = |z_1 - z_2|$

(b) $\arg(z - z_1) = \arg(z - z_2)$

[IIT JEE]

(c) $\begin{vmatrix} z - z_1 & \bar{z} - \bar{z}_1 \\ z_2 - z_1 & \bar{z}_2 - \bar{z}_1 \end{vmatrix} = 0$

(d) arg $(z - z_1)$ = arg $(z_2 - z_1)$

Q 2. Match the statements of Column I with these in Column II.

[IIT JEE]

[Note: Here z takes values in the complex plane and Im (z)and Re (z) denote respectively, the imaginary part and the real part of z]

Column I	**Column II**
(A) The set of points z satisfying $\|z - i\|z\|\| = \|z + i\|z\|\|$ is contained in or equal to	(p) an ellipse with eccentricity $4/5$
(B) The set of points z satisfying $\|z + 4\| + \|z - 4\| = 0$ is contained in or equal to	(q) the set of points z satisfying Im $(z) = 0$
(C) If $\|w\| = 2$, then the set of	(r) the set of points z

points $z = w - \frac{1}{w}$ is	satisfying | Im z | ≤ 1
contained in or equal to	
(D) If | w | = 1, then the set of	(s) the set of points
points $z \neq w + \frac{1}{w}$ is	satisfying | Re z | ≤ 2
contained in or equal to	
	(t) the set of points z
	satisfying | z | ≤ 3

Q 3. Let z = x + iy be a complex number where x and y are integers. Then the area of the rectangle whose vertices are the roots of the equation $z\bar{z}^3 + \bar{z}z^3 = 350$ is

[IIT JEE]

(a) 48

(b) 32

(c) 40

(d) 80

Q 4. If P is the point in the Agrand diagram corresponding to the complex number $\sqrt{3}+i$ and if OPQ is an isosceles right angled triangle, right angled at 'O', then Q represents the complex number

[KCET]

(a) $-1+i\sqrt{3}$ or $1-i\sqrt{3}$

(b) $1\pm i\sqrt{3}$

(c) $\sqrt{3} - i$ or $1 - i\sqrt{3}$

(d) $-1 \pm i\sqrt{3}$

Q 5. The centre of a regular hexagon is at the point z = i. If one of its vertices is at 2 + i, then the adjacent vertices of 2 + i are at the points

[Kerala CEE]

(a) $1 \pm 2i$

(b) $i + 1 \pm \sqrt{3}$

(c) $2 + i(1 \pm \sqrt{3})$

(d) $1+i(1\pm\sqrt{3})$

(e) $1-i(1\pm\sqrt{3})$

Q 6. The set $C=\{z: z\bar{z}+\bar{a}z+az+b=0$, b $\in$ R and b < | a $|^2$ } is

[AMU]

(a) a finite set

(b) an infinite set

(c) an empty set

(d) None of these

Q 7. The equation $\bar{z} = \bar{a} + \frac{r^2}{(z-a)}, r > 0$ represents

[BCECE]

(a) an ellipse

(b) a parabola

(c) a circle

(d) a straight line through point $\bar{a}$

Q 8. If the area of triangle on the argand plane formed by the complex numbers − z, iz, z − iz is 600 sq unit, then | z | is equal to

[VITEEE]

(a) 10

(b) 20

(c) 30

(d) 40

Q 9. Let z_1, z_2 and z_3 be the affixes of the vertices of a triangle having the circumcentre at the origin. If z is the affix of its orthocentre, then z is equal to

[UP SEE]

(a) $\frac{z_1 + z_2 + z_3}{3}$

(b) $\frac{z_1 + z_2 + z_3}{2}$

(c) $z_1 + z_2 + z_3$

(d) None of these

Q 10. The equation of the locus of z such that $\left|\frac{z-i}{z+i}\right| = 2$, where z = x + iy is a complex number, is

[BCECE]

(a) $3x^2 + 3y^2 + 10y - 3 = 0$

(b) $3x^2 + 3y^2 + 10y + 3 = 0$

(c) $3x^2 - 3y^2 - 10y - 3 = 0$

(d) $x^2 + y^2 - 5y + 3 = 0$

Q 11. The points representing complex number z for which $|z-3| = |z-5|$ lie on the locus given by

[Manipal]

(a) an ellipse

(b) a circle

(c) a straight line

(d) None of these

Q 12. For three complex numbers $1 - i$, i, $1 + i$ which of the following is true?

[MPPET]

(a) They form a right triangle

(b) They are collinear

(c) They form an equilateral triangle

(d) They form an isosceles triangle

Q 13. The points representing the complex numbers z, for which $|z - a|^2 + |z + a|^2 = b^2$ lie on

[RPET]

(a) a straight line

(b) a circle

(c) a parabola

(d) a hyperbola

Q 14. If $|z + 4| \leq 3$, then the maximum value of $|z + 1|$ is

[AIEEE]

(a) 4

(b) 10

(c) 6

(d) 0

Q 15. If $\left|\frac{z+i}{z-i}\right| = \sqrt{3}$, then radius of the circle is

[DCE]

(a) $\frac{2}{\sqrt{21}}$

(b) $\frac{1}{\sqrt{21}}$

(c) $\sqrt{3}$

(d) $\sqrt{21}$

Q 16. For all complex numbers z_1, z_2 satisfying $|z_1| = 12$ and $|z_2 - 3 - 4i| = 5$, the minimum value of $|z_1 - z_2|$ is

[BITSAT]

(a) 4

(b) 3

(c) 1

(d) 2

Q 17. Let z_1 and z_2 be the roots of the equation $z^2 + pz + q = 0$ where p, q are real. The points represented by z_1, z_2 and the origin form an equilateral triangle, if

[Kerala CEE]

(a) $p^2 = 3q$

(b) $p^2 > 3q$

(c) $p^2 < 3q$

(d) $p^2 = 2q$

(e) $p = 3q$

Q 18. If the complex numbers z_1, z_2, z_3 are in AP, then they lie on

[AMU]

(a) a circle

(b) a parabola

(c) a straight line

(d) an ellipse

Q 19. The locus of the point z = x + iy satisfying

$\left|\frac{z-2i}{z+2i}\right| = 1$ is

[EAMCET]

(a) x−axis

(b) y−axis

(c) $y = 2$

(d) $x = 2$

Q 20. The region of the Argand diagram defined by $|z-1|+|z+1| \leq 4$ is

[OJEE]

(a) interior of an ellipse

(b) exterior of a circle

(c) interior and boundary of an ellipse

(d) None of the above

Q 21. The equation $z\bar{z} + a\bar{z} + \bar{a}z + b = 0$, $b \in R$

represents a circle, if

[WB JEE]

(a) $|a|^2 = b$

(b) $|a|^2 > b$

(c) $|a|^2 < b$

(d) None of these

Q 22. The locus of the points z which satisfy the condition $\arg\left(\frac{z-1}{z+1}\right)=\frac{\pi}{3}$ is

[WB JEE]

(a) a straight line

(b) a circle

(c) a parabola

(d) None of these

Q 23. If $\text{Im}\left(\frac{z-1}{2z+1}\right)=-4$, then locus of z is

[BCECE]

(a) an ellipse

(b) a parabola

(c) a straight line

(d) a circle

Q 24. If $|z^2 - 1| = |z|^2 + 1$, then z lies on

[Jamia Millia Islamia]

(a) the real axis

(b) the imaginary axis

(c) a circle

(d) an ellipse

Q 25. If $w = \frac{z}{z - \frac{1}{3}i}$ and $|w| = 1$, then z lies on

[MPPET]

(a) a parabola

(b) a straight line

(c) a circle

(d) an ellipse

Q 26. The point $(4, 1)$ undergoes the following three transformations successively

(i) reflection about the line y = x

(ii) translation through a distance of 2 unit along the positive direction of x−axis

(iii) rotation through an angle of $\frac{\pi}{4}$ about the origin in the anti−clockwise direction The final position of the point is

[Kerala CEE]

(a) $\left(\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right)$

(b) $\left(-\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right)$

(c) $(-\sqrt{2}, 7\sqrt{2})$

(d) $(\sqrt{2}, 7\sqrt{2})$

(e) $(\sqrt{2}, -7\sqrt{2})$

Q 27. The locus of the point z = x + iy satisfying the equation $\left|\frac{z-1}{z+1}\right| = 1$ is given by

[EAMCET]

(a) x = 0

(b) y = 0

(c) x = y

(d) x + y = 0

Q 28. The radius of the circle $\left|\frac{z-i}{z+i}\right| = 5$ is given by

[Jamia Millia Islamia]

(a) $\frac{13}{12}$

(b) $\frac{5}{12}$

(c) 5

(d) 625

Q 29. If z = x + iy, then area of the triangle whose vertices are points z, iz, z + iz is

[RPET]

(a) $\frac{1}{2}|z|^2$

(b) $\frac{1}{4}|z|^2$

(c) $|z|^2$

(d) $\frac{3}{2}|z|^2$

Q 30. If arg(z − a) $= \frac{\pi}{4}$, where a $\in$ R, then the locus of z $\in$ C is a

[RPET]

(a) hyperbola

(b) parabola

(c) ellipse

(d) straight line

Q 31. If $S = \left\{ z \in C : \arg\left(\frac{z-2}{z+2} \right) = \frac{\pi}{3} \right\}$ then S is

[J&KCET]

(a) an ellipse

(b) a straight line

(c) a circle

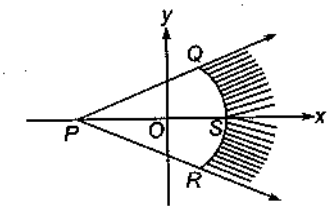
(d) a parabola

Q 32. The shaded region, where

$P \equiv (-1, 0),\ Q \equiv (-1+\sqrt{2}, \sqrt{2})$

$R \equiv (-1+\sqrt{2}, -\sqrt{2})$, $S \equiv (1, 0)$ is represented by

[IIT JEE]



(a) $|z+1|>2, |\arg(z+1)|<\frac{\pi}{4}$

(b) $|z+1|<2, |\arg(z+1)|<\frac{\pi}{2}$

(c) $|z-1|>2, |\arg(z+1)|>\frac{\pi}{4}$

(d) $|z-1|<2, |\arg(z+1)|>\frac{\pi}{2}$

Q 33. If the complex numbers z_1, z_2, z_3 satisfying $\frac{z_1 - z_3}{z_2 - z_3} = \frac{1 - i\sqrt{3}}{2}$, then triangle is

[DCE]

(a) an equilateral triangle

(b) a right angled triangle

(c) a acute angled triangle

(d) an obtuse angled isosceles triangle

Q 34. If magnitude of a complex number $4 - 3i$ is tripled and is rotated anti−clockwise by an angle π, then resulting complex number would be

[DCE]

(a) $-12 + 9i$

(b) $12 + 9i$

(c) $7 - 6i$

(d) $7 + 6i$

Q 35. The points z_1, z_2, z_3, z_4 in the complex plane are the vertices of a parallelogram taken in order, iff

[BCECE]

(a) $z_1 + z_4 = z_2 + z_3$

(b) $z_1 + z_3 = z_2 + z_4$

(c) $z_1 + z_2 = z_3 + z_4$

(d) None of these

Q 36. If $|z + 8| + |z - 8| = 16$, where z is a complex number, then the point z will lie on

[J&K CET]

(a) a circle

(b) an ellipse

(c) a straight line

(d) None of these

Q 37. The locus of point z satisfying $\mathrm{Re}\left(\frac{1}{z}\right)=k$,

where k is a non–zero real number, is

[Kerala CEE]

(a) a straight line

(b) a circle

(c) an ellipse

(d) a hyperbola

(e) None of these

Q 38. If z is a complex number in the Argand plane, then the equation $|z - 2|^2 + |z + 2|^2 = 8$ represents

[OJEE, BCECE]

(a) a parabola

(b) an ellipse

(c) a hyperbola

(d) a circle

Q 39. If $z_1 = 1 + 2i$, $z_2 = 2 + 3i$, $z_3 = 3 + 4i$, then

z_1, z_2, z_3 represents the vertices of a/an

[OJEE]

(a) equilateral triangle

(b) isosceles triangle

(c) right angled triangle

(d) None of these

Q 40. The centre and the radius of the circle

$z\bar{z} + (2 + 3i)\bar{z} + (2 - 3i)z + 12 = 0$ are respectively

[RPET]

(a) $-(2 + 30,(1)$

(b)$(3 + 2i),(1)$

(c) $(3 + 6i),(3)$

(d) None of these

Q 41. The locus represented by the equation $|z - 1| = |z - i|$ is

[J&KCET]

(a) a circle of radius 1

(b) an ellipse with foci at 1 and −i

(c) a line through the origin

(d) a circle on the line joining 1 and −i as diameter

Q 42. The complex number z = x + iy which satisfy the equation $\left|\frac{z-5i}{z+5i}\right| = 1$ lies on

[DCE]

(a) the axis of x

(b) the straight line y = 5

(c) the circle passing through the origin

(d) None of the above

Q 43. If z_1, z_2 and z_3 are any three complex numbers, then the fourth vertex of the parallelogram whose three vertices are z_1, z_2 and z_3 taken in order is

[Kerala CEE]

(a) $z_1 - z_2 + z_3$

(b) $z_1 + z_2 + z_3$

(c) $\frac{1}{3}(z_1 - z_2 + z_3)$

(d) $\frac{1}{3}(z_1 + z_2 - z_3)$

(e) $\frac{1}{3}(z_1 - z_2 - z_3)$

Q 44. A point P which represents a complex number z moves such that $|z - z_1| = |z - z_2|$, then its locus is

(a) a circle with centre z_1

(b) a circle with centre z_2

[Kerala CEE]

(c) a circle with centre z

(d) an ellipse

(e) perpendicular bisector of line joining z_1 and z_2

Q 45. If P, Q, R, S are represented by the complex numbers $4 + i$, $1 + 6i$, $-4 + 3i$, $-1 - 2i$ respectively, then PQRS is a

[OJEE]

(a) a rectangle

(b) a square

(c) a rhombus

(d) a parallelogram

Answers

Topic 1: Algebraic Operations and Equality of Complex Numbers

1. (c) 2, (a) 3. (d) 4. (e) 5. (a) 6. (a) 7. (c) 8. (c)

9. (d) 10. (b)

11. (d) 12. (d) 13. (d) 14. (d) 15. (c) 16. (a) 17.

(b) 18. (d) 19. (b) 20. (b)

21. (b) 22. (a) 23. (d) 24. (c) 25. (a) 26. (b)

Topic 2 : Conjugate, Modulus and Argument

1. (b) 2. (d) 3. (a) 4. (e) 5. (c) 6. (c) 7, (b) 8. (b)

9. (b) 10. (e)

11. (c) 12. (c) 13. (b) 14. (b) 15. (c) 16. (c) 17.

(d) 18. (c) 19. (c) 20. (b)

21. (b) 22. (c) 23. (c) 24. (a) 25. (b) 26. (c) 27.

(a) 28. (c) 29. (c) 30. (b).

31. (d) 32. (b) 33. (b) 34. (d) 35. (b) 36. (a) 37.

(b) 38. (d) 39. (c) 40. (b)

41. (b) 42. (d) 43. (b) 44. (c) 45. (d) 46. (c) 47.

(b) 48. (a) 49. (a) 50. (a)

51. (c) 52. (d) 53. (d) 54. (b) 55. (d) 56. (a) 57.

(b) 58. (a) 59. (a) 60. (d)

61. (a) 62. (a) 63. (d) 64. (c) 65. (b)

Topic 3 : De–Moivre's Theorem, Square Root and Roots of Unity

1. (d) 2. (c) 3. (c) 4. (a) 5. (c) 6. (d) 7. (c) 8. (b)

9. (b) 10. (b)

11. (b) 12. (b) 13. (c) 14. (b) 15. (a) 16. (a) 17.

(d) 18. (d) 19. (b) 20. (d)

21. (a) 22. (b) 23. (c) 24. (c) 25. (a) 26. (b) 27.

(e) 28. (e) 29. (a) 30. (c)

31. (b) 32. (c) 33. (b) 34. (c) 35. (d) 36. (b) 37.

(d) 38. (c) 39. (e) 40. (c)

41. (c) 42. (d) 43. (c) 44. (c) 45. (b) 46. (c) 47.

(d) 48. (d) 49. (d) 50. (c)

51. (a) 52. (d) 53. (c) 54. (a) 55. (a) 56. (a) 57.

(e) 58. (a) 59. (c) 60. (d)

61. (c) 62. (d) 63. (d) 64. (c)

Topic 4 : Applications in Geometry

1. (a,c,d) 2. (*) 3. (a) 4. (a) 5. (d) 6. (b) 7: (c) 8.

(b) 9. (c) 10. (b)

11. (c) 12. (d) 13. (b) 14. (c) 15. (c) 16. (d) 17.

(a) 18. (c) 19. (a) 20. (c)

21. (b) 22. (b) 23. (d) 24. (b) 25. (b) 26. (b) 27.

(a) 28. (b) 29. (a) 30. (d)

31. (c) 32. (a) 33. (a) 34. (a) 35. (b) 36. (c) 37.

(b) 38. (d) 39. (d). 40. (a)

41. (c) 42. (a) 43, (a) 44. (e) 45. (b)

Sequences and Series

Topic 1

Arithmetic Progression and its Mean

Q 1. A person is to count 45000 currency notes.

Let a_n denotes the number of notes he counts in the nth minute. If $a_1 = a_2 = \ldots = 10_{10} = 150$ and a_{10}, $a_{11}, \ldots$ are in AP with common difference -2, then the time taken by him to count all notes, is

[AIEEE]

(a) 24 min

(b) 34 min

(c) 125 min

(d) 135 min

Q 2. The sum of all two digit natural numbers which leave a remainder 5 when they are divided by 7 is equal to

[Kerala CEE]

(a) 715

(b) 702

(c) 615

(d) 602

(e) 589

Q 3. The arithmetic mean of 7 consecutive integers starting with a is m. Then, the arithmetic mean of 11 consecutive integers starting with a + 2 is

[VITEEE]

(a) 2a

(b) $2m$

(c) $a + 4$

(d) $m + 4$

(e) $a + m+2$

Q 4. An AP consists of 23 terms. If the sum of the three terms in the middle is 141 and the sum of the last three terms is 261, then the first term is

[Kerala CEE]

(a) 6

(b) 5

(c) 4

(d) 3

(e) 2

Q 5. If the sum to first n terms of the AP $2, 4, 6,$ … is 240, then the value of n is

[BITSAT]

(a) 14

(b) 15

(c) 16

(d) 17

(e) 18

Q 6. If the sum of first n terms of an AP is cn^2,

then the sum of squares of these n terms is

[IITJEE]

(a) $\frac{n(4n^2-1)c^2}{6}$ (b) $\frac{n(4n^2+1)c^2}{3}$ (c) $\frac{n(4n^2-1)c^2}{3}$ (d)

$\frac{n(4n^2+1)c^2}{6}$

Q 7. The sum of all odd numbers between 1 and

1000 which are divisible by 3, is

[UP SEE]

(a) 83667

(b) 90000

(c) 83660

(d) None of the above

Q 8. A student read common difference of an AP as -3 instead of 3 and obtained the sum of first 10 terms as -30. Then, the actual sum of first 10 terms is equal to

[Kerala CEE]

(a) 240

(b) 120

(c) 300

(d) 180

(e) 480

Q 9. If $a_1, a_2, \ldots, a_n$ are in AP with common difference $d \neq 0$, then $(\sin d)$ [sec a_1 sec $- a_2$ + sec a_2 sec a_1 +.... + seca_{n-1} sec a_n]is equal to

[Kerala CEE]

(a) cot a_n − cot a_1

(b) cot a_1 − cot a_n

(c) tan a_n −tan a_1

(d) $\tan a_1 - \tan a_{n-1}$

(e) $\tan a_1 - \tan a_n$

Q 10. If $a_1, a_2, a_3, \ldots, a_n$ be an AP of non-zero terms, then $\frac{1}{a_1a_2} + \frac{1}{a_2a_3} + \ldots\ldots + \frac{1}{a_{n-1}a_n}$ is equal to

[AMU]

(a) $\frac{n-1}{a_1a_n}$

(b) $\frac{n}{a_1a_n}$

(c) $\frac{n+1}{a_1a_n}$

(d) None of these

Q 11. The arithmetic mean of first n odd natural number is

[OJEE]

(a) n^2

(b) $2n$

(c) n

(d) $3n$

Q 12. If three positive real numbers a,b,c are in AP and abc = 4, then the minimum possible value of b is

(a) $2^{3/2}$

(b) $2^{2/3}$

(c) $2^{1/3}$

(d) $2^{5/2}$

[WBJEE]

Q 13. If $a_1, a_2, \ldots, a_n$ are in arithmetic progression, where $a_i > 0$ for all i.

Then, $\frac{1}{\sqrt{a_1}+\sqrt{a_2}}+\frac{1}{\sqrt{a_2}+\sqrt{a_3}}+\ldots..+\frac{1}{\sqrt{a_{n-1}}+\sqrt{a_n}}$

is equal to

[Jamia Millia Islamia]

(a) $\frac{n^2(n+1)}{2}$

(b) $\frac{n-1}{\sqrt{a_1}+\sqrt{a_n}}$

(c) $\frac{n(n-1)}{2}$

(d) None of these

Q 14. If $\frac{5+9+13+.....+\text{n terms}}{7+9+11+.....+\text{n terms}}=\frac{5}{12}$, then n is equal to

[MPPET]

(a) 5

(b) 6

(c) 9

(d) 12

Q 15. If the first, second and last terms of an arithmetic series are a, b and c respectively, then the number of terms is

[MP PET]

(a) $\frac{b+c-2a}{b-a}$

(b) $\frac{b+c+2a}{b-a}$

(c) $\frac{b+c-2a}{b+a}$

(d) $\frac{b+c+2a}{b+a}$

Q 16. If sum of n terms of an AP is $2n + 3n^2$, then r th term is

[RPET]

(a) $2r + 3r^2$

(b) $3r^2 - 4r + 1$

(c) $6r - 1$

(d) $4r + l$

Q 17. If the sum to $2n$ terms of the AP

$2,5,8,11,\ldots$ is equal to the sum to n terms of the

AP $57, 59, 61, 63,\ldots$, then n is equal to

[J&KCET]

(a) 10

(b) 11

(c) 12

(d) 13

Q 18. If the sum of 12th and 22nd terms of an AP is 100, then the sum of the first 33 terms of the AP is

[Kerala CEE]

(a) 1700

(b) 1650

(c) 3300

(d) 3400

(e) 3500

Q 19. Let a,b,c be in AP. If $0 < a, b, c < 1$, x = $\sum_{n=0}^{\infty} a^n$, $y = \sum_{n=0}^{\infty} b^n$ and $z = \sum_{n=0}^{\infty} c^n$, then

[Kerala CEE]

(a) $2y = x'+z$

(b) $2x = y+z$

(c) $2z = x + y$

(d) $2xz = xy + yz$

(e) $z = \frac{2xy}{x+y}$

Q 20. If a, b, c be in arithmetic progression,

then the value c $(a+2b-c)(2b+c-a)(a+2b+c)$ is

[WBJEE]

(a) $16abc$

(b) $4abc$

(c) $8\ abc$

(d) $3abc$

Q 21. If pth term of an arithmetic progression is

q and the qth term is p, then 10th term is

[J&K CET]

(a) p − q + 10

(b) p + q+ 11

(c) p + q − 9

(d) p + q − 10

Q 22. The sets $S_1, S_2, S_3, \ldots$ are given by

$$S_1 = \left\{\frac{2}{1}\right\}, S_2 = \left\{\frac{3}{2}, \frac{5}{2}\right\}, S_3 = \left\{\frac{4}{3}, \frac{7}{3}, \frac{10}{3}\right\}, S_4 = \left\{\frac{5}{4}, \frac{9}{4}, \frac{13}{4}, \frac{17}{4}\right\}.$$

Then, the sum of the numbers in the set S_{25} is

[Guj. CET]

(a) 320

(b) 322

(c) 324

(d) 326

Q 23. If a^2, b^2, c^2 are in AP, then which of the following is| also an AP?

[DCE]

(a) sin A, sin B, sin C

(b) tan A, tan B, tan C

(c) cot A, cot B, cot C

(d) None of these

Q 24. If $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ are in AP, then

[Kerala CEE]

(a) a, b, c are in AP

(b) c, a, b are in AP

(c) a^2, b^2, c^2 are in AP

(d) a, b, c are in GP

(e) c^2, a^2, b^2 are in AP

Q 25. If the arithmetic mean of a and b is $\frac{a^n + b^n}{a^{n-1} + b^{n-1}}$, then the value of n is

[WBJEE]

(a) -1

(b) 0

(c) 1

(d) None of these

Q 26. Let T_r be the rth term of an AP whose first term is a and common difference is d. If for some positive integers m, n, $m \neq n$, $T_m = \frac{1}{n}$ and $T_n = \frac{1}{m}$, then a − d equals

[Jamia Millia Islamia]

(a) 0

(b) 1

(c) $\frac{1}{mn}$

(d) $\frac{1}{m}+\frac{1}{n}$

Q 27. Let $a_1, a_2, a_3, \ldots$ be terms of an AP. If

$\frac{a_1+a_2+.....+a_p}{a_1+a_2+.....+a_q}=\frac{p^2}{q^2}$, $p \neq q$, then $\frac{a_6}{a_{21}}$ equals

[AIEEE]

(a) $\frac{7}{2}$

(b) $\frac{2}{7}$

(c) $\frac{11}{41}$

(d) $\frac{41}{11}$

Q 28. The sum of n terms of two arithmetic series are in the ratio $2n + 3 : 6n+5$, then the ratio of their 13th terms is

[UP SEE]

(a) $53 : 155$

(b) 27 : 87

(c) 29:83

(d) 31:89

Q 29. In an arithmetic progression, the 24th term is 100. Then, the sum of the first 47 terms of the arithmetic progression is

[Kerala CEE]

(a) 2300

(b) 2350

(c) 2400

(d) 4600

(e) 4700

Q 30. If $1, \log_3\sqrt{(3^{1-x}+2)}, \log_3(4.3^x-1)$ are in AP, then x equals

[OJEE]

(a) $\log_3 4$

(b) $1 - \log_3 4$

(c) $1 - \log_4 3$

(d) $\log_4 3$

Q 31. Three numbers are in AP such that their sum is 18 and sum of their squares is 158. The greatest number among them is

[MPPET]

(a) 10

(b) 11

(c) 12

(d) None of these

Q 32. The sum of the integers from 1 to 100 which are divisible by 3 and 5, is

[RPET]

(a) 2317

(b) 2632

(c) 315

(d) 2489

Q 33. If x, y, z are in AP, then

$\frac{1}{\sqrt{x}+\sqrt{y}}, \frac{1}{\sqrt{z}+\sqrt{x}}, \frac{1}{\sqrt{y}+\sqrt{z}}$ are in

[Kerala CEE]

(a) AP

(b) GP

(c) HP

(d) AP and HP

(e) AP and GP

Q 34. If a, b and c are in AP, then which one of the following is not true?

[MPPET]

(a) $\frac{k}{a}, \frac{k}{b}$ and $\frac{k}{c}$ are in HP

(b) a + k, b + k and c + k are in AP

(c) ka, kb and kc are in AP

(d) a^2, b^2 and c^2 are in AP

Q 35. If twice the 11th term of an AP is equal to 7 times its 21st term, then its 25th term is equal to

[J&K CET]

(a) 24

(b) 120

(c) 0

(d) None of these

Q 36. 7th term of an AP is 40. Then, the sum of first 13 terms is

[Guj. CET]

(a) 520

(b) 53

(c) 2080

(d) 1040

Q 37. Let f(x) be a polynomial function of second degree. If f(1) = f(−l) and a,b,c are in AP, then f'(a),f'(b) and f'(c)are in .

[DCE]

(a) AP

(b) GP

(c) HP

(d) Arithmetico−Geometric Progression

Q 38. If a_1, a_2, a_3, a_4, a_5 and a_6 are six arithmetic means between 3 and 31, then $a_6 - a_5$ and $a_1 + a_6$ are respectively equals to

[Kerala CEE]

(a) 5 and 34

(b) 4 and 35

(c) 4 and 34

(d) 4 and 36

(e) 6 and 36

Q 39. If $a_1, a_2, a_3, \ldots, a_n$ are the n arithmetic means between a and b, then $2\sum_{i=1}^{n} a_i$ equals

[Kerala CEE]

(a) ab

(b) n(a + b)

(c) nab

(d) $\frac{(a+b)}{n}$

(e) $\frac{b(a+b)}{ab}$

Q 40. If $S_n = nP + \frac{1}{2}n(n-1)Q$, where S_n denotes the sum of the first n terms of an AP, then the common difference is

[AMU]

(a) P + Q

(b) $2P + 3Q$

(c) $2Q$

(d) Q

Q 41. The sum of the first and third term of an arithmetic series is 12 and the product of first and second term is 24, then first term is

[MPPET]

(a) 1

(b) 8

(c) 4

(d) 6

Q 42. If $2/3$, it, $5/8$ are in AP, then value of k is

[RPET]

(a) 15

(b) 21

(c) 12

(d) 31/48

Q 43. In an AP the sum of any two terms, such that the distance of one of them from the beginning is same as that of the other from the end, is

[J&KCET]

(a) first term

(b) sum of first and last terms

(c) last term

(d) half of the sum of the series

Topic 2 Geometric Progression and Its Mean

Q 1. The value of n for which $\frac{x^{n+1}+y^{n+1}}{x^n+y^n}$ is the geometric mean of x and y is

[WBJEE]

(a) $n=-\frac{1}{2}$

(b) $n=\frac{1}{2}$

(c) n= 1

(d) n = -1

Q 2. If sum of an infinite geometric series is $\frac{4}{3}$ and its 1st term is $\frac{3}{4}$, then its common ratio is

[BITSAT]

(a) $\frac{7}{16}$

(b) $\frac{9}{16}$

(c) $\frac{1}{9}$

(d) $\frac{7}{9}$

Q 3. If $1 + \sin x + \sin^2 x + \ldots \text{upto } \infty = 4 + 2\sqrt{3}$, $0 < x < \pi$ and $x \neq \frac{\pi}{2}$, then x is equal to

[KCET]

(a) $\frac{\pi}{3}, \frac{5\pi}{6}$

(b) $\frac{2\pi}{3}, \frac{\pi}{6}$

(c) $\frac{\pi}{3}, \frac{2\pi}{3}$

(d) $\frac{\pi}{6}, \frac{\pi}{3}$

Q 4. The value of $\left[(0.16)^{\log 0.25\left(\frac{1}{3}+\frac{1}{3^2}+\frac{1}{3^3}+\ldots\ldots\infty\right)}\right]^{1/2}$ is

[AMU]

(a) 1

(b) -1

(c) 0

(d) None of these

Q 5. If G is the GM of the product of r set of observation with geometric means G_1, G_2,...,G_r respectively, then G is equal to

[Jamia Millia Islamia]

(a) log G_1 + log G_2 +... + log G_n

(b) G_1,G_2,....G_r

(c) logG_1, logG_2,....,logG_n .

(d) None of the above

Q 6. The first two terms of a geometric progression add upto 12. The sum of the third and the fourth terms is 48. If terms of the geometric progression are alternately positive and negative, then the first term is

[AIEEE]

(a) 4

(b) −4

(c) −12

(d) 12

Q 7. If S be the sum, P be the product and R be the sum of the reciprocals of n terms of a GP, then P^2 is equal to

(a) $\left(\frac{S}{R}\right)^n$

(b) $\frac{S}{R}$

(c) $\left(\frac{R}{S}\right)^n$

(d) $\frac{R}{S}$

[UP SEE]

Q 8. If sum of the series $\sum_{n=0}^{\infty} r^n = S$ for $|r| < 1$, then sum of the series $\sum_{n=0}^{\infty} r^{2n}$, is **[UP SEE]**

(a) S^2

(b) $\frac{S^2}{2S+1}$

(c) $\frac{2S}{S^2-1}$

(d) $\frac{S^2}{2S-1}$

Q 9. Geometric mean of $7, 7^2, 7^3, \ldots, 7^n$ is

[OJEE]

(a) $7^{\frac{n+1}{2}}$

(b) 7

(c) $7^{n/2}$

(d) 7^{n}

Q 10. The 5th term of the series $\frac{10}{9}, \frac{1}{3}\sqrt{\frac{20}{3}}, \frac{2}{3}, \ldots\ldots$ is

[WBJEE]

(a) $\frac{1}{3}$

(b) 1

(c) $\frac{2}{5}$

(d) $\sqrt{\frac{2}{3}}$

Q 11. The value of 2.357 is

[Manipal]

(a) $\frac{2355}{999}$

(b) $\frac{2355}{1000}$

(c) $\frac{2355}{1111}$

(d) None of these

Q 12. If the (p + q) th term of a geometric series is m and the (p − q)th term is n, then the pth term is

[MPPET]

(a) $(mn)^{1/2}$

(b) mn

(c) m + n

(d) m− n

Q 13. In a geometric progression consisting of positive terms, each term equals the sum of the next two terms. Then, the common ratio of this progression equals

[AIEEE]

(a) $\frac{1}{2}(1-\sqrt{5})$

(b) $\frac{1}{2}\sqrt{5}$

(c) $\sqrt{5}$

(d) $\frac{1}{2}(\sqrt{5}-1)$

Q 14. If an infinite geometric series the first term is a and common ratio is r. If the sum of the series is 4 and the second term is $3/4$, then (a, r) is

[Kerala CEE]

(a) $(4/7,3/7)$

(b) $(2,3/8)$

(c) $(3/2,1/2)$

(d) $(3,1/4)$

(e) $(4, 3/4)$

Q 15. In a geometric progression (GP) the ratio of the sum of the first three terms and first six terms is $125 : 152$ the common ratio is

[J&K CET]

(a) $\frac{1}{5}$

(b) $\frac{2}{5}$

(c) $\frac{4}{5}$

(d) $\frac{3}{5}$

Q 16. If $x = 1 + a + a^2 + \ldots\infty$ and $y = 1 + b + b^2 + \ldots\infty$ where a and b are proper fractions, then $1 + ab + a^2b^2 + \ldots\infty$ equals

[Guj. CET]

(a) $\frac{xy}{y+x-1}$

(b) $\frac{x+y}{x-y}$

(c) $\frac{x^2+y^2}{x-y}$

(d) None of these

Q 17. If a,b,c,d and p are distinct real number such that $(a^2+b^2+c^2)p^2 - 2(ab + bc + cd)p + (b^2+c^2+d^2) \leq 0$, then a, b, c, d

[DCE]

(a) are in AP

(b) are in GP

(c) are in HP

(d) satisfy ab = cd

Q 18. The value of $0.0\overline{37}$, where $0.0\overline{37}$ stands for the number $0.0373737\ldots\ldots$ ...,is

[UP SEE]

(a) $37/1000$

(b) $37/990$

(c) $1/37$

(d) $1/27$

Q 19. If $a_1, a_2, \ldots, a_{50}$ are in GP, then

$\frac{a_1 - a_3 + a_5 - .... + a_{49}}{a_2 - a_4 + a_6 - .... + a_{50}}$ is equal to

[Kerala CEE]

(a) 0

(b) 1

(c) $\frac{a_1}{a_2}$

(d) $\frac{a_{25}}{a_{24}}$

(e) $\frac{2a_1}{3a_2}$

Q 20. $0.5737373\ldots$ is equal to

[AMU]

(a) $\frac{284}{497}$

(b) $\frac{284}{495}$

(c) $\frac{568}{999}$

(d) $\frac{567}{990}$

Q 21. Fifth term of an GP is 2, then the product of its 9 term is

[OJEE]

(a) 256

(b) 512

(c) 1024

(d) None of these

Q 22. If $\{a_n\}$ is a sequence with $a_0 = p$ and $a_n - a_{n-1} = ra_{n-1}$ for, $n \geq 1$, then the terms of the sequence are in

[J&K CET]

(a) an arithmetic progression

(b) a geometric progression

(c) a harmonic progression

(d) an arithmetico–geometric progression

Q 23. The sum to infinity of the progression

$9-3+1-\frac{1}{3}+.....$ is

[Guj. CET]

(a) 9

(b) 9/2

(c) 27/4

(d) 15/2

Q 24. The value of $9^{1/3} \times 9^{1/9} \times 9^{1/27} \times \ldots \infty$ is

[UP SEE]

(a) 9

(b) 1

(c) 3

(d) None of these

Q 25. If $0 < \phi < \frac{\pi}{2}$, $x = \sum_{n=0}^{\infty} \cos^{2n} \phi$, $y = \sum_{n=0}^{\infty} \sin^{2n} \phi$ and $z = \sum_{n=0}^{\infty} \cos^{2n} \sin^{2n} \phi$, then

[Kerala CEE]

(a) xyz=xz + y

(b) xyz =xy+z

(c) xyz = x + y + z

(d) $xyz = yz + x$

(e) $xyz = x + yz$

Q 26. $i^2 + i^4 + i^6 + \ldots$ upto $(2k + 1)$ terms, $k \in$ N is

[Kerala CEE]

(a) 0

(b) 1

(c) -1

(d) k

(e) k+1

Q 27. The product $(32)\ (32)^{1/6}\ (32)^{1/36}$ ... to ∞

is

[Kerala CEE]

(a) 16

(b) 32

(c) 64

(d) 0

(e) 62

Q 28. If the third term of a GP is P. Then, the product of the first 5 terms of the GP is

[MP PET]

(a) P^3

(b) P^2

(c) P^{10}

(d) P^5

Q 29. If S is the sum of an infinite GP, the first term a, then the common ratio r is given by

[RPET]

(a) $\frac{a-S}{S}$

(b) $\frac{S-a}{S}$

(c) $\frac{a}{1-S}$

(d) $\frac{S-a}{a}$

Q 30. An infinite GP has first term x and sum 5, then

[IITJEE]

(a) $x < -10$

(b) $-10 < x < 0$

(c) $0 < x < 10$

(d) $x > 10$

Q 31. Three numbers are in GP such that their sum is 38 and their product is 1728. The greatest number among them is

[UP SEE]

(a) 18

(b) 16

(c) 14

(d) None of these

Q 32. Which term of the GP $3, 3\sqrt{3}, 9 \ldots\ldots$ is 2187 ?

[Kerala CEE]

(a) 15

(b) 14

(c) 13

(d) 19

(e) 20

Q 33. If $y = 1 + x+x^2 +x^3 + \ldots$, then x is equal to

[BCECE]

(a) $\frac{y-1}{y}$

(b) $\frac{1-y}{y}$

(c) $\frac{y}{a-y}$

(d) None of these

Q 34. If a,b,c are in AP,then 2^{ax+1}, 2^{bx+1}, 2^{cx+1}, x $\neq 0$ are in

[BCECE]

(a) AP

(b) GP only when $x>0$

(c) GP if $x < 0$

(d) GP

Q 35. If x, y, z are in GP and $a^x = b^y = c^z$, then

[BCECE]

(a) $\log_a c = \log_b a$

(b) $\log_b a = \log_c b$

(c) $\log_c b = \log_a c$

(d) None of these

Q 36. The sum of a GP with common ratio 3 is 364 and last term is 243, then the number of terms is

[MP PET]

(a) 6

(b) 5

(c) 4

(d) 10

Q 37. The first term of a GP is 7, the last term is 448 and sum of all term is 889, then the common ratio is

[RPET]

(a) 5

(b) 4

(c) 3

(d) 2

Q 38. If x, $2x + 2$, $3x + 3$,... are in GP, then the fourth term is

[J&KCET]

(a) 27.5

(b) $4x + 5$

(c) -13.5

(d) $4x + 4$

Topic 3 Harmonic Progression and Its Mean

Q 1. If three real numbers a, b, c are in harmonic progression, then which of the following is true?

[WBJEE]

(a) $\frac{1}{a}, b, \frac{1}{c}$ are in AP

(b) $\frac{1}{bc}, \frac{1}{ca}, \frac{1}{ab}$ are in HP

(c) ab, bc, ca are in HP

(d) $\frac{a}{b}, \frac{b}{c}, \frac{c}{a}$ are in HP

Q 2. If the 7th term of HP is $\frac{1}{10}$ and the 12th term is $\frac{1}{25}$, then the 20th term is

[UP SEE]

(a) $\frac{1}{41}$

(b) $\frac{1}{45}$

(c) $\frac{1}{49}$

(d) $\frac{1}{37}$

Q 3. If x, y, z are in HP, then log (x + z) + log (x − 2y + z) is equal to

[AMU]

(a) log(x − z)

(b) 2log(x − z)

(c) 3 log(x − z)

(d) 4log(x − z)

Q 4. If $\frac{1}{b-a}+\frac{1}{b-c}=\frac{1}{a}+\frac{1}{c}$ and b ≠ a + c, then a, b, c are in

[OJEE]

(a) AP

(b) GP

(c) HP

(d) None of these

Q 5. If H is the harmonic mean between P and Q, then the value of $\frac{H}{P}+\frac{H}{Q}$ is

[VITEEE]

(a) 2

(b) $\frac{PQ}{P+Q}$

(c) $\frac{1}{2}$

(d) $\frac{P+Q}{PQ}$

Q 6. If $a_1, a_2, \ldots, a_n$ are in HP, then the expression

$a_1a_2 + a_2a_3 + \ldots + a_{n-1}a_n$ is equal to

[AIEEE]

(a) $(n - 1)(a_1 - a_n)$

(b) na_1a_n

(c) $(n - 1)a_1a_n$

(d) $n(a_1 - a_n)$

Q 7. If $4a^2 + 9b^2 + 16c^2 = 2(3ab + 6bc + 4ca)$,

where, a,b,c are non−zero numbers, then a, b, care

in

[AMU]

(a) AP

(b) GP

(c) HP

(d) None of these

Q 8. If $\frac{a+b}{2}, b, \frac{b+c}{2}$ are in HP, then a, b, care in

[MPPET]

(a) HP

(b) AP

(c) GP

(d) None of these

Q 9. The 4th term of a HP is $3/5$ and 8th term is

$1/3$, then its 6th term is

[RPET]

(a) 1/6

(b) 3/7

(c) 1/7

(d) 3/5

Topic 4 Relation between AP, GP and HP

Q 1. GM and HM of two numbers are 10 and 8 respectively. The numbers are

[WBJEE]

(a) 5,20

(b) 4,25

(c) 2,50

(d) $1,100$

Q 2. Let a be a positive number such that the arithmetic mean of a and 2 exceeds their geometric mean by 1. Then, the value of a is

[Kerala CEE]

(a) 3

(b) 5

(c) 9

(d) 8

(e) 10

Q 3. If a, a_1,a_2,....., a_{2n}, b are in arithmetic progression and a, g_1, g_2,...., g_{2n}, b are in geometric progression and h is the harmonic mean of a and b, then $\frac{a_1 + a_{2n}}{g_1 g_{2n}} + \frac{a_2 + a_{2n-1}}{g_2 g_{2n-1}} + ... + \frac{a_n + a_{n+1}}{g_n g_{n+1}}$ is equal to

[DCE]

(a) 2nh

(b) $\frac{n}{h}$

(c) nh

(d) $\frac{2n}{h}$

Q 4. If a, b, c are in GP and x, y are arithmetic mean of a, b and b, c respectively, then $\frac{1}{x}+\frac{1}{y}$ is equal to

[Kerala CEE]

(a) $\frac{2}{b}$

(b) $\frac{3}{b}$

(c) $\frac{b}{3}$

(d) $\frac{b}{2}$

(e) $\frac{1}{b}$

Q 5. In a sequence of 21 terms, the first 11 terms are in AP with common difference 2 and the last 11 terms are in GP with common ratio 2. If the middle term of AP be equal to the middle term of the GP, then the middle term of the entire sequence is

[AMU]

(a) $-\frac{10}{31}$

(b) $\frac{10}{31}$

(c) $\frac{32}{31}$

(d) $-\frac{32}{31}$

Q 6. Let a, b, c are in GP and $4a$, $5b$, $4c$ are in AP such that a + b + c = 70, then value of b is

[RPET]

(a) 5

(b) 10

(c) 15

(d) 20

Q 7. If the AM of two numbers be A and GM be G, then the numbers will be

[UP SEE]

(a) $A \pm (A^2 - G^2)$

(b) $\sqrt{A} \pm \sqrt{A^2 - G^2}$

(c) $A \pm \sqrt{(A+G)(A-G)}$

(d) $\frac{A \pm \sqrt{(A+G)(A-G)}}{2}$

Q 8. The HM of two numbers is 4. Their AM is A and GM is G. If $2A + G^2 = 27$, then A is equal to

[Kerala CEE]

(a) 9

(b) $\frac{9}{2}$

(c) 18

(d) 27

(e) 15

Q 9. If a is positive and if A and G are the arithmetic mean and the geometric mean of the roots of $x^2 - 2ax + a^2 = 0$ respectively, then

[Kerala CEE]

(a) A = G

(b) $A = 2G$

(c) $2A = G$

(d) $A^2 = G$

(e) $A = G^2$

Q 10. If $a^{1/x} = b^{1/y} = c^{1/z}$ and a, b, c are in geometrical progression, then x, y, z are in

[MP PET]

(a) AP

(b) GP

(c) HP

(d) None of these

Q 11. The difference between two numbers is 48 and the difference between their arithmetic mean and their geometric mean is 18. Then, the greater of two numbers is

[MPPET]

(a) 96

(b) 60

(c) 54

(d) 49

Q 12. Three numbers whose sum is 15 are in AP. If they are added by $1, 4$ and 19 respectively they are in GP. The numbers are

[RPET]

(a) $2,5,8$

(b) $26,5,-16$

(c) 2, 5, 8 and 26,5,−16

(d) None of these

Q 13. If AM of two numbers is twice of their GM, then the ratio of greatest number to smallest number is

[DCE]

(a) $7-4\sqrt{3}$

(b) $7+4\sqrt{3}$

(c) 21

(d) 5

Q 14. If H_1, H_2 are two harmonic means between two positive numbers a and b ($a \neq b$), A and G are the arithmetic and geometric means between a and b, then $\frac{H_2 + H_1}{H_2 H_1}$ is

[Kerala CEE]

(a) $\frac{A}{G}$

(b) $\frac{2A}{G}$

(c) $\frac{A}{2G^2}$

(d) $\frac{A}{G^2}$

(e) $\frac{2A}{G^2}$

Q 15. Let a,b,c be in AP and $|a|<1$, $|b| < 1$, $|c| <$

1. If $x = 1 + a + a^2 + \ldots$ to ∞, $y = 1 + b + b^2 + \ldots$

to ∞,

$z = 1 + c + c^2 + \ldots \infty$, then x, y, z are in

[AMU]

(a) AP

(b) GP

(c) HP

(d) None of these

Q 16. If a, b, c are in AP, b − a, c − b and a are in GP, then a :b:c is

[EAMCET]

(a) 1:2: 3

(b) 1:3:5

(c) 2 : 3 : 4

(d) 1:2:4

Q 17. If $f(x) = \cos^2 x + \sec^2 x$. Then,

[OJEE]

(a) $f(x) < 1$

(b) $f(x) = 1$

(c) $1 < f(x) < 2$

(d) $f(x) \geq 2$

Q 18. If A_1, A_2; G_1, G_2 and H_1, H_2 be two AM's, GM's and HM's between two quantities, then the value of $\frac{G_1 G_2}{H_1 H_2}$ is

[WBJEE]

(a) $\frac{A_1 + A_2}{H_1 + H_2}$

(b) $\frac{A_1 - A_2}{H_1 + H_2}$

(c) $\frac{A_1 + A_2}{H_1 - H_2}$

(d) $\frac{A_1 - A_2}{H_1 - H_2}$

Q 19. If the AM and GM between two numbers are in the ratio m :n, then the numbers are in the ratio

[Guj.CET]

(a) $m + \sqrt{m^2 + n^2} : m - \sqrt{m^2 + n^2}$

(b) $m + \sqrt{n^2 - m^2} : m - \sqrt{n^2 - m^2}$

(c) $m+\sqrt{m^2-n^2} : m-\sqrt{m^2-n^2}$

(d) None of the above

Q 20. If $y = 3^{x-1} + 3^{-x-1}$ (x real), then the least value of y is

[WBJEE]

(a) 2

(b) 6

(c) $2/3$

(d) None of these

Q 21. If p, q, r are in GP and $\tan^{-1} p$, $\tan^{-1} q$, $\tan^{-1} r$ are in AP, then p, q, r satisfies the relation

[DCE]

(a) $p = q = r$

(b) $p \neq q \neq r$

(c) $p + q = r$

(d) None of these

Q 22. Let α, β, γ and δ are four positive real numbers such that their product is unity, then the least value of $(1 + \alpha)(1 + \beta)(1 + \gamma)(1 + \delta)$ is

[BITSAT]

(a) 6

(b) 16

(c) 0

(d) 32

Q 23. If AM and GM of x and y are in the ratio p: q, then x : y is

[Kerala CEE]

(a) $p-\sqrt{p^2+q^2} : p+\sqrt{p^2+q^2}$

(b) $p+\sqrt{p^2-q^2} : p-\sqrt{p^2-q^2}$

(c) p : q

(d) $p+\sqrt{p^2+q^2} : p-\sqrt{p^2+q^2}$

(e) $q+\sqrt{p^2-q^2} : q-\sqrt{p^2-q^2}$

Q 24. The harmonic mean between two numbers is $14\frac{2}{5}$ and the geometric mean is 24. The greater number between them is

[UP SEE]

(a) 72

(b) 54

(c) 36

(d) None of these

Q 25. The number of divisors of $3 \times 7^3, 7 \times 11^2$ and 2×61 are in

[AMU]

(a) AP

(b) GP

(c) HP

(d) None of these

Q 26. If arithmetic mean of two positive numbers is A, their geometric mean is G and harmonic mean is H, then H is equal to

[MP PET]

(a) G^2/A

(b) A^2/G^2

(c) A/G^2

(d) G/A^2

Q 27. If a,b, c are distinct positive real number and $a^2 + b^2 + c^2 = 1$,then $3\ (a^2b^2c^2)^{1/3}$ is

[RPET]

(a) less than 1

(b) equal to 1

(c) greater than 1

(d) any real number

Q 28. If $\alpha \in \left(0, \frac{\pi}{2}\right)$, then $\sqrt{x^2+x} + \frac{\tan^2 \alpha}{\sqrt{x^2+x}}$ is always greater than or equal to

[IITJEE]

(a) $2 \tan \alpha$

(b) 1

(c) 2

(d) $\sec^2 \alpha$

Topic 5 Arithmetico–Geometric Progression

Q 1. The sum to n terms of the infinite series

$1.3^2+2.5^2+3.7^2+\ldots\infty$ is

[Jamia Millia Islamia]

(a) $\frac{n}{6}(n+1)(6n^2+14n+7)$

(b) $\frac{n}{6}(n+1)(2n+1)(3n+1)$

(c) $4n^3+4n^2+n$

(d) None of the above

Q 2. The sum of the series $1 + 3x + 6x^2 + 10x^3 +$

$\ldots\infty$ will be

[DCE]

(a) $\frac{1}{(1-x)^2}$

(b) $\frac{1}{1-x}$

(c) $\frac{1}{(1+x)^2}$

(d) $\frac{1}{(1-x)^3}$

Q 3. If | x | < 1, then the sum of the series $1 + 2x + 3x^2 + 4x^3 + \ldots \infty$ will be

[UPSEE]

(a) $\frac{1}{1-x}$

(b) $\frac{1}{1+x}$

(c) $\frac{1}{1+x^2}$

(d) $\frac{1}{(1-x)^2}$

Q 4. Sum of n terms of the series $\frac{1}{2}+\frac{3}{4}+\frac{7}{8}+\frac{15}{16}+.....$is

[UP SEE, Manipal]

(a) 2^{-n}

(b)$2^{-n}(n-1)$

(c) $2^{n}(n-1)+1$

(d) $2^{-n}+n-1$

Q 5. If |a| < 1,then $1+2a+3a^2+4a^3+\ldots$ is equal

to

[J&K CET]

(a) $\frac{1}{1-a}$

(b) $\frac{1}{1+a}$

(c) $\frac{1}{1+a^2}$

(d) $\frac{1}{(1-a)^2}$

Q 6. $1+\frac{4}{5}+\frac{7}{5^2}+\frac{10}{5^3}+.....\text{to}\ \infty$ is

[BCECE]

(a) $\frac{16}{35}$

(b) $\frac{11}{8}$

(**c**) $\frac{35}{16}$

(d) $\frac{7}{16}$

Q 7. The value of $2^{1/4} \cdot 4^{1/8} \cdot 8^{1/16} \ldots \infty$ is

[OJEE]

(a) 1

(b) 2

(c) $3/2$

(d) 4

Q 8. If the sum of n terms of the series

$1+\frac{4}{5}+\frac{7}{5^2}+\frac{10}{5^3}+....\text{is } 1+\frac{15}{16}\left(1-\frac{1}{5^{n-1}}\right)-\frac{(3n-2)}{4(5^{n-1})}$, then f is

[OJEE]

(a) $\frac{4}{5}$

(b) $\frac{5}{4}$

(c) $\frac{6}{5}$

(d) $\frac{5}{6}$

Topic 6 Special Series, Sum ton Terms and Infinite Number of Terms

Q 1. Sum of n terms of the following series

$1^3+3^3+5^3+7^3+\ldots$is

[WBJEE]

(a) $n^2(2n^2 - 1)$

(b) $n^3(n - 1)$

(c) $n^3 + 8n + 4$

(d) $2n^4 + 3n^2$

Q 2. The sum to the infinity of the series

$1+\frac{2}{3}+\frac{6}{3^2}+\frac{10}{3^3}+\frac{14}{3^4}+.....$ is

[AIEEE]

(a) 3

(b) 4

(c) 6

(d) 2

Q 3. The sum of the products of the numbers ± 1, ± 2, ……, $\pm n$, taken two at a time is

[DCE]

(a) $\frac{-n(n+1)}{2}$

(b) $\frac{n(n+1)(2n+1)}{6}$

(c) $\frac{-n(n+1)(2n+1)}{6}$

(d) 0

Q 4. $\sum_{k=1}^{n}\cdot\sum_{k=1}^{i}\cdot\sum_{k=1}^{j}$ is equal to

[DCE]

(a) $4\frac{n(n+1)(2n+1)}{6}$

(b) $\left[\frac{n(n+1)}{2}\right]^2$

(c) $\frac{n(n+1)}{2}$

(d) $\frac{n(n+1)(n+2)}{6}$

Q 5. $\frac{1}{n!}+\frac{1}{2!(n-2)!}+\frac{1}{4!(n-4)!}+.....\infty$ is

[BITSAT]

(a) $\frac{2^{n-1}}{n!}$

(b) $\frac{2^{n}}{(n+1)!}$

(c) $\frac{2^{n}}{n!}$

(d) $\frac{2^{n-2}}{(n-1)!}$

Q 6. If $y = 3x + 6x^2 + 10x^3 + \ldots$, then the value of x in terms of y is

[UP SEE]

(a) $1 - (1 - y)^{-1/3}$

(b) $1 - (1 + y)^{1/3}$

(c) $1 + (l + y)^{-1/3}$

(d) $1 - (1 + y)^{-1/3}$

Q 7. The value of $\frac{1}{2!} + \frac{2}{3!} + \ldots\ldots + \frac{999}{1000!}$ is equal to

[Kerala CEE]

(a) $\frac{1000!-1}{1000!}$

(b) $\frac{1000!+1}{1000!}$

(c) $\frac{999!-1}{999!}$

(d) $\frac{999!+1}{999!}$

(e) $\frac{10000!-999!}{1000!}$

Q 8. The sum of series $1+\frac{3}{2}+\frac{7}{4}+\frac{15}{8}+\frac{31}{16}+....$ is equal to

[BCECE]

(a) $2(n-1)+\frac{1}{2^{n-1}}$

(b) $2n-\frac{1}{2^n}$

(c) $2+\frac{1}{2^n}$

(d) $2n-1+\frac{1}{2^n}$

Q 9. The value of $1.1! + 2.2! + 3.3! + \ldots + n.n!$ is

[Jamia Millia Islamia]

(a) (n + l)!

(b) (n + 1)! + 1

(c) $(n + 1)! - 1$

(d) None of these

Q 10. For any integer $n \geq t$, the sum $\sum_{k=1}^{n} k(k+2)$ is equal to

[BITSAT, EAMCET]

(a) $\frac{n(n+1)(n+2)}{6}$

(b) $\frac{n(n+1)(2n+1)}{6}$

(c) $\frac{n(n+1)(2n+7)}{6}$

(d) $\frac{n(n+1)(2n+9)}{6}$

Q 11. $\frac{1}{2.5}+\frac{1}{5.8}+\frac{1}{8.11}+....$ upto n terms is equal to

[KCET]

(a) $\frac{n}{4n+6}$

(b) $\frac{1}{6n+4}$

(c) $\frac{n}{6n+4}$

(d) $\frac{n}{3n+7}$

Q 12. The sum of the first n terms of the series

$\frac{1}{\sqrt{2}+\sqrt{5}}+\frac{1}{\sqrt{5}+\sqrt{8}}+\frac{1}{\sqrt{8}+\sqrt{11}}+.....$ **is**

[Kerala CEE]

(a) $\frac{1}{3}(\sqrt{3n+2}-\sqrt{2})$

(b) $\sqrt{3n+2}-\sqrt{2}$

(c) $\sqrt{3n+2}+\sqrt{2}$

(d) $\frac{1}{3}(\sqrt{2}-\sqrt{3n+2})$

(e) $\frac{1}{3}(\sqrt{3n+2}+\sqrt{2})$

Q 13. The sum of the series $(1 + 2) + (1+2 + 2^2)$ $+ (1+2+2^2+2^3) + \ldots$ upto n terms is

[WBJEE]

(a) $2^{n+2} - n - 4$

(b) $2(2^n - 1)-n$

(c) $2^{n+1} - n$

(d) $2^{n+1} - 1$

Q 14. If $S_n = \frac{1}{6.11} + \frac{1}{11.16} + \frac{1}{16.21} + \ldots$ to n terms, then $6S_n$ equals

[Jamia Miliia Islamia]

(a) $\frac{5n-4}{5n+6}$

(b) $\frac{n}{(5n+6)}$

(c) $\frac{2n-1}{5n+6}$

(d) $\frac{1}{(5n+6)}$

Q 15. Let $S_n = \frac{1}{1^3} + \frac{1+2}{1^3+2^3} + ... + \frac{1+2+....+n}{1^3+2^3+....+n^3}$ n = 1, 2, 3,… . Then S_n is not greater than

[Manipal]

(a) $1/2$

(b) 1

(c) 2

(d) 4

Q $16.$ The sum of 24 terms of the following series $\sqrt{2}+\sqrt{8}+\sqrt{18}+\sqrt{32}+.....$ is

[BITSAT]

(a) 300

(b) $200\sqrt{2}$

(c) $300\sqrt{2}$

(d) $250\sqrt{2}$

Q 17. The sum to n terms of the infinite series

$1.3^2 + 2.5^2 + 3.7^2 + \ldots\infty$

[BITSAT]

(a) $\frac{n}{6}(n+1)(6n^2+14n+7)$

(b) $\frac{n}{6}(n+1)(2n+1)(3n+1)$

(c) $4n^3 + 4n^2 + n$

(d) None of the above

Q 18. The sum of all the products of the first n natural numbers taken two at a time is

[AMU]

(a) $\frac{1}{24}n(n-1)(n+1)(3n+2)$

(b) $\frac{n^2}{48}(n-1)(n-2)$

(c) $\frac{1}{6}n(n+1)(n+2)(n+5)$

(d) None of the above

Q 19. The sum of the series

$$\frac{3}{4.8}-\frac{3.5}{4.8.12}+\frac{3.5.7}{4.8.12.16}-.....$$

[EAMCET]

(a) $\sqrt{\frac{3}{2}}-\frac{3}{4}$

(b) $\sqrt{\frac{2}{3}}-\frac{3}{4}$

(c) $\sqrt{\frac{3}{2}}-\frac{1}{4}$

(d) $\sqrt{\frac{2}{3}-\frac{1}{4}}$

Q 20. If $S_n = 1^3..+ 2^3 +...+ n^3$ and $T_n = 1 +$

$2+... + n$, then

(a) $S_n = T_n$

(b) $S_n = T_n^4$

(c) $S_n = T_n^2$

(d) $S_n = T_n^3$

[EAMCET]

Q 21. The sum of n terms of the series l + (l + x) + $(1 + x + x^2)$ + …will be

[WBJEE]

(a) $\frac{1-x^n}{1-x}$

(b) $\frac{x(1-x^n)}{1-x}$

(c) $\frac{n(1-x)-x(1-x^n)}{(1-x)^2}$

(d) None of these

Q 22. The sum of the first n terms of the series

$1^2+2.2^2+3^2+2.4^2+5^2+2.6^2$ + ... is $\frac{n(n+1)^2}{2}$ where n is

even. When n is odd the sum is

[Jamia Millia Islamia]

(a) $\frac{3n(n+1)}{2}$

(b) $\frac{n^2(n+1)}{2}$

(c) $\frac{n(n+1)^2}{4}$

(d) $\left[\frac{n(n+1)}{2}\right]^2$

Q 23. The sum to n term of the series

$\frac{1}{\sqrt{1}+\sqrt{3}}+\frac{1}{\sqrt{3}+\sqrt{5}}+....$ is

[Guj.CET]

(a) $\sqrt{2n+1}$

(b) $\sqrt{2n+1}-1$

(c) $\frac{1}{2}\sqrt{2n+1}$

(d) $\frac{1}{2}(\sqrt{2n+1}-1)$

Q 24. If the sum of first n natural numbers is $1/5$ times the sum of their squares, then the value of n is

[Guj. CET]

(a) 5

(b) 6

(c) 7

(d) 8

Q 25. If the sum of first n natural numbers is $\frac{1}{78}$ times the sum of their cubes, then the value of n is

[BCECE]

(a) 11

(b) 12

(c) 13

(d) 14

Q 26. The sum of the series $1^3 + 2^3 + 3^3$ $+\ldots+15^3$ is

[MPPET]

(a) 22000

(b) 10000

(c) 14400

(d) 15000

Q 27. $\frac{\frac{1}{2}\cdot\frac{2}{2}}{1^3} + \frac{\frac{2}{2}\cdot\frac{3}{2}}{1^3+2^3} + \frac{\frac{3}{2}\cdot\frac{4}{2}}{1^3+2^3+3^3} + \ldots\ldots + n$ terms equals

[BITSAT]

(a) $\left(\frac{n}{n+1}\right)^2$

(b) $\left(\frac{n}{n+1}\right)^3$

(c) $\left(\frac{n}{n+1}\right)$

(d) $\left(\frac{1}{n+1}\right)$

Q 28. $\frac{1}{1!(n-1)!}+\frac{1}{3!(n-3)!}+\frac{1}{5!(n-5)!}+.....$ is equal to

[AMU]

(a) $\frac{2^{n-4}}{n!}$ for even values of n only

(b) $\frac{2^{n-4}+1}{n!}-1$ for odd values of n only

(c) $\frac{2^{n-1}}{n!}$ for all values of n

(d) None of the above

Q 29. The sum of infinite terms of the series

$\frac{1}{(1+a)(2+a)}+\frac{1}{(2+a)(3+a)}+\frac{1}{(3+a)(4+a)}+....+$ to ∞, where a is a constant, is

[AMU]

(a) $\frac{1}{1+a}$

(b) $\frac{2}{1+a}$

(c) ∞

(d) None of these

Q 30. It is given that $\frac{1}{1^4}+\frac{1}{2^4}+\frac{1}{3^4}+.....+\text{to }\infty=\frac{\pi^4}{90}$. Then,

$\frac{1}{1^4}+\frac{1}{3^4}+\frac{1}{5^4}+.....\text{to }\infty$ is equal to

[AMU]

(a) $\frac{\pi^4}{96}$

(b) $\frac{\pi^4}{45}$

(c) $\frac{89}{90}\pi$

(d) None of these

Q 31. If $|a| < 1$, $b = \sum_{k=1}^{\infty} \frac{a^k}{k}$, then a is equal to

[EAMCET]

(a) $\sum_{k=1}^{\infty} \frac{(-1)^k b^k}{k}$

(b) $\sum_{k=1}^{\infty} \frac{(-1)^{k-1} b^k}{k!}$

(c) $\sum_{k=1}^{\infty} \frac{(-1)^k b^k}{(k-1)!}$

(d) $\sum_{k=1}^{\infty} \frac{(-1)^{k-1} b^k}{(k+1)!}$

Trigonometric Identities and Equations

Topic 1

Trigonometrical Ratios of Allied Angles and Basic Fundamental

Q 1. Let $\cos(\alpha + \beta) = \frac{4}{5}$ and let $\sin(\alpha - \beta) = \frac{5}{13}$,

where $0 \leq \alpha, \beta \leq \frac{\pi}{4}$. Then $\tan 2\alpha$ is equal to

[WB JEE]

(a) $\frac{25}{16}$

(b) $\frac{56}{33}$

(c) $\frac{19}{12}$

(d) $\frac{20}{7}$

Q 2. The value of $\frac{\cot 54^{\circ}}{\tan 36^{\circ}}+\frac{\tan 20^{\circ}}{\cot 70^{\circ}}$ is

(a) 0

(b) 2

(c) 3

(d) 1

Q 3. The value of $\frac{\sin 55^o - \cos 55^o}{\sin 10^o}$ is

[VITEEE]

(a) $\frac{1}{\sqrt{2}}$

(b) 2

(c) 1

(d) $\sqrt{2}$

Q 4. The value of $\frac{\cot x - \tan x}{\cot 2x}$ is

[WB JEE]

(a) 1

(b) 2

(c) −1

(d) 4

Q 5. If $\frac{\cos A}{3}=\frac{\cos B}{4}=\frac{1}{5}, -\frac{\pi}{2}<A<0, -\frac{\pi}{2}<B<0$ then value of 2 sin A + 4 sin B is

[BITSAT]

(a) 4

(b) -2

(c) -4

(d) 0

Q 6. If $\alpha, \beta \in \left(0, \frac{\pi}{2}\right)$, $\sin \alpha = \frac{4}{5}$ and $\cos(\alpha + \beta) = -\frac{12}{13}$, then $\sin \beta$ is equal to

[Kerala CEE]

(a) $\frac{63}{65}$

(b) $\frac{61}{65}$

(c) $\frac{3}{5}$

(d) $\frac{5}{13}$

(e) $\frac{8}{65}$

Q 7. If $\tan \alpha = \frac{b}{a}$, $a > b > 0$ and if $0 < \alpha < \frac{\pi}{4}$, then

$\sqrt{\frac{a+b}{a-b}} - \sqrt{\frac{a-b}{a+b}}$ is equal to

[Kerala CEE]

(a) $\frac{2\sin\alpha}{\sqrt{\cos 2\alpha}}$

(b) $\frac{2\cos\alpha}{\sqrt{\cos 2\alpha}}$

(c) $\frac{2\sin\alpha}{\sqrt{\sin 2\alpha}}$

(d) $\frac{2\cos\alpha}{\sqrt{\sin 2\alpha}}$

(e) $\frac{2\tan\alpha}{\sqrt{\cos 2\alpha}}$

Q 8. The value of $\tan 40° + \tan 20° + \sqrt{3}\tan 20°$ $\tan 40°$ is equal to

[BITSAT]

(a) $\sqrt{12}$

(b) $\frac{1}{\sqrt{3}}$

(c) 1

(d) $\frac{\sqrt{3}}{2}$

(e) $\sqrt{3}$

Q 9. If $\tan\left(\frac{x}{2}\right)$ = cosec x – sin x, then the value of $\tan^2\left(\frac{x}{2}\right)$ is

[DCE]

(a) $2-\sqrt{5}$

(b) $2+\sqrt{5}$

(c) $-2-\sqrt{5}$

(d) $-2+\sqrt{5}$

Q 10. The value of

$\cos^4\left(\frac{\pi}{8}\right)+\cos^4\left(\frac{3\pi}{8}\right)+\cos^4\left(\frac{5\pi}{8}\right)+\cos^4\left(\frac{7\pi}{8}\right)$ is

[DCE]

(a) 0

(b) $\frac{1}{2}$

(c) $\frac{3}{2}$

(d) 1

Q 11. IN a ΔABC, $\angle A = \frac{\pi}{2}$, then $\cos^2 B + \cos^2$ C equals

[BITSAT]

(a) -2

(b) -1

(c) 1

(d) 0

Q 12. If cos $20°$ − sin $20°$ = p, then cos $40°$ is equal to

[UP SEE]

(a) $p^2\sqrt{2-p^2}$

(b) $p\sqrt{2-p^2}$

(c) $p+\sqrt{2-p^2}$

(d) $p-\sqrt{2-p^2}$

Q 13. If $\tan x = \frac{b}{a}$, then the value of a cos 2x + b sin 2x is

[UP SEE]

(a) 1

(b) ab

(c) b

(d) a

Q 14. If $S_n = \cos^n \theta + \sin^n \theta$, then the value of $3S_4 - 2S_6$ is given by

[UP SEE]

(a) 4

(b) 0

(c) 1

(d) 7

Q 15. Which one of the following is possible ?

[KCET]

(a) $\sin\theta = \frac{a^2+b^2}{a^2-b^2}, (a \neq b)$

(b) $\sec\theta = \frac{4}{5}$

(c) $\tan\theta = 45$

(d) $\cos\theta = \frac{7}{3}$

Q 16. $3(\sin x - \cos x)^4 + 6(\sin x + \cos x)^2 + 4(\sin^6 x + \cos^6 x)$ is equal to

[KCET]

(a) 12

(b) 13

(c) 14

(d) 11

Q 17. If sin A − $\sqrt{6}$ cos A = $\sqrt{7}$ cos A, then cos A + $\sqrt{6}$ sin A is equal to

[Kerala CEE]

(a) $\sqrt{6}$ sin A

(b) − $\sqrt{7}$ sin A

(c) $\sqrt{6}$ cos A

(d) $\sqrt{7}\cos A$

(e) $\sqrt{42}\cos A$

Q 18. If tan A and tan B are the roots of $abx^2 - c^2x + ab = 0$ where a, b, c are the sides of the triangle ABC, then the value of $\sin^2 A + \sin^2 B + \sin^2 C$ is

[Kerala CEE]

(a) 1

(b) 3

(c) 4

(d) 2

(e) 5

Q 19. If α and β satisfying $2 \sec 2\alpha = \tan \beta + \cot \beta$, then $\alpha + \beta$ is equal to

[OJEE]

(a) $\frac{\pi}{2}$

(b) $\frac{\pi}{3}$

(c) $\frac{\pi}{4}$

(d) π

Q 20. Simplest form of $\frac{2}{\sqrt{2+\sqrt{2+\sqrt{2+2\cos 4x}}}}$ is

[WB JEE]

(a) $\sec\frac{x}{2}$

(b) sec x

(c) cosec x

(d) 1

Q 21. The value of $\cos 15° \cos 7\frac{1}{2}^{\circ} \sin 7\frac{1}{2}^{\circ}$ is

[WB JEE]

(a) $\frac{1}{2}$

(b) $\frac{1}{8}$

(c) $\frac{1}{4}$

(d) $\frac{1}{16}$

Q 22. The value of $\left(1+\cos\frac{\pi}{6}\right)\left(1+\cos\frac{\pi}{3}\right)\left(1+\cos\frac{2\pi}{3}\right)\left(1+\frac{7\pi}{6}\right)$

is

[WB JEE]

(a) $\frac{3}{16}$

(b) $\frac{3}{8}$

(c) $\frac{3}{4}$

(d) $\frac{1}{2}$

Q 23. ΔABC is right angled at C, then tan A + tan B is equal to

[VITEEE]

(a) $\frac{b^2}{ac}$

(b) a + b

(c) $\frac{a^2}{bc}$

(d) $\frac{c^2}{ab}$

Q 24. In a right angled triangle the hypotenuse is $2\sqrt{2}$ times the length of perpendicular drawn from the opposite vertex on the hypotenuse, then the other two angles are

[Jamia Millia Islamia]

(a) $\frac{\pi}{3}, \frac{\pi}{6}$

(b) $\frac{\pi}{4}, \frac{\pi}{4}$

(c) $\frac{\pi}{8}, \frac{3\pi}{8}$

(d) $\frac{\pi}{12}, \frac{5\pi}{12}$

Q 25. The value of sin $50°$ − sin $70°$ + sin $10°$

is

[MP PET]

(a) 0

(b) 1

(c) $\frac{1}{2}$

(d) $\frac{1}{\sqrt{2}}$

Q 26. If $\tan\theta = \frac{1}{\sqrt{7}}$, then $\frac{(\cos ec^2\theta - \sec^2\theta)}{(\cos ec^2\theta + \sec^2\theta)}$ is equal to

[J&K CET]

(a) $\frac{1}{2}$

(b) $\frac{3}{4}$

(c) $\frac{5}{4}$

(d) 2

Q $27.$ $\sin 47° + \sin 61° - \sin 11° - \sin 25°$ is equal to

[J & K CET]

(a) $\sin 7°$

(b) $\cos 7°$

(c) $\sin 36°$

(d) $\cos 36°$

Q 28. In ΔABC, if $\sin^2 A + \sin^2 B + \sin^2 C = 2$,

then the triangle is

[J & K CET]

(a) right angled, but need not be isosceles

(b) right angled and isosceles

(c) isosceles, but need not be right angled

(d) equilateral

Q 29. $\sqrt{3}$ cosec $20°$ − sec $20°$ is equal to

[BITSAT, AMU]

(a) 2

(b) 2 sin $20°$. cosec $40°$

(c) 4

(d) 4 sin $20°$. cosec $40°$

Q 30. The value of $\tan 67\frac{1^{\circ}}{2}+\cot 67\frac{1^{\circ}}{2}$ is

[KCET]

(a) $\sqrt{2}$

(b) $3\sqrt{2}$

(c) $2\sqrt{2}$

(d) $2-\sqrt{2}$

Q 31. If x = h + a sec θ and y = k + b cosec θ.

Then,

[Kerala CEE]

(a) $\frac{a^2}{(x+h)^2}-\frac{b^2}{(y+k)^2}=1$

(b) $\frac{a^2}{(x-h)^2}+\frac{b^2}{(y-k)^2}=1$

(c) $\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}=1$

(d) $\frac{(x-h)^2}{a^2}-\frac{(y-k)^2}{b^2}=1$

(e) $x^2 + y^2 = a^2 + b^2$

Q 32. The value of $\cos^2\frac{\pi}{16}+\cos^2\frac{3\pi}{16}+\cos^2\frac{5\pi}{16}+\cos^2\frac{7\pi}{16}$ is

[OJEE]

(a) 2

(b) 1

(c) 0

(d) None of these

Q 33. The value of $\cos\frac{\pi}{15}\cos\frac{2\pi}{15}\cos\frac{4\pi}{15}\cos\frac{8\pi}{15}$ is

[WB JEE]

(a) $\frac{1}{16}$

(b) $-\frac{1}{16}$

(c) 1

(d) 0

Q 34. If sec θ = m and tan θ = n, then

$\frac{1}{m}\left[(m+n)+\frac{1}{(m+n)}\right]$ is

[Jamia Millia Islamia]

(a) 2

(b) 2m

(c) 2n

(d) mn

Q 35. If $\cos\theta = -\frac{\sqrt{3}}{2}$ and sin α = $-\frac{3}{5}$, where θ does not lie in the third quadrant, then $\frac{2\tan\alpha + \sqrt{3}\tan\theta}{\cot^2\theta + \cos\alpha}$ is equal to

[Manipal]

(a) $\frac{7}{22}$

(b) $\frac{5}{22}$

(c) $\frac{9}{22}$

(d) $\frac{22}{5}$

Q 36. tan 25° + tan 20° + tan 25° tan 20° is equal to

[RPET]

(a) 1

(b) 2

(c) 3

(d) 4

Q 37. If x sin θ = y cos θ = $\frac{2z \tan \theta}{1 - \tan^2 \theta}$, then $4z^2(x^2 + y^2)$ is equal to

[J&K CET]

(a) $(x^2 + y^2)^3$

(b) $(x^2 - y^2)^3$

(c) $(x^2 - y^2)^2$

(d) $(x^2 + y^2)^2$

Q 38. $\sin^2 17.5° + \sin^2 72.5°$ is equal to

[KCET]

(a) $\cos^2 90°$

(b) $\tan^2 45°$

(c) $\cos^2 30°$

(d) $\sin^2 45°$

Q 39. If $x \sin^3 \theta + y \cos^3 \theta = \sin \theta \cos \theta$ and $x \sin \theta = y \cos \theta$, then $x^2 + y^2$ is

[Kerala CEE]

(a) 2

(b) 0

(c) 3

(d) 4

(e) 1

Q 40. $\frac{\tan 80^{\circ} - \tan 10^{\circ}}{\tan 70^{\circ}}$ is equal to

[EAMCET]

(a) 0

(b) 1

(c) 2

(d) 3

Q 41. If θ lies in the first quadrant and $5 \tan \theta =$ 4, then $\frac{5\sin\theta - 3\cos\theta}{\sin\theta + 2\cos\theta}$ is equal to

[EAMCET]

(a) $\frac{5}{14}$

(b) $\frac{3}{14}$

(c) $\frac{1}{14}$

(d) 0

Q 42. The value of sin $36°$ sin $72°$ sin $108°$ sin $144°$ is

[WB JEE]

(a) $\frac{1}{4}$

(b) $\frac{1}{16}$

(c) $\frac{3}{4}$

(d) $\frac{5}{16}$

Q 43. If $\cos x + \cos^2 x = 1$, then the value of $\sin^{12} x + 3 \sin^{10} x + 3 \sin^8 x + \sin^6 x - 1$, is equal to

[VITEEE]

(a) 2

(b) 1

(c) −1

(d) 0

Q 44. Let $\theta \in \left(0, \frac{\pi}{4}\right)$ and $t_1 = (\tan \theta)^{\tan \theta}$, $t_2 = (\tan \theta)^{\cot \theta}$, $t_3 = (\cot \theta)^{\tan \theta}$ and $t_4 = (\cot \theta)^{\tan \theta}$, then

[IIT JEE]

(a) $t_1 > t_2 > t_3 > t_4$

(b) $t_4 > t_3 > t_1 > t_2$

(c) $t_3 > t_1 > t_2 > t_4$

(d) $t_2 > t_3 > t_1 > t_4$

Q 45. If $12 \cot^2 \theta - 31 \operatorname{cosec} \theta + 32 = 0$, then the value of $\sin \theta$ is

[BITSAT, Guj. CET]

(a) $\frac{3}{5}$ or 1

(b) $\frac{2}{3}$ or $-\frac{2}{3}$

(c) $\frac{4}{5}$ or $\frac{3}{4}$

(d) $\pm\frac{1}{2}$

Q 46. If $x + \frac{1}{x} = 2 \cos \theta$, then $x^3 + \frac{1}{x^3}$ is equal to

[UP SEE]

(a) $\sin 3\theta$

(b) $2 \sin 3\theta$

(c) $\cos 3\theta$

(d) $2 \cos 3\theta$

Q 47. If $\sin \theta + \text{cosec } \theta = 2$, the value of $\sin^{10} \theta + \text{cosec}^{10} \theta$ is

[UP SEE, WB JEE]

(a) 2

(b) 2^{10}

(c) 2^{9}

(d) 10

Q 48. The value of $\sin\frac{\pi}{16}\sin\frac{3\pi}{16}\sin\frac{5\pi}{16}\sin\frac{7\pi}{16}$ is

[UP SEE, WB JEE]

(a) $\frac{\sqrt{2}}{16}$

(b) $\frac{1}{8}$

(c) $\frac{1}{16}$

(d) $\frac{\sqrt{2}}{32}$

Q 49. The value of $\frac{\sin 85^\circ - \sin 35^\circ}{\cos 65^\circ}$ is

[KCET]

(a) 2

(b) −1

(c) 1

(d) 0

Q 50. $\frac{1}{\cos 80^{o}} - \frac{\sqrt{3}}{\sin 80^{o}}$ is equal to

[Kerala CEE]

(a) $\sqrt{2}$

(b) 1

(c) 2

(d) 4

(e) $\sqrt{5}$

Q 51. The value of $\cos 480° \cdot \sin 150° + \sin 600° \cdot \cos 390°$ is equal to

[Kerala CEE]

(a) 0

(b) 1

(c) $\frac{1}{2}$

(d) -1

(e) $-\frac{1}{2}$

Q 52. If $\sin x + \sin^2 x = 1$, then $\cos^{12} x + 3 \cos^{10} x + 3 \cos^8 x + \cos^6 x$ is equal to

[AMU, WB JEE]

(a) 1

(b) 2

(c) 3

(d) 0

Q 53. $\sin 120° \cos 150° - \cos 240° \sin 330°$ is equal to

[EAMCET]

(a) 1

(b) -1

(c) $\frac{2}{3}$

(d) $-\left(\frac{\sqrt{3}+1}{4}\right)$

Q 54. $\sin^2\theta = \frac{4xy}{(x+y)^2}$ is true if and only if

[OJEE]

(a) $x + y \neq 0$

(b) $x = y, x \neq 0, y \neq 0$

(c) $x = y$

(d) $x \neq 0, y \neq 0$

Q 55. If $p = \cos 55°$, $q = \cos 65°$ and $r = \cos 175°$, then the value of $\frac{1}{p} + \frac{1}{q} + \frac{r}{pq}$ is

[BCECE]

(a) 0

(b) -1

(c) 1

(d) None of these

Q 56. The value of $\sin 20° (4 + \sec 20°)$ is

[BCECE]

(a) 0

(b) 1

(c) $\sqrt{2}$

(d) $\sqrt{3}$

Q 57. If $\sin^2\theta = \frac{x^2+y^2+1}{2x}$, then x must be

[MP PET]

(a) -3

(b) -2

(c) 1

(d) None of these

Q 58. The expression $\cos\frac{10\pi}{13}+\cos\frac{8\pi}{13}+\cos\frac{3\pi}{13}+\cos\frac{5\pi}{13}$ is equal to

[MP PET]

(a) -1

(b) 0

(c) 1

(d) None of these

Q 59. $\frac{\tan A}{1+\sec A}+\frac{1+\sec A}{\tan A}$ is equal to

[J&K CET]

(a) 2 sin A

(b) 2 cos A

(c) 2 cosec A

(d) 2 sec A

Q $60.$ The value of cos $(270° + \theta)$ cos $(90° - \theta)$ – sin $(270° - \theta)$ cos θ is

[Guj. CET]

(a) 0

(b) -1

(c) $1/2$

(d) 1

Q 61. The value of the expression $\sin^6 \theta + \cos^6 \theta + 3 \sin^2 \theta \cdot \cos^2 \theta$ equals

[DCE]

(a) 0

(b) 2

(c) 3

(d) 1

Q 62. The value of the expression $\cos 1^\circ . \cos 2^\circ . \ldots . \cos 179^\circ$ equals

[DCE]

(a) 0

(b) 1

(c) $1/\sqrt{2}$

(d) -1

Q 63. If θ is an acute angle and $\sin\frac{\theta}{2}=\sqrt{\frac{x-1}{2x}}$, then $\tan\theta$ is equals to

[BITSAT]

(a) $x^2 - 1$

(b) $\sqrt{x^2-1}$

(c) $\sqrt{x^2+1}$

(d) $x^2 + 1$

Q 64. ABC is a right angled isosceles triangle with $\angle B = 90°$. If D is a point on AB so that $\angle CDB = 15°$ and, if AD = 35 cm, then CD is equal to

[Kerala CEE]

(a) $35\sqrt{3}$ cm

(b) $70\sqrt{2}$ cm

(c) $\frac{35\sqrt{3}}{2}$ cm

(d) $35\sqrt{6}$ cm

(e) $\frac{35\sqrt{2}}{2}$ cm

Q 65. sin $12°$ sin $48°$ sin $54°$ is equal to

[BCECE]

(a) $1/16$

(b) $1/32$

(c) $1/8$

(d) ¼

Q 66. If $\sec\theta + \tan\theta = k$, $\cos\theta$ equals to

[MP PET]

(a) $\frac{k^2+1}{2k}$

(b) $\frac{2k}{k^2+1}$

(c) $\frac{k}{k^2+1}$

(d) $\frac{k}{k^2-1}$

Q 67. $\cos 1° + \cos 2° + \cos 3° + \ldots + \cos 180°$ is equal to

[Guj. CET]

(a) 1

(b) 0

(c) 2

(d) -1

Q $68.$ If A = $130°$ and x = sin A + cos A, then

[DCE]

(a) $x > 0$

(b) $x < 0$

(c) $x = 0$

(d) $x \geq 0$

Q 69. The expression $2\cos\frac{\pi}{13}\cos\frac{9\pi}{13}+\cos\frac{3\pi}{13}+\cos\frac{5\pi}{13}$ is equal to

[J&K CET]

(a) -1

(b) 0

(c) 1

(d) None of these

Q 70. If $x+\frac{1}{x}=2\cos\alpha$, then $x^n+\frac{1}{x^n}$ is equal to

[KCET]

(a) $2^n\cos\alpha$

(b) $2^n\cos n\alpha$

(c) $2i\sin n\alpha$

(d) $2 \cos n\alpha$

Q 71. $\sin^2 5° + \sin^2 10° + \sin^2 15° + \ldots + \sin^2 90°$ is equal to

[Kerala CEE]

(a) $8\frac{1}{2}$

(b) 9

(c) $9\frac{1}{2}$

(d) $4\frac{1}{2}$

(e) 0

Q 72. The expression $\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ$ is equal to

[EAMCET]

(a) 4

(b) 3

(c) 2

(d) 1

Q 73. The value of the series $\cos 12° + \cos 84° + \cos 132° + \cos 156°$ is

[EAMCET]

(a) $\frac{1}{2}$

(b) $\frac{1}{4}$

(c) $-\frac{1}{4}$

(d) $-\frac{1}{2}$

Q 74. For $x \in \mathbb{R}$, $3\cos(4x - 5) + 4$ lies in the interval

[EAMCET]

(a) $[1, 7]$

(b) $[4, 7]$

(c) $[0, 7]$

(d) $[2, 7]$

Q 75. If $x \cos\theta = y\cos\left(\theta+\frac{2\pi}{3}\right) = z\cos\left(\theta+\frac{4\pi}{3}\right)$, then the value of $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}$ is equal to

[Jamia Millia Islamia]

(a) 1

(b) 2

(c) 0

(d) $3\cos\theta$

Q 76. The expression $(1 + \tan x + \tan^2 x)(1 - \cot x + \cot^2 x)$ has the positive values for x, given by

[J&K CET]

(a) $0 \leq x \leq \frac{\pi}{2}$

(b) $0 \leq x \leq \pi$

(c) for all $x \in R$

(d) $x \geq 0$

Q 77. If α, β, γ, δ are the smallest positive angles in ascending order of magnitude which have their sines equal to the positive quantity k, then the value of $4\sin\frac{\alpha}{2}+3\sin\frac{\beta}{2}+2\sin\frac{\gamma}{2}+\sin\frac{\delta}{2}$ is equal to

[J & K CET]

(a) $2\sqrt{1-k}$

(b) $2\sqrt{1+k}$

(c) $\frac{\sqrt{1+k}}{2}$

(d) $\sqrt{1+k}$

Q 78. The value of $\tan 1^\circ \tan 2^\circ \tan 3^\circ \ldots \tan 89^\circ$ is equal to

[DCE]

(a) -1

(b) 2

(c) $\frac{\pi}{2}$

(d) 1

Q 79. $\left(1+\cos\frac{\pi}{8}\right)\left(1+\cos\frac{3\pi}{8}\right)\left(1+\cos\frac{5\pi}{8}\right)\left(1+\cos\frac{7\pi}{8}\right)$ is equal to

[DCE]

(a) $\frac{1}{2}$

(b) $\cos\frac{\pi}{8}$

(c) $\frac{1}{8}$

(d) $\frac{1+\sqrt{2}}{2\sqrt{2}}$

Q 80. If sin x + cos x = $\frac{1}{5}$, then tan 2x is

[UP SEE]

(a) $\frac{25}{17}$

(b) $\frac{24}{7}$

(c) $\frac{7}{25}$

(d) $\frac{25}{7}$

Q 81. If $a = \frac{\pi}{18}$ rad, then cos a + cos 2a + ... + cos 18a is equal to

[Kerala CEE]

(a) 0

(b) -1

(c) 1

(d) ± 1

(e) None of these

Q 82. Let n be an odd integer. If $\sin n\theta = \sum_{r=0}^{n} b_r \sin^r \theta$ for every value of θ, then

[Guj. CET]

(a) $b_0 = 1, b_1 = 3$

(b) $b_0 = 0, b_1 = n$

(c) $b_0 = -1, b_1 = n$

(d) $b_0 = 0, b_1 = n^2 - 3n + 3$

Topic – 2

Trigonometrical Ratios of Sum and Difference of Two and Three Angles

Q 1. Let A and B denote the statements

A : $\cos \alpha + \cos \beta + \cos \gamma = 0$

B : $\sin \alpha + \sin \beta + \sin \gamma = 0$

If $\cos(\beta - \gamma) + \cos(\gamma - \alpha) + \cos(\alpha - \beta) = -\frac{3}{2}$, then

[AIEEE]

(a) A is true and B is false

(b) A is false and B is true

(c) Both A and B are true

(d) Both A and B are false

Q 2. If tan α = k cot β, then $\frac{\cos(\alpha-\beta)}{\cos(\alpha+\beta)}$ is equal to

[UP SEE]

(a) $\frac{1+k}{1-k}$

(b) $\frac{1-k}{1+k}$

(c) $\frac{k+1}{k-1}$

(d) $\frac{k-1}{k+1}$

Q 3. If A + B = $45°$, then (cot A – 1) (cot B – 1) is equal to

[Kerala CEE]

(a) 1

(b) $\frac{1}{2}$

(c) -1

(d) -2

(e) 2

Q 4. $\frac{\cos x}{\cos(x-2y)} = \lambda \Rightarrow$ tan(x – y)tan y is equal to

[EAMCET]

(a) $\frac{1+\lambda}{1-\lambda}$

(b) $\frac{1-\lambda}{1+\lambda}$

(c) $\frac{\lambda}{1+\lambda}$

(d) $\frac{\lambda}{1-\lambda}$

Q 5. The value of $1 + \cos 56° + \cos 58° - \cos 66°$ is

[OJEE]

(a) $4 \cos 28° \cos 29° \sin 33°$

(b) $\cos 28° \cos 29° \sin 33°$

(c) $4 \cos 28° \sin 29° \cos 33°$

(d) $4 \cos 28° \sin 29° \sin 33°$

Q 6. Value of $\cos^2(A - B) + \cos^2 B - 2 \cos (A - B)$ cos A cos B is

[OJEE]

(a) sin A

(b) $\sin^2$ A

(c) $\cos^2 A$

(d) cos A

Q 7. If $\sin (x + 3\alpha) = 3 \sin (\alpha - x)$, then

[BCECE]

(a) $\tan x = \tan \alpha$

(b) $\tan x = \tan^2 \alpha$

(c) $\tan x = \tan^3 \alpha$

(d) $\tan x = 3 \tan \alpha$

Q 8. The value of expression $\frac{1}{\cos 290^{o}}+\frac{1}{\sqrt{3}\sin 250^{o}}$ is

equal to

[BCECE]

(a) $\frac{\sqrt{3}}{4}$

(b) $\frac{4}{\sqrt{3}}$

(c) $\frac{2}{\sqrt{3}}$

(d) $\frac{\sqrt{3}}{2}$

Q 9. If $\alpha, \beta, \gamma \in \left[0, \frac{\pi}{2}\right]$, then the value of

$\frac{\sin(\alpha+\beta+\gamma)}{\sin\alpha+\sin\beta+\sin\gamma}$ is

[Jamia Millia Islamia]

(a) < 1

(b) $= -1$

(c) < 0

(d) None of these

Q 10. In a triangle ABC, sin A – cos B = cos C, then angle B is

[Jamia Millia Islamia]

(a) $\frac{\pi}{2}$

(b) $\frac{\pi}{3}$

(c) $\frac{\pi}{4}$

(d) $\frac{\pi}{6}$

Q 11. If $0 < x < \pi$ and $\cos x + \sin x = \frac{1}{2}$, then the value of tan x is

[MP PET]

(a) $\frac{2-\sqrt{7}}{3}$

(b) $-\frac{4+\sqrt{7}}{3}$

(c) $-\frac{1+\sqrt{7}}{3}$

(d) $-\frac{2+\sqrt{7}}{3}$

Q 12. If $\cos\theta = \frac{8}{17}$ and θ lies in the 1st quadrant, then the value of $\cos(30° + \theta) + \cos(45° - \theta) + \cos(120° - \theta)$ is

(a) $\frac{23}{17}\left(\frac{\sqrt{3}-1}{2}+\frac{1}{\sqrt{2}}\right)$

(b) $\frac{23}{17}\left(\frac{\sqrt{3}+1}{2}+\frac{1}{\sqrt{2}}\right)$

(c) $\frac{23}{17}\left(\frac{\sqrt{3}-1}{2}-\frac{1}{\sqrt{2}}\right)$

(d) $\frac{23}{17}\left(\frac{\sqrt{3}+1}{2}-\frac{1}{\sqrt{2}}\right)$

[DCE]

Q 13. If $\tan\theta + \tan\left(\theta+\frac{\pi}{3}\right)+\tan\left(\theta+\frac{2\pi}{3}\right)=3$, then

which of the following is equal to 1 ?

[BITSAT,EAMCET]

(a) $\tan 2\theta$

(b) $\tan 3\theta$

(c) $\tan^2\theta$

(d) $\tan^3\theta$

Q 14. If $\sin\theta = \sin 15° + \sin 45°$, where $0° < \theta < 90°$, then θ is equal to

[Kerala CEE]

(a) $45°$

(b) $54°$

(c) $60°$

(d) $72°$

(e) $75°$

Q 15. If A = $35°$, B = $15°$ and C = $40°$, then tan A . tan B + tan B . tan C + tan C . tan A is equal to

[AMU]

(a) 0

(b) 1

(c) 2

(d) 3

Q 16. If $\alpha + \beta + \gamma = 2\theta$, then $\cos\theta + \cos(\theta - \alpha) + \cos(\theta - \beta) + \cos(\theta - \gamma)$ is equal to

[EAMCET]

(a) $4\sin\frac{\alpha}{2}.\cos\frac{\beta}{2}.\sin\frac{\gamma}{2}$

(b) $4\cos\frac{\alpha}{2}.\cos\frac{\beta}{2}.\cos\frac{\gamma}{2}$

(c) $4\sin\frac{\alpha}{2}.\sin\frac{\beta}{2}.\sin\frac{\gamma}{2}$

(d) $4 \sin\alpha \,.\, \sin\beta \,.\, \sin\gamma$

Q 17. $\frac{1-\tan^2(45^\circ - A)}{1+\tan^2(45^\circ - A)}$ is equal to

[MP PET]

(a) sin $2A$

(b) cos $2A$

(c) tan $2A$

(d) cot $2A$

Q 18. If cos x = 3 cos y, then $2\tan\frac{y-x}{2}$ is equal to

[j&K CET]

(a) $\cot\left(\frac{y-x}{2}\right)$

(b) $\cot\left(\frac{x+y}{4}\right)$

(c) $\cot\left(\frac{y-x}{4}\right)$

(d) $\cot\left(\frac{x+y}{2}\right)$

Q 19. Given tan A and tan B are the roots of x^2 – ax + b = 0. The value of $\sin^2(A + B)$ is

[Guj. CET]

(a) $\frac{a^2}{a^2+(1-b)^2}$

(b) $\frac{a^2}{a^2+b^2}$

(c) $\frac{a^2}{(a+b)^2}$

(d) $\frac{b^2}{a^2+(a-b)^2}$

Q 20. If sin A + cos B = a and sin B + cos A = b, then sin(A + B) is equal to

[BITSAT]

(a) $\frac{a^2+b^2}{2}$

(b) $\frac{a^2-b^2+2}{2}$

(c) $\frac{a^2+b^2-2}{2}$

(d) None of these

Q 21. If sin A = n sin B, then $\frac{n-1}{n+1}\tan\frac{A+B}{2}$ is equal to

[UP SEE]

(a) $\sin\frac{A-B}{2}$

(b) $\tan\frac{A-B}{2}$

(c) $\cot\frac{A-B}{2}$

(d) None of these

Q 22. If $\sin 4A - \cos 2A = \cos 4A - \sin 2A$, $\left(0 < A < \frac{\pi}{4}\right)$, then the value of $\tan 4A$ is

[Kerala CEE]

(a) 1

(b) $\frac{1}{\sqrt{3}}$

(c) $\sqrt{3}$

(d) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$

(e) $\frac{\sqrt{3}+1}{\sqrt{3}-1}$

Q 23. If cos (A – B) = $3/5$ and tan A tan B = 2,

then which one of the following is true ?

[EAMCET]

(a) $\sin(A+B)=\frac{1}{5}$

(b) $\sin(A+B)=-\frac{1}{5}$

(c) $\cos(A-B)=\frac{1}{5}$

(d) $\cos(A+B)=-\frac{1}{5}$

Q 24. sin + sin B = $\sqrt{3}$(cos B – cos A) $\Rightarrow$ sin

3A + sin 3B is equal to

[EAMCET]

(a) 0

(b) 2

(c) 1

(d) −1

Q 25. If tan A = 2 tan B + cot B, then 2 tan(A –

B) is equal to

[OJEE]

(a) tan B

(b) 2 tan B

(c) cot B

(d) 2 cot B

Q 26. If $\sin A = \frac{1}{\sqrt{10}}$ and $\sin B = \frac{1}{\sqrt{5}}$, where A and B are positive acute angles, then A + B is equal to

[WB JEE]

(a) π

(b) $\frac{\pi}{2}$

(c) $\frac{\pi}{3}$

(d) $\frac{\pi}{4}$

Q 27. If $\cos(\theta - \alpha) = a$, $\cos(\theta - \beta) = b$, then

$\sin^2(\alpha - \beta) + 2ab \cos(\alpha - \beta)$ is equal to

[WB JEE]

(a) $a^2 + b^2$

(b) $a^2 - b^2$

(c) $b^2 - a^2$

(d) $-a^2 - b^2$

Q 28. Let α, β be such that $\pi < \alpha - \beta < 3\pi$. If $\sin \alpha + \sin \beta = -\frac{21}{65}$ and $\cos \alpha + \cos \beta = -\frac{27}{65}$, then the value of $\cos\frac{(\alpha - \beta)}{2}$ is

[MP PET]

(a) $-\frac{3}{\sqrt{130}}$

(b) $\frac{3}{\sqrt{130}}$

(c) $\frac{6}{65}$

(d) $-\frac{6}{65}$

Q 29. The value of $\cos^2\left(\frac{\pi}{4}+\theta\right)-\sin^2\left(\frac{\pi}{4}-\theta\right)$ is

[J&K CET]

(a) 0

(b) $\cos 2\theta$

(c) $\sin 2\theta$

(d) $\cos \theta$

Q 30. If $\sin A + \cos A = m$ and $\sin^3 A + \cos^3 A = n$, then

[Guj. CET]

(a) $m^3 - 3m + n = 0$

(b) $n^3 - 3n + 2m = 0$

(c) $m^3 - 3m + 2n = 0$

(d) $m^3 + 3m + 2n = 0$

Q 31. If $A+B=\frac{\pi}{4}$, then $(\tan A + 1)(\tan B + 1)$ equals

[Kerala CEE]

(a) 1

(b) $\sqrt{3}$

(c) 2

(d) $\frac{1}{\sqrt{3}}$

Q 32. If $\frac{\sin(x+y)}{\sin(x-y)}=\frac{a+b}{a-b}$, then $\frac{\tan x}{\tan y}$ is equal to

[Kerala CEE]

(a) $\frac{a^2}{b^2}$

(b) $\frac{a}{b}$

(c) $\frac{b}{a}$

(d) $\frac{a^2+b^2}{a^2-b^2}$

(e) $\frac{a^2-b^2}{a^2+b^2}$

Q 33. cosec 15° + sec 15° is equal to

[EAMCET]

(a) $2\sqrt{2}$

(b) $\sqrt{6}$

(c) $2\sqrt{6}$

(d) $\sqrt{6}+\sqrt{2}$

Q 34. If x = tan $15°$, y = cosec $75°$ and z = 4 sin $18°$, then

[EAMCET]

(a) $x < y < z$

(b) $y < z < x$

(c) $z < x < y$

(d) $x < z < y$

Q 35. If $\tan\alpha = \frac{m}{m+1}$ and $\tan\beta = \frac{1}{2m+1}$, then $\alpha + \beta$ is equal to

[WB JEE,MP PET]

(a) $\frac{\pi}{3}$

(b) $\frac{\pi}{4}$

(c) 0

(d) $\frac{\pi}{2}$

Q 36. If $\sin^3 x \sin 3x = \sum_{m=0}^{n} C_m \cos mx$ is an identity in x, where $C_0, C_1, \ldots\ldots, C_n$ are constant and $C_n \neq 0$, then the value of n equals

[DCE]

(a) 2

(b) 4

(c) 6

(d) 8

Q 37. If $\alpha + \beta = \frac{\pi}{2}$, $\beta + \gamma = \alpha$, then the value of

tan α equals

[DCE]

(a) tan β + tan γ

(b) $2(\tan \beta + \tan \gamma)$

(c) $\tan \beta + 2 \tan \gamma$

(d) $2 \tan \beta + \tan \gamma$

Q 38. $\frac{\cos 9^{o} + \sin 9^{o}}{\cos 9^{o} - \sin 9^{o}}$ is equal to

[Kerala CEE]

(a) $\tan 26°$

(b) $\tan 81°$

(c) $\tan 51°$

(d) tan $54°$

(e) tan $46°$

Q 39. The value of tan $20°$ + 2 tan $50°$ − tan $70°$ is

[AMU]

(a) 1

(b) 0

(c) tan $50°$

(d) None of these

Q 40. If A + C = 2B, then $\frac{\cos C - \cos A}{\sin A - \sin C}$ is equal to

[EAMCET]

(a) cot B

(b) cot 2B

(c) tan 2B

(d) tan B

Q 41. If $y = (1 + \tan A)(1 - \tan B)$, where $A - B = \frac{\pi}{4}$, then $(y + 1)^{y+1}$ is equal to

[J&K CET]

(a) 9

(b) 4

(c) 27

(d) 81

Q 42. The value of $\frac{\tan 70^\circ - \tan 20^\circ}{\tan 50^\circ}$ is equal to

[Guj. CET]

(a) 2

(b) 1

(c) 0

(d) 3

Q 43. If sin x + sin y = a and cos x + cos y = b, then $\tan\left(\frac{x+y}{2}\right)$ is equal to

[OJEE]

(a) $\frac{ab}{a+b}$

(b) $\frac{a}{b}$

(c) $\frac{b}{a}$

(d) None of these

Q 44. If $\sin\theta = \frac{12}{13}, \left(0 < \theta < \frac{\pi}{2}\right)$ and $\cos\phi = -\frac{3}{5}, \left(\pi < \phi < \frac{3\pi}{2}\right)$,

then sin $(\theta + \phi)$ will be

(a) −56/61

(b) $-56/65$

(c) $1/65$

(d) -56

Q 45. If $\cos(\theta + \phi) = m \cos(\theta - \phi)$, then $\tan\theta$ is equal to

(a) $[(1 + m)/(1 - m)] \tan\phi$

(b) $[(1 - m)/(1 + m)] \tan\phi$

(c) $[(1 - m)/(1 + m)] \cot\phi$

(d) $[(1 + m)/(1 - m)]\sec \phi$

Q 46. $\tan 10° + \tan 35° + \tan 10° \tan 35°$ is equal to

[BCECE]

(a) 0

(b) $\frac{1}{2}$

(c) -1

(d) 1

Q 47. The value of $\sin 10° + \sin 20° + \sin 30°$ + …… + $\sin 360°$ is equal to

[Guj. CEE]

(a) 0

(b) 1

(c) $\sqrt{3}$

(d) 2

Q 48. If m tan $(\theta - 30°)$ = n tan $(\theta + 120°)$,

then cos 2θ equals

[Kerala CEE]

(a) $\frac{m+n}{m-n}$

(b) $\frac{m-n}{m+n}$

(c) $\frac{m-n}{2(m+n)}$

(d) $\frac{m+n}{2(m-n)}$

(e) $\frac{2(m+n)}{m-n}$

Q 49. $\cos \alpha \sin (\beta - \gamma) + \cos \beta \sin (\gamma - \alpha) +$ $\cos \gamma \sin (\alpha - \beta)$ is equal to

[EAMCET]

(a) 0

(b) $\frac{1}{2}$

(c) 1

(d) $4 \cos \alpha \cos \beta \cos \gamma$

Q 50. If α, β $(\alpha \neq \beta)$ satisfies the equation a cos θ + b sin θ = c, then the value of $\tan\left(\frac{\alpha+\beta}{2}\right)$ is

[OJEE]

(a) b/a

(b) c/a

(c) a/b

(d) c/b

Q 51. The value of cot $70°$ + 4 cos $70°$ is

[OJEE]

(a) $\frac{1}{\sqrt{3}}$

(b) $\sqrt{3}$

(c) $2\sqrt{3}$

(d) $1/2$

Topic – 3

Trigonometrical Ratios of Multiple and Submultiple Angles

Q 1. If $\sqrt{\frac{1+\cos A}{1-\cos A}} = \frac{x}{y}$, then the value of tan A is equal to

[KCET]

(a) $\frac{x^2+y^2}{x^2-y^2}$

(b) $\frac{2xy}{x^2+y^2}$

(c) $\frac{2xy}{x^2-y^2}$

(d) $\frac{2xy}{y^2-x^2}$

Q 2. If $\cos 2\alpha = \frac{3\cos 2\beta - 1}{3 - \cos 2\beta}$, then tan α is equal to

[OJEE]

(a) $\sqrt{2}\tan\beta$

(b) $\tan\beta$

(c) $\sin 2\beta$

(d) $\sqrt{2}\cot\beta$

Q 3. The value of $\tan\alpha + 2\tan(2\alpha) + 4\tan(4\alpha)$

$+ \ldots + 2^{n-1}\tan(2^{n-1}\alpha) + 2^n\cot(2^n\alpha)$ is

[WB JEE]

(a) $\cot(2^n \alpha)$

(b) $2^n \tan(2^n \alpha)$

(c) 0

(d) $\cot \alpha$

Q 4. If $x + \frac{1}{x} = 2 \cos \theta$, then $x^n + \frac{1}{x^n}$ is equal to

[MP PET]

(a) $2 \sin n\theta$

(b) $2 \cos n\theta$

(c) sin $(2n\theta)$

(d) cos $(2n\theta)$

Q 5. If $\cos\theta = \cos\alpha \cos\beta$, then $\tan\left(\frac{\theta+\alpha}{2}\right)\tan\left(\frac{\theta-\alpha}{2}\right)$ is equal to

[DCE]

(a) $\tan^2\frac{\alpha}{2}$

(b) $\tan^2\frac{\beta}{2}$

(c) $\tan^2 \frac{\theta}{2}$

(d) $\cot^2 \frac{\beta}{2}$

Q 6. If $\tan x = \frac{b}{a}$, then the value of a cos 2x + b sin 2x is

[UP SEE]

(a) a

(b) a− b

(c) a + b

(d) b

Q 7. If $\cos\frac{x}{2}.\cos\frac{x}{2^2}.....\cos\frac{x}{2^n}=\frac{\sin}{2^n\sin\frac{x}{2^n}}$, then

$\frac{1}{2}\tan\frac{x}{2}+\frac{1}{2^2}\tan\frac{x}{2^2}+.....+\frac{1}{2^n}\tan\frac{x}{2^n}$ is

[Kerala CEE]

(a) $\cot x-\cot\frac{x}{2^n}$

(b) $\frac{1}{2}\cot\left(\frac{x}{2^n}\right)-\cot x$

(c) $\frac{1}{2^n}\tan\left(\frac{x}{2^n}\right)-\tan x$

(d) $\frac{1}{2}\cot x-\frac{1}{2^n}\cot\left(\frac{x}{2^n}\right)$

(e) $\cot\left(\frac{x}{2^n}\right)-\cot x$

Q 8. If $0 < x < \pi$ and $\cos x + \sin x = \frac{1}{2}$, then $\tan x$ is

[AIEEE]

(a) $\frac{(4-\sqrt{7})}{3}$

(b) $-\frac{(4+\sqrt{7})}{3}$

(c) $\frac{(1+\sqrt{7})}{4}$

(d) $\frac{(1-\sqrt{7})}{4}$

Q 9. Suppose $0 < t < \pi$ and $\sin t + \cos t = \frac{1}{5}$.

Then, $\tan\frac{t}{2}$ is equal to

[Kerala CEE]

(a) 2

(b) 3

(c) $\frac{1}{3}$

(d) 5

(e) $\frac{1}{5}$

Q 10. If $5 \cos 2\theta + 2 \cos^2 \frac{\theta}{2} + 1 = 0$, $-\pi < \theta < \pi$, then θ is equal to

[AMU]

(a) $\frac{\pi}{3}$

(b) $\frac{\pi}{3}, \cos^{-1}\left(\frac{3}{5}\right)$

(c) $\cos^{-1}\left(\frac{3}{5}\right)$

(d) $\frac{\pi}{3}, \pi - \cos^{-1}\left(\frac{3}{5}\right)$

Q 11. $\cos^4\theta - \sin^4\theta$ is equal to

[WB JEE, MP PET]

(a) $1+2\sin^2\left(\frac{\theta}{2}\right)$

(b) $2\cos^2\theta - 1$

(c) $1-2\sin^2\left(\frac{\theta}{2}\right)$

(d) $1 + 2\cos^2\theta$

Q 12. $\tan\alpha + 2\tan 2\alpha + 4\tan 4\alpha + 8\cot 8\alpha$

is equal to

[Jamia Millia Islamia]

(a) tan 16α

(b) 0

(c) cot α

(d) None of these

Q 13. If $\frac{\tan 3A}{\tan A} = a$, then $\frac{\sin 3A}{\sin A}$ is equal to

[EAMCET]

(a) $\frac{2a}{a+1}$

(b) $\frac{2a}{a-1}$

(c) $\frac{a}{a+1}$

(d) $\frac{a}{a-1}$

Q 14. If n = 1, 2, 3,..., then cos α cos 2α cos 4α .... cos $2^{n-1}\ \alpha$ is equal to

[MP PET]

(a) $\frac{\sin 2n\alpha}{2n\sin\alpha}$

(b) $\frac{\sin 2^n \alpha}{2^n \sin 2^{n-1} \alpha}$

(c) $\frac{\sin 4^{n-1} \alpha}{4^{n-1} \sin \alpha}$

(d) $\frac{\sin 2^n \alpha}{2^n \sin \alpha}$

Q 15. For any angle θ, the expression $\frac{2\cos 8\theta + 1}{2\cos \theta + 1}$ is equal to

[RPET]

(a) $(2 \cos \theta + 1)(2 \cos 2\theta + 1)(2 \cos 4\theta + 1)$

(b) $(\cos \theta - 1)(\cos 2\theta - 1)(\cos 4\theta - 1)$

(c) $(2\cos\theta - 1)(2\cos 2\theta - 1)(2\cos 4\theta - 1)$

(d) $(2\cos\theta + 1)(2\cos 2\theta + 1)(2\cos 4\theta + 1)$

Q 16. If $1 + \cos x = k$, where x is acute, then $\sin\frac{x}{2}$ is

[Kerala CEE]

(a) $\sqrt{\frac{1-k}{2}}$

(b) $\sqrt{2-k}$

(c) $\sqrt{\frac{2+k}{2}}$

(d) $\sqrt{\frac{2-k}{2}}$

(e) $\sqrt{\frac{k}{2}}$

Q 17. If $\theta = \frac{2\sin x}{1+\sin x+\cos x}$, then $\frac{1+\sin x-\cos x}{1+\sin x}$ equals

[OJEE]

(a) 0

(b) $-\theta$

(c) θ

(d) $-\theta/2$

Q 18. The value of $\cos\frac{2\pi}{15}\cos\frac{4\pi}{15}\cos\frac{8\pi}{15}\cos\frac{14\pi}{15}$ is

[EAMCET]

(a) $\frac{1}{16}$

(b) $\frac{1}{8}$

(c) $\frac{1}{12}$

(d) $\frac{1}{4}$

Q 19. If $\sin 6\theta = 32\cos^5\theta - 32\cos^3\theta \sin\theta + 3x$, then x is equal to

[EAMCET]

(a) $\cos\theta$

(b) $\cos 2\theta$

(c) $\sin\theta$

(d) $\sin 2\theta$

Q 20. If $\tan A = \frac{1-\cos B}{\sin B}$, then

[BCECE]

(a) $\tan 2A = \tan B$

(b) $\tan 2A = \tan^2 B$

(c) $\tan 2A = \tan^2 B + 2 \tan B$

(d) None of the above

Topic – 4

Maximum–Minimum values and Conditional Trigonometrical Identities

Q 1. The minimum value of $27^{\cos 2x}\ 81^{\sin 2x}$ is

[KCET]

(a) -5

(b) $\frac{1}{5}$

(c) $\frac{1}{243}$

(d) $\frac{1}{27}$

Q 2. The maximum value of $3 \cos x + 4 \sin x + 5$ is

[AMU]

(a) 5

(b) 6

(c) 7

(d) None of these

Q 3. The smallest value of $5 \cos \theta + 12$ is

[WB JEE]

(a) 5

(b) 12

(c) 7

(d) 17

Q 4. The maximum value of $f(x) = \sin x(1 + \cos x)$

is

[MP PET]

(a) $\frac{3\sqrt{3}}{4}$

(b) $\frac{3\sqrt{3}}{2}$

(c) $3\sqrt{3}$

(d) $\sqrt{3}$

Q 5. The maximum value of $\cos^2\left(\frac{\pi}{3}-x\right)-\cos^2\left(\frac{\pi}{3}+x\right)$ is

[DCE]

(a) $-\frac{\sqrt{3}}{2}$

(b) $\frac{1}{2}$

(c) $\frac{\sqrt{3}}{2}$

(d) $\frac{3}{2}$

Q 6. The maximum value of $5 \cos \theta + 3 \cos \left(\theta + \frac{\pi}{3}\right) + 3$ is

[UP SEE]

(a) 5

(b) 11

(c) 10

(d) -1

Q 7. If $5 \cos x + 12 \cos y = 13$, then the maximum value of $5 \sin x + 12 \sin y$ is

[BCECE]

(a) 12

(b) $\sqrt{120}$

(c) $\sqrt{20}$

(d) 13

Q 8. If $A = \left\{ x : \frac{\pi}{6} \le x \le \frac{\pi}{3} \right\}$ and f(x) = cos x – x (1 + x),

then f(A) is equal to

[Manipal]

(a) $\left[-\frac{\pi}{3}, -\frac{\pi}{6} \right]$

(b) $\left[\frac{\pi}{6}, \frac{\pi}{3} \right]$

(c) $\left[\frac{1}{2} - \frac{\pi}{3}\left(1 + \frac{\pi}{3}\right), \frac{\sqrt{3}}{2} - \frac{\pi}{6}\left(1 + \frac{\pi}{6}\right) \right]$

(d) $\left[\frac{1}{2}+\frac{\pi}{3}\left(1-\frac{\pi}{3}\right),\frac{\sqrt{3}}{2}+\frac{\pi}{6}\left(1-\frac{\pi}{6}\right)\right]$

Q 9. The maximum value of $\sin\left(x+\frac{\pi}{6}\right)+\cos\left(x+\frac{\pi}{6}\right)$ in the interval $\left(0,\frac{\pi}{2}\right)$ is attained at

[Manipal]

(a) $x=\frac{\pi}{12}$

(b) $x=\frac{\pi}{6}$

(c) $x=\frac{\pi}{3}$

(d) $x=\frac{\pi}{2}$

Q 10. The expression $\tan^2 \alpha + \cot^2 \alpha$ is

[DCE, WB JEE]

(a) ≥ 2

(b) ≤ 2

(c) ≥ -2

(d) None of these

Q 11. The maximum value of $3 \cos \theta + 4 \sin \theta$ is

[UP SEE]

(a) 3

(b) 4

(c) 5

(d) None of these

Q 12. If A + B + C = $180°$, then $\Sigma \tan\frac{A}{2}\tan\frac{B}{2}$ is

[Kerala CEE]

(a) 0

(b) 1

(c) 2

(d) 3

(e) 4

Q 13. The maximum value of $4 \sin^2 x - 12 \sin x + 7$ is

[AMU]

(a) 25

(b) 4

(c) does not exist

(d) None of these

Q 14. If $\alpha + \beta - \gamma = \pi$, then $\sin^2 \alpha + \sin^2 \beta - \sin^2 \gamma$ is equal to

[WB JEE]

(a) $2 \sin \alpha \sin \beta \cos \gamma$

(b) $2 \cos \alpha \cos \beta \cos \gamma$

(c) $2 \sin \alpha \sin \beta \sin \gamma$

(d) None of the above

Q 15. If $f : R \to S$ defined by $f(x) = \sin x - \sqrt{3} \cos x + 1$, is onto, then the interval of S is

[MP PET]

(a) $[0, 3]$

(b) $[-1, 1]$

(c) $[0, 1]$

(d) $[-1, 3]$

Q 16. If $y = \cos^2 x + \sec^2 x$, then

[UP SEE, WB JEE]

(a) $y \leq 2$

(b) $y \leq 1$

(c) $y \geq 2$

(d) $1 < y < 2$

Q 17. If A + B + C = π, then sin 2A + sin 2B + sin 2C is equal to

[UP SEE]

(a) 4 sin A sin B sin C

(b) 4 cos A cos B cos C

(c) 2 cos A cos B cos C

(d) 2 sin A sin B sin C

Q 18. The maximum of the function 3 cos x – 4 sin x

[AMU]

(a) 2

(b) 3

(c) 4

(d) 5

Q 19. The minimum value of $f(x) = \sin^4 x + \cos^4 x$, $0 \leq x \leq \frac{\pi}{2}$ is

[Jamia Millia Islamia]

(a) $\frac{1}{2\sqrt{2}}$

(b) $\frac{1}{4}$

(c) $\frac{-1}{2}$

(d) $\frac{1}{2}$

Q 20. $A + B = C \Rightarrow \cos^2 A + \cos^2 B + \cos^2 C - 2 \cos A \cos B \cos C$ is equal to

[EAMCET]

(a) 1

(b) 2

(c) 0

(d) 3

Q 21. Maximum value of $\sin\theta + \cos\theta$ in $\left[0, \frac{\pi}{2}\right]$ is

[MHTCET]

(a) $\sqrt{2}$

(b) 2

(c) 0

(d) $-\sqrt{2}$

Q 22. The maximum value of $4 \sin^2 x + 3 \cos^2 x$ is

[Guj. CET]

(a) 4

(b) 3

(c) 7

(d) 5

Q 23. Minimum value of $\frac{1}{3\sin\theta - 4\cos\theta + 7}$ is

[DCE]

(a) $\frac{1}{12}$

(b) $\frac{5}{12}$

(c) $\frac{7}{12}$

(d) $\frac{1}{6}$

Q 24. The equation a cos θ + b sin θ = c has a solution, when a, b and c are real numbers such that

[Kerala CEE]

(a) $a < b < c$

(b) $a = b = c$

(c) $c^2 \leq a^2 + b^2$

(d) $c^2 \leq a^2 - b^2$

Q 25. In a ΔABC, $\cos\left(\frac{B+2C+3A}{2}\right)+\cos\left(\frac{A-B}{2}\right)$ is equal to

[EAMCET]

(a) -1

(b) 0

(c) -1

(d) 2

Q 26. If $p = \sin^2 x + \cos^4 x$, then

[OJEE, BCECE]

(a) $\frac{3}{4} \le p \le 1$

(b) $\frac{3}{16} \le p \le \frac{1}{4}$

(c) $\frac{1}{4} \le p \le 1$

(d) None of these

Q 27. The maximum value of $12 \sin \theta - 9 \sin^2 \theta$ is

[BCECE]

(a) 3

(b) 4

(c) 5

(d) None of these

Q 28. Let A, B and C are the angles of a triangle and $\tan\left(\frac{A}{2}\right)=\frac{1}{3}, \tan\left(\frac{B}{2}\right)=\frac{2}{3}$. Then $\tan\left(\frac{C}{2}\right)$ is equal to

(a) $\frac{1}{3}$

(b) $\frac{2}{3}$

(c) $\frac{2}{9}$

(d) $\frac{7}{9}$

[UP SEE, OJEE]

Q 29. If A + B + C = $270°$, then cos 2A + cos 2B + cos 2C is equal to

[EAMCET]

(a) 4 sin A sin B sin C

(b) 4 cos A cos B cos C

(c) $1 - 4$ sin A sin B sin C

(d) $1 - 4$ cos A cos B cos C

Topic – 5

Hyperbolic Functions

Q 1. $\sinh^{-1} 2 + \sinh^{-1} 3 = x \Rightarrow \cosh x$ is equal to

[EAMCET]

(a) $\frac{1}{2}(3\sqrt{5}+2\sqrt{10})$

(b) $\frac{1}{2}(3\sqrt{5}-2\sqrt{10})$

(c) $\frac{1}{2}(12+2\sqrt{50})$

(d) $\frac{1}{2}(12-2\sqrt{50})$

Q 2. If $\tanh^{-1}(x + iy)$

$= \frac{1}{2}\tanh^{-1}\left(\frac{2x}{1+x^2+y^2}\right) + \frac{i}{2}\tan^{-1}\left(\frac{2y}{1-x^2-y^2}\right)$, x, y $\in$ R, then $\tanh^{-}$

^{1} (iy) is

[RPET]

(a) $i \tanh^{-1}(y)$

(b) $-i \tanh^{-1}(y)$

(c) $i \tan^{-1} y$

(d) $-i \tan^{-1}(y)$

Q 3. $\frac{1+\tanh\frac{x}{2}}{1-\tanh\frac{x}{2}}$ is equal to

[EAMCET]

(a) e^{-x}

(b) e^{x}

(c) $2e^{x/2}$

(d) $2e^{-x/2}$

Q 4. $\text{sech}^{-1}(\sin\theta)$ is equal to

[EAMCET]

(a) $\log\tan\frac{\theta}{2}$

(b) $\log\sin\frac{\theta}{2}$

(c) $\log\cos\frac{\theta}{2}$

(d) $\log\cot\frac{\theta}{2}$

Q 5. $e^{\log(\cosh^{-1}2)}$ is equal to

[EAMCET]

(a) $\log(2-\sqrt{3})$

(b) $\log(\sqrt{3}-2)$

(c) $\log(2+\sqrt{3})$

(d) $\log(2+\sqrt{5})$

Q 6. $2\tanh^{-1}\frac{1}{2}$ is equal to

[EAMCET]

(a) 0

(b) $\log 2$

(c) $\log 3$

(d) $\log 4$

Q 7. $\sec h^{-1}\left(\frac{1}{2}\right)$ is

[RPET]

(a) $\log(\sqrt{3}+\sqrt{2})$

(b) $\log(\sqrt{3}+1)$

(c) $\log(2+\sqrt{3})$

(d) None of these

Q 8. If $x = \log\left[\cot\left(\frac{\pi}{4}+\theta\right)\right]$, then the value of sinh x is

[EAMCET]

(a) tan 2θ

(b) – tan 2θ

(c) cot 2θ

(d) – cot 2θ

Q 9. The value of the series

$$x \log_e a + \frac{x^3}{x!}(\log_e a)^3 + \frac{x^5}{5!}(\log_e a)^5 + \ldots\ldots \text{ is}$$

[EAMCET]

(a) cos h (x $\log_e$ a) (b) coth (x $\log_e$ a) (c) sin h(x $\log_e$ a) (d) tanh (x $\log_e$ a)

Q 10. If $\cosh^{-1} x = \log(2+\sqrt{3})$, then x is equal to

[RPET]

(a) 2

(b) 1

(c) 3

(d) 5

Q 11. If $-\frac{\pi}{2} < x < \frac{\pi}{2}$, then the value of log sec x is

[RPET]

(a) $2\coth^{-1}\left(\cos ec^2 \frac{x}{2} - 1\right)$

(b) $2\coth^{-1}\left(\cos ec^2\frac{x}{2}+1\right)$

(c) $2\cos ech^{-1}\left(\cot^2\frac{x}{2}-1\right)$

(d) $2\cos ech^{-1}\left(\cot^2\frac{x}{2}+1\right)$

Q 12. $\sinh^{-1}(2^{3/2})$ is equal to

[EAMCET]

(a) $\log(3+\sqrt{8})$

(b) $\log(3-\sqrt{8})$

(c) $\log(2+\sqrt{18})$

(d) $\log(\sqrt{8}+\sqrt{27})$

Topic – 6

Solution of Trigonometrical Equations

2010

Q 1. If $\sin 6\theta + \sin 4\theta + \sin 2\theta = 0$, then the general value of θ is

[WB JEE]

(a) $\frac{n\pi}{4}, n\pi \pm \frac{\pi}{3}$

(b) $\frac{n\pi}{4}, n\pi \pm \frac{\pi}{6}$

(c) $\frac{n\pi}{4}, 2n\pi \pm \frac{\pi}{3}$

(d) $\frac{n\pi}{4}, 2n\pi \pm \frac{\pi}{6}$

Q 2. The number of points of intersection of $2y = 1$ and $y = \sin x$, in $-2\pi \leq x \leq 2\pi$ is **[VITEEE]**

(a) 1

(b) 2

(c) 3

(d) 4

Q 3. The value of x in $\left(0, \frac{\pi}{2}\right)$ satisfying the equation

$\sin x \cos x = \frac{1}{4}$ is **[Kerala CEE]**

(a) $\frac{\pi}{6}$

(b) $\frac{\pi}{3}$

(c) $\frac{\pi}{8}$

(d) $\frac{\pi}{4}$

(e) $\frac{\pi}{12}$

Q 4. The number of solutions of $\cos 2\theta = \sin \theta$ in $(0, 2\pi)$ is

[Kerala CEE]

(a) 1

(b) 2

(c) 3

(d) 4

e) 0

Q 5. The solution of the equation [sin x + cos x]$^{1+\sin 2x} = 2$, $-\pi \leq x \leq \pi$ is

[Kerala CEE]

(a) $\frac{\pi}{2}$

(b) π

(c) $\frac{\pi}{4}$

(d) $\frac{3\pi}{4}$

(e) $\frac{\pi}{3}$

Q 6. Equation $\cos 2x + 7 = a(2 - \sin x)$ can have a real solution for

[AMU]

(a) all values of a

(b) $a \in [2, 6]$

(c) $a \in (-\infty, 2)$

(d) $a \in (0, \infty)$

Q 7. Number of solutions of $|x - 1| = \cos x$ is

[AMU]

(a) 2

(b) 3

(c) 4

(d) None of these

Q 8. If $3 \cos x \neq 2 \sin x$, then the general solution

of $\sin^2 x - \cos 2x = 2 - \sin 2x$ is

[EAMCET]

(a) $n\pi + (-1)^n \frac{\pi}{2}$, $n \in Z$ (b) $\frac{n\pi}{2}$, $n \in Z$ (c) $(4n \pm 1)\frac{\pi}{2}$, $n \in Z$ (d) $(2n - 1)\pi$, $n \in Z$

Q 9. cos $2x$ + k sin x = $2k - 7$ has a solution for

[OJEE]

(a) $2 \leq k \leq 6$

(b) $1 < k < 7$

(c) $4 < k < 7$

(d) None of these

Q 10. If $5 \cos 2\theta + 2 \cos^2 \frac{\theta}{2} + 1 = 0$, when $(0 < \theta < \pi)$, then the values of θ are

[WB JEE]

(a) $\frac{\pi}{3} \pm \pi$

(b) $\frac{\pi}{3}, \cos^{-1}\left(\frac{3}{5}\right)$

(c) $\cos^{-1}\left(\frac{3}{5}\right) \pm \pi$

(d) $\frac{\pi}{3}, \pi - \cos^{-1}\left(\frac{3}{5}\right)$

Q 11. General solution of sin x + cos x =

$\min_{a \in R} \{1, a^2 - 4a + 6\}$ is

[WB JEE]

(a) $\frac{n\pi}{2} + (-1)^n \frac{\pi}{4}$

(b) $2n\pi + (-1)^n \frac{\pi}{4}$

(c) $n\pi + (-1)^{n+1} \frac{\pi}{4}$

(d) $n\pi + (-1)^n \frac{\pi}{4} - \frac{\pi}{4}$

Q 12. If $P = \frac{1}{2}\sin^2\theta + \frac{1}{3}\cos^2\theta$, then

[WB JEE]

(a) $\frac{1}{3} \leq P \leq \frac{1}{2}$

(b) $P \geq \frac{1}{2}$

(c) $2 \leq P \leq 3$

(d) $-\frac{\sqrt{13}}{6} \leq P \leq \frac{\sqrt{13}}{6}$

Q 13. Solution of the equation $3 \tan(\theta - 15) =$ $\tan(\theta + 15)$ is

[VITEEE]

(a) $\theta = n\pi - \frac{\pi}{3}$

(b) $\theta = n\pi + \frac{\pi}{3}$

(c) $\theta = n\pi - \frac{\pi}{4}$

(d) $\theta = n\pi + \frac{\pi}{4}$

Q 14. If $2 \sec 2\alpha = \tan \beta + \cot \beta$, then one of the values of $\alpha + \beta$ is

[VITEEE]

(a) $\frac{\pi}{4}$

(b) $\frac{\pi}{2}$

(c) π

(d) $n\pi - \frac{\pi}{4}$, $n \in I$

Q 15. The general value of θ satisfying the equation $2\sin^2\theta - 3\sin\theta - 2 = 0$ is

[Jamia Millia Islamia]

(a) $n\pi + (-1)^{n+1}\frac{\pi}{6}$

(b) $n\pi + (-1)^{n}\frac{\pi}{2}$

(c) $n\pi + (-1)^n \frac{5\pi}{6}$

(d) $n\pi + (-1)^n \frac{7\pi}{6}$

Q 16. The solution of the equation $4 \cos^2 x + 6 \sin^2 x = 5$ are

[MP PET]

(a) $x = n\pi \pm \frac{\pi}{4}$

(b) $x = n\pi \pm \frac{\pi}{3}$

(c) $x = n\pi \pm \frac{\pi}{2}$

(d) $x = n\pi \pm \frac{2\pi}{3}$

Q 17. The number of solutions of the equation

sin x cos 3x = sin 3x cos 5x in $\left[0, \frac{\pi}{2}\right]$ is

[J&K CET]

(a) 3

(b) 4

(c) 5

(d) 6

Q 18. The most general values of θ satisfying

$\tan\theta + \tan\left(\frac{3\pi}{4}+\theta\right) = 2$ are given by

[J&K CET]

(a) $2n\pi \pm \frac{\pi}{3}, n \in Z$

(b) $n\pi + \frac{\pi}{3}, n \in Z$

(c) $2n\pi \pm \frac{\pi}{6}, n \in Z$

(d) $n\pi \pm \frac{\pi}{6}, n \in Z$

Q 19. The root of the equation $1 - \cos\theta = \sin\theta$. $\sin\frac{\theta}{2}$ is

[DCE]

(a) $k\pi, k \in I$

(b) $2k\pi, k \in I$

(c) $k\frac{\pi}{2}, k \in I$

(d) None of these

Q 20. The most general value of θ satisfying the equations $\sin\theta = \sin\alpha$ and $\cos\theta = \cos\alpha$ is

[UP SEE]

(a) $2n\pi + \alpha$

(b) $2n\pi - \alpha$

(c) $n\pi + \alpha$

(d) $n\pi - \alpha$

Q 21. The number of solutions for the equation $\sin 2x + \cos 4x = 2$ is

[KCET]

(a) 0

(b) 1

(c) 2

(d) ∞

Q 22. The general solution of |sin x| = cos x is

(when n $\in$ I)given by

[KCET]

(a) $n\pi + \frac{\pi}{4}$

(b) $2n\pi \pm \frac{\pi}{4}$

(c) $n\pi \pm \frac{\pi}{4}$

(d) $n\pi - \frac{\pi}{4}$

Q 23. $\{x \in R : \cos 2x + 2\cos^2 x = 2\}$ is equal to

[EAMCET]

(a) $\left\{2n\pi + \frac{\pi}{3} : n \in Z\right\}$

(b) $\left\{n\pi \pm \frac{\pi}{6} : n \in Z\right\}$

(c) $\left\{n\pi \pm \frac{\pi}{3} : n \in Z\right\}$

(d) $\left\{2n\pi - \frac{\pi}{3} : n \in Z\right\}$

Q 24. If $2 \sin^2 \theta + \sqrt{3} \cos \theta + 1 = 0$, then the value of θ is

[OJEE]

(a) $\frac{\pi}{6}$

(b) $\frac{2\pi}{3}$

(c) $\frac{3\pi}{6}$

(d) π

Q 25. The solution of the equation $4 \sin^4 x + \cos^4 x = 1$ is

[OJEE]

(a) $x = 2n\pi$

(b) $x = n\pi + 1$

(c) $x = (n + 2)\pi$

(d) None of the above

Q 26. The number of values of x in $[0, 2\pi]$ satisfying the equation $3 \cos 2x - 10 \cos x + 7 = 0$ is

[OJEE]

(a) 1

(b) 2

(c) 3

(d) 4

Q 27. If $\tan\left(\frac{\alpha\pi}{4}\right) = \cot\left(\frac{\beta\pi}{4}\right)$, then

[WB JEE]

(a) $\alpha + \beta = 0$

(b) $\alpha + \beta = 2n$

(c) $\alpha + \beta = 2n + 1$

(d) $\alpha + \beta = 2(2n + 1)$, $\forall n$ is an integer

Q 28. The equation $\sqrt{3}\sin x + \cos x = 4$ has

[WB JEE]

(a) infinitely many solutions

(b) no solution

(c) two solutions

(d) only one solution

Q 29. If $1 + \sin\theta + \sin^2\theta + \ldots\infty = 4 + 2\sqrt{3}$, $0 < \theta < \pi$, $\theta \neq \frac{\pi}{2}$, then

[Manipal]

(a) $\theta = \frac{\pi}{3}$

(b) $\theta = \frac{\pi}{6}$

(c) $\theta = \frac{\pi}{3} \text{or } \frac{\pi}{6}$

(d) $\theta = \frac{\pi}{3} \text{or } \frac{2\pi}{3}$

Q 30. The most general solutions of the equation sec x – 1 = $(\sqrt{2} - 1)$tan x are given by

[Manipal]

(a) $n\pi + \frac{\pi}{8}$

(b) $2n\pi, 2n\pi + \frac{\pi}{4}$

(c) $2n\pi$

(d) None of these

Q 31. The most general solution of $\sqrt{3}\cos\theta +$ $\sin\theta = \sqrt{2}$ is

[MP PET]

(a) $\theta = n\pi \pm \frac{\pi}{4}+\frac{\pi}{6}$

(b) $\theta = n\pi \pm \frac{\pi}{4}-\frac{\pi}{6}$

(c) $\theta = 2n\pi \pm \frac{\pi}{4}+\frac{\pi}{6}$

(d) $\theta = 2n\pi \pm \frac{\pi}{4} - \frac{\pi}{6}$

Q 32. If $\cos x \neq -\frac{1}{2}$, then the solution of cos x + cos $2x$ + cos $3x$ = 0 are

[J&KCET]

(a) $2n\pi \pm \frac{\pi}{4}$, $n \in Z$

(b) $2n\pi \pm \frac{\pi}{3}$, $n \in Z$

(c) $2n\pi \pm \frac{\pi}{6}$, $n \in Z$

(d) $2n\pi \pm \frac{\pi}{2}$, $n \in Z$

Q 33. The number of solutions of sin x = sin 2x between $\frac{-\pi}{2}$ and $\frac{\pi}{2}$ is

[J&K CET]

(a) 3

(b) 2

(c) 1

(d) 0

Q 34. If $\sin 4A - \cos 2A = \cos 4A - \sin 2A$, $\left(0 < A < \frac{\pi}{4}\right)$, then the value of $\tan 4A$ is

[Guj. CET]

(a) 1

(b) $\frac{1}{\sqrt{3}}$

(c) $\sqrt{3}$

(d) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$

Q 35. The number of solutions of the pair of equations $2 \sin^2 \theta - \cos 2\theta = 0$ and $2 \cos^2 \theta - 3 \sin \theta = 0$ in the interval $[0, 2\pi]$ is

[IIT JEE]

(a) zero

(b) one

(c) two

(d) four

Q 36. The number of solutions of the equation $1 + \sin x \sin^2 \frac{x}{2} = 0$, in $[-\pi, \pi]$ is

[BITSAT]

(a) zero

(b) one

(c) two

(d) three

Q 37. Number of solution of $y = e^x$ and $y = \sin x$ is

[BITSAT]

(a) 0

(b) 1

(c) 2

(d) infinite

Q 38. The general value of θ in the equation

$\cos\theta = \frac{1}{\sqrt{2}}$, tan $\theta = -1$ is

[UP SEE]

(a) $2n\pi \pm \frac{\pi}{6}, n \in I$

(b) $2n\pi \pm \frac{7\pi}{4}, n \in I$

(c) $n\pi + (-1)^n \frac{\pi}{3}, n \in I$

(d) $n\pi + (-1)^n \frac{\pi}{4}, n \in I$

Q 39. If sin 3θ = sin θ, how many solutions exist such that $0 < \theta < 2\pi$?

[KCET]

(a) 8

(b) 9

(c) 5

(d) 7

Q 40. The most general solutions of the equation sec x – 1 = $(\sqrt{2}-1)$ tan x are given by

[WB JEE]

(a) $n\pi+\frac{\pi}{8}$

(b) $2n\pi, 2n\pi+\frac{\pi}{4}$

(c) $2n\pi$

(d) None of these

Q 41. The equation $3 \sin^2 x + 10 \cos x - 6 = 0$

is satisfied, if

[WB JEE]

(a) $x = n\pi \pm \cos^{-1}\left(\frac{1}{3}\right)$

(b) $x = 2n\pi \pm \cos^{-1}\left(\frac{1}{3}\right)$

(c) $x = n\pi \pm \cos^{-1}\left(\frac{1}{6}\right)$

(d) $x = 2n\pi \pm \cos^{-1}\left(\frac{1}{6}\right)$

Q 42. The solution of sin x + sin 5x = sin 3x in

$\left(0, \frac{\pi}{2}\right)$

[J&K CET]

(a) $\frac{\pi}{4}, \frac{\pi}{10}$

(b) $\frac{\pi}{6}, \frac{\pi}{3}$

(c) $\frac{\pi}{4}, \frac{\pi}{2}$

(d) $\frac{\pi}{8}, \frac{\pi}{16}$

Q 43. If cot x + cosec x = $\sqrt{3}$, then the principal value of $\left(x-\frac{\pi}{6}\right)$ is

[J&K CET]

(a) $\frac{\pi}{3}$

(b) $\frac{\pi}{4}$

(c) $\frac{\pi}{2}$

(d) $\frac{\pi}{6}$

Q 44. The solution of tan 2θ tan θ = 1 is

[Guj. CET]

(a) $\frac{\pi}{3}$

(b) $(6n \pm 1)\frac{\pi}{6}$

(c) $(4n \pm 1)\frac{\pi}{6}$

(d) $(2n + \pi)\frac{\pi}{6}$

Q 45. The most general value of θ which satisfy both the equations $\cos\theta = \frac{1}{\sqrt{2}}$ and $\tan\theta = 1$, is

[Guj. CET]

(a) $2n\pi + \frac{5\pi}{4}, n \in I$

(b) $2n\pi + \frac{\pi}{4}, n \in I$

(c) $2n\pi + \frac{3\pi}{4}, n \in I$

(d) None of these

Q 46. The set of values of θ satisfying the inequation $2 \sin^2 \theta - 5 \sin \theta + 2 > 0$, where $0 < \theta < 2\pi$, is

[IIT JEE]

(a) $\left(0, \frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, 2\pi\right)$

(b) $\left[0, \frac{\pi}{6}\right] \cup \left[\frac{5\pi}{6}, 2\pi\right]$

(c) $\left[0, \frac{\pi}{3}\right] \cup \left[\frac{2\pi}{3}, 2\pi\right]$

(d) None of these

Q 47. The number of values of x in the interval $[0, 3\pi]$ satisfying the equation $2 \sin^2 x + 5 \sin x - 3 = 0$ is

[AIEEE]

(a) 6

(b) 1

(c) 2

(d) 4

Q 48. The general solution of sin x – cos x = $\sqrt{2}$

, for any integer n is

[BITSAT, Guj. CET]

(a) $n\pi$

(b) $2n\pi + \frac{3\pi}{4}$

(c) $2n\pi$

(d) $(2n + 1)\pi$

Q 49. If $\frac{\tan 3\theta - 1}{\tan 3\theta + 1} = \sqrt{3}$, then the general value of θ is

[UP SEE]

(a) $\frac{n\pi}{3} - \frac{\pi}{12}$

(b) $n\pi + \frac{7\pi}{12}$

(c) $\frac{n\pi}{3} + \frac{7\pi}{36}$

(d) $n\pi + \frac{\pi}{12}$

Q 50. Which one of the following equations has no solution ?

[KCET]

(a) $\operatorname{cosec}\theta - \sec\theta = \operatorname{cosec}\theta \cdot \sec\theta$

(b) $\operatorname{cosec}\theta \cdot \sec\theta = 1$

(c) $\cos\theta + \sin\theta = \sqrt{2}$

(d) $\sqrt{3}\sin\theta - \cos\theta = 2$

Q 51. For all values of θ, the values of $3 - \cos\theta + \cos\left(\theta + \frac{\pi}{3}\right)$ lie in the interval

[EAMCET]

(a) $[-2, 3]$

(b) $[-2, 1]$

(c) $[2, 4]$

(d) $[1, 5]$

Q 52. The solution of equation $\cos^2\theta + \sin\theta + 1 = 0$ lies in the interval

[MP PET]

(a) $\left(-\frac{\pi}{4}, \frac{\pi}{4}\right)$

(b) $\left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$

(c) $\left(\frac{3\pi}{4}, \frac{5\pi}{4}\right)$

(d) $\left(\frac{5\pi}{4}, \frac{7\pi}{4}\right)$

Q 53. If $\tan (k + 1)\theta = \tan \theta$, then θ belongs to the set

[J&K CET]

(a) $\{n\pi : n \in I\}$

(b) $\{n\pi / 2 : n \in I\}$

(c) $\{n\pi / k : n \in I\}$

(d) $\{n\pi / 2k : n \in I\}$

Q 54. If $12 \cot^2 \theta - 31 \operatorname{cosec} \theta + 32 = 0$, then the value of $\sin \theta$ is

[KCET]

(a) $\frac{3}{5}$ or 1

(b) $\frac{2}{3}$ or $-\frac{2}{3}$

(c) $\frac{4}{5}$ or $\frac{3}{4}$

(d) $\pm\frac{1}{2}$

Q 55. If $\cos\theta - 4\sin\theta = 1$, then $\sin\theta + 4\cos\theta$ is equal to

[EAMCET]

(a) ± 1

(b) 0

(c) ± 2

(d) ± 4

Q 56. If $\cos 2x = (\sqrt{2} + 1)\left(\cos x - \frac{1}{\sqrt{2}}\right)$, $\cos x \neq \frac{1}{2}$,

then $x \in I$

[EAMCET]

(a) $\left\{2n\pi \pm \frac{\pi}{3} : n \in Z\right\}$

(b) $\left\{2n\pi \pm \frac{\pi}{6} : n \in Z\right\}$

(c) $\left\{2n\pi \pm \frac{\pi}{2} : n \in Z\right\}$

(d) $\left\{2n\pi \pm \frac{\pi}{4} : n \in Z\right\}$

Q 57. The number of values of θ in the interval $[-\pi, \pi]$ satisfying the equation $\cos\theta + \sin 2\theta = 0$ is

[MP PET]

(a) 1

(b) 2

(c) 3

(d) 4

Q 58. If n is any integer, then the general solution of the equation $\cos\theta - \sin\theta = \frac{1}{\sqrt{2}}$ is

[J&K CET]

(a) $\theta = 2n\pi - \frac{\pi}{12}$ or $\theta = 2n\pi + \frac{7\pi}{12}$

(b) $\theta = 2n\pi + \frac{\pi}{12}$

(c) $\theta = 2n\pi + \frac{\pi}{12}$ or $\theta = 2n\pi - \frac{7\pi}{12}$

(d) $\theta = 2n\pi + \frac{\pi}{12}$ or $\theta = 2n\pi + \frac{7\pi}{12}$

Q 59. The general solution of the equation tan 2θ . tan $\theta = 1$ for $n \in Z$ is

[Guj. CET]

(a) $(2n+1)\frac{\pi}{4}$

(b) $(2n+1)\frac{\pi}{6}$

(c) $(2n+1)\frac{\pi}{2}$

(d) $\frac{1}{1}(2n+1)\frac{\pi}{3}$

Q 60. Given both θ and ϕ are acute angles sin $\theta = \frac{1}{2}$, cos $\phi = \frac{1}{3}$, then the value of $\theta + \phi$ belongs to

[IIT JEE]

(a) $\left(\frac{\pi}{3}, \frac{\pi}{2}\right]$

(b) $\left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$

(c) $\left(\frac{2\pi}{3}, \frac{5\pi}{6}\right]$

(d) $\left(\frac{5\pi}{6}, \pi\right]$

Q 61. The number of real solutions of $2 \sin (e^x) = 5^x + 5^{-x}$ in $[0, 1]$ is/are

[DCE]

(a) 0

(b) 1

(c) 2

(d) 4

Q 62. The number of solutions of the equation $2 \cos (e^x) = 5^x + 5^{-x}$ are

[UP SEE]

(a) no solution

(b) one solution

(c) two solutions

(d) infinitely many solutions

Q 63. If $0 \leq x \leq \pi$ and $81^{\sin^2 x} + 81^{\cos^2 x} = 30$, then x is equal to

[KCET]

(a) $\frac{\pi}{6}$

(b) $\frac{\pi}{2}$

(c) $\frac{\pi}{4}$

(d) $\frac{3\pi}{4}$

Q 64. If $\sin\left(\frac{\pi}{4}\cot\theta\right)=\cos\left(\frac{\pi}{4}\tan\theta\right)$, then θ is equal to

[Kerala CEE]

(a) $2n\pi+\frac{\pi}{4}$

(b) $2n\pi\pm\frac{\pi}{4}$

(c) $2n\pi - \frac{\pi}{4}$

(d) $n\pi - \frac{\pi}{4}$

(e) $n\pi + \frac{\pi}{4}$

Q 65. The number of solutions of the equation tan x + sec x = 2 cos x and cos x $\neq 0$ lying in the interval $(0, 2\pi)$ is

(a) 2

(b) 1

(c) 0

(d) 3

(e) 4

[Kerala CEE]

Q $66.$ If cos $20°$ = k and cos x = $2k^2 - 1$, then the possible values of x between $0°$ and $360°$ are

[BCECE]

(a) $140°$ and $270°$

(b) $40°$ and $140°$

(c) $40°$ and $320°$

(d) $50°$ and $130°$

Q 67. The equation $\sin x + \sin y + \sin z = -3$

for $0 \leq x \leq 2\pi$, $0 \leq y \leq 2\pi$, $0 \leq z \leq 2\pi$ has

[OJEE]

(a) one solution

(b) two sets of solution

(c) four sets of solution

(d) no solution

Q 68. The number of values of x in the interval

$[0, 5\pi]$ satisfying the equation $3 \sin^2 x - 7 \sin x + 2 = 0$ is

(b) 0

(b) 5

(c) 6

(d) 10

[Jamia Millia Islamia]

Q 69. The least positive non-integral solution of

$\sin \pi(x^2 + x) - \sin \pi x^2 = 0$, is

[J&K CET]

(a) rational

(b) irrational of the form $\sqrt{p}$

(c) irrational of the form $\frac{\sqrt{p}-1}{4}$, when p is an odd integer

(d) irrational of the form $\frac{\sqrt{p}+1}{4}$, where p is an even integer

Q 70. The general solution of the equation $2^{\cos 2x} + 1 = 3 \cdot 2^{-\sin x}$ is

[J&K CET]

(a) $n\pi$

(b) $n\pi - \pi$

(c) $n\pi + \pi$

(d) None of these

Q 71. If $\cos\theta + \cos 2\theta + \cos 3\theta = 0$, the general value of θ is

[UP SEE]

(a) $\theta = 2n\pi \pm \frac{\pi}{4}$

(b) $\theta = n\pi + (-1)^n \frac{2\pi}{3}$

(c) $\theta = n\pi + (-1)^n \frac{\pi}{3}$

(d) $\theta = 2n\pi \pm \frac{\pi}{3}$

Q 72. The solution set of $(5 + 4 \cos \theta)(2 \cos \theta + 1) = 0$ in the interval $[0, 2\pi]$ is

[EAMCET]

(a) $\left\{\frac{\pi}{3}, \frac{2\pi}{3}\right\}$

(b) $\left\{\frac{\pi}{3}, \pi\right\}$

(c) $\left\{\frac{2\pi}{3}, \frac{4\pi}{3}\right\}$

(d) $\left\{\frac{2\pi}{3}, \frac{5\pi}{3}\right\}$

Answers

Topic 1 : Trigonometrical Ratios of Allied Angles and Basic Fundamental

1 (b) 2 (b) 3. (d) 4. (b) 5, (c) 6. (a) 7. (a) 8. (e)

9. (d) 10. (c) 11. (c) 12 (b) 13. (d) 14. c) 15. (c)

16. (b) 17. (b) 18. (d) 19.(c) 20. (a) 21 (b) 22.

(a) 23. (d) 24. (c) 25. (a) 26. (b) 27. (b) 28. (a)

29. (c) 30. (c) 31. (b) 32 (a) 33. (b) 34. (a) 35.

(b) 36. (a) 37. (c) 38. (b) 39. (e) 40. (c) *41.* (a)

42. (d) 43. (d) 44. (b) 45. (c) 46. (d) 47. (a) 48.

(a) 49. (c) 50. (d) 51. (d) 52. (a) 53. (b) 54. (b)

55. (a) 56. (d) 57. (d) 58. (b) 59. (c) 60. (d)

61 (d) 62. (a) 63. (b) 64. (a) 65. (c) 66. (b) 67.

(d) 68. (a) 69. (b) 70. (d) 71. (c) 72. (a) 73. (d)

74. (a) 75. (c) 76. (c) 77. (b) 78. (d) 79.(c) 80.

(b) 81. (b) 82. (b)

Topic 2 : Trigonometrical Ratios of Sum and Difference of Two and Three Angles

1 *(c)* 2 (a) 3. (e) 4. (b) 5. (a) 6. (b) 7. (c) 8. (b)

9. (a) 10. (a) 11. (b) 12. *(a)* 13. (b) 14. (e) 15. (b)

16. (b) 17. (a) 18. (d) 19. (a) 20. (c) 21. (b) 22

(c) 23. (d) 24. (a) 25. (c) 26. (d) 27. (a) 28. (a)

29. (a) 30. (c) 31 (c) 32 (b) 33. (c) 34. (a) 35. (b)

36. (c) 37. (c) 38. (d) 39. (b) 40. (d) 41.(c) 42.

(a) 43. (b) 44. (b) 45. (c) 46. (d) 47. (a) 48. (d)

49. (a) 50. (a) 51. (b)

Topic 3 : Trigonometrical Ratios of Multiple and Submultiple Angles

1. *(c) 2.* (a) 3 (d) 4. *(b)* 5. (b) 6. (a) 7. (b) 8. (b)

9. (a) 10. (d) 11. (b) 12. (c) 13: (b) 14. (d) 15. (c)

16. (d) 17. (c) 18. (a) 19. (d) 20. (a)

Topic 4 : Maximum–Minimum Values and Conditional Trigonometrical Identities

1 *(c) 2* (d) 3. (c) 4. (a) 5. (c) 6. (c) 7. (b) 8. (c) 9.

(a) 10. (a) 11. (c) 12. (b) 13. (d) 14. (a) 15. (d)

16, (c) 17. (a) 18. (d) 19. (d) 20. (a) 21. (a) 22.

(a) 23. (a) 24. (c) 25. (b) 26. (a) 27. (b) 28. (d)

29. (c)

Topic 5 : Hzperbolic Functions

1. (c) 2. (c) 3. (b) 4. (d) 5. (c) 6. (c) 7. (c) 8. (b)

9. (c) 10. (a) 11. (a) 12. (a)

Topic 6 : Solution of Trigonometrical Equations

1 (a) 2 (d) 3. (e) 4. (c) 5. (c) 6. (b) 7. (a) 8. (c) S.

(a) 10.. (d) 11. (d) 12 (a) 13. (d) 14. (a) 15. (a)

16. (a) 17. (c) 18. (b) 19. (b) 20. (c) 21. *(a)* 22.

(b) 23. (b) 24. (c) 25. (d) 28. (d) 27. (d) 28. (b)

29. (d) 30. (b) 31 (d) 32 (a) 33. (a) 34. (c) 35.

(c) 36. (a) 37. (d) 38. (b) 39. (c) 40. (b) 41 (b)

42. (b) 43. (d) 44. (b) 45. (a) 46. (a) 47. (d) 48.

(b) 49. (c) 50. (b) 51. (c) 52. (d) 53.(c) 54.(c) 55.

(d) 56. (d) 57. (d) 58. (c) 59. (b) 60. (b). 61. (a)

62. (a) 63. (a) 64. (e) 65. (a) 66.(0 67. (a) 68. (c)

69. (c) 70. (a) 71. (d) 72. (c)

Height And Distance

Q 1. The angle of elevation of the top of a hill from a point is α. After walking h metres towards the top up a slope inclined at an angle β to the horizon, the angle of elevation of the top becomes γ. Then, the height of the

[DCE]

(a) $\frac{b \sin \alpha \sin(\gamma - \beta)}{\sin(\gamma - \alpha)}$

(b) $\frac{b \sin \alpha \sin(\gamma - \alpha)}{\sin(\gamma - \beta)}$

(c) $\frac{b \sin(\gamma - \beta)}{\sin(\gamma - \alpha)}$

(d) $\frac{\sin(\gamma - \beta)}{b \sin \alpha \sin(\gamma - \alpha)}$

Q 2. From the top of a cliff 50 m high, the angles of depression of the top and bottom of a tower are observed to be $30°$ and $45°$. The height of tower is

(a) 50 m

(b) $50\sqrt{3}$

(c) $50(\sqrt{3}-1)\text{m}$

(d) $50\left(1-\frac{\sqrt{3}}{3}\right)\text{m}$

Q 3. From the top of a tower, the angle of depression of a point on the ground is $60°$. If the

distance of this point from the tower is $\frac{1}{\sqrt{3}+1}$ m, then the height of the tower is

[Kerala CEE]

(a) $\frac{4\sqrt{3}}{2}$ m

(b) $\frac{\sqrt{3}+3}{2}$ m

(c) $\frac{3-\sqrt{3}}{2}$ m

(d) $\frac{\sqrt{3}}{2}$ m

(e) $\sqrt{3}+1$ m

Q 4. P is a point on the segment joining the feet of two vertical poles of heights a and b. The angles of elevation of the tops of the poles from P are $45°$ each. Then, the square of the distance between the tops of the poles is

[EAMCET]

(a) $\frac{a^2 + b^2}{2}$

(b) $a^2 + b^2$

(c) $2(a^2 + b^2)$

(d) $4(a^2 + b^2)$

Q 5. ABC is a triangular park with AB = AC = 100 m. A clock tower is situated at the mid point of BC. The angle of elevation, if the top of the tower at A and B are $\cot^{-1} 3.2$ and $\text{cosec}^{-1} 2.6$ respectively. The height of the tower is

[Jamia Millia Islamia]

(a) 16 m

(b) 25 m

(c) 50 m

(d) None of these

Q 6. AB is a vertical pole with B at the ground level and A at the top. A man finds that the angle of elevation of the point A from a certain point C on the ground is $60°$. He moves away from the pole along the line BC to a point D such that CD = 7 m.

From D the angle of elevation of the point A is $45°$.

Then the height of the pole is

[AIEEE]

(a) $\frac{7\sqrt{3}}{2}\left(\frac{1}{\sqrt{3}+1}\right)\text{m}$

(b) $\frac{7\sqrt{3}}{2}\left(\frac{1}{\sqrt{3}-1}\right)\text{m}$

(c) $\frac{7\sqrt{3}}{2}(\sqrt{3}+1)\text{m}$

(d) $\frac{7\sqrt{3}}{2}(\sqrt{3}-1)\text{m}$

Q 7. A house subtends a right angle at the window of an opposite house and the angle of elevation of the window from the bottom of the first house is $60°$. If the distance between the two houses be 6 m, then the height of the first house is

[DCE]

(a) $8\sqrt{3}$ m

(b) $6\sqrt{3}$ m

(c) $4\sqrt{3}$ m

(d) None of these

Q 8. From the top of a hill h metres high, the angles of depressions of the top and the bottom of a pillar are α and β respectively. The height (in metres) of the pillar is

[BITSAT, AMU]

(a) $\frac{h(\tan\beta - \tan\alpha)}{\tan\beta}$

(b) $\frac{h(\tan\alpha - \tan\beta)}{\tan\alpha}$

(c) $\frac{h(\tan\beta + \tan\alpha)}{\tan\beta}$

(d) $\frac{h(\tan\beta + \tan\alpha)}{\tan\alpha}$

Q 9. The angle of elevation of the top of a TV tower from three points A, B and C in a straight line through the foot of the tower are α, 2α and 3α respectively. If AB = a, then height of the tower is

[KCET]

(a) a tan α

(b) a sin α

(c) a sin 2α

(d) a sin 3α

Q 10. A flagpole stands on a building of height 450 ft and an observer on a level ground is 300 ft from the base of the building. The angle of elevation of the bottom of the flagpole is $30°$ and the height of the flagpole is 50 ft. If θ is the angle of elevation of the top of the flagpole, then tan θ is equal to

[Kerala CEE]

(a) $\frac{4}{3\sqrt{3}}$

(b) $\frac{\sqrt{3}}{2}$

(c) $\frac{9}{2}$

(d) $\frac{\sqrt{3}}{5}$

(e) $\frac{4\sqrt{3}+1}{6}$

Q 11. The elevation of an object on a hill is observed from a certain point in the horizontal

plane through its base, to be $30°$. After walking 120 m towards it on level ground the elevation is found to be $60°$. Then the height of the object (in metres) is

[BCECE]

(a) 120

(b) $60\sqrt{3}$

(c) $120\sqrt{3}$

(d) 60

Q 12. From an aeroplane flying, vertically above a horizontal road, the angles of depression of two consecutive stones on the same side of the aeroplane are observed to be $30°$ and $60°$ respectively. The height at which the aeroplane is flying in km, is

[Jamia Millia Mallila]

(a) $\frac{4}{\sqrt{3}}$

(b) $\frac{\sqrt{3}}{2}$

(c) $\frac{2}{\sqrt{3}}$

(d) 2

Q 13. The base of a cliff is circular. From the extremities of a diameter of the base angles of elevation of the top of the cliff are $30°$ and $60°$. If the height of the cliff be 500 m, then the diameter of the base of the cliff is

[Manipal]

(a) $\frac{2000}{\sqrt{3}}$ m

(b) $\frac{1000}{\sqrt{3}}$ m

(c) $\frac{2000}{\sqrt{2}}$ m

(d) $1000\sqrt{3}$ m

Q 14. A round balloon of radius r subtends an angle α at the eye of the observer, while the angle of elevation of its centre is β. The height of the centre of balloon is

[Manipal]

(a) r cosec α sin $\frac{\beta}{2}$

(b) r sin α cosec $\frac{\beta}{2}$

(c) r sin $\frac{\alpha}{2}$cosec β

(d) r cosec $\frac{\alpha}{2}$sin β

Q 15. The shadow of tower standing on a level ground is x metres long when the sun's altitude is $30°$, while it is y metres long when the altitude is $60°$. If the height of the tower is $45.\frac{\sqrt{3}}{2}$m, then x − y is

[MP PET]

(a) 45 m

(b) $45\sqrt{3}\,\text{m}$

(c) $\frac{45}{\sqrt{3}}\text{m}$

(d) $45.\frac{\sqrt{3}}{2}\text{m}$

Q 16. A tower stands at the centre of a circular park. A and B are two points on the boundary of

the park such that AB(=a) subtends an angle of $60°$ at the foot of the tower and the angle of elevation of the top of the tower from A or B is $30°$. The height of the tower is

[AIEEE]

(a) $\frac{2a}{\sqrt{3}}$

(b) $2a\sqrt{3}$

(c) $\frac{a}{\sqrt{3}}$

(d) $\sqrt{3}$

Q 17. The angle of elevation of top of a tower from a point on the ground is $30°$ and it is $60°$ when it is viewed from a point located 40 m away from the initial point towards the tower. The height of the tower is

[DCE]

(a) $-20\sqrt{3}\,\text{m}$

(b) $\frac{\sqrt{3}}{20}\text{m}$

(c) $-\frac{\sqrt{3}}{20}$ m

(d) $20\sqrt{3}$ m

Q 18. The horizontal distance between two towers is 60 m and the angle of depression of the top of the first tower as seen from the top of the second is $30°$. If the height of the second tower be 150 m, then the height of the first tower is

[BITSAT]

(a) 90 m

(b) $(150-60\sqrt{3})$m

(c) $(150+20\sqrt{3})$m

(d) None of the above

Q 19. The angle of elevation of the sun, if the length of the shadow of a tower is $\sqrt{3}$ times the height of the pole, is

(a) $150°$

(b) $30°$

(c) $60°$

(d) $45°$

Q 20. The angle of elevation of an object from a point a level ground is α. Moving d metres on the ground towards the object, the angle of elevation is found to be β. Then the height (in metres) of the object is

(a) d tan α

(b) d cot β

(c) $\frac{d}{\cot\alpha + \cot\beta}$

(d) $\frac{d}{\cot\alpha - \cot\beta}$

Q 21. A house of height 100 m subtends a right angle at the window of an opposite house. If the height of the window be 64 m, then the distance between the two houses is

[WBJEE]

(a) 48 m

(b) 36 m

(c) 54 m

(d) 72 m

Q 22. On one bank of river there is a tree. On another bank, an observer makes an angle of elevation of $60°$ at the top of the tree. The angle of elevation of the top of the tree at a distance 20 m away from the bank is $30°$. The width of the river is

[BCECE]

(a) 20 m

(b) 10 m

(c) 5 m

(d) 1 m

Q 23. The angle of elevation of the top of the tower observed from each of the three point A,B,C, on the ground forming a triangle is the same angle α. If R is the circumradius of the triangle ABC, then the height of the tower is

[DCE]

(a) R sin α

(b) R cos α

(c) R cot α

(d) R tan α

Q 24. A flag is standing vertically on a tower of height b. On a point at a distance a from the foot of the tower, the flag and the tower subtend equal angles. The height of the flag is

(a) b. $\frac{a^2+b^2}{a^2-b^2}$

(b) a. $\frac{a^2-b^2}{a^2+b^2}$

(c) b. $\frac{a^2-b^2}{a^2+b^2}$

(d) a. $\frac{a^2+b^2}{a^2-b^2}$

Q 25. A tower subtends an angle a at a point A in the plane of its base and the angle of depression

of the foot of the tower at a point b feet just above

A is β. Then, the height of the tower is

[Kerala CEE]

(a) b tan α cot β

(b) b cot α tan β

(c) b cot α cot β

(d) b tan α tan β

(e) b $\tan^2 \alpha$ cot β

Q 26. Two pillars of equal height stand on either side of a road-way which is 60 m wide. At a point in the road-way between the pillars, the elevation of the top of pillars are $60°$ and $30°$. The height of the pillars is

[MP PET]

(a) $15\sqrt{3}$ m

(b) $\frac{15}{\sqrt{3}}$ m

(c) 15 m

(d) 20 m

Q 27. The shadow of a tower is found to be 60 m shorter when the sun's altitude changes from $30°$ to $60°$. The height of the tower from the ground is approximately equal to

[Kerala CEE]

(a) 62 m

(b) 301 m

(c) 101 m

(d) 75 m

(e) 52 m

Q 28. ABCD is a rectangular field. A vertical lamp post of height 12 m stands at the corner A. If the angle of elevation of its top from B is $60°$ and from C is $45°$, then the area of the field is

[Kerala CEE]

(a) $48\sqrt{2}$ sq m

(b) $48\sqrt{3}$ sq m

(c) 48 sq m

(d) $12\sqrt{2}$ sq m

(e) $12\sqrt{3}$ sq m

Q 29. A tower of x metres high, has a flagstaff at its top. The tower and the flagstaff subtend equal angles at a point distant y metres from the foot of the tower. Then the length of the flagstaff (in metres), is

[EAMCET]

(a) $\frac{y(x^2-y^2)}{(x^2+y^2)}$

(b) $\frac{x(y^2+x^2)}{(y^2-x^2)}$

(c) $\frac{x(x^2+y^2)}{(x^2-y^2)}$

(d) $\frac{x(x^2-y^2)}{(x^2+y^2)}$

Q 30. A person standing on the bank of a river, observes that the angle of elevation of the top of a

tree on the opposite bank of the river is $60°$ and when he retires 40 m away from the tree the angle of elevation becomes $30°$. The breadth of the river is

[AIEEE]

(a) 20 m

(b) 30 m

(c) 40 m

(d) 60 m

Q 31. An aeroplane flying with uniform speed horizontally one kilometre above the ground is observed at an elevation of $60°$. After 10 s, if the elevation is observed to be $30°$, then the speed of the plane (in km/h) is

[EAMCET]

(a) $\frac{240}{\sqrt{3}}$

(b) $200\sqrt{3}$

(c) $240\sqrt{3}$

(d) $\frac{120}{\sqrt{3}}$

Q 32. The angle of depression of a ship from the top of a tower 30 m high is $60°$. Then the distance of ship from the base of tower is

[Jamia Millia Islamia]

(a) 30 m

(b) $30\sqrt{3}$ m

(c) $10\sqrt{3}$ m

(d) 10 m

Q 33. The angle of depression of a boat in a river is $30°$ from the top of a tower, 87 m high and the speed of the boat is 5.8 km/h. The time taken by the boat to reach at the base of the tower is

[J&K GET]

(a) 9 min

(b) $\frac{9\sqrt{3}}{10}$ min

(c) 25 min

(d) 15 min

Q 34. The angular depression of the top and the foot of the chimney as seen from the top of a second chimney which is 150 m high and standing on the same level as the first are θ and ϕ respectively. The distance between their tops when $\tan\theta = \frac{4}{3}$ and $\tan\phi = \frac{5}{2}$ is equal to

[Guj. CET]

(a) 50 m

(b) 100 m

(c) 150 m

(d) None of these

Q 35. The upper $\left(\frac{3}{4}\right)$th portion of a vertical pole subtends an angle $\tan^{-1}\left(\frac{3}{5}\right)$ at a point in the horizontal lane through its foot and at a distance 40 m from the foot. A possible height of the vertical pole is

(a) 20 m

(b) 40 m

(c) 60 m

(d) 80 m

[AIEEE]

Q 36. A person observes the angle of elevation of a building as $30°$. The person proceeds towards the building with a speed of $25(\sqrt{3}-1)$ m/h. After two

hours, he observes the angle of elevation as $45°$.

The height of the building (in metres) is

[UP SEE]

(a) $50(\sqrt{3}-1)$

(b) $50(\sqrt{3}+1)$

(c) 50

(d) 100

Q 37. A tower subtends angles α, 2α and 3α respectively at points A, B and C, all lying on a

horizontal line through the foot of the tower, then

$\frac{AB}{BC}$ is equal to

[EAMCET]

(a) $\frac{\sin 3\alpha}{\sin 2\alpha}$

(b) $1 + 2 \cos 2\alpha$

(c) $2 \cos 2\alpha$

(d) $\frac{\sin 2\alpha}{\sin \alpha}$

Q 38. The angle of elevation of the cloud at a point 2500 m high from the lake is $15°$ and the angle of depression of its reflection to the lake is $45°$. Then the height of cloud from the foot of lake is

[MP PET]

(a) $2500\sqrt{3}$ m

(b) 2500 m

(c) $500\sqrt{3}$ m

(d) None of these

Answers

1. (a) 2. (d) 3. (c) 4. (c) 5. (b) 6. (c) 7. (a) 8, (a)

9. (c) 10. (a) 11. (b) 12. (b) 13. (a) 14. (d) 15. (a)

16. (c) 17. (d) 18. (c) 19. (b) 20. (c) 21. (a) 22.

(b) 23. (d) 24. (a) 25. (a) 26. (a) 27. (e) 28. (a)

29. (b) 30. (a) 31. (c) 32. (c) 33. (b) 34. (b) 35.

(b) 36. (c) 37. (b) 38. (a)

Inverse Trigonometric Functions

Topic 1

Solutions of Inverse Trigonometrical Functions

Q 1. Value of $\tan^{-1}\left(\frac{\sin 2x - 1}{\cos 2}\right)$ is

[WB JEE]

(a) $\frac{\pi}{2} - 1$

(b) $1 - \frac{\pi}{4}$

(c) $2-\frac{\pi}{2}$

(d) $\frac{\pi}{4}-1$

Q 2. The value of $\sin^{-1}\{\cos(4095°)\}$ is equal to

[Kerala CEE]

(a) $-\frac{\pi}{3}$

(b) $\frac{\pi}{6}$

(c) $-\frac{\pi}{4}$

(d) $\frac{\pi}{4}$

(e) $\frac{\pi}{2}$

Q 3. If $a_1, a_2, a_3, \ldots, a_n$ are in AP with common difference 5 and if $a_i a_j \neq -1$ for i, j = $1,2,\ldots,$ n then

$$\tan^{-1}\left(\frac{5}{1+a_1a_2}\right)+\tan^{-1}\left(\frac{5}{1+a_2a_3}\right)+.....+\tan^{-1}\left(\frac{5}{1+a_{n-1}a_n}\right)$$

is equal to

[Kerala CEE]

(a) $\tan^{-1}\left(\frac{5}{1+a_na_{n-1}}\right)$

(b) $\tan^{-1}\left(\frac{5a_1}{1+a_na_1}\right)$

(c) $\tan^{-1}\left(\frac{5n-5}{1+a_n a_1}\right)$

(d) $\tan^{-1}\left(\frac{5n-5}{1+a_1 a_{n+1}}\right)$

(e) $\tan^{-1}\left(\frac{5n}{1+a_1 a_n}\right)$

Q 4. If $\tan^{-1}(x+2) + \tan^{-1}(x-2) - \tan^{-1}\left(\frac{1}{2}\right) = 0$,

then one of the values of x is equal to

[Kerala CEE]

(a) -1

(b) 5

(c) $\frac{1}{2}$

(d) 1

(e) $-\frac{1}{2}$

Q 5. If cot $(\cos^{-1} x)$ = sec $\left(\tan^{-1}\frac{a}{\sqrt{b^2-a^2}}\right)$, then x is equal to

[UP SEE]

(a) $\frac{b}{\sqrt{2b^2-a^2}}$

(b) $\frac{a}{\sqrt{2b^2 - a^2}}$

(c) $\frac{\sqrt{2b^2 - a^2}}{a}$

(d) $\frac{\sqrt{2b^2 - a^2}}{b}$

Q 6. Number of solutions of the equation

$\tan^{-1}\left(\frac{1}{2x+1}\right) + \tan^{-1}\left(\frac{1}{4x+1}\right) = \tan^{-1}\left(\frac{2}{x^2}\right)$ is

[UP SEE]

(a) 1

(b) 2

(c) 3

(d) 4

Q 7. The value of $\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{1}{3}+\tan^{-1}\frac{7}{8}$ is

(a) $\tan^{-1}\frac{7}{8}$

(b) $\cot^{-1}15$

(c) $\tan^{-1}15$

(d) $\tan^{-1}\frac{25}{24}$

[UP SEE]

Q 8. $\cot^{-1}(2 \cdot 1^2) + \cot^{-1}(2 \cdot 2^2 j + \cot^{-1}(2 \cdot 3^2) +$

…up to ∞ is equal to

[KCET]

(a) $\frac{\pi}{4}$

(b) $\frac{\pi}{3}$

(c) $\frac{\pi}{2}$

(d) $\frac{\pi}{5}$

Q 9. If x takes negative permissible value, then $\sin^{-1} x$ is equal to

[KCET]

(a) $-\cos^{-1}\sqrt{1-x^2}$

(b) $\cos^{-1}\sqrt{x^2-1}$

(c) $\pi-\cos^{-1}\sqrt{1-x^2}$

(d) $\cos^{-1}\sqrt{1-x^2}$

Q 10. The value of $\cos[\tan^{-1}\{\sin(\cot^{-1} x)\}]$ is

[Kerala CEE]

(a) $\sqrt{\frac{x^2+1}{x^2-1}}$

(b) $\sqrt{\frac{1-x^2}{x^2+2}}$

(c) $\sqrt{\frac{1-x^2}{1+x^2}}$

(d) $\sqrt{\frac{x^2+1}{x^2+2}}$

(e) $\sqrt{\frac{1-x^2}{2-x^2}}$

Q 11. If $-\frac{\pi}{2} < \sin^{-1} x < \frac{\pi}{2}$, then tan $(\sin^{-1} x)$. is equal to

[Kerala CEE]

(a) $\frac{x}{1-x^2}$

(b) $\frac{x}{1+x^2}$

(c) $\frac{x}{\sqrt{1-x^2}}$

(d) $\frac{1}{\sqrt{1-x^2}}$

(e) $\frac{x}{\sqrt{x^2-1}}$

Q 12. If $\cos^{-1} x + \cos^{-1} y + \cos^{-1} z + \cos^{-1} t = 4\pi$, then the value of $x^2 + y^2 + z^2 + t^2$ is

[AMU]

(a) xy + zy + zt

(b) $1 - 2$xyzt

(c) 4

(d) 6

Q 13. $\cos^{-1}\left(\frac{-1}{2}\right) - 2\sin^{-1}\left(\frac{1}{2}\right) + 3\cos^{-1}\left(\frac{-1}{\sqrt{2}}\right)$ $4\ \tan^{-1}(-1)$

equals

[EAMCET]

(a) $\frac{19\pi}{12}$

(b) $\frac{35\pi}{12}$

(c) $\frac{47\pi}{12}$

(d) $\frac{43\pi}{12}$

Q 14. $\tan\left[\frac{\pi}{4}+\frac{1}{2}\cos^{-1}\left(\frac{a}{b}\right)\right]+\tan\left[\frac{\pi}{4}-\frac{1}{2}\cos^{-1}\left(\frac{a}{b}\right)\right]$ is equal to

[WB JEE]

(a) $\frac{2a}{b}$

(b) $\frac{2b}{a}$

(c) $\frac{a}{b}$

(d) $\frac{b}{a}$

Q 15. If $\sin^{-1}\frac{1}{3}+\sin^{-1}\frac{2}{3} = \sin^{-1} x$, then the value of x is

[VITEEE]

(a) 0

(b) $\frac{(\sqrt{5}-4\sqrt{2})}{9}$

(c) $\frac{(\sqrt{5}+4\sqrt{2})}{9}$

(d) $\frac{\pi}{2}$

Q 16. The value of $\cos(2\cos^{-1}0.8)$ is

[Jamia Millia Islamia]

(a) 0.48

(b) 0.96

(c) 0.6

(d) None of these

Q 17. If $\tan^{-1} 2$ and $\tan^{-1} 3$ are two angles of a triangle, then the third angle is

[J&K CET]

(a) $\frac{\pi}{2}$

(b) $\frac{\pi}{3}$

(c) $\frac{\pi}{4}$

(d) $\frac{\pi}{6}$

Q 18. If $0 < x < 1$, then $\sqrt{1+x^2}$ [{x cos(cot^{-1} x) + sin(cot^{-1}x)}^{2} – 1]$^{1/2}$ is equal to

[IITJEE]

(a) $\frac{x}{\sqrt{1+x^2}}$

(b) x

(c) $x\sqrt{1+x^2}$

(d) $\sqrt{1+x^2}$

Q 19. The value of cot $\left(\cos ec^{-1}\frac{5}{3}+\tan^{-1}\frac{2}{3}\right)$ is

[AIEEE]

(a) $\frac{5}{17}$

(b) $\frac{6}{17}$

(c) $\frac{3}{17}$

(d) $\frac{4}{17}$

Q 20. If $\sin^{-1}\left(\frac{3}{x}\right)+\sin^{-1}\left(\frac{4}{x}\right)=\frac{\pi}{2}$, then x is equal to

[BITSAT]

(a) 3

(b) 5

(c) 7

(d) 11

Q 21. If $\tan^{-1}(x-1) + \tan^{-1} x + \tan^{-1}(x+1) = \tan^{-1} 3x$, then x is

[UP SEE]

(a) $\pm\frac{1}{2}$

(b) $0, \frac{1}{2}$

(c) $0, -\frac{1}{2}$

(d) $0, \pm\frac{1}{2}$

Q 22. $\sin\left(2\sin^{-1}\sqrt{\frac{63}{65}}\right)$ is equal to

[KCET]

(a) $\frac{2\sqrt{126}}{65}$

(b) $\frac{4\sqrt{65}}{65}$

(c) $\frac{8\sqrt{63}}{65}$

(d) $\frac{\sqrt{63}}{65}$

Q 23. $5\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)+7\sin^{-1}\left(\frac{2x}{1+x^2}\right)-4\tan^{-1}\left(\frac{2x}{1-x^2}\right)$ – $\tan^{-1}$

$x = 5\pi$, then x is equal to

[Kerala CEE]

(a) 3

(b) $-\sqrt{3}$

(c) $\sqrt{2}$

(d) 2

(e) $\sqrt{3}$

Q 24. If $(\tan^{-1} x)^2 + (\cot^{-1} x)^2 = \frac{5\pi^2}{8}$, then x is equal to

[Kerala CEE]

(a) 0

(b) 2

(c) 1

(d) -1

(e) $2\sqrt{2}$

Q 25. The principal value of $\sin^{-1} \tan\left(\frac{-5\pi}{4}\right)$ is

[WBJEE]

(a) $\frac{\pi}{4}$

(b) $-\frac{\pi}{4}$

(c) $\frac{\pi}{2}$

(d) $-\frac{\pi}{2}$

Q 26. The simplified expression of $\sin(\tan^{-1} x)$, for any real number x is given by

[VITEEE]

(a) $\frac{1}{\sqrt{1+x^2}}$

(b) $\frac{x}{\sqrt{1+x^2}}$

(c) $-\frac{1}{\sqrt{1+x^2}}$

(d) $-\frac{x}{\sqrt{1+x^2}}$

Q 27. If $\cos^{-1} x = \alpha$, $(0 < x < 1)$ and

$\sin^{-1}(2x\sqrt{1-x^2}) + \sec^{-1}\left(\frac{1}{2x^2-1}\right) = \frac{2\pi}{3}$, then $\tan^{-1}(2x)$ equals

[Jamia Millia Islamia]

(a) $\pi/6$

(b) $\pi/4$

(c) $\pi/3$

(d) $\pi/2$

Q 28. If $\alpha = \sin^{-1}\frac{\sqrt{3}}{2} + \sin^{-1}\frac{1}{3}$ and $\beta = \cos^{-1}\frac{\sqrt{3}}{2} + \cos^{-1}\frac{1}{3}$ then

[Manipal]

(a) $\alpha > \beta$

(b) $\alpha = \beta$

(c) $\alpha < \beta$

(d) $\alpha + \beta = 2\pi$

Q 29. The solution of $\tan^{-1} x + 2 \cot^{-1} x = \frac{2\pi}{3}$ is

[Manipal]

(a) $-\frac{1}{\sqrt{3}}$

(b) $\frac{1}{\sqrt{3}}$

(c) $-\sqrt{3}$

(d) $\sqrt{3}$

Q 30. If $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \pi$, then the value of $x + y + z$ is

[MPPET]

(a) $-xyz$

(b) xyz

(c) $\frac{1}{xyz}$

(d) 0

Q 31. If $\tan^{-1}\left(\frac{1}{3}\right)+\tan^{-1}\left(\frac{3}{4}\right)-\tan^{-1}\left(\frac{x}{3}\right)=0$, then x is equal to

[MPPET]

(a) $\frac{7}{3}$

(b) 3

(c) $\frac{11}{3}$

(d) $\frac{13}{3}$

Q 32. $\tan^{-1}\frac{x}{\sqrt{a^2-x^2}}$ is equal to

[J&KCET]

(a) $2\sin^{-1}\frac{x}{a}$

(b) $\sin^{-1}\frac{2x}{a}$

(c) $\sin^{-1}\frac{x}{a}$

(d) $\cos^{-1}\frac{x}{a}$

Q 33. The solution of $\tan^{-1} 2\theta + \tan^{-1} 3\theta = \frac{\pi}{4}$

[J&KCET]

(a) $\frac{1}{\sqrt{3}}$

(b) $\frac{1}{3}$

(c) $\frac{1}{6}$

(d) $\frac{1}{\sqrt{6}}$

Q 34. If $\sec^{-1}\sqrt{1+x^2}+\operatorname{cosec}^{-1}\frac{\sqrt{1+y^2}}{y}+\cot^{-1}\frac{1}{z}=\pi$, then x + y + z is equal to

[Guj. CET]

(a) xyz

(b) 2xyz

(c) xyz^2

(d) x^2yz

Q 35. If $\sin^{-1}\left(\frac{x}{5}\right)+\operatorname{cosec}^{-1}\left(\frac{5}{4}\right)=\frac{\pi}{2}$, then value of x is

[AIEEE]

(a) 1

(b) 3

(c) 4

(d) 5

Q 36. If $\sqrt{3}$ + i = (a + ib)(c + id), then $\tan^{-1}$

$\left(\frac{b}{a}\right) + \tan^{-1}\left(\frac{d}{c}\right)$ has the value

[DCE]

(a) $\frac{\pi}{3} + 2n\pi$, $n \in I$

(b) $n\pi + \frac{\pi}{6}$, $n \in I$

(c) $n\pi - \frac{\pi}{3}$, $n \in I$

(d) $2n\pi - \frac{\pi}{3}$, $n \in I$

Q 37. If $\sin^{-1}\frac{2a}{1+a^2}-\cos^{-1}\frac{1-b^2}{1+b^2}=\tan^{-1}\frac{2x}{1-x^2}$, then the value of x is

[DCE]

(a) a

(b) b

(c) $\frac{a+b}{1-ab}$

(d) $\frac{a-b}{1+ab}$

Q 38. The value of $\sin\left[2\cos^{-1}\frac{\sqrt{5}}{3}\right]$ is

[KCET]

(a) $\frac{\sqrt{5}}{3}$

(b) $\frac{2\sqrt{5}}{3}$

(c) $\frac{4\sqrt{5}}{9}$

(d) $\frac{2\sqrt{5}}{9}$

Q 39. $\tan^{-1}\frac{m}{n}-\tan^{-1}\frac{m-n}{m+n}$ is equal to

[Kerala CEE]

(a) $\tan^{-1}\frac{n}{m}$

(b) $\tan^{-1}\frac{m+n}{m-n}$

(c) $\frac{\pi}{4}$

(d) $\tan^{-1}\left(\frac{1}{2}\right)$

(e) $\frac{\pi}{2}$

Q 40. If $2\ \tan^{-1}(\cos x) = \tan^{-1}(2\ \text{cosec}\ x)$, then the value of x is

[AMU]

(a) $\frac{3\pi}{4}$

(b) $\frac{\pi}{4}$

(c) $\frac{\pi}{3}$

(d) None of these

Q 41. The value of x, where x > 0 and

$\tan\left\{\sec^{-1}\left(\frac{1}{x}\right)\right\} = \sin(\tan^{-1} 2)$ is

[EAMCET]

(a) $\sqrt{5}$

(b) $\frac{\sqrt{5}}{3}$

(c) 1

(d) $\frac{2}{3}$

Q 42. If $\tan^{-1}\left(\frac{a}{x}\right)+\tan^{-1}\left(\frac{b}{x}\right)=\frac{\pi}{2}$, then x is equal to

[WBJEE]

(a) $\sqrt{ab}$

(b) $\sqrt{2ab}$

(c) 2ab

(d) ab

Q 43. If $f(x) = \sin^{-1}\left\{\frac{\sqrt{3}}{2}x - \frac{1}{2}\sqrt{1-x^2}\right\}, -\frac{1}{2} \leq x \leq 1$, then

f(x) is equal to

[WBJEE]

(a) $\sin^{-1}\frac{1}{2} - \sin^{-1}x$

(b) $\sin^{-1}x - \frac{\pi}{6}$

(c) $\sin^{-1}x + \frac{\pi}{6}$

(d) None of these

Q 44. If $\sin^{-1}x - \cos^{-1}x = \frac{\pi}{6}$, then x is

[WBJEE]

(a) $\frac{1}{2}$

(b) $\frac{\sqrt{3}}{2}$

(c) $\frac{-1}{2}$

(d) None of these

Q 45. The number of real solutions of

$\tan^{-1}\sqrt{x(x+1)}+\sin^{-1}\sqrt{x^2+x+1}=\frac{\pi}{2}$ is

[BCECE]

(a) 0

(b) 1

(c) 2

(d) ∞

Q 46. If $\sin^{-1} x + \sin^{-1} y = \frac{\pi}{2}$, then $\cos^{-1} x + \cos^{-1}$

y is equal to

[VITEEE]

(a) $\frac{\pi}{2}$

(b) $\frac{\pi}{4}$

(c) π

(d) $\frac{3\pi}{4}$

Q 47. If $\sin^{-1} x = \frac{\pi}{5}$, for some $x \in (-1, 1)$, then the value of $\cos^{-1} x$ is

[MPPET]

(a) $\frac{3\pi}{10}$

(b) $\frac{5\pi}{10}$

(c) $\frac{7\pi}{10}$

(d) $\frac{9\pi}{10}$

Q 48. The value of $\tan\left\{\cos^{-1}\left(-\frac{2}{7}\right)-\frac{\pi}{2}\right\}$ is

[RPET]

(a) $\frac{2}{3\sqrt{5}}$

(b) $\frac{2}{3}$

(c) $\frac{1}{\sqrt{5}}$

(d) $\frac{4}{\sqrt{5}}$

Q 49. The value of $\cot^{-1} 9 + \operatorname{cosec}^{-1} \frac{\sqrt{41}}{4}$ is

[J&KCET]

(a) $\frac{\pi}{2}$

(b) $\frac{\pi}{4}$

(c) $\frac{\pi}{3}$

(d) π

Q 50. The value of $\cos^{-1}\left(-\frac{1}{2}\right)$ among the following,

is

[J&KCET]

(a) $\frac{9\pi}{3}$

(b) $\frac{8\pi}{3}$

(c) $\frac{5\pi}{3}$

(d) $\frac{11\pi}{3}$

Q 51. If $\sin^{-1}(1-x) - 2\sin^{-1}x = \frac{\pi}{2}$, then x

equals

[Guj. CET]

(a) $0, -\frac{1}{2}$

(b) $0, \frac{1}{2}$

(c) 0

(d) None of these

Q 52. If $\tan^{-1}\frac{1-x}{1+x} = \frac{1}{2}\tan^{-1}x$, then the value of x is

[DCE]

(a) $\frac{1}{2}$

(b) $\frac{1}{\sqrt{3}}$

(c) $\sqrt{3}$

(d) 2

Q 53. The equation $\sin^{-1} x - \cos^{-1} x = \cos^{-1}$

$\left(\frac{\sqrt{3}}{2}\right)$ has

[DCE]

(a) no solution

(b) unique solution

(c) infinite number of solutions

(d) None of the above

Q 54. If a > b > 0, then the value of

$\tan^{-1}\left(\frac{a}{b}\right)+\tan^{-1}\left(\frac{a+b}{a-b}\right)$ depends on

[KCET]

(a) Both a and b

(b) b and not a

(c) a and not b

(d) Neither a nor b

Q 55. The value of cos $[2 \tan^{-1}(-7)]$ is

[Kerala CEE]

(a) $\frac{49}{50}$

(b) $-\frac{49}{50}$

(c) $\frac{24}{25}$

(d) $-\frac{24}{25}$

(e) $\frac{48}{49}$

Q 56. $\sum_{m=1}^{n} \tan^{-1}\left(\frac{2m}{m^4+m^2+2}\right)$ is equal to

[AMU]

(a) $\tan^{-1}\left(\frac{n^2+n}{n^2+n+2}\right)$

(b) $\tan^{-1}\left(\frac{n^2-n}{n^2-n+2}\right)$

(c) $\tan^{-1}(n^2+n+2)$

(d) None of these

Q 57. $\cot^{-1}(\sqrt{\cos\alpha}) - \tan^{-1}(\sqrt{\cos\alpha}) = x$, then sin x is equal to

[OJEE]

(a) $\tan^2\left(\frac{\alpha}{2}\right)$

(b) $\cot^2\left(\frac{\alpha}{2}\right)$

(c) $\tan\alpha$

(d) $\cot\left(\frac{\alpha}{2}\right)$

Q 58. The value of

$\sin\left(\sin^{-1}\frac{1}{3}+\sec^{-1}3\right)+\cos\left(\tan^{-1}\frac{1}{2}+\tan^{-1}2\right)$ is

[WB JEE, MP PET]

(a) 1

(b) 2

(c) 3

(d) 4

Q 59. If α, β are the roots of the equation $6x^2 - 5x + 1 = 0$, then the value of $\tan^{-1}\alpha + \tan^{-1}\beta$ is

[WBJEE]

(a) 0

(b) $\pi/4$

(c) 1

(d) $\pi/2$

Q 60. If $4 \sin^{-1} x + \cos^{-1} x = \pi$, then x is equal to

[BCECE]

(a) 0

(b) $1/2$

(c) $-1/2$

(d) 1

Q 61. $\tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)-\tan^{-1}\left(\frac{2x}{1-x^2}\right)$ is equal to

[J&KCET]

(a) 0

(b) 1

(c) $\tan^{-1} x$

(d) $\tan^{-1} 2x$

Q 62. If $\tan^{-1} x + \tan^{-1} y = \frac{\pi}{4}$, then

[KCET]

(a) $x + y + xy = 1$

(b) $x + y - xy = 1$

(c) $x + y + xy + 1 = 0$

(d) $x + y - xy + 1 = 0$

Q 63. If $\sin\left(\sin^{-1}\frac{1}{5} + \cos^{-1} x\right) = 1$, then the value of x is

[Kerala CEE]

(a) -1

(b) $2/5$

(c) $1/3$

(d) 1

(e) $1/5$

Q 64. $\sin\left[3\sin^{-1}\left(\frac{1}{5}\right)\right]$ is equal to

[Kerala CEE]

(a) $\frac{71}{125}$

(b) $\frac{74}{125}$

(c) $\frac{3}{5}$

(d) $\frac{1}{2}$

(e) $\frac{-3}{5}$

Q 65. If $\angle A = 90°$ in the triangle ABC, then

$\tan^{-1}\left(\frac{c}{a+b}\right)+\tan^{-1}\left(\frac{b}{a+c}\right)$ is equal to

[Kerala CEE]

(a) 0

(b) 1

(c) $\frac{\pi}{4}$

(d) $\frac{\pi}{6}$

(e) $\frac{\pi}{8}$

Q 66. If x, y, z are in AP and $\tan^{-1} x$, $\tan^{-1} y$ and $\tan^{-1} z$ are also in AP, then

[Kerala CEE]

(a) x = y = z

(b) x = y = −z

(c) x = 1, y = 2, z = 3

(d) x = 2,y = 4,z = 6

(e) x = 2, y = 3z

Q 67. The number of positive integral solutions of the equation $\tan^{-1} x + \cos^{-1} \frac{y}{\sqrt{1+y^2}} = \sin^{-1} \frac{3}{\sqrt{10}}$ is

[AMU]

(a) one

(b) two

(c) zero

(d) None of these

Q 68. $\sin^{-1}\frac{4}{5}+2\tan^{-1}\frac{1}{3}$ is equal to

[EAMCET]

(a) $\frac{\pi}{3}$

(b) $\frac{\pi}{4}$

(c) $\frac{\pi}{2}$

(d) 0

Q 69. For the equation $\cos^{-1} x + \cos^{-1} 2x + \pi = 0$, the number of real solutions is

[OJEE]

(a) 1

(b) 2

(c) 0

(d) ∞

Q 70. Which one of the following is true?

[MPPET]

(a) $\sin(\cos^{-1} x) = \cos(\sin^{-1} x)$

(b) $\sec(\tan^{-1} x) - \tan(\sec^{-1} x)$

(c) $\cos(\tan^{-1} x) = \tan(\cos^{-1} x)$

(d) $\tan(\sin^{-1} x) = \sin(\tan^{-1} x)$

Q 71. If $\tan^{-1} a + \tan^{-1} b = \sin^{-1} 1 - \tan^{-1} c$,

then

[RPET]

(a) a + b + c = abc

(b) ab +bc + ca = abc

(c) $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}-\frac{1}{abc}=0$

(d) ab + bc + ca = a + b + c

Q 72. The value of $\cos^{-1}(\cos 12) - \sin^{-1}(\sin 14)$ is

[J&KCET]

(a) 2

(b) $8\pi - 26$

(c) $4\pi + 2$

(d) None of these

Q 73. If $\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = 3\pi$, then

$xy + yz + zx$ is equal to

[Guj. CET]

(a) 1

(b) 0

(c) −3

(d) 3

Q 74. The value of x for which sin $[\cot^{-1}(1 + x)]$ = cos $(\tan^{-1} x)$is

[IITJEE]

(a) $\frac{1}{2}$

(b) 1

(c) 0

(d) $-\frac{1}{2}$

Q 75. $\cos^{-1}\frac{1}{2}+2\sin^{-1}\frac{1}{2}$ is equal to

(a) $\frac{\pi}{4}$

(b) $\frac{\pi}{6}$

(c) $\frac{\pi}{3}$

(d) $\frac{2\pi}{3}$

Q 76. If $\cos^{-1} p + \cos^{-1} q + \cos^{-1} r = 3\pi$, then $p^2 + q^2 + r^2 + 2pqr$ is equal to

[KCET]

(a) 3

(b) 1

(c) 2

(d) -1

Q 77. The sum of the series

$\tan^{-1}\frac{1}{1+1+1^2}+\tan^{-1}\frac{1}{1+2+2^2}+\tan^{-1}\frac{1}{1+3+3^2}+....\infty$ is equal to

[Kerala CEE]

(a) $\frac{\pi}{4}$

(b) $\frac{\pi}{2}$

(c) $\frac{\pi}{3}$

(d) $\frac{\pi}{6}$

(e) π

Q 78. The value of sec is $\left[\tan^{-1}\left(\frac{b+a}{b-a}\right)-\tan^{-1}\left(\frac{a}{b}\right)\right]$

[Kerala CEE]

(a) 2

(b) $\sqrt{2}$

(c) 4

(d) 1

(e) $\frac{a}{b}$

Q 79. If $\sin^{-1} x + \sin^{-1}(1-x) = \cos^{-1} x$, then x belongs to

[AMU, EAMCET]

(a) $\{1,0\}$

(b) $\{-1,1\}$

(c) $\left\{0,\frac{1}{2}\right\}$

(d) $\{2,0\}$

Q 80. The trigonometric equation $\sin^{-1} x = 2$ $\sin^{-1} a$, has a solution for

[AIEEE]

(a) $\frac{1}{2} < |a| < \frac{1}{\sqrt{2}}$

(b) all real values of a

(c) $|a| \leq \frac{1}{2}$

(d) $|a| \geq \frac{1}{\sqrt{2}}$

Q 81. The value of $\cos^{-1}$ is $\left(\cos\frac{5\pi}{3}\right) + \sin^{-1}\left(\cos\frac{5\pi}{3}\right)$

[UP SEE]

(a) $\frac{10\pi}{3}$

(b) 0

(c) $\frac{\pi}{2}$

(d) $\frac{5\pi}{3}$

Q 82. $\sin\left(\frac{1}{2}\cos^{-1}\frac{4}{5}\right)$ is equal to

[KCET]

(a) $-\frac{1}{\sqrt{10}}$

(b) $\frac{1}{\sqrt{10}}$

(c) $-\frac{1}{10}$

(d) $\frac{1}{10}$

Q 83. For the principal value branch of the graph of the function $y = \sin^{-1} x$, $-1 \leq x \leq 1$, which among the bfollowing is a true statement?

[Kerala CEE]

(a) graph is symmetric about the x−axis

(b) graph is symmetric about the y−axis

(c) graph is not continuous

(d) the line x = 1 is a tangent

(e) the line y = 1 is a tangent

Q 84. $\cos\left[\cos^{-1}\left(-\frac{1}{7}\right)+\sin^{1}\left(-\frac{1}{7}\right)\right]$ is equal to

[EAMCET]

(a) $-\frac{1}{3}$

(b) 0

(c) $\frac{1}{3}$

(d) $\frac{4}{9}$

Q 85. If $\theta = \sin^{-1} x + \cos^{-1} x - \tan^{-1} x$, $1 \leq x < \infty$,

then the smallest interval in which θ lies is

[OJEE]

(a) $\frac{\pi}{2} \leq \theta \leq \frac{3\pi}{4}$

(b) $0 \leq \theta \leq \frac{\pi}{4}$

(c) $-\frac{\pi}{4} \leq \theta \leq 0$

(d) $\frac{\pi}{4} \le \theta \le \frac{\pi}{2}$

Q 86. The value of $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right) - \sin^{-1}\left(\frac{1}{2}\right)$ is

[MPPET]

(a) 45°

(b) 90°

(c) 15°

(d) 30°

Q 87. The relation $\tan^{-1}\left(\frac{1+x}{1-x}\right)=\frac{\pi}{4}+\tan^{-1}x$ holds true for all

[J&K CET]

(a) $x \in R$

(b) $x \in (-\infty, 1)$

(c) $x \in (-1, \infty)$

(d) $x \in (-\infty, -1)$

Answers

Topic 1: Solutions of Inverse Trigonometrical Functions

1. (b) 2.(c) 3. (c) 4. (d) 5. (a) 6. (b) 7. (c) 8. (a) 9. (a) 10. (d) 11. (c) 12. (c) 13. (d) 14. (b) 15. (c) 16. (d) 17. (c) 18. (c) 19. (b) 20. (b) 21. (d) 22. (a) 23. (e) 24. (d) 25. (d) 26. (b) 27. (c) 28. (c) 29. (d) 30.(b) 31. (d) 32. (c) 33. (c) 34. (a) 35. (b) 36. (b) 37. (d) 38. (c) 39. (c) 40. (b) 41. (b) 42. (a) 43. (b) 44. (b) 45. (c) 46. (a) 47.

(a) 48. (a) 49. (b) 50. (b) 51. (c) 52. (b)

53. (b) 54. (d) 55. (d) 56. (a) 57. (a) 58.

(a) 59. (b) 60. (b) 61. (c) 62. (a) 63. (e)

64. (a) 65. (c) 66. (a) 67. (a) 68. (c) 69. (c)

70. (a) 71. (c) 72. (b) 73. (d) 74. (d) 75. (d)

76. (b) 77. (a) 78. (b) 79. (c) 80. (c) 81.

(c) 82. (b) 83. (d) 84. (b) 85. (b) 86. (d)

87. (b)

Binomial Theorem and Mathematical Induction

Topic 1

Expansion of Binomial Theorem and General Term

Q 1. If in the expansion of $(a - 2n)^n$, the sum of the 5th and 6th term is zero, then the value of $\frac{a}{b}$ is

[WBJEE]

(a) $\frac{n-4}{5}$

(b) $\frac{2(n-4)}{5}$

(c) $\frac{5}{n-4}$

(d) $\frac{2(n-4)}{5}$

Q 2. The remainder left out when $8^{2n} - (62)^{2n+1}$ is divided by 9, is

[AIEEE]

(a) 0

(b) 2

(c) 7

(d) 8

Q 3. $7^9 + 9^7$ is divisible by

[DCE]

(a) 128

(b) 24

(c) 64

(d) 72

Q 4. Sum of the infinite series

$$1+\frac{2}{3}.\frac{1}{2}+\frac{2}{3}.\frac{5}{6}.\frac{1}{2^2}+\frac{2}{3}.\frac{5}{6}.\frac{8}{9}.\frac{1}{2^3}+.....\infty$$ is

[BITSAT]

(a) $2^{1/3}$

(b) $4^{1/3}$

(c) $8^{1/3}$

(d) $2^{1/5}$

Q 5. The total number of terms in the expansion of

$(x + y)^{m100} + (x - y)^{100}$ after simplification is

[KCET]

(a) 51

(b) 202

(c) 100

(d) 50

Q 6. Let [x] denote the greatest integer less than or

equal to x. If $x = (\sqrt{3}+1)^5$, then [x] is equal to

[Kerala CEE]

(a) 75

(b) 50

(c) 76

(d) 51

(e) 152

Q 7. If $|x| < 1$, then the coefficient of x^6 in the expansion of $(1 + x + x^2)^{-3}$ is

[Kerala CEE]

(a) 3

(b) 6

(c) 9

(d) 12

(e) 15

Q 8. If x^{2r} occurs in $\left(x+\frac{2}{x^2}\right)^n$, then n − 2r must be of the form

[AMU]

(a) 3k − 1

(b) 3k

(c) 3k + 1

(d) 3k + 2

Q 9. If the rth term in the expansion of $\left(\frac{x}{3}-\frac{2}{x^2}\right)^{10}$ contains x^4, then r is equal to

[OJEE]

(a) 3

(b) 0

(c) −3

(d) 5

Q 10. If the coefficients of x^2 and x^3 in the expansion of $(3 + ax)^9$ be same, then the value of a is

[WB JEE]

(a) 3/7

(b) 7/3

(c) 7/9

(d) 9/7

Q 11. If $(1 + 2x + x^2)^5 = \sum_{k=0}^{15} a_k x^k$, then $\sum_{k=0}^{7} = a_{2k}$ is equal to

[BITSAT, AMU]

(a) 128

(b) 256

(c) 512

(d) 1024

Q 12. In the expansion of $\left(x-\frac{1}{x}\right)^6$, the coefficient of x^0 is

[UP SEE]

(a) 20

(b) −20

(c) 30

(d) −30

Q 13. If the expansion of $\left(\frac{3\sqrt{x}}{7}-\frac{5}{2x\sqrt{x}}\right)^{13\pi}$ contains a

term independent of x in 14th term, then n should be

[Kerala CEE]

(a) 10

(b) 5

(c) 6

(d) 4

(e) 11

Q 14. The coefficient of x in the expansion of $(1 + x)(1 + 2x)(1 + 3x).....(1 + 100x)$ is

[Kerala CEE]

(a) 5050

(b) 10100

(c) 5151

(d) 4950

(e) 1100

Q 15. If $\alpha = \frac{5}{2!3} + \frac{5.7}{3!3^2} + \frac{5.7.9}{4!3^2} + ...,$ then $\alpha^2 + 4\alpha$ is equal to

[EAMCET]

(a) 21

(b) 23

(c) 25

(d) 27

Q 16. The coefficient of x^n in the expansion of $(1 + 2x + 3x^2 + ...)^{1/2}$ is

[OJEE]

(a) -1

(b) 0

(c) 2

(d) 1

Q 17. The coefficient of x^{-10} in $\left(x^2-\frac{1}{x^3}\right)^{10}$ is

[WBJEE]

(a) -252

(b) 210

(c) $-(5!)$

(d) -120

Q 18. If the magnitude of the coefficient of x^7 in the expansion of $\left(ax^2+\frac{1}{bx}\right)^8$, where a, b, are positive numbers, is equal to the magnitude of the coefficient of x^{-7} in the expansion of $\left(ax-\frac{1}{bx^2}\right)^8$, then a and b are connected by the relation

[WBJEE]

(a) ab = l

(b) ab = 2

(c) $a^2b = 1$

(d) $ab^2 = 2$

Q 19. $1+\frac{2}{4}+\frac{2.5}{4.8}+\frac{2.5.8}{4.8.12}+\frac{2.5.8.11}{4.8.12.16}+....$ is

[BCECE]

(a) $4^{-2/3}$

(b) $\sqrt[3]{16}$

(c) $\sqrt[3]{4}$

(d) $4^{3/2}$

Q 20. The two consecutive terms in the expansion of $(3 + 2x)^{74}$ whose coefficients are equal, are

[Manipal]

(a) 11,12

(b) 7,8

(c) 30,31

(d) None of these

Q 21. The coefficient of x^r in the expansion of $(1 - x)^{-2}$ is

[J&KCET]

(a) r

(b) r + 1

(c) r + 3

(d) $r - 1$

Q 22. In the binomial expansion of $(a - b)^n$, $n \geq 5$, the sum of 5th and 6th terms is zero, then $\frac{a}{b}$ equals

[AIEEE, OJEE]

(a) $\frac{5}{n-4}$

(b) $\frac{6}{n-5}$

(c) $\frac{n-5}{6}$

(d) $\frac{n-4}{5}$

Q 23. The coefficient of x^{20} in the expansion of

$(1 + 3x + 3x^2 + x^3)^{20}$ is

[DCE]

(a) $^{60}C_{40}$

(b) $^{30}C_{20}$

(c) $^{15}C_2$

(d) None of these

Q 24. $49^n + 16n - 1$ is divisible by

[BITSAT]

(a) 3

(b) 29

(c) 19

(d) 64

Q 25. The ninth term of the expansion $\left(3x - \frac{1}{2x}\right)^8$ is

[KCET]

(a) $\frac{1}{512x^9}$

(b) $\frac{-1}{512x^9}$

(c) $\frac{-1}{256x^8}$

(d) $\frac{1}{256x^8}$

Q 26. In the expansion of $(1 + x + x^2 + x^3)^6$, the coefficient of x^{14} is

[Kerala CEE]

(a) 130

(b) 120

(c) 128

(d) 125

(e) 115

Q 27. The coefficient of x^{53} in the expansion of

$\sum_{m=0}^{100} {}^{100}C_m \ (x - 3)^{100-m} \cdot 2^m$ is

[AMU]

(a) ${}^{100}C_{47}$

(b) ${}^{100}C_{53}$

(d) $-{}^{100}C_{53}$

(d) $-{}^{100}C_{100}$

Q 28. In the expansion of $(2-3x^3)^{20}$, if the ratio of 10th term to 11th term is $45/22$, then x is equal to

[OJEE]

(a) $-\frac{2}{3}$

(b) $\frac{-3}{2}$

(c) $-\sqrt[3]{\frac{2}{3}}$

(d) $-\sqrt[3]{\frac{3}{2}}$

Q 29. If the coefficient of rth and (r + l)th terms in the expansion of $(3 + 7x)^{29}$ are equal, then r equals

[WBJEE]

(a) 15

(b) 21

(c) 14

(d) None of these

Q 30. If in the expansion of $(1 + x)^m (1 - x)^n$,

the coefficient of x and x^2 are 3 and − 6

respectively, then m is

[WB JEE]

(a) 6

(b) 9

(c) 12

(d) 24

Q 31. The larger of $99^{50} + 100^{50}$ and 101^{50} is

[WB JEE]

(a) $99^{50} + 100^{50}$

(b) both are equal

(c) 101^{50}

(d) None of these

Q 32. The coefficient of x^5 in the expansion of $(1+x^2)(1+x)^4$ is

[WB JEE]

(a) 30

(b) 60

(c) 40

(d) None of these

Q 33. The expression $(x + (x^3 - 1)^{1/2}\}^5 + \{x - (x^3 - 1)^{1/2}\}^5$ is a polynomial of degree

[BCECE]

(a) 5

(b) 6

(c) 7

(d) 8

Q 34. The coefficient of x^n in the expansion of $(1 + x)(1 - x)^n$ is

[Jamia Millia Islamia, MP PET]

(a) $(n - 1)$

(b) $(-1)^n(1 - n)$

(c) $(-1)^{n-1}(n - 1)^2$

(d) $(-1)^{n-1}\ n$

Q 35. The coefficient of t^{24} in the expansion of

$(1 + t^2)^{12}(1 + t^{12})(1 + t^{24})$ is

[MPPET]

(a) ${}^{12}C_6 + 2$

(b) ${}^{12}C_5$

(c) ${}^{12}C_6$

(d) ${}^{12}C_7$

Q 36. If the expansion in powers of x of the function $\frac{1}{(1-ax)(1-bx)}$ is $a_0 + a_1x + a_2x^2 + a_3x^3 + \ldots$, then a_n is

[AIEEE]

(a) $\frac{a^n - b^n}{b - a}$

(b) $\frac{a^{n+1} - b^{n+1}}{b - a}$

(c) $\frac{b^{n+1} - a^{n+1}}{b - a}$

(d) $\frac{b^n - a^n}{b - a}$

Q 37. If the second term in the expansion

$\left[\sqrt[13]{a} + \frac{a}{\sqrt{a^{-1}}}\right]^n$ is $14a^{5/2}$, then the value of $\frac{{}^nC_3}{{}^nC_2}$ is

[DCE]

(a) 4

(b) 3

(c) 12

(d) 6

Q 38. The digit at the unit place in the number

$19^{20O5} + 11^{20O5} - 9$ is

[Kerala CEE]

(a) 2

(b) 1

(c) 0

(d) 8

(e) 9

Q 39. For all integers $n \geq 1$, which of the following is divisible by 9?

[EAMCET]

(a) $8^n + 1$

(b) $4^n - 3n - 1$

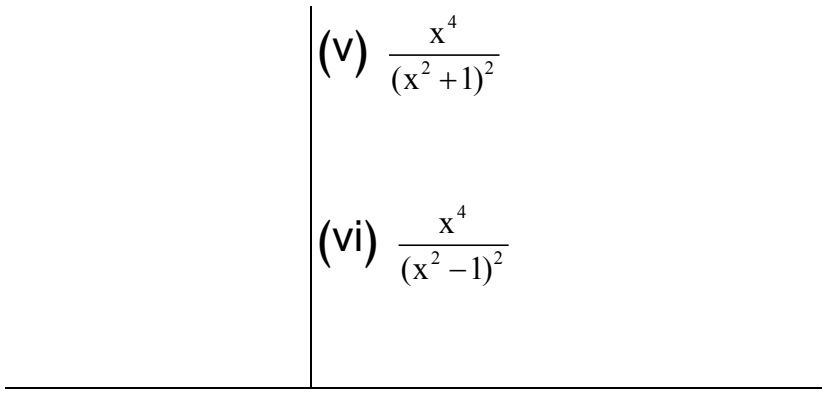
(c) $3^{2n} + 3n + 1$

(d) $10^n + 1$

Q 40. The correct matching of List I from List II is

List I	List II
(A) $(1 - x)^{-N}$	$\frac{x}{x+1}$
(B) $(1 + x)^{-n}$	(ii) $1 - nx + \frac{n(n+1)}{2!}x^2 - \ldots$

	if $\|x\| < 1$
(C) If $x > 1$,	(iii) $1 + nx + \frac{n(n+1)}{2!}x^2 + \ldots$
then $1 + \frac{1}{x} + \frac{1}{x^2}$	if $\|x\| < 1$
$+\ldots$ is	
(D) If $\|x\| > 1$,	(iv) $\frac{x}{x-1}$
then	
$1 - \frac{2}{x^2} + \frac{3}{x^4} - \frac{4}{x^6}$	
$+\ldots$is	



[EAMCET]

(a) (i) (iii) (iv) (v)

(b) (ii) (iii) (iv) (v)

(c) (iii) (ii) (iv) (v)

(d) (ii) (iii) (i) (v)

Q 41. The coefficient of x^4 in the expansion of

$\left(\frac{x}{2}-\frac{3}{x^2}\right)^{10}$ is

[WB JEE, Jamia Millia Islamia]

(a) $\frac{504}{259}$

(b) $\frac{450}{263}$

(c) $\frac{405}{256}$

(d) None of these

Q 42. The coefficient of x^2 term in the binomial expansion of $\left(\frac{1}{3}x^{1/2}+x^{-1/4}\right)^{10}$ is

[BCECE]

(a) $\frac{70}{243}$

(b) $\frac{60}{423}$

(c) $\frac{50}{13}$

(d) None of these

Q 43. The coefficient of x^{100} in the expansion of

$\sum_{j=0}^{200}(1+x)^j$ is

[MPPET]

(a) $\binom{200}{100}$

(b) $\binom{201}{102}$

(c) $\binom{200}{101}$

(d) $\binom{201}{100}$

Q 44. Coefficient of x in the polynomial $(x-1)(x-2)\ldots(x-20)$ is equal to

[MPPET]

(a) 210

(b) -210

(c) $20!$

(d) None of these

Q 45. If in the expansion of $(1 + x)^{21}$, the coefficients of x^r and x^{r+1} be equal, then r is equal to

[RPET]

(a) 9

(b) 10

(c) 11

(d) 12

Q 46. The coefficient of x^{-17} in the expansion of $\left(x^4 - \frac{1}{x^3}\right)^{15}$ is

[J&KCET]

(a) $^{15}C_{11}$

(b) $^{15}C_{12}$

(c) $-\ ^{15}C_{11}$

(d) $-\ ^{15}C_3$

Q 47. If x is so small that x^3 and higher powers of x may be neglected, then $\frac{(1+x)^{3/2}-\left(1+\frac{1}{2}x\right)^3}{(1-x)^{1/2}}$ may be approximated as

[AIEEE]

(a) $\frac{x}{2}-\frac{3}{8}x^2$

(b) $-\frac{3}{8}x^2$

(c) $3x+\frac{3}{8}x^2$

(d) $1-\frac{3}{8}x^2$

Q 48. If the coefficient of x^7 in $\left(ax^2+\frac{1}{bx}\right)^{11}$ equals the coefficient of x^{-7} in $\left(ax-\frac{1}{bx^2}\right)^{11}$, then a and b satisfy the relation

[AIEEE]

(a) $ab = 1$

(b) $\frac{a}{b}=1$

(c) $a + b = 1$

(d) $a - b = 1$

Q 49. If $(1 + x)^{15} = a_0 + a_1 x + \ldots + a_{15} x^{15}$, then

$\sum_{r=1}^{15} r \frac{a_r}{a_{r-1}}$ is equal to

[EAMCET]

(a) 110

(b) 115

(c) 120

(d) 135

Q 50. If $|x| < \frac{1}{2}$, then the coefficient of x^r in the expansion of $\frac{1+2x}{(1-2x)^2}$, is

[EAMCET]

(a) $r2^r$

(b) $(2r - 1)2^r$

(c) $r2^{2r + 1}$

(d) $(2r + 1)2^r$

Q 51. If

$x = \frac{[729 + 6(2)(243) + 15(4)(81) + 20(8)(27) + 15(16)(9) + 6(32)3 + 64]}{1 + 4(4) + 6(16) + 4(64) + 256}$, then

$\sqrt{x} - \frac{1}{\sqrt{x}}$ is equal to

[RPET]

(a) 0.2

(b) 4.8

(c) 1.02

(d) 5.2

Q 52. The coefficient of x^{32} in the expansion of $\left(x^4 - \frac{1}{x^3}\right)^{15}$ is

[Guj. CET]

(a) ${}^{-15}C_3$

(b) ${}^{15}C_4$

(c) ${}^{-15}C_5$

(d) ${}^{15}C_2$

Q 53. The sum of the rational terms in the expansion of $(\sqrt{2}+3^{1/5})^{10}$ is

[Kerala CEE, UP SEE]

(a) 41

(b) 32

(c) 18

(d) 9

(e) 82

Q 54. Coefficient of x in the expansion of

$\left(x^2+\frac{a}{x}\right)^5$ is

[OJEE]

(a) $9a^2$

(b) $10a^3$

(c) $10a^2$

(d) $10a$

Q 55. The coefficient of $1/x$ in. the expansion of $\left(\frac{1}{x}+1\right)^n (1+x)^n$ is

[OJEE]

(a) ${}^{2n}C_n$

(b) ${}^{2n}C_{n-1}$

(c) ${}^{2n}C_1$

(d) ${}^{n}C_{n-1}$

Q 56. The coefficient of x^{-7} in the expansion of $\left[ax - \frac{1}{bx^2}\right]^{11}$ will be

[Jamia Millia Islamia]

(a) $\frac{462a^6}{b^5}$

(b) $\frac{462a^5}{b^6}$

(c) $-\frac{462a^5}{b^6}$

(d) $-\frac{462a^6}{b^5}$

Q 57. If the coefficient of $(2r + 1)$th term and $(r + 2)$th term in the expansion of $(1 + x)^{43}$ are equal, then r is equal to

[EAMCET]

(a) 12

(b) 14

(c) 16

(d) 18

Q 58. The sum of the series $1+\frac{1}{5}+\frac{1.3}{5.10}+\frac{1.3.5}{5.10.15}+....$ is equal to

[BCECE]

(a) $\frac{1}{\sqrt{5}}$

(b) $\frac{1}{\sqrt{2}}$

(c) $\sqrt{3}$

(d) $\sqrt{\frac{5}{3}}$

Q 59. If | x | < 1, then $1+n\left(\frac{2x}{1+x}\right)+\frac{n(n+1)}{2!}\left(\frac{2x}{1+x}\right)^2+....$ is equal to

[J&KCET]

(a) $\left(\frac{2x}{1+x}\right)^n$

(b) $\left(\frac{1+x}{2x}\right)^n$

(c) $\left(\frac{1-x}{1+x}\right)^n$

(d) $\left(\frac{1+x}{1-x}\right)^n$

Q 60. If the fourth term in the expansion of

$\left(ax+\frac{1}{x}\right)^n$ is $\frac{5}{2}$, then

[J&K CET]

(a) a = $1/2$,and n = 6

(b) a = $1/3$ and n = 5

(c) a = 2 and n = 3

(d) a = $1/4$ and n = 1

Topic 2

Independent Term, Middle Term and Greatest Term

2010

Q 1. The coefficient of the middle term in the expansion of $(x+2y)^6$ is

[Kerala CEE]

(a) ${}^6C_3$

(b) $8({}^6C_3)$

(c) $8({}^6C_5)$

(d) ${}^6C_4$

(d) $8({}^{6}C_{4})$

Q 2. The coefficient of the term independent of x in the expansion of $\left[\frac{(x+1)}{x^{2/3}-x^{1/3}+1}-\frac{(x-1)}{x-x^{1/2}}\right]^{10}$ is

[DCE]

(a) 210

(b) 105

(c) 70

(d) 112

Q 3. For |x| < 1, the constant term in the expansion of $\frac{1}{(x-1)^2(x-2)}$ is

[EAMCET]

(a) 2

(b) 1

(c) 0

(d) $-\frac{1}{2}$

Q 4. If in the expansion of $\left(3x-\frac{2}{x^2}\right)^{15}$ rth term is independent of x, then value of r is

[RPET]

(a) 6

(b) 10

(c) 9

(d) 12

Q 5. The greatest term in the expansion of

$\sqrt{3}\left(1+\frac{1}{\sqrt{3}}\right)^{20}$ is

(a) $\frac{26840}{9}$

(b) $\frac{24840}{9}$

(c) $\frac{25840}{9}$

(d) None of these

[DCE]

Q 6. The greatest term in the expansion of $(1 + 3x)^{54}$, where x= $1/3$ is

[DCE]

(a) T_{28}

(b) T_{25}

(c) T_{26}

(d) T_{24}

Q 7. In the expansion of $\left(2x^2 - \frac{1}{x}\right)^{12}$, the term

independent of x is

[UP SEE]

(a) 8th

(b) 7th

(c) 9th

(d) 10th

Q 8. The sum of coefficients of the expansio

$\left(\frac{1}{x}+2x\right)^n$ in 6561. The coefficient of term independent of x is

[BCECE]

(a) $16\ {}^8C_4$

(b) ${}^8C_4$

(c) ${}^8C_5$

(d) None of these

Q 9. The coefficient of the middle term in the

binomial expansion in powers of x of $(1 + ax)^4$ and of $(1 - ax)^6$ is the same, if a equals

[Jamia Millia Islamia, MP PET]

(a) $-\frac{5}{3}$

(b) $\frac{10}{3}$

(c) $-\frac{3}{10}$

(d) $\frac{3}{5}$

Q 10. The term independent of x in the expansion of $\left(\frac{2\sqrt{x}}{5}-\frac{1}{2x\sqrt{x}}\right)^{11}$ is

[J&KCET]

(a) 5th term

(b) 6th term

(c) 11th term

(d) no term

Q 11. The largest coefficient in the expansion of $(1 + x)^{24}$ is

[GUJ.CET]

(a) ${}^{24}C_{24}$

(b) ${}^{24}C_{13}$

(c) ${}^{24}C_{12}$

(d) ${}^{24}C_{11}$

Q 12. The term independent of x in the expansion of $\left(x-\frac{1}{x}\right)^4\left(x+\frac{1}{x}\right)^3$ is

[Guj.CET]

(a) -3

(b) 0

(c) 3

(d) 1

Q 13. The middle term in the expansion of

$\left(x-\frac{1}{x}\right)^{18}$ is

[BITSAT, Guj. CET]

(a) ${}^{18}C_9$

(b) $-{}^{18}C_9$

(c) ${}^{18}C_{10}$

(d) $-{}^{18}C_{10}$

Q 14. The greatest coefficient in the expansion of $(1 + x)^{2n}$ is

[WBJEE]

(a) ${}^{2n}C_n$

(b) ${}^{2n}C_{n+1}$

(c) ${}^{2n}C_{n-1}$

(d) ${}^{2n}C_{2n-1}$

Q 15. The term independent of x in the expansion of $\left(x^3+\frac{2}{x^2}\right)^{15}$ is

[J&KCET]

(a) T_7

(b) T_8

(c) T_9

(d) T_{10}

Q 16. The middle term in the expansion of $(x - a)^8$ is

[BCECE]

(a) $^{-8}C_4x^4a^4$

(b) $^{8}C_4x^4a^4$

(c) $^{8}C_3\ x^5a^3$

(d) $^{-8}C_5x^2a^5$

Q 17. The coefficient of term independent of x in $\left[\sqrt{\left(\frac{x}{3}\right)}+\frac{\sqrt{3}}{x^2}\right]^{10}$ is

[DCE]

(a) $\frac{5}{3}$

(b) $\frac{4}{5}$

(c) 6

(d) $\frac{1}{2}$

Q 18. Coefficient of the term independent of x in the expansion $\left(x+\frac{1}{x^2}\right)^6$ is equal to

[DCE]

(a) 10

(b) 15

(c) 16

(d) None of the above

Q 19. The constant term in the expansion of

$\left(x^2 - \frac{1}{x}\right)^9$ is

[MPPET]

(a) 80

(b) 72

(c) 84

(d) 82

Topic 3 Properties of Binomial Coefficients

2010

Q 1. For r = 0,l,...,10 let A_r, B_r and C_r denotes, respectively, the coefficient of x^r in the expansions of $(1 + x)^{10}$,$(1 + x)^{20}$, and $(1 + x)^{30}$. Then $\sum_{r=1}^{10} A_r (B_{10}B_r - C_{10}A_r)$is equal to

[IIT JEE]

(a) $B_{l0} - C_{10}$

(b) $A_{10}(B_{10}^2 - C_{10}A_{10})$

(c) 0

(d) $C_{10} - B_{10}$

Q 2. Let $S_1 = \sum_{j=1}^{10} j(j-1)\ {}^{10}C_j$, $S_2 = \sum_{j=1}^{10} j\,{}^{10}C_j$ and $S_3 = \sum_{j=1}^{10} j^2\,{}^{10}C_j$

[AIEEE]

Statement I $S_3 = 55 \times 2^9$

Statement II $S_1 = 90x2^8$ and $S_2 = 10 \times 2^8$

(a) Statement I is true, Statement II is true;

Statement II is a correct explanation for Statement I

(b) Statement I is true, Statement II is true;

Statement II is not a correct explanation for

Statement I

(c) Statement I is true, Statement II is false (d)

Statement I is false, Statement II is true

Q 3. If $(1 - x + x^2)^n = a_0 + a_1 x + \ldots + a_{2n} x^{2n}$ then

the value of $a_0 + a_2 + a_4 + \ldots, + a_{2n}$ is

[WB JEE]

(a) $3^n + \frac{1}{2}$

(b) $3^n - \frac{1}{2}$

(c) $\frac{3^n - 1}{2}$

(d) $\frac{3^n + 1}{2}$

Q 4. Sum of the last 30 coefficients in the expansion of $(1 + x)^{59}$, when expanded in ascending powers of x is

[BITSAT]

(a) 2^{59}

(b) 2^{58}

(c) 2^{30}

(d) 2^{29}

Q 5. If the sum of the coefficients in the expansion of $(a^2x^2 - 6ax + 11)^{10}$, where a is constant is 1024, then the value of a is

[Kerala CEE]

(a) 5

(b) 1

(c) 2

(d) 3

(e) 4

Q 6. Let $(1 + x)^n = 1 + a_1x + a_2x^2 + \ldots + a_nx^n$. If a_1, a_2 and a_3 are in AP, then the value of n is

[BITSAT]

(a) 4

(b) 5

(c) 6

(d) 7

(e) 8

Q 7. ${}^{n-2}C_r + 2\,{}^{n-2}C_{r-1} + {}^{n-2}C_{r-2}$ equals

[BITSAT]

(a) ${}^{n+1}C_r$

(b) ${}^{n}C_r$

(c) ${}^{n}C_{r+1}$

(d) ${}^{n-1}C_r$

Q 8. ${}^{15}C_0 \cdot {}^{5}C_5 + {}^{15}C_1 \cdot {}^{5}C_4 + {}^{15}C_2 \cdot {}^{5}C_3 + {}^{15}C_3 \cdot {}^{5}C_2 + {}^{15}C_4 \cdot {}^{5}C_1$ is equal to

[Kerala CEE]

(a) $2^{20} - 2^5$

(b) $\frac{20!}{5!15!}$

(c) $\frac{20!}{5!15!} - 1$

(c) $\frac{20!}{5!15!} - \frac{15!}{5!10!}$

(e) $\frac{15!}{5!10!}$

Q 9. The value of ${}^{15}C_8 + {}^{15}C_9 - {}^{15}C_6 - {}^{15}C_7$ is

[OJEE]

(a) -1

(b) 0

(c) 1

(d) None of these

Q 10. $\frac{{}^{8}C_{0}}{6} - {}^{8}C_{1} + {}^{8}C_{2} \cdot 6 - {}^{8}C_{3} - 6^{2} + \ldots + {}^{8}C_{8} \cdot$

6^{7} is equal to

[J&K CET]

(a) 0

(b) 6^{7}

(c) 6^8

(d) $\frac{5^8}{6}$

Q 11. **Statement I** $\sum_{r=0}^{n}(r+1).^{n}C_{r} = (n+2)2^{n-1}$

Statement II $\sum_{r=0}^{n}(r+1)^{n}C_{r}.x^{r} = (1+x)^{n} + nx(1+x)^{n-1}$

[AEEEE]

(a) Statement I is true, Statement II is true;

Statement II is a correct explanation for Statement I

(b) Statement I is true, Statement II is true;

Statement II is not a correct explanation for

StatementI

(c) Statement I is true, Statement II is false

(d) Statement I is false, Statement II is true

Q 12. $\frac{C_0}{1}+\frac{C_2}{3}+\frac{C_4}{5}+\frac{C_6}{7}+.....$ is equal to

[DCE]

(a) $\frac{2^{n+1}}{n+1}$

(b) $\frac{2^{n+1}-1}{n+1}$

(c) $\frac{2^n}{n+1}$

(d) None of these

Q 13. The value of $\frac{1}{81^n}-\frac{10}{81^n}{}^{2n}C_1+\frac{10^2}{81^n}{}^{2n}C_2-\frac{10^3}{81^n}$

${}^{2n}C_3$ + ... + $\frac{10^{2n}}{81^n}$ is

[UP SEE]

(a) 2

(b) 0

(c) $\frac{1}{2}$

(d) 1

Q 14. The coefficient of $a^5b^6c^7$ in the expansion of $(bc + ca + ab)^9$ is

[Kerala CEE]

(a) 100

(b) 120

(c) 720

(d) 1260

(e) 1440

Q 15. The value of $({}^{7}C_0 + {}^{7}C_1) + ({}^{7}C_1 + {}^{7}C_2) +$

$\ldots + ({}^{7}C_6 + {}^{7}C_7)$ is

[Kerala CEE]

(a) $2^8 - 1$

(b) $2^8 + 1$

(c) 2^8

(d) $1 - 2^8$

(e) $2^8 - 2$

Q 16. If $(1 + x)^n = C_0 + C_1x + C_2x^2 + \ldots + C_nx^n$.

Then, $C_0C_1 + C_1C_2 + \ldots + C_{n-1}C_n$ is equal to

[OJEE]

(a) $\frac{(2n)!}{(n-1)!(n+1)!}$

(b) $\frac{(2n-1)!}{(n-1)!(n+1)!}$

(c) $\frac{2n!}{(n+2)!(n+1)!}$

(d) None of these

Q 17. The value of the sum of the series

$3.\ {}^{n}C_0 - 8\ .\ {}^{n}C_1 + 13\ {}^{n}C_2 - 18\ .\ {}^{n}C_3 + \ldots$ upto $(n + 1)$

terms is

[Manipal]

(a) 0

(b) 3^n

(c) 5^n

(d) None of these

Q 18. If C_0, C_1, C_2,,.....C_n denote the binomial coefficients in the expansion of $(1 + x)^n$, then

$C_0 + \frac{C_1}{2} + \frac{C_2}{3} + ... + \frac{C_n}{n+1}$ is equal to

[J&KCET]

(a) $\frac{2^{n+1} - 1}{n+1}$

(b) $\frac{2^n - 1}{n}$

(c) $\frac{2^{n-1}-1}{n-1}$

(d) $\frac{2^{n+1}-1}{n+2}$

Q 19. If $(1 + x - 3x^2)^{10} = 1 + a_1x + a_2x^2 + \ldots + a_{20}x^{20}$, then $a_2 + a_4 + a_6 + \ldots + a_{20}$ is equal to

[Guj. CET]

(a) $\frac{3^{10}+1}{2}$

(b) $\frac{3^{9}+1}{2}$

(c) $\frac{3^{10}-1}{2}$

(d) $\frac{3^9 - 1}{2}$

Q 20. The sum of the series ${}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \ldots + {}^{20}C_{10}$ is

[AIEEE]

(a) $-{}^{20}C_{10}$

(b) $1/2\ {}^{20}C_{10}$

(c) 0

(d) $^{20}C_{10}$

Q 21. The sum of the last eight coefficients in the expansion of $(1 + x)^{15}$ is

[DCE]

(a) 2^{16}

(b) 2^{15}

(c) 2^{14}

(d) None of these

Q 22. The

(a) $\frac{2^{50}}{51}$ (b) $\frac{2^{50}-1}{51}$ **[Kerala CEE]**

(c) $\frac{2^{50}-1}{50}$ (d) $\frac{2^{51}-1}{51}$

(e) $\frac{2^{51}-1}{50}$

value of $\left(\frac{{}^{50}C_0}{1}+\frac{{}^{50}C_2}{3}+\frac{{}^{50}C_4}{5}+....+\frac{{}^{50}C_{50}}{51}\right)$ is

(a) $\frac{2^{50}}{51}$

(b) $\frac{2^{50}-1}{51}$

(c) $\frac{2^{50}-1}{50}$

(d) $\rightarrow\frac{2^{51}-1}{51}$

(e) $\frac{2^{51}-1}{50}$

Q 23. If n = 5, then $({}^{n}C_0)^2 + ({}^{n}C_1)^2 + ({}^{n}C_2)^2 + \ldots +$ $({}^{n}C_5)^2$ is equal to

[Kerala CEE]

(a) 250

(b) 254

(c) 245

(d) 252

(e) 258

Q 24. For natural numbers m,n if $(1 - y)^m (1 + y)^n = 1 + a_1 y + a_2 y^2 + \ldots$ and $a_1 = a_2 = 10$, then (m, n) is

[AIEEE]

(a) (35, 20)

(b) (45, 35)

(c) (35,45)

(d) (20,45)

Q 25. If ${}^{n}C_{l2} = {}^{n}C_{6}$, then ${}^{n}C_{2}$ is equal to

[BITSAT, Guj. CET]

(a) 72

(b) 153

(c) 306

(d) 2556

Q 26. If ${}^{18}C_{15} + 2({}^{18}C_{16}) + {}^{17}C_{16} + 1 = {}^{n}C_{3}$, then n is equal to **[Kerala CEE]**

(a) 19

(b) 20

(c) 18

(d) 24

(e) 21

Q 27. If $(2x^2 - x - 1)^5 = a_0 + a_1x + a_2x^2 + \ldots + a_{10}x^{10}$, then $a_2 + a_4 + a_6 + a_8 + a_{l0}$ is equal to

[Kerala CEE]

(a) 15

(b) 30

(c) 16

(d) 32

(e) 17

Q 28. The value of $1^2 \cdot C_1 + 3^2 \cdot C_3 + 5^2 \cdot C_5$ +…is

[AMU]

(a) $n(n-1)^{n-2} + n \cdot 2^{n-1}$

(b) $n(n-1)2^{n-2}$

(c) $n(n-1) \cdot 2^{n-3}$

(d) None of these

Q 29. $\binom{30}{0}\binom{30}{10}-\binom{30}{1}\binom{30}{11}+....\binom{30}{20}\binom{30}{30}$ is equal to

[IITJEE]

(a) $^{30}C_{11}$

(b) $^{60}C_{10}$

(c) $^{30}C_{10}$

(d) ${}^{65}C_{55}$

Q 30. The value of ${}^{50}C_4 + \sum_{r=1}^{6} {}^{56-r}C_3$ is

[AEEEE]

(a) ${}^{56}C_4$

(b) ${}^{56}C_3$

(c) ${}^{55}C_3$

(d) ${}^{55}C_4$

Q 31. If the coefficients of r th, (r + 1) th and (r

+ 2) th terms in the binomial expansion of $(1 + y)^m$

are in AP, then m and r satisfy the equation

[AIEEE]

(a) $m^2 - m(4r - 1) + 4r^2 + 2 = 0$

(b) m^2

$m(4r + 1) + 4r^2 - 2 = 0$

(c) $m^2 - m(4r + 1) + 4r^2 + 2 = 0$

(d) $m^2 - n(4r - 1) + 4r^2 - 2 = 0$

Q 32. $\left(1+\frac{C_1}{C_0}\right)\left(1+\frac{C_2}{C_1}\right)\left(1+\frac{C_3}{C_2}\right).....\left(1+\frac{C_n}{C_{n-1}}\right)$ is equal to

[Kerala CEE]

(a) $\frac{n+1}{n!}$

(b) $\frac{(n+1)^n}{(n-1)!}$

(c) $\frac{(n-1)^n}{n!}$

(d) $\frac{(n+1)^n}{n!}$

(e) $\frac{n-1}{n!}$

Q 33. What is the sum of the coefficients of $(x^2 - x - 1)^{99}$?

[OJEE]

(a) 1

(b) 0

(c) −1

(d) None of these

Q 34. Let C_1, C_2, C_3 are the usual binomial coefficients. Let $S = C_1 + 2C_2 + 3C_3 + \ldots + nC_n$, then S equals

[OJEE]

(a) $n2^n$

(b) 2^{n-1}

(c) $n2^{n-1}$

(d) 2^{n+1}

Q 35. If n is an integer greater than 1, then $a - {}^{n}C_{1}(a-1) + {}^{n}C_{2}(a-2) + \ldots + (-1)^{n}(a-n)$ is equal to

[RPET]

(a) a

(b) 0

(c) a^2

(d) 2^n

Q 36. ${}^{20}C_{4} + 2 \cdot {}^{20}C_{3} + {}^{20}C_{2} - {}^{22}C_{18}$ is equal to

[RPET]

(a) 0

(b) 1242

(c) 7315

(d) 6345

Q 37. The sum of the coefficients in the expansion of $(1 + x - 3x^2)^{3148}$ is

[Guj.CET]

(a) 8

(b) 7

(c) 1

(d) −1

Q 38. If ${}^{n-1}C_r = (k^2 - 3)\,{}^nC_{r+1}$, then $k \in$

[IITJEE]

(a) $(-\infty, -2]$

(b) $[2, \infty)$

(c) $[-\sqrt{3}, \sqrt{3}]$

(d) $(\sqrt{3}, 2]$

Q 39. In the expansion of $(1 + x)^{30}$, the sum of the coefficients of odd powers of x is

[DCE]

(a) 2^{30}

(b) 2^{31}

(c) 0

(d) 2^{29}

Q 40. If ${}^{n}C_{o}$, ${}^{n}C_{1}$, ${}^{n}C_{2}$,..., ${}^{n}C_{n}$ denote the coefficients of the binomial expansion $(1 + x)^{n}$, then the value of $C_1 + 3C_3 + 5C_5 + \ldots$ is

[DCE]

(a) $n2^{n-2}$

(b) $n2^{n-1}$

(c) $(n + 1)2^n$

(d) $(n + 2)2^{n-1}$

Q 41. The binomial coefficients which are in decreasing order, are

[EAMCET]

(a) ${}^{15}C_5, {}^{15}C_6, {}^{15}C_7$

(b) ${}^{15}C_{10}, {}^{15}C_9, {}^{15}C_8$

(c) ${}^{15}C_6, {}^{15}C_7, {}^{15}C_8$

(d) ${}^{15}C_7$, ${}^{15}C_{6r}$ ${}^{15}C_{5r}$

Q 42. $\sum_{k=0}^{10} {}^{20}C_k$ is equal to

[OJEE]

(a) $2^{19} + \frac{1}{2}{}^{20}C_{10}$

(b) 2^{19}

(c) ${}^{20}C_{10}$

(d) None of these

Q 43. In the expansion of $(1 + x)^{50}$, the sum of the coefficients of odd powers of x is

[RPET]

(a) zero

(b) 2^{49}

(c) 2^{50}

(d) 2^{51}

Q 44. If $(1 + x - 2x^2)^6 = 1 + a_1x + a_2x^2 + \ldots + a_{12}x^{12}$, then the value of $a_2 + a_4 + \ldots + a_{12}$, is

[UP SEE]

(a) 31

(b) 32

(c) 64

(d) 1024

Q 45. If the sum of the coefficients in the

expansion of $(x + y)^n$ is 1024, then the value of the greatest coefficient in the expansion is

[OJEE]

(a) 356

(b) 252

(c) 210

(d) 120

Q 46. ${}^{47}C_4 + \sum_{r=1}^{5} {}^{52-r}C_3$ is equal to

[BCECE]

(a) ${}^{47}C_6$

(b) ${}^{52}C_5$

(c) ${}^{52}C_4$

(d) None of these

Topic 4 Multinomial Theorem

Q 1. If $(1 + 2x + 3x^2)^{10} = a_0 + a_1x + a_2x^2 + \ldots +$

$a_{20}x^{20}$, then a_1 equals

[Jamia Millia Islamia]

(a) 10

(b) 20

(c) 210

(d) None of these

Q 2. If a_k is the coefficient of x^k in the expansion of

$(1 + x + x^2)^n$ for k = 0, 1, 2, …,2n then

[EAMCET]

(a) $-a_0$

(b) 3^n

(c) $n.3^{n+1}$

(d) $n.3^n$

Q 3. In the expansion of $(1 + 3x + 2x^2)^6$ the coefficient of x^{11} is

[Kerala CEE]

(a) 144

(b) 288

(c) 216

(d) 576

(e) $3 \cdot 2^{11}$

Q 4. The coefficient of $x^3y^4z^5$ in the expansion of $(xy+yz+xz)^6$ is

[EAMCET]

(a) 70

(b) 60

(c) 50

(d) None of these

Q 5. The number of terms in the expansion of $(a + b + c)^{10}$ is

[MPPET]

(a) 11

(b) 21

(c) 55

(d) 66

Q 6. Coefficient of $x^2y^3 z^4$ in $(ax + by + cz)^9$ is

[Kerala CEE]

(a) 1060 $a^2b^3c^4$

(b) 1160 $a^2b^3c^4$

(c) $1260\ a^2b^3c^4$

(d) $960a^2b^3c^4$

(e) $1080a^2b^3c^4$

Q 7. If$(1 + x + x^2)^n = \sum_{r=0}^{2n} a_r x^r$ then, $a_1 - 2a_2 + 3a_3 \ldots$

$- 2na_{2n}$ is equal to

[EAMCET]

(a) n

(b) –n

(c) 0

(d) $2n$

Q 8. The coefficient of a^3b^4c in the expansion of $(1 + a - b + c)^9$ is equal to

[Kerala CEE]

(a) $\frac{9!}{3!6!}$

(b) $\frac{9!}{4!5!}$

(c) $\frac{9!}{3!5!}$

(d) $\frac{9!}{3!4!}$

(e) $\frac{9!}{5!2!}$

Q 9. The number of dissimilar terms in the expansion of $(a + b)^n$ is $n + 1$;'therefore number of dissimilar terms of the expansion $(a + b + c)^{12}$ is

[J&KCET]

(a) 13

(b) 39

(c) 78

(d) 91

Topic 5 Mathematical Induction

2010

Q 1. $(2^{3n} - 1)$ will be divisible by $(\forall\ n \in N)$

[WB JEE]

(a) 25

(b) 8

(c) 7

(d) 3

Q 2. If n is a positive integer, then $n^3 + 2n$ is divisible by

[KCET]

(a) 2

(b) 6

(c) 15

(d) 3

Q 3. If n is a positive integer, then $5^{2n+2} - 24n -$ 25 is divisible by

[Kerala CEE]

(a) 574

(b) 575

(c) 675

(d) 674

(e) 576

Q 4. If $a_1 = 1$ and $a_n = na_{n-1}$ for all positive integer $n \geq 2$, then a_5 is equal to

[Kerala CEE]

(a) 125

(b) 120

(c) 100

(d) 24

(e) 6

Q 5. Using mathematical induction, the numbers

a_n's are defined by $a_0 = 1$, $a_{n+1} = 3n^2 + n + a_n$, $(n \geq 0)$.

Then, a_n is equal to

[EAMCET]

(a) n^3+n^2+1

(b) $n^3 - n^2 + 1$

(c) $n^3 - n^2$

(d) $n^3 + n^2$

Q 6. Statement I For every natural number $n \geq 2$.

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + ... + \frac{1}{\sqrt{n}} > \sqrt{n}\ .$$

Statement II For every natural number $n \geq 2$.

$$\sqrt{n(n+1)} < n+1$$

[AIEEE]

(a) Statement I is true, Statement II is true;

Statement II is a correct explanation for Statement I

(b) Statement I is true, Statement II is true;

Statement II is not a correct explanation for

StatementI

(c) Statement I is true, Statement II is false

(d) Statement I is false, Statement II is true

Q 7. If P(n) : $2 + 4 + 6 + \ldots + (2n)$, n $\in$ N, then

$P(k) = k(k + 1) + 2$ implies $P(k + 1) = (k + 1)(k + 2)$

$+ 2$ is true for all $k \in N$.

So, statement $P(n) - n(n + 1) + 2$ is true for

[Manipal]

(a) $n \geq 1$ (b) $n \geq 2$ (c) $n \geq 3$ (d)

None of these

Q 8. For $n \in N$, $10^{n-2} \geq 8ln$, is

[Manipal]

(a) $n > 5$

(b) $n \geq 5$

(c) $n < 5$

(d) $n > 8$

Q 9. $2^{3n} - 7n - 1$ is divisible by

[AIEEE]

(a) 64

(b) 36

(c) 49

(d) 25

Q 10. If $A=\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}$ and $I=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$, then which one of the following holds for all $n \geq 1$, by the principle of mathematical induction ?

[AIEEE]

(a) $A^n = 2^{n-1} A + (n - 1)I$

(b) $A^n = nA + (n - 1)I$

(c) $A^n = 2^{n-1} A - (n - 1)I$

(d) $A^n = nA - (n - 1)I$

Q 11. $10^n + 3(4^{n+2}) + 5$ is divisible by $(n \in N)$

[Kerala CEE]

(a) 7

(b) 5

(c) 9

(d) 17

(e) 13

Q 12. Matrix A is such that $A^2 = 2A - I$ where

I is the identity matrix, then for $n \geq 2$, A^n is equal to

[BCECE]

(a) $nA - (n - 1)I$

(b) $n A - I$

(c) $2^{n-1} A - (n - 1)I$

(d) 2^{n-1} A – I

Q 13. Let S(k) = $1 + 3 + 5 + \ldots + (2k - 1) = 3$ + k^2. Then, which of the following is true ?

[AIEEE]

(a) S(1) is correct

(b) S(k) $\Rightarrow$ S(k + 1)

(c) S(k) $\not\Rightarrow$ S(k + 1)

(d) Principle of mathematical induction can be used

to prove the formula

Properties of Triangle

Topic 1

Relation between sides and Angles

Q 1. If the angles A, B and C of a triangle are in an arithmetic progression and if a, b and c denote the lengths of the sides opposite to A, B and C respectively, then the value of the expression

$\frac{a}{c}\sin 2C+\frac{c}{a}\sin 2A$ is

[IIT JEE]

(a) $\frac{1}{2}$

(b) $\frac{\sqrt{3}}{2}$

(c) 1

(d) $\sqrt{3}$

Q 2. Let ABC be a triangle such that $\angle ACB = \frac{\pi}{6}$ and let a, b and c denote the lengths of the sides opposite to A, B and C respectively. The value (s)

of x for which $a = x^2 + x + 1$, $b = x^2 - 1$ and $c = 2x + 1$ is (are)

(a) $-(2+\sqrt{3})$

(b) $1+\sqrt{3}$

(c) $2+\sqrt{3}$

(d) $4\sqrt{3}$

Q 3. In a ΔABC, $2ac\sin\frac{A-B+C}{2}$ is equal to

[WB JEE]

(a) $a^2 + b^2 - c^2$

(b) $c^2 + a^2 - b^2$

(c) $b^2 - a^2 - c^2$

(d) $c^2 - a^2 - b^2$

Q 4. If angle A, B and C are in AP, then $\frac{a+c}{b}$ is equal to

[VITEEE]

(a) $2\sin\frac{A-C}{2}$

(b) $2\cos\frac{A-C}{2}$

(c) $\cos\frac{A-C}{2}$

(d) $\sin\frac{A-C}{2}$

Q 5. The sides BC, CA and AB of a triangle ABC are of lengths a, b and c respectively. If D is the mid point of BC and AD is perpendicular to AC, then the value of cos A cos C is

[DCE]

(a) $\frac{3(a^2 - c^2)}{2ac}$

(b) $\frac{2(a^2 - c^2)}{3bc}$

(c) $\frac{(a^2 - c^2)}{3ac}$

(d) $\frac{2(c^2 - a^2)}{3ac}$

Q 6. If one side of a triangle is double the other and the angles opposite to these sides differ by $60°$, then the triangle is

[KCET]

(a) obtuse angled

(b) acute angled

(c) isosceles

(d) right angled

Q 7. In triangle ABC, the value of $\frac{\cos\frac{A}{2}\cot\frac{B}{2}-1}{\cot\frac{A}{2}\cot\frac{B}{2}}$ is

[Kerala CEE]

(a) $\frac{a}{a+b+c}$

(b) $\frac{c}{a+b+c}$

(c) $\frac{2a}{a+b+c}$

(d) $\frac{2b}{a+b+c}$

(e) $\frac{2c}{a+b+c}$

Q 8. In a triangle ABC, if $\angle A = 60°$, a = 5, b = 4,

then c is a root of the equation

[Kerala CEE]

(a) $c^2 - 5c - 9 = 0$

(b) $c^2 - 4c - 9 = 0$

(c) $c^2 - 10c + 25 = 0$

(d) $c^2 - 5c - 41 = 0$

(e) $c^2 - 4c - 41 = 0$

Q 9. In a ΔABC, $\frac{(a+b+c)(b+c-a)(c+a-b)(a+b-c)}{4b^2c^2}$ equals

[EAMCET]

(a) $\cos^2 A$

(b) $\cos^2 B$

(c) $\sin^2 A$

(d) $\sin^2 B$

Q 10. IN triangle ABC, a = 2, b = 3 and $\sin A = \frac{2}{3}$,

then B is equal to

[WB JEE]

(a) $30°$

(b) $60°$

(c) $90°$

(d) $120°$

Q 11. In a triangle ABC, if sin A sin B = $\frac{ab}{c^2}$, then the triangle is

[WB JEE]

(a) equilateral

(b) isosceles

(c) right angled

(d) obtuse angled

Q 12. If a = $2\sqrt{2}$, b = 6, A = 45°, then

[WB JEE]

(a) no triangle is possible

(b) one triangle is possible

(c) two triangles are possible

(d) either no triangle or two triangles are possible

Q 13. In a ΔABC, $2a^2 + 4b^2 + c^2 = 4ab + 2ac$,

then cos B is equal to

[BCECE]

(a) 0

(b) $\frac{1}{8}$

(c) $\frac{3}{8}$

(d) $\frac{7}{8}$

Q 14. In ΔABC, if $\sin^2\frac{A}{2}, \sin^2\frac{B}{2}, \sin^2\frac{C}{2}$ be in HP.

Then, a, b, c will be in

[DCE]

(a) AP

(b) GP

(c) HP

(d) None of these

Q 15. In any triangle ABC, $\frac{\tan\frac{A}{2}-\tan\frac{B}{2}}{\tan\frac{A}{2}+\tan\frac{B}{2}}$ is equal to

[DCE]

(a) $\frac{a-b}{a+b}$

(b) $\frac{a-b}{c}$

(c) $\frac{a-b}{a+b+c}$

(d) $\frac{c}{a+b}$

Q 16. Angles of a triangle are in the ratio $4 : 1 :$ 1. The ratio between its greatest side and perimeter is

[UP SEE]

(a) $\frac{3}{2+\sqrt{3}}$

(b) $\frac{1}{2+\sqrt{3}}$

(c) $\frac{\sqrt{3}}{\sqrt{3}+2}$

(d) $\frac{2}{2+\sqrt{3}}$

Q 17. If in a ΔABC, $2b^2 = a^2 + c^2$, then $\frac{\sin 3B}{\sin B}$ is equal to

[UP SEE]

(a) $\frac{c^2 - a^2}{2ca}$

(b) $\frac{c^2 - a^2}{ca}$

(c) $\left(\frac{c^2 - a^2}{ca}\right)^2$

(d) $\left(\frac{c^2 - a^2}{2ca}\right)^2$

Q 18. The angles A, B and C of a ΔABC are in AP. If b : c = $\sqrt{3} : \sqrt{2}$, then the angle A is

[KCET]

(a) 30°

(b) 15°

(c) 75°

(d) $45°$

Q 19. The area of the triangle whose sides are

$6, 5, \sqrt{13}$ (in square unit) is

[Kerala CEE]

(a) $5\sqrt{2}$

(b) 9

(c) $6\sqrt{2}$

(d) 11

(e) 13

Q 20. In a triangle ABC, $\frac{\cos A}{a} = \frac{\cos B}{b} = \frac{\cos C}{c}$. If $a = \frac{1}{\sqrt{6}}$, then the area of the triangle (in square unit) is

[Kerala CEE]

(a) $\frac{1}{24}$

(b) $\frac{\sqrt{3}}{24}$

(c) $\frac{1}{8}$

(d) $\frac{1}{\sqrt{3}}$

(e) $\frac{5}{12\sqrt{3}}$

Q 21. In any triangle ABC, $c^2 \sin 2B + b^2 \sin 2C$ is equal to

[Kerala CEE]

(a) $\frac{\Delta}{2}$

(b) Δ

(c) 2Δ

(d) 3Δ

(e) 4Δ

Q 22. In ΔABC, if $\frac{1}{b+c}+\frac{1}{c+a}=\frac{3}{a+b+c}$, then C is equal to

[AMU]

(a) $90°$

(b) $60°$

(c) $45°$

(d) $30°$

Q 23. Observe the following statements

I. In ΔABC, $b\cos^2\frac{C}{2}+c\cos^2\frac{B}{2}=s$

II. In ΔABC, $\cot\frac{A}{2}=\frac{b+c}{2}\Rightarrow B = 90°$

Which of the following is correct ?

[EAMCET]

(a) Both I and II are true

(b) I is true, II is false

(c) I is false, II is true

(d) Both I and II are false

Q 24. If the sides of a right angle triangle form an AP, the 'sin' of the acute angles are

[VITEEE]

(a) $\left(\frac{3}{4}, \frac{4}{5}\right)$

(b) $\left(\sqrt{3}, \frac{1}{\sqrt{3}}\right)$

(c) $\left(\sqrt{\frac{\sqrt{5}-1}{2}}, \sqrt{\frac{\sqrt{5}-1}{2}}\right)$

(d) $\left(\sqrt{\frac{\sqrt{3}-1}{2}}, \sqrt{\frac{\sqrt{3}-1}{2}}\right)$

Q 25. If the angles of a triangle are in the ratio

$3 : 4 : 5$, then the sides are in the ratio

[Jamia Millia Islamia]

(a) $2 : \sqrt{6} : \sqrt{3}+1$

(b) $\sqrt{2} : \sqrt{6} : \sqrt{3}+1$

(c) $2 : \sqrt{3} : \sqrt{3}+1$

(d) $3 : 4 : 5$

Q 26. In ΔABC, $(a-b)^2 \cos^2\frac{C}{2}+(a+b)^2\sin^2\frac{C}{2}$ is equal to

[Manipal]

(a) a^2

(b) b^2

(c) c^2

(d) None of these

Q 27. The sides of a triangle are respectively 7 cm, $4\sqrt{3}$ cm and $\sqrt{13}$ cm, then the smallest angle of the triangle is

[MP PET]

(a) $\frac{\pi}{6}$

(b) $\frac{\pi}{3}$

(c) $\frac{\pi}{4}$

(d) $\frac{\pi}{5}$

Q 28. If in a ΔABC, a = 6 cm, b = 8 cm, c = 10 cm, then the value of sin 2A is

[RPET]

(a) $\frac{6}{25}$

(b) $\frac{8}{25}$

(c) $\frac{10}{25}$

(d) $\frac{24}{25}$

Q 29. In any ΔABC under usual notation, a(b cos C – c cos B) is equal to

[J&K CET]

(a) $b^2 - c^2$

(b) $c^2 - b^2$

(c) $\frac{b^2 - c^2}{2}$

(d) $\frac{c^2 - b^2}{2}$

Q 30. If in a ΔABC, $4 \sin A = 4 \sin B = 3 \sin C$, then cos C is equal to

[J&K CET]

(a) $\frac{1}{3}$

(b) $\frac{1}{9}$

(c) $\frac{1}{27}$

(d) $\frac{1}{18}$

Q 31. In a ΔABC, if $(\sqrt{3}-1)a = 2b$, $A = 3B$, then $\angle C$ is

[Guj. CET]

(a) $60°$

(b) $120°$

(c) $30°$

(d) $45°$

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Q 32. In a ΔABC, (b + c)(bc) cos A + (a + C)(ac) cos B + (a + b)(ab) cos C is

[Kerala CEE]

(a) $a^2 + b^2 + c^2$

(b) $a^3 + b^3 + c^3$

(c) $(a + b + c)(a^2 + b^2 + c^2)$

(d) (a + b + c)(ab + bc + ca) (e) abc

Q 33. If in ΔABC, $\sin\frac{A}{2}\sin\frac{C}{2} = \sin\frac{B}{2}$ and 2s is the perimeter of the triangle, then s is

[Kerala CEE]

(a) $2b$

(b) b

(c) $3b$

(d) $4b$

(e) $3b/2$

Q 34. In a ΔABC, if $\tan\frac{A}{2}=\frac{5}{6}, \tan\frac{C}{2}=\frac{2}{5}$, then

[Kerala CEE]

(a) a, c, b are in AP

(b) a, b, c are in GP

(c) b, a, c are in AP

(d) a, b, c are in AP

(e) a, c, b are in GP

Q 35. If two angles of ΔABC are $45°$ and $60°$, then the ratio of the smallest and the greatest sides are

[EAMCET]

(a) $(\sqrt{3} - 1) : 1$

(b) $\sqrt{3} : \sqrt{2}$

(c) $1 : \sqrt{3}$

(d) $\sqrt{3} : 1$

Q 36. In ΔABC, $(a + b + c)\left(\tan\frac{A}{2} + \tan\frac{B}{2}\right)$ is equal to

[EAMCET]

(a) $2c\cot\frac{C}{2}$

(b) $2a\cot\frac{A}{2}$

(c) $2b\cot\frac{B}{2}$

(d) $\tan\frac{C}{2}$

Q 37. The sides of triangle are in the ratio $1:\sqrt{3}:2$, then the angles of the triangle are in ratio

[WB JEE]

(a) 1 : 3 : 5

(b) 2 : 3 : 1

(c) $3 : 2 : 1$

(d) $1 : 2 : 3$

Q 38. Points D, E are taken on the side BC of the ΔABC, such that BD = DE = EC. If $\angle$BAD = x, $\angle$DAE = y, $\angle$EAC = z, then the value of $\frac{\sin(x+y)\sin(y+z)}{\sin x \sin z}$ is equal to

[BCECE]

(a) 1

(b) 2

(c) 4

(d) None of these

Q 39. In ΔABC, if cot A, cot B, cot C be in AP, then a^2, b^2, c^2 are in

[Manipal]

(a) HP

(b) GP

(c) HP

(d) None of these

Q 40. In ΔABC, if $(a + b + c)(a - b + c) = 3ac$,

then

[RPET]

(a) $\angle B = 60°$

(b) $\angle B = 30°$

(c) $\angle C = 60°$

(d) $\angle A + \angle C = 90°$

Q 41. If in a ΔABC, $\frac{b+c}{11}=\frac{c+a}{12}=\frac{a+b}{13}$, then cos A is equal to

[Guj. CET]

(a) $\frac{1}{5}$

(b) $\frac{5}{7}$

(c) $\frac{19}{35}$

(d) None of these

Q 42. A triangular park is enclosed on two sides by a fence and on the third side by a straight river bank. The two sides having fence are of same length x. The maximum area enclosed by the park is

[AIEEE]

(a) $\sqrt{\frac{x^3}{8}}$

(b) $\frac{1}{2}x^2$

(c) πx^2

(d) $\frac{3}{2}x^2$

Q 43. In a ΔABC, a, c, A are given and b_1, b_2 are two values, if the third side b such that $b_2 = 2b_1$, then sin A is equal to

[DCE]

(a) $\frac{\sqrt{9a^2 - c^2}}{8a^2}$

(b) $\frac{\sqrt{9a^2 - c^2}}{8c^2}$

(c) $\frac{\sqrt{9a^2 + c^2}}{8a^2}$

(d) None of these

Q 44. In a ΔABC, if the sides are a = 3, b = 5 and c = 4, then $\sin\frac{B}{2}+\cos\frac{B}{2}$ is equal to

[BITSAT]

(a) $\sqrt{2}$

(b) $\frac{\sqrt{3}+1}{2}$

(c) $\frac{\sqrt{3}-1}{2}$

(d) 1

Q 45. The area of the ΔABC, in which a = 1, b = 2, $\angle$C = $60°$, is

[UP SEE]

(a) 4 sq units

(b) $\frac{1}{2}$ sq unit

(c) $\frac{\sqrt{3}}{2}$ sq unit

(d) $\sqrt{3}$ sq units

Q 46. In a ΔABC, b = $\sqrt{3}$, c = 1 and $\angle A = 30°$, then the largest angle of the triangle is

[UP SEE]

(a) $60°$

(b) $135°$

(c) $90°$

(d) $120°$

Q 47. If in the ΔABC, $\angle B = 45°$, then $a^4 + b^4 + c^4$ is equal to

[Kerala CEE]

(a) $2a^2(b^2 + c^2)$

(b) $2c^2(a^2 + b^2)$

(c) $2(a^2b^2 + b^2c^2 + c^2a^2)$

(d) $2a^2b^2 + 2b^2c^2 + 2a^2c^2$

Q 48. If in a ΔABC, $a = 15$, $b = 36$, $c = 39$, then $\sin\frac{C}{2}$ is equal to

[Kerala CEE]

(a) $\frac{\sqrt{3}}{2}$

(b) $\frac{1}{2}$

(c) $\frac{1}{\sqrt{2}}$

(d) $-\frac{1}{\sqrt{2}}$

(e) $-\frac{1}{2}$

Q 49. The angles of a triangle are in the ratio 3 : 5 : 10. Then, the ratio of the smallest side the greatest side is

(a) 1 : sin $10°$

(b) 1 : 2 sin $10°$

(c) 1 : cos $10°$

(d) 1 : 2 cos $10°$

Q $50.$ In a ΔABC, 2ca sin $\frac{A-B+C}{2}$ is equal to

[OJEE]

(a) $a^2 + b^2 - c^2$

(b) $c^2 + a^2 - b^2$

(c) $b^2 - c^2 - a^2$

(d) $c^2 - a^2 - b^2$

Q 51. In a ΔABC, if b + c = 2a and $\angle$A = $60°$,

then ΔABC is

[WB JEE]

(a) equilateral

(b) right angled

(c) isosceles

(d) scalene

Q 52. The side of a triangle are 4 cm, 5 cm and 6 cm. The area of the triangle is equal to

[MP PET]

(a) $\frac{15}{4}\text{cm}^2$

(b) $\frac{15}{4}\sqrt{7}\,\text{cm}^2$

(c) $\frac{4}{15}\sqrt{7}\,\text{cm}^2$

(d) None of these

Q 53. If sin A : sin B : sin C = $3 : 4 : 5$, then cos A : cos B is equal to

[J&K CET]

(a) $4 : 3$

(b) $5 : 3$

(c) $3 : 4$

(d) $3 : 5$

Q 54. If two angles of a triangle are $45°$ and $\tan^{-1}(2)$, then the third angle is

[J&K CET]

(a) $60°$

(b) $75°$

(c) $\tan^{-1} 3$

(d) $90°$

Q 55. In a ΔABC, among the following which one is true ?

[IIT JEE]

(a) $(b+c)\cos\frac{A}{2}=a\sin\left(\frac{B+C}{2}\right)$

(b) $(b+c)\cos\left(\frac{B+C}{2}\right)=a\sin\frac{A}{2}$

(c) $(b-c)\cos\left(\frac{B-C}{2}\right)=a\cos\frac{A}{2}$

(d) $(b-c)\cos\frac{A}{2}=a\sin\left(\frac{B-C}{2}\right)$

Q 56. If in a ΔABC, the altitudes from the vertices A, B, C on opposite sides are in HP, then sin A, sin B, sin C are in

[AIEEE]

(a) HP

(b) Arithmetico–Geometric Progression

(c) AP

(d) GP

Q 57. If the sides of the triangle are p, q, $\sqrt{p^2+q^2+pq}$, then the greatest angle is

[Kerala CEE]

(a) $\frac{\pi}{2}$

(b) $\frac{5\pi}{4}$

(c) $\frac{2\pi}{3}$

(d) $\frac{7\pi}{4}$

(e) $\frac{5\pi}{3}$

Q 58. If $\Delta = a^2 - (b - c)^2$, where Δ is the area of ΔABC, then tan A is equal to

[Kerala CEE]

(a) $\frac{15}{16}$

(b) $\frac{8}{17}$

(c) $\frac{8}{15}$

(d) $\frac{1}{2}$

(e) $\frac{11}{15}$

Q 59. If in a ΔABC a = 5, b = 4, $A = \frac{\pi}{2} +$ B, then

C

[Kerala CEE]

(a) is $\tan^{-1}\left(\frac{1}{9}\right)$

(b) is $\tan^{-1}\left(\frac{9}{40}\right)$

(c) cannot be evaluated

(d) is $2\tan^{-1}\left(\frac{1}{9}\right)$

(e) is $2\tan^{-1}\left(\frac{1}{40}\right)$

Q 60. In ΔABC, a($\cos^2$ B + $\cos^2$ C) + cos A (c cos C + b cos B) is equal to

[EAMCET]

(a) a

(b) b

(c) c

(d) a + b + c

Q 61. In a ΔABC, $\Sigma(b + c) \tan\frac{A}{2}\tan\left(\frac{B-C}{2}\right)$ is equal to

[EAMCET]

(a) a

(b) b

(c) c

(d) 0

Q 62. Two sides of a triangle are given by the roots of the equation $x^2 - 5x + 6 = 0$ and the angle between the sides is $\frac{\pi}{3}$. Then, the perimeter of the triangle is

[EAMCET]

(a) $5+\sqrt{2}$

(b) $5+\sqrt{3}$

(c) $5+\sqrt{5}$

(d) $5+\sqrt{7}$

Q 63. If a, b and c are the sides of a triangle such that $a^4 + b^4 + c^4 = 2c^2(a^2 + b^2)$, then the angles opposite to the side C is

[J & K CET]

(a) $45°$ or $90°$

(b) $30°$ or $135°$

(c) $45°$ or $135°$

(d) $60°$ or $120°$

Q 64. The sides of a triangle are $\sin\alpha$, $\cos\alpha$ and $\sqrt{1+\sin\alpha\cos\alpha}$ for some $0 < \alpha < \frac{\pi}{2}$. Then, the greatest angle of the triangle is

[AIEEE]

(a) $60°$

(b) $90°$

(c) $120°$

(d) $150°$

Q 65. The perimeter of a ΔABC is 6 times the arithmetic mean of the sine ratios of its angles. If a = 1, then A is equal to

[Kerala CEE]

(a) $\frac{\pi}{6}$

(b) $\frac{\pi}{3}$

(c) $\frac{\pi}{2}$

(d) $\frac{2\pi}{3}$

(e) $\frac{3\pi}{4}$

Q 66. The base angle of triangle are $22\frac{1}{2}^{\circ}$ and $112\frac{1}{2}^{\circ}$. If b is the base and h is the height of the triangle, then

[Kerala CEE]

(a) b = 2h

(b) $b = 3h$

(c) $b = (1 + \sqrt{3})h$

(d) $b = (2 + \sqrt{3})h$

(e) $2b = 3h$

Q 67. In ΔABC, if $a = 3$, $b = 4$, $c = 5$, then the value of $\sin 2B$ is

[J&K CET]

(a) $\frac{4}{5}$

(b) $\frac{3}{20}$

(c) $\frac{24}{25}$

(d) $\frac{1}{50}$

Q 68. If in a ΔABC, $a\cos^2\left(\frac{C}{2}\right)+c\cos^2\left(\frac{A}{2}\right)=\frac{3b}{2}$, then the sides a, b and c

[AIEEE]

(a) are in AP

(b) are in GP

(c) are in HP

(d) satisfy a + b = c

Q 69. In a ΔABC, medians AD and BE are drawn. If AD = 4, $\angle$DAB = $\frac{\pi}{6}$ and $\angle$ABE = $\frac{\pi}{3}$, then the area of the ΔABC is

[AIEEE]

(a) $\frac{8}{3}$ sq units

(b) $\frac{16}{3}$ sq units

(c) $\frac{32}{3\sqrt{3}}$ sq units

(d) $\frac{64}{3}$ sq units

Q 70. In a triangle vertex angles are A, B, C and side BC are given. The area of ΔABC is

[DCE]

(a) $\frac{s(s-a)(s-b)(s-c)}{2}$

(b) $\frac{b^2 \sin C \sin A}{\sin B}$

(c) ab sin C

(d) $\frac{1}{2}.\frac{a^2 \sin B \sin C}{\sin A}$

Q 71. Angles A, B and C of a triangle are in AP with common difference 15 degree, then angle A is equal to

[DCE]

(a) 45°

(b) 60°

(c) 75°

(d) $30°$

Q 72. In a ΔABC, $\angle B = \frac{\pi}{3}$ and $\angle C = \frac{\pi}{4}$. If D divides BC internally in ratio $1 : 3$, then the value of $\frac{\sin \angle BAD}{\sin \angle CAD}$ is

[UP SEE]

(a) $\frac{1}{\sqrt{3}}$

(b) $\frac{1}{\sqrt{6}}$

(c) $\sqrt{\frac{2}{3}}$

(d) $\frac{1}{3}$

Q 73. In ΔABC, if $\begin{vmatrix} 1 & a & b \\ 1 & c & a \\ 1 & b & c \end{vmatrix} = 0$, then

$\sin^2 A + \sin^2 B + \sin^2 C$ is equal to

[KCET]

(a) $\frac{4}{9}$

(b) $\frac{9}{4}$

(c) $3\sqrt{3}$

(d) 1

Q 74. In a triangle, if b = 20, c = 21 and $\sin A = \frac{3}{5}$, then a is equal to

[EAMCET]

(a) 12

(b) 13

(c) 14

(d) 15

Q 75. In ΔABC, $\frac{1+\cos(A-B)\cos C}{1+\cos(A-C)\cos B}$ is equal to

[MP PET]

(a) $\frac{a-b}{a+b}$

(b) $\frac{a+b}{a+c}$

(c) $\frac{a^2-b^2}{a^2-c^2}$

(d) $\frac{a^2+b^2}{a^2+c^2}$

Q 76. In a ΔABC, a = 5, b = 4 and $\cos(A+B)=\frac{31}{32}$.

In this triangle, c is equal to

[J&K CET]

(a) $\sqrt{6}$

(b) 36

(c) 6

(d) None of these

Answers

Topic 1 : Relation between sides and Angles

1. (d) 2. (b) 3. (b) 4. (b) 5. (d) 6. (d) 7. (e) 8. (b)

9. (c) 10. (c) 11. (c) 12. (a) 13. (d) 14. (c) 15. (b)

16. (c) 17. (d) 18. (c) 19. (b) 20. (b) 21. (e) 22.

(b) 23. (b) 24. (a) 25. (a) 26. (c) 27. (a) 28. (d)

29. (a) 30. (b) 31. (b) 32. (b) 33. (a) 34. (d) 35.

(a) 36. (a) 37. (d) 38. (c) 39. (c) 40. (a) 41. (a)

42. (b) 43. (b) 44. (a) 45. (c) 46. (d) 47. (c) 48.

(c) 49. (d) 50. (b) 51. (a) 52. (b) 53. (a) 54. (c)

55. (d) 56. (c) 57. (c) 58. (c) 59. (b) 60. (a)

61. (d) 62. (d) 63. (c) 64. (c) 65. (a) 66. (a) 67.

(c) 68. (a) 69. (c) 70. (d) 71. (b) 72. (b) 73. (b)

74. (b) 75. (d) 76. (d)

Topic 2 : Circle Connected with Triangle

1. (c) 2. (e) 3. (c) 4. (a) 5. (c) 6. (b) 7. (c) 8. (d)

9. (a) 10. (a) 11. (b) 12. (a) 13. (c) 14. (b) 15. (d)

16. (d) 17. (c) 18. (c) 19. (c) 20. (c) 21. (b) 22.

(d) 23. (a) 24. (c) 25. (b) 26. (a)

Mathematical Logic and Bollean Algebra

Useful Results & Formulae

1. Table for basic Logical Connections

p	q	~ p	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$
T	T	F	T	T	T	T
T	F	F	F	T	F	F

F	T	T	F	T	T	F
F	F	T	F	F	T	T

2. (i) $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p) \equiv q \leftrightarrow p$

(ii) $\sim(p \rightarrow q) \equiv p \wedge \sim q$

(iii) $\sim(p \leftrightarrow q) \equiv (p \wedge \sim q) \vee (q \wedge \sim p)$

3. (i) $p \vee p = p$

(ii) $p \wedge p = p$

4. (i) $p \vee q = q \vee p$

(ii) $p \wedge q = q \wedge p$

5. (i) $(p \vee q) \vee r = p \vee (q \vee r)$

(ii) $(p \wedge q) \wedge r = p \wedge (q \wedge r)$

6. (i) $p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$

(ii) $p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$

7. (i) $\sim(p \wedge q) = \sim p \vee \sim q$

(ii) $\sim(p \vee q) = \sim p \wedge \sim q$

8. (i) $p \wedge T = p$

(ii) $p \vee T = T$

(iii) $p \vee F = p$

(iv) $p \wedge F = F$

9. (i) $p \wedge \sim p = F$

(ii) $p \vee \sim p = T$

10. (i) Contrapositive law $p \rightarrow q = \sim q \rightarrow \sim p$

(ii) Converse law $p \rightarrow q \equiv q \rightarrow p$

(iii) Inverse of $p \rightarrow q$ is $\sim p \rightarrow \sim q$

11. For dual replace $\vee$ by $\wedge$ and $\wedge$ by $\vee$.

12. (i) $a + b = b + a$

(ii) $a \cdot b = b \cdot a$

(iii) $a + (b \cdot c) = (a + b) \cdot (a + c)$

(Distribute Laws)

(iv) $a(b + c) = (a \cdot b) + (a \cdot c)$

(Distribute Laws)

(v) $a + 0 = 0 + a = a$

(vi) $a \cdot 1 = 1 \cdot a = a$

13. (i) a + a'= 1 (ii) a . a' = 0 (iii) 0' = 1 (iv) l' = 0

14. (i) (a + b)' = a' . b'

(De–Morgan's Laws)

(ii) (a . b) = a' + b'

(De–Morgan's Laws)

15. (i) a + (a . b) = a

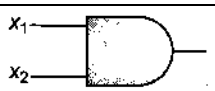
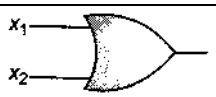
(Absorption Laws)

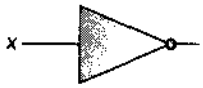
(ii) a . (a + b) = a

(Absorption Laws)

16. For dual replace '+' by '.', '.' by '+', 0 by '1' and 1 by '0'.

17.

Gate	Figure	Representation
AND		$x_1 . x_2$
OR		$x_1 + x_2$

NOT		x'

Topic 1 Mathematical Logic

Q 1. Let S be a non-empty subset of R.

Consider the following statement

P : There is a rational number x $\in$ S such that x > 0. Which of the following statements is the negation of the statement P ?

[A1EEE]

(a) There is a rational number $x \in S$ such that $x \leq 0$

(b) There is no rational number x **e** S such that $x \ 0$

(c) Every rational number $x \in S$ satisfies $x \leq 0$

(d) $x \in S$ and $x \leq 0 \Rightarrow$ x is not rational

Q 2. If 5 (p, q, r) = (~ p) $\vee$ [~ (q $\wedge$ r)] is a compound statement, then S (~ p,~ q,~ r) is

[Kerala CEE]

(a) ~ S (P, q, r)

(b) S (p, q, r)

(c) $p \vee (q \wedge r)$

(d) $p \vee (q \vee r)$

(e) S (p, q, ~ r)

Q 3. For any two statements p and q, ~ (p $\vee$ q)$\vee$ (~ p $\wedge$ q) is logically equivalent to

[VITEEE]

(a) p

(b) ~ p

(c) q

(d) ~q

(e) P $\vee$ q

Q 4. Let p: 7 is not greater than 4 and

[Kerala CEE]

q : Paris is in France be two statements.

Then, ~ (p $\vee$ q) is the statement

(a) 7 is greater than 4 or Paris is not in France

(b) 7 is not greater than 4 and Paris is not in

France

(c) 7 is greater than 4 and Paris is in France

(d) 7 is not greater than 4 or Paris is not in France

(e) 7 is greater than 4 and Paris is not in France

Q 5. **Statement I** ~ $(p \leftrightarrow \sim q)$ is equivalent to $p \leftrightarrow q$.

Statement II ~ $(p \leftrightarrow \sim q)$ is a tautology.

[AIEEE]

(a) Statement I is true, Statement II is true;

Statement II is a correct explanation for Statement I.

(b) StatementI is true, Statement II is true;

Statement II is not a correct explanation for

StatementI.

(c) Statement I is true, Statement II is false.

(d) Statement I is false, Statement II is true.

Q 6. Which of the following statements is a

tautology ?

[DCE]

(a) $(\sim q \wedge p) \wedge q$

(b) $(\sim q \wedge p) \wedge (p \wedge \sim p)$

(c) $(\sim q \wedge p) \vee (p \vee \sim p)$

(d) $(p \wedge q) \wedge (\sim(p \wedge q))$

Q 7. The negation of $p \wedge (q \rightarrow \sim r)$ is

[KCET]

(a) $\sim p \wedge (q \wedge r)$

(b) $p \vee (q \vee r)$

(c) $p \vee (q \wedge r)$

(d) ~ p $\vee$ (q $\wedge$ r)

Q 8. Identify the false statement

[Kerala CEE]

(a) ~ [p $\vee$ (~ q)] $\equiv$ (~ p) $\wedge$ q

(b) [p $\vee$ q] $\vee$ (~ p) is a tautology

(c) [p $\wedge$ q] $\wedge$ (~ p) is a contradiction

(d) ~ [p $\wedge$ (~ p)] is a tautology

(e) ~(p $\vee$ q) = (~ p)$\vee$ (~ q)

Q 9. $\sim (\sim p \rightarrow q) \equiv$

[MHT CET]

(a) $p \wedge \sim q$

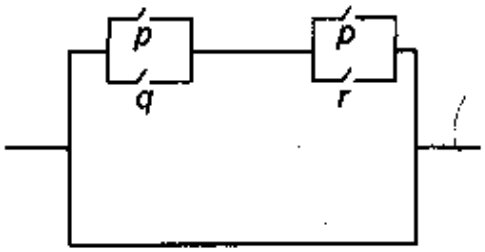
(b) $\sim p \wedge q$

(c) $\sim p \wedge \sim q$

(d) $\sim p \vee \sim q$

Q 10. Simplify the following circuit and find the boolean polynomial.

[MHT CET]



(a) $p \vee (q \wedge r)$

(b) $p \wedge (q \vee r)$

(c) $p \vee (q \vee r)$

(d) $p \wedge (q \wedge r)$

Q 11. Simplify $(p \vee q) \wedge (p \vee \sim q)$

[MHT CET]

(a) p

(b) T

(c) F

(d) q

Q 12. Let p be the statement "x is an irrational number", q be the statement "y is a transcendental

number" and r be the statement "x is a rational

number iff y is a transcendental number".

Statement I r is equivalent to either q or p.

Statement II r is equivalent to ~ (p $\leftrightarrow$ ~ q).

[AIEEE]

(a) Statement I is true, Statement II is true;

Statement II is a correct explanation for Statement I

(b) StatementI is true, StatementII is true;

StatementII is NOT a correct explanation for

StatementI

(c) Statement I is true, Statement II is false

(d) Statement I is false, Statement II is true

Q 13. The statement $p \rightarrow (q \rightarrow p)$ is equivalent

to

[AIEEE]

(a) $p \rightarrow (p \leftrightarrow q)$

(b) $p \rightarrow (p \rightarrow q)$

(c) $p \rightarrow (p \vee q)$

(d) $p \rightarrow (p \wedge q)$

Q 14. $\sim (p \vee q) \vee (\sim p \wedge q)$ is logically equivalent to

[DCE]

(a) $\sim p$

(b) p

(c) q

(d) ~ q

Q 15. The converse of the contrapositive of the conditional $p \rightarrow \sim q$ is

[KCET]

(a) $p \rightarrow q$

(b) $\sim p \rightarrow \sim q$

(c) $\sim q \rightarrow p$

(d) $\sim p \rightarrow q$

Q 16. Let p be the statement 'Ravi races' and let q be the statement 'Ravi wins'. Then, the verbal translation of $\sim (p \vee (\sim q))$ is

[Kerala CEE]

(a) Ravi does not race and Ravi does not win

(b) It is not true that Ravi races and that Ravi does not win

(c) Ravi does not race and Ravi wins

(d) It is not true that Ravi races or that Ravi does not win

(e) It is not true that Ravi does not race and Ravi does not win

Q 17. If p, q and r are simple propositions with truth values T, F, T, then the truth value of $(\sim p \vee q) \wedge \sim q \rightarrow p$ is

[BCECE]

(a) true

(b) false

(c) true, if r is false

(d) None of these

Q 18. Negation of the conditional, "If it rains, I shall go to school" is

[MHT CET]

(a) It rains and I shall go to school

(b) It rains and I shall not go to school

(c) It does not rains and I shall go to school

(d) None of the above

Q 19. Dual of $(x' \; y')' = x \wedge y$ is

[MHT CET]

(a) $(x' \vee y') = x \vee y$

(b) $(x' \wedge y')' = x \vee y$

(c) $(x' \wedge y')' = x \wedge y$

(d) None of the above

Q 20. Negation of "Paris is in France and London is in England" is

[Manipal]

(a) Paris is in England and London is in France

(b) Paris is not in France or London is not in England

(c) Paris is in England or London is in France

(d) None of the above

Q 21. The contrapositive of $(p \vee g) \rightarrow r$ is

[Manipal]

(a) $-r \rightarrow (p \vee q)$

(b) $r \rightarrow (p \vee q)$

(c) $\sim r \rightarrow (\sim p \wedge \sim q)$

(d) $p \rightarrow (q \vee r)$

Q 22. The negation of the proposition "If 2 is prime, then 3 is odd" is

[KCET]

(a) if 2 is not prime, then 3 is not odd

(b) 2 is prime and 3 is not odd

(c) 2 is not prime and 3 is odd

(d) if 2 is not prime, then 3 is odd

Q 23. If p: 4 is an even prime number, q : 6 is a divisor of 12 and r: the HCF of 4 and 6 is 2, then which one of the following is true ?

[Kerala CEE]

(a) $(p \wedge q)$

(b) $(p \vee q) \wedge \sim r$

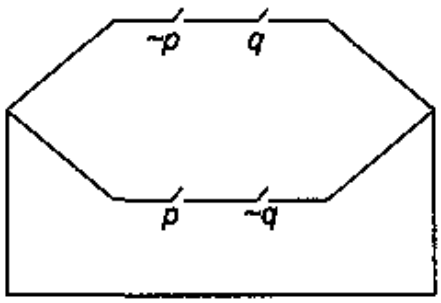
(c) $\sim (q \wedge r) \vee p$

(d) $\sim p \vee (q \wedge r)$

(e) $p \leftrightarrow (q \wedge r)$

Q 24. For the circuit shown below, the Boolean polynomial is

[BCECE]



(a) (~ p $\vee$ q) $\vee$ (p $\vee$ ~ q)

(b) (~ p $\wedge$ q) $\wedge$ (p $\wedge$ q)

(c) (~ p $\wedge$ ~ q) $\wedge$ (q $\wedge$ p)

(d) (~ p $\wedge$ q) $\vee$ (p $\wedge$ ~q)

Q 25. The dual of the statement [p $\vee$ (~ q)] $\wedge$ (~ p) is

[MHT CET]

(a) $p \vee (\sim q) \vee \sim p$

(b) $(p \wedge \sim q) \vee \sim p$

(c) $p \wedge \sim (q \vee \sim p)$

(d) None of these

Q 26. Which of the following statement has the truth value 'F'?

[MHT CET]

(a) A quadratic equation has always a real root

(b) The number of ways of seating 2 persons in two chairs out of n persons is P(n, 2)

(c) The cube roots of unity are in GP

(d) None of the above

Q 27. If$(p \wedge \sim r) \rightarrow (\sim p \vee q)$ is false, then the truth values of p, q and r are respectively

[KCET]

(a) T, F and F

(b) F, F and T

(c) F, T and T

(d) T,F and T

Q 28. The statement ~ $(p \rightarrow q)$ is equivalent to

[Kerala CEE]

(a) $p \wedge (\sim q)$

(b) $\sim p \wedge q$

(c) $p \wedge q$

(d) $\sim p \wedge \sim q$

(e) $p \vee q$

Q 29. $\sim p \wedge q$ is logically equivalent to

[AMU]

(a) $p \rightarrow q$

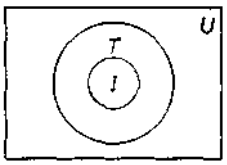
(b) $q \rightarrow p$

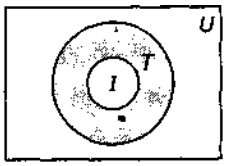
(c) $-(p \rightarrow q)$

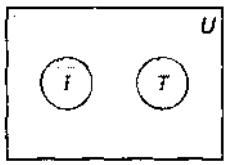
(d) ~(q $\rightarrow$ p)

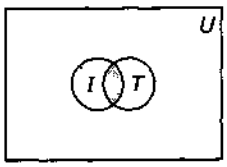
Q 30. Some triangles are not isosceles. Identify the Venn diagram

[MHT CET]

(a) 

(b) 

(c) 

(d) 

Q 31. The negation of the statement "he is rich and happy" is given by

[MHT CET]

(a) he is not rich and not happy

(b) he is not rich or not happy

(c) he is rich and happy

(d) he is not rich and happy

Q 32. The contrapositive of "If two triangles are identical, then these are similar" is

[KCET]

(a) If two triangles are not similar, then these are not identical

(b) If two triangles are not identical, then these are not similar

(c) If two triangles are not identical, then these are similar

(d) If two triangles are not similar, then these are identical

Q 33. Which of the following is not a statement in logic ?

[KCET]

1. Earth is planet. 2. Plants are living objects.

3. $\sqrt{-3}$ is a rational number. 4. $x^2 - 5x + 6 < 0$, when $x \in -R$.

(a) 1

(b) 3

(c) 2

(d) 4

Q 34. ~[p $\leftrightarrow$ q] is

[MHT CET]

(a) tautology

(b) contradiction

(c) neither (a) nor (b)

(d) either (a) or (b)

Q 35. $\sim[(p \wedge q) \rightarrow (\sim p \vee q)]$ is

[MHTCET]

(a) tautology

(b) contradiction

(c) neither (a) nor (b)

(d) either (a) or (b)

Q 36. $(p \wedge \sim q) \wedge (\sim p \wedge q)$ is

[MHT CET]

(a) a tautology

(b) a contradiction

(c) tautology and contradiction

(d) neither a tautology nor a contradiction

Q 37. Which of the following is the inverse of the proposition : "If a number is a prime, then it is odd" ?

[KCET]

(a) If a number is not a prime, then it is odd

(b) If a number is not a prime, then it is not odd

(c) If a number is not odd, then it is not a prime

(d) If a number is odd, then it is a prime

Q 38. If p: A man is happy q : A man is rich.

Then, the statement, "If a man is not happy, then he is not rich" is written as

[MHT CET]

(a) ~ p $\rightarrow$ ~q

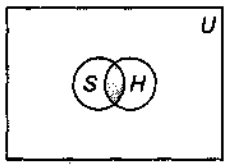
(b) $\sim q \rightarrow p$

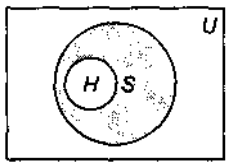
(c) $\sim q \rightarrow \sim p$

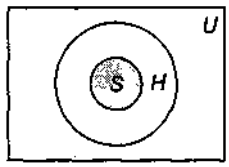
(d) $q \rightarrow \sim p$

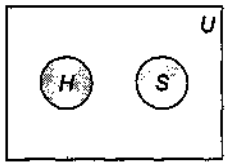
Q 39. H : Set of holidays, S : Set of Sundays and U : Set of day's. Then, the Venn diagram of statement, 'Every Sunday implies holiday' is

[MHT CET]

(a) 

(b) 

(c) 

(d) 

Q 40. Which of the following is not a proposition ?

[KCET]

(a) 3 is prime

(b) $\sqrt{2}$ is irrational

(c) Mathematics is interesting

(d) 5 is an even integer

Q 41. Let p and q be two statements, then $(p \vee q) \vee \sim p$ is

[BCECE]

(a) tautology

(b) contradiction

(c) Both (a) and (b)

(d) None of these

Topic 2 Boolean Algebra

Q 1. In a boolean algebra B with respect to '+' and '•', x denotes the negation of $x \in B$. Then,

[Kerala CEE]

(a) $x - x' = 1$ and $xx' = 1$

(b) $x + x' = 1$ and $x \cdot x' = 0$

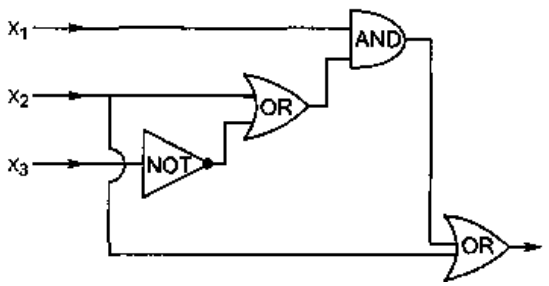
(c) $x + x' = 0$ and $xx' = 1$

(d) $x + x' = 0$ and $x \cdot x' = 0$

(e) $x - x' = 0$ and $xx' = 0$

Q 2. The boolean expression corresponding to the combinational circuit is

[Kerala CEE]



(a) $(x_1 + x_2 - x'_3)x_2$

(b) $(x_1 \cdot (x_2 + x_3)) + x_2$

(c) $(x_1 \cdot (x_2 + x_3)) + x_2$

(d) $(x_1 \cdot (x_2 + x_3')) + x_3$

(e) $(x_1 + x'_2 \cdot x_3) \cdot x_2$

Q 3. Find the function $f(x_1, x_2, x_3)$ satisfying

$f(x_1, x_2, x_3) = 1$ at $x_1 = 1, x_2 = x_3 = 0$.

[MHTCET]

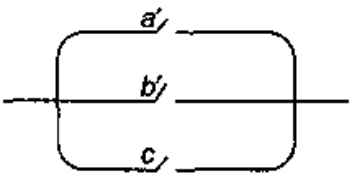
(a) $x_1 - x_2$

(b) $x_1 \cdot x_2'$

(c) $(x_1 + x_2 + x_3)' \cdot x_2$

(d) $(x'_1 + x_3) \cdot x_3$

Q 4. Consider the switching circuit given below



The logical expression corresponding to the

complementary to the above circuit is

[J&K CET]

(a) $a' \cdot b' \cdot c$

(b) $a + b + c'$

(c) $a \cdot b \cdot c'$

(d) $a' + b' + c$

Q 5. Let B be a Boolean algebra. If $x, y \in B$, then $(x \cdot y)'$ is equal to

[Kerala CEE]

(a) x . y

(b) x . y'

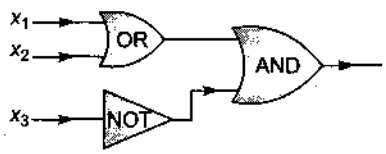
(c) x' . y'

(d) (x' − y')'

(e) x'+ y'

Q 6. The output of the circuit is

[Kerala CEE]



(a) $x_3 \cdot (x_1' + x_2)$

(b) $(x_3' + x_2) \cdot x_1$

(c) $x_3' \cdot (x_1 + x_2)$

(d) $(x_1 + x_2) \cdot x_3$

(e) $(x_1' + x_2') \cdot x_3$

Q 7. The dual of $x + (y \cdot x) = x$ is

[BCECE]

(a) $(x + y) \cdot (x + x) = x$

(b) $x \cdot (y + x) = x$

(c) $x \cdot (y \cdot x) = x$

(d) None of the above

Q 8. If B is a Boolean algebra and $a, b \in B$, then $a \cdot (a + b)$ is equal to

[J&K CET]

(a) a

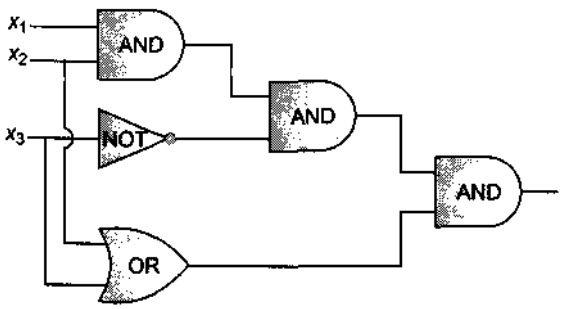
(b) b

(c) 1

(d) a'

Q 9. The output of the circuit is

[Kerala CEE]



(a) $(x_2 + x_3)$. $[(x_1 . x_2)$. $x_3']$

(b) $(x_2 + x_3') \cdot [(x_1 \cdot x_2) \cdot x_3']$

(c) $(x_2 + x_3) + [(x_1 \cdot x_2) \cdot x_3']$

(d) $(x_2 \cdot x_3) + [(x_1 \cdot x_2) \cdot x_3']$

(e) $(x_1 + x_3) \cdot [(x_1 \cdot x_2) \cdot x_3']$

Q 10. Let a be any element in a Boolean algebra B. If $a + x = 1$ and $ax = 0$, then

[AMU]

(a) x = I

(b) $x = 0$

(c) $x = a$

(d) $x = a'$

Q 11. Dual of $(x + y) . (x + 1) = x + x . y + y$ is

[AMU]

(a) $(x . y) + (x . 0) = x . (x + y) . y$

(b) $(x + y) + (x . 1) = x . (x + y) . y$

(c) $(x . y)(x . 0) = x . (x + y) . y$

(d) None of the above

Q 12. In Boolean algebra, the unit element '1'

[BCECE]

(a) has two values

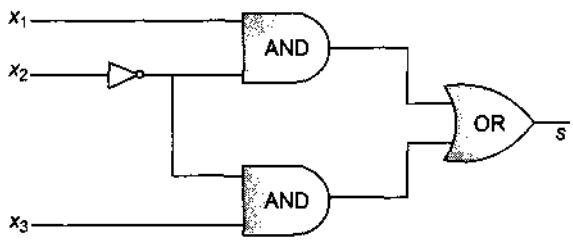
(b) is unique

(c) has at least two values

(d) None of these

Q 13. The output s as a Boolean expression in the inputs x_1, x_2 and x_3 for the logic circuit in the following figure is

[Kerala CEE]



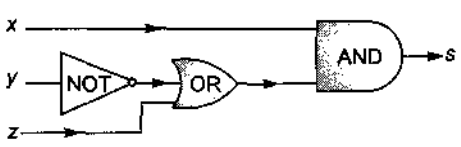
(a) $x_1x_2' + x_2' + x_3$

(b) $x_1 + x_2'x_3 + x_3$

(c) $(x_1x_2)' + x_1x_2'x_3$

(d) $x_1 + x_2' + x_3$

(e) $x_1x_2' + x_2'x_3$

Q 14. 

then s is equal to

[Kerala CEE]

(a) x . (y' + z)

(b) x . (y' + z')

(c) x . (y+z) (d) (x + y) . z

(e)x . y+z

Q 15. If flow values of switches x_1, x_2 and x_3 are respectively 0, 0 and 1, then the flow value of the circuit s = (x_1' . x_2' . x_3) + (x_1 . x_2' . x_3') + (x_1' . x_2 . x_3') is

[AMU]

(a) 0

(b) 1

(c) 2

(d) None of these

Q 16. In a Boolean algebra $a \vee (a' \wedge b)$ is equal to

[AMU]

(a) $a \vee b$

(b) $a \wedge b$

(c) a'

(d) b’

Q 17. The Boolean function of the input/output table as given below

[AMU]

	Input		**Output**
x_1	x_2	x_3	x_4
1	1	1	1
1	1	0	**1**

1	**0**	1	1
1	0	0	0
0	1	0	0
0	0	0	1

is

(a) $f(x_1,x_2,x_3) = x_1 \cdot x_2 \cdot x_3 + x_1 \cdot x_2 \cdot x_3' + x_1 \cdot x_2' \cdot x_3$

$+ x_1' \cdot x_2' \cdot x_3'$

(b) $f(x_1,x_2,x_3) = x_1' \cdot x_2' \cdot x_3' + x_1' \cdot x_2' \cdot x_3 + x_1' \cdot x_2 \cdot x_3' + x_1 \cdot x_2 \cdot x_3$

(c) $f(x_1,x_2,x_3) = x_1 \cdot x_2' \cdot x_3' + x_1' \cdot x_2 \cdot x_3'$

(d) $f(x_1, x_2, x_3) = x_1' \cdot x_2 \cdot x_3 + x_1 \cdot x_2' \cdot x_3$

Q 18. An OR gate is the Boolean function defined of

[BCECE]

(a) $f(x_1, x_2) = x_1x_2; x_1,x_2 \in \{0,1\}$

(b) $f(x_1,x_2) = x_1 + x_2; x_1,x_2 \in \{0,1\}$

(c) $f(x_1, x_2) = x_1; x_1, x_2 \in \{0,1\}$

(d) $f(x_1, x_2) = x_2; x_1, x_2 \in \{0,1\}$

Answers

Topic 1 : Mathematical Logic

1- (c) 2. (d) 3. (b) 4. (e) 5. (c) 6. (c) 7. (d) 8. (e)

9. (c) 10. (a) 11. (a) 12. (*) 13. (c) 14. (a) 15. (d)

16. (c) 17. (a) 18. (b) 19. (b) 20. (b) 21. (c) 22.

(b) 23. (d) 24. (d) 25. (b) 26. (a) 27. (a) 28. (a)

29. (d) 30. (b) 31. (b) 32. (a) 33. (c) 34. (c) 35.

(b) 36. (b) 37. (b) 38. (a) 39. (c) 40. (c) 41. (a)

Topic 2 : Boolean Algebra

1- (b) 2. (c) 3. (b) 4. (d) 5. (e) 6. (c) 7. (b) 8. (a)

9. (a) 10. (d) 11. (a) 12. (b) 13. (e) 14. (a) 15. (b)

16. (a) 17. (a) 18. (b)

Permutations and Combinations

Topic 1

Number of Permutations, Conditional Permutations

Q 1. The number of permutations by taking all letters and keeping the vowels of the word COMBINE in the odd places is

[WBJEE]

(a) 96

(b) 144

(c) 512

(d) 576

Q 2. If ${}^{56}P_{r+6} : {}^{54}P_{r+3} = 30800 : 1$, then the value of r is

[Kerala CEE]

(a) 40

(b) 51

(c) 41

(d) 510

(e) 101

Q 3. The number of seven digit integers, with sum of the digits equal to 10 and formed by using the digits 1, 2 and 3 only, is

[ITT JEE]

(a) 55

(b) 66

(c) 77

(d) 88

Q 4. If ${}^{2n+1}P_{n-1} : {}^{2n-1}P_n = 3:5$, then the value of n is equal to

[Kerala CEE]

(a) 4

(b) 3

(c) 2

(d) 1

(e) 5

Q 5. If P_m stands for ${}^{m}P_{m}$, then $1 + 1P_1 + 2P_2 + 3P_3 + \ldots\ldots + n \cdot P_n$ is equal to

[AMU]

(a) n!

(b) (n + 3)!

(c) $(n+2)!$

(d) $(n + 1)!$

Q 6. The number of ways of arranging letters of the word HAVANA so that V and N do not appear together is

[BCECE]

(a) 60

(b) 80

(c) 100

(d) 120

Q 7. Eight chair are numbered 1 to 8. Two women and three men wish to occupy one chair each. First the women choose the chairs from amongst the chair marked 1 to 4; and then the men select the chairs from amongst the remaining. The number of possible arrangements is

[Jamia Millia Islamia]

(a) ${}^{6}C_{3} \times {}^{4}C_{2}$

(b) ${}^4P_2 \times {}^6P_3$

(c) ${}^4C_2 + {}^4P_3$

(d) None of these

Q 8. The number of ways in which 5 boys and 5 girls can be seated for a photograph so that no two girls sit next to each other is

[J&K CET]

(a) $6! - 5!$

(b) $(5!)^2$

(c) $\frac{10!}{(5!)}$

(d) $\frac{10!}{(5!)^2}$

Q 9. How many numbers lying between 999 and 10000 can be formed with the help of the digits $0,2,3,6,7,8$ when the digits are not be repeated ?

[DCE]

(a) 100

(b) 200

(c) 300

(d) 400

Q 10. 9 balls are to be placed in 9 boxes and 5 of the balls cannot fit into 3 small boxes. The number of ways of arranging one ball in each of the boxes is

[BITSAT, EAMCET]

(a) 18720

(b) 18270

(c) 17280

(d) 12780

Q 11. How many 10 digits numbers can be written by using digits 9 and 2 ?

[UP SEE]

(a) ${}^{10}C_1 + {}^{9}C_2$

(b) 2^{10}

(c) ${}^{10}C_2$

(d) $10!$

Q 12. The ten's digit in $1!+4!+7!+10!+12!+13!+15!+16!+17!$ is divisible by

[KCET]

(a) 4

(b) $3!$

(c) 5

(d) 7

Q 13. All the words that can be formed using alphabets A, H, L, U, R are written as in a dictionary (no alphabet is repeated). Then, the rank of the word RAHUL is

[Kerala CEE]

(a) 70

(b) 71

(c) 72

(d) 73

(e) 74

Q 14. Eight different letters of an alphabet are given. Words of four letters from these are formed. The number of such words with at least one letter repeated is

[EAMCET, BCECE]

(a) $\binom{8}{4} - {}^{8}P_4$

(b) $8^4 + \binom{8}{4}$

(c) $8^4 - {}^8P_4$

(d) $8^4 - \binom{8}{4}$

Q 15. The number of natural numbers less than 1000, in which no two digits are repeated, is

[BCECE]

(a) 738

(b) 792

(c) 837

(d) 720

Q 16. In how many different ways can the letters of the word 'MATHEMATICS' be arranged ?

(a) $11!$

(b) $-11!/2!$

(c) $11!/(2!)^2$

(d) $11!/(2!)^3$

[MPPET]

Q 17. The number of permutations of the letters of the word 'CONSEQUENCE' in which all the three E's are together, is

[Guj.CET]

(a) $9!3!$

(b) $\frac{9!}{2!2!}$

(c) $\frac{9!}{2!2!3!}$

(d) $\frac{9!}{2!3!}$

Q 18. The sum of all five digit numbers that can be formed using the digits $1, 2, 3, 4, 5$ when repetition of digits is not allowed, is

[AMU]

(a) 366000

(b) 660000

(c) 360000

(d) 3999960

Q 19. How many ways are there to arrange the letters in the word 'GARDEN' with the vowels in alphabetical order ?

[Jamia Millia Islamia]

(a) 120

(b) 240

(c) 360

(d) 480

Q 20. If the letters of the word 'SACHIN' are arranged in all possible ways and these words are written out as in dictionary, then the word 'SACHIN' appears at serial number

[MP PET]

(a) 602

(b) 603

(c) 600

(d) 601

Q 21. If all permutations of the letters of the word AGAIN are arranged as in dictionary, then fifteenth word is

[Guj. CET]

(a) NAAGI

(b) NAGAI

(c) NAAIG

(d) NAIAG

Q 22. If r, s, t are prime numbers and p, q are the positive integers such that LCM of p, q is $r^2s^4t^2$, then the number of ordered pairs (p, q) is

[IIT JEE]

(a) 252

(b) 254

(c) 225

(d) 224

Q 23. How many numbers of 6 digits can be formed from the digits of the number 112233 ?

[AMU]

(a) 30

(b) 60

(c) 90

(d) 120

Q 24. How many numbers greater than 40000 can be formed from the digits 2, 4, 5, 5, 7 ?

(a) 12

(b) 24

(c) 36

(d) 48

Q 25. There are 10 true-false questions in an examination. Then, these questions can be answered in

[DCE]

(a) 240 ways

(b) 20 ways

(c) 1024 ways

(d) 100 ways

Q 26. How many words can be made from the letters of the word 'COMMITTEE'?

[BCECE]

(a) $\frac{9!}{(2!)^2}$

(b) $\frac{9!}{(2!)^3}$

(c) $\frac{9!}{2!}$

(d) $9!$

Q 27. The number of words that can be formed out of the letters of the word 'ARTICLE' so that the vowels occupy even places, is

[Guj. CET]

(a) 574

(b) 36

(c) 754

(d) 144

Q 28. The number of different words that can be formed from the letters of the word 'PENCIL' so that no two vowels are together, is

[DCE]

(a) 120

(b) 260

(c) 240

(d) 480

Q 29. How many four, digit numbers can be formed using the digits $1, 2, 3, 4,, 5$ such that at least one of the digit is repeated?

[DCE]

(a) $4^4 - 5!$

(b) $4^5 - 4!$

(c) $5^4 - 4!$

(d) $5^4 - 5!$

Q 30. If there are n number of seats and m number of people have to be seated, then how many ways are possible to do this $(m < n)$?

[BITSAT]

(a) ${}^nP_m$

(b) ${}^nC_m$

(c) ${}^{n}C_{n} \times (m - 1)!$

(d) ${}^{n-1}P_{m-1}$

Topic 3

Number of Combinations, Division into Groups and Dearrangements

Q 1. If ${}^{n-1}C_{3} + {}^{n-1}C_{4} > {}^{n}C_{3}$, then n is just greater than integer

[WB JEE]

(a) 5

(b) 6

(c) 4

(d) 7

Q 2. From 12 books, the difference between number of ways a selection of 5 books when one specified book is always excluded and one specified book is always included, is

[Kerala CEE]

(a) 64

(b) 118

(c) 132

(d) 330

(e) 462

Q 3. There are two urns. Urn A lias 3 distinct red balls and urn B has 9 distinct blue balls. From each urn two balls are taken out at random and then transferred to the other. The number of ways in which this can be done, is

[AIEEE]

(a) 3

(b) 36

(c) 66

(d) 108

Q 4. From 6 different novels and 3 different dictionaries, 4 novels and 1 dictionary are to be selected and arranged in a row on the shelf so that

the dictionary is always in the middle. Then the number of such arrangements is

[AIEEE]

(a) at least 500 but less than 750

(b) at least 750 but less than 1000

(c) at least 1000

(d) less than 500

Q 5. In a cricket championship there are 36 matches. The number of teams, if each plays 1 match with other are

[BITSAT]

(a) 9

(b) 10

(c) 8

(d) 12

Q 6. Four dice are rolled. The number of possible outcomes in which at least one dice shows 2 is

[AMU]

(a) 625

(b) 671

(c) 1023

(d) 1296

Q 7. Three straight lines L_1, L_2, L_3 are parallel and lie in the same plane, A total of m points are taken

on L_1, n points on L_2,k points on L_3. The maximum number of triangles formed with vertices at these points are

[AMU]

(a) ${}^{m+n+k}C_3$

(b) ${}^{m+n+k}C_3 - {}^{m}C_3 - {}^{n}C_3$

(c) ${}^{m+n+ lc}C_3 + {}^{m}C_3 + {}^{n}C_3$

(d) None of the above

Q 8. The number of subsets of $\{1, 2, 3, \ldots, 9\}$ containing at least one odd number is

[EAMCET]

(a) 324

(b) 396

(c) 496

(d) 512

Q 9. p points are chosen on each of the three coplanar lines. The maximum number of triangles formed with vertices at these points is

[EAMCET]

(a) $p^3 + 3p^2$

(b) $\frac{1}{2}(p^3 + p)$

(c) $\frac{p^2}{2}(5p - 3)$

(d) $p^2(4p - 3)$

Q 10. A binary sequence is an array of 0's and 1's. The number of n −digit binary sequences which contain even number of 0's is

[EAMCET]

(a) 2^{n-1}

(b) $2^n - 1$

(c) $2^{n-1} - 1$

(d) 2^n

Q 11. 7 relatives of a man comprises 4 ladies and 3 gentlemen his wife has also 7 relatives, 3 of them are ladies and 4 gentlemen. In how many ways can they invite a dinner party of 3 ladies and 3 gentlemen so that there are 3 of man's relative and 3 of the wife's relative ?

[Jamia Millia Islamia]

(a) 485

(b) 500

(c) 486

(d) 102

Q 12. If ${}^{n+2}C_8 : {}^{n-2}P_4 = \frac{57}{16}$, then n is equal to

[VITEEE]

(a) 19

(b) 2

(c) 20

(d) 5

Q 13. The number of diagonals of a polygon of 20 sides is

[J&KCET]

(a) 210

(b) 190

(c) 180

(d) 170

Q 14. In a shop there are five types of ice-creams available. A child buys six ice-creams.

Statement I The number of different ways the child can buy the six ice-creams is ${}^{10}C_5$.

Statement II The number of different ways the child can buy the six ice-creams is equal to the number of different ways of arranging 6 A's and 4 B 's in a row.

[AIEEE]

(a) Statement I is true, Statement II is true;

Statement II is a correct explanation for Statement I

(b) Statement I is true, Statement II is true;

Statement II is not a correct explanation for

Statement I

(c) Statement I is true, Statement II is false (d)

Statement I is false, Statement II is true

Q 15. How many different words can be formed by jumbling the letters in the word MISSISSIPPI in which no two S are adjacent ?

[AIEEE]

(a) $7 . {}^{6}C_{4} . {}^{8}C_{4}$

(b) $8 . {}^{6}C_{4} . {}^{7}C_{4}$

(c) $6 . 7 . {}^{8}C_{4}$

(d) $6 . 8 . {}^{7}C_{4}$

Q 16. All possible two factors products are formed from numbers $1,2,3,4,\ldots, 200$. The number of factors out of the total obtained which are multiples of 5, is

[DCE]

(a) 5040

(b) 7180

(c) 8150

(d) None of these

Q 17. If ${}^{n}P_{r} = 30240$. and ${}^{n}C_{r} = 252$, then the ordered pair (n, r) is equal to

[EAMCET]

(a) (12, 6)

(b) (10, 5)

(c) (9, 4)

(d) (16, 7)

Q 18. The number of four–letter words that can be formed (the words need not be meaningful)

using the letters of the word MEDITERRANEAN

such that the first letter is E and the last letter is R,

is

[Kerala CEE]

(a) $\frac{11!}{2!2!2!}$

(b) 59

(c) 56

(d) $\frac{11!}{3!2!2!}$ (e) $\frac{11!}{3!3!2!}$

Q 19. If ${}^{16}C_r = {}^{16}C_{r+1}$, then the value of ${}^{r}P_{r-3}$ is

[WBJEE]

(a) 31

(b) 120

(c) 210

(d) 840

Q 20. If ${}^{189}C_{35} + {}^{189}C_X = {}^{190}C_x$, then x is equal to

[MP PET]

(a) 34

(b) 35

(c) 36

(d) 37

Q 21. In a Mathematics paper there are three sections containing 4,5 and 6 questions respectively. From each section 3 questions are to be answered. In how many ways can the selection of questions be made ?

[MPPET]

(a) 34

(b) 800

(c) 1600

(d) 9600

Q 22. The number of permutations of 4 letters that can be made out of the letters of the word EXAMINATION is

[J&KCET]

(a) 2454

(b) 2452

(c) 2450

(d) 1806

Q 23. The letters of the word COCHIN are permuted and all the permutations are arranged in an alphabetical order as in an English dictionary.

The number of words that appear before the word COCHIN is

[IITJEE]

(a) 360

(b) 192

(c) 96

(d) 48

Q 24. The set S = {1,2,3,..., 12} is to be partitioned into three sets A, B, C of equal size.

Thus, $A \cup B \cup C = S$

$A \cap B = B \cap C = A \cap C = \phi$

The number of ways to partition S is

[AIEEE]

(a) $12!/3!(4!)^3$

(b) $12!/3!(3!)^4$

(c) $12!/(4!)^3$

(d) $12!/(3!)^4$

Q 25. Out of 6 boys and 4 girls, a group of 7 is to be formed. In how many ways can this be done, if the group is to have a majority of boys ?

[UP SEE]

(a) 120

(b) 80

(c) 90

(d) 100

Q 26. If m = ${}^{n}C_{2}$, then ${}^{m}C_{2}$ is equal to

[Kerala CEE]

(a) $3\ {}^{n}C_4$

(b) ${}^{n+1}C_4$

(c) $3.{}^{n+1}C_4$

(d) $3.{}^{n+1}C_3$

(e) $3.{}^{n+\ 1}C_2$

Q 27. There are 5 letters and 5 different envelopes. The number of ways in which all the letters can be put in wrong envelope, is

[AMU]

(a) 119

(b) 44

(c) 59

(d) 40

Q 28. The number of selecting at least 4 candidates from 8 candidates is

[OJEE]

(a) 270

(b) 70

(c) 163

(d) None of these

Q 29. The number of times the digit 5 will be written when listing the integers from 1 to 1000, is

[WB JEE]

(a) 271

(b) 272

(c) 300

(d) None of these

Q 30. Number of divisors of the form $(4n + 2)$, $n \geq 0$ of the integer 240 is

[BCECE]

(a) 4

(b) 8

(c) 10

(d) 3

Q 31. Everybody in a room shakes hands with everybody else. The total number of hand shakes is 66. The total number of persons in the room is

[VITEEE]

(a) 9

(b) 12

(c) 10

(d) 14

Q 32. If, $s_n = \sum_{r=0}^{n} \frac{1}{{}^nC_r}$ and $t_n = \sum_{r=0}^{n} \frac{r}{{}^nC_r}$, then $\frac{t_n}{s_n}$ is equal to

[Jamia Millia Islamia]

(a) $\frac{n}{2}$

(b) $\frac{n}{2} - 1$

(c) n – 1

(d) $\frac{2n-1}{2}$

Q 33. The value of ${}^{50}C_4 + \sum_{r=1}^{6} {}^{56-r}C_3$ is

(a) ${}^{56}C_4$

(b) ${}^{56}C_3$

(c) ${}^{55}C_3$

(d) ${}^{55}C_4$

Q 34. A father with 8 children takes 3 at a time to the zoological garden, as often as he can without taking the same 3 children together more than

once. The number of times he will go to the garden, is

[Guj.CET]

(a) 112

(b) 56

(c) 336

(d) None of these

Q 35. There are 10 lamps in a hall. Each one of them can be switched on independently. The

number of ways in which the hall can be illuminated, is

[Guj. CET]

(a) 2^{10}

(b) $10!$

(c) 1023

(d) 10^2

Q 36. At an election, a voter may vote for any number of candidates not greater than the number to be elected. There are 10 candidates and 4 are to be elected. If a voter votes for at least one Candidate, then the number of ways in which he can vote, is

[AIEEE]

(a) 6210

(b) 385

(c) 1110

(d) 5040

Q 37. If $\frac{1}{{}^{4}C_{n}} = \frac{1}{{}^{5}C_{n}} + \frac{1}{{}^{6}C_{n}}$, then n is equal to

[Kerala CEE]

(a) 3

(b) 2

(c) 1

(d) 0

(e) 4

Q 38. The number of ways in which one can select three distinct integers between 1 and 30, both inclusive, whose sum is even, is

[Kerala CEE]

(a) 455

(b) 1575

(c) 1120

(d) 2030

(e) 1930

Q 39. There are 5 roads leading to a town from a village. The number of different ways in which a villager can go to the town and return back, is

[MPPET]

(a) 20

(b) 25

(c) 5

(d) 10

Q 40. If $P(n, r) = 1680$ and $C(n, r) = 70$, then

$69n + r!$ is equal to

[Kerala CEE]

(a) 128

(b) 576

(c) 256

(d) 625

(e) 1152

Q 41. A student is to answer 10 out of 13 questions in an examination such that he must choose at least 4 from the first five questions. The number of choices available to him is

[Kerala CEE]

(a) 140

(b) 196

(c) 280

(d) 346

(e) 265

Q 42. $\sum_{r=0}^{m} {}^{n+r}C_n$ is equal to

[Guj.CET]

(a) ${}^{n+m+1}C_{n+1}$

(b) ${}^{n+m+2}C_n$

(c) ${}^{n+m+3}C_{n-1}$

(d) None of these

Q 43. In a club election the number of contestants is one more than the number of maximum candidates for which a voter can vote. If the total number of ways in which a voter can vote be 126, then the number of contestants is

[AMU]

(a) 4

(b) 5

(c) 6

(d) 7

Q 44. A three digit number n is such that the last two digits of it are equal and differ from the first. The number of such n's is

[EAMCET]

(a) 64

(b) 72

(c) 81

(d) 900

Q 45. In how many ways can 21 English and 19 Hindi books be placed in a row so that no two Hindi books are together ?

[BCECE]

(a) 1540

(b) 1450

(c) 1504

(d) 1405

Q 46. The number of ways of distributing 8 identical balls in 3 distinct boxes, so that none of the boxes is empty, is

[AIEEE]

(a) 5

(b) 21

(c) 3^8

(d) ${}^8C_3$

Q 47. ${}^nP_r = 3024$ and ${}^nC_r = 126$, then r is

[Kerala CEE]

(a) 5

(b) 4

(c) 3

(d) 2

(e) 1

Q 48. In a chess tournament where the participants were to play one game with one another, two players fell ill having played 6 games

each, without playing among themselves. If the total number of games is 117, then the number of participants at the beginning was

[AMU]

(a) 15

(b) 16

(c) 17

(d) 18

Q 49. The number of positive odd divisors of 216 is

[EAMCET]

(a) 4

(b) 6

(c) 8

(d) 12

Q 50. $S_1, S_2,\ldots,S_{10}$ are the speakers in a conference. If S_1 addresses only after S_2, then the number of ways the speakers address is

[EAMCET]

(a) $10!$

(b) $9!$

(c) $10 \times 8!$

(d) $10! \times 2!$

Q 51. A lady gives a dinner party for six guest. The number of ways in which they may be selected from among ten friends, if two of the friends will not, attends the party together is

[BCECE]

(a) 112

(b) 140

(c) 164

(d) None of these

Q 52. A student is allowed to select at most n books from a collection of $(2n+1)$ books. If the total number of ways in which he can select one book is 63, then the value of n is equal to

[Jamia Millia Islamla]

(a) 2

(b) 3

(c) 4

(d) 1

Q 53. The number of ways in which a committee can be formed of 5 members from 6 men and 4 women if the committee has at least one woman, is

[RPET]

(a) 186

(b) 246

(c) 252

(d) 244

Q 54. If a man and his wife enter in a bus, in which five seats are vacant, then the number of different ways in which they can be seated, is

[Guj. CET]

(a) 2

(b) 5

(c) 20

(d) 40

Q 55. If ${}^{n}C_{r}$ denotes the number of combinations of n things taken r at a time, then the expression ${}^{n}C_{r+1} + {}^{n}C_{r-1} + 2 \times {}^{n}G_{r}$ equals

[AIEEE]

(a) ${}^{n+2}C_{r}$

(b) ${}^{n+2}C_{r+1}$

(c) ${}^{n+1}C_{r}$

(d) ${}^{n+1}C_{r+1}$

Q 56. The number of ways in which 6 rings can be worn on four fingers of one hand, is

[AMU]

(a) 4^6

(b) ${}^6C_4$

(c) 6^4

(d) 24

Q 57. The number of ways in which 9 persons can be divided into three equal groups is

[OJEE]

(a) 1680

(b) 840

(c) 560

(d) 280

Q 58. There are n different books and m copies of each in a college library. The number of ways in which a student can make a selection of one or more books is

[J&KCET]

(a) $(m + 1)^n$

(b) $\frac{(mn)!}{(m!)^n}$

(c) ${}^{mn}C_n \times {}^{n}C_1$

(d) $(m + 1)^n - 1$

Q 59. If ${}^{n}C_{r-1} = 36$, ${}^{n}C_r = 84$ and ${}^{n}C_{r+1} = 126$, then the value of r is

[Guj.CET]

(a) 1

(b) 2

(c) 3

(d) None of these

Topic 4 Used in Geometrical Problem

Q 1. Out of 8 given points, 3 are collinear. How many different straight lines can be drawn by joining any two points from those 8 points ?

[WB JEE]

(a) 26

(b) 28

(c) 27

(d) 25

Q 2. There is a set of m parallel lines intersecting a set of another n parallel lines in a plane. The number of parallelograms formed, is

[DCE]

(a) ${}^{m-1}C_2 \cdot {}^{n-1}C_2$

(b) ${}^{m}C_2 \cdot {}^{n}C_2$

(c) ${}^{m-1}C_2 \cdot {}^{n}C_2$

(d) ${}^{m}C_2 \cdot {}^{n-1}C_2$

Q 3. If a polygon of n sides has 275 diagonals,

then n is equal to

[EAMCET]

(a) 25

(b) 35

(c) 20

(d) 15

Q 4. The number of diagonals in a octagon will be

[MP PET]

(a) 28

(b) 20

(c) 10

(d) 16

Q 5. The number of triangles which can be formed by using the vertices of a regular polygon of $(n + 3)$ sides is 220. Then, n is equal to

[Kerala CEE]

(a) 8

(b) 9

(c) 10

(d) 11

(e) 12

Q 6. Let T_n denote the number of triangles which can be formed using the vertices of a regular

polygon of n sides. If $T_{n+1} - T_n = 21$, then n

equals

[OJEE]

(a) 5

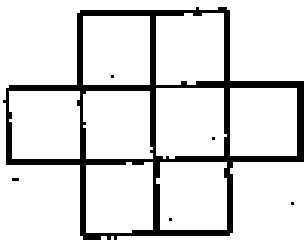
(b) 7

(c) 6

(d) 4

Q 7. Six x have to be placed in the squares of the figure given, such that each row contains at least one x, this can be done in

[Kerala CEE]



(a) 24 ways

(b) 28 ways

(c) 26 ways

(d) 36 ways

(e) 45 ways

Q 8. The number of diagonals that can be drawn in a polygon of 15 sides, is

[RPET]

(a) 16

(b) 60

(c) 90

(d) 80

Q 9. A polygon has 44 diagonals, then the number of its sides are

[DCE]

(a) 11

(b) 7

(c) 8

(d) None of these

Q 10. A polygon has 54 diagonals. Number of sides of this polygon is

[Kerala CEE]

(a) 12

(b) 15

(c) 16

(d) 9

(e) 14

Q 11. Let l_1 and l_2 be two lines intersecting at P. If A_1, B_1,C_1 are points on l_1 and A_2,B_2,C_2,D_2, E_2 are points on l_2 and if none of these coincides with P, then the number of triangles formed by these eight points, is

[EAMCET]

(a) 56

(b) 55

(c) 46

(d) 45

Answers

Topic 1: Number of Permutations, Conditional Permutations

1. (d) 2. (c) 3. (c) 4. (a) 5. (d) 6. (b) 7. (b) 8. (a) 9. (c) 10. (c) 11. (b)

12. (b) 13. (e) 14. (c) 15. (a) 16. (d) 17. (b)

18. (d) 19. (c) 20. (d) 21. (c) 22. (c) 23. (c) 24.

(d) 25. (c) 26. (b) 27. (d) 28. (d) 29. (d) 30. (a)

31. (d) 32. (c) 33. (d) 34. (b) 35. (c) 36. (c) 37.

(c) 38. (d) 39. (b) 40. (b) 41. (a) 42. (c)

Topic 2 : Circular Permutations

1. (a) 2. (b) 3. (d) 4. (b) 5. (a) 6. (a) 7. (d)

Topic 3 : Number of Combinations, Division into Groups and De–arrangements

1. (d) 2. (c) 3. (d) 4. (c) 5. (a) 6. (b) 7. (d) 8. (c)

9. (d) 10. (a) 11. (a) 12. (a) 13. (d) i4. (d) 15.

(a) 16. (b) 17. (b) 18. (b) 19. (*) 20. (c) 21. (b)

22. (a) 23. (c) 24. (c) 25. (d) 26. (c) 27. (b) 28.

(c) 29. (c) 30. (a) 31. (b) 32. (a) 33. (a) 34. (b)

35. (c) 36. (b) 37. (b) 38. (d) 39. (b) 40. (b) 41.

(b) 42. (a) 43. (d) 44. (c) 45. (a) 46. (b) 47. (b)

48. (a) 49. (a) 50. (d) 51. (b) 52. (b) 53. (b) 54.

(c) 55. (b) 56. (a) 57. (a) 58. (d) 59. (c)

Topic 4 : Used in Geometrical Problem

1. (a) 2. (b) 3. (a) 4. (b) 5. (b) 6. (b) 7. (c) 8. (c)

9. (a) 10. (a) 11. (d)

Note'*'None of the given option is correct.

Straight Lines and Pair of Straigt lines

Useful Results & Formulae

1. General equation of straight line is ax + by + c = 0 whose slope is m = $-\frac{a}{b}$.

2. Equations of straight lines in different forms are

(i) Slope intercept form y = mx + c

(ii) Point slope form $y - y_1 = m(x - x_1)$

(iii) Intercept form $\frac{x}{a}+\frac{y}{b}=1$

(iv) Two points form $(y - y_1)= \frac{y_2-y_1}{x_2-x_1}(x-x_1)$

(v) Normal form $x \cos \alpha + y \sin \alpha = p$

(vi) Distance form $\frac{x-x_1}{\cos\theta}=\frac{y-y_1}{\sin\theta}=r$

3. (i) Equation of a line which is parallel to $ax + by + c = 0$ is $ax + by + \lambda = 0$.

(ii) Equation of line which is perpendicular to $ax + by + c = 0$ is $bx - ay + \lambda = 0$.

4. If the equation of line be $a \sin \theta + b \cos \theta = c$, then line

(i) parallel to it, is $a \sin \theta + b \cos \theta = d$

(ii) perpendicular to it, is $a \sin\left(\frac{\pi}{2}+\theta\right) + b \cos\left(\frac{\pi}{2}+\theta\right) = d$

5. Point of intersection of two lines $a_1x + b_1y + c = 0$ and $a_2x + b_2y + c_2 = 0$ is $\left(\frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}, \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}\right)$

6. (i) Angle between the lines $y = m_1x + c_1$ and $y = m_2x + c_2$ is $\theta = \tan^{-1}\left|\frac{m_1 - m_2}{1 + m_1m_2}\right|$

If $m_1m_2 = -1$, then lines are perpendicular and if m_1

$= m_2$, then lines are parallel,

(ii) Angle between the lines $a_1x + b_1y + c_1 = 0$ and

$a_2x + b_2y + c_2 = 0$ is given by $\tan\theta = \left|\frac{a_2b_1 - a_1b_2}{a_1a_2 + b_1b_2}\right|$

7. Two lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2$

$= 0$ are

(i) coincident, if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

(ii) parallel, if $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

(iii) perpendicular, if $a_1a_2 + b_1b_2 = 0$

(iv) intersecting, if $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$

8. Equation of line passing through the point of intersection of two lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ is

$(a_1x + b_1y + c_1) + \lambda(a_2x + b_2y + c_2) = 0$.

9. Equation of bisectors of the angle between the lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$$

(i) If $a_1a_2 + b_1b_2 > 0$, then '+ ve' sign gives obtuse angle bisector and'– ve' sign gives acute angle bisector,

(ii) If $a_1a_2 + b_1b_2 < 0$, then '+ ve' sign gives acute angle bisector and '– ve' sign gives obtuse angle bisector.

10. (i) Length of perpendicular from the point (x_1, y_1) to the line $ax + by + c = 0$ is $\frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$

(ii) Length of perpendicular from the point (x_1, y_1) to the line $x \cos \alpha + y \sin \alpha = p$ is

$|x_1 \cos \alpha + y_1 \sin \alpha - p|$.

(iii) Distance between two parallel lines $ax + by + c_1 = 0$ and $ax + by + c_2 = 0$ is $\frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$.

11. Three lines $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ and $a_3x + b_3y + c_3 = 0$ are concurrent, if

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0.$$

12. The foot of the perpendicular (h,k) from (x_1, y_1) to the line ax + by + c =0 is given by

$$\frac{h-x_1}{a}=\frac{k-y_1}{b}=-\frac{(ax_1+by_1+c)}{a^2+b^2}$$

13. If the lines represented by $ax^2 + 2hxy + by^2 = 0$ are $y-m_1x = 0$ and $y - m_2x = 0$, then $m_1+ m_2 = -\frac{2h}{b}$ and $m_1m_2 = \frac{a}{b}$.

14. If $ax^2 + 2hxy + by^2 + 2gx + + 2fy + c = 0$ represents a pair of straight lines, then

$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$ or $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$

15. Angle between the lines represented by $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

or $ax^2 + 2hxy + by^2 = 0$ is given by $\tan\theta = \left|\frac{2\sqrt{h^2 - ab}}{a + b}\right|$.

Lines are parallel if $h^2 = ab$ and $af^2 = bg^2$ or $\frac{a}{h} = \frac{h}{b} = \frac{g}{f}$

Lines are perpendicular if $a + b = 0$.

16. Equation of bisector of the angles between the lines $ax^2 + 2hxv + by^2 = 0$ is $\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$.

17. Distance between the lines represented by $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is $2\sqrt{\frac{g^2 - ac}{a(a + b)}}$ or $2\sqrt{\frac{f^2 - bc}{b(a + b)}}$

18. The product of perpendiculars drawn from to the line $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is $\frac{c}{\sqrt{(a - b)^2 + 4h^2}}$.

19. To eliminate the first degree terms from the line $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$. The point to which origin is to be shifted is given by

$$\left(\frac{bg - fh}{h^2 - ab}, \frac{af - gh}{h^2 - ab}\right).$$

20. If bisectors of the angles between the lines $ax^2 + 2hxy + by^2 = 0$ and $a' x^2 + 2h' xy + b' y^2 = 0$ are same, then $\frac{h'}{h} = \frac{a' - b'}{a - b}$.

Topic 1 Equation of Line and Pair of Lines

Q 1. The equations of the lines through $(1, 1)$ and making angles of $45°$ with the line $x + y = 0$ are

[WB JEE]

(a) $x - 1 = 0, x - y = 0$

(b) $x - y = 0, y - 1 = 0$

(c) $x + y - 2 = 0, y - 1 = 0$

(d) $x - 1 = 0, y - 1 = 0$

Q 2. If the sum of distances from a point P on two mutually perpendicular straight lines is 1 unit, then the locus of P is

[WBJEE]

(a) a parabola

(b) a circle

(c) an ellipse

(d) a straight line

Q 3. The straight line $3x + y = 9$ divides the line segment joining the points $(1,3)$ and $(2,7)$ in the ratio

[BITSAT]

(a) $3:4$ externally

(b) $3:4$ internally

(c) $4:5$ internally

(d) $5:6$ externally

Q 4. The equations $y = \pm\sqrt{3}x$, $y = 1$ are the sides of

[BITSAT]

(a) an equilateral triangle

(b) a right angled triangle

(c) an isosceles triangle

(d) an obtuse angled triangle

Q 5. The equation of a straight line which passes through the point $(a\cos^3\theta, a\sin^3\theta)$ and perpendicular to $x\sec\theta + y\,\mathrm{cosec}\,\theta = a$ is

[Kerala CEE]

(a) $\frac{x}{a}+\frac{y}{a}=a\cos\theta$

(b) $x\cos\theta - y\sin\theta = -a\cos 2\theta$

(c) $x\cos\theta + y\sin\theta = a\cos 2\theta$

(d) $x\cos\theta + y\sin\theta - a\cos 2\theta = 1$

(e) $x\cos\theta - y\sin\theta + a\cos 2\theta = -1$

Q 6. If the line px − qy = r intersects the coordinate axes at $(a, 0)$ and $(0, b)$, then the value of a + b is equal to

[Kerala CEE]

(a) $r\left(\frac{q+p}{pq}\right)$

(b) $r\left(\frac{q-p}{pq}\right)$

(c) $r\left(\frac{p-q}{pq}\right)$

(d) $r\left(\frac{p+q}{p-q}\right)$

(e) $r\left(\frac{p-q}{p+q}\right)$

Q 7. If the slope of one of the lines represented by $ax^2 + 2hxy + by^2 = 0$ be the square of the other, then $\frac{a+b}{h}+\frac{8h^2}{ab}$ is

[AMU]

(a) 3

(b) 4

(c) 5

(d) 6

Q 8. The pairs of straight lines $x^2 - 3xy + 2y^2 = 0$ and $x^2 - 3xy + 2y^2 + x - 2 = 0$ form a

[EAMCET]

(a) square but not rhombus

(b) rhombus

(c) parallelogram

(d) rectangle but not a square

Q 9. Joint equation of pair of lines through $(3, -2)$

and parallel to $x^2 - 4xy + 3y^2 = 0$ is

[MHT CET]

(a) $x^2 + 3y^2 - 4xy - 14x + 24y + 45 = 0$

(b) $x^2 + 3y^2 + 4xy - 14x + 24y + 45 = 0$

(c) $x^2 + 3y^2 + 4xy - 14x + 24y - 45 = 0$

(d) $x^2 + 3y^2 + 4xy - 14x - 24y - 45 = 0$

Q 10. If the lines $kx - 2y - 1 = 0$ and $6x - 4y - m = 0$ are identical (coincident) lines, then the values of k and m are

[Guj. CET]

(a) $k = 3, m = 2$

(b) $k = -3, m = 2$

(c) $k = -3, m = -2$

(a) $k = 3, m = -2$

Q 11. Let a and b be non-zero and real numbers. Then, the equation $(ax^2 + by^2 + c)(x^2 - 5xy + 6y^2) = 0$ represents

[IIT JEE]

(a) Four straight lines, when $c = 0$ and a, b are of the same sign

(b) Two straight lines and a circle, when a = b and c is of sign opposite to that of a

(c) Two straight lines and a hyperbola, when a and b are of the same sign and c is of sign opposite to that of a

(d) A circle and an ellipse, when a and b are of the same sign and c is of sign opposite to that of a

Q 12. The perpendicular bisector of the line segment joining P(1, 4) and Q (k, 3) has y− intercept −4. Then, a possible value of k is

[AIEEE]

(a) -4

(b) 1

(c) 2

(d) -2

Q $13.$ The equation of line through the point $(1,1)$ and making angles of $45°$ with the line x + y = 0 are

[DCE]

(a) x − 1 = 0, x − y = 0

(b) $x - 1 = 0, y - 1 = 0$

(c) $x - y = 0, y - 1 = 0$

(d) $x + y - 2 = 0, y - 1 = 0$

Q 14. The point $P(a, b)$ lies on the straight line $3x + 2y = 13$ and the point $Q(b,a)$ lies on the straight line $4x - y = 5$, then equation of the line PQ is

[DCE]

(a) $x - 5 = 5$

(b) $x + y = 5$

(c) $x + y = -5$

(d) $x - y = -5$

Q 15. The equation of the line bisecting perpendicularly the segment joining the points $(-4,6)$ and $(8, 8)$ is

[UP SEE]

(a) $6x + y - 19 = 0$

(b) $y = 7$

(c) $6x + 2y - 19 = 0$

(d) $x + 2y - 7 = 0$

Q 16. The slopes of the lines represented by $x^2 + 2hxy + 2v^2 = 0$ are in the ratio $1 : 2$, then h equals

[UP SEE]

(a) $\pm\frac{1}{2}$

(b) $\pm\frac{3}{2}$

(c) ± 1

(d) ± 3

Q 17. If $3x + xy - y^2 - 3x + 6y + k = 0$

represents a pair of lines, then k is equal to

[KCET]

(a) 0

(b) 9

(c) 1

(d) -9

Q 18. The equation of a line through the point $(1, 2)$ whose distance from the point $(3,1)$ has the greatest value, is

[Kerala CEE]

(a) $y = 2x$

(b) $y = x + 1$

(c) $x + 2y = 5$

(d) $y = 3x - 1$

(e) $y = x + 1$

Q 19. If a line with y−intercept 2, is perpendicular to the line $3x - 2y = 6$, then its x−intercept is

[Kerala CEE]

(a) 1

(b) 2

(c) -4

(d) 4

(e) 3

Q 20. If the lines ax+ ky + $10 = 0$,bx + (k +1)y + $10 = 0$ and cx + (k + 2)y + $10 = 0$ are concurrent, then

[Kerala CEE]

(a) a, b, c are in GP

(b) a, b, c are in HP

(c) a, b, c are in AP

(d) $(a + b)^2 = c$

(e) a + b = c

Q 21. The value of k such that the lines $2x-3y + k = 0$, $3x - 4y - 13 = 0$ and $8x - 11y - 33 = 0$ are concurrent, is

[AMU]

(a) 20

(b) −7

(c) 7

(d) −20

Q 22. The value of λ, such that $\lambda x^2 - 10xy + 12y^2 + 5x - 16y - 3 = 0$ represents a pair of straight lines, is

[EAMCET]

(a) 1

(b) -1

(c) 2

(d) -2

Q 23. The equation of straight line equally inclined to the axes and equidistant from the points $(1,-2)$ and $(3, 4)$ is ax + by + c = 0, where

[OJEE]

(a) a = 1, b = -1, c = 3

(b) a= 1, b = -1, c = -3

(c) a= 1, b = 1, c = -3

(d) None of these

Q 24. The line which is parallel to x−axis and crosses the curve $y = \sqrt{x}$ at an angle $45°$, is

[WB JEE]

(a) $y = \frac{1}{4}$

(b) $y = \frac{1}{2}$

(c) $y = 1$

(d) $y = 4$

Q 25. The equation of the line passing through the point of intersection of the lines $x - 3y + 2 = 0$ and $2x + 5y - 7 = 0$ and perpendicular to the line $3x+2y + 5 = 0$, is

[BCECE]

(a) $2x - 3y + 1 = 0$

(b) $6x - 9y + 11 = 0$

(c) $2x - 3y + 5 = 0$

(d) $3x - 2y + l = 0$

Q 26. If the lines $x + 3y - 9 = 0$, $4x + by - 2 = 0$ and $2x - y - 4 = 0$ are concurrent, then b equals

[Jamia Millia Islamia]

(a) -5

(b) 5

(c) 1

(d) 0

Q 27. The equation $12x^2 + 7xy + ay^2 + 13x - y + 3 = 0$ represents a pair of perpendicular lines.

Then, the value of 'a' is

[MHTCET]

(a) $7/2$

(b) -19

(c) -12

(d) 12

Q 28. The equation of perpendicular bisectors of sides AB and AC of a A ABC are $x - y + 5 = 0$ and $x + 2y = 0$ respectively. If the coordinates of vertex A are $(1, -2)$, then equation of BC is

[Manipal]

(a) $14x + 23y - 40 = 0$

(b) $14x - 23y + 40 = 0$

(c) $23x + 14y - 40 = 0$

(d) $23x - 14y + 40 = 0$

Q 29. The line passing through the point of intersection of $x + y = 2$, $x - y = 0$ and is parallel to $x + 2y = 5$, is

[J&KCET]

(a) $x + 2y = 1$

(b) $x + 2y = 2$

(c) $x + 2y = 4$

(d) $x + 2y = 3$

Q 30. The straight line $3x + 4y - 5 = 0$ and $4x = 3y + 15$ intersect at the point P. On these lines the points Q and R are chosen so that PQ = PR. The slopes of the lines QR passing through $(1,2)$ are

[Guj. CET]

(a) $-7,1/7$

(b) $7,1/7$

(c) $7,-1/7$

(d) $3, -1/3$

Q 31. A line passes through the point of intersection of the lines $3x + y + 1 = 0$ and $2x - y + 3 = 0$ and makes equal intercepts with axes. Then, equation of the line is

[DCE]

(a) $5x + 5y - 3 = 0$

(b) $x + 5y - 3 = 0$

(c) $5x - y - 3 = 0$

(d) $5x + 5y + 3 = 0$

Q 32. If the lines $3x + 4y + 1 = 0$, $5x + \lambda y + 3 = 0$ and $2x + y - 1 = 0$ are concurrent, then λ is equal to

[UP SEE]

(a) -8

(b) 8

(c) 4

(d) -4

Q 33. The equation of the lines through the point $(3, 2)$ which makes an angle of $45°$ with the line $x - 2y = 3$, are

[Kerala CEE]

(a) $3x - y = 7$ and $x + 3y = 9$

(b) $x - 3y = 7$ and $3x + y = 9$

(c) $x - y = 3$ and $x + y = 2$

(d) $2x + y = 7$ and $x - 2y = 9$

(e) $2x - y = 7$ and $x + 2y = 9$

Q 34. The equation of the line which is such that the portion of line segment intercepted between the coordinate axes is bisected at $(4, -3)$, is

[Kerala CEE]

(a) $3x + 4y = 24$

(b) $3x - 4y = 12$

(c) $3x - 4y = 24$

(d) $4x - 3y = 24$

(e) $4x - 3y = 12$

Q 35. A straight line through the point A$(3, 4)$ is such that its intercept between the axes is bisected at A. Its equation is

[AMU]

(a) $3x - 4y + 7 = 0$

(b) $4x + 3y = 24$

(c) $3x + 4y = 25$

(d) $x + y = 7$

Q 36. The equation of straight line through the intersection of the lines $x - 2y = 1$ and $x + 3y = 2$ and parallel to $3x + 4y = 0$, is

[AMU]

(a) $3x + 4y + 5 = 0$

(b) $3x + 4y - 10 = 0$

(c) $3x + 4y - 5 = 0$

(d) $3x + 4y + 6 = 0$

Q 37. If the lines $x^2 + 2xy - 35y^2 - 4x + 44y - 12 = 0$ and $5x + \lambda y - 8 = 0$ are concurrent, then the value of λ is

[EAMCET]

(a) 0

(b) 1

(c) -1

(d) 2

Q 38. In order to eliminate the first degree terms from the equation $2x^2 + 4xy + 5y^2 - 4x - 22y + 7 = 0$, the point to which origin is to be shifted, is

[EAMCET]

(a) $(1,-3)$

(b) $(2,3)$

(c) $(-2,3)$

(d) $(1,3)$

Q 39. The equation of the pair of straight lines parallel to x−axis and touching the circle $x^2 + y^2 - 6x - 4y - 12 = 0$ is

[WBJEE]

(a) $y^2 - 4y - 21 = 0$

(b) $y^2 + 4y - 21 = 0$

(c) $y^2 - 4y + 21 = 0$

(d) $y^2 + 4y + 21 = 0$

Q 40. The equation of the straight line passing through the point $(4, 3)$ and making intercepts on the coordinate axes whose sum is -1, is

[Jamia Millia Islamia]

(a) $\frac{x}{2}+\frac{y}{3}=-1$ and $\frac{x}{-2}+\frac{y}{1}=-1$

(b) $\frac{x}{2}-\frac{y}{3}=-1$ and $\frac{x}{-2}+\frac{y}{1}=-1$

(c) $\frac{x}{2}+\frac{y}{3}=1$ and $\frac{x}{-2}+\frac{y}{1}=1$

(d) $\frac{x}{2}-\frac{y}{3}=1$ and $\frac{x}{-2}+\frac{y}{1}=1$

Q 41. If one of the lines given by $6x^2 - xy + 4cy^2 = 0$ is $3x + 4y = 0$, then c equals

[MP PET]

(a) 1

(b) −1

(c) 3

(d) −3

Q 42. If non-zero numbers a, b, c are in HP, then the straight line $\frac{x}{a}+\frac{y}{b}+\frac{1}{c}=0$ always passes through a fixed point. That point is

[RPET]

(a) $\left(1,-\frac{1}{2}\right)$

(b) $(1,-2)$

(c) $(-1,-2)$

(d) $(-1,2)$

Q 43. The point of concurrence of the lines ax + by + c = 0 and a,b,c satisfy the relation 3a + 2b + 4c = 0 is

[J&K CET]

(a) $\left(\frac{3}{2},\frac{1}{4}\right)$

(b) $\left(\frac{3}{4},\frac{1}{4}\right)$

(c) $\left(\frac{3}{4},\frac{1}{2}\right)$

(d) $\left(\frac{3}{2},\frac{1}{2}\right)$

Q 44. The equation to the bisecting the join of $(3,-4)$ and $(5, 2)$ and having its intercepts on the x−axis and the y−axis in the ratio $2 : 1$ is

[BITSAT, Guj. CET]

(a) $x + y - 3 = 0$

(b) $2x - y = 9$

(c) $x + 2y = 2$

(d) $2x + y = 7$

Q 45. The three straight lines $ax + by = c$, $bx + cy = a$ and $cx + ay = b$ are collinear, if

[UP SEE, WB JEE]

(a) $b + c = a$

(b) $c + a = b$

(c) $a + b + c = 0$

(d) $a + b = c$

Q 46. The equation of pair of lines joining origin to the points of intersection of $x^2 + y^2 = 9$ and $x + y = 3$ is

[UP SEE]

(a) $x^2+(3-x)^2 = 9$

(b) $xy = 0$

(c) $(3 + y)^2 + y^2 = 9$

(d) $(x - y)^2 = 9$

Q 47. The equation of the line passing through the origin and the point of intersection of the lines $\frac{x}{a}+\frac{y}{b}=1$ and $\frac{x}{b}+\frac{y}{a}=1$ is

[Kerala CEE]

(a) bx − ay = 0

(b) x + y = 0

(c) ax − by = 0

(d) x− y = 0

(e) ax + by = 0

Q 48. If $ax^2 - y^2 + 4x - y = 0$ represents a pair of lines, then a is equal to

[AMU]

(a) -16

(b) 16

(c) 4

(d) -4

Q 49. The centroid of the triangle formed by the pair of straight lines $12x^2 - 20xy + 7y^2 = 0$ and the line $2x - 3y + 4 = 0$ is

[EAMCET]

(a) $\left(-\frac{7}{3}, \frac{7}{3}\right)$

(b) $\left(-\frac{8}{3}, \frac{8}{3}\right)$

(c) $\left(\frac{8}{3}, \frac{8}{3}\right)$

(d) $\left(\frac{4}{3}, \frac{4}{3}\right)$

Q 50. The lines represented by the equation $x^2 - y^2 - x + 3y - 2 = 0$ are

[EAMCET]

(a) $x + y - 1 = 0, x - y + 2 = 0$

(b) $x - y - 2 = 0, x + y + 1 = 0$

(c) $x + y + 2 = 0, x - y - 1 = 0$

(d) $x - y + 1 = 0, x + y - 2 = 0$

Q 51. A straight line through the point $(2, 2)$ intersects the lines $\sqrt{3}x + y = 0$ and $\sqrt{3}x - y = 0$ at

the points A and B. The equation of the line AB, so that the ΔOAB is equilateral, is

[OJEE]

(a) $x - 2 = 0$

(b) $y - 2 = 0$

(c) $x + y - 4 = 0$

(d) None of these

Q 52. The value of λ, for which the equation $x^2 - y^2 - x + \lambda y - 2 = 0$ represents a pair of straight lines, are

[WB JEE]

(a) $-3, 1$

(b) $-1, 1$

(c) $3, -3$

(d) $3, 1$

Q 53. The straight line whose sum of the intercepts on the axes is equal to half of the product of the intercepts, passes through the point

[BCECE]

(a) $(1,1)$

(b) $(2, 2)$

(c) $(3, 3)$

(d) $(4, 4)$

Q 54. The value of 'p' for which the equation $x^2 + pxy + y^2 - 5x - 7y + 6 = 0$ represents a pair of straight lines, is

[Jamia Millia Islamia,RPET]

(a) $5/2$

(b) 5

(c) 2

(d) $2/5$

Q 55. A straight line through the point $(1,1)$ meets the x−axis at 'A' and y−axis at 'B'. The locus of the mid point of AB is

[MP PET]

(a) $2xy + x + y = 0$

(b) $x + y - 2xy = 0$

(c) $x + y + 2 = 0$

(d) $x + y - 2 = 0$

Q 56. The equation of the pair of straight lines perpendicular to the pair $2x^2 + 3xy + 2y^2 + 10x + 5y = 0$ and passing through the origin, is

[J&KCET]

(a) $2x^2 + 5xy + 2y^2 = 0$

(b) $2x^2 - 3xy + 2y^2 = 0$

(c) $2x^2 + 3xy + y^2 = 0$

(d) $2x^2 - 5xy + 2y^2 = 0$

Q 57. The line parallel to the x−axis and passing through the intersection of the lines $ax+2by+3b=0$ and $bx-2ay-3a=0$, where $(a, b) \neq (0,0)$ is

[AIEEE]

(a) above the x−axis at a distance of $(2/3)$ from it

(b) above the x−axis at a distance of $(3/2)$ from it

(c) below the x−axis at a distance of $(2/3)$ from it

(d) below the x−axis at a distance of $(3/2)$ from it

Q 58. If $\frac{x^2}{a}+\frac{y^2}{b}+\frac{2xy}{h}=0$ represents pair of straight lines such that slope of one line is twice the other.

Then, ab : h^2 is

[DCE]

(a) 9 : 8

(b) 8 : 9

(c) 1 : 2

(d) 2 : 1

Q 59. If $(\sin\theta, \cos\theta)$ and $(3, 2)$ lies on the same side of the line $x + y = 1$, then θ lies between

[DCE]

(a) $\left(0, \frac{\pi}{2}\right)$

(b) $(0, \pi)$

(c) $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

(d) $\left(0, \frac{\pi}{4}\right)$

Q 60. The equation of a line passing through $(-2,-4)$ and perpendicular to the line $3x - y + 5 = 0$ is

[UP SEE]

(a) $3y + x - 8 = 0$

(b) $3x + y + 6 = 0$

(c) $x + 3y + 14 = 0$

(d) None of these

Q 61. If the point (a, a) falls between the lines | x + y | = 4, then

[AMU]

(a) | a | = 2

(b) | a | = 3

(c) | a | < 2

(d) | a | < 3

Q 62. The equation of the straight line perpendicular to 5x − 2y = 7 and passing through

the point of intersection of the lines $2x + 3y = 1$ and $3x + 4y = 6$, is

[EAMCET]

(a) $2x + 5y + 17 = 0$

(b) $2x + 5y - 17 = 0$

(c) $2x - 5y + 17 = 0$

(d) $2x - 5y = 17$

Q 63. If PM is the perpendicular from P$(2, 3)$ onto the line x + y = 3, then the coordinates of M are

[EAMCET]

(a) $(2,1)$

(b) $(-1,4)$

(c) $(1,2)$

(d) $(4,-1)$

Q 64. If the equation $4x^2 + hxy + y^2 = 0$

represent coincident lines, then h is equal to

[MHTCET]

(a) 1

(b) 3

(c) 2

(d) 4

Q 65. A straight line through P$(1, 2)$ is such that its intercept between the axes is bisected at P. Its equation is

[MP PET]

(a) $x + y = -1$

(b) $x + y = 3$

(c) $x + 2y = 5$

(d) $2x + y = 4$

Q 66. If the equation $kx^2 - 2xy - y^2 - 2x + 2y = 0$ represents a pair of lines, then k is equal to

[MP PET]

(a) 2

(b) −2

(c) −5

(d) 3

Q 67. Two of the lines represented by the equation $ay^4 + bxy^3 + cx^2y^2 + dx^3y + ex^4 = 0$ will be perpendicular, then

[RPET]

(a) $(b + d)(ad + be) + (e - a)^2(a + c + e) = 0$

(b) $(b + d)(ad + be) + (e + a)^2(a + c + e) = 0$

(c) $(b - d)(ad - be) + (e - a)^2(a + c + e) = 0$

(d) $(b - d)(ad - be) + (e + a)^2(a + c + e) = 0$

Q 68. The points $(1, 1)$, $(-5,5)$ and $(13, \lambda)$ lie on the same straight line, if λ is equal to

[RPET]

(a) 7

(b) −7

(c) ± 7

(d) 0

Q 69. The equation of line bisecting perpendicularly the segment joining the points $(-4,6)$ and $(8, 8)$, is

[Guj. CET]

(a) $y = 7$

(b) $6x + y - 19 = 0$

(c) $x + 2y - 7 = 0$

(d) $6x + 2y - 19 = 0$

Q 70. If the slopes of one of the lines given by $ax^2 + 2hxy + by^2 = 0$ is 5 times the other, the

[Guj. CET]

(a) $5h^2 = 9ab$

(b) $5h^2 = ab$

(c) $h^2 = ab$

(d) $9h^2 = 5ab$

Q 71. The equation $y^2 - x^2 + 2x - 1 = 0$

represents

[UP SEE]

(a) a hyperbola

(b) an ellipse

(c) a pair of straight lines

(d) a rectangular hyperbola

Q 72. Equation of the straight line making equal intercepts on the axes and passing through the point $(2,4)$, is

[KCET]

(a) $4x - y - 4 = 0$

(b) $2x + y - 8 = 0$

(c) $x + y - 6 = 0$

(d) $x + 2Y - 10 = 0$

Q 73. If x_1, x_2, x_3 as well as y_1, y_2, y_3 are in GP with the same common ratio, then the points (x_1, y_1),(x_2, y_2) and (x_3, y_3)

[Kerala CEE]

(a) lie on a parabola

(b) lie on an ellipse

(c) lie on a circle

(d) are the vertices of a triangle

(e) lie on a straight line

Q 74. If the straight line ax + by + c = 0 always passes through $(1, -2)$ then a, b, care in

[AMU]

(a) AP

(b) HP

(c) GP

(d) None of these

Q 75. Suppose A, B are two points on $2x - y + 3 = 0$ and $P(1, 2)$, is such that PA = PB. Then, the mid point of AB is

[EAMCET]

(a) $\left(-\frac{1}{5}, \frac{13}{5}\right)$

(b) $\left(-\frac{7}{5}, \frac{9}{5}\right)$

(c) $\left(\frac{7}{5}, \frac{-9}{5}\right)$

(d) $\left(\frac{-7}{5}, \frac{-9}{5}\right)$

Q 76. The polar equation $\cos\theta + 7\sin\theta = \frac{1}{r}$

represents a

[EAMCET]

(a) circle

(b) parabola

(c) straight line

(d) hyperbola

Q 77. Given the four lines with equations x + 2

y = 3,3x + 4y = 7, 2x + 3y = 4 and 4x + 5y = 6,

then these lines are

[Jamia Millia Islamia]

(a) concurrent

(b) perpendicular

(c) the sides of a rectangle

(d) None of the above

Q 78. The equation to a pair of opposite sides of a parallelogram are $x^2 - 5x + 6 = 0$ and $y^2 - 6y + 5 = 0$, the equation to its diagonals are

[Jamia Millia Islamia]

(a) $x + 4y = 13$ and $y = 4x - 7$

(b) $4x + y = 13$ and $4y = x - 7$

(c) $4x + y - 13$ and $y - 4x - 7$

(d) $y - 4x = 13$ and $y + 4x = 7$

Q 79. The sum of slopes of lines $3x^2 + 5xy - 2y^2 = 0$ is

[MHT CET]

(a) $-\frac{5}{3}$

(b) $\frac{5}{2}$

(c) $-\frac{5}{2}$

(d) $-\frac{2}{3}$

Q 80. A square of side a lies above the x–axis and has one vertex at the origin. The side passing through the origin makes an angle $\alpha\left(0<\alpha<\frac{\pi}{4}\right)$ with the positive direction of x–axis. The equation of its diagonal not passing through the origin is

[AIEEE]

(a) $y(\cos\alpha - \sin\alpha) - x(\sin\alpha - \cos\alpha) = a$

(b) $y(\cos\alpha + \sin\alpha) + x(\sin\alpha - \cos\alpha) = a$

(c) $y(\cos\alpha + \sin\alpha) + x(\sin\alpha + \cos\alpha) = a$

(d) $y(\cos \alpha + \sin \alpha) + x(\cos \alpha - \sin \alpha) = a$

Q 81. y–intercept of line passes through $(2, 2)$ and is perpendicular to the line $3x + y = 3$, is

[DCE]

(a) $\frac{1}{3}$

(b) $\frac{2}{3}$

(c) 1

(d) $\frac{4}{3}$

Q 82. Two consecutive sides of a parallelogram are $4x + 5y = 0$ and $7x+2y = 0$. One diagonal of the parallelogram is $11x + 7y = 9$. If the other diagonal is $ax + by + c = 0$, then

[Kerala CEE]

(a) $a = -1, b = -1, c = 2$

(b) $a = 1, b = -1, c = 0$

(c) $a = -1, b = -1, c = 0$

(d) $a = 1, b = 1, c = 0$

(e) a = −1, b = −1, c = 1

Q 83. The equation of the sides of a triangle are $x - 3y = 0$, $4x + 3y = 5$ and $3x + y = 0$. The line $3x - 4y = 0$ passes through

[Kerala CEE]

(a) the incentre

(b) the centroid

(c) the orthocentre

(d) the circumcentre

(e) None of these

Q 84. Given four lines with equations $x + 2y - 3 = 0$, $2x + 3y - 4 = 0$, $3x + 4y - 5 = 0$, $4x + 5y - 6 = 0$ These lines are

[Kerala CEE]

(a) concurrent

(b) the sides of a quadrilateral

(c) the sides of a parallelogram

(d) the sides of a square

(e) the sides of a rhombus

Q 85. If $(-4,5)$ is one vertex and $7x - y + 8 = 0$ is one diagonal of a square, then the equation of the second diagonal is

[Kerala CEE]

(a) $x + 3y = 21$

(b) $2x - 3y = 7$

(c) $x + 7Y = 31$

(d) $2x + 3y = 21$

(e) $x - 3y = 21$

Q 86. Coordinates of the foot of the perpendicular drawn from $(0, 0)$ to the line joining $(a \cos \alpha, a \sin \alpha)$ and $(a \cos \beta, a \sin \beta)$ are

[AMU]

(a) $\left(\frac{a}{2}, \frac{b}{2}\right)$

(b) $\left[\frac{a}{2}(\cos\alpha + \cos\beta), \frac{a}{2}(\sin\alpha + \sin\beta)\right]$

(c) $\left[\cos\frac{\alpha+\beta}{2}, \sin\frac{\alpha+\beta}{2}\right]$

(d) $\left(0, \frac{b}{2}\right)$

Q 87. If the lines $4x + 3y - 1 = 0$, $x - y + 5 = 0$ and $kx + 5y - 3 = 0$ are concurrent, then k is equal to

[EAMCET]

(a) 4

(b) 5

(c) 6

(d) 7

Q 88. A ray of light passing through the point $(1, 2)$ is reflected on the x−axis at a point P and passes through the point $(5,3)$, then the abscissa of a point P is

[OJEE]

(a) 3

(b) 13/3

(c) 13/5

(d) $13/4$

Q 89. The equation $4x^2 - 24xy + 11\ y^2 = 0$

represents

[OJEE]

(a) two parallel lines

(b) two perpendicular lines

(c) two lines through the origin

(d) a circle

Q 90. If the equations, $12x^2 - 10xy + 2y^2 + 11x - 5y + k = 0$ represents two straight lines, then the value of k is

[MPPET]

(a) 1

(b) 2

(c) 0

(d) 3

Q 91. Equation of straight line cutting off an intercept 2 from the negative direction of the axes of y and inclined at $30°$ to the positive direction of axis of x, is

[RPET]

(a) $y + x - \sqrt{3} = 0$

(b) $y - x + 2 = 0$

(c) $y - \sqrt{3}x - 2 = 0$

(d) $\sqrt{3}y - x + 2\sqrt{3} = 0$

Topic 2 Distance between Lines and Foot of Perpendicular

Q 1. The line L given by $\frac{x}{5}+\frac{y}{b}=1$ passes through the point $(13, 32)$. The line K is parallel to L and has the equation $\frac{x}{c}+\frac{y}{3}=1$. Then, the distance between L and K is

[AIEEE]

(a) $\frac{23}{\sqrt{15}}$

(b) $\sqrt{17}$

(c) $\frac{17}{\sqrt{15}}$

(d) $\frac{23}{\sqrt{17}}$

Q 2. The equation of one of the lines parallel to $4x - 3y = 5$ and at a unit distance from the point $(-1, -4)$ is

[Kerala CEE]

(a) $3x + 4y - 3 = 0$

(b) $3x + 4y + 3 = 0$

(c) $4x - 3y + 3 = 0$

(d) $4x - 3y - 3 = 0$

(e) $4x - 3y - 4 = 0$

Q 3. A line through the point A $(2,0)$ which makes an angle of $30°$ with the positive direction of x−axis is rotated about A in clockwise direction through an angle of $15°$. Then, the equation of the straight line in the new position is

[WB JEE]

(a) $(2-\sqrt{3})x+y-4+2\sqrt{3}=0$

(b) $(2-\sqrt{3})x-y-4+2\sqrt{3}=0$

(c) $(2-\sqrt{3})x-y+4+2\sqrt{3}=0$

(d) $(2-\sqrt{3})x+y+4+2\sqrt{3}=0$

Q 4. A line has slope m and y−intercept 4.

The distance between the origin and the line is

equal to

[Kerala CEE]

(a) $\frac{4}{\sqrt{1-m^2}}$

(b) $\frac{4}{\sqrt{m^2-1}}$

(c) $\frac{4}{\sqrt{m^2+1}}$

(d) $\frac{4m}{\sqrt{1+m^2}}$

(e) $\frac{4m}{\sqrt{m-1}}$

Q 5. The distance of the point $(1, 2)$ from the line x + y + 5 = 0 measured along the line parallel to 3x − y = 7 is equal to

[Kerala CEE]

(a) $4\sqrt{10}$

(b) 40

(c) $\sqrt{40}$

(d) $10\sqrt{2}$

(e) $2\sqrt{20}$

Q 6. The number of points on the line x + y = 4 which are unit distance apart from the line 2x + 2y = 5 is

[WB JEE]

(a) 0

(b) 1

(c) 2

(d) ∞

Q 7. The distance between the pair of lines represented by the equation $x^2 - 6xy + 9y^2 + 3x - 9y - 4 = 0$ is

[UP SEE]

(a) $\frac{15}{\sqrt{10}}$

(b) $\frac{1}{2}$

(c) $\sqrt{\frac{5}{2}}$

(d) $\frac{1}{\sqrt{10}}$

Q 8. The distance between the lines $5x - 12y + 65 = 0$ and $5x - 12y - 39 = 0$ is

[WBJEE]

(a) 4

(b) 16

(c) 2

(d) 8

Q 9. The coordinates of the foot of perpendicular from $(a, 0)$ on the line $y = mx + \frac{a}{m}$ are

[WBJEE]

(a) $\left(0, \frac{a}{m}\right)$

(b) $\left(0, -\frac{a}{m}\right)$

(c) $\left(\frac{a}{m}, 0\right)$

(d) $\left(-\frac{a}{m}, 0\right)$

Q 10. The length of the perpendicular from the origin to the line $\frac{x \sin \alpha}{b} - \frac{y \cos \alpha}{a} - 1 = 0$ is

[J&KCET]

(a) $\frac{|ab|}{\sqrt{a^2 \cos^2 \alpha - b^2 \sin^2 \alpha}}$

(b) $\frac{|ab|}{\sqrt{a^2\cos^2\alpha + b^2\sin^2\alpha}}$

(c) $\frac{|ab|}{\sqrt{a^2\sin^2\alpha - b^2\cos^2\alpha}}$

(d) $\frac{|ab|}{\sqrt{a^2\sin^2\alpha + b^2\cos^2\alpha}}$

Q 11. The distance between the parallel lines y

= x + a,y = x + b is

[J&KCET]

(a) $\frac{|b-a|}{\sqrt{2}}$

(b) |a − b|

(c) $|a + b|$

(d) $\frac{|b+a|}{\sqrt{2}}$

Q 12. The distance between the lines $3x + 4y = 9$ and $6x + 8y = 15$ is

[UPSEE]

(a) $\frac{3}{2}$

(b) $\frac{3}{10}$

(c) 6

(d) None of these

Q 13. The coordinates of the foot of the perpendicular drawn from the point $(3, 4)$ on the line $2x + y - 7 = 0$ is

[KCET]

(a) $\left(\frac{9}{5}, \frac{17}{5}\right)$

(b) $(1,5)$

(c) $(-5,1)$

(d) $(1,-5)$

Q 14. If A $(2,-1)$ and B$(6, 5)$ are two points, then the ratio in which the foot of the perpendicular from $(4, 1)$ to AB divides it, is

[EAMCET]

(a) $8 : 15$

(b) $5 : 8$

(c) $-5 : 8$

(d) $-8 : 5$

Q 15. If the foot of the perpendicular from the origin to a straight line is at the point $(3, -4)$. Then, the equation of the line is

[WB JEE]

(a) $3x - 4y = 25$

(b) $3x - 4Y + 25 = 0$

(c) $4x + 3y - 25 = 0$

(d) $4x - 3y + 25 = 0$

Q 16. If $3, 4$ are intercepts of a line L = 0, then the distance of L = 0 from the origin is

[J&K CET]

(a) 5 units

(b) 12 units

(c) $\frac{5}{12}$ units

(d) $\frac{12}{5}$ units

Q 17. The distance between the pair of parallel lines $x + 2xy + y^2 - 8ax - 8ay - 9a^2 = 0$ is

[BITSAT, Guj. CET]

(a) $2\sqrt{5}a$

(b) $\sqrt{10}\,a$

(c) $10a$

(d) $5\sqrt{2}\,a$

Q 18. The length of perpendicular from the point $(a\cos\alpha, a\sin\alpha)$ upon the straight line

$y = x \tan \alpha + c$, $c > 0$, is

[UP SEE, WB JEE]

(a) c

(b) $c \sin^2 \alpha$

(c) $c \cos \alpha$

(d) $c \sec \alpha$

Q 19. The distance between the pair of parallel lines given by $x^2 - 1005 x + = 0$ is

[J&K CET]

(a) 1001

(b) 1000

(c) 1005

(d)

Q 20. The equation $x^2 + 2\sqrt{2}xy + 2y^2 + 4x + 4\sqrt{2}y + 1 = 0$ represents a pair of lines which are parallel to each other. The distance between them is

[DCE]

(a) 4 units

(b) $2\sqrt{3}$ units

(c) $4\sqrt{3}$ units

(d) 2 units

Q 21. If the equation of base of an equilateral triangle is $2x - y = 1$ and the vertex is $(-1, 2)$, then the length of the side of the triangle is

[Kerala CEE]

(a) $\sqrt{\frac{20}{3}}$

(b) $\frac{2}{\sqrt{15}}$

(c) $\sqrt{\frac{8}{15}}$

(d) $\sqrt{\frac{15}{2}}$

(e) $\sqrt{5}$

Q 22. The image of the origin with reference to the line $4x+3y-25=0$ is

[Kerala CEE]

(a) $(-8,6)$

(b) $(8,6)$

(c) $(-3,4)$

(d) $(8,-6)$

(e) $(-4,-3)$

Q 23. L is a variable line such that the algebraic sum of the distances of the points $(1, 1)$, $(2, 0)$ and $(0, 2)$ from the line is equal to zero. The line L will always pass through

[AMU]

(a) $(1, 1)$

(b) $(2, 1)$

(c) $(1, 2)$

(d) $(2, 2)$

Q 24. The product of the perpendicular distances from the origin on the pair of straight lines

$12x^2 + 25xy + 12y^2 + 10x + 11y + 2 = 0$ is

[EAMCET]

(a) $\frac{1}{25}$

(b) $\frac{2}{25}$

(c) $\frac{3}{25}$

(d) $\frac{4}{25}$

Q 25. Let a be the distance between lines $-x + y = 2$ and $x - y = 2$ and β be the distance between the lines $4x - 3y = 5$ and $6y - 8x = 1$, then

[J&K CET]

(a) $20\sqrt{2}\beta = 11\alpha$

(b) $20\sqrt{2}\alpha = 11\beta$

(c) $11\sqrt{2}\beta = 20\alpha$

(d) None of these

Q 26. If p is the length of the perpendicular from the origin on the line whose intercepts on the axes are a and b, then

[Guj. CET]

(a) $p^2 = a^2 + b^2$

(b) $p^2 = a^2 - b^2$

(c) $\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}$

(d) $\frac{1}{p^2} = \frac{1}{a^2} - \frac{1}{b^2}$

Q 27. The distance between the pair of parallel lines $x^2 + 4xy + 4y^2 + 3x + 6y - 4 = 0$ is

[Kerala CEE]

(a) $\sqrt{5}$

(b) $\frac{2}{\sqrt{5}}$

(c) $\frac{1}{\sqrt{5}}$

(d) $\frac{\sqrt{5}}{2}$

(e) $\sqrt{\frac{5}{2}}$

Q 28. If the line $\frac{x}{a}+\frac{y}{b}=1$ moves such that

$\frac{1}{a^2}+\frac{1}{b^2}=\frac{1}{c^2}$ where c is a constant, then the locus of

the foot of the perpendicular from the origin to the line is

[AMU]

(a) straight line

(b) circle

(c) parabola

(d) ellipse

Q 29. Distance between the two parallel lines $y = 2x + 7$ and $y = 2x + 5$ is

[OJEE]

(a) $\sqrt{5}/2$

(b) $2/5$

(c) $2/\sqrt{5}$

(d) $1/\sqrt{5}$

Q 30. The orthocentre of a triangle formed by the lines $x + y = 1$, $2x + 3y = 6$ and $4x - y + 4 = 0$ lies in the

[J&K CET]

(a) Ist quadrant

(b) IInd quadrant

(c) IIIrd quadrant

(d) IVth quadrant

Q 31. The equation of the base of an equilateral triangle is $x + y = 2$ and the vertex is $(2, -1)$, then the length of the side of the triangle is

[AMU]

(a) $\sqrt{3/2}/\sqrt{2/3}$

(b) $\sqrt{2}$

(c) $\sqrt{2/3}$

(d) $\sqrt{3/2}$

Q 32. Consider the fourteen lines in the plane given by $y = x + r$, $y = -x + r$, where $r \in \{0,1,2,3,4,5,6\}$. The number of squares formed by these lines, whose sides are of length $\sqrt{2}$, is

[EAMCET]

(a) 9

(b) 16

(c) 25

(d) 36

Q 33. The point on the axis of x, whose perpendicular distance from the straight line $\frac{x}{a}+\frac{y}{b}=1$ is a, are

[MP PET]

(a) $\frac{b}{a}(a \pm \sqrt{a^2 + b^2}, 0)$

(b) $\left(\frac{a}{b}(b \pm \sqrt{a^2 + b^2}), 0\right)$

(c) $\frac{b}{a}(a + b, 0)$

(d) $\frac{a}{b}(a \pm \sqrt{a^2 + b^2}, 0)$

Topic 3 Angle between Intersection Lines, Condition for Parallel and Perpendicular Lines

Q 1. The slopes of the lines which make an angle $45°$ with the line $3x - y = -5$ are

[Kerala CEE]

(a) $1, -1$

(b) $\frac{1}{2}, -1$

(c) $1, \frac{1}{2}$

(d) $2, -\frac{1}{2}$

(e) $-2, \frac{1}{2}$

Q 2. The locus of the orthocentre of the triangle formed by the lines $(1 + p)\, x - py + p\,(l + p) = 0$,

$(1 + q)\, x - qy + q\,(1 + q) = 0$ and $y = 0$, where $p \neq q$, is

[IIT JEE]

(a) a hyperbola

(b) a parabola

(c) an ellipse

(d) a straight line

Q 3. The lines $p(p^2 + 1)x - y + q = 0$ and $(p^2 + 1)^2x + (p^2 + 1)y + 2q = 0$ are perpendicular to a common line for

[AIEEE]

(a) exactly one value of p

(b) exactly two values of p

(c) more than two values of p

(d) no values of p

Q 4. All chords of the curve $3x^2 - y^2 - 2x + 4y = 0$ which subtend a right angle at the origin, pass through the fix point

[BITSAT]

(a) $(1, 2)$

(b) $(1, -2)$

(c) $(-1, 2)$

(d) $(-1, -2)$

Q 5. The pair of lines joining origin to the points of intersection of the two curves $ax^2 + 2hxy + by^2 + 2gx = 0$ and $a'x^2 + 2h'xy + b' y^2 + 2g'x = 0$ will be at right angles, if

[UP SEE]

(a) $(a' + b')g' = (a + b)g$

(b) $(a + b)g' = (a' + b')g$

(c) $h^2 - ab = h'^2 - a'b'$

(d) $a + b + h^2 = a' + b' + h'^2$

Q 6. If the lines $y = 3x + 1$ and $2y = x + 3$ are equally inclined to the line $y = mx + 4$, $\left(\frac{1}{2} < m < 3\right)$, then the values of m are

[Kerala CEE]

(a) $\frac{1}{7}(1 \pm 5\sqrt{3})$

(b) $\frac{1}{7}(1 \pm 5\sqrt{5})$

(c) $\frac{1}{7}(1 \pm 5\sqrt{2})$

(d) $\frac{1}{7}(1 \pm 2\sqrt{5})$

(e) $\frac{1}{7}(1 \pm 3\sqrt{2})$

Q 7. The area (in square unit) of the triangle formed by $x + y + 1 = 0$ and the pair of straight lines $x^2 - 3xy + 2y^2 = 0$ is

[EAMCET]

(a) $\frac{7}{12}$

(b) $\frac{5}{12}$

(c) $\frac{1}{12}$

(d) $\frac{1}{6}$

Q 8. A line passes through point $(2, 2)$ and perpendicular to the line $3x + y = 3$. Then, y–intercept is

[OJEE]

(a) 2

(b) $\frac{5}{3}$

(c) $\frac{4}{3}$

(d) $\frac{3}{4}$

Q 9. If the lines $px^2 - qxy - y^2 = 0$ make the angles α and β with x–axis, then the value of tan $(\alpha + \beta)$ is

[VITEEE]

(a) $\frac{-q}{1+p}$

(b) $\frac{q}{1+p}$

(c) $\frac{p}{1+q}$

(d) $\frac{-p}{1+q}$

Q 10. If 'θ' is the angle between the lines $ax^2 + 2hxy + by^2 = 0$, then angle between $x^2 + 2xy \sec\theta + y^2 = 0$ is

[MHT CET]

(a) θ

(b) 2θ

(c) $\frac{\theta}{2}$

(d) 3θ

Q 11. The lines $(a+2b)x + (a-3b)y = a-b$ for different values of a and b pass through the fixed point whose coordinates are

[Kerala CEE]

(a) $\left(\frac{2}{5}, \frac{2}{5}\right)$

(b) $\left(\frac{3}{5}, \frac{3}{5}\right)$

(c) $\left(\frac{1}{5}, \frac{1}{5}\right)$

(d) $\left(\frac{3}{5}, \frac{2}{5}\right)$

(e) $\left(\frac{2}{5}, \frac{3}{5}\right)$

Q 12. A line passes through the point of intersection of the lines $100x + 50y - 1 = 0$ and

$75x+25y + 3 = 0$ and makes equal intercept on the axes. Its equation is

[Kerala CEE]

(a) $25X + 25Y - 1 = 0$

(b) $5x - 5Y + 3 = 0$

(c) $25x + 25y - 4 = 0$

(d) $25x - 25y + 6 = 0$

(e) $5x - 5y + 7 = 0$

Q 13. A pair of perpendicular straight lines passes through the origin and also through the point of intersection of the curve $x^2 + y^2 = 4$ with $x + y = a$. The set containing the value of 'a' is

[AMU]

(a) $\{-2, 2\}$

(b) $\{-3, 3\}$

(c) $\{-4, 4\}$

(d) $\{-5, 5\}$

Q 14. The lines represents by $ax^2 + 2hxy + by^2 = 0$ are perpendicular to each other, if

[Jamia Millia Manila]

(a) $h^2 = a + b$

(b) $a + b = 0$

(c) $h^2 = ab$

(d) $h = 0$

Q 15. The equation of the bisector of the obtuse angle between the lines $3x - 4y + 7 = 0$

and $-12x - 5y + 2 = 0$, is

[Manipal]

(a) $21x + 77y - 101 = 0$

(b) $99x - 27y + 81 = 0$

(c) $21x - 77y + 101 = 0$

(d) None of these

Q 16. The acute angle between the lines joining the origin to the points of intersection of the line $\sqrt{3}x + y = 2$ and the circle $x^2 + y^2 = 4$, is

[Kerala CEE]

(a) $\pi/2$

(b) $\pi/3$

(c) $\pi/4$

(d) $\pi/6$

(e) $\pi/12$

Q 17. The angle between the line joining the points $(1,-2)$, $(3,2)$ and the line $x+2y-7=0$ is

[EAMCET]

(a) π

(b) $\pi/2$

(c) $\pi/3$

(d) $\pi/6$

Q 18. The angle between the pair of straight lines formed by joining the points of intersection of

$x^2 + y^2 = 4$ and $y = 3x + c$ to the origin is a right angle. Then, c^2 is equal to

[EAMCET]

(a) 20

(b) 13

(c) 1/5

(d) 5

Q 19. The angle between lines joining origin and intersection points of line $2x + y = 1$ and curve $3x^2 + 4yx - 4x + 1 = 0$ is

[WB JEE]

(a) $\pi/2$

(b) $\pi/3$

(c) $\pi/4$

(d) $\pi/6$

Q 20. The angle between the straight lines

$x - y\sqrt{3} = 5$ and $\sqrt{3}x + y = 7$ is

[WBJEE]

(a) $90°$

(b) $60°$

(c) $75°$

(d) $30°$

Q 21. The angle between the pair of lines $(x^2 + y^2)\sin^2 \alpha = (x \cos \theta - y \sin \theta)^2$ is

[J&KCET]

(a) θ

(b) 2θ

(c) α

(d) 2α

Q 22. If (a, a^2) falls inside the angle made by the lines $y = \frac{x}{2}$, $x > 0$ and $y = 3x$, $x > 0$, then a belongs to

[AIEEE]

(a) $(3, \infty)$

(b) $\left(\frac{1}{2}, 3\right)$

(c) $\left(-3, -\frac{1}{2}\right)$

(d) $\left(0, \frac{1}{2}\right)$

Q 23. The angle between the lines $\sqrt{3}x - y - 2 = 0$ and $x - \sqrt{3}y + 1 = 0$ is

[Kerala CEE]

(a) $90°$

(b) $60°$

(c) $45°$

(d) $15°$

(e) $30°$

Q 24. The angle between the lines represented by the equation $2x^2 + 3xy - 5y^2 = 0$, is [MHT CET]

(a) $\frac{\pi}{3}$

(b) $\frac{\pi}{2}$

(c) $\tan^{-1}\left|\frac{12}{5}\right|$

(d) $\tan^{-1}\left|\frac{7}{3}\right|$

Q 25. A line passing through origin and is perpendicular to two given lines $2x + y + 6 = 0$ and $4x + 2y - 9 = 0$. The ratio in which the origin divides this line, is

[DCE]

(a) $1 : 2$

(b) $2 : 1$

(c) $4 : 2$

(d) $4 : 3$

Q 26. The angle between the pair of straight lines $y^2 \sin^2\theta - xy \sin^2\theta + x^2(\cos^2\theta - 1) = 0$ is

[Kerala CEE]

(a) $\pi/3$

(b) $\pi/4$

(c) $\pi/6$

(d) $\pi/2$

(e) π

Q 27. A straight line through the point $(2, 2)$ intersects the lines $\sqrt{3}x + y = 0$ and $\sqrt{3}x - y = 0$ at the points A and B. The equation to the line AB so that the ΔOAB is equilateral is

[Jamia Millia Manila]

(a) $x - 2 = 0$

(b) $y - 2 = 0$

(c) $x + y - 4 = 0$

(d) None of these

Q 28. The angle between the lines in $x^2 - xy - 6y^2 - 7x + 31y - 18 = 0$ is

[Guj.CET]

(a) $60°$

(b) $45°$

(c) $30°$

(d) $90°$

Q 29. $ax + by - a^2 = 0$, where a,b are non-zero, is the equation to the straight line perpendicular to a line l and passing through the point where l crosses the x-axis. Then, equation to the line l is

[Kerala CEE]

(a) $\frac{x}{b}-\frac{y}{a}=1$

(b) $\frac{x}{a}-\frac{y}{b}=1$

(c) $\frac{x}{b}+\frac{y}{a}=ab$

(d) $\frac{x}{a}-\frac{y}{b}=ab$

(e) $\frac{x}{a}+\frac{y}{b}=ab$

Q 30. If the pair of straight lines given by Ax^2 $+2Hxy + By^2 = 0$ $(H^2 > AB)$ forms an equilateral

triangle with line ax + by + c = 0, then (A+3B) (3 A + B) is equal to

[EAMCET]

(a) H^2

(b) $-H^2$

(c) $2H^2$

(d) $4H^2$

Q 31. The line passing through $\left(-1, \frac{\pi}{2}\right)$ and perpendicular to $\sqrt{3}\sin\theta + 2\cos\theta = \frac{4}{r}$ is

[EAMCET]

(a) $2 = \sqrt{3}r\cos\theta - 2r\sin\theta$

(b) $5 = -2\sqrt{3}r\sin\theta + 4r\cos\theta$

(c) $2 = \sqrt{3}r\cos\theta + 2r\sin\theta$

(d) $5 = 2\sqrt{3}r\sin\theta + 4r\cos\theta$

Q 32. The lines $(lx + my)^2 - 3(mx - ly)^2 = 0$ and $lx + my + n = 0$ form

[MP PET]

(a) an isosceles triangle

(b) a right angled triangle

(c) an equilateral triangle (d) None of these

Topic 4 Position of a Point, Bisectors of the Lines

Q 1. The vertices of a triangle are A $(3, 7)$, B $(3, 4)$ and C $(5, 4)$. The equation of the bisector of the angle ABC is

[Kerala CEE]

(a) $y = x + 1$

(b) $y = x - 1$

(c) $y = 3x - 5$

(d) $y = x$

(e) $y = -x$

Q 2. If P is a point (x, y) on the line $y = -3x$ such that P and the point $(3, 4)$ are on the opposite sides of the line $3x - 4y - 8 = 0$, then

[AMU]

(a) $x > \frac{8}{15}, y < -\frac{8}{5}$

(b) $x > \frac{8}{5}, y < -\frac{8}{15}$

(c) $x = \frac{8}{15}, y = -\frac{8}{5}$

(d) None of these

Q 3. The bisector of the acute angle formed between the lines $4x - 3y + 7 = 0$ and $3x - 4y + 14 = 0$ has the equation

[DCE]

(a) $x + y + 3 = 0$

(b) $x - y - 3 = 0$

(c) $x - y + 3 = 0$

(d) $3x + y - 7 = 0$

Q 4. If the bisectors of angles represented by $ax^2 + 2hxy+by^2 = 0$ and $a'x^2 +2h'xy + b' y^2 = 0$ are same, then

[OJEE]

(a) $(a - b)h' = (a' - b')h$

(b) $(a - b)h = (a' - b')h'$

(c) $(a + b)h' = (a' - b')h$

(d) $(a - b)h' = (a' + b')h$

Q 5. If pairs straight lines $x^2 - 2pxy - y^2 = 0$ and $x^2 - 2qxy - y^2 = 0$ be such that each pair bisects the angle between the other pair, then

[MP PET]

(a) $pq = 1$

(b) $pq = -1$

(c) $pq = 2$

(d) $pq = -2$

Q 6. Lines $L_1 : y - x = 0$ and $L_2 : 2x + y = 0$ intersect the line $L_3 : y + 2 = 0$ at P and Q, respectively. The bisector of the acute angle between L_1 and L_2 intersects L_3 at R.

[IIT JEE]

Statement I The ratio PR : PQ equals $2\sqrt{2} : \sqrt{5}$.

Because Statement II In any triangle, bisector of an angle divides the triangle into two similar triangles.

(a) Statement I is true, Statement II is true;

Statement II is a correct explanation for Statement I

(b) Statement 1 is true, Statement II is true;

Statement II is not a correct explanation for

Statement I

(c) Statement I is true, Statement II is false

(d) Statement I is false, Statement II is true

Q 7. If one of the lines of $my^2 + (1 - m^2)xy - mx^2 = 0$ is a bisector of the angle between the lines $xy = 0$, then m is/are

[AIEEE]

(a) $-\frac{1}{2}$

(b) -2

(c) ± 1

(d) 2

Q 8. Let P = $(-1, 0)$, Q = $(0, 0)$ and R= $(3, 3\sqrt{3})$ be three points. The equation of the bisector of the angle PQR is

[AIEEE]

(a) $\sqrt{3}x + y = 0$

(b) $x + \frac{\sqrt{3}}{2} y = 0$

(c) $\frac{\sqrt{3}}{2} x + y = 0$

(d) $x + \sqrt{3} y = 0$

Q 9. The equation of bisectors of the angles between the lines | x | − | y | are

[WB JEE]

(a) $y = \pm x$ and $x = 0$

(b) $x = \frac{1}{2}$ and $y = \frac{1}{2}$

(c) $y = 0$ and $x - 0$

(d) None of these

Q 10. The equation of the bisector of the acute angle between the lines $3x - 4y + 7 = 0$ and

$12x + 5y - 2 = 0$ is

[WBJEE]

(a) $99x - 27y - 81 = 0$

(b) $11x - 3y + 9 = 0$

(c) $21x + 77y - 101 = 0$

(d) $21x + 77y + 101 = 0$

Q 11. The line $2x - y = 1$ bisects angle between two lines. If equation of one line is $y = x$, then the equation of the other line is

[Kerala CEE]

(a) $7x - y - 6 = 0$

(b) $x - 2y + 1 = 0$

(c) $3x - 2y - 1 = 0$

(d) $x - 7y + 6 = 0$

(e) $2x - 3y + 1 = 0$

Q 12. The position of reflection of the point $(4, 1)$ about the line $y = x - 1$ is

[Kerala CEE]

(a) $(1, 2)$

(b) $(3, 4)$

(c) $(-1, 0)$

(d) $(-2, -1)$

(e) $(2, 3)$

Q 13. Three straight lines $2x + 11y - 5 = 0$, $24x + 7y - 20 = 0$ and $4x - 3y - 2 = 0$

[OJEE]

(a) from a triangle

(b) are only concurrent

(c) are concurrent with one line bisecting the angle between the other two

(d) None of the above

Q 14. The image of the origin with reference to the line $4x + 3y - 25 = 0$, is

[MHT CET]

(a) $(-8,6)$

(b) $(8,6)$

(c) $(-3,4)$

(d) $(8,-6)$

Q 15. The bisector of the acute angle formed between the lines $4x - 3y + 7 = 0$ and $3x - 4y + 14 = 0$ has the equation

[Guj. CET]

(a) $x + y + 3 = 0$

(b) $x - y - 3 = 0$

(c) $x - y + 3 = 0$

(d) $3x + y - 7 = 0$

Q 16. The equation of the line bisecting perpendicularly the segment joining the points $(-4,6)$ and $(8, 8)$ is

[KCET]

(a) $y = 7$

(b) $6x + y - 19 = 0$

(c) $x + 2y - 7 = 0$

(d) $6x + 2y - 19 = 0$

Answers

Topic 1 : Equation of Line and Pair of Lines

1. (d) 2. (d) 3. (b) 4. (a) 5. (b) 6. (b) 7. (d) 8. (c)

9. (a) 10. (a) 11. (b) 12. (a) 13. (b) 14. (b) 15. (a)

16. (b) 17. (d) 18. (c) 19. (e) 20. (c) 21. (b) 22.

(c) 23. (c) 24. (b) 25. (a) 26. (a) 27. (c) 28. (a)

29. (d) 30. (a) 31. (a) 32. (b) 33. (a) 34. (c) 35.

(b) 36. (c) 37. (d) 38. (c) 39. (a) 40. (d) 41. (d)

42. (b) 43. (c) 44. (c) 45. (c) 46. (b) 47. (d) 48.

(b) 49. (c) 50. (d) 51. (b) 52. (c) 53. (b) 54. (a)

55. (b) 56. (b) 57. (d) 58. (a) 59. (a) 60. (c)

61. (a) 62. (a) 63. (c) 64. (d) 65. (d) 66. (d) 67.

(a) 68. (b) 69. (b) 70. (a) 71. (c) 72. (c) 73. (e)

74. (a) 75. (a) 76. (c) 77. (d) 78. (c) 79. (b) 80.

(d) 81. (d) 82. (b) 83. (c) 84. (a) 85. (c) 86. (b)

87. (c) 88. (c) 89. (c) 90. (b) 91. (d)

Topic 2 : Distance between Lines and Foot of Perpendicular

1. (d) 2. (d) 3. (b) 4. (c) 5. (c) 6. (a) 7. (c) 8. (d)

9. (a) 10. (d) 11. (a) 12. (b) 13. (a) 14. (b) 15. (a)

16. (d) 17. (d) 18. (c) 19. (a) 20. (d) 21. (a) 22.

(b) 23. (a) 24. (b) 25. (a) 26. (c) 27. (a) 28. (b)

29. (c) 30. (a) 31. (c) 32. (c) 33. (b)

Topic 3 : Angle between Intersection Lines, Condition for Parallel and Perpendicular Lines

1. (e) 2. (d) 3. (a) 4. (b) 5. (b) 6. (c). 7. (c) 8. (c)

9. (a) 10. (a) 11. (e) 12. (c) 13. (a) 14. (b) 15. (a)

16. (b) 17. (b) 18. (a) 19. (a) 20. (a) 21. (d) 22.

(b) 23. (e) 24. (d) 25. (d) 26. (d) 27. (b) 28. (b)

29. (b) 30. (d) 31. (a) 32. (c)

Topic 4 : Position of a Point, Bisectors of the Lines

1. (a) 2. (a) 3. (c) 4. (a) 5. (b) 6. (b) 7. (c) 8. (a)

9. (c) 10. (c) 11. (b) 12. (e) 13. (c) 14. (b) 15. (c)

16. (b)

Circle and System of Circles

Useful Results & Formulae

1. General equation of circle is $x^2 + y^2 + 2gx + 2gy + c = 0$, whose radius $= \sqrt{g^2 + f^2 - c}$ and centre $= -(-g, -f)$

2. Different equations of circle are

(i) $(x - h)^2 + (y - k)^2 = r^2$

(ii) $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$

(when end points of a diameters are given)

3. The lengths of intercepts made by the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ on x and y axes are $2\sqrt{g^2 - c}$ and $2\sqrt{f^2 - c}$ respectively.

4. The equation of circle in polar form, whose centre (c, α) and radius a is $r^2 - 2cr\cos(\theta - \alpha) = a^2 - c^2$

5. (i) Equation of tangent at (x_1, y_1) to circle $x^2 + y^2 = a^2$ is $xx_1 + yy_1 = a^2$.

(ii) Equation of tangent at (x_1, y_1) to circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is

$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$.

(iii) Equation of tangent at $(a \cos \theta, a \sin \theta)$ to circle $x^2 + y^2 =$ or is $x \cos \theta + y \sin \theta = a$.

6. Line $y = mx + c$ is a tangent to the circle $x^2 + y^2 = a^2$, if $c^2 = a^2 (1 + m^2)$ and point of contact **is**

$$\left(\frac{\pm m}{\sqrt{1+m^2}}, \frac{\mp a}{\sqrt{1+m^2}} \right).$$

7. The length of intercept cut off from the line $y = mx + c$ by the circle $x^2 + y^2 = a^2$ is $2\sqrt{\frac{a^2(1+m^2)-c^2}{1+m^2}}$

8. Length of tangent from external point (x_1, y_1) to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is

$$\sqrt{x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c} = \sqrt{S_1}$$

9. (i) Equation of pair of tangent is $SS_1 = T^2$ where,

$S = x^2 + y^2 + 2gx + 2fy + c$

$S_1 = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$ and $T = xx_1 + yy_1 + g(x + x_l) + f(y + y_1) + c$.

(ii) Angle between the two tangents from (α, β) to the circle $x^2 + y^2 = r^2$ is $2 \tan^{-1} \frac{r}{\sqrt{S_1}}$.

10. (i) Equation of normal at (x_1, y_1) to circle $x^2 + y^2 = a^2$ is $\frac{x}{x_1} = \frac{y}{y_1}$.

(ii) Equation of normal at (x_1, y_1) to circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $\frac{x - x_1}{x_1 + g} = \frac{y - y_1}{y_1 + f}$.

(iii) Equation of normal at $(a \cos \theta, a \sin \theta)$ to circle $x^2 + y^2 = a^2$ is $\frac{x}{a \cos \theta} = \frac{y}{a \sin \theta}$.

11. If a line is normal to the circle, then it is passes through the centre of circle.

12. Equation of chord of contact from outside point (x_1, y_1) to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is

$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$. If $P(x_1, y_1)$ is the mid point of chord, then equation of chord is

$T = S_1$

i.e., $xx_1 + yy_1 + g(x + x_1) + f(y + y_1)$ $x_1^2 + y_1^2$ $+ 2gx_1 + 2fy_1$

13. Equation of common chord of two circles is $S_1 - S_2 = 0$.

14. If two perpendicular tangents to the circle $x^2 + y^2 = a^2$ meet at P, then the point P lies on a director circle, therefore locus is $x^2 + y^2 = 2a^2$.

15. (i) One circle lies inside the other if $| C_1C_2 | < | r_1 - r_2 |$. There is no common tangent,

(ii) Two circles touch each other internally, if $| C_1 C_2 | = | r_1 - r_2 |$. There is one common tangent.

(iii) Two circles intersect each other, if $| C_1C_2 | < r_1 + r_2$. There are two direct common tangents.

(iv) Two circles touch each other externally if | C_1 C_2 | = r_1 + r_2. There are three common tangents and point of contact is $\left(\frac{r_1x_2 + r_2x_1}{r_1 + r_2}, \frac{r_1y_2 + r_2y_1}{r_1 + r_2}\right)$.

(v) Two circles neither touch nor intersect each other, if | C_1 C_2 | > r_1 + r_2. There are two direct and two transverse common tangents,

16. Let two circles be $x^2 + y^2 + 2g_1 x + 2f_1y + c_1 = 0$ and $x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$.

(i) Angle of intersection of these two circles is

$$\theta = \cos^{-1}\left|\frac{r_1^2 + r_2^2 - d^2}{2r_1r_2}\right|$$

or $$\theta = \cos^{-1}\left|\frac{2(g_1g_2 + f_1f_2) - (c_1 + c_2)}{2\sqrt{g_1^2 + f_1^2 - c_1}\sqrt{g_2^2 + f_2^2 - c_2}}\right|$$

where d is the distance between the centres.

(ii) Two circles cut orthogonally, if

$$2g_1g_2 + 2f_1f_2 = c_1 + c_2.$$

17. (i) Equation of circle passing through the point of intersection of line(L) and circle(S) is $S + \lambda L = 0$.

(ii) Equation of circle passing through the points of intersections of circles $S_1 = 0$ and $S_2 = 0$ is

$S_1 + \lambda S_2 = 0$

18. (i) Equation of radical axis of two circles is $S_1 - S_2 = 0$.

(ii) Equation of coaxial system of circles is $S_1 + \lambda(S_1 - S_2) = 0$ or $S_2 + \lambda (S_1 - S_2) = 0$.

Topic 1 Circle and Its Equations

Q 1. The circle $x^2 + y^2 = 4x + 8y + 5$ intersects the line $3x - 4y = m$ at two distinct points, if

[AIEEE]

(a) $-85 < m < -35$

(b) $-35 < m < 15$

(c) $15 < m < 65$

(d) $35 < m < 85$

Q 2. The straight line $x + y - 1 = 0$ meets the circle $x^2 + y^2 - 6x - 8y = 0$ at A and B. Then, the equation of the circle of which AB is a diameter is

[WB JEE]

(a) $x^2+y^2 - 2y - 6 = 0$

(b) $x^2 +y^2 +2y - 6 = 0$

(c) $2(x^2 + y^2)+2y - 6 = 6$

(d) $3(x^2 + y^2)+ 2y - 6 = 0$

Q 3. The equation of family of circles with centre at (h, k) touching the x−axis is given by [Kerala CEE]

(a) $x^2 + y^2 - 2hx + h^2 = 0$

(b) $x^2 + y^2 - 2hx - 2ky + h^2 = 0$

(c) $x^2 + y^2 - 2hx - 2ky - h^2 = 0$

(d) $x^2 + y^2 - 2hx - 2ky = 0$

(e) $x^2 + y^2 + 2hx + 2ky = 0$

Q 4. The distance of the mid point of line joining two points $(4, 0)$ and $(0, 4)$ from the centre of the circle $x^2 + y^2 = 16$ is

[VITEEE]

(a) $\sqrt{2}$

(b) $2\sqrt{2}$

(c) $3\sqrt{2}$

(d) $2\sqrt{3}$

(e) $\sqrt{3}$

Q 5. The line segment joining the points $(4, 7)$ and $(-2, -1)$ is a diameter of a circle. If the circle intersects the x−axis at A and B, then AB is equal to

[BITSAT]

(a) 4

(b) 5

(c) 6

(d) 7

(e) 8

Q 6. A circle passes through $(0, 0)$, $(a, 0)$ and $(0, b)$ the coordinates of its centre are

[BITSAT]

(a) $\left(\frac{b}{2}, \frac{a}{2}\right)$

(b) $\left(\frac{a}{2}, \frac{b}{2}\right)$

(c) (b, a)

(d) (a, b)

Q 7. The coordinates of the centre of the smallest circle passing through the origin and having y = x +1 as a diameter are

[KCET]

(a) $\left(\frac{1}{2}, -\frac{1}{2}\right)$

(b) $\left(\frac{1}{2}, \frac{1}{3}\right)$

(c) $(-1, 0)$

(d) $\left(-\frac{1}{2},\frac{1}{2}\right)$

Q 8. Area of the equilateral triangle inscribed in the circle $x^2 + y^2 - 7x + 9y + 5 = 0$ is

[Kerala CEE]

(a) $\frac{155}{8}\sqrt{3}$ sq units

(b) $\frac{165}{8}\sqrt{3}$ sq units

(c) $\frac{175}{8}\sqrt{3}$ sq units

(d) $\frac{185}{8}\sqrt{3}$ sq units

(e) $\frac{195}{8}\sqrt{3}$ sq units

Q 9. The area (in square unit) of the circle which touches the lines $4x + 3y = 15$ and $4x + 3y = 5$ is

[EAMCET]

(a) 4π

(b) 3π

(c) 2π

(d) π

Q 10. The equations of the circle which pass through the origin and makes intercepts of lengths 4 and 8 on the x and y −axes respectively are

[EAMCET]

(a) $x^2 + y^2 \pm 4x \pm 8y = 0$

(b) $x^2 + y^2 \pm 2x \pm 4y = 0$

(c) $x^2 + y^2 \pm 8x \pm 16y = 0$

(d) $x^2 + y^2 \pm x \pm y = 0$

Q 11. The locus of centre of a circle which passes through the origin and cuts off a length of 4 unit from the line x = 3 is

[EAMCET]

(a) $y^2 + 6x = 0$

(b) $y^2 + 6x = 13$

(c) $y^2 + 6x = 10$

(d) $x^2 + 6y = 13$

Q 12. The diameters of a circle are along $2x + y - 7 = 0$ and $x + 3y - 11 = 0$. Then, the equation of this circle, which also passes through $(5, 7)$, is

[EAMCET]

(a) $x^2 + y^2 - 4x - 6y - 16 = 0$

(b) $x^2 + y^2 - 4x - 6y - 20 = 0$

(c) $x^2 + y^2 - 4x - 6y - 12 = 0$

(d) $x^2 + y^2 + 4x + 6y - 12 = 0$

Q 13. The equation of circle touches the line x = y at origin and passes through the point $(2, 1)$ is $x^2 + y^2 + px + qy = 0$. Then p, q are

[OJEE]

(a) $-5,-5$

(b) $-5,5$

(c) $5,-5$

(d) None of these

Q 14. If the tangent at point P on the circle $x^2 + y^2 + 6x + 6y - 2 = 0$ meets the straight line $5x - 2y + 6 = 0$ at a point Q on the y–axis, then length PQ is

[BCECE]

(a) 4

(b) $2\sqrt{5}$

(c) 5

(d) $3\sqrt{5}$

Q 15. The line $ax + by + c = 0$ is normal to the circle $x^2 + y^2 + 2gx + 2fy + d = 0$, if

[MPPET]

(a) $ag + bf + c = 0$

(b) $ag + bf - c = 0$

(c) $ag - bf + c = 0$

(d) $ag - bf - c = 0$

Q 16. The equation of circle which touches the x−axis and y−axis at the points $(1, 0)$ and $(0, 1)$ respectively, is

[RPET]

(a) $x^2 + y^2 - 4y + 3 = 0$

(b) $x^2 + y^2 - 2y - 2 = 0$

(c) $x^2 + y^2 - 2x - 2y + 2 = 0$

(d) $x^2 + y^2 - 2x - 2y + 1 = 0$

Q 17. If the equation $\lambda x^2 + (2\lambda - 3)y^2 - 4x - 1 = 0$ represents a circle, then its radius is

[J&K CET]

(a) $\frac{\sqrt{11}}{3}$

(b) $\frac{\sqrt{13}}{3}$

(c) $\frac{\sqrt{7}}{3}$

(d) $\frac{1}{3}$

Q 18. The point diametrically opposite to the point P$(1, 0)$ on the circle $x^2 + y^2 + 2x + 4y - 3 = 0$ is

[AIEEE]

(a) $(3,4)$

(b) $(3,-4)$

(c) $(-3,4)$

(d) $(-3,-4)$

Q 19. The radius of the circle with the polar equation $r^2 - 8r(\sqrt{3}\cos\theta + \sin\theta) + 15 = 0$ is

[BITSAT]

(a) 8

(b) 7

(c) 6

(d) 5

Q 20. If the lines $2x - 3y = 5$ and $3x - 4y = 7$ are two diameters of a circle of radius 7, then the equation of the circle is

[BITSAT]

(a) $x^2 + y^2 + 2x - 4y - 47 = 0$

(b) $x^2 + y^2 = 49$

(c) $x^2 + y^2 - 2x + 2y - 47 = 0$

(d) $x^2 + y^2 = 17$

Q 21. The equation of the smallest circle passing through the points $(2, 2)$ and $(3, 3)$ is

[KCET]

(a) $x^2 + y^2 + 5x + 5y + 12 = 0$

(b) $x^2 + y^2 - 5x - 5y + 12 = 0$

(c) $x^2 + y^2 + 5x - 5y + 12 = 0$

(d) $x^2 + y^2 - 5x + 5y - 12 = 0$

Q 22. The point $(5, -7)$ lies outside the circle

[KCET]

(a) $x^2 + y^2 - 8x = 0$

(b) $x^2 + y^2 - 5x + 7y = 0$

(c) $x^2 + y^2 - 5x + 7y - 1 = 0$

(d) $x^2 + y^2 - 8x + 7y - 2 = 0$

Q 23. If $(3, -2)$ is the centre of a circle and $4x + 3y + 19 = 0$ is a tangent to the circle, then the equation of the circle is

[Kerala CEE]

(a) $x^2 + y^2 - 6x + 4y + 25 = 0$

(b) $x^2 + y^2 - 6x + 4y + 12 = 0$

(c) $x^2 + y^2 - 6x + 4y - 12 = 0$

(d) $x^2 + y^2 - 6x + 4y + 13 = 0$

(e) $x^2 + y^2 - 6x + 4y + 9 = 0$

Q 24. The image of the centre of the circle $x^2 + y^2 = a^2$ with respect to the mirror $x + y = 1$ is

[Kerala CEE]

(a) $\left(\frac{1}{\sqrt{2}}, \sqrt{2}\right)$

(b) $\left(\sqrt{2}, \sqrt{2}\right)$

(c) $\left(\sqrt{2}, 2\sqrt{2}\right)$

(d) $\left(-\sqrt{2}, 2\right)$

(e) None of these

Q 25. The equation $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$ represents a circle whose centre is

[WB JEE]

(a) $\left(\frac{x_1 - x_2}{2}, \frac{y_1 - y_2}{2}\right)$

(b) $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

(c) (x_1, y_1)

(d) (x_2, y_2)

Q 26. The equation of the circle having $x - y - 2 = 0$ and $x - y + 2 = 0$ as two tangents and $x - y = 0$ as a diameter is

[Jamia Millia Islamia]

(a) $x^2 + y^2 + 2x - 2y + 1 = 0$

(b) $x^2 + y^2 - 2x + 2y - 1 = 0$

(c) $x^2 + y^2 = 2$

(d) $x^2 + y^2 = 1$

Q 27. If one end of the diameter is $(1, 1)$ and the other end lies on the line $x + y = 3$, then locus of centre of circle is

[MHT CET]

(a) $x + y = 1$

(b) $2(x - y) = 5$

(c) $2x + 2y = 5$

(d) None of these

Q 28. The circle $x + y - 4x - 4y + 4 = 0$ is inscribed in a triangle which has two of its sides along the coordinate axes. If the locus of the circumcentre of the triangle is $x + y - xy + k\sqrt{x^2 + y^2} = 0$, then the value of k is equal to

[Manipal]

(a) 2

(b) 1

(c) -2

(d) 3

Q 29. If the line $y = 7x - 25$ meets the circle $x^2 + y^2 = 25$ in the points A, B, then the distance between A and B is

[J&K CET]

(a) $\sqrt{10}$

(b) 10

(c) $5\sqrt{2}$

(d) 5

Q 30. Centre of circle whose normals are $x^2 - 2xy - 3x + 6y = 0$, is

[BITSAT]

(a) $\left(3, \frac{3}{2}\right)$

(b) $\left(3, -\frac{3}{2}\right)$

(c) $\left(\frac{3}{2}, 3\right)$

(d) None of these

Q 31. The equation of two circles which touch the y−axis at $(0, 3)$ and make an intercept of 8 unit on x−axis, are

[KCET]

(a) $x^2 + y^2 \pm 10x - 6y + 9 = 0$

(b) $x^2 + y^2 \pm 6x - 10y + 9 = 0$

(c) $x^2 + y^2 - 8x \pm 10y + 9 = 0$

(d) $x^2 + y^2 + 10x \pm 6y + 9 = 0$

Q 32. Two diameters of the circle $3x^2 + 3y^2 - 6x - 18y - 7 = 0$ are along the lines $3x + y = c_1$ and $x - 3y = c_2$. Then, the value of c_1c_2 is

[Kerala CEE]

(a) -48

(b) 80

(c) −72

(d) 54

(e) 24

Q 33. The equation of the circle of radius 3 that lies in the fourth quadrant and touching the lines x = 0 and y = 0 is

[EAMCET]

(a) $x^2 + y^2 - 6x + 6y + 9 = 0$

(b) $x^2 + y^2 - 6x - 6y + 9 = 0$

(c) $x^2 + y^2 + 6x - 6y + 9 = 0$

(d) $x^2 + y^2 + 6x + 6y + 9 = 0$

Q 34. One of the diameter of the circle $x^2 + y^2 - 12x + 4y + 6 = 0$ is given by

[BCECE]

(a) $x + y = 0$

(b) $x+3y = 0$

(c) $x = y$

(d) $3x + 2y = 0$

Q 35. The intercept on the line $y = x$ by the circle $x + y - 2x = 0$ is AB. Equation of the circle on AB as a diameter is

[Jamia Millia Islamia]

(a) $x^2 + y^2 - x - y = 0$

(b) $x^2 + y^2 - x + y = 0$

(c) $x^2 + y^2 + x + y = 0$

(d) $x^2 + y^2 + x - y = 0$

Q 36. The area of the circle centred at $(1, 2)$ and passing through $(4, 6)$, is

[MHT CET]

(a) 5π sq units

(b) 10π sq units

(c) 25π sq units

(d) None of these

Q 37. If the points $(2, 0)$, $(0, 1)$, $(4, 5)$ and $(0, c)$ are concyclic, then the value of c is **[MHT CET]**

(a) 1

(b) $\frac{14}{3}$

(c) 5

(d) None of these

Q 38. A variable circle passes through the fixed point A(p, q) and touches x−axis. The locus of the other end of the diameter through A is

[MPPET]

(a) $(x - p)^2 = 4qy$

(b) $(x - q)^2 = 4py$

(c) $(y - p)^2 = 4qx$

(d) $(y - q)^2 = 4px$

Q 39. The other end of the diameter through the point $(-1, 1)$ on the circle $x^2 + y^2 - 6x + 4y - 12 = 0$ is

[J&K CET]

(a) $(-7, 5)$

(b) $(-7, -5)$

(c) $(7, -5)$

(d) $(7, 5)$

Q 40. If the lines $3x - 4y - 7 = 0$ and $2x - 3y - 5 = 0$ are two diameters of a circle of area 49π sq unit, the equation of the circle is

[AIEEE]

(a) $x^2 + y^2 + 2x - 2y - 62 = 0$

(b) $x^2 + y^2 - 2x + 2y - 62 = 0$

(c) $x^2 + y^2 - 2x + 2y - 41 = 0$

(d) $x^2 + y^2 + 2x - 2y - 47 = 0$

Q 41. The equation of the circle concentric to the circle $2x^2 + 2y^2 - 3x + 6y + 2 = 0$ and having area double the area of this circle, is

[DCE]

(a) $8x^2 + 8y^2 - 24x + 48y - 13 = 0$

(b) $16x^2 + 16y^2 + 24x - 48y - 13 = 0$

(c) $16x^2 + 16y^2 - 24x + 48y - 13 = 0$

(d) $8x^2 + 8y^2 + 24x - 48y - 13 = 0$

Q 42. The equation of the circle with centre

$(2,1)$ and touching the line $3x + 4y = 5$ is

[BITSAT]

(a) $x^2 + y^2 - 4x - 2y + 5 = 0$

(b) $x^2 + y^2 - 4x - 2y - 5 = 0$

(c) $x^2 + y^2 - 4x - 2y + 4 = 0$

(d) $x^2 + y^2 - 4x - 2y - 4 = 0$

Q 43. The equation of the circumcircle of the triangle formed by the lines $x = 0$, $y = 0$, $2x + 3y = 5$ is

[UP SEE]

(a) $6(x^2 + y^2) + 5(3x - 2y) = 0$

(b) $x^2 + y^2 - 2x - 3y + 5 = 0$

(c) $x^2 + y^2 + 2x - 3y - 5 = 0$

(d) $6(x^2 + y^2) - 5(3x + 2y) = 0$

Q 44. For an equilateral triangle the centre is the origin and the length of altitude is a. Then, the equation of the circumcircle is

[Kerala CEE]

(a) $x^2 + y^2 = a^2$

(b) $3x^2 + 3y^2 = 2a^2$

(c) $x^2 + y^2 = 4a^2$

(d) $3x^2 + 3y^2 = a^2$

(e) $9x^2 + 9y^2 = 4a^2$

Q 45. Suppose a circle passes through $(2, 2)$ and $(9, 9)$ and touches the x−axis at P. If O is the origin, then OP is equal to

[Kerala CEE]

(a) 4

(b) 5

(c) 6

(d) 9

(e) 11

Q 46. The circle $x^2 + y^2 - 8x + 4y + 4 = 0$

touches

[AMU]

(a) x−axis

(b) y−axis

(c) both axes

(d) neither x−axis nor y −axis

Q 47. Observe the following statements

[EAMCET]

I. The circle $x^2 + y - 6x - 4y - 7 = 0$ touches y−axis.

II. The circle $x^2 + y + 6x + 4y - 7 = 0$ touches x−axis.

Which of the following is a correct statement ?

(a) Both I and II are true

(b) Neither I nor II is true

(c) I is true, II is false

(d) I is false, II is true

Q 48. The Polar equation of the circle with centre $\left(2, \frac{\pi}{2}\right)$ and radius 3 units is

[EAMCET]

(a) $r^2 - 4r \cos\theta = 5$

(b) $r^2 + 4r \sin\theta = 5$

(c) $r^2 - 4r \sin \theta = 5$

(d) $r^2 - 4r \cos \theta = 5$

Q 49. The greatest distance of the point P$(10,$

$7)$ from the circle $x^2 + y^2 - 4x - 2y - 20 = 0$ is

[OJEE]

(a) 10

(b) 15

(c) 5

(d) None of these

Q 50. The equation of the circle passing through $(4, 5)$ and having the centre $(2,2)$, is

[WB JEE]

(a) $x^2 + y^2 + 4x + 4y - 5 = 0$

(b) $x^2 + y^2 - 4x - 4y - 5 = 0$

(c) $x^2 + y^2 - 4x = 13$

(d) $x^2 + y^2 - 4x - 4y + 5 = 0$

Q 51. The centre of the circle $x = 2 + 3\cos\theta$, $y = 3\sin\theta - 1$ is

[Guj.CET]

(a) $(3,3)$

(b) $(2,-1)$

(c) $(-2,1)$

(d) $(1-2)$

Q 52. A point P moves in such a way that the ratio of its distance from two coplanar points is always a fixed number $(\neq 1)$. Then, its locus is a

[J&K CET]

(a) parabola

(b) circle

(c) hyperbola

(d) pair of straight lines

Q 53. If $2x - 4y = 9$ and $6x - 12y + 1 = 0$ are common tangents to the circle, then radius of the circle is

[DCE]

(a) $\frac{\sqrt{3}}{5}$

(b) $\frac{17}{6\sqrt{5}}$

(c) $\frac{\sqrt{2}}{3}$

(d) $\frac{17}{3\sqrt{5}}$

Q 54. Let P(x_1, y_1) and Q(x_2, y_2) are two points such that their abscissa x_1 and x_2 are the roots of the equation $x + 2x - 3 = 0$ while the ordinates y_1 and y_2 are the roots of the equation $y^2 + 4y - 12 = 0$. The centre of the circle with PQ as diameter is

[BITSAT, OJEE]

(a) $(-1,-2)$

(b) $(1,2)$

(c) $(1,-2)$

(d) $(-1,2)$

Q 55. The radius of the circle $x^2 + y^2 + 4x + 6y + 13 = 0$ is

[KCET]

(a) $\sqrt{26}$

(b) $\sqrt{13}$

(c) $\sqrt{23}$

(d) 0

Q 56. The locus of the centre of the circle for which one end of a diameter is $(1, 1)$ while the other end is on the line $x + y = 3$, is

[AMU]

(a) $x + y = 1$

(b) $2(x - y) = 5$

(c) $2x + 2y = 5$

(d) None of these

Q 57. If $x - y + 1 = 0$ meets the circle $x^2 + y^2 + y - 1 = 0$ at A and B, then the equation of the circle with AB as diameter is

[EAMCET]

(a) $2(x^2 + y^2) + 3x - y + 1 = 0$

(b) $2(x^2 + y^2) + 3x - y + 2 = 0$

(c) $2(x^2 + y^2) + 3x - y + 3 = 0$

(d) $x^2 + y^2 + 3x - y + 1 = 0$

Q 58. Which of the following equations gives circle ?

[EAMCET]

(a) $r = 2 \sin \theta$

(b) $r^2 \cos 2\theta = 1$

(c) $r(4 \cos \theta + 5 \sin \theta) = 3$

(d) $5 = r(1 + \sqrt{2} \cos \theta)$

Q 59. The circle $x^2 + y^2 + 4x - 4y + 4 = 0$ touches

[BCECE]

(a) x−axis

(b) y−axis

(c) x−axis and y−axis

(d) None of these

Q 60. Equation of a circle passing through the origin and making intercept by the line $4x + 3y = 12$ with coordinate axes, is

[MHT CET]

(a) $x^2 + y^2 + 3x + 4y = 0$

(b) $x^2 + y^2 + 3x - 4y = 0$

(c) $x^2 + y^2 - 3x + 4y = 0$

(d) $x^2 + y^2 - 3x - 4y = 0$

Q 61. The radius of any circle touching the lines $3x - 4y + 5 = 0$ and $6x - 8y - 9 = 0$ is

[MP PET, RPET]

(a) 1.9

(b) 0.95

(c) 2.9

(d) 1.45

Q 62. If the lines $2x + 3y + 1 = 0$ and $3x - y - 4 = 0$ lie along diameters of a circle of circumference 10π, then the equation of the circle is

[AIEEE]

(a) $x^2 + y^2 - 2x + 2y - 23 = 0$

(b) $x^2 + y^2 - 2x - 2y - 23 = 0$

(c) $x^2 + y^2 + 2x + 2y - 23 = 0$

(d) $x^2 + y^2 + 2x - 2y - 23 = 0$

Q 63. The equation of the circle touching $x = 0$, $y = 0$ and $x = 4$ is

[UP SEE]

(a) $x^2 + y^2 - 4x - 4y + 16 = 0$

(b) $x^2 + y^2 - 8x - 8y + 16 = 0$

(c) $x^2 + y^2 + 4x + 4y - 4 = 0$

(d) $x^2 + y^2 - 4x - 4y + 4 = 0$

Q 64. The radius of the circle passing through the point P$(6, 2)$ and two of whose diameter are $x + y = 6$ and $x + 2y = 4$, is

[KCET]

(a) 4

(b) 6

(c) 20

(d) $\sqrt{20}$

Q 65. Let AB be the intercept of the line y = x by the circle $x^2 + y^2 - 2x = 0$. Then, the equation of the circle with AB as its diameter is

[Kerala CEE]

(a) $x^2 + y^2 - x - y = 0$

(b) $x^2 + y^2 + x + y = 0$

(c) $x^2 + y^2 + 2(x - y) = 0$

(d) $x^2 + y^2 - 2x + y = 0$

(e) $x^2 + y^2 + 2x - y = 0$

Q 66. If P_1, P_2, P_3 are the perimeters of the three circles $x^2 + y^2 + 8x - 6y = 0$, $4x^2 + 4y^2 - 4x - 12y - 186 = 0$ and $x^2 + y^2 - 6x + 6y - 9 = 0$ respectively, then

[EAMCET]

(a) $P_1 < P_2 < P_3$

(b) $P_1 < P_3 < P_2$

(c) $P_3 < P_2 < P_1$

(d) $P_2 < P_3 < P_1$

Q 67. The centre of the circle $r^2 - 4r(\cos\theta + \sin\theta) - 4 = 0$ in cartesian coordinates is

[EAMCET]

(a) $(1,1)$

(b) $(-1,-1)$

(c) $(2, 2)$

(d) $(-2,-2)$

Q 68. The radius of the circle r = $\sqrt{3}\sin\theta + \cos\theta$ is

[EAMCET]

(a) 1

(b) 2

(c) 3

(d) 4

Q 69. The area of the circle whose centre is at $(2, 3)$ and passing through $(4,6)$, is

[BCECE]

(a) 5π sq units

(b) $10\ \pi$ sq units

(c) $13\ \pi$ sq units

(d) None of these

Q 70. A circle of radius 5 touches another circle $x^2 + y^2 - 2x - 4y - 20 = 0$ at $(5, 5)$, then its equation is

[Jamia Millia Islamia]

(a) $x^2 + y^2 + 18x + 16y + 120 = 0$

(b) $x^2 + y^2 - 18x - 16y + 120 = 0$

(c) $x^2 + y^2 - 18x + 16y + 120 = 0$

(d) None of the above

Q 71. The equation of circle is $x^2 + y^2 - 2x = 0$.

The point $P(-1, 0)$ lies

[MHT CET]

(a) on the circle

(b) inside the circle

(c) outside the circle

(d) on the centre of the circle

Q 72. Equation of the circle which of the mirror image of the circle $x^2 + y^2 - 2x = 0$ in the line $x + y = 2$ is

[J&KCET]

(a) $x^2 + y^2 - 2x + 4y + 3 = 0$

(b) $2(x^2 + y^2) + x + y + 1 = 0$

(c) $x^2 + y^2 - 4x - 2y + 4 = 0$

(d) None of the above

Q 73. The equation of the circle of radius 5 in the first quadrant which touches x−axis and the line $4y = -3x$ is

[J&K CET]

(a) $x^2 + y^2 - 24x - y - 25 = 0$

(b) $x^2 + y^2 - 30x - 10y + 225 = 0$

(c) $x^2+y^2 - 16x - 18y + 64 = 0$

(d) $x^2 + y^2 - 20x - 12y + 144 = 0$

Q 74. The equation of a circle with origin as a centre and passing through an equilateral triangle whose median is of length $3a$, is

[J&K CET]

(a) $x^2 + y^2 = 9a^2$

(b) $x^2 + y^2 = 16a^2$

(c) $x^2 + y^2 = 4a^2$

(d) $x^2 + y^2 = a^2$

Q 75. If the area of the circle $4x^2 + 4y^2 - 8x + 16y + k = 0$ is 9π sq unit, then the value of k is

[Kerala CEE]

(a) 4 units

(b) 16 units

(c) -16 units

(d) $+16$ units

(e) None of these

Q 76. If $5x - 12y + 10 = 0$ and $12y - 5x + 16 = 0$ are two tangents to a circle, then the radius of the circle is

[EAMCET]

(a) 1

(b) 2

(c) 4

(d) 6

Q 77. If $g^2 + f^2 = c$, then the equation $x^2 + y^2 + 2gx + 2fy + c = 0$ will represent

[MP PET]

(a) a circle of radius g

(b) a circle of radius f

(c) a circle of diameter $\sqrt{c}$

(d) a circle of radius zero

Q 78. ABCD is a square whose side is a. If AB and AD are axes of coordinates, the equation of the circle circumscribing the square will be

[MP PET]

(a) $x^2 + y^2 = a^2$

(b) $x^2 + y^2 = a(x + y)$

(c) $x^2 + y^2 = 2a(x + y)$

(d) $x^2 + y^2 = \frac{a^2}{4}$

Q 79. The locus of the centre of a circle of radius 2 which rolls on the outside of the circle, is $x^2+y^2 + 3x - 6y - 9 = 0$ is

[RPET]

(a) $x^2 + y^2 + 3x - 6y + 5 = 0$

(b) $x^2 +y^2 +3x - 6y - 31 = 0$

(c) $x^2 + y^2 +3x - 6y+\frac{29}{4} = 0$

(d) None of the above

Topic 2 Tangent and Normal to a Circle

Q 1. The equation of normal of $x^2 + y^2 - 2x + 4y - 5 = 0$ at $(2, 1)$ is

[WB JEE]

(a) $y = 3x - 5$

(b) $2y = 3x - 4$

(c) $y = 3x + 4$

(d) $y = x + 1$

Q 2. The equation of the tangent from the point $(0, 1)$ to the circle $x^2 + y^2 - 2x - 6y + 6 = 0$, is

[VITEEE]

(a) $y - 1 = 0$

(b) $4x + 3y + 3 = 0$

(c) $4x - 3y - 3 = 0$

(d) $y + 1 = 0$

Q 3. Locus of the point of intersection of perpendicular tangents to the circle $x^2 + y^2 = 16$ is

[MHT CET]

(a) $x^2 + y^2 = 8$

(b) $x^2 + y^2 = 32$

(c) $x^2 + y^2 = 64$

(d) $x^2 + y^2 = 16$

Q 4. The angle between the tangents drawn at the points $(5, 12)$ and $(12, -5)$ to the circle $x^2 + y^2 = 169$ is

[MPPET]

(a) $45°$

(b) $60°$

(c) $30°$

(d) $90°$

Q 5. If the line $y \cos \alpha = x \sin \alpha + a \cos \alpha$ be a tangent to the circle $x^2 + y^2 = a^2$, then

[DCE]

(a) $\sin^2 \alpha = 1$

(b) $\cos^2 \alpha = 1$

(c) $\sin^2 \alpha = a^2$

(d) $\cos^2 \alpha = a^2$

Q 6. If θ is the angle between the tangents from $(-1,0)$ to the circle $x^2 + y^2 - 5x + 4y - 2 = 0$, then θ is equal to

[EAMCET]

(a) $2\tan^{-1}\left(\frac{7}{4}\right)$

(b) $\tan^{-1}\left(\frac{7}{4}\right)$

(c) $2\cot^{-1}\left(\frac{7}{4}\right)$

(d) $\cot^{-1}\left(\frac{7}{4}\right)$

Q 7. The equations of the tangents to the circle $x^2 + y^2 = 13$ at the points whose abscissa is 2, are

[MHT CET]

(a) $2x + 3y = 13, 2x - 3y = 13$

(b) $3x + 2y = 13, 2x - 3y = 13$

(c) $2x + 3y = 13, 3x - 2y = 13$

(d) None of the above

Q 8. Tangents are drawn from the point $(17, 7)$ to the circle $x^2+y^2 = 169$.

Statement I The tangents are mutually perpendicular.

Statement II The locus of the points from which mutually perpendicular tangents can be drawn to the given circle is $x^2 + y^2 = 338$

[IITJEE]

(a) Statement I is true, Statement II is true;

Statement II is a correct explanation for Statement I

(b) Statement I is true, Statement II is true;

Statement II is not a correct explanation for

StatementI

(c) Statement I is true, Statement II is false

(d) Statement I is false, Statement II is true

Q 9. From the point P(16, 7) tangents PQ and PR

are drawn to the circle $x^2 + y^2 - 2x - 4y - 20 = 0$. If

C be the centre of the circle, then area of

quadrilateral PQCR is

[UP SEE]

(a) 450 sq units

(b) 15 sq units

(c) 50 sq units

(d) 75 sq units

Q 10. If $3x + y + k = 0$ is a tangent to the circle

$x^2 + y^2 = 10$, the values of k are

[KCET]

(a) ± 7

(b) ± 5

(c) ± 10

(d) ± 9

Q 11. Length of the tangents from the point $(1,2)$ to the circles $x^2 + y^2 + x + y - 4 = 0$ and $3x^2 + 3y^2 - x - y - k = 0$ are in the ratio $4:3$, then k is equal to

[Kerala CEE]

(a) $37/2$

(b) $4/37$

(c) 12

(d) 7

(e) $39/4$

Q 12. The condition for a line $y = 2x + c$ to touch the circle $x^2 + y^2 = 16$ is

[BITSAT, UP SEE]

(a) $c = 10$

(b) $c^2 = 80$

(c) $c = 12$

(d) $c^2 = 64$

Q 13. The equation of the common tangent of the two touching circles, $y^2 + x^2 - 6x - 12y + 37 = 0$ and $x^2 + y^2 - 6y + 7 = 0$ is

[KCET]

(a) $x + y - 5 = 0$

(b) $x - y + 5 = 0$

(c) $x - y - 5 = 0$

(d) $x + y + 5 = 0$

Q 14. The locus of the point (l, m) so that lx + my = 1 touches the circle $x^2 + y^2 = a^2$, is

[Kerala CEE]

(a) $x^2 + y^2 - ax = 0$

(b) $x^2 + y^2 = \frac{1}{a^2}$

(c) $y^2 = 4ax$

(d) $x^2 + y^2 - ax - ay + a^2 = 0$

(e) $x^2 - y^2 = a^2$

Q 15. The values of λ so that the line $3x - 4y = \lambda$ touches $x^2 + y^2 - 4x - 8y - 5 = 0$ are

[Kerala CEE]

(a) $-35, 15$

(b) $3, -5$

(c) $35, -15$

(d) $-3, 5$

(e) $20, 15$

Q 16. The length of the tangent drawn to the circle $x^2 + y^2 - 2x + 4y - 11 = 0$ from the point $(1,3)$ is

[EAMCET]

(a) 1

(b) 2

(c) 3

(d) 4

Q 17. The equation of the tangent to the circle $x^2 + y^2 + 4x - 4y + 4 = 0$ which makes equal intercepts on the positive coordinate axes, is

[OJEE]

(a) $x + y = 2$

(b) $x + y = 2\sqrt{2}$

(c) $x + y = 4$

(d) $x + y = 8$

Q 18. The equation of tangent drawn from the origin to the circle $x^2 + y^2 - 2rx - 2hy + h^2 = 0$ are

[WB JEE, Jamia Millia Islamia]

(a) $x = 0, y = 0$

(b) $x = 1, y = 0$

(c) $(h^2 - r^2)x - 2rhy = 0. y = 0$

(d) $(h - r^2)x - 2rhy = 0, x=0$

Q 19. If the equation of tangent to the circle $x^2 + y^2 - 2x + 6y - 6 = 0$ parallel to $3x - 4y + 7 = 0$ is

$3x - 4y + k = 0$, then the value of k are

[MHT CET]

(a) $5, -35$

(b) $-5, 35$

(c) $7, -32$

(d) $-7, 32$

Q 20. If $\frac{x}{\alpha}+\frac{y}{\beta}=1$ touches the circle $x^2 + y^2 = a^2$,

then point $(1/\alpha, 1/\beta)$ lies on a /an

[BITSAT, OJEE]

(a) straight line

(b) circle

(c) parabola

(d) ellipse

Q 21. If y = 3x is a tangent to a circle with centre $(1, 1)$, then the other tangent drawn through $(0, 0)$ to the circle is

[EAMCET]

(a) $3y = x$

(b) $y = -3x$

(c) $y = 2x$

(d) $y = -2x$

Q 22. The locus of a point which moves so that the ratio of the length of the tangents to the circles $x^2 + y^2 + 4x + 3 = 0$ and $x^2 + y^2 - 6x + 5 = 0$ is $2 : 3$, is

[Kerala CEE]

(a) $5x^2 + 5y^2 - 60x + 7 = 0$

(b) $5x^2 + 5y^2 + 60x - 7 = 0$

(c) $5x^2 + 5y^2 - 60x - 7 = 0$

(d) $5x^2 + 5y^2 + 60x + 7 = 0$

(e) $5x^2 + 5y^2 + 60x + 12 = 0$

Q 23. Two tangents to the circle $x^2 + y^2 = 4$ at the points A and B meet at $P(-4, 0)$. The area of the quadrilateral PAOB, where O is the origin, is

[Jamia Millia Islamia]

(a) 4 sq units

(b) $6\sqrt{2}$ sq units

(c) $4\sqrt{3}$ sq units

(d) None of these

Q 24. Angle between tangent drawn to circle x + y = 20, from the point $(6,2)$ is

[MHT CET]

(a) $\frac{\pi}{2}$

(b) π

(c) $\frac{\pi}{4}$

(d) 2π

Q 25. The equations of the tangent to circle $5x$ $+5y = 1$. parallel to line $3x + 4y = 1$ are

[J&K CET]

(a) $3x + 4y = \pm 2\sqrt{5}$

(b) $6x + 8y = \pm \sqrt{5}$

(c) $3x + 4y = \pm\sqrt{5}$

(d) None of these

Q 26. The equations of the tangents to the circle $x^2 + y^2 - 6x + 4y - 12 = 0$ which are parallel to line $4x + 3y + 5 = 0$ are

[Kerala CEE]

(a) $4x + 3y + 11 = 0$ and $4x + 3y + 8 = 0$

(b) $4x + 3y - 9 = 0$ and $4x + 3y + 7 = 0$

(c) $4x + 3y + 19 = 0$ and $4x + 3y - 31 = 0$

(d) $4x + 3y - 10 = 0$ and $4x + 3y + 12 = 0$

(e) $4x + 3y + 3 = 0$ and $4x + 3y - 1 = 0$

Q 27. The angle between the tangent drawn from the origin to the circle $(x - 7)^2 + (y + 1)^2 = 25$ is

[BCECE]

(a) $\frac{\pi}{3}$

(b) $\frac{\pi}{6}$

(c) $\frac{\pi}{2}$

(d) $\frac{\pi}{8}$

Q 28. The area of the triangle formed by the tangent at $(3,4)$ to the circle $x^2 + y^2 = 25$ and the coordinate axes is

[Guj. CET]

(a) $\frac{24}{25}$

(b) 0

(c) $\frac{625}{24}$

(d) $-\left(\frac{24}{25}\right)$

Q 29. The slope of the tangent at the point (h, h) to the circle $x^2 + y^2 = a^2$ is

[DCE]

(a) 0

(b) 1

(c) −1

(d) will depend on h

Q 30. The line $3x - 2y = k$ meets the circle $x^2 + y^2 = 4r^2$ at only one point if k^2 is equal to

[KCET]

(a) $52r^2$

(b) $20r^2$

(c) $\frac{20}{9}r^2$

(d) $\frac{52}{9}r^2$

Q 31. If P is a point such that the ratio of the square of the lengths of the tangents from P to the circles $x^2+y^2 + 2x - 4y - 20 = 0$ and $x^2 + y^2 - 4x + 2y - 44 = 0$ is $2 : 3$, then the locus of P is a circle with centre

[EAMCET]

(a) $(7,-8)$

(b) $(-7,8)$

(c) $(7, 8)$

(d) $(-7,-8)$

Q 32. The equation of the tangent to the circle $x^2 + y^2 = 4$, which are parallel to $x + 2y + 3 = 0$, are

[MP PET]

(a) $x - 2y = 2$

(b) $x+2y = \pm 2\sqrt{3}$

(c) $x + 2y = \pm 2\sqrt{5}$

(d) $x - 2y = \pm 2\sqrt{5}$

Topic 3

Chord of Contact of Tangent, Pole and Polar

Q 1. The equation of the chord of the circle $x + y = 81$, which is bisected at the point $(-2, 3)$, is

[Kerala CEE]

(a) $3x - y = 13$

(b) $3x - 4y = 13$

(c) $2x - 3y = 13$

(d) $3x - 3y = 13$

(e) $2x - 3y = -13$

Q 2. The radius of the circle, which is touched by the line y =x and has its centre on the positive direction of x−axis and also cuts−off a chord of length 2 unit along the line $\sqrt{3}y - x = 0$, is

[DCE]

(a) $\sqrt{5}$

(b) $\sqrt{3}$

(c) $\sqrt{2}$

(d) 1

Q 3. The locus of the mid points of the chords of the circle $x^2 + y^2 = 4$ which subtend a right angle at the origin is

[KCET]

(a) $x^2 + y^2 = 1$

(b) $x^2 + y^2 = 2$

(c) $x + y = 1$

(d) $x + y = 2$

Q 4. The length of the chord joining the points $(4 \cos \theta, 4 \sin \theta)$ and $(4 \cos (\theta + 60°), 4 \sin (\theta + 60°))$ of the circle $x^2 + y^2 = 16$ is

[KCET]

(a) 4

(b) 8

(c) 16

(d) 2

Q 5. If two chords having lengths $a^2 - 1$ and $3(a + 1)$, where a is a constant of a circle bisect each other, then the radius of the circle is

[Kerala CEE]

(a) 6

(b) $\frac{15}{2}$

(c) 8

(d) $\frac{19}{2}$

(e) 10

Q 6. The inverse of the point $(1,2)$ with respect to the circle $x^2 + y^2 - 4x - 6y + 9 = 0$, is

[EAMCET]

(a) $\left(1, \frac{1}{2}\right)$

(b) $(2, 1)$

(c) $(0, 1)$

(d) $(1,0)$

Q 7. The length of the common chord of the circles $x^2 + y^2 + 2x + 3y + 1 = 0$ and $x^2 + y^2 + 4x + 3y + 2 = 0$ is

[DCE]

(a) $\frac{9}{2}$

(b) $2\sqrt{2}$

(c) $3\sqrt{2}$

(d) $\frac{3}{2}$

Q 8. If the circles $x^2 + y^2 - 2x - 2y - 7 = 0$ and $x^2 + y^2 + 4x + 2y + k = 0$ cut orthogonally, then the length of the common chord of the circle is

[KCET]

(a) $\frac{12}{\sqrt{13}}$

(b) 2

(c) 5

(d) 8

Q 9. If the circle $x^2 + y^2 + 4x + 22y + c = 0$ bisects the circumference of the circle $x^2 + y^2 - 2x + 8y - d = 0$, then c + d is equal to

[Kerala CEE]

(a) 30

(b) 50

(c) 40

(d) 56

(e) 52

Q 10. Which of the following is a point on the common chord of the circle $x + y^2 + 2x - 3y + 6 = 0$ and $x^2 + y^2 + x - 8y - 13 = 0$?

[AMU]

(a) $(1,-2)$

(b) $(1,4)$

(c) $(1, 2)$

(d) $(1,-4)$

Q 11. A variable chord is drawn through the origin to the circle $x^2 + y^2 - 2ax = 0$. The locus of the centre of the circle drawn on this chord as diameter is

[BCECE]

(a) $x^2 + y^2 + ax = 0$

(b) $x^2 + y^2 - ax = 0$

(c) $x^2 + y^2 + ay = 0$

(d) $x^2 + y^2 - ay = 0$

Q 12. The locus of the mid point of the chord of the circle $x^2 + y^2 - 2x - 2y - 2 = 0$ which makes an angle of $120°$ at the centre, is

[MP PET]

(a) $x^2 + y^2 - 2x - 2y - 1 = 0$

(b) $x^2 + y^2 + x + y - 1 = 0$

(c) $x^2 + y^2 - 2x - 2y + 1 = 0$

(d) None of the above

Q 13. Let C be the circle with centre $(0, 0)$ and radius 3 unit. The equation of the locus of the mid points of the chords of the circle C that subtend an angle of $\frac{2\pi}{3}$ at its centre, is

[AIEEE]

(a) $x^2 + y^2 = 1$

(b) $x^2 + y^2 = \frac{27}{4}$

(c) $x^2 + y^2 = \frac{9}{4}$

(d) $x^2 + y^2 = \frac{3}{2}$

Q 14. The equation of the circle whose diameter is the common chord of the circles $x^2 + y^2 + 2x + 3y + 2 = 0$ and $x^2 + y^2 + 2x - 3y - 4 = 0$ is

[EAMCET]

(a) $x^2 + y^2 + 2x + 2y + 2 = 0$

(b) $x^2 + y^2 + 2x + 2y - 1 = 0$

(c) $x^2 + y^2 + 2x + 2y + 1 = 0$

(d) $x^2 + y^2 + 2x + 2y + 3 = 0$

Q 15. If one of the diameters of the circle $x^2 + y^2 - 2x - 6y + 6 = 0$ is a chord to the circle with centre $(2,1)$, then the radius of the circle is

[IIT JEE]

(a) $\sqrt{3}$

(b) $\sqrt{2}$

(c) 3

(d) 2

Q 16. If $(1, a)$, $(b,2)$ are conjugate points with respect to the circle $x^2 + y^2 = 25$, then $4a + 2b$ is equal to

[EAMCET]

(a) 25

(b) 50

(c) 100

(d) 150

Q 17. If the chord of contact of tangents from a point on the circle $x^2 + y^2 = r^2$ to the circle $x^2 + y^2 = r_2^2$ touches the circle $x^2 + y^2 = r^2$, then r_1, r_2, r_3 are in

[OJEE]

(a) AP

(b) HP

(c) GP

(d) AGP

Q 18. The equation of the chord of the circle, $x^2 + y^2 = a^2$ having (x_1, y_1) as its mid point, is

[Jamia Millia Islamia]

(a) $xy_1 + yx_1 = a^2$

(b) $x_1 + y_1 = a$

(c) $xx_1 + yy_1 = x_1^2 + y_1^2$

(d) $xx_1 + yy_1 = a^2$

Q 19. If the circle $x^2 + y^2 + 6x - 2y + k = 0$ bisects the circumference of the circle $x^2 + y^2 + 2x - 6y - 15 = 0$, then k is equal to

[EAMCET]

(a) 21

(b) −21

(c) 23

(d) -23

Topic 4 System of Circles

Q 1. If the two circles $(x + 7)^2 + (y - 3)^2 = 36$ and $(x-5)^2 + (y + 2)^2 = 49$ touch each other externally, then the point of contact is

[Kerala CEE]

(a) $\left(\frac{-19}{13}, \frac{19}{13}\right)$

(b) $\left(\frac{-19}{13}, \frac{9}{13}\right)$

(c) $\left(\frac{17}{13}, \frac{9}{13}\right)$

(d) $\left(\frac{-17}{13}, \frac{9}{13}\right)$

(e) $\left(\frac{19}{13}, \frac{19}{13}\right)$

Q 2. Tangents drawn from the point P $(1,8)$ to the circle $x^2 + y^2 - 6x - 4y - 11 = 0$ touch the circle at the points A and B. The equation of the circumcircle of the triangle PAB is

[IIT JEE]

(a) $x^2 + y^2 + 4x - 6y + 19 = 0$

(b) $x^2 + y^2 - 4x - 10y + 19 = 0$

(c) $x^2 + y^2 - 2x + 6y - 29 = 0$

(d) $x^2 + y^2 - 6x - 4y + 19 = 0$

Q 3. If P and Q are the points of intersection of the circles $x^2 + y^2 + 3x + 7y + 2p - 5 = 0$ and $x^2 + y^2 + 2x + 2y - p^2 = 0$, then there is a circle passing through P, Q and $(1,1)$ and

[AIEEE]

(a) all values of p

(b) all except one value of p

(c) all except two values of p

(d) exactly one value of p

Q 4. The number of common tangents to the circles $x^2 + y^2 - y = 0$ and $x^2 + y^2 + y = 0$ is

[KCET]

(a) 2

(b) 3

(c) 0

(d) 1

Q 5. The equation of the circle which cuts orthogonally the circle $x^2 + y^2 - 6x + 4y - 3 = 0$, passes through $(3, 0)$ and touches the axis of y is

[AMU]

(a) $x^2 + y^2 + 6x - 6y + 9 = 0$

(b) $x^2 + y^2 - 6x + 6y - 9 = 0$

(c) $x^2 + y^2 - 6x - 6y + 9 = 0$

(d) None of the above

Q 6. The point $(3, -4)$ lies on both the circles

$x^2 + y^2 - 2x + 8y + 13 = 0$ and $x^2 + y^2 - 4x + 6y + 11 = 0$. Then, the angle between the circles is

[EAMCET]

(a) $60°$

(b) $\tan^{-1}\left(\frac{1}{2}\right)$

(c) $\tan^{-1}\left(\frac{3}{5}\right)$

(d) $45°$

Q 7. The equation of the circle which passes through the origin and cuts orthogonally each of the circles $x^2 + y^2 - 6x + 8 = 0$ and $x^2 + y^2 - 2x - 2y - 7 = 0$ is

[EAMCET]

(a) $3x^2 + 3y^2 - 8x - 13y = 0$

(b) $3x^2 + 3y^2 - 8x + 29y = 0$

(c) $3x^2 + 3y^2 + 8x + 29y = 0$

(d) $3x^2 + 3y^2 - 8x - 29y = 0$

Q 8. The centres of three circles $x^2 + y^2 = 1$, $x^2 + y^2 + 6x - 2y - 1$, $x^2 + y^2 - 12x + 4y = 1$ are

[OJEE]

(a) collinear

(b) non−collinear

(c) nothing to be said

(d) None of these

Q 9. For the two circles $x^2 + y^2 = 16$ and $x + y^2 - 2y = 0$ there is/are

[WBJEE]

(a) one pair of common tangents

(b) only one common tangent

(c) three common tangents

(d) no common tangent

Q 10. The equation of the circle passing through the point $(1, 1)$ and through the points of intersection of the circles $x^2 + y^2 = 6$ and $x^2 + y^2 - 6y + 8 = 0$ is

[J&K CET]

(a) $x^2 + y^2 + 3y - 13 = 0$

(b) $x^2 + y^2 - 3y + 1 = 0$

(c) $x^2 + y^2 - 3x + 1 = 0$

(d) $5x^2 + 5y^2 + 6y + 16 = 0$

Q 11. If the circles $x^2 + y^2 + 4x + 8y = 0$ and $x^2 + y^2 + 8x + 2ky = 0$ touch each oilier, then k is equal to

[Guj. CET]

(a) 12

(b) 8

(c) −8

(d) 4

Q 12. For the given circles $x^2 + y^2 - 6x - 2y + l = 0$ and $x^2 + y^2 + 2x - 8y + 13 = 0$, which of the following is true?

[DCE]

(a) One circle lies inside the other

(b) One circle lies completely outside the other

(c) Two circles intersect in two points

(d) They touch each other externally

Q 13. If the circles $x^2 + y^2 = 9$ and $x^2 + y^2 + 2\alpha x + 2y + 1 = 0$ touch each other internally, then α is equal to

[KCET]

(a) $\pm\frac{4}{3}$

(b) 1

(c) $\frac{4}{3}$

(d) $-\frac{4}{3}$

Q 14. The circles $x^2 + y^2 - 4x - 6y - 12 = 0$ and

$x^2 + y^2 + 4x + 6y + 4 = 0$

[Kerala CEE]

(a) touch externally

(b) do not intersect

(c) intersect at two points

(d) are concentric

Q 15. If the circles $x^2 + y^2 + 2ky + 2x + 6 = 0$ and $x^2 + y^2 + 2ky + k = 0$ intersects orthogonally. Then, the value of k is

[OJEE]

(a) $\frac{3}{2}$

(b) -2

(c) $-\frac{3}{2}$

(d) $\frac{1}{2}$

Q 16. The equation of the circle which passes through the points of intersection of the circles $x^2 + y^2 - 6x = 0$ and $x^2 + y^2 - 6y = 0$ and has its centre at $\left(\frac{3}{2}, \frac{3}{2}\right)$, is

[WBJEE]

(a) $x^2 + y^2 + 3x + 3y + 9 = 0$

(b) $x^2 + y^2 + 3x + 3y = 0$

(c) $x^2 + y^2 - 3x - 3y = 0$

(d) $x^2 + y^2 - 3x - 3y + 9 = 0$

Q 17. The circles $x^2 + y^2 + 6x + 6y = 0$ and $x^2 + y^2 - 12x - 12y = 0$

[WBJEE]

(a) cut orthogonally

(b) touch each other internally

(c) intersect two points

(d) touch each other externally

Q 18. The number of common tangents to the two circles $x^2 + y^2 - 8x + 2y = 0$ and $x^2 + y^2 - 2x - 16y + 25 = 0$ is

[BCECE]

(a) 1

(b) 2

(c) 3

(d) 4

Q 19. If the length of the tangent from any point on the circle $(x - 3)^2 + (y + 2)^2 = 5r^2$ to the circle $(x - 3)^2 + (y + 2)^2 = r^2$ is 16 units, then the area between the two circles in sq units is

[Jamia Millia Islamia]

(a) 32π

(b) 4π

(c) 8π

(d) 256π

Q 20. The circles $ax^2 + ay^2 + 2g_1x + 2f_1 y + c_1 = 0$ and $bx^2 + by^2 + 2g_2x + 2f_2 y + c_2 = 0$ ($a \neq 0$ and $b \neq 0$) cut orthogonally, if

[Jamia Millia Islamia]

(a) $g_1g_2 + f_1f_2 = ac_1 + bc_2$

(b) $2(g_1g_2 + f_1 f_2) = bc_1 + ac_2$

(c) $bg_1g_2 + af_1f_2 = -bc_1 + ac_2$

(d) $g_1g_2 + f_1f_2 = c_1 + c_2$

Q 21. If a circle passes through the point $(1, 2)$ and cuts the circle $x^2 + y^2 = 4$ orthogonally, then the equation of the locus of its centre is

[Manipal]

(a) $x^2 + y^2 - 3x - 8y + 1 = 0$

(b) $x^2 + y^2 - 2x - 6y - 7 = 0$

(c) $2x + 4y - 9 = 0$

(d) $2x + 4y - 1 = 0$

Q 22. The limiting points of coaxial−system determined by the circles $x^2 + y^2 + 5x + y + 4 = 0$ and $x^2 + y^2 + 10x - 4y - 1 = 0$ are

[MPPET]

(a) $(0, 3)$ and $(2, 1)$

(b) $(0,-3)$ and $(-2,-1)$

(c) $(0, 3)$ and $(1, 2)$

(d) $(0,-3)$ and $(2, 1)$

Q 23. Two circles $x^2 + y^2 - 2x - 3 = 0$ and $x^2 + y^2 - 4x - 6y - 8 = 0$ are such that

[MP PET]

(a) they touch internally

(b) they touch externally

(c) they intersect at two points

(d) they are non-intersecting

Q 24. Consider a family of circles, which are passing through the point $(-1,1)$ and are tangent to x–axis. If (h, k) are the coordinates of the centre of the circles, then the set of values of k is given by the interval

[AIEEE]

(a) $0 < k < \frac{1}{2}$

(b) $k \geq \frac{1}{2}$

(c) $-\frac{1}{2} \leq k \leq \frac{1}{2}$

(d) $k \leq \frac{1}{2}$

Q 25. The value of k so that $x^2 + y^2 + kx + 4y + 2 = 0$ and $2(x + y^2) - 4x - 3y + k = 0$ cut orthogonally, is

[DCE, WB JEE]

(a) $\frac{10}{3}$

(b) $-\frac{8}{3}$

(c) $-\frac{10}{3}$

(d) $\frac{8}{3}$

Q 26. The radical centre of the circles $x^2 + y^2 - 16x + 60 = 0$, $x^2 + y^2 - 12x + 27 = 0$ and $x^2 + y^2 - 12x + 8 = 0$ is

[UP SEE]

(a) $\left(13, \frac{33}{4}\right)$

(b) $\left(\frac{33}{4}, -13\right)$

(c) $\left(\frac{33}{4}, 13\right)$

(d) None of these

Q 27. $x^2 + y^2 - 6x - 6y + 4 = 0$, $x^2 + y^2 - 2x - 4y + 3 = 0$, $x^2 + y^2 + 2kx + 2y + 1 = 0$. If the radical centre of the above three circles exists, then which of the following cannot be the value of k

[KCET]

(a) 2

(b) 1

(c) 5

(d) 4

Q 28. The number of common tangents to the circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 6x - 8y + 24 = 0$ is

[KCET]

(a) 3

(b) 4

(c) 2

(d) 1

Q 29. If $(-3,2)$ lies on the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ which is concentric with the circle $x^2 + y^2 + 6x + 8y - 5 = 0$, then c is equal to

[AMU]

(a) 11

(b) −11

(c) 24

(d) 100

Q 30. The condition for the coaxial system $x^2 + y^2 + 2\lambda x + c = 0$, where λ is a parameter and c is a constant to have distinct limiting points, is

[EAMCET]

(a) $c = 0$

(b) $c < 0$

(c) $c = -1$

(d) $c > 0$

Q 31. C_1 is a circle of radius 2 touching the x–axis and the y–axis. C_2 is another circle of radius > 2 and touching the axes as well as the circle C_1. Then, the radius of C_2 is

[AMU]

(a) $6-4\sqrt{2}$

(b) $6+4\sqrt{2}$

(c) $6-4\sqrt{3}$

(d) $6+4\sqrt{3}$

Q 32. If the radical axis of the circles $x^2 + y^2 + 2gx + 2fy + c = 0$ and $2x^2 + 2y^2 + 3x + 8y + 2c = 0$, touches the circle $x^2 + y^2 + 2x + 2y + 1 = 0$, then

[Guj.CET]

(a) $g = \frac{3}{4}$ and $f \neq 2$

(b) $g \neq \frac{3}{4}$ and $f = 2$

(c) $g = \frac{3}{4}$ or $f = 2$

(d) None of these

Q 33. The two circles $x^2 + y^2 - 5 = 0$ and $x^2 + y^2 - 2x - 4y - 15 = 0$

[Guj.CET]

(a) touch each other externally

(b) touch each other internally

(c) cut each other orthogonally

(d) do not intersect

Q 34. The two circles $x^2 + y^2 - 2 - x + 22y + 5$ $=0$ and $x^2 + y^2 + 14x + 6y + k = 0$ intersect orthogonally provided k is equal to

[BITSAT, Guj. CET]

(a) 47

(b) −47

(c) 49

(d) −49

Q 35. The two circles $x^2 + y^2 - 2x + 6y + 6 = 0$ and $x^2 + y^2 - 5x + 6y + 15 = 0$ touch each other

[MHT CET]

(a) externally

(b) internally

(c) coincide

(d) None of these

Q 36. The circles $x^2 + y^2 - 10x + 16 = 0$ and $x^2 + y^2 = r^2$ intersect each other at two distinct points, if

[MP PET]

(a) $r < 2$

(b) $r > 8$

(c) $2 < r < 8$

(d) $2 < r < 8$

Q 37. The radical axis of the coaxial system of circles with limiting points $(1,2)$ and $(-2,1)$ is

[J&K CET]

(a) $x + 3y = 0$

(b) $3x + y = 0$

(c) $2x + 3y = 0$

(d) $3x + 2y = 0$

Q 38. If the circles $x^2 + y^2 + 2ax + cy + a = 0$ and $x^2 + y^2 - 3ax + dy - 1 = 0$ intersect in two distinct points P and Q, then the line $5x + by - a = 0$ passes through P and Q for

(a) exactly two values of a

(b) infinitely many values of a

(c) no value of a

(d) exactly one value of a

Q 39. If a circle passes through the point (a, b) and cuts the circle $x^2 + y^2 = 4$ orthogonally, then the locus of its centre is

(a) $2ax+2by + (a^2 + b^2 + 4) = 0$

(b) $2ax+2by - (a^2 + b^2 +4) = 0$

(c) $2ax - 2by + (a^2 + b^2 + 4) = 0$

(d) $2ax - 2by - (a^2 +b^2 +4)=0$

Q 40. The number of common tangents to circles $x^2 +y^2 +2x + 8y - 23 = 0$ and $x^2 + y^2 - 4x - 10y + 9 = 0$, is

[OJEE]

(a) 1

(b) 3

(c) 2

(d) None of these

Q $41.$ If two circles of the same radius r and centres at $(2, 3)$ and $(5, 6)$ respectively cut orthogonally, then the value of r is

[RPET]

(a) 3

(b) 2

(c) 1

(d) 5

Q 42. If The circles $x^2 + y^2 + 2gx + 2fy = 0$ and

$x^2 + y^2 + 2g\,x + 2f'y = 0$ touch each other, then

[Guj.CET]

(a) ff' = gg'

(b) fg = f'g'

(c) $(fg)^2 = (f'g')^2$

(d) $fg' = f'g$

Q 43. How many common tangents can be drawn to the following circles $x^2 + y^2 = 6x$ and $x^2+y^2 + 6x + 2y+ 1 = 0$?

[DCE]

(a) 4

(b) 3

(c) 2

(d) 1

Q 44. The coaxial system of circles given by $x^2 + y^2 + 2gx + c = 0$ for $c < 0$ represents

[KCET]

(a) intersecting circles

(b) non−intersecting circles

(c) touching circles

(d) touching or non−intersecting circles

Q 45. The circle $x^2 + y^2 + 8y - 4 = 0$, cuts the real circle $x^2 + y^2 + gx + 4 = 0$ orthogonally, if

[Kerala CEE]

(a) for any real value of g

(b) for no real value of g

(c) $g = 0$

(d) $g < -2, g > 2$

(e) $g > 0$

Q 46. The value of λ, for which the circle $x^2 + y^2 + 2\lambda x + 6y + l = 0$ intersects the circle $x^2 + y^2 + 4x + 2y = 0$ orthogonally, is

[RPET]

(a) $\frac{11}{8}$

(b) -1

(c) $\frac{-5}{4}$

(d) $\frac{5}{2}$

Q 47. If the two circles $(x-1)^2 + (y-3)^2 = r^2$ and $x^2 + y^2 - 8x + 2y + 8 = 0$ intersect in two distinct points, then

[AIEEE, AMU]

(a) $2 < r < 8$

(b) $r < 2$

(c) $r = 2$

(d) $r > 2$

Q 48. The limiting points of the coaxial system of circles $x^2 + y^2 + 2\lambda y + 4 = 0$ are

[KCET]

(a) $(0, \pm 4)$

(b) $(\pm 2, 0)$

(c) $(0, \pm 1)$

(d) $(0, \pm 2)$

Q 49. The number of common tangents to two circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 8x + 12 = 0$ is

[Kerala CEE]

(a) 1

(b) 2

(c) 5

(d) 4

(e) 3

Q 50. The equation of the circle, which cuts orthogonally each of three circles

[RPET]

$x^2 + y^2 - 2x + 3y - 7 = 0$, $x^2 + y^2 + 5x - 5y + 9 = 0$ and $x^2 + y^2 + 7x - 9y + 29 = 0$

(a) $x^2 + y^2 - 16x - 18y - 4 = 0$

(b) $x^2 + y^2 = a^2$

(c) $x^2 + y^2 - 16x = 0$

(d) $y^2 - x^2 + 2x = 0$

Answers

Topic 1 : Circle and Its Equations

1. (b) 2. (a) 3. (b) 4. (b) 5. (e) 6. (b) 7. (d) 8. (b)

9. (d) 10. (a) 11. (b) 12. (c) 13. (b) 14. (c) 15. (b)

16. (d) 17. (c) 18. (d) 19. (b) 20. (c) 21. (b) 22.

(a) 23. (c) 24. (e) 25. (b) 26. (c) 27. (c) 28. (b)

29. (c) 30. (a) 31. (a) 32. (a) 33. (a) 34. (b) 35.

(a) 36. (c) 37. (b) 38. (a) 39. (c) 40. (c) 41. (c)

42. (c) 43. (d) 44. (e) 45. (c) 46. (b) 47. (b) 48.

(c) 49. (b) 50. (b) 51. (b) 52. (b) 53. (b) 54. (a)

55. (d) 56. (c) 57. (a) 58. (a) 59. (c) 60. (d)

61. (b) 62. (a) 63. (d) 64. (d) 65. (a) 66. (b) 67.

(c) 68. (a) 69. (c) 70. (b) 71. (c) 72. (c) 73. (b)

74. (c) 75. (c) 76. (a) 77. (d) 78. (b) 79. (b)

Topic 2 : Tangent and Normal to a Circle

1. (a) 2. (a) 3. (b) 4. (d) 5. (b) 6. (a) 7. (a) 8. (a)

9. (d) 10. (c) 11. (e) 12. (b) 13. (c) 14. (b) 15. (a)

16. (c) 17. (b) 18. (d) 19. (a) 20. (b) 21. (a) 22.

(d) 23. (a) 24. (a) 25. (c) 26. (c) 27. (c) 28. (c)

29. (c) 30. (a) 31. (b) 32. (c)

Topic 3 : Chord of Contact of Tangent, Pole and Polar

1. (e) 2. (c) 3. (b) 4. (a) 5. (b) 6. (c) 7. (b) 8. (a)

9. (b) 10. (d) 11. (b) 12. (c) 13. (c) 14. (c) 15. (c)

16. (b) 17. (c) 18. (c) 19. (d)

Topic 4 : System of Circles

1. (b) 2. (b) 3. (b) 4. (b) 5. (c) 6. (d) 7. (b) 8. (a)

9. (d) 10. (b) 11. (b) 12. (d) 13. (a) 14. (c) 15. (c)

16. (c) 17. (d) 18. (b) 19. (d) 20. (b) 21. (c) 22.

(b) 23. (c) 24. (b) 25. (c) 26. (d) 27. (c) 28. (b)

29. (b) 30. (d) 31. (b) 32. (c) 33. (b) 34. (a) 35.

(b) 36. (c) 37. (b) 38. (c) 39. (b) 40. (c) 41. (a)

42. (d) 43. (a) 44. (a) 45. (a) 46. (c) 47. (a) 48.

(d) 49. (e) 50. (a)

Statistics

Useful Results & Formulae

1. (i) Arithmetic mean $\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$

$\overline{x} = A + \frac{\sum d}{n}$, where d = x – A

$$\overline{x} = \frac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i}$$

(ii) Weighted arithmetic mean

$$= \frac{w_1 x_1 + w_2 x_2 + ... + w_n x_n}{w_1 + w_2 + ... + w_n}$$

(iii) Combined mean, $\bar{x}_{12} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$, where $\bar{x}_1$ and n_1 are mean and size of first data and $\bar{x}_2$ and n_2 are mean and size of second data.

2. Geometric mean, GM = $(x_1 \cdot x_2 \cdot x_3 \ldots \cdot x_n)^{1/n}$

$$GM = (x_1^{f_1} . x_2^{f_2} . x_3^{f_3} ..... x_n^{f_n})^{1/N},$$

where $N = f_1 + f_2 + \ldots + f_n$

3. Harmonic mean, $HM = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + ... + \frac{1}{x_n}} = \frac{n}{\sum_{i=1}^{n} \frac{1}{x_i}}$

$HM = \frac{N}{\sum_{i=1}^{n} \left(\frac{f_i}{x_i} \right)}$, where $N = f_1 + f_2 + \ldots + f_n$

4. AM $\geq$ GM $\geq$ HM

5. (i) Median of individual series

(a) If n is odd, then median = value of $\left(\frac{n+1}{2}\right)$th

observation

(b) If n is even, then median $= \frac{1}{2}\left[\text{value of}\left\{\frac{n}{2}\text{th} + \left(\frac{n}{2}+1\right)\text{th}\right\}\right]$

observations

(ii) Median of continuous series

Median = $I + \left(\frac{\left(\frac{n}{2}\right) - C}{f}\right) \times h$

where l = lower limit, N = total frequency h = width of median class C = cumulative frequency of the class preceding the median class.

6. (i) Mode of individual series :

The maximum number of times, the observation repeated is a mode

(ii) Mode of continuous series:

Mode = $I_1 + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right] \times h$

where l_1 = lower limit and

h = width of modal class f_1 = frequency of the

modal class f_0 = frequency of the class preceding

the modal class f_2 = frequency of the class

succeeding the modal class

7. Mode = 3 Median − 2 Mean

8. (i) Range = $X_{max} - X_{min}$

(ii) Coefficient of range $= \frac{X_{max} - X_{min}}{X_{max} + X_{min}}$

(iii) Quartile deviation, QD $= \frac{1}{2}(Q_3 - Q_1)$

(iv) Coefficient of QD = $\frac{Q_3 - Q_1}{Q_3 + Q_1}$

9. (i) Mean deviation for ungrouped data (or individual series)

$$= \frac{\sum |x - M|}{n}$$

where M = mean (mean, median or mode)

n = number of terms

(ii) Mean deviation for continuous series

$$= \frac{\sum f |x - M|}{\sum f}$$

10. (i) Standard deviation

$$\sigma = \sqrt{\frac{\sum(x - \bar{x})^2}{n}}$$

Standard deviation $= \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2}$

where d = x − A (A = assumed mean)

(ii) Standard deviation is not depend on change of origin but it depends on change of scale.

11. (i) Variance = σ^2

(ii) Coefficient of variance $= \frac{\sigma}{\bar{x}} \times 100$

12. (i) In a symmetrical distribution,

Mean = Median = Mode

(ii) When the distribution is not symmetrical, it is called asymmetrical or skewed distribution

Mean $\neq$ Median $\neq$ Mode

(iii) Absolute measures of skewness (S_k)

S_k = Mean − Median = Mean − Mode

(iv) Karl Pearson's coefficient of skewness

$$S_k = \frac{\text{Mean} - \text{Mode}}{\text{Standard deviation}} = \frac{3(\text{Mean} - \text{Median})}{\text{Standard deviation}}$$

13. Covariance, cov (x, y) $= \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{n}$

$$= \left(\frac{\sum x_i y_i}{n} - \bar{x}\ \bar{y} \right)$$

14. (i) Karl Pearson's correlation coefficient

$$r = \frac{\text{cov}(x, y)}{\sqrt{\text{var}(x).\text{var}(y)}} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 . \sum(y_i - \bar{y})^2}}$$

$$= \frac{n\sum xy - \sum x \sum y}{\sqrt{\{n\sum x^2 - (\sum x)^2\}}\sqrt{\{n\sum y^2 - (\sum y)^2\}}}$$

(ii) Coefficient of rank correlation for

$$R = 1 - \frac{6\sum d_i^2}{n(n^2 - 1)}$$

where d_i, is rank difference.

(iii) $-1 \leq r \leq 1$

15. (i) Standard Error, $SE = \frac{1-r^2}{\sqrt{n}}$

(ii) Probable Error, PE = $0.6745 \left(\frac{1-r^2}{\sqrt{n}}\right)$

16. (i) Regression line of y on x

$$y - \bar{y} = \frac{cov(x, y)}{\sigma_x^2}(x - \bar{x})$$

or $y - \bar{y} = b_{yx}(x - \bar{x})$,

where $b_{yx} = r\frac{\sigma_y}{\sigma_x}$ = regression coefficient

(ii) Regression line of x on y

$$x - \bar{x} = \frac{\text{cov}(x, y)}{\sigma_y^2}(y - \bar{y})$$

or $x - \bar{x} = b_{xy}(y - \bar{y})$,

where $b_{xy} = r\frac{\sigma_x}{\sigma_y}$ = regression coefficient

17. Angle between two lines of regression is given by

$$\tan\theta = \left(\frac{1 - r^2}{r}\right)\frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2}$$

18. $r = \pm\sqrt{b_{yx}.b_{xy}}$, if both b_{yx} and b_{xy} are positive, then r will be positive and if both are negative, then r will be negative.

19. The interesection of two regression lines is on mean ie, $(\bar{x}.\bar{y})$

Topic 1 Mean, Median and Mode

Q 1. The mean of the values 0, 1, 2, 3, ..., n with the corresponding weights ${}^{n}C_{o}$, ${}^{n}C_{1}$,...., ${}^{n}C_{n}$ respectively, is

[Kerala CEE]

(a) $\frac{n+1}{2}$

(b) $\frac{n-1}{2}$

(c) $\frac{2^{n}-1}{2}$

(d) $\frac{2^{n}+1}{2}$

(e) $\frac{n}{2}$

Q 2. The mean age of a combined group of men and women is 25 yr. If the mean age of the group of men is 26 and that of the group of women is 21, then the percentage of men and women in the group is

[AMU]

(a) 46, 60

(b) 80, 20

(c) 20, 80

(d) 60, 40

Q 3. Median of ${}^{2n}C_0$, ${}^{2n}C_1$, ${}^{2n}C_2$, ${}^{2n}C_3$,..., ${}^{2n}C_n$

(where n is even) is

[UP SEE]

(a) ${}^{2n}C_{\frac{n}{2}}$

(b) ${}^{2n}C_{\frac{n+1}{2}}$

(c) ${}^{2n}C_{\frac{n-1}{2}}$

(d) None of these

Q 4. The average of the four-digit numbers that can be formed using each of the digits $3,5,7$ and 9 exactly once in each number, is

[Kerala CEE]

(a) 4444

(b) 5555

(c) 6666

(d) 7777

(e) 8888

Q 5. The values of mean, median and mode coincide, then the distribution is

[Jamia Millia Islamia]

(a) positive skewness

(b) symmetric distribution

(c) negative skewness

(d) All of the above

Q 6. The average marks of boys in a class is 52 and that of girls is 42. The average marks of boys and girls combined is 50. The percentage of boys in the class is

[AIEEE]

(a) 40%

(b) 20%

(c) 80%

(d) 60%

Q 7. In a class of 100 students there are 70 boys whose average marks in a subject are 75. If the average marks of the complete class is 72, then what is the average of the girls ?

[OJEE]

(a) 73

(b) 65

(c) 68

(d) 74

Q 8. If in a frequency distribution, the mean and median are 21 and 22 respectively, then its mode is approximately

[AIEEE]

(a) 24.0

(b) 25.5

(c) 20.5

(d) 22.0

Q 9. Mean marks scored by the students of a class is 53. The mean marks of the girls is 55 and the mean marks of the boys is 50. What is the percentage of girls in the class ?

[Kerala CEE]

(a) 60%

(b) 40%

(c) 50%

(d) 45%

(e) 55%

Q 10. The weight (in kilogram) of 15 students are as follows

31, 35, 27, 29, 32, 43, 37, 41, 34, 28, 36, 44, 45, 42, 30. If the weight 44 kg is replaced by 46 kg and 27 kg is by 25 kg, then new median is

[J&K CET]

(a) 32

(b) 33

(c) 34

(d) 35

Q 11. The median of a set of 9 distinct observations is 20.5. If each of the largest 4 observations of the set is increased by 2, then the median of the new set

[AIEEE]

(a) is increased by 2

(b) is decreased by 2

(c) is two times the original median

(d) remains the same as that of the original set

Q 12. The median from the table

Value	7	8	10	9	11	12	13
Frequency	9	1	4	5	6	1	3

is

[AMU]

(a) 100

(b) 10

(c) 110

(d) 1110

Q 13. The mode of the series 3,4,2,6,1,7,6,7,6, 8,9,5 is

[AMU]

(a) 5

(b) 6

(c) 7

(d) 8

Topic 2 Measure of Dispersion

Q 1. For two data sets, each of size 5, the variances are given to be 4 and 5 and the corresponding means are given to be 2 and 4, respectively. The variance of the combined data set is

[AIEEE]

(a) $\frac{5}{2}$

(b) $\frac{11}{2}$

(c) 6

(d) $\frac{13}{2}$

Q 2. The means and variance of n observations x_1, x_2 ,x_3,..., x_n are 5 and 0 respectively. If $\sum_{i=1}^{n} x_i^2 = 400$, then the value of n is equal to

[AIEEE]

(a) 80

(b) 25

(c) 20

(d) 16

(e) 4

Q 3. If the mean deviation of number $1, 1+ d, 1 + 2d, \ldots, 1 + 100c$? from their mean is 255, then the d is equal to

[AIEEE]

(a) 10.0

(b) 20.0

(c) 10.1

(d) 20.2

Q 4. **Statement I** The variance of first n even natural numbers is $\frac{n^2-1}{4}$.

[AIEEE]

Statement II The sum of first n natural numbers is $\frac{n(n+1)}{2}$ and the sum of squares of first n natural numbers is $\frac{n(n+1)(2n+1)}{6}$.

(a) Statement I is true, Statement II is true;

Statement II is a correct explanation for Statement I.

(b) Statement I is true, Statement II is true;

Statement II is not a correct explanation for Statement I.

(c) Statement I is true, Statement II is false.

(d) Statement I is false Statement II is true.

Q 5. If the variable takes the values $0, 1, 2,$ …, n with frequencies proportional to the binomial coefficients C(n, 0), C(n, 1), C(n, 2),…, C(n, n) respectively, then the variance of the distribution is

[DCE]

(a) n

(b) $\frac{\sqrt{n}}{2}$

(c) $\frac{n}{2}$

(d) $\frac{n}{4}$

Q 6. If the variance of $1,2,3, 4,5,\ldots, 10$ is $\frac{99}{12}$, then the standard deviation of $3, 6, 9, 12,\ldots, 30$ is

[Kerala CEE]

(a) $\frac{297}{4}$

(b) $\frac{3}{2}\sqrt{33}$

(c) $\frac{3}{2}\sqrt{99}$

(d) $\sqrt{\frac{99}{12}}$

(e) $3\sqrt{\frac{3}{2}}$

Q 7. If the coefficient of variation is 45% and the mean is 12 then its standard deviation is

[AMU]

(a) 5.2

(b) 5.3

(c) 5.4

(d) None of these

Q 8. The mean deviation from the mean of the set of observations, $-1, 0, 4$ is

[OJEE]

(a) 3

(b) 1

(c) -2

(d) 2

Q 9. The variance of first n numbers is

[BCECE]

(a) $\frac{n^2+1}{12}$

(b) $\frac{n^2-1}{12}$

(c) $\frac{(n+1)(2n+1)}{6}$

(d) $\left[\frac{n(n+1)}{2}\right]^2$

Q 10. If $x_1, x_2, \ldots x_{18}$ are observation such that $\sum_{j=1}^{18}(x_j - 8) = 9$ and $\sum_{j=1}^{18}(x_j - 8)^2 = 45$, then these standard deviation of these observations is

[J&K CET]

(a) $\sqrt{\frac{81}{34}}$

(b) 5

(c) $\sqrt{5}$

(d) $\frac{3}{2}$

Q 11. The mean of the numbers a, b, 8, 5, 10 is 6 and the variance is 6.80. Then, which one of the following gives possible values of a and b ?

[AIEEE]

(a) a = 3, b = 4

(b) a = 0, b = 7

(c) a = 5, b = 2

(d) a =1, b = 6

Q 12. If the standard deviation of the observations $-5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5$ is $\sqrt{10}$. The standard deviation of observations $15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25$ will be

[DCS]

(a) $\sqrt{10}+20$

(b) $\sqrt{10}+10$

(c) $\sqrt{10}$

(d) None of these

Q 13. The standard deviation for the scores 1, 2, 3, 4, 5, 6 and 7 is 2. Then, the standard deviation of

12, 23, 34, 45, 56, 67 and 78 is

[Kerala CEE]

(a) 2

(b) 4

(c) 22

(d) 11

(e) 44

Q 14. Mean and standard deviation from the following observations of marks of 5 students of a tutorial group (marks out of 25) 8, 12, 13, 15, 22 are respectively

[VITEEE]

(a) 14, 4.604

(b) 15, 4.604

(c) 14, 5.604

(d) None of these

Q 15. The standard deviation of a variable x is 10. Then, the standard deviation of $50 + 5x$ is

[OJEE]

(a) 50

(b) 550

(c) 10

(d) 500

Q 16. The standard deviation of the numbers

31, 32, 33, ..., 46,47 is

[Guj. CET]

(a) $\sqrt{\frac{17}{12}}$

(b) $\sqrt{\frac{47^2-1}{12}}$

(c) $2\sqrt{6}$

(d) $4\sqrt{3}$

Q 17. The quartile deviation for the following data is

X	2	3	4	5	6
F	3	4	8	4	1

[Kerala CEE]

(a) 0

(b) $\frac{1}{4}$

(c) $\frac{1}{2}$

(d) 1

(e) $\frac{1}{5}$

Q 18. If the mean of n observations $1^2, 2^2, 3^2,$ …, n^2 is $\frac{46n}{11}$, then n is equal to

[AMU]

(a) 11

(b) 12

(c) 23

(d) 22

Q 19. What is the standard deviation of the following series

Measurements	0−10	10−20	20−30	30−40
Frequency	1	3	4	2

[EAMCET]

(a) 81

(b) 7.6

(c) 9

(d) 2.26

Q 20. Suppose a population A has 100 observations 101, 102, ..., 200 and another population B has 100 observations 151, 152, ...,

250. If V_A and V_B represent the variances of the two populations respectively, then $\frac{V_A}{V_B}$ is

[AIEEE]

(a) $\frac{9}{4}$

(b) $\frac{4}{9}$

(c) $\frac{2}{3}$

(d) 1

Q 21. The standard deviation of n observations $x_1, x_2, \ldots, x_n$ is 2. If $\sum_{i=1}^{n} x_i = 20$ and $\sum_{i=1}^{n} x_i^2 = 100$, then n is

[Kerala CEE]

(a) 10 or 20

(b) 5 or 10

(c) 5 or 20

(d) 5 or 15

(e) 25

Q 22. For the arithmetic progression a, (a + d), (a + 2d), (a + 3d), ..., (a + 2nd), the mean deviation from mean is

[Kerala CEE]

(a) $\frac{n(n+1)d}{2n-1}$

(b) $\frac{n(n+1)d}{2n+1}$

(c) $\frac{n(n-1)d}{2n+1}$

(d) $\frac{(n+1)d}{2}$

(e) $\frac{n(n-1)d}{2n-1}$

Q 23. If σ is the standard deviation of a random variable x, then the standard deviation of the random variable ax + b, where a, b $\in$ R is

[J&K CET]

(a) $a\sigma + b$

(b) $|a|\ \sigma$

(c) $|a|\ \sigma + b$

(d) $a^2\ \sigma$

Q 24. Let $x_1, x_2, \ldots, x_x$, be n observations such that $\sum x_i^2 = 400$ and $\sum x_i = 80$. Then, a possible value of n among the following is

[AIEEE]

(a) 12

(b) 9

(c) 18

(d) 15

Q 25. Consider the following statements

(1) Mode can be computed from histogram

(2) Median is not independent of change of scale

(3) Variance is independent of change of origin

and scale Which of these is/are correct ?

[AIEEE]

(a) Only (1)

(b) Only (2)

(c) Only (1) and (2)

(d) Only (1), (2) and (3)

Q 26. In a series of 2n observations, half of them equal a and remaining half equal − a. If the standard deviation of the observations is 2, then | a | equals

[AIEEE]

(a) $\frac{1}{n}$

(b) $\sqrt{2}$

(c) 2

(d) $\frac{\sqrt{2}}{n}$

Q 27. In an experiment with 15 observations on x, the following results were available $\sum x^2 = 2830$, $\sum x = 170$. One observation that was 20, was found to be wrong and was replaced by the correct value 30. Then, the corrected variance is

[AIEEE]

(a) 78.0

(b) 188.66

(c) 177.33

(d) 8.33

Q 28. Standard deviation of the first $2n+1$ natural numbers is equal to

[Kerala CEE]

(a) $\sqrt{\frac{n(n+1)}{2}}$

(b) $\sqrt{\frac{n(n+1)(2n+1)}{3}}$

(c) $\sqrt{\frac{n(n+1)}{3}}$

(d) $\sqrt{\frac{n(n-1)}{2}}$

(e) $2n+1$

Q 29. If the standard deviation of a variable x is σ, then the standard deviation of another variable $\frac{ax+b}{c}$ is

[AMU]

(a) $\frac{\sigma a + b}{c}$

(b) $\frac{\sigma a}{c}$

(c) σ

(d) None of these

Q 30. The, coefficient of SD and coefficient of variance from the given data

Class interval	0–10	10–20	20–30	30–40	40–50

Frequency	2	10	8	4	6

is

[AMU]

(a) 50,48.1

(b) 51.9,48.1

(c) 0.481,48.1

(d) 0.481,51.8

Q 31. The mean-deviation and coefficient of mean deviation from the data. Weight (in kg) 54, 50,40, 42, 51,45,47, 55, 57 is

[AMU]

(a) 0.0900

(b) 0.0956

(c) 0.0056

(d) 0.0946

Topic 3 Correlation and Rank Correlation

Correlation

Q 1. If Z = aX + bY and r be the correlation coefficient between X and Y, then σ_Z^2 is equal to

[Jamia Millia Islamia]

(a) $a^2\sigma_X^2 + b^2\sigma_Y^2 + 2abr\,\sigma_X\sigma_Y$

(b) $a^2\sigma_X^2 + b^2\sigma_Y^2 - 2abr\,\sigma_X\sigma_Y$

(c) $2abr\,\sigma_X\sigma_Y$

(d) None of the above

Q 2. For the given data, the calculation corresponding to all values of variates (x, y) is following $\sum(x-\bar{x})^2 = 36,\ \sum(y-\bar{y})^2 = 25,\ \sum(x-\bar{x})(y-\bar{y}) = 20$

[Jamia Millia Islamia]

The Karl Pearson's correlation coefficient is

(a) 0.2

(b) 0.5

(c) 0.66

(d) 0.33

Q 3. The correlation coefficient between x and y from the following data $\Sigma x = 40$, $\Sigma y = 50$, $\Sigma xy = 220$, $\Sigma x^2 = 200$, $\Sigma y^2 = 262$, $n = 10$ is

[MP PET]

(a) 0.89

(b) 0.76

(c) 0.91

(d) 0.98

Q 4. Two numbers within the brackets denote the ranks of 10 students of a class in two subjects $(1, 10)$, $(2, 9)$, $(3, 8)$, $(4,7)$, $(5,6)$, $(6, 5)$, $(7,4)$, $(8,3)$, $(9,2)$, $(10,1)$, then rank correlation coefficient is

[MP PET]

(a) 0

(b) -1

(c) 1

(d) 0.5

Q 5. If $\bar{x} = \bar{y} = 0$, $\Sigma x_i y_i = 12$, $\sigma_x = 2$, $\sigma_y = 3$ and $n = 10$, then the coefficient of correlation is

[Jamia Millia Islamia]

(a) 0.1

(b) 0.3

(c) 0.2

(d) 0.1

Q 6. If var (x) = 8.25, var (y) = 33.96 and cov (x, y) = 102 then the correlation coefficient is

[AMU]

(a) 0.89

(b) -0.98

(c) 0.61

(d) −0.16

Q 7. If $\Sigma x = 15$, $\Sigma y = 36$, $\Sigma xy = 110$, $n = 5$,

then cov (x, y) equals

[AMU]

(a) 1/5

(b) −1/5

(c) 2/5

(d) $-2/5$

Q 8. If r is Karl Pearson's coefficient of correlation between two sets of variates, then

[J&K CET]

(a) $r < 1$

(b) $r > 1$

(c) $r < -1$

(d) $|r| \leq 1$

Topic 4 Regression

Q 1. If the variance of x = 9 and regression equations are $4x - 5y + 33 = 0$ and $20 - 9y - 10 = 0$, then the coefficient of correlation between x and y and the variance of y respectively are

[Jamia Millia Islamia]

(a) 0.6; 16

(b) 0.16; 16

(c) $0.3;4$

(d) $0.6;\ 4$

Q 2. The correlation coefficient of two variables x and y is 0.8. The regression coefficient of y on x is 0.2, then the regression coefficient of x on y is

[MP PET]

(a) 3.2

(b) -3.2

(c) 4

(d) 0.16

Q 3. Angle between two lines of regression is given by

[DCE]

(a) $\tan^{-1}\left(\frac{b_{xy}-\frac{1}{b_{yx}}}{1-\frac{b_{xy}}{b_{yx}}}\right)$

(b) $\tan^{-1}\left(\frac{b_{yx}.b_{xy}-1}{b_{yx}+b_{xy}}\right)$

(c) $\tan^{-1}\left(\dfrac{b_{xy}-\dfrac{1}{b_{yx}}}{1+\dfrac{b_{xy}}{b_{yx}}}\right)$

(d) $\tan^{-1}\left(\dfrac{b_{yx}-b_{xy}}{1+b_{yx}.b_{xy}}\right)$

Q 4. If θ is the angle between two regression lines with correlation coefficient γ, then

[UP SEE]

(a) $\sin\theta \geq 1 - \gamma^2$

(b) $\sin\theta \leq 1 - \gamma^2$

(c) $\sin\theta \leq \gamma^2 + 1$

(d) $\sin \theta \leq \gamma^2 - 1$

Q 5. In a bivariate data $\Sigma x = 30$, $\Sigma y = 400$, $\Sigma x^2 = 196$, $\Sigma xy = 850$ and $n = 10$. The regression coefficient of y on x is

[MPPET]

(a) -3.1

(b) -3.2

(c) -3.3

(d) -3.4

Q 6. If the lines of regression are $3x + 12y = 19$ and $3y + 9x = 46$, then r_x will be

[MP PET]

(a) 0.289

(b) -0.289

(c) 0.209

(d) None of these

Q 7. If two lines of regression are $3\bar{x} - 2\bar{y} + 1 = 0$ and $2\bar{x} - \bar{y} - 2 = 0$, then $(\bar{x}, \bar{y})$ is

[DCE]

(a) (8, 5)

(b) (5, 8)

(c) (5, 5)

(d) (8, 8)

Q 8. If there exists a linear statistical relationship between two variables x and y, then the regression coefficient of y on x is

[BCECE]

(a) $\frac{\text{cov}(x, y)}{\sigma_x \sigma_y}$

(b) $\frac{\text{cov}(x, y)}{\sigma_y^2}$

(c) $\frac{\text{cov}(x, y)}{\sigma_x^2}$

(d) None of these

Q 9. When the origin is changed, then the coefficient of correlation

[BCECE]

(a) becomes zero

(b) varies

(c) remains fixed

(d) None of these

Q 10. If the two lines of regression are $4x + 3y + 7 = 0$ and $3x + 4y + 8 = 0$, then the means of x and yare

[BCECE]

(a) $\frac{-4}{7}, \frac{-11}{7}$

(b) $\frac{-4}{7}, \frac{11}{7}$

(c) $\frac{4}{7}, \frac{-11}{7}$

(d) 4, 7

Q 11. If the values of regression coefficients are -0.33 and -1.33, then the value of coefficients of correlation between the two variables, is

[MP PET]

(a) 0.2

(b) -0.66

(c) 0.4

(d) -0.4

Q 12. The intersecting point of two regression lines is

[Kerala CEE]

(a) $(\bar{x}, 0)$

(b) $(0, \bar{y})$

(c) (b_{xy}, b_{yx})

(d) $(0,0)$

(e) $(\bar{x}, \bar{y})$

Q 13. The regression coefficient of y on x is $2/3$ and that of x on y is $4/3$. The acute angle between the two regression lines is $\tan^{-1} k$, where k is equal to

[Kerala CEE]

(a) $1/9$

(b) $2/9$

(c) $1/18$

(d) $1/3$

(e) $8/9$

Q 14. The two regression lines are $2x - 7y + 6 = 0$ and $7x - 2y + 1 = 0$. The correlation coefficient between x and y is

[BCECE]

(a) $-\frac{2}{3}$

(b) $\frac{2}{7}$

(c) $\frac{4}{9}$

(d) None of these

Q 15. If the two lines of regression are $x + 4y = 3$ and $3x + y = 15$, then value of x for $y = 3$ is

[BCECE]

(a) 4

(b) −9

(c) −4

(d) None of these

Q 16. If both the regression lines intersect perpendicularly, then

[Jamia Millia Islamia]

(a) $r < -1$

(b) $r = -1$

(c) $r = 0$

(d) $r = \frac{1}{2}$

Q 17. If the regression coefficients are 0.8 and 0.2, then the value of coefficient of correlation is

[DCE]

(a) 0.16

(b) 0.4

(c) 0.04

(d) 0.164

Q 18. The coefficient of correlation (r) and the two regression coefficients b_{yx}, b_{xy} are related as

(a) $r = \frac{b_{xy}}{b_{yx}}$

(b) $r = b_{xy} \times b_{yx}$

(c) $r = b_{xy} + b_{yx}$

(d) $r = (\text{sign } b_{yx}) \sqrt{b_{xy} b_{yx}}$

Q 19. The two lines of regression are given by $3x + 2y = 26$ and $6x + y = 31$. The coefficient of correlation between x and y is

[AMU]

(a) $-1/3$

(b) $1/3$

(c) $-1/2$

(d) $1/2$

Answers

Topic 1: Mean, Median and Mode

1. (e) 2. (b) 3. (a) 4. (c) 5. (b) 6. (c) 7. (b) 8. (a)

9. (a) 10. (d) 11. (d) 12. (b) 13. (b)

Topic 2 : Measure of Dispersion

1. (b) 2. (d) 3. (c) 4. (d) 5. (d) 6. (b) 7. (c) 8. (d)

9. (b) 10. (d) 11. (a) 12. (e) 13. (c) 14. (a) 15. (a)

16. (c) 17. (d) 18. (a) 19. (c) 20. (d) 21. (c) 22.

(b) 23. (b) 24. (c) 25. (c) 26. (c) 27. (a) 28. (c)

29. (b) 30. (c) 31. (b)

Topic 3 : Correlation and Rank Correlation

1. (a) 2. (c) 3. (c) 4. (b) 5. (c) 6. (c) 7. (c) 8. (d)

Topic 4 : Regression

1. (a) 2. (a) 3. (b) 4. (b) 5. (c) 6. (b) 7. (b) 8. (c)

9. (c) 10. (a) 11. (b) 12. (e) 13. (c) 14. (b) 15. (a)

16. (c) 17. (b) 18. (d) 19. (c)

Conic Section

Useful Results & Formulae

1. The second degree equation

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, represents

(i) a parabola if $\Delta \neq 0$, $ab - h^2 = 0$, $e = 1$

(ii) an ellipse if $\Delta \neq 0$, $ab - h^2 > 0$, $e < 1$

(iii) a hyperbola if $\Delta \neq 0$, $ab - h^2 < 0$, $e > 1$

(iv) a rectangular hyperbola, if $\Delta \neq 0$, ab – $h^2 < 0$, a + b = 0, e = $\sqrt{2}$

2.

Important Terms	**Parabola** $y^2 = 4ax$	**Ellipse** $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, **a > b**	**Hyperbola** $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$
Centre	$(0, 0)$	$(0, 0)$	$(0, 0)$

Focus/Foci	$(a, 0)$	$(\pm ae, 0)$	$(\pm ae, 0)$
Vertex/Vertices	$(0, 0)$	$(\pm a, 0)$	$(\pm a, 0)$
Equation of directrix/directrices	$x + a = 0$	$x = \pm \frac{a}{e}$	$x = \pm \frac{a}{e}$
Length of latusrectum	$4a$	$\frac{2b^2}{a}$	$\frac{2b^2}{a}$

Ends of latusrectum	(a, ±2a)	$\left(\pm ae, \pm \frac{b^2}{a}\right)$	$\left(\pm ae, \pm \frac{b^2}{a}\right)$
Parametric equation	$x = at^2$, $y = 2at$	$x = a \cos \alpha$, $y = b \sin \alpha$	$x = a \sec \alpha$, $y = b \tan \alpha$
Distance between foci	−	$2ae$	$2ae$
Equation of tangent at	$yy_1 = 2a(x +$	$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$	$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$

point (x_1, y_1)	$x_1)$		
Equation of tangent in slope form	$y = mx + \frac{a}{m}$	$y = mx \pm \sqrt{a^2m^2 + b^2}$	$y = mx \pm \sqrt{a^2m^2 - b^2}$
Equation of tangent in parametric form	$ty = x + at^2$	$\frac{x}{a}\cos\alpha + \frac{y}{b}\sin\alpha = 1$	$\frac{x}{a}\sec\alpha - \frac{y}{b}\tan\alpha = 1$

Line y = mx + c touches if	$c = \frac{a}{m}$	$c^2 = a^2m^2 + b^2$	$c^2 + a^2m^2 - b^2$
Point of contact of y = mx + c and curve	$\left(\frac{a}{m^2}, \frac{2a}{m}\right)$, $m \neq 0$	$\left(\frac{\pm a^2m}{\sqrt{a^2m^2 + b^2}}, \frac{\pm$	$\left(\frac{\pm a^2m}{\sqrt{a^2m^2 - b^2}}, \frac{\pm$
Equation of	$y - y_1 = \frac{-y_1}{2a}(x$	$\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$	$\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2$

normal at point (x_1, y_1)			
Equation of normal in parametric form	$y + tx = 2at + at^3$	ax sec α − by cosec α = $a^2 - b^2$	ax cos α − by cot α = $a^2 + b^2$
Equation of normal in	$y = mx - 2am -$	$y = mx \pm \frac{m(a^2 - b^2)}{\sqrt{a^2 + b^2 m^2}}$	$y = mx \mp \frac{m(a^2 + b^2)}{\sqrt{a^2 - b^2 m^2}}$

slope form	am^3		
Condition for normally	c = – $2am$ – am^3	$c = \pm\frac{m(a^2-b^2)}{\sqrt{a^2+b^2m^2}}$	$c = \mp\frac{m(a^2-b^2)}{\sqrt{a^2-m^2b^2}}$
Equation of pair of tangents $SS_1 = T^2$	$S = y^2$ – $4ax$ $S_1 = y_1^2$ – $4ax_1$	$S = \frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ $S_1 = \frac{x_1^2}{a^2}+\frac{y_1^2}{b^2}-1$ $T = \frac{xx_1}{a^2}+\frac{yy_1}{b^2}-1$	$S = \frac{x^2}{a^2}-\frac{y^2}{b^2}-1$ $S_1 = \frac{x_1^2}{a^2}-\frac{y_1^2}{b^2}-1$ $T = \frac{xx_1}{a^2}-\frac{yy_1}{b^2}-1$

where	$T = yy_1 - 2a(x + x_1)$		
Equation of chord of contact	$yy_1 = 2a(x + x_1)$	$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$	$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$
Equation of chord with	$T = S_1$	$T = S_1$	$T = S_1$

mid point (x_1, y_1)			
Equation of diameter	$y = \frac{2a}{m}$	$y = -\frac{b^2}{a^2 m}x$	$y = \frac{b^2}{a^2 m}x$
Equation of polar	$yy_1 =$ $2a(x +$ $x_1)$	$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$	$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$
Coordinates	$\left(\frac{n}{l}, -\frac{2am}{l}\right)$	$\left(-\frac{a^2 l}{n}, -\frac{b^2 m}{n}\right)$	$\left(-\frac{a^2 l}{n}, \frac{b^2 m}{n}\right)$

of pole (line lx + my + n = 0)			
Auxiliary circle	–	$x^2 + y^2 = a^2$	$x^2 + y^2 = a^2$
Director circle	–	$x^2 + y^2 = a^2$ $+ b^2$	$x^2 + y^2 = a^2$ $- y^2$
Eccentricity	e = 1	$e = \sqrt{\left(\frac{a^2 - b^2}{a^2}\right)}$	$e = \sqrt{\left(\frac{a^2 + b^2}{a^2}\right)}$

3. (i) Point P(x_1, y_1) lies outside, on or inside the parabola $y^2 = 4ax$ according as $y_1^2 - 4ax_1 >, =, < 0$.

(ii) The point of intersection of tangent at two points P($at_1^2, 2at_1$) and Q($at_2^2, 2at_2$) on the parabola $y^2 = 4ax$ is $(at_1t_2, a(t_1 + t_2))$.

(iii) If the normal at P($at_1^2, 2at_1$), then $t_2 = -t_1 - \frac{2}{t_1}$

(iv) Equation of chord joining P($at_1^2, 2at_1$), Q($at_2^2, 2at_2$)

$$y - 2at_1 = \frac{2}{t_1 + t_2}(x - at_1^2)$$

If it passes through the focus, that $t_2 = -\frac{1}{t_1}$

$$\Rightarrow \quad t_1 t_2 = -1$$

4. Point P(x_1, y_1) lies outside, on or inside the ellipse according as $\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1$ >, =, < 0.

5. (i) Point (x_1, y_1) lies outside, on or inside the hyperbola according as <, =, > 0.

(ii) The equation of two asymptotes of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are $y = \pm \frac{b}{a} x$ or $\frac{x}{a} \pm \frac{y}{b} = 0$.

6. (i) For rectangular hyperbola $xy = c^2$, the coordinates of foci are $(\pm c\sqrt{2}, \pm c\sqrt{2})$ and directrices are $x + y = \pm c\sqrt{2}$.

(ii) Equation of tangent to the hyperbola $xy = c^2$ at (x_1, y_1) is $xy_1 + yx_1 = 2c^2$ and at $\left(ct, \frac{c}{t}\right)$ is $\frac{x}{t} + yt = 2c$.

(iii) Equation of normal to the hyperbola $xy = c^2$ at (x_1, y_1) is $xx_1 - yy_1 = x_1^2 - y_1^2$ and at $\left(ct, \frac{c}{t}\right)$ is $xt^3 - ty - ct^4 + c = 0$.

Topic 1 Basic Terms of Conics

Q 1. For different values of α, the locus of the point of intersection of the two straight lines $\sqrt{3}x - y - 4\sqrt{3}\alpha = 0$ and $\sqrt{3}\alpha x + \alpha y - 4\sqrt{3} = 0$ is

[WBJEE]

(a) a hyperbola with eccentricity 2

(b) an ellipse with eccentricity $\sqrt{\frac{2}{3}}$

(c) an hyperbola with eccentricity $\sqrt{\frac{19}{16}}$

(d) an ellipse with eccentricity $\frac{3}{4}$

Q 2. S and T are the foci of an ellipse and B is end point of the minor axis. If STB is an equilateral triangle, the eccentricity of the ellipse is

[WB JEE]

(a) $\frac{1}{4}$

(b) $\frac{1}{3}$

(c) $\frac{1}{2}$

(d) $\frac{2}{3}$

Q 3. If t_1 and t_2 be the parameters of the end points of a focal chord for the parabola $y^2 = 4ax$, then which one is true?

[VITEEE]

(a) $t_1t_2 = 1$

(b) $\frac{t_1}{t_2} = 1$

(c) $t_1t_2 = -1$

(d) $t_1 + t_2 = -1$

Q 4. The distance between the foci of the conic $7x^2 - 9y^2 = 63$ is equal to

[Kerala CEE]

(a) 8

(b) 4

(c) 3

(d) 7

(e) 12

Q 5. One of the points on the parabola $y^2 = 12x$ with focal distance 12, is

[Kerala CEE]

(a) $(3, 6)$

(b) $(9, 6\sqrt{3})$

(c) $(7, 2\sqrt{21})$

(d) $(8, 4\sqrt{6})$ (e) $(1, \sqrt{12})$

Q 6. If a point P (x, y) moves along the ellipse

$\frac{x^2}{25}+\frac{y^2}{16}=1$ and if C is the centre of the ellipse, then

the sum of maximum and minimum values of CP is

[Kerala CEE]

(a) 25

(b) 9

(c) 4

(d) 5

(e) 16

Q 7. If the length of the major axis of an ellipse is $\frac{17}{8}$ times the length of the minor axis, then the eccentricity of the ellipse is

[Kerala CEE]

(a) $\frac{8}{17}$

(b) $\frac{15}{17}$

(c) $\frac{9}{17}$

(d) $\frac{2\sqrt{2}}{17}$

(e) $\frac{13}{17}$

Q 8. If 1 denotes the semi-latusrectum of the parabola $y^2 = 4ax$ and SP and SQ denote the segments of any focal chord PQ, S being the focus, then SP, l and SQ are in the relation.

[DCE]

(a) AP

(b) GP

(c) HP

(d) $l^2 = SP^2 + SQ^2$

Q 9. If distance between directrices of a rectangular hyperbola is 10, then distance between its foci will be

[UP SEE]

(a) $10\sqrt{2}$

(b) 5

(c) $5\sqrt{2}$

(d) 20

Q 10. The sum of the distances of a point $(2, -3)$ from the foci of an ellipse $16\,(x-2)^2 + 25(y+3)^2 = 400$ is

[Kerala CEE]

(a) 8

(b) 6

(c) 50

(d) 32

(e) 10

Q 11. If e_1 is the eccentricity of the ellipse

$\frac{x^2}{16}+\frac{y^2}{7}=1$ and e_2 is the eccentricity of the hyperbola

$\frac{x^2}{9}-\frac{y^2}{7}=1$, then $e_1 + e_2$ is equal to

[Kerala CEE]

(a) $\frac{16}{7}$

(b) $\frac{25}{4}$

(c) $\frac{25}{12}$

(d) $\frac{16}{9}$

(e) $\frac{23}{16}$

Q 12. In an ellipse, if the lines joining focus to

the extremities of the minor axis form an equilateral triangle with the minor axis, then the eccentricity of the ellipse is

[AMU]

(a) $\frac{\sqrt{3}}{2}$

(b) $\frac{\sqrt{3}}{4}$

(c) $\frac{1}{\sqrt{2}}$

(d) $\sqrt{\frac{2}{3}}$

Q 13. The coordinates of the focus of the parabola described parametrically by $x = 5t^2 + 2$, $y = -10t + 4$ are

[WB JEE]

(a) $(7, 4)$

(b) $(3, 4)$

(c) $(3, -4)$

(d) $(-7, 4)$

Q 14. The focal distance of a point P on the parabola $y^2 = 12x$, if the ordinate of P is 6, is

[MHT CET]

(a) 12

(b) 6

(c) 3

(d) 9

Q 15. The equation $y^2 - 8y - x + 19 = 0$ represents

[RPET]

(a) a parabola whose focus is $\left(\frac{1}{4}, 0\right)$ and directrix is $x = \frac{-1}{4}$

(b) a parabola whose vertex is $(3, 4)$ and directrix is $x = \frac{11}{4}$

(c) a parabola whose focus is $\left(\frac{13}{4}, 4\right)$ and vertex is $(0,0)$

(d) a curve which is not a parabola

Q 16. If in a hyperbola, the distance between the foci is 10 and the transverse axis has length 8, then the length of its latusrectum is

[J&K CET]

(a) 9

(b) $\frac{9}{2}$

(c) $\frac{32}{3}$

(d) $\frac{64}{3}$

Q 17. Equation of the directrix of parabola $2x^2$ $=14y$ is equal to

[Guj. CET]

(a) $y = -\frac{7}{4}$

(b) $x = -\frac{7}{4}$

(c) $y = \frac{7}{4}$

(d) $x = \frac{7}{4}$

Q 18. A parabola has the origin as its focus and the line x = 2 as the directrix. Then, the vertex of the parabola is at

[AIEEE]

(a) $(2,0)$

(b) $(0,2)$

(c) $(1, 0)$

(d) $(0, 1)$

Q 19. A focus of an ellipse is at the origin. The directrix is the line x = 4 and the eccentricity is $\frac{1}{2}$, then length of semi major axis is

[AIEEE]

(a) 5/3

(b) 8/3

(c) 2/3

(d) 4/3

Q 20. Eccentricity of the ellipse $x^2 + 2y^2 - 2x + 3y + 2 = 0$ is

[UP SEE]

(a) $\frac{1}{\sqrt{2}}$

(b) $\frac{1}{2}$

(c) $\frac{1}{2\sqrt{2}}$

(d) $\frac{1}{\sqrt{3}}$

Q 21. If e_1 and e_2 are the eccentricities of a hyperbola $3x^2 - 3y^2 = 25$ and its conjugate, then

[KCET]

(a) $e_1^2 + e_2^2 = 2$

(b) $e_1^2 + e_2^2 = 4$

(c) $e_1 + e_2 = 4$

(d) $e_1 + e_2 = \sqrt{2}$

Q 22. A conic section is defined by the

equations x = -1 + sec t, y = 2 + 3 tan t. The coordinates of the foci are

[Kerala CEE]

(a) $(-1 - \sqrt{10}, 2)$ and $(-1 + \sqrt{10}, 2)$

(b) $(-1 - \sqrt{8}, 2)$ and $(-1 + \sqrt{8}, 2)$

(c) $(-1, 2 - \sqrt{8})$ and $(-1, 2 + \sqrt{8})$

(d) $(-1, 2 - \sqrt{10})$ and $(-1, 2 + \sqrt{10})$

(e) $(\sqrt{10}, 0)$ and $(-\sqrt{10}, 0)$

Q 23. If the lines joining the foci of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, where a > b and an extremity of its minor axis are inclined at an angle $60°$, then the eccentricity of the ellipse is

[Kerala CEE]

(a) $-\frac{\sqrt{3}}{2}$

(b) $\frac{1}{2}$

(c) $\frac{\sqrt{5}}{2}$

(d) $\frac{\sqrt{7}}{3}$

(e) $\sqrt{3}$

Q 24. The distance between the foci of the hyperbola $x^2 - 3y^2 - 4x - 6y - 11 = 0$ is

[EAMCET]

(a) 4

(b) 6

(c) 8

(d) 10

Q 25. The sum of the focal distances from any point on the ellipse $9x^2 + 16y^2 = 144$ is

[OJEE]

(a) 3

(b) 6

(c) 8

(d) 4

Q 26. The equation of the ellipse whose foci are at $(\pm 2,0)$ and eccentricity is $\frac{1}{2}$, is $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. Then,

[OJEE]

(a) $a^2 = 16, b^2 = 12$

(b) $a^2 = 12, b^2 = 16$

(c) $a^2 = 16, b^2 = 4$

(d) $a^2 = 4, b^2 = 16$

Q 27. The vertex of the parabola $x^2 + 2y = 8x - 7$ is

[WB JEE]

(a) $\left(\frac{9}{2}, 0\right)$

(b) $\left(4, \frac{9}{2}\right)$

(c) $\left(2, \frac{9}{2}\right)$

(d) $\left(4, \frac{7}{2}\right)$

Q 28. The radius of the circle passing through the foci of the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ and having its centre $(0, 3)$ is

[MP PET]

(a) 4

(b) $\frac{3}{7}$

(c) $\sqrt{12}$

(d) $\frac{7}{2}$

Q 29. Equation of the latusrectum of the ellipse

$9x^2 + 4y^2 - 18x - 8y - 23 = 0$ are

[BCECE]

(a) $y = \pm\sqrt{5}$

(b) $x = \pm\sqrt{5}$

(c) $y = 1 \pm \sqrt{5}$

(d) $x = -1 \pm \sqrt{5}$

Q 30. The directrix of the parabola $y^2 + 4x + 3 = 0$ is

[VITEEE]

(a) $x - \frac{4}{3} = 0$

(b) $x + \frac{1}{4} = 0$

(c) $x - \frac{3}{4} = 0$

(d) $x - \frac{1}{4} = 0$

Q 31. If the distance between the foci and the distance between the directrices of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are in the ratio $3 : 2$, then a : b is

[Jamia Millia Islamia]

(a) $\sqrt{2} : 1$

(b) $\sqrt{3} : \sqrt{2}$

(c) $1 : 2$

(d) $2 : 1$

Q 32. The focal distance of a point on the parabola $y^2 = 16x$ whose ordinate is twice the abscissa, is

[MHT CET]

(a) 6

(b) 8

(c) 10

(d) 12

Q 33. If e and e' be the eccentricities of a hyperbola and its conjugate, then $\frac{1}{e^2}+\frac{1}{(e')^2}$ is equal to

[MP PET]

(a) 0

(b) 1

(c) 2

(d) 3

Q 34. The eccentricity of the ellipse $9x^2 + 5y^2 - 30y = 0$ is

[RPET]

(a) $1/3$

(b) $2/3$

(c) $3/4$

(d) $4/5$

Q 35. If OAB is an equilateral triangle inscribed in the parabola $y^2 = 4ax$ with O as the vertex, then the length of the side of the ΔOAB is

[RPET]

(a) $8a\sqrt{3}$

(b) $4a\sqrt{3}$

(c) $2a\sqrt{3}$

(d) $a\sqrt{3}$

Q 36. One of the directrices of the ellipse $8x^2 + 6y^2 - 16x + 12y + 13 = 0$ is

[J&KCET]

(a) $3y - 3 = \sqrt{6}$

(b) $3y + 3 = \sqrt{6}$

(c) $y + 1 = \sqrt{3}$

(d) $y - 1 = -\sqrt{3}$

Q 37. Statement I The curve $y = -\frac{x^2}{2} + x + 1$ is symmetric with respect to the line $x = 1$. Because

Statement II A parabola is symmetric about its axis.

[IIT JEE]

(a) Statement I is true, Statement II is true;

Statement II is a correct explanation for Statement I

(b) Statement I is true, Statement II is true;

Statement II is not a correct explanation for

StatementI

(c) Statement I is true, Statement II is false

(d) Statement I is false, Statement II is true

Q 38. For the hyperbola $\frac{x^2}{\cos^2\alpha}-\frac{y^2}{\sin^2\alpha}=1$, which of the following remains constant when α varies ?

[AIEEE]

(a) Eccentricity

(b) Directrix

(c) Abscissae of vertices

(d) Abscissae of foci

Q 39. The foci of the conic section $25x^2 + 16y^2 - 150x = 175$ are

[DCE]

(a) $(0, \pm 3)$

(b) $(0, \pm 2)$

(c) $(3, \pm 3)$

(d) $(0, \pm 1)$

Q 40. The focus of the parabola $y^2 - x - 2y + 2 = 0$ is

[UP SEE]

(a) $\left(\frac{1}{4}, 0\right)$

(b) $(1, 2)$

(c) $\left(\frac{5}{4}, 1\right)$

(d) $\left(\frac{3}{4}, \frac{5}{2}\right)$

Q 41. If P is any point on the ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$

and Sand S' are the foci, then PS + PS' is equal to

[KCET]

(a) 4

(b) 8

(c) 10

(d) 12

Q 42. The eccentricity of the conic $\frac{(x+2)^2}{7} + (y - 1)^2 = 14$ is

[Kerala CEE]

(a) $\sqrt{\frac{7}{8}}$

(b) $\sqrt{\frac{6}{17}}$

(c) $\frac{\sqrt{3}}{2}$

(d) $\sqrt{\frac{6}{11}}$

(e) $\sqrt{\frac{6}{7}}$

Q 43. If for the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, y−axis is the minor axis and the length of the latusrectum is one half of the length of its minor axis, then its eccentricity is

[Kerala CEE]

(a) $\frac{1}{\sqrt{2}}$

(b) $\frac{1}{2}$

(c) $\frac{\sqrt{3}}{2}$

(d) $\frac{3}{4}$

(e) $\frac{3}{5}$

Q 44. For the parabola $y^2 + 6y - 2x + 5 = 0$

(I) The vertex is $(-2,-3)$

(II) The directrix is $y + 3$

$= 0$ Which of the following is correct ?

[EAMCET]

(a) Both I and II are true

(b) I is true, II is false

(c) I is false, II is true

(d) Both I and II are false

Q 45. Coordinates of foci of hyperbola are $(-5,3)$ and $(7,3)$ and eccentricity is $3/2$. Then, length of its latusrectum is

[OJEE]

(a) 20

(b) 10

(c) 40

(d) None of these

Q 46. Coordinates of the foci of the ellipse $5x^2 + 9y^2 + 10x - 36y - 4 = 0$, are

[OJEE]

(a) $(1, 2)$ and $(-3, 2)$

(b) $(2, 1)$ and $(-3, 2)$

(c) $(1, 2)$ and $(3, 2)$

(d) None of these

Q 47. The equation of a directrix of the ellipse

$\frac{x^2}{16}+\frac{y^2}{25}=1$ is

[VITEEE]

(a) $3y = \pm 5$

(b) $y = \pm 5$

(c) $3y = \pm 25$

(d) $y = \pm 3$

Q 48. The equation of the latusrectum of the parabola $x^2 + 4x + 2y = 0$, is equal to

[MHT CET]

(a) $2y + 3 = 0$

(b) $3y = 2$

(c) $2y = 3$

(d) $3y + 2 = 0$

Q 49. In an ellipse, the distances between its foci is 6 and minor axis is 8. Then, its eccentricity is

[AIEEE]

(a) $\frac{1}{2}$

(b) $\frac{4}{5}$

(c) $\frac{1}{\sqrt{5}}$

(d) $\frac{3}{5}$

Q 50. The ends of the latusrectum of the conic

$x^2 + 10x - 16y + 25 = 0$ are

[BITSAT, Guj. CET]

(a) $(3,-4), (13, 4)$

(b) $(-3,-4), (13,-4)$

(c) $(3, 4), (-13, 4)$

(d) $(5, -8), (-5, 8)$

Q 51. The ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ and the hyperbola $\frac{x^2}{25}-\frac{y^2}{16}=1$ have in common

[KCET]

(a) centre only

(b) centre, foci and directrices

(c) centre, foci and vertices

(d) centre and vertices only

Q 52. The locus of the point which moves such

that the ratio of its distance from two fixed point in the plane is always a constant k (< 1) is

[KCET]

(a) hyperbola

(b) ellipse

(c) straight line

(d) circle

Q 53. The eccentricity of the hyperbola in the standard form $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, passing through $(3, 0)$ and $(3\sqrt{2}, 2)$ is

[Kerala CEE]

(a) $\frac{13}{3}$

(b) $\sqrt{13}$

(c) $\sqrt{3}$

(d) $\frac{\sqrt{13}}{3}$

(e) $\frac{5}{3}$

Q 54. If the eccentricities of the ellipse $\frac{x^2}{4}+\frac{y^2}{3}=1$ and the hyperbola $\frac{x^2}{64}-\frac{y^2}{b^2}=1$ are reciprocals of each other, then b^2 is equal to

[Kerala CEE]

(a) 192

(b) 64

(c) 16

(d) 32

(e) 128

Q 55. If the vertex of the parabola $y = x^2 - 16x +$ k lies on x−axis, then the value of k is

[Kerala CEE]

(a) 16

(b) 8

(c) 64

(d) -64

(e) -8

Q 56. For the ellipse $24x^2 + 9y^2 - 120x - 90y + 225 = 0$, the eccentricity is equal to

[AMU]

(a) $\frac{2}{5}$

(b) $\frac{3}{5}$

(c) $\sqrt{\frac{15}{24}}$

(d) $\frac{1}{5}$

Q 57. If the eccentricity of a hyperbola is $\sqrt{3}$, then the eccentricity of its conjugate hyperbola is

[EAMCET]

(a) $\sqrt{2}$

(b) $\sqrt{3}$

(c) $\sqrt{\frac{3}{2}}$

(d) $2\sqrt{3}$

Q 58. If b and c are the lengths of the segments of any focal chord of a parabola $y^2 = 4ax$, then the length of the semilatusrectum is

[EAMCET]

(a) $\frac{bc}{b+c}$

(b) $\sqrt{bc}$

(c) $\frac{b+c}{2}$

(d) $\frac{2bc}{b+c}$

Q 59. The equation of the directrix of parabola

$y^2 + 4y + 4x + 2 = 0$ is

[OJEE]

(a) $x = -1$

(b) $x = 1$

(c) $x = -\frac{3}{2}$

(d) $x = \frac{3}{2}$

Q 60. The eccentricity of the conic $4x^2 + 16y^2 - 24x - 32y = 1$ is

[WBJEE]

(a) $\frac{1}{2}$

(b) $\sqrt{3}$

(c) $\frac{\sqrt{3}}{2}$

(d) $\frac{\sqrt{3}}{4}$

Q 61. The latusrectum of the parabola $y^2 = 4ax$,

whose focal chord is PSQ, such that SP = 3 and SQ = 2 is given by

[BCECE]

(a) $\frac{24}{5}$

(b) $\frac{12}{5}$

(c) $\frac{6}{5}$

(d) $\frac{1}{5}$

Q 62. The eccentricity of the hyperbola $9x^2 - 16y^2 - 18x - 64y - 199 = 0$ is

[Jamia Millia Islamia, MP PET]

(a) $\frac{16}{9}$

(b) $\frac{5}{4}$

(c) $\frac{25}{16}$

(d) zero

Q 63. Suppose S and S 'are foci of the ellips

$\frac{x^2}{25}+\frac{y^2}{16}=1$. If P is a variable point on the ellipse and if Δ is area of the triangles PSS', then the maximum value Δ is

[MHTCET]

(a) 8

(b) 12

(c) 16

(d) 20

Q 64. The equation of the directrix of the parabola $x^2 + 8y - 2x = 7$ is

[MPPET]

(a) $y = 3$

(b) $y = -3$

(c) $y = 2$

(d) $y = 0$

Q 65. The eccentricity of the hyperbola conjugate to $x^2 - 3y^2 = 2x + 8$ is

[MPPET]

(a) $\frac{2}{\sqrt{3}}$

(b) $\sqrt{3}$

(c) 2

(d) None of these

Q 66. If $(-1, -2\sqrt{2})$ is one of extremity of a focal chord of the parabola $y^2 = -8x$, then the other extremity is

[J&K CET]

(a) $(-1,-\sqrt{2})$

(b) $(2\sqrt{2},-1)$

(c) $(-4,4\sqrt{2})$

(d) $(4,4,\sqrt{2})$

Q 67. The sum of the focal distances of any point on the conic $\frac{x^2}{25}+\frac{y^2}{16}=1$ is

[Guj. CET]

(a) 10

(b) 9

(c) 41

(d) 18

Q 68. An ellipse has OB as semi minor axis, F and F' its foci and the angle FBF' is a right angle. Then, the eccentricity of the ellipse is

[AIEEE]

(a) $\frac{1}{\sqrt{3}}$

(b) $\frac{1}{4}$

(c) $\frac{1}{2}$

(d) $\frac{1}{\sqrt{2}}$

Q 69. The eccentricity of the hyperbola $\frac{x^2}{16}-\frac{y^2}{25}=1$

is

[KCET]

(a) $\frac{3}{4}$

(b) $\frac{3}{5}$

(c) $\frac{\sqrt{41}}{4}$

(d) $\frac{\sqrt{41}}{5}$

Q 70. The eccentricity of the ellipse $25x^2 + 16y^2 - 150x - 175 = 0$ is

[Kerala CEE]

(a) $\frac{2}{5}$

(b) $\frac{2}{3}$

(c) $\frac{4}{5}$

(d) $\frac{3}{4}$

(e) $\frac{3}{5}$

Q 71. Distance between foci is 8 and distance between directrices is 6 of hyperbola, then length of latusrectum is

[MHT CET]

(a) $4\sqrt{3}$

(b) $\frac{4}{\sqrt{3}}$

(c) $\sqrt{\frac{3}{4}}$

(d) None of these

Q 72. In the standard form of an ellipse sum of the focal distances of a point is

[MHT CET]

(a) 1

(b) $-2a$

(c) $2a$

(d) None of these

Q 73. The distance between the directrices of the hyperbola $x = 8 \sec \theta$, $y = 8 \tan \theta$ is

[MP PET]

(a) $8\sqrt{2}$

(b) $16\sqrt{2}$

(c) $4\sqrt{2}$

(d) $6\sqrt{2}$

Q 74. The locus of a point which moves such that the difference of its distances from two fixed points is ways a constant, is

[RPET]

(a) a circle

(b) a straight line

(c) a hyperbola

(d) an ellipse

Q 75. If e and e_1 are the eccentricities of hyperbolas $xy = c^2$ and $x^2 - y^2 = c^2$, then $e^2 + e_1^2$ is

[DCE]

(a) 1

(b) 4

(c) 6

(d) 8

Q 76. The eccentricity of the hyperbola $5x^2 - 4y^2 + 20x + 8y = 4$ is

[UP SEE]

(a) $\sqrt{2}$

(b) $\frac{3}{2}$

(c) 2

(d) 3

Q 77. For the ellipse $25x^2 + 9y^2 - 150x - 90y + 225 = 0$, the eccentricity e is equal to

[KCET]

(a) $\frac{2}{5}$

(b) $\frac{3}{5}$

(c) $\frac{4}{5}$

(d) $\frac{1}{5}$

Q 78. The centre of the ellipse $9x^2 + 25y^2 - 18x - 100y - 166 = 0$ is

[Kerala CEE]

(a) $(1,1)$

(b) $(-1, 2)$

(c) $(-1, 1)$

(d) $(2, 2)$

(e) $(1, 2)$

Q 79. The latusrectum of the ellipse $9x^2 + 16y^2 = 144$ is

[Kerala CEE]

(a) 4

(b) $\frac{11}{4}$

(d) $\frac{9}{2}$

(e) $\frac{10}{3}$

Q 80. Any point on the hyperbola $\frac{(x+1)^2}{16} - \frac{(y-2)^2}{4} = 1$ is of the form

[Kerala CEE]

(a) $(4 \sec \theta, 2 \tan \theta)$

(b) $(4 \sec \theta + 1, 2 \tan \theta - 2)$

(c) $(4 \sec \theta - 1, 2 \tan \theta - 2)$

(d) $(\sec \theta - 4, \tan \theta = 2)$ (e) $(4 \sec \theta - 1, 2 \tan \theta + 2)$

Q 81. The eccentricity of the hyperbola $x^2 - y^2 =$ is

[Kerala CEE]

(a) $\sqrt{3}$

(b) 2

(c) $2\sqrt{2}$

(d) $\sqrt{2}$

(e) 1.5

Q 82. The equation of the directrix of $(x - 1)^2 = 2(y - 2)$ is

[Kerala CEE]

(a) $2y + 3 = 0$

(b) $2x + 1 = 0$

(c) $2x - 1 = 0$

(d) $2y - 1 = 0$

(e) $2y - 3 = 0$

Q 83. The eccentricity of the conic $36x^2 + 144y^2 - 36x - 96y - 119 = 0$ is

[EAMCET]

(a) $\frac{\sqrt{3}}{2}$

(b) $\frac{1}{2}$

(c) $\frac{\sqrt{3}}{4}$

(d) $\frac{1}{\sqrt{3}}$

Q 84. If $x^2 + 6x + 20y - 51 = 0$, then axis of parabola is

[OJEE]

(a) $x + 3 = 0$

(b) $x - 3 = 0$

(c) $x = 1$

(d) $x + 1 = 0$

Q 85. The difference in focal distances of any point on the hyperbola $\frac{x^2}{16}-\frac{y^2}{9}=1$ is

[OJEE]

(a) 8

(b) 9

(c) 0

(d) 6

Q 86. P is any point on the ellipse $81\ x^2 + 144y^2 = 1944$, whose foci are S and S'. Then, SP + S'P equals

[BCECE]

(a) 3

(b) $4\sqrt{6}$

(c) 36

(d) 324

Q 87. The point $P(9/2, 6)$ lies on the parabola $y^2 = 4ax$, then parameter of the point P is

[MHTCET]

(a) $\frac{3a}{2}$

(b) $\frac{2}{3a}$

(c) $\frac{2}{3}$

(d) $\frac{3}{2}$

Q 88. The focus of the parabola $x^2 + 2y + 6x = 0$ is

[RPET]

(a) $(-3,4)$

(b) $(3, 4)$

(c) $(3,-4)$

(d) $(-3,-4)$

Q 89. The point on the curve $3x^2 - 4y^2 = 72$,

which is nearest to the line $3x + 2y - 1 = 0$, is

[J&K CET]

(a) $(6, 3)$

(b) $(6, -3)$

(c) $(6, 6)$

(d) $(6, 5)$

Q 90. The latusractum of the hyperbola $9x^2 - 16y^2 + 72x - 32y - 16 = 0$ is

[J&K CET]

(a) $\frac{9}{2}$

(b) $-\frac{9}{2}$

(c) $\frac{32}{3}$

(d) $-\frac{32}{3}$

Q 91. The foci of the ellipse $\frac{x^2}{16}+\frac{y^2}{b^2}=1$ and the hyperbola $\frac{x^2}{144}-\frac{y^2}{81}=\frac{1}{25}$ coincide. Then, the value of b^2

is

[AIEEE]

(a) 1

(b) 5

(c) 7

(d) 9

Q 92. Length of major axis of ellipse $9x^2 + 7y^2 =$ 63 is **[DCE]**

(a) 3

(b) 9

(c) 6

(d) $2\sqrt{7}$

Q 93. The directrix of the hyperbola $\frac{x^2}{9}-\frac{y^2}{4}=1$ is

[UP SEE]

(a) $y=\frac{6}{\sqrt{13}}$

(b) $x=\frac{6}{\sqrt{13}}$

(c) $y = \frac{9}{\sqrt{13}}$

(d) $x = \frac{9}{\sqrt{13}}$

Q 94. The directrix of the parabola $x^2 - 4x - 8y + 12 = 0$ is

[KCET]

(a) $y = 0$

(b) $x = 1$

(c) $y = -1$

(d) $x = -1$

Q 95. If a focal chord of the parabola $y^2 = ax$ is $2x - y - 8 = 0$, then the equation of the directrix is

[Kerala CEE]

(a) $x + 4 = 0$

(b) $x - 4 = 0$

(c) $y - 4 = 0$

(d) $y + 4 = 0$

(e) None of these

Q 96. Vertex of the parabola $9x^2 - 6x + 36y + 9 = 0$ is

[AMU]

(a) $(1/3, -2/9)$

(b) $(-1/3, -1/2)$

(c) $(-1/3, 1/2)$

(d) $(1/3, 1/2)$

Q 97. If the foci and vertices of an ellipse be $(\pm 1,0)$ and $(\pm 2,0)$ then the minor axis of the ellipse is

[AMU]

(a) $2\sqrt{5}$

(b) 2

(c) 4

(d) $2\sqrt{3}$

Q 98. The eccentricity of the ellipse $9x^2 + 5y^2 - 18x - 207 - 16 = 0$ is

[EAMCET]

(a) $\frac{1}{2}$

(b) $\frac{2}{3}$

(c) $\frac{3}{2}$

(d) 2

Q 99. The point of the parabola $y^2 = 18x$, for which the ordinate is three times the abscissa is

[MP PET]

(a) $(6,2)$

(b) $(-2,-6)$

(c) $(3,18)$

(d) $(2,\ 6)$

Q 100. If transverse and conjugate axes of hyperbola are equal, then its eccentricity is

[RPET]

(a) $\sqrt{3}$

(b) $\sqrt{2}$

(c) $\frac{1}{\sqrt{2}}$

(d) 2

Topic 2 Equation of Conics and Intersection of Two Curves

Q 1. Passage The circle $x^2 + y^2 - 8x = 0$ and hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ intersect at the points A and B.

[IIT JEE]

I. Equation of a common tangent with positive slope to the circle as well as to the hyperbola is

(a) $2x - \sqrt{5}y - 20 = 0$

(b) $2x - \sqrt{5}y + 4 = 0$

(c) $3x - 4y + 8 = 0$

(d) $4x - 3y + 4 = 0$

II. Equation of the circle with AB as its diameter is

(a) $x^2 + y^2 - 12x + 24 = 0$

(b) $x^2 + y^2 + 12x + 24 = 0$

(c) $x^2 + y^2 + 24x - 12 = 0$

(d) $x^2 + y^2 - 24x - 12 = 0$

Q 2. The ellipse $x^2 + 4y^2 = 4$ is inscribed in a rectangle aligned with the coordinate axes, which is turn in inscribed in another ellipse that passes through the point $(4, 0)$. Then, the equation of the ellipse is

[AIEEE]

(a) $x^2 + 12y^2 = 16$

(b) $4x^2 + 48y^2 = 48$

(c) $4x^2 + 64y^2 = 48$

(d) $x^2 + 16y^2 = 16$

Q 3. If the circle $x^2 + y^2 = a^2$ intersects the hyperbola $xy = c^2$ in four points (x_i, y_i), for $i = 1, 2, 3$ and 4, then $y_1 + y_2 + y_3 + y_4$ equals

[EAMCET]

(a) 0

(b) c

(c) a

(d) c^4

Q 4. Focus of hyperbola is $(\pm 3, 0)$ and equation of tangent is $2x + y - 4 = 0$, find the equation of hyperbola

[MHT CET]

(a) $4x^2 - 5y^2 = 20$

(b) $4x^2 - 4y^2 = 20$

(c) $4x^2 - 5y^2 = 1$

(d) $5x^2 - 4y^2 = 1$

Q 5. Consider the two curves $C_1 : y^2 = 4x$

$C_2 : x^2 + y^2 - 6x + 1 = 0$, then

[IIT JEE]

(a) C_1 and C_2 touch each other only at one point

(b) C_1 and C_2 touch each other exactly at two points

(c) C_1 and C_2 intersect (but do not touch) at exactly two point

(d) C_1 and C_2 intersect nor touch each other

Q 6. The equation of ellipse whose distance between the foci is equal to 8 and distance between the directrix is 18, is

[DCE]

(a) $5x^2 - 9y^2 = 180$

(b) $9x^2 + 5y^2 = 180$

(c) $9x^2 - 5y^2 = 180$

(d) $5x^2 + 9y^2 = 180$

Q 7. The locus of the equation $x^2 - y^2 = 0$, is

[UP SEE]

(a) a circle

(b) a hyperbola

(c) a pair of lines

(d) a pair of lines at right angles

Q 8. The equation $\frac{x^2}{2-\lambda}-\frac{y^2}{\lambda-5}-1=0$, represent an ellipse, if

[KCET]

(a) $\lambda > 5$

(b) $\lambda < 2$

(c) $2 < \lambda < 5$

(d) $2 > \lambda > 5$

Q 9. The locus of the mid point of the line joining the focus and any point on the parabola $y^2 = 4ax$ is a parabola with the equation of directrix as

[KCET]

(a) $x + a = 0$

(b) $2x + a = 0$

(c) $x = 0$

(d) $x = \frac{a}{2}$

Q 10. For an ellipse with eccentricity $1/2$ the centre is at the origin. If one directrix is x = 4, then the equation of the ellipse is

[AMU]

(a) $3x^2 + 4y^2 = 1$

(b) $3x^2 + 4y^2 = 12$

(c) $4x^2 + 3y^2 = 1$

(d) $4x^2 + 3y^2 = 12$

Q 11. Length of the straight line $x - 3y = 1$ intercepted by the hyperbola $x^2 - 4y^2 = 1$ is

[OJEE]

(a) $\frac{3}{5}\sqrt{10}$

(b) $\frac{6}{5}\sqrt{10}$

(c) $\frac{5}{3}\sqrt{10}$

(d) $\frac{5}{6}\sqrt{10}$

Q 12. The two parabolas $x^2 = 4y$ and $y^2 = 4x$ meet in two distinct points. One of these is the origin and the other is

[WB JEE]

(a) $(2, 2)$

(b) $(4, -4)$

(c) $(4, 4)$

(d) $(-2, 2)$

Q 13. The equation of the ellipse having vertices at $(\pm 5,0)$ and foci $(\pm 4,0)$ is

[WBJEE]

(a) $\frac{x^2}{25}+\frac{y^2}{16}=1$

(b) $9x^2+25y^2=225$

(c) $\frac{x^2}{9}+\frac{y^2}{25}=1$

(d) $4x^2+5y^2=20$

Q 14. If t is a parameter, then $x = a\left(t + \frac{1}{t}\right), y = b\left(t - \frac{1}{t}\right)$ represents

[WB JEE]

(a) an ellipse

(b) a circle

(c) a pair of straight lines

(d) a hyperbola

Q 15. Let O be the origin and A be a point on the curve $y^2 = 4x$. Then, the locus of the mid point of OA, is

[BCECE]

(a) $x^2 = 4y$

(b) $x^2 = 2y$

(c) $x^2 = 16y$

(d) $y^2 = 2x$

Q 16. $x = 4(1 + \cos\theta)$ and $y = 3(1 + \sin\theta)$

are the parametric equations of

[Jamia Millia Islamia]

(a) $\frac{(x-3)^2}{9} + \frac{(y-4)^2}{16} = 1$

(b) $\frac{(x+4)^2}{16} + \frac{(y+3)^2}{9} = 1$

(c) $\frac{(x-4)^2}{16} - \frac{(y-3)^2}{9} = 1$

(d) $\frac{(x-4)^2}{16} + \frac{(y-3)^2}{9} = 1$

Q 17. A parabola is drawn with its focus at $(3,4)$ and vertex at the focus of the parabola $y^2 - 12x - 4y + 4 = 0$. The equation of the parabola is

[Manipal]

(a) $y^2 - 8x - 6y + 25 = 0$

(b) $y^2 - 6x + 8y - 25 = 0$

(c) $x^2 - 6x - 8y + 25 = 0$

(d) $x^2 + 6x - 8y - 25 = 0$

Q 18. Locus of the point which divides double ordinate of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ in the ratio $1:2$ internally, is

[Manipal]

(a) $\frac{x^2}{a^2}-\frac{9y^2}{b^2}=\frac{1}{9}$

(b) $\frac{x^2}{a^2}+\frac{9y^2}{b^2}=1$

(c) $\frac{9x^2}{a^2}+\frac{9y^2}{b^2}=1$

(d) None of these

Q 19. The equation of the parabola whose focus is $(3, -4)$ and directrix $6x - 7y + 5 = 0$, is

[MP PET]

(a) $(7x + 6y)^2 - 570x + 750y + 2100 = 0$

(b) $(7x + 6y)^2 + 570x - 750y + 2100 = 0$

(c) $(7x - 6y)^2 - 570x + 750y + 2100 = 0$

(d) $(7x - 6y)^2 + 570x - 750y + 2100 = 0$

Q 20. A point P moves so that sum of its distances from $(-ae, 0)$ and $(ae, 0)$ is $2a$. Then, the locus of P is

[J&K CET]

(a) $\frac{x^2}{a^2}-\frac{x^2}{a^2(1-e^2)}=1$

(b) $\frac{x^2}{a^2}+\frac{y^2}{a^2(1-e^2)}=1$

(c) $\frac{x^2}{a^2}+\frac{y^2}{a^2(1+e^2)}=1$

(d) $\frac{x^2}{a^2}-\frac{y^2}{a^2(1+e^2)}=1$

Q 21. A hyperbola, having the transverse axis of length $2 \sin \theta$, is confocal with the ellipse $3x^2 + 4y^2 = 12$. Then, its equation is

[IIT JEE]

(a) $x^2 \operatorname{cosec}^2 \theta - y^2 \sec^2 \theta = 1$

(b) $x^2 \sec^2 \theta - y^2 \operatorname{cosec}^2 \theta = 1$

(c) $x^2 \sin^2 \theta - y^2 \cos^2 \theta = 1$

(d) $x^2 \cos^2 \theta - y^2 \sin^2 \theta = 1$

Q 22. Equation of the parabola with its vertex at $(1, 1)$ and focus $(3, 1)$ is

[BITSAT]

(a) $(x-1)^2 = 8(y-1)$

(b) $(y-1)^2 = 8(x-3)$

(c) $(y-1)^2 = 8(x-1)$

(d) $(x-3)^2 = 8(y-1)$

Q 23. If $\frac{x^2}{36} - \frac{y^2}{k^2} = 1$, is a hyperbola, then which of the following statements can by true ?

[KCET]

(a) $(-3,1)$ lies on the hyperbola

(b) $(3,1)$ lies on the hyperbola

(c) $(10, 4)$ lies on the hyperbola

(d) $(5, 2)$ lies on the hyperbola

Q 24. The parametric representation of a point of the ellipse whose foci are $(3, 0)$ and $(-1, 0)$ and eccentricity $2/3$ is

[Kerala CEE]

(a) $(1 + 3 \cos \theta, \sqrt{3} \sin \theta)$

(b) $(1 + 3 \cos \theta, 5 \sin \theta)$

(c) $(1 + 3 \cos \theta, 1 + \sqrt{5} \sin \theta)$

(d) $(1 + 3 \cos \theta, 1 + 5 \sin \theta)$

(e) $(1 + 3 \cos \theta, \sqrt{5} \sin \theta)$

Q 25. The line $x = at^2$ meets the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

in the real points, iff

[AMU]

(a) $|t| < 2$

(b) $|t| \leq 1$

(c) $|t| > 1$

(d) None of these

Q 26. The curve represented by $x = 3(\cos t + \sin t)$, $y = 4(\cos t - \sin t)$ is

[WB JEE]

(a) ellipse

(b) parabola

(c) hyperbola

(d) circle

Q 27. The curve described parametrically by x = $t^2 + 2t - 1$, y = $3t + 5$ represents

[VITEEE]

(a) an ellipse

(b) a hyperbola

(c) a parabola

(d) a circle

Q 28. The parametric representation of a point on the ellipse whose foci are $(-1,0)$ and $(7, 0)$ and eccentricity $1/2$ is

[MHT CET]

(a) $(3 + 8 \cos \theta, 4 \sqrt{3} \sin \theta)$

(b) $(8 \cos \theta, 4 \sqrt{3} \sin \theta)$

(c) $(3 + 4 \sqrt{3} \cos \theta, 8 \sin \theta)$

(d) None of these

Q 29. The equation of an ellipse whose eccentricity is $\frac{1}{2}$ and the vertices are $(4,0)$ and $(10,0)$ is

[MP PET]

(a) $3x^2 + 4y^2 - 42x + 120 = 0$

(b) $3x^2 + 4y^2 + 42x + 120 = 0$

(c) $3x^2 + 4y^2 + 42x - 120 = 0$

(d) $3x^2 + 4y^2 - 42x - 120 = 0$

Q 30. Let P be the point $(1, 0)$ and Q a point on the locus of $y^2 = 8x$. The locus of mid point of PQ is

[RPET]

(a) $x^2 - 4y + 2 = 0$

(b) $x^2 + 4y + 2 = 0$

(c) $y^2 + 4x + 2 = 0$

(d) $y^2 - 4x + 2 = 0$

Q 31. If e_1 is the eccentricity of the ellipse $\frac{x^2}{16}+\frac{y^2}{25}=1$ and e_2 is the eccentricity of the hyperbola passing through the foci of the ellipse and $e_1\ e_2$ = 1, then equation of the hyperbola is

[IIT JEE]

(a) $\frac{x^2}{9}-\frac{y^2}{16}=1$

(b) $\frac{x^2}{16}-\frac{y^2}{9}=-1$

(c) $\frac{x^2}{9}-\frac{y^2}{25}=1$

(d) None of these

Q 32. Axis of a parabola is y − x and vertex and focus are at a distance $\sqrt{2}$ and $2\sqrt{2}$ respectively from the origin. Then, equation of the parabola is

[IIT JEE]

(a) $(x - y)^2 = 8(x + y - 2)$

(b) $(x + y)^2 = 2(x + y - 2)$

(c) $(x - y)^2 = 4(x + y - 2)$

(d) $(x + y)^2 = 2(x - y + 2)$

Q 33. The locus of the vertices of the family of parabolas $y = \frac{a^3x^2}{3} + \frac{a^2x}{2} - 2a$ is

[AIEEE]

(a) $xy = \frac{3}{4}$

(b) $xy = \frac{35}{16}$

(c) $xy = \frac{64}{105}$

(d) $xy = \frac{105}{64}$

Q 34. The equation to the hyperbola having its eccentricity 2 and the distance between its foci is 8, is

[BITSAT]

(a) $\frac{x^2}{12}-\frac{y^2}{4}=1$

(b) $\frac{x^2}{4}-\frac{y^2}{12}=1$

(c) $\frac{x^2}{8}-\frac{y^2}{2}=1$

(d) $\frac{x^2}{16}-\frac{y^2}{9}=1$

Q 35. The equation of the parabola with vertex at $(-1,1)$ and focus $(2,1)$ is

[KCET]

(a) $y^2 - 2y - 12x - 11 = 0$

(b) $x^2 + 2x - 12y + 13 = 0$

(c) $y^2 - 2y + 12x + 11 = 0$

(d) $y^2 - 2y - 12x + 13 = 0$

Q 36. The equation of the hyperbola whose vertices are at $(5, 0)$ and $(-5, 0)$ and one of the directrices is $x = \frac{25}{7}$, is

[Kerala CEE]

(a) $\frac{x^2}{25} - \frac{y^2}{24} = 1$

(b) $\frac{x^2}{24} - \frac{y^2}{25} = 1$

(c) $\frac{x^2}{16} - \frac{y^2}{25} = 1$

(d) $\frac{x^2}{25} - \frac{y^2}{16} = 1$

(e) $\frac{x^2}{25}-\frac{y^2}{24}=-1$

Q 37. If $(0, 6)$ and $(0, 3)$ are respectively the vertex and focus of a parabola, then its equation is

[AMU]

(a) $x^2 + 12y = 72$

(b) $x^2 - 12y = 72$

(c) $y^2 - 12x = 72$

(d) $y^2 + 12x = 72$

Q 38. The equation of the ellipse whose foci are $(\pm 2, 0)$ and eccentricity $\frac{1}{2}$ is

[OJEE]

(a) $\frac{x^2}{12}+\frac{y^2}{16}=1$

(b) $\frac{x^2}{16}+\frac{y^2}{12}=1$

(c) $\frac{x^2}{16}+\frac{y^2}{8}=1$

(d) None of these

Q 39. Equation of the ellipse whose foci are $(2,$ $2)$ and $(4,\ 2)$ and the major axis is of length 10 is

[WB JEE]

(a) $\frac{(x+3)^2}{24}+\frac{(y+2)^2}{25}=1$

(b) $\frac{(x-3)^2}{24}+\frac{(y-2)^2}{25}=1$

(c) $\frac{(x+3)^2}{25}+\frac{(y+2)^2}{24}=1$

(d) $\frac{(x-3)^2}{25}+\frac{(y-2)^2}{24}=1$

Q 40. The equation of parabola with focus $(0, 0)$ and directrix $x + y = 4$, is [Jamia Millia Islamia]

(a) $x^2 + y^2 - 2xy + 8x + 8y - 16 = 0$

(b) $x^2 + y^2 - 2xy + 8x + 8y = 0$

(c) $x^2 + y^2 + 8x + 8y - 16 = 0$

(d) $x^2 - y^2 + 8x + 8y - 16 = 0$

Q 41. The equation of the hyperbola in the standard form (with transverse axis along the .x-

axis) having the length of the latusrectum = 9 unit and eccentricity = $\frac{5}{4}$, is

[MHT CET]

(a) $\frac{x^2}{16}-\frac{y^2}{18}=1$

(b) $\frac{x^2}{36}-\frac{y^2}{27}=1$

(c) $\frac{x^2}{64}-\frac{y^2}{36}=1$

(d) $\frac{x^2}{36}-\frac{y^2}{64}=1$

Q 42. The curve with parametric equation $x = e^t + e^{-t}$ $y = e^t - e^{-t}$ and is

[J&K CET]

(a) a circle

(b) an ellipse

(c) a hyperbola

(d) a parabola

Q 43. The equation of a parabola which passes through the intersection of a straight line $x + y = 0$ and the circle $x^2 + y^2 + 4y = 0$ is

[BITSAT, O JEE]

(a) $y^2 = 4x$

(b) $y^2 = x$

(c) $y^2 = 2x$

(d) None of these

Q 44. The point $(4,-3)$ with respect to the ellipse $4x^2 + 5y^2 = 1$ is

[BITSAT, O JEE]

(a) lies on the curve

(b) is inside the curve

(c) is outside the curve

(d) is focus of the curve

Q 45. The foci of an ellipse are $(0, \pm 4)$ and the equations for the directrices are $y = \pm 9$. The equation for the ellipse is

[UP SEE]

(a) $5x^2 + 9y^2 = 4$

(b) $2x^2 - 6y^2 = 28$

(c) $6x^2 + 3y^2 = 45$

(d) $9x^2 + 5y^2 = 180$

Q 46. The parabola with directrix $x+2y-1=0$ and focus $(1,0)$ is

[EAMCET]

(a) $4x^2 - 4xy + y^2 - 8x + 4y + 4 = 0$

(b) $4x^2 + 4xy + y^2 - 8x + 4y + 4 = 0$

(c) $4x^2 + 5xy + y^2 + 8x - 4y + 4 = 0$

(d) $4x^2 - 4xy + y^2 - 8x - 4y + 4 = 0$

Q 47. If a point P moves such that its distances from the point A $(1, 1)$ and the line $x + y + 2 = 0$ are equal, then the locus of P is

[EAMCET]

(a) a straight line

(b) a pair of straight lines

(c) a parabola

(d) an ellipse

Q 48. The length intercepted by the curve $y^2 = 4x$ on the line satisfying $dy/dx = 1$ and passing through point $(0, 1)$, is given by

[OJEE]

(a) 1

(b) 2

(c) 0

(d) None of these

Q 49. The equation of the parabola with vertex at the origin and directrix $y = 2$ is

[MPPET]

(a) $y^2 = 8x$

(b) $y^2 = -8x$

(c) $y^2 = \sqrt{8}x$

(d) $x^2 = -8y$

Q 50. If the foci of an ellipse are $(\pm\sqrt{5}, 0)$ and its eccentricity is $\sqrt{5}/3$, then the equation of the ellipse is

[J&K CET]

(a) $9x^2 + 4y^2 = 36$

(b) $4x^2 + 9y^2 = 36$

(c) $36x^2 + 9y^2 = 4$

(d) $9x^2 + 36y^2 = 4$

Q 51. If $a \neq 0$ and the line $2bx + 3cy + 4d = 0$, passes through the points of intersection of the parabolas $y^2 = 4ax$ $x^2 = 4ay$, then

[AIEEE]

(a) $d^2 + (26 + 3c)^2 = 0$

(b) $d^2 + (36 + 2c)^2 = 0$

(c) $d^2 + (2b - 3c)^2 = 0$

(d) $d^2 + (3b - 2c)^2 = 0$

Q 52. The length of major and minor axis of an ellipse are 10 and 8 respectively and its major axis along the y−axis the equation of the ellipse referred to its centre as origin is

[RPET]

(a) $\frac{x^2}{25}+\frac{y^2}{16}=1$

(b) $\frac{x^2}{16}+\frac{y^2}{25}=1$

(c) $\frac{x^2}{100}+\frac{y^2}{64}=1$

(d) $\frac{x^2}{64}+\frac{y^2}{100}=1$

Q 53. The equation $|\sqrt{x^2+(y-1)^2}-\sqrt{x^2+(y+1)^2}|=k$ will represent a hyperbola for

[J&K CET]

(a) $k \in (0, 2)$

(b) $k \in (0, 1)$

(c) $k \in (1, \infty)$

(d) $k \in R^+$

Q 54. The curve represented by the equation

$4x^2 + 16y^2 - 24x - 32y - 12 = 0$ is

[Kerala CEE]

(a) a parabola

(b) a pair of straight lines

(c) an ellipse with eccentricity ½

(d) an ellipse with eccentricity $\sqrt{3}/2$

(e) a hyperbola with eccentricity $3/2$

Q 55. The equation $13[(x-1)^2 + (y-2)^2] = 3(2x + 3y - 2)^2$ represents

[AMU]

(a) parabola

(b) ellipse

(c) hyperbola

(d) None of these

Q 56. The one which does not represent a hyperbola is

[AMU]

(a) $xy = 1$

(b) $x^2 - y^2 = 5$

(c) $(X - 1)(Y - 3) = 0$

(d) $x^2 - Y^2 = 0$

Topic 3 Tangent and Normal of Conic

Q 1. Let A and B be two distinct points on the parabola $y^2 = 4x$. If the axis of the parabola touches a circle of radius r having AB as its diameter, then the slope of the line joining A and B can be

[IIT JEE]

(a) $-\frac{1}{r}$

(b) $\frac{1}{r}$

(c) $\frac{2}{r}$

(d) $-\frac{2}{r}$

Q 2. Passage Tangents are drawn from the point P $(3, 4)$ to the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ touching the ellipse at points A and B.

[IIT JEE]

I. The coordinates of A and B are

(a) $(3, 0)$ and $(0,2)$

(b) $\left(-\frac{8}{5}, \frac{2\sqrt{161}}{15}\right) \text{and} \left(-\frac{9}{5}, \frac{8}{5}\right)$

(c) $\left(-\frac{8}{5}, \frac{2\sqrt{161}}{15}\right) \text{and} (0,2)$

(d) $(3,0) \text{and} \left(-\frac{9}{5}, \frac{8}{5}\right)$

II. The orthocentre of the ΔPAB is

(a) $\left(5, \frac{8}{7}\right)$

(b) $\left(\frac{7}{5}, \frac{25}{8}\right)$

(c) $\left(\frac{11}{5}, \frac{8}{5}\right)$

(d) $\left(\frac{8}{25}, \frac{7}{5}\right)$

III. The equation of the locus of the point whose distance from the point P and the line AB are equal, is

(a) $9x^2 + y^2 - 6xy - 54x - 62y + 241 = 0$

(b) $x^2 + 9y^2 + 6xy - 54x + 62y - 241 = 0$

(c) $9x^2 + 9y^2 - 6xy - 54x - 62y - 241 = 0$

(d) $x^2 + Y^2 - 2XY + 27X + 31Y - 120 = 0$

Q 3. If two tangents drawn from a point P to the parabola $y = 4x$ are at right angles, then the locus of P is

[AIEEE]

(a) $x = 1$

(b) $2x + 1 = 0$

(c) $x = -1$

(d) $2x - 1 = 0$

Q 4. The normal at a point P on the ellipse $x^2 + 4y^2 = 16$ meets the x–axis at Q. If M is the mid point of the line segment PQ, then the locus of M intersects the latusrectum of the given ellipse at the points

[IIT JEE]

(a) $\left(\pm\frac{3\sqrt{5}}{2}, \pm\frac{2}{7}\right)$

(b) $\left(\pm\frac{3\sqrt{5}}{2}, \pm\frac{\sqrt{19}}{4}\right)$

(c) $\left(\pm 2\sqrt{3}, \pm \frac{1}{7}\right)$

(d) $\left(\pm 2\sqrt{3}, \pm \frac{4\sqrt{3}}{7}\right)$

Q 5. If tangents at extremities of a focal chord AB of the parabola $y^2 = 4ax$ intersect at a point C, then $\angle ACB$ is equal to

[UP SEE]

(a) $\frac{\pi}{4}$

(b) $\frac{\pi}{3}$

(c) $\frac{\pi}{2}$

(d) $\frac{\pi}{6}$

Q 6. The number of values of c such that the line $y = 4x + c$ touches the curve $\frac{x^2}{4} + y^2 = 1$ is

[KCET]

(a) 1

(b) 2

(c) ∞

(d) 0

Q 7. The number of normals drawn to the parabola $y^2 = 4x$ from the point $(1, 0)$ is

[EAMCET]

(a) 0

(b) 1

(c) 2

(d) 3

Q 8. The locus of the foot of the perpendicular from the centre of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ on any tangent is given by $(x^2 + y^2)^2 = lx^2 + my^2$ where

[OJEE]

(a) $l = a^2$, $m = b^2$

(b) $l = b^2$, $m = a^2$

(c) $l = m = a$

(d) $l = m = b$

Q 9. The equation of the tangent to the conic

$x^2 - y^2 - 8x + 2y + 11 = 0$ at $(2, 1)$ is

[WB JEE]

(a) $x + 2 = 0$

(b) $2x + 1 = 0$

(c) $x + y + 1 = 0$

(d) $x - 2 = 0$

Q 10. The total number of tangents through the points $(3,5)$ that can be drawn to the ellipses $3x^2 + 5y^2 = 32$ and $25x^2 + 9y^2 = 450$ is

[WBJEE]

(a) 0

(b) 2

(c) 3

(d) 4

Q 11. A line is drawn through the point $P(3,11)$ to cut the circle $x^2 + y^2 = 9$ at A and B. Then, PA . PB is equal to

[BCECE]

(a) 9

(b) 121

(c) 205

(d) 139

Q 12. The line $x + y = 6$ is a normal to the parabola $y^2 = 8x$ at the point

[BCECE]

(a) $(18, -12)$

(b) $(4, 2)$

(c) $(2, 4)$

(d) $(8, 8)$

Q 13. The tangent at $(1, 7)$ to the curve $x^2 = y - 6$ touches the circle $x^2 + y^2 + 16x + 12y + c = 0$ at

[Jamia Millia Islamia]

(a) $(6, 7)$

(b) $(-6, 7)$

(c) $(6, -7)$

(d) $(-6, -7)$

Q 14. If $4x - 3y + k = 0$ touches the ellipse $5x^2 + 9y^2 = 45$, then k is equal to

[MHT CET]

(a) $\pm 3\sqrt{21}$

(b) $3\sqrt{21}$

(c) $-3\sqrt{21}$

(d) $2\sqrt{21}$

Q 15. If m_1 and m_2 are the slopes of tangents to the circle $x^2 + y^2 = 4$ from the point $(3, 2)$, then $m_1 - m_2$ is equal to

[MHT CET]

(a) $\frac{5}{12}$

(b) $\frac{12}{5}$

(c) $\frac{3}{2}$

(d) 0

Q 16. Tangent to the ellipse $\frac{x^2}{32}+\frac{y^2}{18}=1$ having slope $-\frac{3}{4}$ meet the coordinate axes in A and B. Find the area of the Δ AOB, where O is the origin

[MHT CET]

(a) 12 sq unit

(b) 8 sq unit

(c) 24 sq unit

(d) 32 sq unit

Q 17. Equation of tangent to the parabola y^2 $=16x$ at $P(3,6)$ is

[MHT CET]

(a) $4x - 3Y + 12 = 0$

(b) $3Y - 4x - 12 = 0$

(c) $4x - 3y - 24 = 0$

(d) $3y - x - 24 = 0$

Q 18. If the line $y = 2x + \lambda$ be a tangent to the hyperbola $36x^2 - 25y^2 = 3600$, then λ is equal to

[DCE]

(a) 16

(b) -16

(c) ± 16

(d) None of these

Q 19. The equation of the normal to the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ at $(-4, 0)$ is

[KCET]

(a) $2x - 3y = 1$

(b) $x = 0$

(c) $x = 1$

(d) $y = 0$

Q 20. Three normals to the parabola $y = x$ through point $(a, 0)$. Then,

[OJEE]

(a) $a = \frac{1}{2}$

(b) $a = \frac{1}{4}$

(c) $a > \frac{1}{2}$

(d) $a < \frac{1}{2}$

Q 21. The tangent to the parabola $y^2 = 16x$, which is perpendicular to a line $y - 3x - 1 = 0$, is

[OJEE]

(a) $3y + x + 36 = 0$

(b) $3y - x - 36 = 0$

(c) $x + y - 36 = 0$

(d) $x - y + 36 = 0$

Q 22. The equation of the line which is tangent to both the circle $x^2 + y^2 = 5$ and the parabola $y^2 = 40x$ is

[Jamia Millia Islamia]

(a) $2x - y \pm 5 = 0$

(b) $2x - y + 5 = 0$

(c) $2x - y - 5 = 0$

(d) $2x + y + 5 = 0$

Q 23. If the line $x + y - 1 = 0$ is a tangent to the parabola $y^2 - y + x = 0$, then the point of contact is

[MHT CET]

(a) $(0, 1)$

(b) $(1,0)$

(c) $(0, -1)$

(d) $(-1,0)$

Q 24. If the line x cos α + y sin α = p be normal to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, then

[MHTCET]

(a) $p^2(a^2\cos^2\alpha+b^2\sin^2\alpha) = a^2 - b^2$

(b) $p^2(a^2\cos^2\alpha+ b^2\sin^2\alpha) = (a^2- b^2)^2$

(c) $p^2(a^2\sec^2\alpha+b^2\,\mathrm{cosec}^2\alpha) = a^2 - b^2$

(d) $p^2(a^2\sec^2\alpha+b^2\mathrm{cosec}^2\alpha) = (a^2 - b^2)^2$

Q 25. The equation the tangent parallel to $y - x + 5 = 0$, drawn to $\frac{x^2}{3} - \frac{y^2}{2} = 1$ is

[MHT CET]

(a) $x - y - 1 = 0$

(b) $x - y + 2 = 0$

(c) $x + y - 1 = 0$

(d) $x + y + 2 = 0$

Q 26. The line $3x + 5y = 15\sqrt{2}$ is a tangent to the ellipse $\frac{x^2}{25}+\frac{y^2}{9}=1$, at a point whose eccentric angle is

[MP PET]

(a) $\pi/6$

(b) $\pi/4$

(c) $\pi/3$

(d) $2\pi/3$

Q 27. The value of c, for which the line $y = 2x + c$, is tangent to the parabola $y^2 = 4a(x + a)$, is

[MPPET]

(a) a

(b) $\frac{3a}{2}$

(c) $2a$

(d) $\frac{5a}{2}$

Q 28. The common tangent of the parabolas $y^2 = 4x$ and $x^2 = -8y$, is

[DCE]

(a) $y = x + 2$

(b) $y = x - 2$

(c) $y = 2x + 3$

(d) None of these

Q 29. The equation of normal at the point $(0,3)$ of the ellipse $9x^2 + 5y^2 = 45$, is

[UPSEE]

(a) x−axis

(b) y−axis

(c) $y + 3 = 0$

(d) $y - 3 = 0$

Q 30. Equation of tangents to the ellipse

$\frac{x^2}{9}+\frac{y^2}{4}=1$, which are perpendicular to the line $3x + 4y = 7$, are

[UP SEE]

(a) $4x-3y = \pm 6\sqrt{5}$

(b) $4x - 3y = \pm \sqrt{12}$

(c) $4x - 3y = \pm \sqrt{2}$

(d) $4x-3y = \pm 1$

Q 31. The slope of tangents drawn form a point

$(4, 10)$ to the parabola $y^2 = 9x$ are

[AMU]

(a) $\frac{1}{4}, \frac{3}{4}$

(b) $\frac{1}{4}, \frac{9}{4}$

(c) $\frac{1}{4}, \frac{1}{3}$

(d) None of these

Q 32. If the line lx + my = 1 is a normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1,$ then $\frac{a^2}{l^2} - \frac{b^2}{m^2}$ is equal to

[EAMCET]

(a) $a^2 - b^2$

(b) $a^2 + b^2$

(c) $(a^2 + b^2)^2$

(d) $(a^2 - b^2)^2$

Q 33. A common tangent to $9x^2 - 16y^2 = 144$ and $x^2 + y^2 = 9$, is

[WBJEE, MHTCET]

(a) $y = \frac{3}{\sqrt{7}}x + \frac{15}{\sqrt{7}}$

(b) $y = 3\sqrt{\frac{2}{7}}x + \frac{15}{\sqrt{7}}$

(c) $y = 2\sqrt{\frac{3}{7}}x + 15\sqrt{7}$

(d) none of these

Q 34. The equation of the tangents to the ellipse $4x^2 + 3y^2 = 5$, which are parallel to the line $y = 3x + 7$ are

[WB JEE]

(a) $y = 3x \pm \sqrt{\frac{155}{3}}$

(b) $y = 3x \pm \sqrt{\frac{155}{12}}$

(c) $y = 3x \pm \sqrt{\frac{95}{12}}$

(d) None of these

Q 35. The angle between the tangent drawn from the point $(1, 4)$ to the parabola $y^2 = 4x$ is

[BCECE]

(a) $\frac{\pi}{6}$

(b) $\frac{\pi}{4}$

(c) $\frac{\pi}{3}$

(d) $\frac{\pi}{2}$

Q 36. If the normal at $(ap , 2ap)$ on the parabola $y^2 = 4ax$, meets the parabola again at $(aq^2, 2aq)$, then

[VITEEE]

(a) $p^2 + pq + 2 = 0$

(b) $p^2 - pq + 2 = 0$

(c) $q^2 + pq + 2 = 0$

(d) $p^2 + pq + 1 = 0$

Q 37. The equation to the line touching both the parabolas $y^2 = 4x$ and $x^2 = -32y$, is

[MHT CET]

(a) $x + 2y + 4 = 0$

(b) $2x + y - 4 = 0$

(c) $x - 2y - 4 = 0$

(d) $x - 2y + 4 = 0$

Q 38. The locus of a point P(α, β) moving under the condition that the line $y = \alpha x + \beta$ is a tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, is

[MP PET]

(a) a hyperbola

(b) a parabola

(c) a circle

(d) an ellipse

Q 39. The equation of a tangent parallel to y = x drawn to $\frac{x^2}{3}-\frac{y^2}{2}=1$, is

[Guj.CET]

(a) $x - y + 1 = 0$

(b) $x - y + 2 = 0$

(c) $x + y - 1 = 0$

(d) $x - y + 2 = 0$

Q 40. Two tangents are drawn from the point $(-2, -1)$ to the parabola $y^2 = 4x$. If α is the angle between these tangents, then $\tan \alpha$ is equal to

[Guj. CET]

(a) 3

(b) $1/3$

(c) 2

(d) $1/2$

Q 41. The value of m, for which the line y = mx +2 becomes a tangent to the conic $4x^2 - 9y^2 = 36$ are

[UP SEE]

(a) $\pm\frac{2}{3}$

(b) $\pm\frac{2\sqrt{2}}{3}$

(c) $\pm\frac{8}{9}$

(d) $\pm\frac{4\sqrt{2}}{3}$

Q 42. The value of m, for which the line y= mx + $\frac{25\sqrt{3}}{3}$ is a normal to the conic $\frac{x^2}{16}-\frac{y^2}{9}=1$, is

[UP SEE, WB JEE]

(a) $\pm\frac{2}{\sqrt{3}}$

(b) $\pm\sqrt{3}$

(c) $\pm\frac{\sqrt{3}}{2}$

(d) None of these

Q 43. The number of maximum normals which can be drawn from a point to ellipse is

[WBJEE]

(a) 4

(b) 2

(c) 1

(d) 3

Q 44. A line touches the circle $x^2 + y^2 - 2$ and the parabola $y^2 = 8x$, then equation of tangent is

[MHT CET]

(a) $y = x + 3$

(b) $y = x+2$

(c) $y = x + 4$

(d) $y = x + 1$

Q 45. If the parabola $y^2 = 4ax$ passes through the point $(1, -2)$, then tangent at this point is

[RPET]

(a) $x - y - 1 = 0$

(b) $x + y + 1 = 0$

(c) $x - y + 1 = 0$

(d) None of these

Q 46. Tangent at a point of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is drawn which cuts the coordinate axes at A and B. The minimum area of the ΔOAB is (O being the origin)

[IIT JEE]

(a) ab

(b) $\frac{a^3 + ab + b^3}{3}$

(c) $a^2 + b^2$

(d) $\frac{(a^2 + b^2)}{4}$

Q 47. The line among the following which touches the parabola $y^2 = 4ax$, is

[EAMCET]

(a) $x + my + am^3 = 0$

(b) $x - my + am^2 = 0$

(c) $x + my - am^2 = 0$

(d) $y + mx + am^2 = 0$

Q 48. The equation of the normal at the point $(a \sec\theta, b \tan\theta)$ of the curve $b^2x^2 - a^2y^2 = a^2b^2$ is

[RPET]

(a) $\frac{ax}{\cos\theta} + \frac{by}{\sin\theta} = a^2 + b^2$

(b) $\frac{ax}{\tan\theta}+\frac{by}{\sec\theta}=a^2+b^2$

(c) $\frac{ax}{\sec\theta}+\frac{by}{\tan\theta}=a^2+b^2$

(d) $\frac{ax}{\sec\theta}+\frac{by}{\tan\theta}=a^2-b^2$

Q 49. From the point $(-1, -6)$ two tangents are drawn to the parabola $y^2 = 4x$. Then, the angle between the two tangents is

[J&K CET]

(a) $30°$

(b) $45°$

(c) $60°$

(d) $90°$

Q 50. If the line $2x + \sqrt{6}y = 2$ touches the hyperbola $x^2 - 2y^2 = 4$, then the point of contact is

[IIT JEE]

(a) $(-2, \sqrt{6})$

(b) $(-5, 2\sqrt{6})$

(c) $\left(\frac{1}{2}, \frac{1}{\sqrt{6}}\right)$

(d) $(4, -\sqrt{6})$

Q 51. If tangents are drawn to the ellipse $x^2 + 2y^2 = 2$, then the locus of the mid point of the intercept made by the tangents between the coordinate axes is

[IIT JEE]

(a) $\frac{1}{2x^2}+\frac{1}{4y^2}=1$

(b) $\frac{1}{4x^2}+\frac{1}{2y^2}=1$

(c) $\frac{x^2}{2}+\frac{y^2}{4}=1$

(d) $\frac{x^2}{4}+\frac{y^2}{2}=1$

Q 52. How many real tangents can be drawn to the ellipse $5x^2+9y^2=32$ from the point $(2,3)$?

[DCE]

(a) 2

(b) 1

(c) 0

(d) 3

Q 53. The conditions that ax + by + c = 0 is tangent to the parabola $y^2 = 4ax$, is

[Kerala CEE]

(a) $a^2 = b^2 = c^2$

(b) a = b

(c) $b^2 = c$

(d) $b^2 = a$

(e) $a^2=b$

Q 54. On the ellipse $4x^2 + 9y^2 = 1$ the point at which the tangent are parallel to $8x = 9y$ are

[KeralaCEE]

(a) $\left(\frac{2}{5},\frac{1}{5}\right) \text{or} \left(-\frac{2}{5},\frac{1}{5}\right)$

(b) $\left(-\frac{2}{5},\frac{1}{5}\right) \text{or} \left(\frac{2}{5},-\frac{1}{5}\right)$

(c) $\left(-\frac{2}{5},-\frac{1}{5}\right)$

(d) $\left(-\frac{3}{5},-\frac{2}{5}\right)\text{or}\left(\frac{3}{5},\frac{2}{5}\right)$

(e) $\left(-\frac{3}{5},\frac{2}{5}\right)\text{or}\left(\frac{3}{5},-\frac{2}{5}\right)$

Q 55. The line y = mx +1 is a tangent to the parabola $y^2 = 4x$, if

[BCECE]

(a) m = 1

(b) m = 2

(c) $m = 4$

(d) $m = 3$

Q 56. The point of intersection of tangents at the ends of the latusrectum of the parabola $y^2 = 4x$, is equal to

[Jamia Millia Islamia]

(a) $(1,0)$

(b) $(-1, 0)$

(c). $(0, 1)$

(d) $(0, -1)$

Q 57. The equation of tangent to the ellipse $x^2 + 4y^2 = 5$ at $(-1,1)$, is

[MHT CET]

(a) $x + 4y + 5 = 0$

(b) $x - 4y - 5 = 0$

(c) $x + 4y - 5 = 0$

(d) $x - 4y + 5 = 0$

Q 58. The angle between the pair of tangents drawn to the ellipse $3x^2 + 2y^2 = 5$ from the point $(1, 2)$ is

[MP PET]

(a) $\tan^{-1}\left(\frac{12}{5}\right)$

(b) $\tan^{-1}(6\sqrt{5})$

(c) $\tan^{-1}\left(\frac{12}{\sqrt{5}}\right)$

(d) $\tan^{-1}(12\sqrt{5})$

Q 59. The value of λ, for which the line

$2x - \frac{8}{3}\lambda y = -3$ is a normal to the conic $x^2 + \frac{y^2}{4} = 1$ is

[MP PET]

(a) $-\frac{\sqrt{3}}{2}$

(b) $\frac{1}{2}$

(c) -3

(d) $\pm\frac{\sqrt{3}}{2}$

Q 60. The straight line x + y = c will be tangent to the ellipse $\frac{x^2}{9}+\frac{y^2}{16}=1$, c is equal to

[RPET]

(a) 8

(b) ±5

(c) +10

(d) ±6

Q 61. The area of the quadrilateral formed by the tangents at the end points of latus rectum to the ellipse $\frac{x^2}{9}+\frac{y^2}{5}=1$, is

[IITJEE]

(a) $27/4$ sq units

(b) 9 sq units

(c) $27/2$ sq units

(d) 27 sq units

Q 62. The normal at the point $(bt^2, 2bt_1)$ on a parabola $y^2 = 4bv$ meets the parabola again in the point $(bt_2^2, 2bt_2)$, then

[AIEEE]

(a) $t_2 = -t_1 - \frac{2}{t_1}$

(b) $t_2 = -t_1 + \frac{2}{t_1}$

(c) $t_2 = t_1 - \frac{2}{t_1}$

(d) $t_2 = t_1 + \frac{2}{t_1}$

Q 63. If the straight line $lx + my + n = 0$ touches the parabola, $y^2 = 4ax$, then

[**BCECE**]

(a) $nm = al^2$

(b) $nl = am^2$

(c) $nl = am$

(d) $ml = an$

Topic 4 Chord of Contact, Pole and Polar, Director Circle

Q 1. The middle point of the chord $x + 3y = 2$ of the conic $x^2 + xy - y^2 = 1$ is

[UPSEE]

(a) $(5,-1)$

(b) $(1,1)$

(c) $(2, 0)$

(d) $(-1,1)$

Q 2. The mid point of the chord $4x - 3y = 5$ of the hyperbola $2x^2 - 3y^2 = 12$ is

[EAMCET]

(a) $\left(0, -\frac{5}{3}\right)$

(b) $(2, 1)$

(c) $\left(\frac{5}{4}, 0\right)$

(d) $\left(\frac{11}{4}, 2\right)$

Q 3. The equation of the chord of the circle $x^2 + y^2 - 4x = 0$, whose mid point is $(1,0)$ is

[WB JEE]

(a) $y = 2$

(b) $y = 1$

(c) $x = 2$

(d) $x = 1$

Q 4. AB is a chord of the parabola $y^2 = 4ax$ with vertex at A. BC is drawn perpendicular to AB

meeting the axes at C. The projection of BC on the axis of the parabola is

[Jamia Millia Islamia]

(a) 2

(b) 2 a

(c) 4a

(d) 8a

Q 5. If $\tan \theta_1$, $\tan \theta_2 = -\frac{a^2}{b^2}$, then the chord joining two points θ_1 and θ_2 on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ will subtend a right angle at

[Jamia Millia Islamia]

(a) focus

(b) centre

(c) end of the major axis

(d) end of the minor axis

Q 6. Locus of mid point of any focal chord of $y^2 = 4ax$ is

[UP SEE]

(a) $y^2 = a(x - 2a)$

(b) $y^2 = 2a(x - 2a)$

(c) $y^2 = 2a(x - a)$

(d) None of these

Q 7. The locus of the middle points of the chords of the parabola $y^2 = 4ax$, which passes through the origin is

[OJEE]

(a) $y^2 = ax$

(b) $y^2 = 2ax$

(c) $y^2 = 4ax$

(d) $x^2 = 4ay$

Q 8. If $P(at^2, 2at)$ be one end of a focal chord of the parabola $y^2 = 4ax$, then the length of the chord is

[WB JEE]

(a) $a\left(t-\frac{1}{t}\right)^2$

(b) $a\left(t-\frac{1}{t}\right)$

(c) $a\left(t+\frac{1}{t}\right)$

(d) $a\left(t+\frac{1}{t}\right)^2$

Q 9. The length of the common chord of the parabolas $y^2 = x$ and $x^2 = y$ and is

[WB JEE]

(a) $2\sqrt{2}$

(b) 1

(c) $\sqrt{2}$

(d) $\frac{1}{\sqrt{2}}$

Q 10. The equation of the director circle of the hyperbola $9x^2 - 16y^2 = 144$ is

[MP PET]

(a) $x^2 + y^2 = 7$

(b) $x^2 + y^2 = 9$

(c) $x^2 + y^2 = 16$

(d) $x^2 + y^2 = 25$

Q 11. Let a focal chord of parabola $y^2 = 16x$ cuts it at points (f, g) and (h, k). Then, $f \cdot h$ is equal to

[OJEE]

(a) 12

(b) 16

(c) 14

(d) None of these

Q 12. The locus of middle points of chords of hyperbola $3x^2 - 2y^2 + 4x - 6y = 0$ parallel to $y = 2x$ is

[WB JEE]

(a) $3x - 4y = 4$

(b) $3y - 4x + 4 = 0$

(c) $4x - 3y = 3$

(d) $3x - 4y = 2$

Q 13. In the parabola $y^2 = 4ax$, the length of the chord passing through the vertex inclined to the axis at $\frac{\pi}{4}$ is

[J&K CET]

(a) $4a\sqrt{2}$

(b) $2a\sqrt{2}$

(c) $a\sqrt{2}$

(d) a

Q 14. The equation of the chord joining two points (x_1, y_1) and (x_2, y_2) on the rectangular hyperbola $xy = c^2$ is

[OJEE]

(a) $\frac{x}{x_1 + x_2} + \frac{y}{y_1 + y_2} = 1$

(b) $\frac{x}{x_1 - x_2} + \frac{y}{y_1 - y_2} = 1$

(c) $\frac{x}{y_1 + y_2} + \frac{y}{x_1 + x_2} = 1$

(d) $\frac{x}{y_1 - y_2} + \frac{y}{x_1 - x_2} = 1$

Q 15. If the chords of contact of tangents from two points (x_1, y_1) and (x_2, y_2) to the hyperbola $4x^2 - 9y^2 - 36 = 0$ are at right angles, then $\frac{x_1x_2}{y_1y_2}$ is equal to

[BCECE]

(a) $\frac{9}{4}$

(b) $-\frac{9}{4}$

(c) $\frac{81}{16}$

(d) $-\frac{81}{16}$

Q 16. The length of the chord of the parabola y^2 = 4ax, which passes through the vertex and makes an angle α with the axis of the parabola is

[RPET]

(a) $4a \cos \alpha \operatorname{cosec}^2 \alpha$

(b) $4 \cos \alpha \operatorname{cosec}^2 \alpha$

(c) $a \cos \alpha \operatorname{cosec}^2 \alpha$

(d) a $\cos^2 \alpha$ cosec α

Q 17. If a focal chord of the parabola $y^2 = ax$ is $2x - y - 8 = 0$, then the equation of the directrix is

[MPPET]

(a) $x + 4 = 0$

(b) $x - 4 = 0$

(c) $y - 4 = 0$

(d) $y + 4 = 0$

Q 18. The equation of the director circle of the hyperbola $\frac{x^2}{16} - \frac{y^2}{4} = 1$, is given by

[KCET]

(a) $x^2 + y^2 = 16$

(b) $x^2 + y^2 = 4$

(c) $x^2 + y^2 = 20$

(d) $x^2 + y^2 = 12$

Q 19. If (x_1, y_1) and (x_2, y_2) are the ends of a focal chord of $y^2 = 4ax$, then $x_1, x_2 + y_1 y_2$ is equal to

[Kerala CEE]

(a) $-3a^2$

(b) $3a^2$

(c) $-4a^2$

(d) $4a^2$

(e) $2a^2$

Q 20. The focal chord to $y^2 = 16x$ is tangent to $(x - 6)^2 + y^2 = 2$, then the possible values of the slope of this chord, are

[IIT JEE]

(a) $\{-1,1\}$

(b) $\{-2,2\}$

(c) $\{-2,1/2\}$

(d) $\{2,-1/2\}$

Q 21. Which of the following is a point on the common chord of the circles $x^2 + y^2 + 2x - 3y + 6 = 0$ and $x^2 + y^2 + x - 8y - 13 = 0$

[KCET]

(a) $(1,4)$

(b) $(1,-2)$

(c) $(1,-4)$

(d) $(1,2)$

Q 22. The locus of the point of intersection n of the perpendicular tangents to ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ is

[KCET]

(a) $x^2 + y^2 = 4$

(b) $x^2 + y^2 = 9$

(c) $x^2 + y^2 = 5$

(d) $x^2 + y^2 = 13$

Topic 5 Diameter, Auxiliary Circle and Asymptotes

Q 1. The line passing through the extremity A of the major axis and extremity B of the minor axis of the ellipse $x^2 + 9y^2 = 9$ meets its auxiliary circle at the point M. Then, the area of the triangle with vertices at A, M and the origin O is

[IIT JEE]

(a) $\frac{31}{10}$

(b) $\frac{29}{10}$

(c) $\frac{21}{10}$

(d) $\frac{27}{10}$

Q 2. Asymptotes of a hyperbola $\frac{x^2}{25}-\frac{y^2}{16}=1$ are

[Guj.CET]

(a) $x=\pm\frac{25}{16}y$

(b) $x=\pm\frac{4}{5}y$

(c) $y=\pm\frac{5}{4}x$

(d) $y = \pm \frac{4}{5} x$

Q 3. The product of perpendiculars drawn from any point of a hyperbola to its asymptotes is

[UP SEE]

(a) $\frac{a^2 b^2}{a^2 + b^2}$

(b) $\frac{a^2 + b^2}{a^2 b^2}$

(c) $\frac{ab}{\sqrt{a} + \sqrt{b}}$

(d) $\frac{ab}{a^2+b^2}$

Q 4. If 2 y = x and 3 y + 4x = 0 are the equations of a pair of conjugate diameters of an ellipse, then the eccentricity of the ellipse is

[WBJEE]

(a) $\sqrt{\frac{2}{3}}$

(b) $\sqrt{\frac{2}{5}}$

(c) $\sqrt{\frac{1}{3}}$

(d) $\sqrt{\frac{1}{2}}$

Q 5. If the area of the auxiliary circle of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ (a > b) is twice the area of the ellipse, then the eccentricity of the ellipse is

[KCET]

(a) $\frac{1}{\sqrt{2}}$

(b) $\frac{\sqrt{3}}{2}$

(c) $\frac{1}{\sqrt{3}}$

(d) $\frac{1}{2}$

Q 6. The value of k, if $(1,2)$, (k, -1) are conjugate points with respect to the ellipse $2x^2 + 3y^2 = 6$, is

[EAMCET]

(a) 2

(b) 4

(c) 6

(d) 8

Q 7. Equation of asymptotes of xy = 7x + 5y are

[DCE]

(a) x = 7,y = 5

(b) x = 5,y = 7

(c) $xy = 35$

(d) None of these

Q 8. The product of the lengths of perpendiculars drawn from any point on the hyperbola $x^2 - 2y^2 - 2 = 0$ to its asymptotes is

[EAMCET]

(a) $1/2$

(b) $2/3$

(c) 3/2

(d) 20

Answers

Topic 1 : Basic Terms of Conics

1. (a) 2. (c) 3. (c) 4. (a) 5. (b) 6. (b) 7. (b) 8. (c)

9. (d) 10. (b)11. (c) 12. (a) 13. (a) 14. (b) 15. (b)

16. (b) 17. (a) 18. (c) 19. (b) 20. (a) 21. (b) 22.

(a) 23. (b) 24. (c) 25. (c) 26. (a) 27. (b) 28. (a)

29. (c) 30. (d) 31. (a) 32. (b) 33. (b) 34. (b) 35.

(a) 36. (b) 37. (a) 38. (d) 39. (c) 40. (c) 41. (d)

42. (e) 43. (c) 44. (b) 45. (b) 46. (a) 47. (c) 48.

(c) 49. (d) 50. (c) 51. (d) 52. (b) 53. (d) 54. (a)

55. (c) 56. (c) 57. (c) 58. (d) 59. (d) 60. (c)

61. (a) 62. (b) 63. (b) 64. (a) 65. (c) 66. (c) 67.

(a) 68. (d) 69. (c) 70. (e) 71. (b) 72. (c) 73. (a)

74. (c) 75. (b) 76. (b) 77. (c) 78. (e) 79. (d) 80.

(e) 81. (d) 82. (e) 83. (a) 84. (a) 85. (a) 86. (b)

87. (d) 88. (a) 89. (b) 90. (a) 91. (c) 92. (c) 93.

(d) 94. (c) 95. (a) 96. (a) 97. (d) 98. (b) 99. (d)

100. (b)

Topic 2 : Equation of Conics and Intersection of Two Curves

1. (i) b (ii) a 2. (a) 3. (a) 4. (a) 5. (b) 6. (d) 7. (b)

8. (b) 9. (c) 10. (b) 11. (b) 12. (c) 13. (b) 14. (d)

15. (d) 16. (d) 17. (c) 18. (b) 19. (a) 20. (b)

21. (a) 22. (c) 23. (c) 24. (e) 25. (b) 26. (a) 27.

(c) 28. (a) 29. (a) 30. (d)

31. (b) 32. (a) 33. (d) 34. (b) 35. (a) 36. (a) 37.

(a) 38. (b) 39. (d) 40. (a) 41. (c) 42. (c) 43. (c)

44. (c) 45. (d) 46. (a) 47. (c) 48. (c) 49. (d) 50.

(b) 51. (a) 52. (b) 53. (a) 54. (d) 55. (c) 56. (d)

Topic 3 : Tangent and Normal of Conic

1. (c,d) 2. (i) d (ii) c (iii) a 3. (c) 4. (c) 5. (c) 6. (b)

7. (b) 8. (a) 9. (d) 10. (c) 11. (b) 12. (c) 13. (d)

14. (a) 15. (b) 16. (c) 17. (b) 18. (c) 19. (d) 20.

(c) 21. (a) 22. (a) 23. (a) 24. (d) 25. (a) 26. (b)

27. (d) 28. (d) 29. (b) 30. (a) 31. (b) 32. (c) 33.

(b) 34. (b) 35. (c) 36. (a) 37. (d) 38. (a) 39. (a)

40. (a) 41. (b) 42. (a) 43. (a) 44. (b) 45. (b) 46.

(a) 47. (b) 48. (c) 49. (d) 50. (d) 51. (a) 52. (a)

53. (c) 54. (b) 55. (a) 56. (b) 57. (d) 58. (c) 59.

(d) 60. (b) 61. (d) 62. (a) 63. (b)

Topic 4 : Chord of Contact, Pole and Polar, Director Circle

1. (d) 2. (b) 3. (d) 4. (c) 5. (b) 6. (c) 7. (b) 8. (d)

9. (c) 10. (a) 11. (b) 12. (a) 13. (a) 14. (a) 15. (d)

16. (b) 17. (a) 18. (d) 19. (a) 20. (a) 21. (c) 22.

(d)

Topic 5 : Diameter, Auxiliary Circle and Asymptotes

1. (d) 2. (d) 3. (a) 4. (c) 5. (b) 6. (c) 7. (b) 8. (b)

Equations and Inequalities

Surds and Indices

Q 1. If $x = \sqrt{7} - \sqrt{5}$ and $y = \sqrt{13} - \sqrt{11}$, then

[UP SEE]

(a) x > y

(b) x < y

(c) x = y

(d) None of these

Q 2. One of the square roots of $6+4\sqrt{3}$ is

[UP SEE]

(a) $\sqrt{3}(\sqrt{3}+1)$

(b) $-\sqrt{3}(\sqrt{3}-1)$

(c) $\sqrt{3}(-\sqrt{3}+1)$

(d) None of these

Q 3. Given that a, b $\in \{0,1,2,\ldots,9\}$ with a + $6 \neq 0$

and that $\left(a+\frac{b}{10}\right)^{x}=\left(\frac{a}{10}+\frac{b}{100}\right)^{y}=1000$. Then, $\frac{1}{x}-\frac{1}{y}$ is equal to

[EAMCET]

(a) 1

(b) $\frac{1}{2}$

(c) $\frac{1}{3}$

(d) $\frac{1}{4}$

Q 4. If $x=\frac{1}{2}\left(\sqrt{7}+\frac{1}{\sqrt{7}}\right)$, then $\frac{\sqrt{x^2-1}}{x-\sqrt{x^2-1}}$ is equal to

[EAMCET]

(a) 1

(b) 2

(c) 3

(d) 4

Q 5. If $x = \sqrt{\frac{2+\sqrt{3}}{2-\sqrt{3}}}$, then $x^2 (x - 4)^2$ is equal to

[BCECE]

(a) 7

(b) 4

(c) 2

(d) 1

Q 6. $\sqrt{2+\sqrt{5}-\sqrt{6-3\sqrt{5}+\sqrt{14-6\sqrt{5}}}}$ is equal to

[EAMCET]

(a) 1

(b) 2

(c) 3

(d) 4

Q 7. The greatest number among $\sqrt[3]{9}, \sqrt[4]{11}, \sqrt[6]{17}$ is

[UP SEE]

(a) $\sqrt[3]{9}$

(b) $\sqrt[4]{11}$

(c) $\sqrt[6]{17}$

(d) Cannot be determined

Q 8. $\sqrt{12-\sqrt{68+48\sqrt{2}}}$ is equal to

[EAMCET]

(a) $\sqrt{2}-3$

(b) $2+\sqrt{2}$

(c) $2-\sqrt{2}$

(d) $6-2\sqrt{8}$

Q 9. Solution of the equation $4.9^{x-1} = 3\sqrt{(2^{2x+1})}$ is

[UP SEE]

(a) 3

(b) 2

(c) $\frac{3}{2}$

(d) $\frac{2}{3}$

Q 10. If $x = 8 + 3\sqrt{7}$ and xy = 1, then the value of

$\frac{1}{x^2} + \frac{1}{y^2}$ is

[Kerala CEE]

(a) 254

(b) 192

(c) 292

(d) 66

(e) 62

Q 11. If $x = \frac{1}{2}\left(\sqrt{3} + \frac{1}{\sqrt{3}}\right)$, then $\frac{\sqrt{x^2 - 1}}{x - \sqrt{x^2 - 1}}$ is equal to

[EAMCET]

(a) 1

(b) 2

(c) 3

(d) ½

Q 12. The value of $\sqrt{42+\sqrt{42+\sqrt{42+.....}}}$ is equal to

[EAMCET]

(a) 7

(b) -6

(c) 5

(d) 4

Q 13. If $x = \sqrt{3018 + \sqrt{36} + \sqrt{169}}$, then the value of x is

[J&KCET]

(a) 55

(b) 44

(c) 63

(d) 42

Topic 2 Logarithm and Its Properties

2010

Q 1. In a right-angled triangle, the sides are a, b and c, with c as hypotenuse, and $c - b \neq 1$, $c + b \neq 1$. Then the value of $(\log_{c+b} a + \log_{c-b} a)/(2 \log_{c+b} a \times \log_{c-b} a)$ will be

[WBJEE]

(a) 2

(b) -1

(c) $\frac{1}{2}$

(d) 1

Q 2. The value of $\frac{\log_3 5 \times \log_{25} 27 \times \log_{49} 7}{\log_{81} 3}$ is

[WBJEE]

(a) 1

(b) 6

(c) $\frac{2}{3}$

(d) 3

2009

Q 3. If $x = \log_a bc$, $y = \log_b ca$, $z = \log_c ab$, then the value of $\frac{1}{1+x}+\frac{1}{1+y}+\frac{1}{1+z}$ will be

[WB JEE]

(a) x + y + z

(b) 1

(c) ab + bc + ca

(d) abc

Q 4. The approximate value of $\sqrt[3]{28}$ is

[MHTCET]

(a) 3.0037

(b) 3.037

(c) 3.0086

(d) 3.37

Q 5. If $2^x \cdot 3^{x+4} = 7^x$, then x is equal to

[MPPET]

(a) $\frac{4\log_e 3}{\log_e 7 - \log_e 6}$

(b) $\frac{4\log_e 3}{\log_e 6 - \log_e 7}$

(c) $\frac{2\log_e 3}{\log_e 7 - \log_e 6}$

(d) $\frac{3\log_e 4}{\log_e 6 - \log_e 7}$

Q 6. The number of solutions of $\log_4 (x - 1) = \log_2 (x - 3)$ is

[UP SEE]

(a) 3

(b) 1

(c) 2

(d) 0

Q 7. If $\log_{10} 7 = 0.8451$, then the position of the first significant figure of 7^{-20} is

[KCET]

(a) 16

(b) 17

(c) 20

(d) 15

Q 8. The value of $\log_2 20 \log_2 80 - \log_2 5 \log_2 320$ is equal to

[Kerala CEE]

(a) 5

(b) 6

(c) 7

(d) 8

(e) 10

Q 9. If $\log_5 \log_5 \log_2 x = 0$, then the value of x is

[WBJEE]

(a) 32

(b) 125

(c) 625

(d) 25

Q 10. If $\log_4 2 + \log_4 .4 + \log_4 x + \log_4 \mathbf{16} = 6$,

then the value of x is

[Jamia Millia Islamia]

(a) 64

(b) 4

(c) 8

(d) 32

Q 11. The value of $7 \log_2 \frac{16}{15} + 5\log\frac{25}{24} + 3\log_2 \frac{81}{80}$ is

[MPPET]

(a) 1

(b) $\log_2 (105)$

(c) $\log_2\left(\frac{9}{8}\right)$

(d) $\log_2\left(\frac{8}{9}\right)$

Q 12. If $a = \log_2 3$, $b = \log_2 5$, $c = \log_7 2$, then $\log_{140} 63$ in terms of a, b, c is

[BITSAT]

(a) $\frac{2ac+1}{2c+abc+1}$

(b) $\frac{2ac+1}{2a+c+a}$

(c) $\frac{2ac+1}{2c+ab+a}$

(d) None of these

Q 13. $7^{2\log_7 5}$ is equal to

[KCET]

(a) $\log_7 35$

(b) 5

(c) 25

(d) $\log_7 25$

Q 14. If $\log_{\sqrt{3}} 5 = a$ and $\log_{\sqrt{3}} 2 = b$, then $\log_{\sqrt{3}} 300$ is equal to

[Kerala CEE]

(a) $2(a + b)$

(b) $2(a + b + 1)$

(c) $2(a + b + 2)$

(d) $a + b + 4$

(e) $a + b + 1$

Q 15. If a, b, c are distinct positive numbers

each being different from 1 such that

[Kerala CEE]

$(\log_b a \cdot \log_c a - \log_a a) + (\log_a b \log_c b - \log_b b) +$

$(\log_a c \cdot \log_b c - \log_c c) = 0$, then abc is

(a) 0

(b) e

(c) 1

(d) 2

(e) 3

Q 16. If $\log_e \left(\frac{a+b}{2}\right) = \frac{1}{2}(\log_e a + \log_e b)$, then

[AMU]

(a) a = b

(b) $a = \frac{b}{2}$

(c) 2a = b

(d) $a = \frac{b}{3}$

Q 17. If $a^x = b^y = c^z = d^w$, then the, value of x

$\left(\frac{1}{y} + \frac{1}{z} + \frac{1}{w}\right)$ is

[EAMCET]

(a) $\log_a (abc)$

(b) $\log_a (bcd)$

(c) $\log_b (cda)$

(d) $\log_c (dab)$

Q 18. If $y = 2^{1/\log_x (8)}$, then x is equal to

[WBJEE]

(a) y

(b) y^2

(c) y^3

(d) None of these

Q 19. If $x > 0$ and $\log_3 x + \log_3 (\sqrt{x}) + \log_3 (\sqrt[3]{x}) + \log_3 (\sqrt[8]{x}) + \log_3 (\sqrt[16]{x}) + \ldots = 4$, then x equals

(a) 9

(b) 81

(c) 1

(d) 27

[VTTEEE]

Q 20. If $\log_{0.3}(x-1) < \log_{0.09}(x-1)$, then x lies in the interval

[DCE]

(a) $(2, \infty)$

(b) $(1,2)$

(c) $(-2,-1)$

(d) None of these

Q 21. The value of $\log_2 \log_2 \log_4 256 + {}_{2\log_{\sqrt{2}} 2}$ is

[Kerala CEE]

(a) 1

(b) 2

(c) 3

(d) 4

(e) 5

Q 22. Let n = !. Then $\frac{1}{\log_2 n}+\frac{1}{\log_3 n}+.....+\frac{1}{\log_{2006} n}$ is

equal to

[Kerala CEE]

(a)

(b)

(c) !

(d) 1

(e) 0

Q 23. If $\frac{\log x}{a-b} = \frac{\log y}{b-c} = \frac{\log z}{c-a}$, then xyz is equal to

[AMU]

(a) 0

(b) 1

(c) −1

(d) 2

Q 24. The solution of $\log_{99} \{\log_2 (\log_3 x)\} = 0$ is

[BCECE]

(a) 4

(b) 9

(c) 44

(d) 99

Q 25. The solution set of the equation

$$\left[4\left(1-\frac{1}{3}+\frac{1}{9}-\frac{1}{27}+.....\right)\right]^{\log_2 x}=\left[54\left(1+\frac{1}{3}+\frac{1}{9}+\frac{1}{27}+....\right)\right]^{\log_x 2}$$ is

[BCECE]

(a) $\left\{4, \frac{1}{4}\right\}$

(b) $\left\{2, \frac{1}{2}\right\}$

(c) $\{1, 2\}$

(d) $\left\{8, \frac{1}{8}\right\}$

Q 26. The value of $\log_2 \log_3 .... \log_{100} 100^{99^{98^{\cdots^{2^1}}}}$ is equal to

[AMU]

(a) 0

(b) 1

(c) 2

(d) $100!$

Q 27. If a, b, c $\neq 0$ and belong to the set $\{0, 1, 2, 3, \ldots\ldots, 9\}$, then $\log_{10}\left(\frac{a+10b+10^2c}{10^{-4}a+10^{-3}b+10^{-2}c}\right)$ is equal to

(a) 1

(b) 2

(c) 3

(d) 4

[EAMCET]

Q 28. If x, y, z are in GP and $a^x = b^y = c^z$, then

[BCECE]

(a) $\log_a c = \log_b a$

(b) $\log_b a = \log_c b$

(c) $\log_c 6 = \log_a c$

(d) None of the above

Q 29. If $\log_{27}(\log_3 x) = \frac{1}{3}$, then the value of x is

[EAMCET]

(a) 3

(b) 6

(c) 9

(d) 27

Q 30. The solution set of the equation $x^{\log_x (1-x)^2} = 9$

is

[Jamia Millia Islamia]

(a) $\{-2,4\}$

(b) $\{4\}$

(c) $\{0, -2, 4\}$

(d) None of these

Q 31. If correlations = , evaluate

[J & K CET]

$$\frac{1}{\log_2 n!}+\frac{1}{\log_3 n!}+\frac{1}{\log_4 n!}+.....+\frac{1}{\log_{2002} n!}$$

(a) 1

(b) 2

(c) 3

(d) 4

Q 32. If $x = \log_b a$, $y = \log_c b$, $z = \log_a c$, then xyz is

[UP SEE]

(a) 0

(b) 1

(c) 3

(d) None of the above

Q 33. If $\log_2 [\log_3 \{\log_4 (\log_5 x)\}] = 0$, then the value of x is

[J&KCET]

(a) 5^{24}

(b) 1

(c) 2^{25}

(d) 5^{64}

Topic 3 Relation between Roots and Coefficients

2010

Q 1. Let p and q be real numbers such that $p \neq 0$,

$p^3 \neq q$ and $p^3 \neq -q$. If α and β are non-zero

complex numbers satisfying $\alpha + \beta = -p$ and $\alpha^3 +$

$\beta^3 = q$, then a quadratic equation having $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its

foots is

[IIT JEE]

(a) $(p^3 + q)x^2 - (p^3 + 2q)x + (p^3 + q) = 0$ (b) $(p^3 +$

$q)x^2 - (p^3 - 2q)x + (p^3 + q) = 0$

(c) $(p^3 - q)x^2 - (5p^3 - 2q)x + (p^3 - q) = 0$ (d) $(p^3$

$- q)x^2 - (5p^3 + 2q)x + (p^3 - q) = 0$

Q 2.If α and β are the roots of the equation $x^2 - x + 1 = 0$ then $\alpha^{2009} + \beta^{2009}$ is equal to

[AIEEE]

(a) -2

(b) -1

(c) 1

(d) 2

Q 3. If α, β be the roots of the quadratic equation

$x^2 + x + 1 = 0$, then the equation whose roots are

α^{19}, β^7 is

[WB JEE]

(a) $x^2 - x + 1 = 0$

(b) $x^2 - x - 1 = 0$

(c) $x^2 + x - 1 = 0$

(d) $x^2 + x + 1 = 0$

Q 4. If one root of the equation $x^2 + px + q = 0$ is

$2+\sqrt{3}$, then the value of p and q are respectively

(a) $-4,1$

(b) $4,-1$

(c) $2,\sqrt{3}$

(d) $-2,-\sqrt{3}$

[VITEEE]

Q 5. If a and bare the roots of the equation x +ax + b = 0, a $\neq 0$, b $\neq 0$, then the values of a and b are respectively

[Kerala CEE]

(a) 2 and -2

(b) 2 and -1

(c) 1 and -2

(d) 1 and 2

(e) -1 and 2

Q 6. If α and β are the roots of the equation ax

$+bx + c = 0, (c \neq 0)$, then the equation whose roots

are $\frac{1}{a\alpha+b}$ and $\frac{1}{a\beta+b}$ is

[BITSAT]

(a) $acx^2 - bx + 1 = 0$

(b) $x^2 - acx + bc + 1 = 0$

(c) $acx^2 + bx - 1 = 0$

(d) $x^2 + acx - bc + 11 = 0$

(e) $acx^2 - bx - 11 = 0$

Q 7. If the roots of the equation $x^2 - bx + c = 0$ are two consecutive integers, then $b^2 - 4c$ is

[Kerala CEE]

(a) -1

(b) 0

(c) 1

(d) 2

(e) 3

Q 8. If $x + px + q = 0$ has the roots then α and β, the value of $(\alpha - \beta)^2$ is equal to

[VITEEE]

(a) $p^2 - 4q$

(b) $(p^2 - 4q)^2$

(c) $p^2 + 4q$

(d) $(p^2 + 4q)^2$

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Q 9. If α, β are the roots of the equation $\lambda(x^2 - x)$ + x + 5 = 0 and if λ_1 and λ_2 are two values of λ obtained from $\frac{\alpha}{\beta}+\frac{\beta}{\alpha}=\frac{4}{5}$, then $\frac{\lambda_1}{\lambda_2^2}+\frac{\lambda_2}{\lambda_1^2}$ is equal to

[DCE]

(a) 4192

(b) 4144

(c) 4096

(d) 4048

Q 10. If the sum of the roots of the equation $ax^2 + 2x + 3a = 0$ is equal to their product, then value of a is

[BITSAT]

(a) $-\frac{2}{3}$

(b) -3

(c) 4

(d) $-\frac{1}{2}$

Q 11. If one root of equation $x^2 + ax + 12 = 0$ is 4 while the equation $x^2 + ax + b = 0$ has equal roots, then the value of b is

[UP SEE]

(a) $\frac{4}{49}$

(b) $\frac{49}{4}$

(c) $\frac{7}{4}$

(d) $\frac{4}{7}$

Q 12. If the roots of the equation $\frac{1}{x+p}+\frac{1}{x+q}=\frac{1}{r}$, ($x \neq -p$, $x \neq -q$, $r \neq 0$) are equal in magnitude but opposite in sign, then p + q is equal to

[Kerala CEE]

(a) r

(b) $2r$

(c) r^2

(d) $\frac{1}{r}$

(e) $\frac{2}{r}$

Q 13. If $2 - i$ is a root of the equation $ax^2 + 12x + b = 0$ (where a and b are real), then the value of ab is equal to

[Kerala CEE]

(a) 45

(b) 15

(c) −15

(d) −45

(e) 25

Q 14. If one root of the equation $lx^2 + mx + n = 0$ is $\frac{9}{2}$(l, m and n are positive integers) and $\frac{m}{4n} = \frac{1}{m}$, then l + n is equal to

[Kerala CEE]

(a) 80

(b) 85

(c) 90

(d) 95

(e) 100

Q 15. If α and β are the roots of $x^2 - 2x + 4 = 0$, then the value of $\alpha^6 + \beta^6$ is

[EAMCET]

(a) 32

(b) 64

(c) 128

(d) 256

Q 16. If α, β be the roots of $x^2 - a(x-1) + b = 0$, then the value of $\frac{1}{\alpha^2 - a\alpha} + \frac{1}{\beta^2 - a\beta} + \frac{2}{a+b}$ is

[WBJEE]

(a) $\frac{4}{a+b}$

(b) $\frac{1}{a+b}$

(c) 0

(d) -1

Q 17. If α, β, γ are such that $\alpha + \beta + \gamma = 2$, $\alpha^2 + \beta^2 + \gamma^2 = 6$, $\alpha^3 + \beta^3 + \gamma^3 = 8$, then' $\alpha^4 + \beta^4 + \gamma^4$ is equal to

[BCECE]

(a) 7

(b) 12

(c) 18

(d) 36

Q 18. If x_1, x_2, x_3, x_4 are roots of the equation $x^4 - x^3 \sin 2\beta + x^2 \cos 2\beta - x \cos \beta - \sin \beta = 0$, then $\tan^{-1} x_1 + \tan^{-1} x_2 + \tan^{-1} x_3 + \tan^{-1} x_4$ is equal to

[Jamia Millia Islamia]

(a) β

(b) $\frac{\pi}{2} - \beta$

(c) $\pi - \beta$

(d) $-\beta$

Q 19. Let the two numbers have arithmetic mean 9 and geometric mean 4. Then, these numbers are the roots of the quadratic equation

[DCE]

(a) $x^2 - 18x - 16 = 0$

(b) $x^2 - 18x + 16 = 0$

(c) $x^2 + 18x - 16 = 0$

(d) $x^2 + 18x + 16 = 0$

Q 20. If $\alpha + \beta = -2$ and $\alpha^3 + \beta^3 = -56$, then the quadratic equation whose roots are α and β is

[BITSAT, EAMCET]

(a) $x^2 + 2x - 16 = 0$

(b) $x^2 + 2x + 15 = 0$

(c) $x^2 + 2x - 12 = 0$

(d) $x^2 + 2x - 8 = 0$

Q 21. If α, β are the roots of the equation $ax^2 + bx + c = 0$, then $\frac{\alpha}{a\beta + b} + \frac{\beta}{a\alpha + b}$ is equal to

(a) $\frac{2}{a}$

(b) $\frac{2}{b}$

(c) $\frac{2}{c}$

(d) $-\frac{2}{a}$

Q 22. Suppose that two persons A and B solve the equation $x^2 + ax + b = 0$. While solving A commits a mistake in the coefficient of x was taken as 15 in

place of -9 and finds the roots as -7 and -2.

Then, the equation is

[Kerala CEE]

(a) $x^2 + 9x + 14 = 0$

(b) $x^2 - 9x + 14 = 0$

(c) $x^2 + 9x - 14 = 0$

(d) $x^2 - 9x - 14 = 0$

(e) None of these

Q 23. The cubic equation whose roots are thrice to each of the roots of $x^3 + 2x^2 - 4x + 1 = 0$ is

[AMU]

(a) $x^3 - 6x^2 + 36x + 27 = 0$

(b) $x^3 + 6x^2 + 36x + 27 = 0$

(c) $x^3 - 6x^2 - 36x + 27 = 0$

(d) $x^3 + 6x^2 - 36x + 27 = 0$

Q 24. Let α and β be the roots of the quadratic equation $ax^2 + bx + c = 0$. Observe the lists given below

[EAMCET]

List I	List II
(i) $\alpha = \beta$	(A) $(ac^2)^{1/3} + (a^2c)^{1/3} + b = 0$
(ii) $\alpha = 2\beta$	(B) $2b^2 = 9ac$
(iii) $\alpha = 3\beta$	(C) $b^2 = 6ac$

(iv) $\alpha = \beta^2$	(D) $3b^2 = l6ac$
	(E) $b^2 = 4ac$
	(F) $(ac^2)^{1/3} + (a^2c)^{1/3} = b$

The correct match of List I from List II is

(i) (ii) (iii) (iv)

(a) E B D F

(b) E B A D

(c) E D B F

(d) E B D A

Q 25. Let α, β be the roots of $x^2 - 2x \cos \phi + 1 = 0$, then the equation whose roots are α^n, β^n is

[WBJEE]

(a) $x^2 - 2x \cos n\phi - 1 = 0$

(b) $x^2 - 2x \cos n\phi + 1 = 0$

(c) $x^2 - 2x \sin n\phi + 1 = 0$

(d) $x^2 + 2x \sin n\phi - 1 = 0$

Q 26. If α, β, γ are the roots of the equation $x^3 - 6x^2 + 11x + 6 = 0$, then $\Sigma\alpha^2\beta + \Sigma\alpha\beta^2$ is equal to

[BCECE, EAMCET]

(a) 80

(b) 84

(c) 90

(d) −84

Q 27. Let α, β be the roots of the equation $x^2 - ax + b = 0$ and $A_n = \alpha^n + \beta^n$. Then, $A_{n+1} - a A_n + bA_{n-1}$ is equal to

[VITEEE]

(a) $-a$

(b) b

(c) 0

(d) $a - b$

Q 28. If α, β and γ are the roots of the equation

$x^3 - 8x + 8 = 0$, then $\sum\alpha^2$ and $\sum\frac{1}{\alpha\beta}$ are respectively

[Jamia Millia Islamia]

(a) 0 and -16

(b) 16 and 18

(c) -16 and 0

(d) 16 and 0

Q 29. If α, β are the roots of the equation $x^2 - (1 + n^2)x + \frac{1}{2}(1 + n^2 + n^4)$ then $\alpha^2 + \beta^2$ is

[MPPET]

(a) n^2

(b) $-n^2$

(c) n^4

(d) $-n^4$

Q 30. If α, β are the roots of the equation $x^2 + ax + b = 0$, then $\frac{1}{\alpha^2}+\frac{1}{\beta^2}$ is equal to

[RPET]

(a) $\frac{a^2-2b}{b^2}$

(b) $\frac{b^2-2a}{b^2}$

(c) $\frac{a^2+2b}{b^2}$

(d) $\frac{b^2+2a}{b^2}$

Q 31. If α and β are the roots of the equation

$ax^2 + bx + c = 0$, $\alpha\beta = 3$ and a, b, c are in AP,

then $\alpha + \beta$ is equal to

[Guj.CET]

(a) -4

(b) 1

(c) 4

(d) -2

Q 32. Let α, β be the roots of the equation $x^2 - px + r = 0$ and $\frac{\alpha}{2}, 2\beta$ be the roots of the equation $x^2 - qx + r = 0$. Then the value of r is

[IITJEE]

(a) $\frac{2}{9}(p-q)(2q-p)$

(b) $\frac{2}{9}(q-p)(2p-q)$

(c) $\frac{2}{9}(q-2p)(2q-p)$

(d) $\frac{2}{9}(2p-q)(2q-p)$

Q 33. If one root of the quadratic equation $ax^2 + bx + c = 0$ is equal to nth power of the other root, then the value of $(ac^n)^{1/(n+1)} + (a^n c)^{1/(n+1)}$ is equal to

[BITSAT]

(a) b

(b) –b

(c) $\frac{1}{b^{n+1}}$

(d) $-b^{\frac{1}{n+1}}$

Q 34. If α, β are the roots of the equation $lx^2 + mx + n = 0$, then the equation whose roots are $\alpha^3\beta$ and $\alpha\beta^3$, is

[UP SEE]

(a) $l^4 x^2 - nl(m^2 - 2nl)x + n^4 = 0$

(b) $l^4x^2 + nl(m^2 - 2nl)x + n^4 = 0$

(c) $l^4x^2 + nl(m^2 - 2nl)x - n^4 = 0$

(d) $l^4x^2 - nl(m^2 + 2nl)x + n^4 = 0$

Q 35. If sin α and cos α are the roots of the equation $px^2 + qx + r = 0$, then

[UPSEE]

(a) $p^2 + q^2 - 2pr = 0$

(b) $p^2 - q^2 + 2pr = 0$

(c) $p^2 - q^2 - 2pr = 0$

(d) $p^2 + q^2 + 2qr = 0$

Q 36. If sec α and cosec α are the roots of the equation $x^2 - px + q = 0$, then

[Kerala CEE]

(a) $p^2 = p + 2q$

(b) $q^2 = p + 2q$

(c) $p^2 = q(q+2)$

(d) $q^2 = p(p+2)$

(e) $p^2 = q(q-2)$

Q 37. If α and β are the roots of the equation $x^2 + px + q = 0$ and if the sum … exists, then it is equal to

(a) $\log(x^2 + px + q)$

(b) $\log(x^2 - px + q)$ (c) $\log(1 + px + qx^2)$

(d) $\log(1 - px + qx^2)$

(e) $\log(x^2 + qx + p)$

[Kerala CEE]

Q 38. If one root is square of the other root of the equation $x^2 + px + q = 0$, then the relation between p and q is

[AMU]

(a) $p^3 - (3p - 1)q + q^2 = 0$

(b) $p^3 - q(3p + 1) + q^2 = 0$

(c) $p^3 + q(3p - 1) + q^2 = 0$

(d) $p^3 + q(3p + 1) + q^2 = 0$

Q 39. If α and β are the roots of the equation

$ax^2 + bx + c = 0$ and, if $px^2 + qx + r = 0$ has roots

$\frac{1-\alpha}{\alpha}$ and $\frac{1-\beta}{\beta}$, then r is equal to

[EAMCET]

(a) a +2b

(b) a + b + c

(c) ab + bc + ca

(d) abc

Q 40. If α, β, γ are the roots of $x^3 - 2x^2 + 3x - 4 = 0$, then the value of $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2$ is

[EAMCET]

(a) -7

(b) -5

(c) -3

(d) 0

Q 41. Let α, α^2 be the roots of $x^2 + x + 1 = 0$, then the equation whose roots are α^{31}, α^{62} is

[WBJEE]

(a) $x^2 - x + 1 = 0$

(b) $x^2 + x - 1 = 0$

(c) $x^2 + x + 1 = 0$

(d) $x^{60} + x^{30} + 1 = 0$

Q 42. If α and β are the roots of $x^2 + 5x + 4 = 0$,

then equation whose roots are $\frac{\alpha+2}{3}, \frac{\beta+2}{3}$ is

[MPPET]

(a) $9x^2 + 3x + 2 = 0$

(b) $9x^2 - 3x - 2 = 0$

(c) $9x^2 + 3x - 2 = 0$

(d) $9x^2 - 3x + 2 = 0$

Q 43. If the difference between the roots of $x^2 + ax - b = 0$ is equal to the difference between the roots of $x^2 - px + q = 0$, then $a^2 - p^2$ in terms of b and q is

[RPET]

(a) $-4(b + q)$

(b) $4(b + q)$

(c) $4(b - q)$

(d) $4(q - b)$

Q 44. If α, β, γ are the roots of the equation $x^3 - 7x + 7 = 0$, then $\frac{1}{\alpha^4}+\frac{1}{\beta^4}+\frac{1}{\gamma^4}$ is

[J&KCET]

(a) $7/3$

(b) $3/7$

(c) $4/7$

(d) $7/4$

Q 45. If α, β are the roots of the equation $ax^2 + bx + c = 0$, then the equation whose roots are $2\alpha + 3\beta$ and $3\alpha + 2\beta$, is

[Guj. CET]

(a) $acx^2 + (a + c)bx - (a + c)^2 = 0$

(b) $acx^2 - (a + c)bx + (a + c)^2 = 0$

(c) $abx^2 - (a + b)cx + (a + b)^2 = 0$

(d) None of the above

Q 46. If the roots of the quadratic equation $x^2 + px + q = 0$ are $\tan 30°$ and $\tan 15°$ respectively, then the value of $2 + q - p$ is

[AIEEE]

(a) 3

(b) 0

(c) 1

(d) 2

Q 47. If α, β, γ are the roots of the equation $2x^3 - 3x^2 + 6x + 1 = 0$, then $\alpha^2 + \beta^2 + \gamma^2$ is equal to

[BITSAT, Guj. CET]

(a) $-\frac{15}{4}$

(b) $\frac{15}{4}$

(c) $\frac{9}{4}$

(d) 4

Q 48. If α, β are the roots of the equation $6x^2 - 5x + 1 = 0$, then the value of $\tan^{-1}\alpha + \tan^{-1}\beta$ is

(a) 0

(b) $\pi/4$

(c) 1

(d) $\pi/2$

[UPSEE]

Q 49. If are α_1, α_2, α_3, α_4 the roots of the equation $x^4 + (2-\sqrt{3})x^2 + 2 + \sqrt{3} = 0$, then the value of $(1 - \alpha_1)(1 - \alpha_2)(1 - \alpha_3)(1 - \alpha_4)$ is

[Kerala CEE]

(a) 1

(b) 4

(c) $2+\sqrt{3}$

(d) 5

(e) 0

Q 50. If a, b and care in geometric progression and the roots of the equation $ax^2 + 2bx + c = 0$ are α and β and those of $cx^2 + 2bx + a = 0$ are γ and δ, then

[Kerala CEE]

(a) $\alpha \neq \beta \neq \gamma \neq \delta$

(b) $\alpha \neq \beta$ and $\gamma \neq \delta$

(c) $a\alpha = a\beta = c\gamma = c\delta$

(d) $\alpha = \beta$ and $\gamma \neq \delta$

(e) $\alpha \neq \beta$ and $\gamma \neq \delta$

Q 51. If α and β are the roots of the equation $x^2 - 7x + 1 = 0$, then the value of $\frac{1}{(\alpha-7)^2} + \frac{1}{(\beta-7)^2}$ is

[Kerala CEE]

(a) 45

(b) 47

(c) 49

(d) 50

(e) 51

Q 52. The quadratic equation whose roots are $\sin^2 18°$ and $\cos^2 36°$, is

[EAMCET]

(a) $16x^2 - 12x + 1 = 0$

(b) $16x^2 + 12x + 1 = 0$

(c) $16x^2 - 12x - 1 = 0$

(d) $16x^2 + 10x + 1 = 0$

Q 53. If $\alpha \neq \beta$ and $\alpha^2 = 5\alpha - 3$, $\beta^2 = 5\beta - 3$, then the equation having α/β and β/α as its roots is

[OJEE]

(a) $3x^2 + 19x + 3 = 0$

(b) $3x^2 - 19x + 3 = 0$

(c) $3x^2 - 19x - 3 = 0$

(d) $x^2 - 16x + 1 = 0$

Q 54. If ω and ω^2 are the two imaginary cube roots of unity, then the equation whose roots are $a\omega^{317}$ and $a\omega^{382}$, is

[Jamia Millia Islamia]

(a) $x^2 + ax + a^2 = 0$

(b) $x^2 + a^2x + a = 0$

(c) $x^2 - ax + a^2 = 0$

(d) $x^2 - a^2x + a = 0$

Q 55. The value of a for which the sum of the squares of the roots of the equation $x^2 - (a - 2)x - a - 1 = 0$ assumes the least value, is

[BCECE]

(a) 0

(b) 1

(c) 2

(d) 3

Q 56. If are α, β, γ the roots of $x^3 + bx + c = 0$, then $\alpha^2\beta + \alpha\beta^2 + \beta^2\gamma + \beta\gamma^2 + \gamma^2\alpha + \gamma\alpha^2$ is equal to

(a) c

(b) $-c$

(c) $-3c$

(d) $3c$

[RPET]

Q 57. If a, $c \neq 0$ and α, β are the roots of the equation $ax^2 + bx + c = 0$, then the quadratic equation with $1/\alpha$ and $1/\beta$ as its root is

[J&KCET]

(a) $x^2/a + x/b + 1/c = 0$

(b) $cx^2 + bx + a = 0$

(c) $bx^2 + cx + a = 0$

(d) $ax^2 + cx + b = 0$

Q 58. If $\sin\theta + \cos\theta = h$, then the quadratic equation having $\sin\theta$ and $\cos\theta$ as its roots, is

[J&KCET]

(a) $x^2 - hx + (h^2 - 1) = 0$

(b) $2x^2 - 2hx + (h^2 - 1) = 0$

(c) $x^2 - hx + 2(h^2 - 1) = 0$

(d) $x^2 - 2hx + (h^2 - 1) = 0$

Q 59. Let $f(x) = ax^2 + bx + c$, $a \neq 0$ and $\Delta = b^2 - 4ac$. If $\alpha + \beta$, $\alpha^2 + \beta^2$ and $\alpha^3 + \beta^3$ are in GP, then

(a) $\Delta \neq 0$

(b) $b\Delta = 0$

(c) $c\Delta = 0$

(d) $bc \neq 0$

[IITJEE]

Q 60. In a triangle PQR, $\angle R = \frac{\pi}{2}$. If

$\tan\left(\frac{P}{2}\right)$ and $\tan\left(\frac{Q}{2}\right)$ are the roots of $ax^2 + bx + c = 0$, $a \neq 0$, then

(a) b = a + c

(b) b = c

(c) c = a + b

(d) a = b + c

[AIEEE]

Q 61. The condition that $x^3 - px^2 + qx - r = 0$ may have two of its roots equal in magnitude but of opposite sign, is

[DCE]

(a) $r = pq$

(b) $r = 2p^3 + pq$

(c) $r = p^2q$

(d) None of the above

Q 62. If $3p^2 = 5p + 2$ and $3q^2 - 5q + 2$ where p $\neq$ q, then the equation whose roots are $3p - 2q$ and $3q - 2p$ is

[Kerala CEE]

(a) $3x^2 - 5x - 100 = 0$

(b) $5x^2 + 3x + 100 = 0$

(c) $3x^2 - 5x + 100 = 0$

(d) $3x^2 + 5x - 100 = 0$

(e) $5x^2 - 3x - 100 = 0$

Q 63. If α and β are the roots of the equation $x^2 - 6x + a = 0$ and satisfy the relation $3\alpha + 2\beta = 16$, then the value of a is

[Kerala CEE]

(a) -8

(b) 8

(c) -16

(d) 9

(e) None of these

Q 64. If α, β, γ are the roots of $x^3 + 2x^2 - 3x - 1 = 0$, then $\alpha^{-2} + \beta^{-2} + \gamma^{-2}$ is equal to

[OJEE]

(a) 12

(b) 13

(c) 14

(d) 15

Q 65. Let α, β are the roots of equation $2x^2-(p+1)x+(p-1)=0$. If $\alpha-\beta=\alpha\beta$, then what is the value of p ?

[OJEE]

(a) 1

(b) 2

(c) 3

(d) -2

Q 66. If the sum of two of the roots of $x^3 + px^2 - qx + r = 0$ is zero, then pq is equal to

[BCECE]

(a) $-r$

(b) r

(c) $2r$

(d) $-2r$

Q 67. If the sum of the roots of the equation $ax^2 + bx + c = 0$ be equal to the sum of the reciprocal of their squares, then bc^2, ca^2, ab^2 will be in

[BCECE]

(a) AP

(b) GP

(c) HP

(d) None of these

Q 68. If α and β are the solutions of the quadratic equation $ax^2 + bx + c = 0$ such that $\beta = \alpha^{1/3}$, then

[MPPET]

(a) $(ac)^{1/3} + (ab)^{1/3} + c = 0$

(b) $(a^3b)^{1/4} + (ab^3)^{1/4} + c = 0$

(c) $(a^3c)^{1/4} + (ac^3)^{1/4} + b = 0$

(d) $(a^4c)^{1/3} + (ac^4)^{1/3} + b = 0$

Q 69. $\sqrt{4}, \sqrt[4]{4}, \sqrt[8]{4}\text{m}, \sqrt[16]{4}, \ldots\ldots\ldots \text{to } \infty$ are root of the equation

[RPET]

(a) $x^2 - 4 = 0$

(b) $x^2 - 4x + 6 = 0$

(c) $x^2 - 5x + 4 = 0$

(d) $x^2 - 3x + 2 = 0$

Q 70. If α, β and γ are the roots of equation $x^3 - 3x^2 + x + 5 = 0$, then $y = \Sigma\alpha^2 + \alpha\beta\gamma$ satisfies the equation

[J&K CET]

(a) $y^3 + y + 2 = 0$

(b) $y^3 - y^2 - y - 2 = 0$

(c) $y^3 + 3y^2 - y - 3 = 0$

(d) $y^2 + 4y^2 + 5y + 20 = 0$

Q 71. The quadratic equation in x such that the arithmetic mean of its roots is 5 and geometric mean of the roots is 4, is given by

[Kerala CEE]

(a) $x^2 + 20x + 16 = 0$

(b) $x^2 - 10x + 16 = 0$

(c) $x^2 + 10x + 16 = 0$

(d) $x^2 - 10x - 16 = 0$

(e) $x^2 + 20x + 32 = 0$

Q 72. If one of the roots of the equation $x^2 + bx + 3 = 0$ is thrice the other, then b is equal to

[Kerala CEE]

(a) ± 3

(b) ± 2

(c) 0

(d) ± 4

(e) ± 1

Q 73. If α, β are the roots of the equation $ax^2 + bx + c = 0$, then the value of $\frac{1}{a\alpha + b} + \frac{1}{a\beta + b}$ is equal to

[Kerala CEE]

(a) $\frac{ac}{b}$

(b) 1

(c) $\frac{ab}{c}$

(d) $\frac{bc}{a}$

(d) $\frac{b}{ac}$

Q 74. If the roots of the equation $4x^3 - 12x^2 + 11x + k = 0$ are in arithmetic progression, then k is equal to

[EAMCET]

(a) -3

(b) 1

(c) 2

(d) 3

Q 75. Let α, β, γ are the roots of the equation $x^3 - 10x^2 + 7x + 8 = 0$. Match the following and choose the correct answer.

[EAMCET]

(A) $\alpha + \beta + \gamma$	(1) $-\frac{43}{4}$
(B) $\alpha^2 + \beta^2 + \gamma^2$	(2) $-\frac{7}{8}$
(C) $\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}$	(3) 86
(D) $\frac{\alpha}{\beta\gamma}+\frac{\beta}{\gamma\alpha}+\frac{\gamma}{\alpha\beta}$	(4) 0

(5) 10

	A	B	C	D
(a)	5	3	1	2
(b)	4	3	1	2
(c)	5	3	2	1
(d)	5	2	3	1

Q 76. If α and β are the roots of the equation $ax^2 + bx + c = 0$, then $(1 + \alpha + \alpha^2)(1 + \beta + \beta^2)$ is equal to '

[BCECE]

(a) zero

(b) positive

(c) negative

(d) None of these

Q 77. If α, β, γ are the roots of the equation $x^3 + x + 1 = 0$, then the value of $\alpha^3 + \beta^3 + \gamma^3$ is

[MPPET]

(a) 0

(b) 3

(c) -3

(d) -1

Q 78. Given that tan A and tan B are' the roots of $x^2 - px + q = 0$, then the−value of $\sin^2 (A + B)$ is

(a) $\frac{p^2}{p^2 + (1-q)^2}$

(b) $\frac{q^2}{p^2 + q^2}$

(c) $\frac{q^2}{p^2-(1-q^2)}$

(d) $\frac{p^2}{p^2+q^2}$

[RPET]

Q 79. For the equation $\frac{1}{x+a}-\frac{1}{x+b}=\frac{1}{x+c}$, if the product of the roots is zero, then the sum of the roots is

[J&K CET]

(a) 0

(b) $\frac{2ab}{b+c}$

(c) $\frac{2bc}{b+c}$

(d) $-\frac{2bc}{b+c}$

Q 80. If the roots of the equation $3x^2 - 6x + 5 = 0$ are α and β, then the equation whose roots are $\alpha + \beta$ and $\frac{2}{\alpha+\beta}$ will be

[J&KCET]

(a) $x^2 + 3x - 1 = 0$

(b) $x^2 + 3x - 2 = 0$

(c) $x^2 + 3x + 2 = 0$

(d) $x^2 - 3x + 2 = 0$

Q 81. If the sum of the roots of the quadratic equation $ax + bx + c = 0$ is equal to the sum of the squares of their reciprocals, then $\frac{a}{c}, \frac{b}{a} \text{ and } \frac{c}{b}$ are in

[AIEEE]

(a) arithmetic progression

(b) geometric progression

(c) harmonic progression

(d) arithmetico–geometric progression

Q 82. The value of a for which one root of the quadratic equation $(a^2 - 5a + 3)x^2 + (3a - 1)x + 2 = 0$ is twice as large as the other, is

[AIEEE]

(a) $2/3$

(b) $-2/3$

(c) $1/3$

(d) $-1/3$

Q 83. If α, β, γ are the roots of the equation $x^3 + 4x - 1 = 0$, then $(\alpha + \beta)^{-1} + (\beta + \gamma)^{-1} + (\gamma + \alpha)^{-1}$ is equal to

[UP SEE, EAMCET]

(a) 2

(b) 3

(c) -4

(d) 5

Q 84. If $\alpha + \beta = 4$ and $\alpha^3 + \beta^3 = 44$, then α, β are the roots of the equation

[Kerala CEE]

(a) $2x^2 - 7x + 6 = 0$

(b) $3x^2 + 9x + 11 = 0$

(c) $9x^2 - 27x + 20 = 0$

(d) $4x^2 + 22x + 15 = 0$

(e) $3x^2 - 12x + 5 = 0$

Q 85. If α and β are imaginary cube roots of unity, then $\alpha^4 + \beta^4 + \frac{1}{\alpha\beta}$ is equal to

[AMU]

(a) 3

(b) 0

(c) 1

(d) 2

Topic 4 Solution of Equations, Nature of Roots

2010

Q 1. The roots of the quadratic equation

$x^2 - 2\sqrt{3}x - 22 = 0$ are

[WBJEE]

(a) imaginary

(b) real, rational and equal

(c) real, irrational and unequal

(d) real, rational and unequal

Q 2. The quadratic equation $x^2 + 15|x| + 14 = 0$ has

[BITSAT]

(a) only positive solutions

(b) only negative solutions

(c) no solution

(d) both positive and negative solution

Q 3. If $\frac{3}{2}+\frac{7}{2}i$ is a solution of the equation $ax^2 - 6x + b = 0$, where a and b are real numbers, then the value of a + b is equal to

[Kerala CEE]

(a) 10

(b) 22

(c) 30

(d) 29

(e) 31

Q 4. The value of a for which the equation

$2x^2 + 2\sqrt{6}x + a = 0$ has equal roots, is

[VITEEE]

(a) 3

(b) 4

(c) 2

(d) $\sqrt{3}$

(e) $\sqrt{2}$

2009

Q 5. The solution of the equation

$(3+2\sqrt{2})^{x^2-8}+(3+2\sqrt{2})^{8-x^2}=6$ are

[Kerala CEE]

(a) $3\pm 2\sqrt{2}$

(b) ± 1

(c) $\pm 3\sqrt{3}, \pm 2\sqrt{2}$

(d) $\pm 7, \pm\sqrt{3}$

(e) $\pm 3, \pm\sqrt{7}$

Q 6. If p, q, r are positive and are in AP, then roots of the quadratic equation $px^2 + qx + r = 0$ are complex for

[AMU]

(a) $\left|\frac{r}{p} - 7\right| \geq 4\sqrt{3}$

(b) $\left|\frac{p}{r} - 7\right| < 4\sqrt{3}$

(c) all p and r

(d) no p and r

Q 7. The roots of $(x - a)(x - a - 1) + (x - a - 1)$

$(x - a - 2) + (x - a)(x - a - 2) = 0$, $a \in R$ are

always

[EAMCET]

(a) equal

(b) imaginary

(c) real and distinct

(d) rational and equal

Q 8. Let $f(x) = x^2 + ax + b$, where $a, b \in R$. If $f(x) = 0$ has all its roots imaginary, then the roots of $f(x) + f'(x) + f''(x) = 0$ are

[EAMCET]

(a) real and distinct

(b) imaginary

(c) equal

(d) rational and equal

Q 9. If x satisfies $|x^2 - 3x + 2| + |x - 1| = x - 3$,

then

[VITEEE]

(a) $x \in \phi$

(b) $x \in [1, 2]$

(c) $x \in [3, \infty)$

(d) $x \in (-\infty, \infty)$

Q 10. The equation $\frac{3}{4}(\log_2 x)^2 + \log_2 x - \frac{5}{4} = \log_x \sqrt{2}$ has

[Jamia Millia Islamia]

(a) at least one real solutions

(b) exactly three real solutions

(c) exactly one irrational solution

(d) complex roots

Q 11. If the roots of the quadratic equation $x^2 - 4x - \log_3 a = 0$ are real, then the least value of a is

[Jamia Millia Islamia]

(a) 81

(b) 1/81

(c) 1/64

(d) None of these

Q 12. If the equation $(a + 1)x^2 - (a + 2)x + (a + 3) = 0$ has roots equal in magnitude but opposite in signs, then the roots of the equation are

[J&KCET]

(a) $\pm a$

(b) $\pm\frac{1}{2}a$

(c) $\pm\frac{3}{2}a$

(d) $\pm 2a$

Q 13. The real root of the equation $x^3 - 6x + 9 = 0$ is

[KCET]

(a) -6

(b) -9

(c) 6

(d) -3

Q 14. If one root of the equation $x^2 + (1 - 3i)x - 2(1 + i) = 0$ is $-1 + i$, then the other root is

[WB JEE]

(a) $-1 - i$

(b) $\frac{-1-i}{2}$

(c) i

(d) $2i$

Q 15. The equation $x^2 - 3|x| + 2 = 0$ has

[WBJEE]

(a) no real root

(b) one real root

(c) two real roots

(d) four real roots

Q 16. The number of real roots of the equation

$x^4 + \sqrt{x^4 + 20} = 22$ is

[VITEEE]

(a) 4

(b) 2

(c) 0

(d) 1

Q 17. If a + b + c = 0, then the roots of the equation $4ax^2 + 3bx + 2c = 0$ are

[Manipal]

(a) equal

(b) imaginary

(c) real

(d) None of these

Q 18. A value of k for which the quadratic equation $x^2 - 2x(1 + 3k) + 7(2k + 3) = 0$ has equal roots, is

(a) 1

(b) 2

(c) 3

(d) 4

[J&K CET]

Q 19. The number of real roots of $f(x) = 0$, where $f(x) = (x-1)(x-2)(x-3)(x-4)$ lying in the interval $(1, 3)$ is

[J&KCET]

(a) 1

(b) 2

(c) 3

(d) 4

Q 20. If the difference between the roots of the equation $x^2 + ax + l = 0$ is less than $\sqrt{5}$, then the set of possible values of a is

[AIEEE]

(a) $(-3, 3)$

(b) $(-3, \infty)$

(c) $(3, \infty)$

(d) $(-\infty, -3)$

Q 21. Let $(\sin a)x^2 + (\sin a)\, x + (1 - \cos a) = 0$. The value of a for which roots of this equation are real and distinct, is

[DCE]

(a) $(0, 2 \tan^{-1} 1/4)$

(b) $(0, 2\pi/3)$

(c) $(0, \pi)$

(d) $(0, 2\pi)$

Q 22. If p, q, r are real and $p \neq q$, then the roots of the equation $(p - q)x^2 + 5(p + q)x - 2(p - q) = r$, are

(a) real and equal

(b) unequal and rational

[DCE]

(c) unequal and irrational

(d) nothing can be said

Q 23. Both the roots of the given equation (x –

(a) $(x - b) + (x - b)(x - c) + (x - c)(x - a) = 0$

are always

[UP SEE, BCECE]

(a) positive

(b) negative

(c) real

(d) imaginary

Q 24. If one root of the equation $x^2 + px + 12 = 0$ is 4, while the equation $x^2 + px + q = 0$ has equal roots, then the value of q is

[Kerala CEE]

(a) 4

(b) 12

(c) 3

(d) $\frac{29}{4}$

(e) $\frac{49}{4}$

Q 25. The number of real solutions of the equation $\left(\frac{9}{10}\right) = -3 + x - x^2$ is

[AMU]

(a) 0

(b) 1

(c) 2

(d) None of these

Q 26. If $1, 2, 3$ and 4 are the roots of the equation $x^4 + ax^3 + bx^2 + cx + d = 0$, then $a+2b + c$ is equal to

[EAMCET]

(a) -25

(b) 0

(c) 10

(d) 24

Q 27. If $e^{\cos x} - e^{-\cos x} = 4$, then the value of the cos x is

[WBJEE]

(a) $\log(2+\sqrt{5})$

(b) $-\log(2+\sqrt{5})$

(c) $\log(-2+\sqrt{5})$

(d) None of these

Q 28. If $x^{2/3} - 7x^{1/3} + 10 = 0$, then the value of x is

[BCECE]

(a) $\{125\}$

(b) $\{8\}$

(c) ϕ

(d) $\{125,8\}$

Q 29. If $b^2 \geq 4ac$ for the equation $ax^4 + bx^2 + c = 0$, then all the roots of the equation will be real positive if

[VITEEE]

(a) $b > 0, a < 0, c > 0$

(b) $b < 0, a > 0, c > 0$

(c) $b > 0, a > 0, c > 0$

(d) $b > 0, a > 0, c < 0$

Q 30. The number of real roots of the equation

$\left(x+\frac{1}{x}\right)^3 + x + \frac{1}{x} = 0$ is

[VITEEE]

(a) 0

(b) 2

(c) 4

(d) 6

Q 31. If $(1 - p)$ is a root of quadratic equation

$x^2 + px + (1 - p) = 0$, then its roots are

[Jamia Millia Islamia, MP PET]

(a) $0,1$

(b) $-1,1$

(c) $0,-1$

(d) $-1,2$

Q 32. The maximum number of real roots of the equation $x^{2n} - 1 = 0$ is

[RPET]

(a) 2

(b) 3

(c) n

(d) $2n$

Q 33. The solution of the equation $2x^3 - x^2 - 22x - 24 = 0$ when two of the roots are in the ratio $3 : 4$, is

[J&K CET]

(a) $3, 4, \frac{1}{2}$

(b) $-\frac{3}{2}, -2, 4$

(c) $-\frac{1}{2}, \frac{3}{2}, 2$

(d) $\frac{3}{2}, 2, \frac{5}{2}$

Q 34. If a, b, c are the sides of a triangle ABC such that $a \neq b \neq c$ and $x^2 - 2(a + b + c)x + 3\lambda (ab + bc + ca) = 0$ has real roots, then

[IITJEE]

(a) $\lambda < \frac{4}{3}$

(b) $\lambda > \frac{5}{3}$

(c) $\lambda \in \left(\frac{4}{3}, \frac{5}{3}\right)$

(d) $\lambda \in \left(\frac{1}{3}, \frac{5}{3}\right)$

Q 35. The roots of the equation $(q - r)\, x^2 + (r - p)\, x + (p - q) = 0$ are

[Kerala CEE]

(a) $\frac{r-p}{q-r}, 1$

(b) $\frac{p-q}{q-r}, 1$

(c) $\frac{p-r}{q-r}, 2$

(d) $\frac{q-r}{p-q}, 2$

(e) $\frac{r-p}{p-q}, 1$

Q 36. If $f(x) = 2x^3 + mx^2 - 13x + n$ and $2, 3$ are roots of the equation $f(x) = 0$, then the values of m and n are

[AMU]

(a) $-5, -30$

(b) $-5, 30$

(c) 5, 30

(d) None of these

Q 37. The number of solutions for the equation

$x^2 - 5|x| + 6 = 0$ is

[AMU]

(a) 4

(b) 3

(c) 2

(d) 1

Q 38. The difference between two roots of the equation $x^3 - 13x^2 + 15x + 189 = 0$ is 2. Then the roots of the equation are

[EAMCET]

(a) $-3,7,9$

(b) $-3,-7,-9$

(c) $3,-5,7$

(d) $-3,-7,9$

Q 39. The real roots of the equation $x^{2/3} + x^{1/3} - 2 = 0$ are **[MPPET]**

(a) 1,8

(b) −1,−8

(c)−1,8

(d) 1,−8

Q 40. If the roots of $x^3 - 3x^2 - 6x + 8 = 0$ are in arithmetic progression, then the roots of the equation are

(a) 3, 4, 5

(b) 4, 7, 10

(c) −2,1,4

(d) 1,4,7

[J&K CET]

Q 41. If the roots of the equation $x^2 - bx + c = 0$ be two consecutive integers, then $b^2 - 4c$ equals

(a) 1

(b) 2

(c) 3

(d) -2

[AIEEE]

Q 42. If the cube roots of unity are $1, \omega, \omega^2$, then the roots of the equation $(x - 1)^3 + 8 = 0$, are

[AIEEE]

(a) $-1, 1 + 2\omega, 1 + 2\omega)^2$

(b) $-1, 1-2\omega, 1-2\omega^2$

(c) $-1,-1,-1$

(d) $-1,-1+2\omega, -1-2\omega)^2$

Q 43. The roots of the equation $x^4 - 2x^3 + x =$ 380 are

[DCE]

(a) $5,-4,\frac{1\pm 5\sqrt{-3}}{2}$

(b) $-5,4,\frac{-1\pm 5\sqrt{-3}}{2}$

(c) $5, 4, \frac{-1 \pm 5\sqrt{-3}}{2}$

(d) $-5, -4, \frac{1 \pm 5\sqrt{3}}{2}$

Q 44. sin A, sin B, cos A are in GP. Roots of x^2 + 2x cot B + 1 = 0 are always

[BITSAT, OJEE]

(a) real

(b) imaginary

(c) greater than 1

(d) equal

Q 45. The solution set of the equation $pqx^2 - (p + q)^2x + (p + q)^2 = 0$ is

[Kerala CEE]

(a) $\left\{\frac{p}{q}, \frac{q}{p}\right\}$

(b) $\left\{pq, \frac{q}{p}\right\}$

(c) $\left\{\frac{q}{p}, pq\right\}$

(d) $\left\{\frac{p+q}{p}, \frac{p+q}{q}\right\}$

(e) $\left\{\frac{p-q}{p}, \frac{p-q}{q}\right\}$

Q 46. The coefficients of x in the quadratic equation $x^2 + bx + c = 0$ was taken as 17 in place of 13, its roots were found to be -2 and -15. The correct roots of the original equation are

[AMU]

(a) $-10, -3$

(b) $-9, -4$

(c) $-8, -5$

(d) $-7,-6$

Q 47. The polynomial $(ax^2 + bx + c)(ax^2 - dx - c)$, $ac \neq 0$ has

[AMU]

(a) four real roots

(b) at least two real roots

(c) at most two real roots

(d) no real roots

Q 48. E_1 : a + b + c = 0, if 1 is a root of ax^2 + bx + c = 0,

E_2 : $b^2 - a^2 = 2ac$, if sin θ, cos θ are the roots of $ax^2 + bx + c = 0$.

49. Which of the following is true ?

[EAMCET]

(a) E_1 is true, E_2 is true

(b) E_1 is true, E_2 is false

(c) E_1 is false, E_2 is true

(d) E_1 is false, E_2 is false

Q 49. The roots of the equation $x^3 - 3x - 2 = 0$ are

[EAMCET]

(a) $-1,-1, 2$

(b) $-1,1,-2$

(c) $-1,2,-3$

(d) $-1,-1,-2$

Q 50. Let a, b be the solutions of $x^2 + px + 1 = 0$ and c, d be the solutions, of $x^2 + qx + 1 = 0$. If $(a - c)(b - c)$ and $(a + d)(b + d)$ are the solutions of $x^2 + ax + \beta = 0$, then β equals

[MP PET]

(a) $p + q$

(b) $p - q$

(c) $p^2 + q^2$

(d) $q^2 - p^2$

Q 51. If a, b and c are distinct positive real numbers in AP, then the roots of the equation $ax^2 + 2bx + c = 0$ are

[RPET]

(a) imaginary

(b) rational and equal

(c) rational and distinct

(d) irrational

Q 52. If one of the roots of equation $x^2 + ax + 3 = 0$ is 3 and one of the roots of the equation $x^2 + ax + b = 0$ is three times the other root, then the value of b is equal to

[J&K CET]

(a) 3

(b) 4

(c) 2

(d) 1

Q 53. The equation $x^2 - 2\sqrt{2}kx + 2e^{2\log k} - 1 = 0$ has the product of roots equal to 31, then for what value of k it has real roots ?

[DCE]

(a) 1

(b) 2

(c) 3

(d) 4

Q 54. If α, β are the roots of the equation $(x - a)(x - b) = 5$, then the roots of the equation $(x - \alpha)(x - \beta) + 5 = 0$ are

[Kerala CEE]

(a) a, 5

(b) b, 5

(c) a, α

(d) a, β

(e) a, b

Q 55. If $(x - 2)$ is a common factor of the expressions $x^2 + ax + b$ and $x^2 + cx + d$, then $\frac{b-d}{c-a}$ is equal to

(a) -2

(b) -1

(c) 1

(d) 2

[EAMCET]

Q 56. If f(x) is a polynomial of degree n with rational coefficients and $1 + 2i$, $2 - \sqrt{3}$ and 5 are three roots of f(x) = 0, then the least value of n is

[EAMCET]

(a) 5

(b) 4

(c) 3

(d) 6

Q 57. Real roots of the equation k, $x^2 + 5|x| + 4 = 0$ are

[OJEE]

(a) $1, -1$

(b) $2, 0$

(c) $0, 1$

(d) None of these

Q 58. For how many values of k, $x^2 + x + 1 + 2k(x^2 - x - 1) = 0$ is a perfect square ?

[OJEE]

(a) 2

(b) 0

(c) 1

(d) 3

Q 59. The roots of the equation $|x^2 - x - 6| = x + 2$ are

[BCECE]

(a) $-2, 1, 4$

(b) $0, 2, 4$

(c) $0, 1, 4$

(d) $-2,2,4$

Q 60. Number of roots of the equation

$x - \frac{2}{x-1} = 1 - \frac{2}{x-1}$ is

[Jamia Millia Islamia, Guj. CET]

(a) one

(b) two

(c) infinite

(d) None of these

Q 61. If $2+i\sqrt{3}$ is a root of the equation $x^2 + px +$ $q = 0$, then the value of (p, q) is

[MP PET]

(a) $(-7, 4)$

(b) $(-4, 7)$

(c) $(4, -7)$

(d) $(7, -4)$

Q 62. If the roots of the given equation $(\cos p - 1) x^2 + (\cos p) x + \sin p = 0$ are real, then

[J&K CET]

(a) $p \in (-\pi, 0)$

(b) $p \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(c) $p \in (0, \pi)$

(d) $p \in (0, 2\pi)$

Q 63. If $a \leq 0$, then one of the roots of $x^2 - 2a|x - a| - 3a^2 = 0$is

[DCE]

(a) $(-1+\sqrt{6})a$

(b) $(\sqrt{6}-1)a$

(c) a

(d) None of these

Q 64. The values of x and y such that y satisfy the equation (x, y $\in$ real numbers) $x^2 - xy + y^2 - 4x - 4y + 16 = 0$ is

[DCE]

(a) 4,4

(b) 3,3

(c) 2,2

(d) None of these

Q 65. The roots of $|x - 2|^2 + |x - 2| - 6 = 0$ are

[UPSEE]

(a) 4, 2

(b) 0, 4

(c) −1, 3

(d) 5, 1

Q 66. The roots of $4x^2 + 6px + 1 = 0$ are equal,

then the value of p is

[RPET]

(a) $4/5$

(b) $1/3$

(c) $\pm 2/3$

(d) $4/3$

Q 67. If the cube roots of unity are $1, \omega, \omega^2$, then the roots of the equation $(x - 2)^3 + 27 = 0$ are

[Guj. CET]

(a) $-1, -1, -1$

(b) $-1, -\omega, -\omega^2$

(c) $-1, 2 + 3\omega, 2 + 3\omega^2$

(d) $-1, 2 - 3\omega, 2 - 3\omega^2$

Topic 5 Condition for Common Roots, Quadratic Expressions and Position of Roots

2009

Q 1. If the roots of the equation $bx^2 + cx + a = 0$ be imaginary, then for all real values of x, the expression $3b^2x^2 + 6bcx + 2c^2$ is

[AIEEE]

(a) greater than $4ab$

(b) less than 4 ab

(c) greater than $-4ab$

(d) less than $-4ab$

Q 2. If a, b, c are in GP, then the equation $ax^2 + 2bx + c = 0$ and $dx^2 + 2ex + f = 0$ have a common root, if $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in

[UP SEE]

(a) AP

(b) HP

(c) GP

(d) None of these

Q 3. If $ax^2 + bx + c = 0$ and $2x^2 + 3x + 4 = 0$ have a common root where a, b, c $\in$ N (set of natural numbers), the least value of a + b + c is

[AMU]

(a) 13

(b) 11

(c) 7

(d) 9

Q 4. If x is complex, the expression $\frac{x^2+34x-71}{x^2+2x-7}$ takes all which lie in the interval (a, b) where

[AMU]

(a) a = -1, b = 1

(b) a = 1, b = -1

(c) a = 5, b = 9

(d) a = 9, b = 5

Q 5. If α, β, γ are the roots of $x^3 + 4x + 1 = 0$, then the equation whose roots are $\frac{\alpha^2}{\beta+\gamma}, \frac{\beta^2}{\gamma+\alpha}, \frac{\gamma^2}{\alpha+\beta}$ is

[EAMCET]

(a) $x^3 - 4x - 1 = 0$

(b) $x^3 - 4x + 1 = 0$

(c) $x^3 + 4x - 1 = 0$

(d) $x^3 + 4x + 1 = 0$

Q 6. The quadratic equation whose roots are three times the roots of $3ax^2 + 3bx + c = 0$ is

[WB JEE]

(a) $ax^2 + 3bx + 3c = 0$

(b) $ax^2 + 3bx + c = 0$

(c) $9ax^2 + 9bx + c = 0$

(d) $ax^2 + bx + 3c = 0$

Q 7. The equation whose roots are reciprocal of the roots of the equation $ax^2 + bx + c = 0$ is

[RPET]

(a) $bx^2 + cx + a = 0$

(b) $bx^2 + ax + c = 0$

(c) $cx^2 + ax + b = 0$

(d) $cx^2 + bx + a = 0$

Q 8. If α and β are roots of the quadratic equation $x^2 + 4x + 3 = 0$, then the equation whose roots are $2\alpha + \beta$ and $\alpha + 2\beta$ is

[J&K CET]

(a) $x^2 - 12x + 35 = 0$

(b) $x^2 + 12x - 33 = 0$

(c) $x^2 - 12x - 33 = 0$

(d) $x^2 + 12x + 35 = 0$

Q 9. The quadratic equations $x^2 - 6x + a = 0$ and $x^2 - cx + 6 = 0$ have one root in common. The

other roots of the first and second equations are integers in the ratio $4:3$. Then the common root is

(a) 2

(b) 1

(c) 4

(d) 3

[AIEEE]

Q 10. Let a, b, c be real. If $ax^2 + bx + c = 0$ has two real roots α and β, where $\alpha < -1$ and $\beta > 1$, then $1+\frac{c}{a}+\left|\frac{b}{a}\right|$ is

[BCECE]

(a) < 0

(b) > 0

(c) ≤ 0

(d) None of these

Q 11. If x is real, then $\frac{x^2-2x+4}{x^2+2x+4}$ takes values in the interval

[Guj. CET]

(a) $\left[\frac{1}{3},3\right]$

(b) $\left(\frac{1}{3},3\right)$

(c) $(3, 3)$

(d) $\left(-\frac{1}{3},3\right)$

Q 12. All the values of m for which both roots of the equation $x^2 - 2mx + m^2 - 1 = 0$ are greater than -2 but less than 4 lie in the interval

[AIEEE]

(a) $m > 3$

(b) $-1 < m < 3$

(c) $1 < m < 4$

(d) $-2 < m < 0$

Q 13. The values of k for which the equations $x^2 - kx - 21 = 0$ and $x^2 - 3kx + 35 = 0$ will have a common roots are

[Jamia Millia Islamia]

(a) $k = \pm 4$

(b) $k = \pm 1$

(c) $k = \pm 3$

(d) $k = 0$

Q 14. If both the roots of the quadratic equation $x^2 - 2kx + k^2 + k - 5 = 0$ are less than 5, then k lies in the interval

[AEEEE]

(a) $[4,5]$

(b) $(-\infty, 4)$

(c) $(6, \infty)$

(d) $(5,6]$

Q 15. If x is real, then the minimum value of

$\frac{x^2 - x + 1}{x^2 + x + 1}$, is

[EAMCET]

(a) $\frac{1}{3}$

(b) 3

(c) $\frac{1}{2}$

(d) 1

Q 16. If x is real, then the value of $\frac{x+2}{2x^2+3x+6}$ is equal to

[BCECE]

(a) $\left(\frac{1}{13},\frac{1}{3}\right)$

(b) $\left(-\frac{1}{13},\frac{1}{3}\right)$

(c) $\left(-\frac{1}{3},\frac{1}{13}\right)$

(d) None of these

Q 17. If $(x^2 - 3x + 2)$ is a factor of $x^4 - px^2 + q = 0$, then the values of p and q are

[DCE]

(a) $-5, 4$

(b) $5, 4$

(c) $5, -4$

(d) $-5, -4$

Q 18. If two equations $x^2 + a^2 = 1 - 2ax$ and $x^2 + b^2 = 1 - 2bx$ have only one common root, then

[DCE]

(a) $a - b = 1$

(b) $a - b = -1$

(c) $a - b = 2$

(d) $|a - b| = 1$

Q 19. For $a \neq b$, if the equation $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$ have a common root, then the value of $a + b$ equals

[Kerala CEE]

(a) -1

(b) 0

(c) 1

(d) 2

(e) -2

Q 20. Let a, b, cbe real, if $ax^2 + bx + c = 0$ has two real roots α and β, where $\alpha < -2$ and $\beta > 2$, then

[DCE]

(a) $4-\frac{2b}{a}+\frac{c}{a}<0$

(b) $4+\frac{2b}{a}-\frac{c}{a}<0$

(c) $4-\frac{2b}{a}+\frac{c}{a}=0$

(d) $4+\frac{2b}{a}+\frac{c}{a}=0$

Topic 6 Types of Partial Fractions

2009

Q 1. If a real valued function / of a real variable x is such that $\frac{1}{(1+x)(1+x^2)} = \frac{A}{1+x} + \frac{f(x)}{1+x^2}$, then f(x) is equal to

[UP SEE]

(a) $\frac{1-x}{2}$

(b) $\frac{x^2+1}{2}$

(c) 1 − x

(d) None of these

Q 2. If $\frac{x^2+x+1}{x^2+2x+1} = A + \frac{B}{x+1} + \frac{C}{(x+1)^2}$, then A − B is equal to

[EAMCET]

(a) 4C

(b) 4C + 1

(c) 3C

(d) 2C

Q 3. $\frac{3x^2+1}{x^2-6x+8}$ is equal to

[Manipal]

(a) $3+\frac{49}{2(x-4)}-\frac{13}{2(x-2)}$

(b) $\frac{49}{2(x-4)}-\frac{13}{2(x-2)}$

(c) $\frac{-49}{2(x-4)}+\frac{13}{2(x-2)}$

(d) $\frac{49}{2(x-4)}+\frac{13}{2(x-2)}$

Q 4. The partial fractions of $\frac{3x^3-8x^2+10}{(x-1)^4}$ is

[KCET]

(a) $\frac{3}{(x-1)}+\frac{1}{(x-1)^2}+\frac{7}{(x-1)^3}+\frac{5}{(x-1)^4}$

(b) $\frac{3}{(x-1)}+\frac{1}{(x-1)^2}-\frac{7}{(x-1)^3}+\frac{5}{(x-1)^4}$

(c) $\frac{3}{(x-1)}+\frac{1}{(x-1)^2}-\frac{7}{(x-1)^3}-\frac{5}{(x-1)^4}$

(d) None of the above

Q 5. If $\frac{3x}{(x-a)(x-b)}=\frac{2}{x-a}+\frac{1}{x-b}$, then a :b is equal to

[EAMCET]

(a) 1:2

(b) $-2:1$

(c) $1:3$

(d) $3:1$

Q 6. If $\frac{3x+2}{(x+1)(2x^2+3)} = \frac{A}{x+1} + \frac{Bx+C}{2x^2+3}$, then A + C − B is equal to

[EAMCET]

(a) 0

(b) 2

(c) 3

(d) 5

Q 7. If $\frac{x^3}{(2x-1)(x+2)(x-3)} = A + \frac{B}{2x-1} + \frac{C}{x+2} + \frac{D}{x-3}$, then A is equal to

[EAMCET]

(a) $\frac{1}{2}$

(b) $-\frac{1}{50}$

(c) $-\frac{8}{25}$

(d) $\frac{27}{25}$

Q 8. If $\frac{(x+1)}{(2x-1)(3x+1)} = \frac{A}{(2x-1)} + \frac{B}{(3x+1)}$, then $16A + 9B$ is equal to

[EAMCET]

(a) 4

(b) 5

(c) 6

(d) 8

Q 9. If $\frac{x-4}{x^2-5x+6}$ can be expanded in the ascending powers of x, then the coefficient of x^3 is

[EAMCET]

(a) $\frac{-73}{648}$

(b) $\frac{73}{648}$

(c) $\frac{71}{648}$

(d) $\frac{-71}{648}$

Q 10. Let a, b and c be such that

$\frac{1}{(1-x)(1-2x)(1-3x)} = \frac{a}{1-x} + \frac{b}{1-2x} + \frac{c}{1-3x}$, then $\frac{a}{1} + \frac{b}{3} + \frac{c}{5}$ is equal to

[EAMCET]

(a) 1/15

(b) $1/6$

(c) $1/5$

(d) $1/3$

Topic 7 Solution of Inequalities

2010

Q 1. If $3 < 3t - 18 \leq 18$, then which one of the following is true ?

[Kerala CEE]

(a) $15 \leq 2t + 1 \leq 20$

(b) $8 \leq t < 12$

(c) $8 \leq t + 1 \leq 13$

(d) $21 \leq 3t \leq 24$

(e) $t \leq 7$ or $t \geq 12$

Q 2. The minimum value of $f(x) = |3 - x| + 7$ is

[VITEEE]

(a) 0

(b) 6

(c) 7

(d) 8

Q 3. The solution set of the inequation $\frac{x+11}{x-3} > 0$ is

[Kerala CEE]

(a) $(-\infty, 11) \cup (3, \infty)$

(b) $(-\infty, -10) \cup (2, \infty)$

(c) $(-100, -11) \cup (1, \infty)$

(d) $(-5, 0) \cup (3, 7)$

(e) $(0, 5) \cup (-1, 0)$

2009

Q 4. If $\log_{10}(x^3 + y^3) - \log_{10}(x^2 + y^2 - xy) \leq 2$, then the maximum value of xy, for all $x \geq 0$, $y \geq 0$ is

(a) 2500

(b) 3000

(c) 1200

(d) 3500

[DCE]

Q 5. If a,b,c are sides of triangle, then $\frac{(a+b+c)^2}{(ab+bc+ca)}$

always belongs to

[DCE]

(a) [1,2]

(b) [2,3]

(c) [3,4]

(d) [4,5]

Q 6. If x satisfies the inequations $2x - 7 < 11$, $3x + 4 < -5$, then x lies in the interval

[Kerala CEE]

(a) $(-\infty, 3)$

(b) $(-\infty, 2)$

(c) $(-\infty, -3)$

(d) $(-\infty, \infty)$

(e) $(3, \infty)$

Q 7. The set of all real x satisfying the inequality

$$\frac{3-|x|}{4-|x|} > 0$$

[Kerala CEE]

(a) $[-3, 3] \cup (-\infty, -4) \cup (4, \infty)$

(b) $(-\infty, -4) \cup (4, \infty)$

(c) $(-\infty, -3) \cup (4, \infty)$

(d) $(-\infty, -3) \cup (3, \infty)$

(e) $[-3, 3] \cup (4, \infty)$

Q 8. If $x^2 + 4ax + 2 > 0$ for all values of x, then a

lies in the interval

[Kerala CEE]

(a) $(-2,4)$

(b) $(1, 2)$

(c) $(-\sqrt{2}, \sqrt{2})$

(d) $\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

(e) $(-4, 2)$

Q 9. For $\frac{|x-1|}{x+2} < 1$, x lies in the interval

[AMU]

(a) $(-\infty, -2) \cup \left(-\frac{1}{2}, \infty\right)$

(b) $(-\infty, 1) \cup [2, 3]$

(c) $(-\infty, -4)$

(d) $\left[-\frac{1}{2}, 1\right]$

Q 10. Solution set of inequality $\log_e \frac{x-2}{x-3}$ i

[AMU]

(a) $(2, \infty)$

(b) $(-\infty, 2)$

(c) $(-\infty, \infty)$

(d) $(3, \infty)$

Q 11. If a, b, c > 0 and if abc = 1, then the value of a + b + c + ab + bc + ca lies in the interval

[AMU]

(a) $(\infty, -6]$

(b) $(-6, 0)$

(c) $(0, 6)$

(d) $(6, \infty)$

Q 12. The number of positive integers satisfying the inequality ${}^{n+1}C_{n-2} - {}^{n+1}C_{n-1} \leq 50$ is

[AMU]

(a) 9

(b) 8

(c) 7

(d) 6

Q 13. If a,b,c > 0, the minimum value of

$\frac{a}{b+c}+\frac{b}{c+a}+\frac{c}{a+b}$ is

[BCECE]

(a) 1

(b) $\frac{3}{2}$

(c) 2

(d) $\frac{5}{2}$

Q 14. The largest interval for which $x^{12} - x^9 + x^4 - x + 1 > 0$ is

[Jamia Millia Islamia]

(a) $-4 < x < 0$

(b) $0 < x < 1$

(c) $-100 < x < 100$

(d) $-\infty < x < \infty$

Q 15. If $x^2 + 2ax + 10 - 3a > 0$ for all $x \in R$,

then

[DCE]

(a) $-5 < a < 2$

(b) $a < -5$

(c) $a > 5$

(d) $2 < a < 5$

Q 16. If $x^2 + 2x + n > 10$ for all real numbers x, then which of the following conditions is true ?

[Kerala CEE]

(a) $n < 11$

(b) $n = 10$

(c) $n = 11$

(d) $n > 11$

(e) $n < -11$

Q 17. Suppose a, b and c are real numbers such that $\frac{a}{b} > 1$ and $\frac{a}{c} < 0$. Which one of the following is true?

E**[Kerala CEE]**

(a) $a + b - c > 0$

(b) $a > b$

(c) $(a - c)(b - c) > 0$

(d) $a + b + c > 0$

(e) abc > 0

Q 18. The set of admissible values of x such that $\frac{2x+3}{2x-9} < 0$ is

[Kerala CEE]

(a) $\left(-\infty, -\frac{3}{2}\right) \cup \left(\frac{9}{2}, \infty\right)$

(b) $(-\infty, 0) \cup \left(\frac{9}{2}, \infty\right)$

(c) $\left(-\frac{3}{2}, 0\right)$

(d) $\left(0, \frac{9}{2}\right)$

(e) $\left(-\frac{3}{2},\frac{9}{2}\right)$

Q 19. If $\sqrt{9x^2+6x+1}<(2-x)$, then

[BCECE]

(a) $x \in \left(-\frac{3}{2},\frac{1}{4}\right)$

(b) $x \in \left(-\frac{3}{2},\frac{1}{4}\right]$

(c) $x \in \left[-\frac{3}{2},\frac{1}{4}\right)$

(d) $x<\frac{1}{4}$

Q 20. If $\frac{2x}{2x^2+5x+2} > \frac{1}{x+1}$, then

[DCE]

(a) $-2 > x > -1$ (b) $-2 \geq x \geq -1$ (c) $-2 < x < -1$ (d) $-2 < x \leq -1$

Q 21. $(x-1)(x^2-5x+7) < (x-1)$, then x belongs to **[BITSAT]**

(a) $(1, 2) \cup (3, \infty)$

(b) $(-\infty, 1) \cup (2, 3)$

(c) $(2, 3)$

(d) None of these

Q 22. Number of integral solutions of $\frac{x+2}{x^2+1} > \frac{1}{2}$ is

[Kerala CEE]

(a) 0

(b) 1

(c) 2

(d) 3

(e) 4

Q 23. If r is a real number such that $|r| < 1$ and if $a = 5(1 - r)$, then

[Kerala CEE]

(a) $0 < a < 5$

(b) $-5 < a < 5$

(c) $0 < a < 10$

(d) $0 \leq a < 10$

(e) $-10 < a < 10$

Q 24. The set of values of x for which the inequalities $x^2 - 2x - 10 < 0$, $10x - x^2 - 16 > 0$ hold simultaneously, is

[EAMCET]

(a) $(-2,5)$

(b) $(2,8)$

(c) $(-2,8)$

(d) $(2,5)$

Q 25. The set of all x satisfying the inequality

$\frac{4x-1}{3x+1} \geq 1$ is

[Kerala CEE]

(a) $\left(-\infty, -\frac{1}{3}\right) \cup \left[\frac{1}{4}, \infty\right)$

(b) $\left(-\infty, -\frac{2}{3}\right) \cup \left[\frac{5}{4}, \infty\right)$

(c) $\left(-\infty, -\frac{1}{3}\right) \cup [2, \infty)$

(d) $\left(-\infty, -\frac{2}{3}\right) \cup [4, \infty)$

(e) $\left(-\infty, -\frac{1}{3}\right) \cup \left[\frac{1}{2}, \infty\right)$

Q 26. The set of values of x satisfying $2 \leq |x - 3| < 4$ is

[Kerala CEE]

(a) $(-1, 1] \cup [5, 7)$

(b) $-4 \leq x \leq 2$

(c) $-1 < x < 7$ or $x \geq 5$

(d) $x < 7$ or $x \geq 5$

(e) $-\infty < x \leq 1$ or $5 \leq x < \infty$

Q 27. The set of all solutions of the inequation

$x^2 - 2x + 5 < 0$ in R is

[EAMCET]

(a) $R - (-\infty, -5)$

(b) $R - (5, \infty)$

(c) ϕ

(d) $R - (-\infty, -4)$

Q 28. If $f(x) = x^2 + 2bx + 2c^2$ and $g(x) = -x^2 - 2cx + b^2$ such that min f(x) > max g(x), then the relation between b and c, is

[IIT JEE]

(a) no real value of b and c

(b) $0 < c < b\sqrt{2}$

(c) $|c| < |b|\sqrt{2}$

(d) $|c| > |b|\sqrt{2}$

Q 29. The solution set contained in R of the inequation $3^x + 3^{1-x} - 4 < 0$, is

[EAMCET]

(a) $(1,3)$

(b)$(0, 1)$

(c) $(1,2)$

(d) $(0,2)$

Q 30. Let $x = \left[\frac{a+2b}{a+b}\right]$ and $y = \frac{a}{b}$, where a and b are positive integers. If $y^2 > 2$, then

[OJEE]

(a) $x^2 \leq 2$

(b) $x^2 < 2$

(c) $x^2 > 2$

(d) $x^2 \geq 2$

Answers

Topic 1: Surds and Indices

1. (a) 2. (d) 3. (c) 4.(c) 5. (d) 6. (b) 7. (a) 8. (c) 9. (c) 10. (a) 11. (a) 12. (a) 13. (a)

Topic 2 : Logarithm and Its Properties

1. (d) 2. (d) 3. (b) 4. (b) 5. (a) 6. (b) 7. (b) 8. (d) 9. (a) 10. (d) 11. (a) 12. (d) 13. (c) 14. (b)

15. (c) 16. (a) 17. (b) 18. (c) 19. (a)

20. (a) 21. (e) 22. (d) 23. (b) 24. (b)

25. (a) 26. (b) 27. (d) 28. (b)

29. (d) 30. (b) 31. (a) 32. (b) 33. (d)

Topic 3 : Relation between Roots and Coefficients

1. (b) 2. (c) 3. (d) 4. (a) 5. (c) 6. (a) 7. (c). 8. (a) 9. (d) 10. (a) 11. (b)

12. (b) 13. (a) 14. (b)

15. (c) 16. (c) 17. (c) 18. (b) 19. (b)

20. (d) 21. (d) 22. (b) 23. (d) 24. (d)

25. (b) 26. (b) 27. (c) 28. (d)

29. (a) 30. (a) 31. (d) 32. (d) 33. (b)

34. (a) 35. (b) 36. (c) 37. (d) 38. (a)

39. (b) 40. (a) 41. (c) 42. (c)

43. (a) 44. (b) 45. (d) 46. (a) 47. (a)

48. (b) 49. (d) 50. (c) 51. (b) 52. (a)

53. (b) 54. (a) 55. (b) 56. (d)

57. (b) 58. (b) 59. (c) 60. (c) 61. (a)

62. (a) 63. (b) 64. (b) 65. (b) 66. (a)

67. (a) 68. (c) 69. (c) 70. (b)

71. (b) 72. (d) 73. (e) 74. (a) 75. (c)

76. (b) 77. (c) 78. (a) 79. (d) 80. (d)

81. (c) 82. (a) 83. (c) 84. (e)

85. (b)

Topic 4 : Solution of Equations, Nature of Roots

1. (d) 2. (c) 3. (e) 4. (a) 5. (e) 6. (b) 7. (c) 8. (b) 9. (a) 10. (b) 11. (b) 12. (b) 13. (d) 14. (d)

15. (d) 16. (b) 17. (c) 18. (b) 19. (a)

20. (a) 21. (a) 22. (d) 23. (c) 24. (e)

25. (a) 26. (c) 27. (a) 28. (d)

29. (b) 30. (a) 31. (c) 32. (a) 33. (b)

34. (a) 35. (b) 36. (b) 37. (a) 38. (a)

39. (d) 40. (c) 41. (a) 42. (b)

43. (a) 44. (a) 45. (d) 46. (a) 47. (b)

48. (a) 49. (a) 50. (d) 51. (c) 52. (a)

53. (d) 54. (e) 55. (d) 56. (a)

57. (d) 58. (a) 59. (d) 60. (d) 61. (b)

62. (c) 63. (b) 64. (a) 65. (b) 66. (c)

67. (d)

Topic 5 : Condition for Common Roots, Quadratic Expressions and Position of Roots

1. (c) 2. (a) 3. (d) 4. (c) 5. (c) 6. (a) 7. (d) 8. (d) 9. (a) 10. (a) 11. (a)

12. (b) 13. (a) 14. (b)

15. (a) 16. (b) 17. (b) 18. (c) 19. (a)

20. (a)

Topic 6 : Types of Partial Fractions

1. (a) 2. (d) 3. (a) 4. (b) 5. (b) 6. (b) 7. (a) 8. (c) 9. (a) 10. (a)

Topic 7 : Solution of Inequalities

1. (c) 2. (c) 3. (a) 4. (a) 5. (c) 6. (c) 7. (a) 8.(d) 9. (a) 10. (b) 11. (d) 12. (c) 13. (b) 14. (d)

15. (a) 16. (d) 17. (c) 18. (e) 19. (d)

20. (c) 21. (b) 22. (d) 23. (c) 24. (d)

25. (c) 26. (a) 27. (c) 28. (d)

29. (b) 30. (b)

Rectangular Cartesian Coordinates

Topic – 1

System of Coordinates and Distance between Points

Q 1. If the three points $(0, 1)$, $(0, -1)$ and $(x, 0)$ are vertices of an equilateral triangle, then the values of x are

[Kerala CEE]

(a) $\sqrt{3}, \sqrt{2}$

(b) $\sqrt{3}, -\sqrt{3}$

(c) $-\sqrt{5}, \sqrt{3}$

(d) $\sqrt{2}, -\sqrt{2}$

(e) $\sqrt{5}, -\sqrt{5}$

Q 2. If the distance between $(2, 3)$ and $(-5, 2)$ is equal to the distance between $(x, 2)$ and $(1, 3)$, then the values of x are

[BITSAT]

(a) $-6, 8$

(b) $6,8$

(c) $-8,6$

(d) $-7, 7$

(e) $-8, -6$

Q 3. Point Q is symmetric to P $(4, -1)$ with respect to the bisector of the first quadrant. The length of PQ is

[DCE]

(a) $3\sqrt{2}$

(b) $5\sqrt{2}$

(c) $7\sqrt{2}$

(d) $9\sqrt{2}$

Q 4. The point on the line $3x + 4y = 5$, which is equidistant from $(1,2)$ and $(3,4)$ is

[EAMCET]

(a) $(7,-4)$

(b) $(15,-10)$

(c) $\left(\frac{1}{7},\frac{8}{7}\right)$

(d) $\left(0,\frac{5}{4}\right)$

Q 5. If C is a point on the line segment joining A $(-3, 4)$ and B $(2,1)$ such that AC = 2 BC, then the coordinate of C is

[WB JEE]

(a) $\left(\frac{1}{3},2\right)$

(b) $\left(2, \frac{1}{3}\right)$

(c) $(2, 7)$

(d) $(7, 2)$

Q 6. If C is the reflection of A $(2, 4)$ in x−axis and B is the reflection of C in y−axis, then | AB | is

[WB JEE]

(a) 20

(b) $2\sqrt{5}$

(c) $4\sqrt{5}$

(d) 4

Q 7. Let A (k, 2) and B ($3,5$) are points. The point

(t, t) divide $\overline{AB}$ from A's side in the ratio of k, the: k

= ....k $\in$ R $-\{0,-1\}$

[Guj. CET]

(a) -4

(b) -2

(c) 4

(d) 2

Q 8. Consider three points

$P = (-\sin(\beta - \alpha), -\cos\beta)$,

$Q = (\cos(\beta - \alpha), \sin\beta)$

and $R = (\cos(\beta - \alpha + \theta), \sin(\beta - \theta))$,

where $0 < \alpha, \beta, \theta < \frac{\pi}{4}$. Then,

[IITJEE]

(a) P lies on the line segment RQ

(b) Q lies on the line segment PR

(c) R lies on the line segment QP

(d) P, Q, R are non-collinear

Q 9. If the sum of the distance of a point P from two perpendicular lines in a plane is 1, then the locus of P is a

[EAMCET]

(a) rhombus

(b) circle

(c) straight line

(d) pair of straight lines

Q 10. One possible condition for the three points (a, b), (b, a) and (a^2, $-b^2$) to be collinear, is

[WB JEE]

(a) $a - b = 2$

(b) $a + b = 2$

(c) a= 1 + b

(d) a = 1 − b

Q 11. The mid point of the line joining the points $(-10,8)$ and $(-6,12)$ divides the line joining the points $(4,-2)$ and $(-2,4)$ in the ratio

[Kerala CEE]

(a) $1:2$ internally

(b) $1:2$ externally

(c) $2:1$ internally

(d) $2:1$ externally

(e) $2:3$ externally

Q $12.$ The image of the centre of the circle $x^2 + y^2 = a^2$ with respect to the mirror image $x + y = 1$, is

[OJEE]

(a) $\left(\frac{1}{\sqrt{2}}, \sqrt{2}\right)$

(b) $(\sqrt{2},\sqrt{2})$

(c) $(\sqrt{2},2\sqrt{2})$

(d) None of these

Q 13. If P(l, 2), Q(4,6), R(5,7) and S (a, b) are the vertices of a parallelogram PQRS, then

[AMU]

(a) a = 2,b = 4

(b) a = b, b = 4

(c) a = 2, b = 3

(d) a = 3, b = 5

Q 14. The point P is equidistant from A$(1,3)$, B$(-3,5)$ and C$(5, -1)$, then PA is equal to

[RPET]

(a) 5

(b) $5\sqrt{5}$

(c) 25

(d) $5\sqrt{10}$

Q 15. The mid points of the sides of a triangle are D $(6,1)$, E $(3,5)$ and F $(-1, -2)$, then the vertex opposite to D is

[J&K CET]

(a) $(-4,2)$

(b) $(-4,5)$

(c) $(2,5)$

(d) $(10,8)$

Q 16. $(0, -1)$ and $(0, 3)$ are two opposite vertices of a square. The other two vertices are

[BITSAT]

(a) $(0, 1), (0, -3)$

(b) $(3,-1), (0,0)$

(c) $(2, 1), (-2, 1)$

(d) $(2, 2), (1, 1)$

Q 17. Point $\left(\frac{1}{2}, -\frac{13}{4}\right)$ divides the line joining the points $(3, -5)$ and $(-7, 2)$ in the ratio of

[Kerala CEE]

(a) $1:3$ internally

(b) $3:1$ internally

(c) $1:3$ externally

(d) $3:1$ externally

(e) None of these

Q 18. The ratio in which the line x + y = 4 divides the line joining the points $(1, -1)$ and $(5,7)$ is

[Kerala CEE]

(a) 1:2

(b) 2 : 1

(c) 1 : 3

(d) 3 : 1

(e) 3 : 2

Q 19. If $A(3, 5)$, $B(-5, -4)$, $C(7, 10)$ are the vertices of a parallelogram, taken in the order, then the coordinates of the fourth vertex are

[MHTCET]

(a) $(10, 19)$

(b) $(15, 19)$

(c) $(19, 10)$

(d) $(19, 15)$

Q 20. The intercepts on the straight line y = mx by the line y = 2 and y = 6 is less than 5, then m belongs to

[DCE]

(a) $\left]-\frac{4}{3},\frac{4}{3}\right[$

(b) $\left]\frac{4}{3},\frac{3}{8}\right[$

(c) $\left]-\infty,-\frac{4}{3}\right[\cup\left]\frac{4}{2},\infty\right[$

(d) $\left]\frac{4}{3},\infty\right[$

Q 21. If the distance between the points $(a \cos \theta, a \sin \theta)$ and $(a \cos \phi, a \sin \phi)$ is $2a$, then B is equal to

[EAMCET]

(a) $2n\pi \pm \pi + \phi$, $n \in Z$

(b) $n\pi + \frac{\pi}{2} + \phi$, $n \in Z$

(c) $n\pi - \phi$, $n \in Z$

(d) $2n\pi + \phi$, $n \in Z$

Q 22. The points $(1,3)$ and $(5,1)$ are the opposite vertices of a rectangle. The other two vertices lie on the line y = 2x + c, then the value of c will be

[Jamia Millia Islamia]

(a) 4

(b) -4

(c) 2

(d) -2

Q 23. The centre of circle inscribed in square formed by the lines $x^2 - 8x + 12 = 0$ and $y^2 - 14y + 45 = 0$, is

[IITJEE]

(a) (4,7)

(b) (7,4)

(c) (9, 4)

(d) (4, 9)

Q 24. The vertex of an equilateral triangle is $(2,-1)$ and the equation of its base is x + 2y = 1, the length of its sides is

[UP SEE]

(a) $\frac{2}{\sqrt{15}}$

(b) $\frac{4}{3\sqrt{3}}$

(c) $\frac{1}{\sqrt{5}}$

(d) $\frac{4}{\sqrt{5}}$

Topic 2 Orthocentre, Circumcentre and Incentre

Q 1. Three distinct points A, B and C given in the two dimensional coordinate plane such that the ratio of the distance of any one of them from the point $(1, 0)$ to the distance from the point $(-1,0)$ is equal to $\frac{1}{3}$. Then, the circumcentre of the triangle ABC is at the point

[AIEEE]

(a) $\left(\frac{5}{4}, 0\right)$

(b) $\left(\frac{5}{2}, 0\right)$

(c) $\left(\frac{5}{3}, 0\right)$

(d) $(0, 0)$

Q 2. The vertices of a triangle are $(6,0)$, $(0,6)$ and $(6,6)$. The distance between its circumcentre and centroid is

[KCET]

(a) 2

(b) $\sqrt{2}$

(c) 1

(d) $2\sqrt{2}$

Q 3. The coordinates of the incentre of the triangle having sides $3x - 4y = 0$, $5x + 12y = 0$ and $y - 15 = 0$ are

[OJEE]

(a) $-1,8$

(b) $1, -8$

(c) $2,6$

(d) None of these

Q 4. The circumcentre of a triangle formed by the lines $xy + 2x+2y + 4 = 0$ and $x + y + 2 = 0$ is

[DCE]

(a) $(-1,-1)$

(b) $(0,-1)$

(c) $(1, 1)$

(d) $(-1,0)$

Q 5. The circumcentre of the triangle formed by the lines y = x, y = 2x and y = 3x + 4 is

[UP SEE]

(a) $(6, 8)$

(b) $(6,-8)$

(c) $(3,4)$

(d) $(-3,-4)$

Q 6. Circumcentre of triangle whose vertices are

$(0,0)$, $(3,0)$ and $(0,4)$ is

[UP SEE]

(a) $\left(\frac{3}{2},2\right)$

(b) $\left(2,\frac{3}{2}\right)$

(c) $(0,0)$

(d) None of these

Q 7. The centroid of the triangle ABC, where A $\equiv$ $(2, 3)$, B $\equiv$ $(8, 10)$ and C $\equiv$ $(5,5)$ is

[KCET]

(a) $(5, 6)$

(b) $(6, 5)$

(c) $(6,6)$

(d) $(15,18)$

Q 8. The circumcentre of the triangle with vertices $(0, 30)$, $(4,0)$ and $(30,0)$ is

[Kerala CEE]

(a) $(10, 10)$

(b) $(10, 12)$

(c) $(12, 12)$

(d) $(15, 15)$

(e) $(17, 17)$

Q 9. The coordinates of the orthocentre of the triangle formed by $(0,0)$, $(8,0)$, $(4,6)$ is

[OJEE]

(a) $(4, 0)$

(b) $(6, 3)$

(c) $(6,0)$

(d) None of these

Q 10. The orthocentre of the triangle with vertices O$(0, 0)$, A$\left(0, \frac{3}{2}\right)$, B$(-5, 0)$ is

[Manipal]

(a) $\left(\frac{5}{2}, \frac{3}{4}\right)$

(b) $\left(\frac{-5}{2}, \frac{3}{4}\right)$

(c) $\left(-5, \frac{3}{2}\right)$

(d) $(0, 0)$

Q 11. The vertices P,Q,R of a triangle are $(2, 1)$, $(5, 2)$ and $(3, 4)$ respectively. Then, the circumcentre is

[Guj. CET]

(a) $\left(\frac{13}{4}, -\frac{9}{4}\right)$

(b) $\left(-\frac{13}{4}, \frac{9}{4}\right)$

(c) $\left(-\frac{13}{4}, -\frac{9}{4}\right)$

(d) $\left(\frac{13}{4}, \frac{9}{4}\right)$

Q 12. Let O$(0,0)$, P$(3,4)$, Q$(6,0)$ be the vertices of the triangle OPQ. The point R inside the triangle OPQ is such that the triangles OPR,PQR,OQR are of equal area. The coordinates of R are

[IITJEE]

(a) $\left(\frac{4}{3},3\right)$

(b) $\left(3,\frac{2}{3}\right)$

(c) $\left(3, \frac{4}{3}\right)$

(d) $\left(\frac{4}{3}, \frac{2}{3}\right)$

Q 13. If a vertex of a triangle is $(1, 1)$ and the mid points of two sides through the vertex are $(-1, 2)$ and $(3, 2)$, then the centroid of the triangle is

[BITSAT]

(a) $\left(1, \frac{7}{3}\right)$

(b) $\left(\frac{1}{3}, \frac{7}{3}\right)$

(c) $\left(-\frac{1}{3},\frac{7}{3}\right)$

(d) $\left(-1,\frac{7}{3}\right)$

Q 14. In Δ ABC, G is the centroid, D is the mid point of BC. If A = $(2,3)$ and G = $(7,5)$, then the point D is

[KCET]

(a) $\left(\frac{9}{2},4\right)$

(b) $\left(\frac{19}{2},6\right)$

(c) $\left(\frac{11}{2}, \frac{11}{2}\right)$

(d) $\left(8, \frac{13}{2}\right)$

Q 15. In the triangle with vertices at A$(6,3)$,B$(-6,3)$ and C$(-6, -3)$, the median through A meets BC at P, the line AC meets the x−axis at Q, while R and S respectively denote the orthocentre and centroid of the triangle. Then the correct matching of the coordinates of points in List−I to List−II is

[EAMCET]

List–I	**List–II**
(i) P	(A) $(0,0)$
(ii) Q	(B) $(6,0)$
(iii) R	(C) $(-2,l)$
(iv) S	(D) $(-6,0)$
(E) $(-6,-3)$	(F) $(-6,3)$

(i)	(ii)	(iii)	(iv)
(a) D	A	E	C

(b) D B E C

(c) D A F C

(d) B A F C

Q 16. Let ABC be a triangle, two of whose vertices are $(15, 0)$ and $(0, 10)$. If the orthocentre is $(6, 9)$, then the third vertex is

[OJEE]

(a) $(15, 10)$

(b) $(10, -15)$

(c) $(0, 0)$

(d) None of these

Q 17. If orthocentre and circumcentre of a triangle are respectively $(1,1)$ and $(3,2)$, then the coordinates of its centroid are

[OJEE]

(a) $\left(\frac{7}{3},\frac{5}{3}\right)$

(b) $\left(\frac{5}{3},\frac{7}{3}\right)$

(c) $(7, 5)$

(d) None of these

Q 18. If the centroid of the triangle formed by the points $(0,0)$, $(\cos\theta, \sin\theta)$ and $(\sin\theta, -\cos\theta)$ lies on the line $y = 2x$, then θ is equal to

[J&K CET]

(a) $\tan^{-1} 2$

(b) $\tan^{-1} 3$

(c) $\tan^{-1} (-3)$

(d) $\tan^{-1}(-2)$

Q 19. The incentre of the triangle with vertices $(1,\sqrt{3})$, $(0,0)$ and $(2,0)$ is

[OJEE]

(a) $\left(1,\frac{\sqrt{3}}{2}\right)$

(b) $\left(\frac{2}{3},\frac{1}{\sqrt{3}}\right)$

(c) $\left(\frac{2}{3},\frac{\sqrt{3}}{2}\right)$

(d) $\left(1, \frac{1}{\sqrt{3}}\right)$

Q 20. The orthocentre of the ΔOAB, where O is the origin, A $(6,0)$ and $B(3, 3\sqrt{3})$ is

[J&K CET]

(a) $(9/2, \sqrt{3}/2)$

(b) $(3, \sqrt{3})$

(c) $(\sqrt{3}, 3)$

(d) $(3, -\sqrt{3})$

Q 21. ABC is a triangle with vertices A$(-1, 4)$, B$(6,-2)$ and C$(-2, 4)$. D,E and F are the points which divide each AB, BC and CA respectively in the ratio $3 : 1$ internally. Then, the centroid of the triangle DEF is

[Kerala CEE]

(a) $(3, 6)$

(b) $(1, 2)$

(c) $(4,8)$

(d) $(-3,6)$

(e) $(-1,2)$

Q 22. The equations of the three sides of a triangle are $x = 2$, $y + 1 = 0$ and $x + 2y = 4$. The coordinates of the-circumcentre of the triangle are

[AMU]

(a) $(4,0)$

(b) $(2,-1)$

(c) $(0,4)$

(d) $(-1,2)$

Q 23. The x−coordinate of the incentre of the triangle where the mid points of the sides are $(0,$ $1)$, $(1,\ 1)$ and $(1,0)$, is

[MP PET]

(a) $2+\sqrt{2}$

(b) $1+\sqrt{2}$

(c) $2-\sqrt{2}$

(d) $1-\sqrt{2}$

Q 24. The orthocentre of the triangle with vertices $(-2,-6)$, $(-2,4)$ and $(l,3)$ is

[J&K CET]

(a) $(3, 1)$

(b) $(1, 1/3)$

(c) $(1,3)$

(d) None of these

Q 25. The incentre of a triangle with vertices (7, l),(−l,5)and $(3+2\sqrt{3},3+4\sqrt{3})$ is

[J&K CET]

(a) $\left(3+\frac{2}{\sqrt{3}},3+\frac{4}{\sqrt{3}}\right)$

(b) $\left(1+\frac{2}{3\sqrt{3}},1+\frac{4}{3\sqrt{3}}\right)$

(c) (7, 1)

(d) None of the above

Q 26. Orthocentre of the triangle formed by the lines $x + y = 1$ and $xy = 0$ is

[OJEE]

(a) $(0, 0)$

(b) $(0, 1)$

(c) $(1, 0)$

(d) $(-1, 1)$

Q 27. Orthocentre of triangle with vertices $(0, 0)$, $(3, 4)$ and $(\mathbf{4,0})$ is

[IITJEE]

(a) $\left(3, \frac{5}{4}\right)$

(b) $(3, 12)$

(c) $\left(3, \frac{3}{4}\right)$

(d) $(3, 9)$

Q 28. The incentre of the triangle formed by $(0,$ $0)$, $(5, 12)$, $(16,12)$ is

[AMU]

(a) $(7, 9)$

(b) $(9, 7)$

(c) $(-9,7)$

(d) $(-7,9)$

Topic 3 Area of Some Geometrical Figures

2010

Q 1. If the three points $(3q, 0)$, $(0, 3p)$ and $(1,1)$ are collinear then which one is true?

[WB JEE]

(a) $\frac{1}{p}+\frac{1}{q}=0$

(b) $\frac{1}{p}+\frac{1}{q}=1$

(c) $\frac{1}{p}+\frac{1}{q}=3$

(d) $\frac{1}{p}+\frac{3}{q}=1$

Q 2. The area of the triangle formed by the points

$(2,2)$, $(5, 5)$, $(6, 7)$ is equal to (in square unit)

[Kerala CEE]

(a) $\frac{9}{2}$

(b) 5

(c) 10

(d) $\frac{3}{2}$

(e) 14

Q 3. The vertices of a family of triangles have integer coordinates. If two of the vertices of all the triangles are $(0, 0)$ and $(6, 8)$, then the least value of areas of the triangles is

[Kerala CEE]

(a) 1

(b) $\frac{3}{2}$

(c) 2

(d) $\frac{5}{2}$

(e) 3

Q 4. The area between the curve y = 1 − j x j and the x−axis is equal to

[Kerala CEE]

(a) 1 sq unit

(b) $\frac{1}{2}$ sq unit

(c) $\frac{1}{3}$ sq unit

(d) 2 sq units

(e) 3 sq units

Q 5. Area of the triangle formed by the lines v =

2x, y = 3x and y = 5 is equal to (in square unit)

[Kerala CEE]

(a) $\frac{25}{6}$

(b) $\frac{25}{12}$

(c) $\frac{5}{6}$

(d) $\frac{17}{12}$

(e) 6

Q 6. Triangle ABC has vertices $(0, 0)$, $(11, 60)$ and $(91, 0)$. If the line y = kx cuts the triangle into two triangles of equal area, then k is equal to

[Kerala CEE]

(a) $\frac{30}{51}$

(b) $\frac{4}{7}$

(c) $\frac{7}{4}$

(d) $\frac{30}{91}$

(e) $\frac{27}{37}$

Q 7. One side of length 3 a of a triangle of area a square unit lies on the line x = a. Then, one of the lines on which the third vertex lies, is

[Kerala CEE]

(a) $x = -a^2$

(b) $x = a^2$

(c) $x = -a$

(d) $x = \frac{a}{3}$

(e) $x = -\frac{a}{3}$

Q 8. If t_1,t_2 and t_3 are distinct points $(t_1, 2at_1 + at_1^3)$.

$(t_2, 2at_2 + at_2^3)$ and $(t_3, 2at_3 + at_3^3)$ are collinear, if

[Jamia Millia Islamia]

(a) $t_1t_2t_3 = 1$

(b) $t_1 + t_2 + t_3 = t_1t_2t_3$

(c) $t_1 + t_2 + t_3 = 0$

(d) $t_1 + t_2 + t_3 = -1$

Q 9. If A$(0, 0)$, B$(12,0)$, C$(12, 2)$, D$(6, 7)$ and E$(0, 5)$ are the vertices of the pentagon ABCDE, then its area in square units, is

[Kerala CEE]

(a) 58

(b) 60

(c) 61

(d) 62

(e) 63

Q 10. If a > 0, b > 0 the maximum area of the triangle formed by the points O(0,0), A(a cos θ, b sin θ) and B (a cos 6, − bsin 0) is (in sq unit)

[Manipal]

(a) $\frac{ab}{2}$ when $\theta = \frac{\pi}{4}$

(b) $\frac{3ab}{4}$ when $\theta = \frac{\pi}{4}$

(c) $\frac{ab}{2}$ when $\theta = -\frac{\pi}{2}$

(d) a^2b^2

Q 11. The x−axis, p−axis and a line passing through the point A $(6,0)$ form a triangle ABC, If

$\angle A = 30°$, then the area of the triangle, in sq units is

[Guj. CET]

(a) $6\sqrt{3}$

(b) $12\sqrt{3}$

(c) $4\sqrt{3}$

(d) $8\sqrt{3}$

Q 12. Let A(h, k), B(1, 1) and C(2,l) be the vertices of a angled triangle with AC as its hypotenuse. If the area of the triangle is 1, then the set of values which 'k' can take is given by

[AIEEE]

(a) $\{1,3\}$

(b) $\{0,2\}$

(c) $\{-1,3\}$

(d) $\{-3,-2\}$

Q 13. If the points (a, b), (a', b') and (a − a', b − b') are collinear, then

[BITSAT]

(a) ab' = ab

(b) ab = a'b'

(c) aa' = bb'

(d) $a^2 + b^2 = 1$

Q 14. ABC is a triangle with $\angle A = 30°$, BC = 10 cm. The area of the circumcircle of the triangle is

[KCET]

(a) $100\ \pi$ sq cm

(b) 5 sq cm

(c) 25 sq cm

(d) $\frac{100\pi}{3}$ sq m

Q 15. ABC is a right angled triangle with $\angle$B = $90°$, a = 6 cm. If the radius of the circumcircle is 5 cm. Then the area of ΔABC is

[Kerala CEE]

(a) 25 cm^2

(b) 30 cm^2

(c) 36 cm^2

(d) 24 cm^2

(e) 48 cm^2

Q 16. The area of an equilateral triangle that can be inscribed in the circle $x^2 + y^2 - 4x - 6y - 12 = 0$, is

[Kerala CEE]

(a) $\frac{25\sqrt{3}}{4}$ sq units

(b) $\frac{35\sqrt{3}}{4}$ sq units

(c) $\frac{55\sqrt{3}}{4}$ sq units

(d) $\frac{75\sqrt{3}}{4}$ sq units

(e) $\frac{25}{4}$ sq units

Q 7. The area (in square unit) of the triangle formed by the points with polar coordinates $(1, 0)$, $\left(2, \frac{\pi}{3}\right)$ and $\left(3, \frac{2\pi}{3}\right)$ is

[EAMCET]

(a) $\frac{11\sqrt{3}}{4}$

(b) $\frac{5\sqrt{3}}{4}$

(c) $\frac{5}{4}$

(d) $\frac{11}{4}$

Q 18. If A$(-5,0)$ and B$(3, 0)$ are two vertices of a triangle ABC. Its area is 20 sq cm. The vertex C lies on the line x − y = 2. The coordinates of C are

[Guj.CET]

(a) $(-7,-5)$or $(3, 5)$

(b) $(-3,-5)$ or $(-5,7)$

(c) $(7, 5)$ or $(3, 5)$

(d) $(-3,-5)$ or $(7, 5)$

Q 19. The area of the segment of a circle of radius a subtending an angle of 2a at the centre is

[UP SEE, WB JEE]

(a) $a^2\left(\alpha+\frac{1}{2}\sin 2\alpha\right)$

(b) $\frac{1}{2}a^2\sin 2\alpha$

(c) $a^2\left(\alpha-\frac{1}{2}\sin 2\alpha\right)$

(d) $a^2\alpha$

Q 20. Area (in sq unit) enclosed by $y = 1$, $2x + y = 2$ and $x + y = 2$ is

[Kerala CEE]

(a) $\frac{1}{2}$ sq unit

(b) $\frac{1}{4}$ sq unit

(c) 1 sq unit

(d) 2 sq units

(e) 4 sq units

Q 21. If the area of the triangle with vertices (x, 0), (1, 1) and (.0. 2) is 4 sq unit, then the value of x is

[AMU]

(a) −2

(b) −4

(c) −6

(d) 8

Q 22. The area enclosed within the curve | x | + | y | = 1 is

[Jamia Millia Islamia]

(a) 1 sq unit

(b) $2\sqrt{2}$ sq units

(c) $\sqrt{2}$ sq units

(d) 2 sq units

Q 23. The straight lines $x + y = 0$, $3x + y - 4 = 0$ and $x + 3y - 4 = 0$ form a triangle which is

[UP SEE]

(a) right angled

(b) equilateral

(c) isosceles

(d) None of these

Q 24. The area (in square unit) of the triangle formed by the lines $x = 0$, $y = 0$ and $3x + 4y = 12$, is

[EAMCET]

(a) 3

(b) 4

(c) 6

(d) 12

Q 25. Three points are A$(6, 3)$, B$(-3,5)$, C$(4,-2)$ and P(x, y) is a point, then the ratio of area of ΔPBC and ΔABC is

[BCECE]

(a) $\left|\frac{x+y-2}{7}\right|$

(b) $\left|\frac{x-y+2}{2}\right|$

(c) $\left|\frac{x-y-2}{7}\right|$

(d) None of these

Q 26. A triangle with vertices $(4, 0)$, $(-1,-1)$, $(3, 5)$ is

[Jamia Millia Islamia]

(a) isosceles and right angled

(b) isosceles but not right angled

(c) right angled but not isosceles

(d) neither right angled nor isosceles

Q 27. Area of triangle formed by the lines x + y = 3 and angle bisectors of the pair of straight lines $x^2 - y^2 + 2y = 1$ is

[ITT JEE]

(a) 2 sq units

(b) 4 sq units

(c) 6 sq units

(d) 8 sq units

Q 28. Area of the triangle formed by the lines

$3x^2 - 4xy + y^2 = 0$, $2x - y = 6$ is

[EAMCET]

(a) 16 sq units

(b) 25 sq units

(c) 36 sq units

(d) 49 sq units

Q 29. The triangle formed by $x^2 - 3y^2 = 0$ and

$x = 4$ is

[OJEE]

(a) isosceles

(b) equilateral

(c) right angled

(d) None of these

Q 30. The area of triangle formed by the points

(a, b + c), (b, c + a), (c, a + b) is equal to

[Jamia Millia Islamia]

(a) abc

(b) $a^2 + b^2 + c^2$

(c) ab + bc + ca

(d) 0

Q 31. If the points $(1, 1)$, $(-1,-1)$, $(-\sqrt{3},\sqrt{3})$ are the vertices of a triangle, then this triangle is

[MPPET]

(a) right angled

(b) isosceles

(c) equilateral

(d) None of these

Q 32. Area of quadrilateral whose vertices are

$(2, 3)$, $(3, 4)$, $(4,5)$ and $(5,6)$ is equal to

[DCE]

(a) 0

(b) 4

(c) 6

(d) None of the above

Q 33. The points $(1, 1)$, $(-5, 5)$ and $(13, \lambda)$ lie on the same straight line, if λ is equal to

[Kerala CEE]

(a) 7

(b) −7

(c) ± 7

(d) 0

(e) 14

Topic 4 Transformation of Axes and Locus

Q 1. If the sum of the distances from two perpendicular lines in a plane is 1, then its locus is

[DCE]

(a) a square

(b) a circle

(c) a straight line

(d) two intersecting lines

Q 2. A variable line through the point $\left(\frac{1}{5},\frac{1}{5}\right)$ cuts the coordinate axes in the points A and B. If the point P divides AB internally in the ratio $3:1$, then the locus of P is

[AMU]

(a) $3y + x = 20xy$

(b) $y + 3x = 20xy$

(c) $x + y = 20xy$

(d) $3x + 3y = 20xy$

Q 3. The locus of a points which moves such that the sum of the squares of its distance from three vertices of the triangle is constant is a/an

[Jamia Millia Islamia]

(a) circle

(b) straight line

(c) ellipse

(d) None of the above

Q 4. The transformed equation of $3x^2 + 3y^2 + 2xy = 2$, when the coordinate axes are rotated through an angle of $45°$, is

[BITSAT, AMU]

(a) $x^2 + 2y^2 = 1$

(b) $2x^2 + y^2 = 1$

(c) $x^2 + y^2 = 1$

(d) $x^2 + 3y^2 = 1$

Q 5. A variable line $\frac{x}{a}+\frac{y}{b}=1$ is such that a + b = 4.

The locus of the mid point of the portion of the line

intercepted between the axes is

[KCET]

(a) x + y = 4

(b) x + y = 8

(c) x + y = l

(d) $x + y = 2$

Q 6. If a point $P(4, 3)$ is shifted by a distance $\sqrt{2}$ unit parallel to the line $y = x$, then coordinates of P in new position are

[AMU]

(a) $(5, 4)$

(b) $(5+\sqrt{2}, 4+\sqrt{2})$

(c) $(5-\sqrt{2}, 4-\sqrt{2})$

(d) None of these

Q 7. The pair of lines $\sqrt{3}x^2 - 4xy + \sqrt{3}y^2 = 0$ are rotated about the origin by $n/6$ in the anti-clockwise sense. The equation of the pair in the new position is

[OJEE]

(a) $\sqrt{3}y^2 - xy = 0$

(b) $\sqrt{3}x^2 - xy = 0$

(c) $x^2 - y^2 = 0$

(d) $\sqrt{3}x^2 + xy = 0$

Q 8. The locus of a point P which moves such thr

$2PA = 3PB$, where $A(0, 0)$ and $B(4, -3)$ are

points, is

(a) $5x^2 - 5y^2 - 72x + 54y + 225 = 0$

(b) $5x^2+5y^2 - 72x + 54y + 225=0$

[WB JEE]

(c) $5x^2 +5y^2 +72x - 54y+ 225=0$

(d) $5x^2 +5y^2 -72x - 54y - 225 =0$

Q 9. Let A$(2, -3)$ and B$(-2, 1)$ be vertices of a triangle ABC If the centroid of this triangle moves on the line $2x + 3y = l$, then the locus of the vertex C is the line

[Jamia Millia Islamia]

(a) $2x + 3y = 9$

(b) $2x - 3y = 7$

(c) $3x + 2y = 5$

(d) $3x - 2y = 3$

Q 10. What is the equation of the locus of a point which moves such that 4 times its distance from the x−axis is the square of its distance from the origin ?

[AMU]

(a) $x^2 - y^2 - 4y = 0$

(b) $x^2 + y^2 - 4|y| = 0$

(c) $x^2 + y^2 - 4x = 0$

(d) $x^2 + y^2 - 4|x| = 0$

Q 11. The transformed equation of $x^2 + 6xy + 8y^2 = 10$ when the axes are rotated through an angle $\frac{\pi}{4}$ is

(a) $15x^2 - 14xy + 3y^2 = 20$

(b) $15x^2 + 14xy - 3y^2 = 20$

[EAMCET]

(c) $15x^2 + 14xy + 3y^2 = 20$

(d) $15x^2 - 14xy - 3y^2 = 20$

Q 12. Let P be the point $(1,0)$ and Q be the point on $y^2 = 8x$. The locus of mid point of PQ is

[AIEEE]

(a) $x^2 - 4y + 2 = 0$

(b) $x^2 + 4y + 2 = 0$

(c) $y^2+4x+2 = 0$

(d) $y^2 - 4x+2 = 0$

Q 13. If the distance of any point P from the points $A(a + b, a - b)$ and $B(a - b, a + b)$ are equal, then the locus of P is

[GujCET]

(a) $ax + by = 0$

(b) $x - y = 0$

(c) $x + y = 0$

(d) $bx - ay = 0$

Q 14. The locus of the point (x, y) which is equidistant from the points (a + b,b − a) and (a − b,a + b) is

[Kerala CEE]

(a) ax = by

(b) ax + by = 0

(c) bx + ay = 0

(d) bx−ay = 0

(e) $\frac{x}{a}+\frac{y}{b}=1$

Q 15. If A and B are two fixed points, then the locus of a point which moves in such a way that the angle, APB is a right angle is

[RPET]

(a) a circle

(b) an ellipse

(c) a parabola

(d) None of these

Q 16. The equation of the locus of a point equidistant from the points (a_1, b_1) and (a_2, b_2) is $(a_1 - a_2)x + (b_1 - b_2)y + c = 0$, then the value of c is

[J&K CET]

(a) $\sqrt{(a_1^2 + b_1^2 + c_1^2)}$

(b) $a_1^2 - b_1^2 - c_1^2$

(c) $\frac{1}{2}(a_2^2 + b_2^2 - a_1^2 - b_1^2)$

(d) None of the above

Q 17. Locus of centroid of the triangle whose vertices are $(a \cos t, a \sin t)$, $(b \sin t, -b \cos t)$ and $(1,0)$, where t is a parameter, is

[AIEEE]

(a) $(3x - 1)^2+(3y)^2 = a^2-b^2$

(b) $(3x - 1)^2 +(3y)^2 =a^2 +b^2$

(c) $(3x + l)^2+(3y)^2 = a^2 + b^2$

(d) $(3x + l)^2+(3y)^2 = a^2 - b^2$

Q 18. If P = $(1,0)$, Q = $(-1,0)$ and R = $(2,0)$ are three given points, then the locus of a point S satisfying the relation $SQ^2 + SR^2 = 2SP^2$ is

[AMU]

(a) a straight line parallel to x−axis

(b) a circle through origin

(c) a circle with centre at the origin

(d) a straight line parallel to y−axis

Q 19. Given the points A $(0,4)$ and B $(0, -4)$, then the equation of the locus of the point P(x, y) such that, $|AP - BP| = 6$, is

[AMU]

(a) $\frac{x^2}{7}+\frac{y^2}{9}=1$

(b) $\frac{x^2}{9}+\frac{y^2}{7}=1$

(c) $\frac{x^2}{7}-\frac{y^2}{9}=1$

(d) $\frac{y^2}{9}-\frac{x^2}{7}=1$

Q 20. The coordinate axes are rotated through an angle $135°$. If the coordinates of a point P in the new system are known to be $(4,-3)$, then the coordinates of P in the original system are

[EAMCET]

(a) $\left(\frac{1}{\sqrt{2}},\frac{7}{\sqrt{2}}\right)$

(b) $\left(\frac{1}{\sqrt{2}},-\frac{7}{\sqrt{2}}\right)$

(c) $\left(-\frac{1}{\sqrt{2}},-\frac{7}{\sqrt{2}}\right)$

(d) $\left(-\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right)$

Q 21. The locus of the point of intersection of the lines x cot θ + y cosec θ = 2 and x cosec θ + y cot θ = 6 is

[J&K GET]

(a) a straight line

(b) circle

(c) a hyperbola

(d) an ellipse

Answers

Topic 1 : System of Coordinates and Distance between Points

1. (b) 2. (a) 3, (b) 4. (b) 5. (a) 6. (c) 7. (b) 8. (d)

9. (a) 10. (c) 11. (d) 12. (d) 13. (c) 14. (d) 15. (a)

16. (c) 17. (a) 18. (a) 19. (b) 20. (c) 21. (a) 22.

(b) 23. (a) 24. (a)

Topic 2 : Orthocentre, Circumcentre and Incentre

1. (a) 2. (b) 3. (a) 4. (a) 5. (b) 6. (a) 7. (a) 8. (e)

9. (d) 10. (d) 11. (d) 12. (c) 13. (a) 14. (b) 15. (c)

16. (c) 17. (a) 18. (c) 19. (d) 20. (b) 21. (b) 22.

(a) 23. (c) 24. (c) 25. (a) 2$. (a) 27. (c) 28. (a)

Topic 3 : Area of Some Geometrical Figures

1. (c) 2. (d) 3. (a) 4. (a) 5. (b) 6. (a) 7. (d) 8. (c)

9. (e) 10. (a) 11. (a) 12. (c) 13. (a) 14. (a) 15. (d)

16. (d) 17. (b) 18. (d) 19. (d) 20. (b) 21. (c) 22.

(d) 23. (c) 24. (c) 25. (a) 26. (a) 27. (a) 28.,(c)

29. (b) 30. (d) 31. (c) 32. (a) 33. (b)

Topic 4 : Transformation of Axes and Locus

1. (a) 2. (b) 3. (a) 4. (b) 5. (d) 6. (a) 7. (a) 8. (b)

9. (a) 10. (b) 11. (c) 12. (d) 13. (b) 14. (d) 15. (a)

16. (c) 17. (b) 18. (d) 19. (d) 20. (d) 21. (c)

Differentiation

Topic – 1

Differentiation of Algebraic and Transcendental Functions

Q 1. If f : $(-1, 1) \rightarrow$ R be a differentiable function with f(0) = -1 and f'(0) = 1. Let g(x) = [f$(2$f(x) = $2)]^2$.

Then, g'(0) is equal to

[AIEEE]

(a) 4

(b) − 4

(c) 0

(d) −2

Q 2. If $y = (1 + x)(1 + x^2)(1 + x^4) \ldots (1 + x^{2n})$, then the value of $\left(\frac{dy}{dx}\right)_{x=0}$ is

[WB JEE]

(a) 0

(b) -1

(c) 1

(d) 2

Q 3. If $y = \tan^{-1}x + \cot^{-1}x + \sec^{-1} x + \operatorname{cosec}^{-1} x$,

then $\frac{dy}{dx}$ is equal to

[Kerala CEE]

(a) $\frac{x^2-1}{x^2+1}$

(b) π

(c) 0

(d) 1

Q 4. If $y = \sin^{-1}\sqrt{1-x}$, then $\frac{dy}{dx}$ is equal to

[BITSAT]

(a) $\frac{1}{\sqrt{1-x}}$

(b) $\frac{-1}{2\sqrt{1-x}}$

(c) $\frac{1}{\sqrt{x}}$

(d) $\frac{-1}{2\sqrt{x}\sqrt{1-x}}$

(e) $\frac{1}{\sqrt{x}\sqrt{1-x}}$

Q 5. If $f(x) = |x - 2| + |x + 1| = x$, then $f'(-10)$ is

equal to

[Kerala CEE]

(a) -3

(b) -2

(c) -1

(d) 0

(e) 1

Q 6. If $x^y = e^{2(x-y)}$, then $\frac{dy}{dx}$ is equal to

[Kerala CEE]

(a) $\frac{2(1+\log x)}{(2+\log x)^2}$

(b) $\frac{1+\log x}{(2+\log x)^2}$

(c) $\frac{2}{2+\log x}$

(d) $\frac{2(1-\log x)}{(2+\log x)^2}$

(e) $\frac{2+\log x}{(2-\log x)^2}$

Q 7. If $y = \tan^{-1}\left(\frac{\cos x}{1+\sin x}\right)$, then $\frac{dy}{dx}$ is equal to

[Kerala CEE]

(a) $\frac{1}{2}$

(b) 2

(c) -2

(d) $-\frac{1}{2}$

(e) -1

Q 8. Let y be an implicit function of x defined by x^{2x} $- 2x^{x}$ cot y $- 1 = 0$. Then, y'(1) equals

[AIEEE]

(a) -1

(b) 1

(c) log 2

(d) $-$ log 2

Q 9. If $y = (\log_{\cos x} \sin x)(\log_{\sin x} \cos x) + \sin^{-1} \frac{2x}{1+x^2}$,

then $\frac{dy}{dx}$ at $x = \frac{\pi}{2}$ is equal to

[DCE]

(a) $\frac{8}{(4+\pi^2)}$

(b) 0

(c) $-\frac{8}{(4+\pi^2)}$

(d) 1

Q 10. Let $f(x) = 2^{2x-1}$ and $\phi(x) = -2^x + 2x \log 2$.

If $f'(x) > \phi'(x)$, then

[BITSAT]

(a) $0 < x < 1$

(b) $0 \leq x < 1$

(c) $x > 0$

(d) $x \geq 0$

Q 11. If $f(x) = \log_{x^2}(\log_e x)$, then $f'(x)$ at $x = e$ is

[KCET]

(a) 1

(b) $\frac{1}{e}$

(c) $\frac{1}{2e}$

(d) 0

Q 12. If $y = \sin^n x \cos nx$, then $\frac{dy}{dx}$ is

[KCET]

(a) $n \sin^{n-1} x \sin (n + 1)x$

(b) $n \sin^{n-1} x \cos (n-1)x$

(c) $n \sin^{n-1} x \cos nx$

(d) $n \sin^{n-1} x \cos (n+1)x$

Q 13. If $f(x) = \frac{g(x)+g(-x)}{2} + \frac{2}{[h(x)+h(-x)]^{-1}}$ where g and h are differentiable function, then $f'(0)$ is

(a) 1

(b) $\frac{1}{2}$

(c) $\frac{3}{2}$

(d) 0

[KCET]

Q 14. If $y = \log_2 \log_2(x)$, then $\frac{dy}{dx}$ is equal to

[Kerala CEE]

(a) $\frac{\log_2 e}{\log_e x}$

(b) $\frac{\log_2 e}{x \log_x 2}$

(c) $\frac{\log_2 x}{\log_e 2}$

(d) $\frac{\log_2 e}{\log_2 x}$

(e) $\frac{\log_2 e}{x \log_e x}$

Q 15. If $y = \sin[\cos^{-1}\{\sin(\cos^{-1}x)\}]$, then

$\frac{dy}{dx}$ at $x = \frac{1}{2}$ is equal to

[Kerala CEE]

(a) 0

(b) −1

(c) $\frac{2}{\sqrt{3}}$

(d) $\frac{1}{\sqrt{3}}$

Q 16. If $x^2 + y^2 = t - \frac{1}{t}$ and $x^4 + y^4 = t^2 + \frac{1}{t^2}$,

then $\frac{dy}{dx}$ is equal to

[Kerala CEE]

(a) $\frac{1}{x^2y^3}$

(b) $\frac{1}{xy^3}$

(c) $\frac{1}{x^2y^2}$

(d) $\frac{1}{x^3y}$

(e) $\frac{-1}{x^3y}$

Q 17. If $y = \sec^{-1}[\text{cosec } x] + \text{cosec}^{-1}[\sec x] + \sin^{-1}[\cos x] + \cos^{-1}[\sin x]$, then $\frac{dy}{dx}$ is equal to

[Kerala CEE]

(a) 0

(b) 2

(c) −2

(d) −4

(e) 1

Q 18. If $y = e^x . e^{x^2} . e^{x^3} ..... e^{x^n} ....$, for $0 < x < 1$,

then $\frac{dy}{dx}$ at $x = \frac{1}{2}$ is

[Kerala CEE]

(a) e

(b) 4e

(c) 2e

(d) 3e

(e) 5e

Q 19. The derivative of $\tan^{-1}\left(\frac{2x}{1-x^2}\right)$ with respect to $\cos^{-1}\sqrt{1-x^2}$ is

[Kerala CEE]

(a) $\frac{\sqrt{1-x^2}}{1+x^2}$

(b) $\frac{1}{\sqrt{1-x^2}}$

(c) $\frac{2}{\sqrt{1-x^2}(1+x^2)}$

(d) $\frac{2}{1+x^2}$

(e) $\frac{2\sqrt{1-x^2}}{1+x^2}$

Q 20. Let $f(x) = (x - 7)^2 (x - 2)^7$, $x \in [2, 7]$ The value of $\theta \in (2, 7)$ such that $f'(\theta) = 0$ is equal to

[Kerala CEE]

(a) $\frac{49}{4}$

(b) $\frac{53}{4}$

(c) $\frac{53}{7}$

(d) $\frac{49}{9}$

(e) $\frac{45}{7}$

Q 21. If $y = \cos^{-1}(\cos x)$, then $\frac{dy}{dx}$ is

[AMU]

(a) 1 in the whole plane

(b) -1 in the whole plane

(c) 1 in the 2nd and 3rd quadrants of the plane

(d) -1 in the 3rd and 4th quadrants of the plane

Q 22. If $f'(x) = g(x)$ and $g'(x) = -f(x)$ for all x and $f(2) = 4 = f'(2)$, then $f^2(4) + g^2(4)$ is

[AMU]

(a) 8

(b) 16

(c) 32

(d) 64

Q 23. $x = \frac{1-\sqrt{y}}{1+\sqrt{y}} \Rightarrow \frac{dy}{dx}$ is equal to

[EAMCET]

(a) $\frac{4}{(x+1)^2}$

(b) $\frac{4(x-1)}{(1+x)^3}$

(c) $\frac{x-1}{(1+x)^3}$

(d) $\frac{4}{(x+1)^3}$

Q 24. $x=\cos^{-1}\left(\frac{1}{\sqrt{1+t^2}}\right), y=\sin^{-1}\left(\frac{t}{\sqrt{1+t^2}}\right) \Rightarrow \frac{dy}{dx}$ is equal to

[EAMCET]

(a) 0

(b) tan t

(c) 1

(d) sin t cos t

Q 25. If $2f(x) = f'(x)$ and $f(0) = 3$, then $f(2)$ is equal to

[OJEEE]

(a) $3e^4$

(b) $3e^2$

(c) e^4

(d) None of these

Q 26. If $y = \tan^{-1}\sqrt{\frac{1-\sin x}{1+\sin x}}$, then the value of $\frac{dy}{dx}$ at $x = \frac{\pi}{6}$ is

[WB JEE]

(a) $-\frac{1}{2}$

(b) $\frac{1}{2}$

(c) 1

(d) -1

Q 27. If $8f(x) + 6f\left(\frac{1}{x}\right) = x + 5$ and $y = x^2 f(x)$,

then $\frac{dy}{dx}$ at $x = -1$ is equal to

[VITEEE]

(a) 0

(b) $\frac{1}{14}$

(c) $-\frac{1}{14}$

(d) 1

Q 28. If $x^2 + y^2 = t + \frac{1}{t}$ and $x^4 + y^4 = t^2 + \frac{1}{t^2}$,

then $\frac{dy}{dx}$ is equal to

[VITEEE]

(a) $\frac{y}{x}$

(b) $-\frac{y}{x}$

(c) $\frac{x}{y}$

(d) $-\frac{x}{y}$

Q 29. If $y = x - x^2$, then the derivatives of y^2 w.r.t. x^2 is

[VITEEE]

(a) $2x^2 + 3x - 1$

(b) $2x^2 - 3x + 1$

(c) $2x^2 + 3x + 1$

(d) $2x^2 - 3x - 1$

Q 30. If $y = \sqrt{x \log_e x}$, then $\frac{dy}{dx}$ at x = e is

[VITEEE]

(a) $\frac{1}{e}$

(b) $\frac{1}{\sqrt{e}}$

(c) $\sqrt{e}$

(d) e^2

Q 31. Find $\frac{dy}{dx}$, if $x = 2\cos\theta - \cos 2\theta$ and $y = 2\sin\theta - \sin 2\theta$.

[MHTCET]

(a) $\tan\frac{3\theta}{2}$

(b) $-\tan\frac{3\theta}{2}$

(c) $\cot\frac{3\theta}{2}$

(d) $-\cot\frac{3\theta}{2}$

Q 32. Find the derivative of y with respect to x if

$e^x + e^y = e^{x+y}$

[MHTCET]

(a) $-e^{x-y}$

(b) e^{x-y}

(c) $-e^{y-x}$

(d) e^{y-x}

Q 33. If $xy = \tan^{-1}(xy) + \cot^{-1}(xy)$, then $\frac{dy}{dx}$ is equal to

[MHTCET]

(a) $\frac{y}{x}$

(b) $-\frac{y}{x}$

(c) $\frac{x}{y}$

(d) $-\frac{x}{y}$

Q 34. The derivative of $\cos^3 x$ w.r.t. $\sin^3 x$ is

[MHTCET]

(a) – cot x

(b) cot x

(c) tan x

(d) – tan x

Q 35. If $y = \log x^x$, then the value of $\frac{dy}{dx}$ is

[MPPET]

(a) $x^x(1 + \log x)$

(b) log (ex)

(c) $\log\left(\frac{e}{x}\right)$

(d) $\log\left(\frac{x}{e}\right)$

Q 36. The derivative of $\tan^{-1}\left(\frac{\sqrt{1+x}-\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}}\right)$ is

[MPPET]

(a) $\sqrt{1-x^2}$

(b) $\frac{1}{\sqrt{1-x^2}}$

(c) $\frac{1}{2\sqrt{1-x^2}}$

(d) x

Q 37. If $x^y = e^{x-y}$, $x > 0$ then the value of $\frac{dy}{dx}$ at

$(1, 1)$ is

[RPET]

(a) 0

(b) $\frac{1}{2}$

(c) 1

(d) 2

Q 38. The value of differentiation of e^{x^2} with respect to e^{2x-1} at $x = 1$ is **[RPET]**

(a) e

(b) 0

(c) e^{-1}

(d) 1

Q 39. If $y = \log(\sin(x^2))$, $0 < x < \frac{\pi}{2}$, then $\frac{dy}{dx}$ at $x = \frac{\sqrt{\pi}}{2}$ is

[J&K CET]

(a) 0

(b) 1

(c) $\frac{\pi}{4}$

(d) $\sqrt{\pi}$

Q 40. $\frac{d}{dx}[\log_e e^{\sin(x^2)}]$ is equal to

[Guj. CET]

(a) $2\cos(x^2)$

(b) $2\cos x$

(c) $2x \cdot \cos x$

(d) $2x \cos(x^2)$

Q 41. $\frac{d}{dx}[\cos x°]$ is equal to

[Guj CET]

(a) $-\frac{\pi}{180}\sin x^{o}$

(b) – sin x°

(c) $\frac{\pi}{180}\sin x^{o}$

(d) $-\frac{\pi x}{180}\sin x$

Q 42. $\frac{d}{dx}[x^x + x^a + a^x + a^a] = \ldots.$, a is constant

[Guj CET]

(a) $x^x(1 + \log x) + a \cdot x^{a-1}$

(b) $x^x(1 + \log x) + a \cdot x^{a-1} + a^x \log a$

(c) $x^x(1 + \log x) + (a)\ (1 + \log a)$

(d) $x^x\ (1 + \log x) + a^a\ (1 + \log a) + ax^{a-1} + (a)\ (1 +$

$\log a)$

Q 43. If $y = \cot^{-1}\ (\cos 2x)^{1/2}$, then the value of

$\frac{dy}{dx}$ at $x = \frac{\pi}{6}$ will be

[UP SEE]

(a) $\left(\frac{2}{3}\right)^{1/2}$

(b) $\left(\frac{1}{3}\right)^{1/2}$

(c) $(3)^{1/2}$

(d) $(6)^{1/2}$

Q 44. The derivative of sin (x^3) w.r.t. cos (x^3) is

[KCET]

(a) – tan (x^3)

(b) tan (x^3)

(c) – cot (x^3)

(d) cot (x^3)

Q 45. If $y = \log_a x + \log_x a + \log_x x + \log_a a$,

then $\frac{dy}{dx}$ is equal to

[Kerala CEE]

(a) $\frac{1}{x} + x \log a$

(b) $\frac{\log a}{x} + \frac{x}{\log a}$

(c) $\frac{1}{x \log a} + x \log a$

(d) x log a

Q 46. If $f(x)=\frac{x-1}{4}+\frac{(x-1)^3}{12}+\frac{(x-1)^5}{20}+\frac{(x-1)^7}{28}+....,$ where 0 $< x < 2$, then f'(x) is equal to

[Kerala CEE]

(a) $\frac{1}{4x(2-x)}$

(b) $\frac{1}{4(x-2)^2}$

(c) $\frac{1}{2-x}$

(d) $\frac{1}{2+x}$

(e) $\frac{1}{(x-4)^2}$

Q 47. If $f(x + y) = 2f(x)\, f(y)$, $f'(5) = 1024(\log 2)$ and $f(2) = 8$, then the value of $f'(3)$ is

[Kerala CEE]

(a) $64\ (\log 2)$

(b) $128\ (\log 2)$

(c) 256

(d) $256\ (\log 2)$

(e) $1024\ (\log 2)$

Q 48. If $x = a\left\{\cos\theta + \log\tan\left(\frac{\theta}{2}\right)\right\}$ and y = a sin θ, then

$\frac{dy}{dx}$ is equal to

[AMU]

(a) cot θ

(b) tan θ

(c) sin θ

(d) cos θ

Q 49. The value of $\frac{dy}{dx}$ at x = $\frac{\pi}{2}$, where y is given by $y = x^{\sin x} + \sqrt{x}$, is

[WBJEE]

(a) $1+\frac{1}{\sqrt{2\pi}}$

(b) 1

(c) $\frac{1}{\sqrt{2\pi}}$

(d) $1-\frac{1}{\sqrt{2\pi}}$

Q 50. If $x^y = y^x$, then $x(x - y \log x)\ \frac{dy}{dx}$ is equal to

[BCECE, J&K CET]

(a) y(y – x log y)

(b) y(y + x log y)

(c) x(x + y log x)

(d) x(y – x log y)

Q 51. If $y = 1 + \frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3} + ...\infty$ with |x| > 1, then $\frac{dy}{dx}$ is

[Jamia Millia Islamia]

(a) $\frac{x^2}{y^2}$

(b) x^2y^2

(c) $\frac{y^2}{x^2}$

(d) $-\frac{y^2}{x^2}$

Q 52. The derivative of $a^{\sec x}$ w.r.t. $a^{\tan x}$ $(a > 0)$

is

[Jamia Millia Islamia]

(a) sec x $a^{\sec x - \tan x}$

(b) sin x $a^{\tan x - \sec x}$

(c) $\sin x\, a^{\sec x - \tan x}$

(d) $a^{\sec x - \tan x}$

Q 53. If $y = x^n \log x + x(\log x)^n$, then $\frac{dy}{dx}$ is equal to

[MHTCET]

(a) $x^{n-1}(1 + n \log x) + (\log x)^{n-1}\ [n + \log x]$

(b) $x^{n-2}(1 + n \log x) + (\log x)^{n-1}\ [n + \log x]$

(c) $x^{n-1}\ (1 + n \log x) + (\log x)^{n-1}\ [n - \log x]$

(d) None of the above

Q 54. $\frac{d}{dx}\sin^{-1}(x\sqrt{1-x}+\sqrt{x}\sqrt{1-x^2})$ is equal to

[Manipal]

(a) $-\frac{1}{2x\sqrt{1-x}}-\frac{1}{\sqrt{1-x^2}}$

(b) $\frac{1}{2x\sqrt{1-x}}-\frac{1}{\sqrt{1-x^2}}$

(c) $\frac{1}{2\sqrt{x}\sqrt{1-x}}+\frac{1}{\sqrt{1-x^2}}$

(d) $-\frac{1}{2\sqrt{x}\sqrt{1-x}}+\frac{1}{\sqrt{1-x^2}}$

Q 55. The derivative of f(tan x) w.r.t g(sec x) at $x = \frac{\pi}{4}$, where f'(1) = 2 and g'($\sqrt{2}$) = 4, is

[Manipal]

(a) $\frac{1}{\sqrt{2}}$

(b) $\sqrt{2}$

(c) 1

(d) None of these

Q 56. If $y = \sqrt{\frac{1-x}{1+x}}$, then $(1 - x^2)\frac{dy}{dx} + y$ is equal to

[MPPET]

(a) 1

(b) -1

(c) 2

(d) 0

Q 57. A value of x in the interval $(1, 2)$ such that $f'(x) = 0$, where $f(x) = x^3 - 3x^2 + 2x + 10$ is

[RPET]

(a) $\frac{3+\sqrt{3}}{3}$

(b) $\frac{3+\sqrt{2}}{2}$

(c) $1+\sqrt{2}$

(d) $\sqrt{2}$

Q 58. If $y=\frac{3at^2}{1+t^3}, x=\frac{3at}{1+t^3}$, then $\frac{dy}{dx}$ is equal to

[J&K CET]

(a) $\frac{t(2-t^3)}{(1-2t^3)}$

(b) $\frac{t(2+t^3)}{(1-2t^3)}$

(c) $\frac{t(2-t^3)}{(1+2t^3)}$

(d) $\frac{t(2+t^3)}{(1+2t^3)}$

Q 59. If $\sec\left(\frac{x-y}{x+y}\right)=a$, then $\frac{dy}{dx}$ is

[Guj. CET]

(a) $\frac{y}{x}$

(b) $-\frac{y}{x}$

(c) $\frac{x}{y}$

(d) $-\frac{x}{y}$

Q 60. If $\phi(x)$ is the inverse of the function f(x) and $f'(x) = \frac{1}{1+x^5}$, then $\frac{d}{dx}\phi(x)$ is

[DCE]

(a) $\frac{1}{1+\{\phi(x)\}^5}$

(b) $\frac{1}{1+\{f(x)\}^5}$

(c) $1 + \{\phi(x)\}^5$

(d) $1 + f(x)$

Q 61. The derivative of $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$ with respect to $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ is

[BITSAT]

(a) -1

(b) 1

(c) 2

(d) 4

Q 62. Differential coefficient of $\sqrt{\sec\sqrt{x}}$ is

[UP SEE]

(a) $\frac{1}{4\sqrt{x}}\sec\sqrt{x}\sin\sqrt{x}$

(b) $\frac{1}{4\sqrt{x}}(\sec\sqrt{x})^{3/2}.\sin\sqrt{x}$

(c) $\frac{1}{2}\sqrt{x}\sec\sqrt{x}\sin\sqrt{x}$

(d) $\frac{1}{2}\sqrt{x}(\sec\sqrt{x})^{3/2}.\sin\sqrt{x}$

Q 63. If $y = 2^{\log x}$, then $\frac{dy}{dx}$ is

[KCET]

(a) $\frac{2^{\log x}}{\log 2}$

(b) $2^{\log x} \cdot \log 2$

(c) $\frac{2^{\log x}}{x}$

(d) $\frac{2^{\log x} \cdot \log 2}{x}$

Q 64. If $\sec^{-1}\left(\frac{1+x}{1-y}\right) = a$, then $\frac{dy}{dx}$ is

[KCET]

(a) $\frac{y-1}{x+1}$

(b) $\frac{y+1}{x-1}$

(c) $\frac{x-1}{y-1}$

(d) $\frac{x-1}{y+1}$

Q 65. If $x=\frac{2at}{1+t^3}$ and $y=\frac{2at^2}{(1+t^3)^2}$, then $\frac{dy}{dx}$ is

[Kerala CEE]

(a) ax

(b) a^2x^2

(c) $\frac{x}{a}$

(d) $\frac{x}{2a}$

(e) 2a

Q 66. If $f(x) = (x - 2)(x - 4)(x - 6) \ldots (x - 2n)$,

then $f'(2)$ is

[Kerala CEE]

(a) $(-1)^n 2^{n-1} (n - 1)!$

(b) $(-2)^{n-1}(n-1)!$

(c) $(-2)^n n!$

(d) $(-1)^{n-1} 2^n (n - 1)!$

(e) $2^{n-1} (n - 1)!$

Q 67. If $2x^2 - 3xy + y^2 + x + 2y - 8 = 0$, then

$\frac{dy}{dx}$ is equal to

[EAMCET]

(a) $\frac{3y-4x-1}{2y-3x+2}$

(b) $\frac{3y+4x+1}{2y+3x+2}$

(c) $\frac{3y-4x+1}{2y-3x-2}$

(d) $\frac{3y-4x+1}{2y+3x+2}$

Q 68. If $y=\frac{\sqrt{1-\sin x}+\sqrt{1+\sin x}}{\sqrt{1-\sin x}-\sqrt{1+\sin x}}$, then $\frac{dy}{dx}$ is equal to

[OJEE]

(a) $\frac{1}{2}\text{cosec}^2\frac{x}{2}$

(b) $\frac{1}{2}\text{cosec}\frac{x}{2}$

(c) $\frac{1}{2}\text{cosec}^2 x$

(d) $\text{cosec}^2\frac{x}{2}$

Q 69. If $y = f\left(\frac{2x+3}{3-2x}\right)$ and f(x) = sin (log x), then $\frac{dy}{dx}$ is equal to

[OJEE]

(a) $\frac{12}{9-4x^2}\cos\left\{\log\left(\frac{2x+3}{3-2x}\right)\right\}$

(b) $\frac{12}{4x^2-9}\cos\left\{\log\left(\frac{2x+3}{3-2x}\right)\right\}$

(c) $\frac{12}{9-4x^2}\cos\left\{\log\left(\frac{3-2x}{2x+3}\right)\right\}$

(d) $\frac{12}{9-4x^2}\cos\left\{\log\left(\frac{2x+3}{2x-3}\right)\right\}$

Q 70. Let $f(x) = e^x$, $g(x) = \sin^{-1} x$ and $h(x) = f[g(x)]$, then $\frac{h'(x)}{h(x)}$ is equal to

[WB JEE]

(a) $e^{\sin^{-1} x}$

(b) $\frac{1}{\sqrt{1-x^2}}$

(c) $\sin^{-1} x$

(d) $\frac{1}{(1-x^2)}$

Q 71. If $y = \sin^{-1}\frac{x}{2} + \cos^{-1}\frac{x}{2}$, then the value of $\frac{dy}{dx}$ is

[BCECE]

(a) 1

(b) −1

(c) 0

(d) 2

Q 72. If y is a function of x and $\log(x + y) =$

$2xy$, then the value of $y'(0)$ is equal to

[BCECE]

(a) 1

(b) -1

(c) 2

(d) 0

Q 73. If $x = e^{y+e^{y+\ldots\ldots\infty}}$, $x > 0$ then $\frac{dy}{dx}$ is

[Jamia Millia Islamia]

(a) $\frac{x}{1+x}$

(b) $\frac{1}{x}$

(c) $\frac{1-x}{x}$

(d) $\frac{1+x}{x}$

Q 74. The derivative of log |x| is

[MHT CET]

(a) $\frac{1}{x}$, $x > 0$

(b) $\frac{1}{|x|}$, $x \neq 0$

(c) $\frac{1}{x}$, $x \neq 0$

(d) None of these

Q 75. If $y = \log_{\cos x} \sin x$, then $\frac{dy}{dx}$ is equal to

[MHTCET]

(a) $\frac{(\cot x \log \cos x + \tan x \log \sin x)}{(\log \cos x)^2}$

(b) $\frac{(\tan x \log \cos x + \cot x \log \sin x)}{(\log \cos x)^2}$

(c) $\frac{(\cot x \log \cos x + \tan x \log \sin x)}{(\log \sin x)^2}$

(d) None of the above

Q 76. Derivative of $\log_{10} x$ with respect to x^2 is

[J & K CET]

(a) $2x^2 \log_e 10$

(b) $\frac{\log_{10} e}{2x^2}$

(c) $\frac{\log_e 10}{2x^2}$

(d) $x^2 \log_e 10$

Q 77. The differential coefficient of f(log x), where f(x) = log x is

[Guj. CET]

(a) $\frac{x}{\log x}$

(b) $(x \log x)^{-1}$

(c) $\frac{\log x}{x}$

(d) None of these

Q 78. If $y = \cos^{-1}\left(\frac{1-\log x}{1+\log x}\right)$, then $\frac{dy}{dx}$ at x = e is

[DCE]

(a) $-\frac{1}{e}$

(b) $-\frac{1}{2e}$

(c) $\frac{1}{2e}$

(d) $\frac{1}{e}$

Q 79. For $0 < x < 2$, $\frac{d}{dx}\left(\tan^{-1}\sqrt{\frac{1+\cos\frac{x}{2}}{1-\cos\frac{x}{2}}}\right)$ is equal to

[UP SEE, WB JEE]

(a) $-1/4$

(b) $1/4$

(c) $-1/2$

(d) $1/2$

Q 80. If f(x) and g(x) are two functions with g(x) $= x - \frac{1}{x}$ and fog (x) $= x^3 - \frac{1}{x^3}$, then f'(x) is

[KCET]

(a) $3x^2 + 3$

(b) $x^2 - \frac{1}{x^2}$

(c) $1 + \frac{1}{x^2}$

(d) $3x^2 + \frac{3}{x^4}$

Q 81. Let f be twice differentiable function such that $f''(x) = -f(x)$, and $f'(x) = g(x)$, $h(x) = \{f(x)\}^2 + \{g(x)\}^2$. If $h(5) = 11$, then $h(10)$ is equal to

[Kerala CEE]

(a) 22

(b) 11

(c) 0

(d) 20

(e) None

Q 82. A differentiable function $f(x)$ is defined for all $x > 0$ and satisfies $f(x^3) = 4x^4$ for all $x > 0$. The value of $f'(8)$ is

[Kerala CEE]

(a) $\frac{16}{3}$

(b) $\frac{32}{3}$

(c) $\frac{16\sqrt{2}}{3}$

(d) $\frac{32\sqrt{2}}{3}$

(e) $\frac{32(2)^{1/3}}{3}$

Q 83. If $y=\left(1+\frac{1}{x}\right)\left(1+\frac{2}{x}\right)\left(1+\frac{3}{x}\right)....\left(1+\frac{n}{x}\right)$ and $x \neq 0$, then

$\frac{dy}{dx}$ when $x = -1$ is

[Kerala CEE]

(a) $n!$

(b) $(n-1)!$

(c) $(-1)^n (n-1)!$

(d) $(-1)^n n!$

(e) $(n+1)!$

Q 84. If $\sin y = x \sin(a + y)$, then $\frac{dy}{dx}$ is

[OJEE]

(a) $\frac{\sin a}{\sin^2(a+y)}$

(b) $\frac{\sin^2(a+y)}{\sin a}$

(c) sin a $\sin^2$ (a + y)

(d) $\frac{\sin^2(a+y)}{\sin a}$

Q 85. If $2^x + 2^y = 2^{x+y}$, then $\frac{dy}{dx}$ is equal to

[Jamia Millia Islamia]

(a) $\frac{(2^x+2^y)}{(2^x-2^y)}$

(b) $\frac{(2^x+2^y)}{(1+2^{x+y})}$

(c) $2^{x-y}\left(\frac{2^y-1}{1-2^x}\right)$

(d) $\frac{2^{x+y}-2^{x}}{2^{y}}$

Q 86. If $y = 5^x x^5$, then $\frac{dy}{dx}$ is

[MHTCET]

(a) $5^x (x^5 \log 5 - 5x^4)$

(b) $x^5 \log 5 - 5x^4$

(c) $x^5 \log 5 + 5x^4$

(d) $5^x (x^5 \log 5 + 5x^4)$

Q 87. If $y = \tan^{-1}\left(\frac{a\cos x - b\sin x}{b\cos x + a\sin x}\right)$, then $\frac{dy}{dx}$ is equal to

[MHT CET]

(a) 2

(b) -1

(c) $\frac{a}{b}$

(d) 0

Q 88. Let $y = x^{x^{x^{\cdots\cdots}}}$, then $\frac{dy}{dx}$ is equal to

[MPPET]

(a) yx^{y-1}

(b) $\frac{y^2}{x(1-y\log x)}$

(c) $\frac{y}{x(1+y\log x)}$

(d) None of these

Q 89. If $x = y\sqrt{1-y^2}$, then $\frac{dy}{dx}$ is equal to

[RPET]

(a) yx^{y-1}

(b) $\frac{y^2}{x(1-y\log x)}$

(c) $\frac{y}{x(1+y\log x)}$

(d) None of these

Q 90. If $y = \frac{e^x + e^{-x}}{e^x - e^{-x}}$, then $\frac{dy}{dx}$ is equal to

[Guj. CET]

(a) sec h^2x

(b) cosec h^2x

(c) – sec h^2x

(d) – cosec h^2x

Q 91. If $r = [2\phi + \cos^2(2\phi + \pi/4)]^{1/2}$, then what is the value of the derivative of $dr/d\phi$ at $\phi = \pi/4$?

[BITSAT, OJEE]

(a) $2\left(\frac{1}{\pi+1}\right)^{1/2}$

(b) $2\left(\frac{2}{\pi+1}\right)^{2}$

(c) $\left(\frac{2}{\pi+1}\right)^{1/2}$

(d) $2\left(\frac{2}{\pi+1}\right)^{1/2}$

Q 92. If $y = \sec^{-1}\frac{x+1}{x-1} + \sin^{-1}\frac{x-1}{x+1}$, then $\frac{dy}{dx}$ is

[Kerala CEE]

(a) 1

(b) 0

(c) $\frac{x-1}{x+1}$

(d) $\frac{x+1}{x-1}$

(e) $\frac{x^2+1}{x^2-1}$

Q 93. The derivative of $y = (1 - x)(2 - x) \ldots (n - x)$ at $x = 1$ is equal to

[Kerala CEE]

(a) 0

(b) $(-1)(n-1)!$

(c) $n! - 1$

(d) $(-1)^{n-1}\,(n-1)!$

(e) $(-1)^{n}\,(n-1)!$

Q 94. Let $f(x)$ be a polynomial function of the second degree. If $f(1) = f(-1)$ and a_1, a_2, a_3 are in

AP, then $f'(a_1)$, $f'(a_2)$, $f'(a_3)$ are in

[AMU]

(a) AP

(b) GP

(c) HP

(d) None of these

Q 95. If $f(x) = \cos x \cos 2x \cos 4x \cos 8x \cos 16x$, then $f'\left(\frac{\pi}{4}\right)$ is

[AMU]

(a) $\sqrt{2}$

(b) $\frac{1}{\sqrt{2}}$

(c) 0

(d) $\frac{\sqrt{3}}{2}$

Q 96. If $x\sqrt{1+y}+y\sqrt{1+x}=0$, then $\frac{dy}{dx}$ is equal to

[EAMCET]

(a) $\frac{1}{(1+x)^2}$

(b) $-\frac{1}{(1+x)^2}$

(c) $\frac{1}{1+x^2}$

(d) $\frac{1}{1-x^2}$

Q 97. $\frac{d}{dx}\sqrt{\frac{1-\sin 2x}{1+\sin 2x}}$ is equal to

[BCECE]

(a) $\sec^2 x$

(b) $-\sec^2\left(\frac{\pi}{4}-x\right)$

(c) $\sec^2\left(\frac{\pi}{4}+x\right)$

(d) $\sec^2\left(\frac{\pi}{4}-x\right)$

Q 98. If $y = x^2 + \cfrac{1}{x^2 + \cfrac{1}{x^2 + \cfrac{1}{x^2 + ...\infty}}}$, then $\frac{dy}{dx}$ is equal to

[BCECE]

(a) $\frac{2xy}{2y-x^2}$

(b) $\frac{xy}{y+x^2}$

(c) $\frac{xy}{y-x^2}$

(d) $\frac{2x}{2+\frac{x^2}{y}}$

Q 99. If $f(x)=\frac{1}{1-x}$, then the derivative of the composite function f[f{f(x)}] is equal to

[Jamia Millia Islamia]

(a) 0

(b) $\frac{1}{2}$

(c) 1

(d) 2

Q 100. If $y = a \sin^3 \theta$ and $x = a \cos^3 \theta$, then at $\theta = \frac{\pi}{3}, \frac{dy}{dx}$ is equal to

[MHTCET]

(a) $\frac{1}{\sqrt{3}}$

(b) $-\sqrt{3}$

(c) $\frac{-1}{\sqrt{3}}$

(d) $\sqrt{3}$

Q 101. If $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + ...\infty}}}$, then $(2y-1)\frac{dy}{dx}$ is equal to

[Kerala CEE, MHTCET]

(a) sin x

(b) – cos x

(c) cos x

(d) – sin x

Q 102. For $|x| < 1$, let $y = 1 + x + x^2 + \ldots$ to ∞,

then $\frac{dy}{dx} - y$ is

[MPPET]

(a) $\frac{x}{y}$

(b) $\frac{x^2}{y^2}$

(c) $\frac{x}{y^2}$

(d) xy^2

Q 103. If $y = \log^n x$, where $\log^n$ means log log log ..... (repeated n times), then

$x \log x \log^2 x \log^3 x \ldots\ldots \log^{n-1} x \log^n x \frac{dy}{dx}$ is equal to

[MP PET]

(a) log x

(b) x

(c) $\frac{1}{\log x}$

(d) $\log^n x$

Q 104. If $\sec\left(\frac{x^2-y^2}{x^2+y^2}\right) = e^a$, then $\frac{dy}{dx}$ is equal to

[RPET]

(a) $\frac{y^2}{x^2}$

(b) $\frac{y}{x}$

(c) $\frac{x}{y}$

(d) $\frac{x^2-y^2}{x^2+y^2}$

Q 105. If $\sin^{-1} x + \sin^{-1} y = \frac{\pi}{2}$, then $\frac{dy}{dx}$ is equal to

[RPET]

(a) $\frac{x}{y}$

(b) $-\frac{x}{y}$

(c) $\frac{y}{x}$

(d) $-\frac{y}{x}$

Q 106. If $x^m y^n = (x + y)^{m+n}$, then $(dy/dx)_{x=1,\ y=2}$ is equal to

[J&KCET]

(a) $\cot\frac{\theta}{2}$

(b) $\tan\frac{\theta}{2}$

(c) $\frac{1}{2}\operatorname{cosec}^2\frac{\theta}{2}$

(d) $-\frac{1}{2}\operatorname{cosec}^2\frac{\theta}{2}$

Q 107. If $x = a(\theta - \sin\theta)$, $y = a(1 - \sin\theta)$, then $\frac{dy}{dx}$ is equal to

[J&K CET]

(a) $\cos\frac{\theta}{2}$

(b) $\tan\frac{\theta}{2}$

(c) $\frac{1}{2}\cos ec^2\frac{\theta}{2}$

(d) $-\frac{1}{2}\cos ec^2\frac{\theta}{2}$

Q 108. If f(x) = x $\tan^{-1}$ x, then f'(1) is equal to

[DCE]

(a) $\frac{1}{2}+\frac{\pi}{4}$

(b) $-\frac{1}{2}+\frac{\pi}{4}$

(c) $-\frac{1}{2}-\frac{\pi}{4}$

(d) $\frac{1}{2}-\frac{\pi}{4}$

Q 109. If $y = \tan^{-1}(\sec x - \tan x)$, then $\frac{dy}{dx}$ is

[KCET]

(a) 2

(b) -2

(c) $\frac{1}{2}$

(d) $-\frac{1}{2}$

Q 110. If $y = e^{(1/2)\log(1+\tan^2 x)}$, then $\frac{dy}{dx}$ is equal to

[Kerala CEE]

(a) $\frac{1}{2}\sec^2 x$

(b) $\sec^2 x$

(c) sec x tan x

(d) $e^{1/2\log(1+\tan^2 x)}$

(e) $e^{1/2\log(1+\tan^2 x)} \cdot \frac{1}{2} \cdot \frac{1}{(1+\tan^2 x)}$

Q 111. If $y = x + x^2 + x^3 + \ldots \infty$, where $|x| < 1$, then for $|y| < 1$, $\frac{dy}{dx}$ is equal to

[Kerala CEE]

(a) $y + y^2 + y^3 + \ldots \infty$

(b) $1 - y + y^2 - y^3 + \ldots \infty$

(c) $1 - 2y + 3y^2 - \ldots \infty$

(d) $1 + 2y + 3y^2 + \ldots \infty$

(e) $y - y^2 + y^3 - \ldots \infty$

Q 112. If f'(x) = sin (log x) and $y = f\left(\frac{2x+3}{3-2x}\right)$, then

$\frac{dy}{dx}$ at x = 1 is equal to

[Kerala CEE]

(a) 6 sin log (5)

(b) 5 sin log (6)

(c) 12 sin log (5)

(d) 5 sin log (12)

(e) 12 sin log (6)

Q 113. If $f(x)=\sqrt{1+\cos^2(x^2)}$, then $f'\left(\frac{\sqrt{\pi}}{2}\right)$ is

[OJEE]

(a) $\frac{\sqrt{\pi}}{6}$

(b) $-\sqrt{\frac{\pi}{6}}$

(c) $\frac{1}{\sqrt{6}}$

(d) $\frac{\pi}{\sqrt{6}}$

Q 114. If $x = a(\cos\theta + \theta\sin\theta)$ and $y = a(\sin\theta - \theta\cos\theta)$, then $\frac{dy}{dx}$ is equal to

[BCECE]

(a) cos θ

(b) tan θ

(c) sec θ

(d) cosec θ

Q 115. Differential coefficient of $\sec^{-1}\frac{1}{2x^2-1}$ with respect to $\sqrt{1-x^2}$ at $x=\frac{1}{2}$ is equal to

[Jamia Millia Islamia]

(a) 2

(b) 4

(c) 6

(d) 1

Q 116. The derivative of $\left[\frac{e^x+1}{e^x}\right]$ is equal to

[MHTCET]

(a) 0

(b) $\frac{1}{e^x}$

(c) $-\frac{1}{e^x}$

(d) e^x

Q 117. If $x = \log(1 + t^2)$ and $y = t - \tan^{-1} t$.

Then, $\frac{dy}{dx}$ is equal to

[MHTCET]

(a) $e^x - 1$

(b) $t^2 - x$

(c) $\frac{\sqrt{e^x - 1}}{2}$

(d) $e^x - y$

Q 118. Derivative of sin x w.r.t. cos x is

[MHTCET]

(a) cos x

(b) cot x

(c) – cot x

(d) tan x

Q 119. $\frac{d}{dx}(\log x)^4$ is equal to

[MHTCET]

(a) $4 \log x^3$

(b) $4(\log x)^3$

(c) $\frac{4(\log x)^3}{x}$

(d) $\frac{(4 \log x)^3}{x}$

Q 120. If $y = (\cos x^2)^2$, then $\frac{dy}{dx}$ is equal to

[RPET]

(a) $-4x \sin 2x^2$

(b) $-x \sin x^2$

(c) $-2x \sin 2x^2$

(d) $-x \cos 2x^2$

Q 121. If $f(x) = \log_x (\log_e x)$, then $f'(x)$ at $x = e$ is equal to

[Kerala CEE]

(a) 1

(b) 2

(c) 0

(d) e

(e) $\frac{1}{e}$

Q 122. If $y = \cos(\sin x^2)$, then at $x = \sqrt{\frac{\pi}{2}}, \frac{dy}{dx}$ is equal to

[AMU]

(a) -2

(b) 2

(c) $-2\sqrt{\frac{\pi}{2}}$

(d) 0

Q 123. If $y=\sqrt{\log x+\sqrt{\log x+\sqrt{\log x+\sqrt{\log x+.....\infty}}}}$, then $\frac{dy}{dx}$

is equal to

[AMU]

(a) $\frac{x}{2y-1}$

(b) $\frac{x}{2y+1}$

(c) $\frac{1}{x(2y-1)}$

(d) $\frac{1}{x(1-2y)}$

Differentiation by Substitution, Higher Order Derivatives

Q 1. If x = a $(1 + \cos\theta)$, y = a $(\theta + \sin\theta)$, then

$\frac{d^2y}{dx^2}$ at $\theta = \frac{\pi}{2}$ is

[Kerala CEE]

(a) $-\frac{1}{a}$

(b) $\frac{1}{a}$

(c) -1

(d) -2

(e) $-\frac{2}{a}$

Q 2. The derivative of $\sin^{-1}(2x\sqrt{1-x^2})$ with respect to $\sin^{-1}(3x - 4x^3)$ is

[Kerala CEE]

(a) $\frac{2}{3}$

(b) $\frac{3}{2}$

(c) $\frac{1}{2}$

(d) 1

(e) 0

Q 3. If $y = \sin^{-1}\left(\frac{5x + 12\sqrt{1-x^2}}{13}\right)$, then $\frac{dy}{dx}$ is equal to

[DCE]

(a) $-\frac{1}{\sqrt{1-x^2}}$

(b) $\frac{1}{\sqrt{1-x^2}}$

(c) $\frac{3}{\sqrt{1-x^2}}$

(d) $\frac{x}{\sqrt{1-x^2}}$

Q 4. If $f(x) = 1 + nx + \frac{n(n-1)}{2}x^2 + \frac{n(n-1)(n-2)}{6}x^3 + \ldots +$ x^n, then $f''(1)$ is equal to

[KCET]

(a) $n(n-1)2^{n-1}$

(b) $(n-1)2^{n-1}$

(c) $n(n-1)2^{n-2}$

(d) $n(n-1)2^n$

Q 5. $y = e^{a\sin^{-1}x} \Rightarrow (1-x^2)\, y_{n+2} - (2n+1)xy_{n+1}$ is equal to

[EAMCET]

(a) $-(n^2 + a^2)y_n$

(b) $(n^2 - a^2)y_n$

(c) $(n^2 + a^2)\, y_n$

(d) $-(n^2 - a^2)y_n$

Q 6. The second order derivative of $a \sin^3 t$ with respect to $a \cos^3 t$ at $t = \frac{\pi}{4}$ is

[WBJEE]

(a) 2

(b) $\frac{1}{12a}$

(c) $\frac{4\sqrt{2}}{3a}$

(d) $\frac{3a}{4\sqrt{2}}$

Q 7. Let f be a twice differentiable function such that $f''(x) = -f(x)$ and $f'(x) = g(x)$. If $h'(x) = [f(x)^2 + g(x)^2]$ $h(1) = 8$ and $h(0) = 2$, then $h(2)$ is equal to

[Jamia Millia Islamia]

(a) 1

(b) 2

(c) 3

(d) None of these

Q 8. Let $x = \log_e t$, $t > 0$ and $y + 1 = t^2$. Then, $\frac{d^2x}{dy^2}$ is equal to

[J&KCET]

(a) $4e^{2x}$

(b) $-\frac{1}{2}e^{-4x}$

(c) $-\frac{3}{4}e^{5x}$

(d) $4e^{x}$

Q 9. Let g(x) = log f(x), where f(x) is a twice differentiable positive function on $(0, \infty)$ such that f(x + 1) = xf(x). Then, for N = $1, 2, 3, \ldots$,

$g''\left(N+\frac{1}{2}\right)-g''\left(\frac{1}{2}\right)$ is equal to

[IIT JEE]

(a) $-4\left\{1+\frac{1}{9}+\frac{1}{25}+.....+\frac{1}{(2N-1)^2}\right\}$

(b) $4\left\{1+\frac{1}{9}+\frac{1}{25}+.....+\frac{1}{(2N-1)^2}\right\}$

(c) $-4\left\{1+\frac{1}{9}+\frac{1}{25}+.....+\frac{1}{(2N+1)^2}\right\}$

(d) $4\left\{1+\frac{1}{9}+\frac{1}{25}+.....+\frac{1}{(2N+1)^2}\right\}$

Q 10. If f(x) = be^{ax} + ae^{bx}, then $f''(0)$ is equal to

e**[KCET]**

(a) 0

(b) 2ab

(c) ab(a + b)

(d) ab

Q 11. Let $f(x) = \sin x$, $g(x) = x^2$ and $h(x) = \log_e x$. If $F(x) = (hogof)(x)$, then $F''(x)$ is equal to

[Kerala CEE]

(a) a $csoec^3 x$

(b) $2 \cot x^2 - 4x^2 \operatorname{cosec}^2 x^2$

(c) $2x \cot x^2$

(d) $-2 \operatorname{cosec}^2 x$

(e) $4 \operatorname{cosec}^2 x$

Q 12. If $y = \sin(\log_e x)$, then $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx}$ is equal to

[EAMCET]

(a) $\sin(\log_e x)$

(b) $\cos(\log_e x)$

(c) y^2

(d) $-y$

Q 13. If $y = e^{ax} \sin bx$, then $\frac{d^2y}{dx^2} - 2a\frac{dy}{dx} + a^2y$ is equal to

[OJEE]

(a) 0

(b) 1

(c) $-b^2y$

(d) $-by$

Q 14. If $x = e^t \sin t$, $y = e^t \cot t$, then $\frac{d^2y}{dx^2}$ at $x = \pi$ is

[WB JEE]

(a) $2e^{\pi}$

(b) $\frac{1}{2}e^{\pi}$

(c) $\frac{1}{2e^{\pi}}$

(d) $\frac{2}{e^{\pi}}$

Q 15. $f(x) = e^x \sin x$, then $f''(x)$ is equal to

[BCECE]

(a) $e^{6x} \sin 6x$

(b) $2e^x \cos x$

(c) $8e^x \sin x$

(d) $8e^x \cos x$

Q 16. $\frac{d^2x}{dy^2}$ equals

[IIT JEE]

(a) $\left(\frac{d^2y}{dx^2}\right)^{-1}$

(b) $-\left(\frac{d^2y}{dx^2}\right)^{-1}\left(\frac{dy}{dx}\right)^{-1}$

(c) $\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-2}$

(d) $-\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-3}$

Q 17. If $y = \cos^2\frac{3x}{2} - \sin^2\frac{3x}{2}$, then $\frac{d^2y}{dx^2}$ is

[KCET]

(a) $-3\sqrt{1-y^2}$

(b) $9y$

(c) $-9y$

(d) $3\sqrt{1-y^2}$

Q 18. $x = \cos\theta, y = \sin 5\theta \Rightarrow (1 - x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx}$

is

[EAMCET]

(a) $-5y$

(b) $5y$

(c) $25y$

(d) $-25y$

Q 19. If $y = a^x.b^{2x-1}$, then $\frac{d^2y}{dx^2}$ is

[WB JEE]

(a) $y^2 \log ab^2$

(b) $y \log ab^2$

(c) y^2

(d) $y(\log ab^2)^2$

Q 20. $\frac{d^n}{dx^n}(\log x)$ is equal to

[WB JEE]

(a) $\frac{(n-1)!}{x^n}$

(b) $\frac{n!}{x^n}$

(c) $\frac{(n-2)!}{x^n}$

(d) $(-1)^{n-1}\frac{(n-1)!}{x^n}$

Q 21. If $y^2 = ax^2 + bx + c$, where a, b, c are constants, then $y^3 \frac{d^2y}{dx^2}$ is equal to

[MHTCET]

(a) a constant

(b) a function of x

(c) a function of y

(d) a function of x and y

Q 22. If $y^2 = ax^2 + bx + c$, where a, b, c are constants, then $y^3 \frac{d^2y}{dx^2}$ is equal to

[MHTCET]

(a) a constant

(b) a function of x

(c) a function of y

(d) a function of x and y both

Q 23. If $f''(x) = -f(x)$, where $f(x)$ is a continuous double differentiable function and $g(x) = f'(x)$.

If $F(x) = \left(f\left(\frac{x}{2}\right)\right)^2 + \left(g\left(\frac{x}{2}\right)\right)^2$ and $f(5) = 5$, then $F(10)$ is

[IIT JEE]

(a) 0

(b) 5

(c) 10

(d) 25

Q 24. If $x = \sin t$, $y = \cos pt$, then

[BITSAT]

(a) $(1 - x^2)y_2 + xy_1 + p^2y = 0$

(b) $(1 - x^2)y_2 + xy_1 - p^2y = 0$

(c) $(1 + x^2)y_2 - xy_1 + p^2y = 0$

(d) $(1 - x^2)y_2 - xy_1 + p^2y = 0$

Q 25. If $y = \tan^{-1}\frac{\sqrt{1+x^2}-\sqrt{1-x^2}}{\sqrt{1+x^2}+\sqrt{1-x^2}}$, then $\frac{dy}{dx}$ is equal to

[BITSAT]

(a) $\frac{x^2}{\sqrt{1-x^4}}$

(b) $\frac{x^2}{\sqrt{1+x^4}}$

(c) $\frac{x}{\sqrt{1+x^4}}$

(d) $\frac{x}{\sqrt{1-x^4}}$

Q 26. If $y = x \log\left(\frac{x}{a+bx}\right)$, then $\frac{x^3 d^2y}{dx^2}$ is equal to

[UP SEE]

(a) $x\frac{dy}{dx} - y$

(b) $\left(x\frac{dy}{dx} - y\right)^2$

(c) $y\frac{dy}{dx} - x$

(d) None of these

Q 27. If sin (x + y) + cos (x + y) = log (x + y), then $\frac{d^2y}{dx^2}$ is

[KCET]

(a) $\frac{-y}{x}$

(b) 0

(c) −1

(d) 1

Q 28. Derivative of $\sec^{-1}\left(\frac{1}{1-2x^2}\right)$ w.r.t $\sin^{-1}(3x - 4x^3)$ is

[Kerala CEE]

(a) $\frac{1}{4}$

(b) $\frac{3}{2}$

(c) 1

(d) $\frac{2}{3}$

(e) $-2/3$

Q 29. If x = A cos 4t + B sin 4t, then $\frac{d^2x}{dt^2}$ is

equal to

[AMU]

(a) $-16x$

(b) $16\ x$

(c) x

(d) –x

Q 30. If $y = (x + \sqrt{1+x^2})^n$, then $(1+x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx}$ is

[OJEE]

(a) n^2y

(b) $-n^2y$

(c) $-y$

(d) $2x^2y$

Q 31. Let $y = t^{10} + 1$ and $x = t^8 + 1$, then $\frac{d^2y}{dx^2}$ is

equal to

[MP PET]

(a) $\frac{5}{2}t$

(b) $20\ t^8$

(c) $\frac{5}{16t^6}$

(d) None of these

Q 32. $\frac{d}{dx}\left\{\tan^{-1}\left(\frac{2x}{1-x^2}\right)+\tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)-\tan^{-1}\left(\frac{4x-4x^3}{1-6x^2+x^4}\right)\right\}$ is

equal to

[J&K CET]

(a) $\frac{1}{\sqrt{1-x^2}}$

(b) $-\frac{1}{\sqrt{1-x^2}}$

(c) $\frac{1}{1+x^2}$

(d) $-\frac{1}{1+x^2}$

Q 33. The value of $\frac{d}{dx}\left[\tan^{-1}\left(\frac{\sqrt{x}(3-x)}{1-3x}\right)\right]$ is

[Kerala CEE]

(a) $\frac{1}{2(1+x)\sqrt{x}}$

(b) $\frac{3}{(1+x)\sqrt{x}}$

(c) $\frac{2}{(1+x)\sqrt{x}}$

(d) $\frac{3}{2(1-x)\sqrt{x}}$

(e) $\frac{3}{2(1+x)\sqrt{x}}$

Q 34. If $f : R \to R$ is an even function which is twice differentiable on R and $f''(\pi) = 1$, then $f''(-\pi)$ is equal to

(a) -1

(b) 0

(c) 1

(d) 2

[MHTCET]

Q 35. Observe the following statements

I. $f(x) = ax^{41} + bx^{-40} \Rightarrow \frac{f''(x)}{f(x)} = 1640x^{-2}$

II. $\frac{d}{dx}\tan^{-1}\left(\frac{2x}{1-x^2}\right) = \frac{1}{1+x^2}$

[EAMCET]

Which of the following is correct ?

(a) I is true, but II is false

(b) Both I and II are true

(c) Neither I nor II is true

(d) I is false, but II is true

Q 36. If $f(x) = 10 \cos x + (13 + 2x) \sin x$, then

$f''(x) + f(x)$ is equal to

[EAMCET]

(a) cos x

(b) 4 cos x

(c) sin x

(d) 4 sin x

Q $37.$ nth derivative of $(x + 1)^n$ is equal to

[MHTCET]

(a) $(n - 1)!$

(b) $(n + 1)!$

(c) n!

(d) $n[(n + 1)]^{n-1}$

Q 38. If $y = x^2 e^{mx}$, where m is a constant, then

$\frac{d^3y}{dx^3}$ is equal to

[MPPET]

(a) $me^{mx} (m^2x^2 + 6mx + 6)$

(b) $2m^3 xe^{mx}$

(c) $me^{mx} (m^2x^2 + 2mx + 2)$

(d) None of these

Q 39. If $y = 1 - x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} - \ldots.$, then $\frac{d^2y}{dx^2}$ is equal to

[RPET]

(a) –x

(b) x

(c) y

(d) –y

Q 40. If $y = 2^x \cdot 3^{2x-1}$, then $\frac{d^2y}{dx^2}$ is equal to

[Kerala CEE]

(a) (log 2) (log 3)

(b) (log 18)

(c) $(\log 18^2)y^2$

(d) (log 18)y

(e) $(\log 18)^2 y$

Q 41. If $f : R \to R$ is an even function having derivatives of all orders, then an odd function among the following is

[EAMCET]

(a) f''

(b) f'''

(c) $f' + f''$

(d) $f'' + f'''$

Q 42. $f(x)=\begin{vmatrix} x^3 & x^4 & 3x^2 \\ 1 & -6 & 4 \\ p & p^2 & p^3 \end{vmatrix}$, here p is a constant, then $\frac{d^4f(x)}{dx^4}$ is

[BCECE]

(a) proportional to x^2

(b) proportional to x

(c) proportional to x^3

(d) a constant

Q 43. f(x) is a polynomial of degree 2, $f(0) = 4$, $f'(0) = 3$ and $f''(0) = 4$, then $f(-1)$ is equal to

[MHTCET]

(a) 3

(b) -2

(c) 2

(d) -3

Q 44. If $f(x) = x^n$, then the value of

$$f(1) = \frac{f'(1)}{1!} + \frac{f''(1)}{2!} - \frac{f'''(1)}{3!} + ... + \frac{(-1)^n f^n(1)}{n!}$$ is

[AIEEE]

(a) 2^n

(b) 2^{n-1}

(c) 0

(d) 1

Q 45. If $y = \tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{2}\right)$, then $y'(0)$ is

[DCE]

(a) $1/2$

(b) 0

(c) 1

(d) does not exist

Topic 3

Partial Derivative

Q 1. $z = \tan(y + ax) + \sqrt{y - ax} \Rightarrow z_{xx} - a^2 z_{yy}$ is equal to

[EAMCET]

(a) 0

(b) 2

(c) $z_x + z_y$

(d) $z_x z_y$

Q 2. If $z = \sec^{-1}\left(\frac{x^4 + y^4 - 8x^2y^2}{x^2 + y^2}\right)$, then $x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y}$ is equal to

[EAMCET]

(a) cot z

(b) 2 cot z

(c) 2 tan z

(d) 2 sec z

Q 3. If z = log (tan x + tan y), then (sin 2x)

$\frac{\partial z}{\partial x}+(\sin 2y)\frac{\partial z}{\partial y}$ is equal to

[EAMCET]

(a) 1

(b) 2

(c) 3

(d) 4

Q 4. If $x^x\ y^y\ z^z = c$, then $\frac{\partial z}{\partial x}$ is equal to

[OJEE]

(a) $\left(\frac{1+\log x}{1+\log z}\right)$

(b) $-\left(\frac{1+\log x}{1+\log z}\right)$

(c) $\left(\frac{1+\log z}{1+\log x}\right)$

(d) None of these

Q 5. If $(x + y)\sin u = x^2y^2$, then $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ is equal to

[VITEEE]

(a) sin u

(b) cosec u

(c) 2 tan u

(d) 3 tan u

Q 6. If $u = \sin^{-1}\left(\frac{x^2 + y^2}{x + y}\right)$, then $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ is equal to

[EAMCET]

(a) sin u

(b) tan u

(c) cos u

(d) cot u

Q 7. If $f(x, y) = \frac{\cos(x-4y)}{\cos(x+4y)}$, then $\left.\frac{\partial f}{\partial x}\right|_{y=\frac{\pi}{2}}$ is equal to

[EAMCET]

(a) -1

(b) 0

(c) 1

(d) 2

Q 8. If $u = \sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right)$, then the value of $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ is

[EAMCET]

(a) 0

(b) 1

(c) 2

(d) None of these

Q 9. If $f(x, y) = 2(x - y)^2 - x^4 - y^4$, then $|(f_{xx}f_{yy} - f^2_{xy})|_{(0, 0)}$ is

[EAMCET]

(a) 32

(b) 16

(c) 0

(d) −1

Q 10. If $z = y + f(v)$, where $v = \left(\frac{x}{y}\right)$, then $v\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}$ is

[OJEE]

(a) -1

(b) 1

(c) 0

(d) 2

Q 11. If $u(x, y) = y \log x + x \log y$, then $u_x u_y - u_x \log x - u_y \log y + \log x \log y$ is equal to

[EAMCET]

(a) 0

(b) -1

(c) 1

(d) 2

Q 12. If $u = \log\left(\frac{x^2 + y^2}{x + y}\right)$, then the value of $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ is

[OJEE]

(a) -1

(b) 0

(c) 1

(d) 2

Answers

Topic 1 : Differentiation of Algebraic and Transcendental Functions

1. (b) 2. (c) 3. (c) 4. (d) 5. (a) 6. (a) 7. (d) 8. (a)

9. (*) 10. (c) 11. (c) 12. (d) 13. (d) 14. (e) 15. (e)

16. (d) 17. (d) 18. (b) 19. (e) 20. (b) 21. (d) 22.

(c) 23. (b) 24. (c) 25. (a) 26. (a) 27. (c) 28. (b)

29. (b) 30. (b) 31. (a) 32. (c) 33. (b) 34. (a) 35.

(b) 36. (c) 37. (a) 38. (d) 39. (d) 40. (d), 41. (a)

42. (b) 43. (a) 44. (c) 45. (e) 46. (a) 47. (a) 48.

(b) 49. (a) 50. (a) 51. (d) 52. (c) 53. (a) 54. (c)

55. (a) 56. (d) 57. (a) 58. (a) 59. (a) 60. (c)

61. (b) 62. (b) 63. (d) 64. (a) 65. (c) 66. (b) 67.

(a) 68. (a) 69. (a) 70. (b) 71. (c) 72. (a) 73. (c)

74. (c) 75. (a) 76. (b) 77. (b) 78. (c) 79. (a) 80.

(a) 81. (b) 82. (b) 83. (c) 84. (b) 85. (c) 86. (d)

87. (b) 88. (b) 89. (b) 90. (d) 91. (d) 92. (b) 93.

(b) 94. (a) 95. (c) 96. (b) 97. (b) 98. (a) 99. (c)

100. (b) 101. (c) 102. (d) 103. (d) 104. (b) 105.

(b) 106. (b) 107. (a) 108. (a) 109. (d) 110. (c)

101. (c) 112. (c) 113. (b) 114. (b) 115. (b) 116.

(c) 117. (c) 118. (c) 119. (c) 120. (c) 121. (e)

122. (d) 123. (c)

Topic 2 : Differentiation by Substitution, Higher Order Derivatives

1. (a) 2. (a) 3. (b) 4. (c) 5. (c) 6. (c) 7. (d) 8. (b)

9. (a) 10. (c) 11. (d) 12. (d) 13. (c) 14. (d) 15.

(b> 16. (d) 17. (c) 18. (d) 19. (d) 20. (d)

21. (a) 22. (b) 23. (b) 24. (d) 25. (d) 26. (b) 27.

(b) 28. (d) 29. (a) 30. (a) 31. (c) 32. (c) 33. (e)

34. (c) 35. (a) 36. (b) 37. (c) 38. (a) 39. (c) 40.

(e) 41. (b) 42. (d) 43. (a) 44. (c) 45. (a)

Topic 3 : Partial Derivative

1. (a) 2. (b) 3. (b) 4. (b) 5. (d) 6. (b) 7. (b) 8. (a)

9. (c) 10. (b) 11. (c) 12. (c)

Straight Lines and Pair of Straigt lines

Useful Results & Formulae

1. General equation of straight line is ax + by + c = 0 whose slope is m = $-\frac{a}{b}$.

2. Equations of straight lines in different forms are

(i) Slope intercept form y = mx + c

(ii) Point slope form $y - y_1 = m(x - x_1)$

(iii) Intercept form $\frac{x}{a}+\frac{y}{b}=1$

(iv) Two points form $(y - y_1)= \frac{y_2-y_1}{x_2-x_1}(x-x_1)$

(v) Normal form $x \cos \alpha + y \sin \alpha = p$

(vi) Distance form $\frac{x-x_1}{\cos\theta}=\frac{y-y_1}{\sin\theta}=r$

3. (i) Equation of a line which is parallel to $ax + by + c = 0$ is $ax + by + \lambda = 0$.

(ii) Equation of line which is perpendicular to $ax + by + c = 0$ is $bx - ay + \lambda = 0$.

4. If the equation of line be $a \sin \theta + b \cos \theta = c$, then line

(i) parallel to it, is $a \sin \theta + b \cos \theta = d$

(ii) perpendicular to it, is $a \sin\left(\frac{\pi}{2}+\theta\right) + b \cos\left(\frac{\pi}{2}+\theta\right) = d$

5. Point of intersection of two lines $a_1x + b_1y + c = 0$ and $a_2x + b_2y + c_2 = 0$ is $\left(\frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}, \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}\right)$

6. (i) Angle between the lines $y = m_1x + c_1$ and $y = m_2x + c_2$ is $\theta = \tan^{-1}\left|\frac{m_1 - m_2}{1 + m_1m_2}\right|$

If $m_1m_2 = -1$, then lines are perpendicular and if m_1

$= m_2$, then lines are parallel,

(ii) Angle between the lines $a_1x + b_1y + c_1 = 0$ and

$a_2x + b_2y + c_2 = 0$ is given by $\tan\theta = \left|\frac{a_2b_1 - a_1b_2}{a_1a_2 + b_1b_2}\right|$

7. Two lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2$

$= 0$ are

(i) coincident, if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

(ii) parallel, if $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

(iii) perpendicular, if $a_1a_2 + b_1b_2 = 0$

(iv) intersecting, if $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$

8. Equation of line passing through the point of intersection of two lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ is

$(a_1x + b_1y + c_1) + \lambda(a_2x + b_2y + c_2) = 0$.

9. Equation of bisectors of the angle between the lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$$

(i) If $a_1a_2 + b_1b_2 > 0$, then '+ ve' sign gives obtuse angle bisector and'– ve' sign gives acute angle bisector,

(ii) If $a_1a_2 + b_1b_2 < 0$, then '+ ve' sign gives acute angle bisector and '– ve' sign gives obtuse angle bisector.

10. (i) Length of perpendicular from the point (x_1, y_1) to the line $ax + by + c = 0$ is $\frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$

(ii) Length of perpendicular from the point (x_1, y_1) to the line $x \cos \alpha + y \sin \alpha = p$ is

$|x_1 \cos \alpha + y_1 \sin \alpha - p|$.

(iii) Distance between two parallel lines $ax + by + c_1 = 0$ and $ax + by + c_2 = 0$ is $\frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$.

11. Three lines $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ and $a_3x + b_3y + c_3 = 0$ are concurrent, if

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0.$$

12. The foot of the perpendicular (h,k) from (x_1, y_1) to the line ax + by + c =0 is given by

$$\frac{h-x_1}{a}=\frac{k-y_1}{b}=-\frac{(ax_1+by_1+c)}{a^2+b^2}$$

13. If the lines represented by $ax^2 + 2hxy + by^2 = 0$ are $y-m_1x = 0$ and $y - m_2x = 0$, then $m_1+ m_2 = -\frac{2h}{b}$ and $m_1m_2 = \frac{a}{b}$.

14. If $ax^2 + 2hxy + by^2 + 2gx + + 2fy + c = 0$ represents a pair of straight lines, then

$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$ or $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$

15. Angle between the lines represented by $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

or $ax^2 + 2hxy + by^2 = 0$ is given by $\tan\theta = \left|\frac{2\sqrt{h^2 - ab}}{a + b}\right|$.

Lines are parallel if $h^2 = ab$ and $af^2 = bg^2$ or $\frac{a}{h} = \frac{h}{b} = \frac{g}{f}$

Lines are perpendicular if $a + b = 0$.

16. Equation of bisector of the angles between the lines $ax^2 + 2hxv + by^2 = 0$ is $\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$.

17. Distance between the lines represented by $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is $2\sqrt{\frac{g^2 - ac}{a(a + b)}}$ or $2\sqrt{\frac{f^2 - bc}{b(a + b)}}$

18. The product of perpendiculars drawn from to the line $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is $\frac{c}{\sqrt{(a - b)^2 + 4h^2}}$.

19. To eliminate the first degree terms from the line $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$. The point to which origin is to be shifted is given by

$$\left(\frac{bg - fh}{h^2 - ab}, \frac{af - gh}{h^2 - ab}\right).$$

20. If bisectors of the angles between the lines $ax^2 + 2hxy + by^2 = 0$ and $a' x^2 + 2h' xy + b' y^2 = 0$ are same, then $\frac{h'}{h} = \frac{a' - b'}{a - b}$.

Topic 1 Equation of Line and Pair of Lines

Q 1. The equations of the lines through $(1, 1)$ and making angles of $45°$ with the line $x + y = 0$ are

[WB JEE]

(a) $x - 1 = 0, x - y = 0$

(b) $x - y = 0, y - 1 = 0$

(c) $x + y - 2 = 0, y - 1 = 0$

(d) $x - 1 = 0, y - 1 = 0$

Q 2. If the sum of distances from a point P on two mutually perpendicular straight lines is 1 unit, then the locus of P is

[WBJEE]

(a) a parabola

(b) a circle

(c) an ellipse

(d) a straight line

Q 3. The straight line $3x + y = 9$ divides the line segment joining the points $(1,3)$ and $(2,7)$ in the ratio

[BITSAT]

(a) $3:4$ externally

(b) $3:4$ internally

(c) $4:5$ internally

(d) $5:6$ externally

Q 4. The equations $y = \pm\sqrt{3}x$, $y = 1$ are the sides of

[BITSAT]

(a) an equilateral triangle

(b) a right angled triangle

(c) an isosceles triangle

(d) an obtuse angled triangle

Q 5. The equation of a straight line which passes through the point $(a\cos^3\theta, a\sin^3\theta)$ and perpendicular to $x\sec\theta + y\,\mathrm{cosec}\,\theta = a$ is

[Kerala CEE]

(a) $\frac{x}{a}+\frac{y}{a}=a\cos\theta$

(b) $x\cos\theta - y\sin\theta = -a\cos 2\theta$

(c) $x\cos\theta + y\sin\theta = a\cos 2\theta$

(d) $x\cos\theta + y\sin\theta - a\cos 2\theta = 1$

(e) $x\cos\theta - y\sin\theta + a\cos 2\theta = -1$

Q 6. If the line px − qy = r intersects the coordinate axes at $(a, 0)$ and $(0, b)$, then the value of a + b is equal to

[Kerala CEE]

(a) $r\left(\frac{q+p}{pq}\right)$

(b) $r\left(\frac{q-p}{pq}\right)$

(c) $r\left(\frac{p-q}{pq}\right)$

(d) $r\left(\frac{p+q}{p-q}\right)$

(e) $r\left(\frac{p-q}{p+q}\right)$

Q 7. If the slope of one of the lines represented by $ax^2 + 2hxy + by^2 = 0$ be the square of the other, then $\frac{a+b}{h}+\frac{8h^2}{ab}$ is

[AMU]

(a) 3

(b) 4

(c) 5

(d) 6

Q 8. The pairs of straight lines $x^2 - 3xy + 2y^2 = 0$ and $x^2 - 3xy + 2y^2 + x - 2 = 0$ form a

[EAMCET]

(a) square but not rhombus

(b) rhombus

(c) parallelogram

(d) rectangle but not a square

Q 9. Joint equation of pair of lines through $(3, -2)$

and parallel to $x^2 - 4xy + 3y^2 = 0$ is

[MHT CET]

(a) $x^2 + 3y^2 - 4xy - 14x + 24y + 45 = 0$

(b) $x^2 + 3y^2 + 4xy - 14x + 24y + 45 = 0$

(c) $x^2 + 3y^2 + 4xy - 14x + 24y - 45 = 0$

(d) $x^2 + 3y^2 + 4xy - 14x - 24y - 45 = 0$

Q 10. If the lines $kx - 2y - 1 = 0$ and $6x - 4y - m = 0$ are identical (coincident) lines, then the values of k and m are

[Guj. CET]

(a) $k = 3, m = 2$

(b) $k = -3, m = 2$

(c) $k = -3, m = -2$

(a) $k = 3, m = -2$

Q 11. Let a and b be non-zero and real numbers. Then, the equation $(ax^2 + by^2 + c)(x^2 - 5xy + 6y^2) = 0$ represents

[IIT JEE]

(a) Four straight lines, when $c = 0$ and a, b are of the same sign

(b) Two straight lines and a circle, when a = b and c is of sign opposite to that of a

(c) Two straight lines and a hyperbola, when a and b are of the same sign and c is of sign opposite to that of a

(d) A circle and an ellipse, when a and b are of the same sign and c is of sign opposite to that of a

Q 12. The perpendicular bisector of the line segment joining P(1, 4) and Q (k, 3) has y− intercept −4. Then, a possible value of k is

[AIEEE]

(a) -4

(b) 1

(c) 2

(d) -2

Q $13.$ The equation of line through the point $(1,1)$ and making angles of $45°$ with the line x + y = 0 are

[DCE]

(a) x − 1 = 0, x − y = 0

(b) $x - 1 = 0, y - 1 = 0$

(c) $x - y = 0, y - 1 = 0$

(d) $x + y - 2 = 0, y - 1 = 0$

Q 14. The point $P(a, b)$ lies on the straight line $3x + 2y = 13$ and the point $Q(b,a)$ lies on the straight line $4x - y = 5$, then equation of the line PQ is

[DCE]

(a) $x - 5 = 5$

(b) $x + y = 5$

(c) $x + y = -5$

(d) $x - y = -5$

Q 15. The equation of the line bisecting perpendicularly the segment joining the points $(-4,6)$ and $(8, 8)$ is

[UP SEE]

(a) $6x + y - 19 = 0$

(b) $y = 7$

(c) $6x + 2y - 19 = 0$

(d) $x + 2y - 7 = 0$

Q 16. The slopes of the lines represented by $x^2 + 2hxy + 2v^2 = 0$ are in the ratio $1 : 2$, then h equals

[UP SEE]

(a) $\pm\frac{1}{2}$

(b) $\pm\frac{3}{2}$

(c) ± 1

(d) ± 3

Q 17. If $3x + xy - y^2 - 3x + 6y + k = 0$

represents a pair of lines, then k is equal to

[KCET]

(a) 0

(b) 9

(c) 1

(d) -9

Q 18. The equation of a line through the point $(1, 2)$ whose distance from the point $(3,1)$ has the greatest value, is

[Kerala CEE]

(a) $y = 2x$

(b) $y = x + 1$

(c) $x + 2y = 5$

(d) $y = 3x - 1$

(e) $y = x + 1$

Q 19. If a line with y−intercept 2, is perpendicular to the line $3x - 2y = 6$, then its x−intercept is

[Kerala CEE]

(a) 1

(b) 2

(c) -4

(d) 4

(e) 3

Q 20. If the lines ax+ ky + $10 = 0$,bx + (k +1)y + $10 = 0$ and cx + (k + 2)y + $10 = 0$ are concurrent, then

[Kerala CEE]

(a) a, b, c are in GP

(b) a, b, c are in HP

(c) a, b, c are in AP

(d) $(a + b)^2 = c$

(e) a + b = c

Q 21. The value of k such that the lines $2x-3y + k = 0$, $3x - 4y - 13 = 0$ and $8x - 11y - 33 = 0$ are concurrent, is

[AMU]

(a) 20

(b) −7

(c) 7

(d) −20

Q 22. The value of λ, such that $\lambda x^2 - 10xy + 12y^2 + 5x - 16y - 3 = 0$ represents a pair of straight lines, is

[EAMCET]

(a) 1

(b) -1

(c) 2

(d) -2

Q 23. The equation of straight line equally inclined to the axes and equidistant from the points $(1,-2)$ and $(3, 4)$ is ax + by + c = 0, where

[OJEE]

(a) a = 1, b = -1, c = 3

(b) a= 1, b = -1, c = -3

(c) a= 1, b = 1, c = -3

(d) None of these

Q 24. The line which is parallel to x−axis and crosses the curve $y = \sqrt{x}$ at an angle $45°$, is

[WB JEE]

(a) $y = \frac{1}{4}$

(b) $y = \frac{1}{2}$

(c) $y = 1$

(d) $y = 4$

Q 25. The equation of the line passing through the point of intersection of the lines $x - 3y + 2 = 0$ and $2x + 5y - 7 = 0$ and perpendicular to the line $3x+2y + 5 = 0$, is

[BCECE]

(a) $2x - 3y + 1 = 0$

(b) $6x - 9y + 11 = 0$

(c) $2x - 3y + 5 = 0$

(d) $3x - 2y + l = 0$

Q 26. If the lines $x + 3y - 9 = 0$, $4x + by - 2 = 0$ and $2x - y - 4 = 0$ are concurrent, then b equals

[Jamia Millia Islamia]

(a) -5

(b) 5

(c) 1

(d) 0

Q 27. The equation $12x^2 + 7xy + ay^2 + 13x - y + 3 = 0$ represents a pair of perpendicular lines.

Then, the value of 'a' is

[MHTCET]

(a) $7/2$

(b) -19

(c) -12

(d) 12

Q 28. The equation of perpendicular bisectors of sides AB and AC of a A ABC are $x - y + 5 = 0$ and $x + 2y = 0$ respectively. If the coordinates of vertex A are $(1, -2)$, then equation of BC is

[Manipal]

(a) $14x + 23y - 40 = 0$

(b) $14x - 23y + 40 = 0$

(c) $23x + 14y - 40 = 0$

(d) $23x - 14y + 40 = 0$

Q 29. The line passing through the point of intersection of $x + y = 2$, $x - y = 0$ and is parallel to $x + 2y = 5$, is

[J&KCET]

(a) $x + 2y = 1$

(b) $x + 2y = 2$

(c) $x + 2y = 4$

(d) $x + 2y = 3$

Q 30. The straight line $3x + 4y - 5 = 0$ and $4x = 3y + 15$ intersect at the point P. On these lines the points Q and R are chosen so that PQ = PR. The slopes of the lines QR passing through $(1,2)$ are

[Guj. CET]

(a) $-7,1/7$

(b) $7,1/7$

(c) $7,-1/7$

(d) $3, -1/3$

Q 31. A line passes through the point of intersection of the lines $3x + y + 1 = 0$ and $2x - y + 3 = 0$ and makes equal intercepts with axes. Then, equation of the line is

[DCE]

(a) $5x + 5y - 3 = 0$

(b) $x + 5y - 3 = 0$

(c) $5x - y - 3 = 0$

(d) $5x + 5y + 3 = 0$

Q 32. If the lines $3x + 4y + 1 = 0$, $5x + \lambda y + 3 = 0$ and $2x + y - 1 = 0$ are concurrent, then λ is equal to

[UP SEE]

(a) -8

(b) 8

(c) 4

(d) -4

Q 33. The equation of the lines through the point $(3, 2)$ which makes an angle of $45°$ with the line $x - 2y = 3$, are

[Kerala CEE]

(a) $3x - y = 7$ and $x + 3y = 9$

(b) $x - 3y = 7$ and $3x + y = 9$

(c) $x - y = 3$ and $x + y = 2$

(d) $2x + y = 7$ and $x - 2y = 9$

(e) $2x - y = 7$ and $x + 2y = 9$

Q 34. The equation of the line which is such that the portion of line segment intercepted between the coordinate axes is bisected at $(4, -3)$, is

[Kerala CEE]

(a) $3x + 4y = 24$

(b) $3x - 4y = 12$

(c) $3x - 4y = 24$

(d) $4x - 3y = 24$

(e) $4x - 3y = 12$

Q 35. A straight line through the point A$(3, 4)$ is such that its intercept between the axes is bisected at A. Its equation is

[AMU]

(a) $3x - 4y + 7 = 0$

(b) $4x + 3y = 24$

(c) $3x + 4y = 25$

(d) $x + y = 7$

Q 36. The equation of straight line through the intersection of the lines $x - 2y = 1$ and $x + 3y = 2$ and parallel to $3x + 4y = 0$, is

[AMU]

(a) $3x + 4y + 5 = 0$

(b) $3x + 4y - 10 = 0$

(c) $3x + 4y - 5 = 0$

(d) $3x + 4y + 6 = 0$

Q 37. If the lines $x^2 + 2xy - 35y^2 - 4x + 44y - 12 = 0$ and $5x + \lambda y - 8 = 0$ are concurrent, then the value of λ is

[EAMCET]

(a) 0

(b) 1

(c) -1

(d) 2

Q 38. In order to eliminate the first degree terms from the equation $2x^2 + 4xy + 5y^2 - 4x - 22y + 7 = 0$, the point to which origin is to be shifted, is

[EAMCET]

(a) $(1,-3)$

(b) $(2,3)$

(c) $(-2,3)$

(d) $(1,3)$

Q 39. The equation of the pair of straight lines parallel to x−axis and touching the circle $x^2 + y^2 - 6x - 4y - 12 = 0$ is

[WBJEE]

(a) $y^2 - 4y - 21 = 0$

(b) $y^2 + 4y - 21 = 0$

(c) $y^2 - 4y + 21 = 0$

(d) $y^2 + 4y + 21 = 0$

Q 40. The equation of the straight line passing through the point $(4, 3)$ and making intercepts on the coordinate axes whose sum is -1, is

[Jamia Millia Islamia]

(a) $\frac{x}{2}+\frac{y}{3}=-1$ and $\frac{x}{-2}+\frac{y}{1}=-1$

(b) $\frac{x}{2}-\frac{y}{3}=-1$ and $\frac{x}{-2}+\frac{y}{1}=-1$

(c) $\frac{x}{2}+\frac{y}{3}=1$ and $\frac{x}{-2}+\frac{y}{1}=1$

(d) $\frac{x}{2}-\frac{y}{3}=1$ and $\frac{x}{-2}+\frac{y}{1}=1$

Q 41. If one of the lines given by $6x^2 - xy + 4cy^2 = 0$ is $3x + 4y = 0$, then c equals

[MP PET]

(a) 1

(b) −1

(c) 3

(d) −3

Q 42. If non-zero numbers a, b, c are in HP, then the straight line $\frac{x}{a}+\frac{y}{b}+\frac{1}{c}=0$ always passes through a fixed point. That point is

[RPET]

(a) $\left(1,-\frac{1}{2}\right)$

(b) $(1,-2)$

(c) $(-1,-2)$

(d) $(-1,2)$

Q 43. The point of concurrence of the lines ax + by + c = 0 and a,b,c satisfy the relation 3a + 2b + 4c = 0 is

[J&K CET]

(a) $\left(\frac{3}{2},\frac{1}{4}\right)$

(b) $\left(\frac{3}{4},\frac{1}{4}\right)$

(c) $\left(\frac{3}{4},\frac{1}{2}\right)$

(d) $\left(\frac{3}{2},\frac{1}{2}\right)$

Q 44. The equation to the bisecting the join of $(3,-4)$ and $(5, 2)$ and having its intercepts on the x−axis and the y−axis in the ratio $2 : 1$ is

[BITSAT, Guj. CET]

(a) $x + y - 3 = 0$

(b) $2x - y = 9$

(c) $x + 2y = 2$

(d) $2x + y = 7$

Q 45. The three straight lines $ax + by = c$, $bx + cy = a$ and $cx + ay = b$ are collinear, if

[UP SEE, WB JEE]

(a) $b + c = a$

(b) $c + a = b$

(c) $a + b + c = 0$

(d) $a + b = c$

Q 46. The equation of pair of lines joining origin to the points of intersection of $x^2 + y^2 = 9$ and $x + y = 3$ is

[UP SEE]

(a) $x^2+(3-x)^2 = 9$

(b) $xy = 0$

(c) $(3 + y)^2 + y^2 = 9$

(d) $(x - y)^2 = 9$

Q 47. The equation of the line passing through the origin and the point of intersection of the lines $\frac{x}{a}+\frac{y}{b}=1$ and $\frac{x}{b}+\frac{y}{a}=1$ is

[Kerala CEE]

(a) bx − ay = 0

(b) x + y = 0

(c) ax − by = 0

(d) x− y = 0

(e) ax + by = 0

Q 48. If $ax^2 - y^2 + 4x - y = 0$ represents a pair of lines, then a is equal to

[AMU]

(a) -16

(b) 16

(c) 4

(d) -4

Q 49. The centroid of the triangle formed by the pair of straight lines $12x^2 - 20xy + 7y^2 = 0$ and the line $2x - 3y + 4 = 0$ is

[EAMCET]

(a) $\left(-\frac{7}{3}, \frac{7}{3}\right)$

(b) $\left(-\frac{8}{3}, \frac{8}{3}\right)$

(c) $\left(\frac{8}{3}, \frac{8}{3}\right)$

(d) $\left(\frac{4}{3}, \frac{4}{3}\right)$

Q 50. The lines represented by the equation $x^2 - y^2 - x + 3y - 2 = 0$ are

[EAMCET]

(a) $x + y - 1 = 0, x - y + 2 = 0$

(b) $x - y - 2 = 0, x + y + 1 = 0$

(c) $x + y + 2 = 0, x - y - 1 = 0$

(d) $x - y + 1 = 0, x + y - 2 = 0$

Q 51. A straight line through the point $(2, 2)$ intersects the lines $\sqrt{3}x + y = 0$ and $\sqrt{3}x - y = 0$ at

the points A and B. The equation of the line AB, so that the ΔOAB is equilateral, is

[OJEE]

(a) $x - 2 = 0$

(b) $y - 2 = 0$

(c) $x + y - 4 = 0$

(d) None of these

Q 52. The value of λ, for which the equation $x^2 - y^2 - x + \lambda y - 2 = 0$ represents a pair of straight lines, are

[WB JEE]

(a) $-3, 1$

(b) $-1, 1$

(c) $3, -3$

(d) $3, 1$

Q 53. The straight line whose sum of the intercepts on the axes is equal to half of the product of the intercepts, passes through the point

[BCECE]

(a) $(1,1)$

(b) $(2, 2)$

(c) $(3, 3)$

(d) $(4, 4)$

Q 54. The value of 'p' for which the equation $x^2 + pxy + y^2 - 5x - 7y + 6 = 0$ represents a pair of straight lines, is

[Jamia Millia Islamia,RPET]

(a) $5/2$

(b) 5

(c) 2

(d) $2/5$

Q 55. A straight line through the point $(1,1)$ meets the x−axis at 'A' and y−axis at 'B'. The locus of the mid point of AB is

[MP PET]

(a) $2xy + x + y = 0$

(b) $x + y - 2xy = 0$

(c) $x + y + 2 = 0$

(d) $x + y - 2 = 0$

Q 56. The equation of the pair of straight lines perpendicular to the pair $2x^2 + 3xy + 2y^2 + 10x + 5y = 0$ and passing through the origin, is

[J&KCET]

(a) $2x^2 + 5xy + 2y^2 = 0$

(b) $2x^2 - 3xy + 2y^2 = 0$

(c) $2x^2 + 3xy + y^2 = 0$

(d) $2x^2 - 5xy + 2y^2 = 0$

Q 57. The line parallel to the x−axis and passing through the intersection of the lines $ax+2by+3b=0$ and $bx-2ay-3a=0$, where $(a, b) \neq (0,0)$ is

[AIEEE]

(a) above the x−axis at a distance of $(2/3)$ from it

(b) above the x−axis at a distance of $(3/2)$ from it

(c) below the x−axis at a distance of $(2/3)$ from it

(d) below the x−axis at a distance of $(3/2)$ from it

Q 58. If $\frac{x^2}{a}+\frac{y^2}{b}+\frac{2xy}{h}=0$ represents pair of straight lines such that slope of one line is twice the other.

Then, ab : h^2 is

[DCE]

(a) 9 : 8

(b) 8 : 9

(c) 1 : 2

(d) 2 : 1

Q 59. If $(\sin\theta, \cos\theta)$ and $(3, 2)$ lies on the same side of the line $x + y = 1$, then θ lies between

[DCE]

(a) $\left(0, \frac{\pi}{2}\right)$

(b) $(0, \pi)$

(c) $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

(d) $\left(0, \frac{\pi}{4}\right)$

Q 60. The equation of a line passing through $(-2,-4)$ and perpendicular to the line $3x - y + 5 = 0$ is

[UP SEE]

(a) $3y + x - 8 = 0$

(b) $3x + y + 6 = 0$

(c) $x + 3y + 14 = 0$

(d) None of these

Q 61. If the point (a, a) falls between the lines | x + y | = 4, then

[AMU]

(a) | a | = 2

(b) | a | = 3

(c) | a | < 2

(d) | a | < 3

Q 62. The equation of the straight line perpendicular to 5x − 2y = 7 and passing through

the point of intersection of the lines $2x + 3y = 1$ and $3x + 4y = 6$, is

[EAMCET]

(a) $2x + 5y + 17 = 0$

(b) $2x + 5y - 17 = 0$

(c) $2x - 5y + 17 = 0$

(d) $2x - 5y = 17$

Q 63. If PM is the perpendicular from P$(2, 3)$ onto the line x + y = 3, then the coordinates of M are

[EAMCET]

(a) $(2,1)$

(b) $(-1,4)$

(c) $(1,2)$

(d) $(4,-1)$

Q 64. If the equation $4x^2 + hxy + y^2 = 0$

represent coincident lines, then h is equal to

[MHTCET]

(a) 1

(b) 3

(c) 2

(d) 4

Q 65. A straight line through P$(1, 2)$ is such that its intercept between the axes is bisected at P. Its equation is

[MP PET]

(a) $x + y = -1$

(b) $x + y = 3$

(c) $x + 2y = 5$

(d) $2x + y = 4$

Q 66. If the equation $kx^2 - 2xy - y^2 - 2x + 2y = 0$ represents a pair of lines, then k is equal to

[MP PET]

(a) 2

(b) −2

(c) −5

(d) 3

Q 67. Two of the lines represented by the equation $ay^4 + bxy^3 + cx^2y^2 + dx^3y + ex^4 = 0$ will be perpendicular, then

[RPET]

(a) $(b + d)(ad + be) + (e - a)^2(a + c + e) = 0$

(b) $(b + d)(ad + be) + (e + a)^2(a + c + e) = 0$

(c) $(b - d)(ad - be) + (e - a)^2(a + c + e) = 0$

(d) $(b - d)(ad - be) + (e + a)^2(a + c + e) = 0$

Q 68. The points $(1, 1)$, $(-5,5)$ and $(13, \lambda)$ lie on the same straight line, if λ is equal to

[RPET]

(a) 7

(b) −7

(c) ± 7

(d) 0

Q 69. The equation of line bisecting perpendicularly the segment joining the points $(-4,6)$ and $(8, 8)$, is

[Guj. CET]

(a) $y = 7$

(b) $6x + y - 19 = 0$

(c) $x + 2y - 7 = 0$

(d) $6x + 2y - 19 = 0$

Q 70. If the slopes of one of the lines given by $ax^2 + 2hxy + by^2 = 0$ is 5 times the other, the

[Guj. CET]

(a) $5h^2 = 9ab$

(b) $5h^2 = ab$

(c) $h^2 = ab$

(d) $9h^2 = 5ab$

Q 71. The equation $y^2 - x^2 + 2x - 1 = 0$

represents

[UP SEE]

(a) a hyperbola

(b) an ellipse

(c) a pair of straight lines

(d) a rectangular hyperbola

Q 72. Equation of the straight line making equal intercepts on the axes and passing through the point $(2,4)$, is

[KCET]

(a) $4x - y - 4 = 0$

(b) $2x + y - 8 = 0$

(c) $x + y - 6 = 0$

(d) $x + 2Y - 10 = 0$

Q 73. If x_1, x_2, x_3 as well as y_1, y_2, y_3 are in GP with the same common ratio, then the points (x_1, y_1),(x_2, y_2) and (x_3, y_3)

[Kerala CEE]

(a) lie on a parabola

(b) lie on an ellipse

(c) lie on a circle

(d) are the vertices of a triangle

(e) lie on a straight line

Q 74. If the straight line ax + by + c = 0 always passes through $(1, -2)$ then a, b, care in

[AMU]

(a) AP

(b) HP

(c) GP

(d) None of these

Q 75. Suppose A, B are two points on $2x - y + 3 = 0$ and $P(1, 2)$, is such that PA = PB. Then, the mid point of AB is

[EAMCET]

(a) $\left(-\frac{1}{5}, \frac{13}{5}\right)$

(b) $\left(-\frac{7}{5}, \frac{9}{5}\right)$

(c) $\left(\frac{7}{5}, \frac{-9}{5}\right)$

(d) $\left(\frac{-7}{5}, \frac{-9}{5}\right)$

Q 76. The polar equation $\cos\theta + 7\sin\theta = \frac{1}{r}$

represents a

[EAMCET]

(a) circle

(b) parabola

(c) straight line

(d) hyperbola

Q 77. Given the four lines with equations x + 2

y = 3,3x + 4y = 7, 2x + 3y = 4 and 4x + 5y = 6,

then these lines are

[Jamia Millia Islamia]

(a) concurrent

(b) perpendicular

(c) the sides of a rectangle

(d) None of the above

Q 78. The equation to a pair of opposite sides of a parallelogram are $x^2 - 5x + 6 = 0$ and $y^2 - 6y + 5 = 0$, the equation to its diagonals are

[Jamia Millia Islamia]

(a) $x + 4y = 13$ and $y = 4x - 7$

(b) $4x + y = 13$ and $4y = x - 7$

(c) $4x + y - 13$ and $y - 4x - 7$

(d) $y - 4x = 13$ and $y + 4x = 7$

Q 79. The sum of slopes of lines $3x^2 + 5xy - 2y^2 = 0$ is

[MHT CET]

(a) $-\frac{5}{3}$

(b) $\frac{5}{2}$

(c) $-\frac{5}{2}$

(d) $-\frac{2}{3}$

Q 80. A square of side a lies above the x–axis and has one vertex at the origin. The side passing through the origin makes an angle $\alpha\left(0<\alpha<\frac{\pi}{4}\right)$ with the positive direction of x–axis. The equation of its diagonal not passing through the origin is

[AIEEE]

(a) $y(\cos\alpha - \sin\alpha) - x(\sin\alpha - \cos\alpha) = a$

(b) $y(\cos\alpha + \sin\alpha) + x(\sin\alpha - \cos\alpha) = a$

(c) $y(\cos\alpha + \sin\alpha) + x(\sin\alpha + \cos\alpha) = a$

(d) $y(\cos \alpha + \sin \alpha) + x(\cos \alpha - \sin \alpha) = a$

Q 81. y–intercept of line passes through $(2, 2)$ and is perpendicular to the line $3x + y = 3$, is

[DCE]

(a) $\frac{1}{3}$

(b) $\frac{2}{3}$

(c) 1

(d) $\frac{4}{3}$

Q 82. Two consecutive sides of a parallelogram are $4x + 5y = 0$ and $7x+2y = 0$. One diagonal of the parallelogram is $11x + 7y = 9$. If the other diagonal is $ax + by + c = 0$, then

[Kerala CEE]

(a) $a = -1, b = -1, c = 2$

(b) $a = 1, b = -1, c = 0$

(c) $a = -1, b = -1, c = 0$

(d) $a = 1, b = 1, c = 0$

(e) a = −1, b = −1, c = 1

Q 83. The equation of the sides of a triangle are $x - 3y = 0$, $4x + 3y = 5$ and $3x + y = 0$. The line $3x - 4y = 0$ passes through

[Kerala CEE]

(a) the incentre

(b) the centroid

(c) the orthocentre

(d) the circumcentre

(e) None of these

Q 84. Given four lines with equations $x + 2y - 3 = 0$, $2x + 3y - 4 = 0$, $3x + 4y - 5 = 0$, $4x + 5y - 6 = 0$ These lines are

[Kerala CEE]

(a) concurrent

(b) the sides of a quadrilateral

(c) the sides of a parallelogram

(d) the sides of a square

(e) the sides of a rhombus

Q 85. If $(-4,5)$ is one vertex and $7x - y + 8 = 0$ is one diagonal of a square, then the equation of the second diagonal is

[Kerala CEE]

(a) $x + 3y = 21$

(b) $2x - 3y = 7$

(c) $x + 7Y = 31$

(d) $2x + 3y = 21$

(e) $x - 3y = 21$

Q 86. Coordinates of the foot of the perpendicular drawn from $(0, 0)$ to the line joining $(a \cos \alpha, a \sin \alpha)$ and $(a \cos \beta, a \sin \beta)$ are

[AMU]

(a) $\left(\frac{a}{2}, \frac{b}{2}\right)$

(b) $\left[\frac{a}{2}(\cos\alpha + \cos\beta), \frac{a}{2}(\sin\alpha + \sin\beta)\right]$

(c) $\left[\cos\frac{\alpha+\beta}{2}, \sin\frac{\alpha+\beta}{2}\right]$

(d) $\left(0, \frac{b}{2}\right)$

Q 87. If the lines $4x + 3y - 1 = 0$, $x - y + 5 = 0$ and $kx + 5y - 3 = 0$ are concurrent, then k is equal to

[EAMCET]

(a) 4

(b) 5

(c) 6

(d) 7

Q 88. A ray of light passing through the point $(1, 2)$ is reflected on the x−axis at a point P and passes through the point $(5,3)$, then the abscissa of a point P is

[OJEE]

(a) 3

(b) 13/3

(c) 13/5

(d) $13/4$

Q 89. The equation $4x^2 - 24xy + 11\ y^2 = 0$

represents

[OJEE]

(a) two parallel lines

(b) two perpendicular lines

(c) two lines through the origin

(d) a circle

Q 90. If the equations, $12x^2 - 10xy + 2y^2 + 11x - 5y + k = 0$ represents two straight lines, then the value of k is

[MPPET]

(a) 1

(b) 2

(c) 0

(d) 3

Q 91. Equation of straight line cutting off an intercept 2 from the negative direction of the axes of y and inclined at $30°$ to the positive direction of axis of x, is

[RPET]

(a) $y + x - \sqrt{3} = 0$

(b) $y - x + 2 = 0$

(c) $y - \sqrt{3}x - 2 = 0$

(d) $\sqrt{3}y - x + 2\sqrt{3} = 0$

Topic 2 Distance between Lines and Foot of Perpendicular

Q 1. The line L given by $\frac{x}{5}+\frac{y}{b}=1$ passes through the point $(13, 32)$. The line K is parallel to L and has the equation $\frac{x}{c}+\frac{y}{3}=1$. Then, the distance between L and K is

[AIEEE]

(a) $\frac{23}{\sqrt{15}}$

(b) $\sqrt{17}$

(c) $\frac{17}{\sqrt{15}}$

(d) $\frac{23}{\sqrt{17}}$

Q 2. The equation of one of the lines parallel to $4x - 3y = 5$ and at a unit distance from the point $(-1, -4)$ is

[Kerala CEE]

(a) $3x + 4y - 3 = 0$

(b) $3x + 4y + 3 = 0$

(c) $4x - 3y + 3 = 0$

(d) $4x - 3y - 3 = 0$

(e) $4x - 3y - 4 = 0$

Q 3. A line through the point A $(2,0)$ which makes an angle of $30°$ with the positive direction of x−axis is rotated about A in clockwise direction through an angle of $15°$. Then, the equation of the straight line in the new position is

[WB JEE]

(a) $(2-\sqrt{3})x+y-4+2\sqrt{3}=0$

(b) $(2-\sqrt{3})x-y-4+2\sqrt{3}=0$

(c) $(2-\sqrt{3})x-y+4+2\sqrt{3}=0$

(d) $(2-\sqrt{3})x+y+4+2\sqrt{3}=0$

Q 4. A line has slope m and y−intercept 4.

The distance between the origin and the line is

equal to

[Kerala CEE]

(a) $\frac{4}{\sqrt{1-m^2}}$

(b) $\frac{4}{\sqrt{m^2-1}}$

(c) $\frac{4}{\sqrt{m^2+1}}$

(d) $\frac{4m}{\sqrt{1+m^2}}$

(e) $\frac{4m}{\sqrt{m-1}}$

Q 5. The distance of the point $(1, 2)$ from the line x + y + 5 = 0 measured along the line parallel to 3x − y = 7 is equal to

[Kerala CEE]

(a) $4\sqrt{10}$

(b) 40

(c) $\sqrt{40}$

(d) $10\sqrt{2}$

(e) $2\sqrt{20}$

Q 6. The number of points on the line x + y = 4 which are unit distance apart from the line 2x + 2y = 5 is

[WB JEE]

(a) 0

(b) 1

(c) 2

(d) ∞

Q 7. The distance between the pair of lines represented by the equation $x^2 - 6xy + 9y^2 + 3x - 9y - 4 = 0$ is

[UP SEE]

(a) $\frac{15}{\sqrt{10}}$

(b) $\frac{1}{2}$

(c) $\sqrt{\frac{5}{2}}$

(d) $\frac{1}{\sqrt{10}}$

Q 8. The distance between the lines $5x - 12y + 65 = 0$ and $5x - 12y - 39 = 0$ is

[WBJEE]

(a) 4

(b) 16

(c) 2

(d) 8

Q 9. The coordinates of the foot of perpendicular from $(a, 0)$ on the line $y = mx + \frac{a}{m}$ are

[WBJEE]

(a) $\left(0, \frac{a}{m}\right)$

(b) $\left(0, -\frac{a}{m}\right)$

(c) $\left(\frac{a}{m}, 0\right)$

(d) $\left(-\frac{a}{m}, 0\right)$

Q 10. The length of the perpendicular from the origin to the line $\frac{x \sin \alpha}{b} - \frac{y \cos \alpha}{a} - 1 = 0$ is

[J&KCET]

(a) $\frac{|ab|}{\sqrt{a^2 \cos^2 \alpha - b^2 \sin^2 \alpha}}$

(b) $\frac{|ab|}{\sqrt{a^2\cos^2\alpha + b^2\sin^2\alpha}}$

(c) $\frac{|ab|}{\sqrt{a^2\sin^2\alpha - b^2\cos^2\alpha}}$

(d) $\frac{|ab|}{\sqrt{a^2\sin^2\alpha + b^2\cos^2\alpha}}$

Q 11. The distance between the parallel lines y

= x + a,y = x + b is

[J&KCET]

(a) $\frac{|b-a|}{\sqrt{2}}$

(b) |a − b|

(c) $|a + b|$

(d) $\frac{|b+a|}{\sqrt{2}}$

Q 12. The distance between the lines $3x + 4y = 9$ and $6x + 8y = 15$ is

[UPSEE]

(a) $\frac{3}{2}$

(b) $\frac{3}{10}$

(c) 6

(d) None of these

Q 13. The coordinates of the foot of the perpendicular drawn from the point $(3, 4)$ on the line $2x + y - 7 = 0$ is

[KCET]

(a) $\left(\frac{9}{5}, \frac{17}{5}\right)$

(b) $(1,5)$

(c) $(-5,1)$

(d) $(1,-5)$

Q 14. If A $(2,-1)$ and B$(6, 5)$ are two points, then the ratio in which the foot of the perpendicular from $(4, 1)$ to AB divides it, is

[EAMCET]

(a) $8 : 15$

(b) $5 : 8$

(c) $-5 : 8$

(d) $-8 : 5$

Q 15. If the foot of the perpendicular from the origin to a straight line is at the point $(3, -4)$. Then, the equation of the line is

[WB JEE]

(a) $3x - 4y = 25$

(b) $3x - 4Y + 25 = 0$

(c) $4x + 3y - 25 = 0$

(d) $4x - 3y + 25 = 0$

Q 16. If $3, 4$ are intercepts of a line L = 0, then the distance of L = 0 from the origin is

[J&K CET]

(a) 5 units

(b) 12 units

(c) $\frac{5}{12}$ units

(d) $\frac{12}{5}$ units

Q 17. The distance between the pair of parallel lines $x + 2xy + y^2 - 8ax - 8ay - 9a^2 = 0$ is

[BITSAT, Guj. CET]

(a) $2\sqrt{5}a$

(b) $\sqrt{10}\,a$

(c) $10a$

(d) $5\sqrt{2}\,a$

Q 18. The length of perpendicular from the point $(a\cos\alpha, a\sin\alpha)$ upon the straight line

$y = x \tan \alpha + c$, $c > 0$, is

[UP SEE, WB JEE]

(a) c

(b) $c \sin^2 \alpha$

(c) $c \cos \alpha$

(d) $c \sec \alpha$

Q 19. The distance between the pair of parallel lines given by $x^2 - 1005 x + = 0$ is

[J&K CET]

(a) 1001

(b) 1000

(c) 1005

(d)

Q 20. The equation $x^2 + 2\sqrt{2}xy + 2y^2 + 4x + 4\sqrt{2}y + 1 = 0$ represents a pair of lines which are parallel to each other. The distance between them is

[DCE]

(a) 4 units

(b) $2\sqrt{3}$ units

(c) $4\sqrt{3}$ units

(d) 2 units

Q 21. If the equation of base of an equilateral triangle is $2x - y = 1$ and the vertex is $(-1, 2)$, then the length of the side of the triangle is

[Kerala CEE]

(a) $\sqrt{\frac{20}{3}}$

(b) $\frac{2}{\sqrt{15}}$

(c) $\sqrt{\frac{8}{15}}$

(d) $\sqrt{\frac{15}{2}}$

(e) $\sqrt{5}$

Q 22. The image of the origin with reference to the line $4x+3y-25=0$ is

[Kerala CEE]

(a) $(-8,6)$

(b) $(8,6)$

(c) $(-3,4)$

(d) $(8,-6)$

(e) $(-4,-3)$

Q 23. L is a variable line such that the algebraic sum of the distances of the points $(1, 1)$, $(2, 0)$ and $(0, 2)$ from the line is equal to zero. The line L will always pass through

[AMU]

(a) $(1, 1)$

(b) $(2, 1)$

(c) $(1, 2)$

(d) $(2, 2)$

Q 24. The product of the perpendicular distances from the origin on the pair of straight lines

$12x^2 + 25xy + 12y^2 + 10x + 11y + 2 = 0$ is

[EAMCET]

(a) $\frac{1}{25}$

(b) $\frac{2}{25}$

(c) $\frac{3}{25}$

(d) $\frac{4}{25}$

Q 25. Let a be the distance between lines $-x + y = 2$ and $x - y = 2$ and β be the distance between the lines $4x - 3y = 5$ and $6y - 8x = 1$, then

[J&K CET]

(a) $20\sqrt{2}\beta = 11\alpha$

(b) $20\sqrt{2}\alpha = 11\beta$

(c) $11\sqrt{2}\beta = 20\alpha$

(d) None of these

Q 26. If p is the length of the perpendicular from the origin on the line whose intercepts on the axes are a and b, then

[Guj. CET]

(a) $p^2 = a^2 + b^2$

(b) $p^2 = a^2 - b^2$

(c) $\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}$

(d) $\frac{1}{p^2} = \frac{1}{a^2} - \frac{1}{b^2}$

Q 27. The distance between the pair of parallel lines $x^2 + 4xy + 4y^2 + 3x + 6y - 4 = 0$ is

[Kerala CEE]

(a) $\sqrt{5}$

(b) $\frac{2}{\sqrt{5}}$

(c) $\frac{1}{\sqrt{5}}$

(d) $\frac{\sqrt{5}}{2}$

(e) $\sqrt{\frac{5}{2}}$

Q 28. If the line $\frac{x}{a}+\frac{y}{b}=1$ moves such that

$\frac{1}{a^2}+\frac{1}{b^2}=\frac{1}{c^2}$ where c is a constant, then the locus of

the foot of the perpendicular from the origin to the line is

[AMU]

(a) straight line

(b) circle

(c) parabola

(d) ellipse

Q 29. Distance between the two parallel lines $y = 2x + 7$ and $y = 2x + 5$ is

[OJEE]

(a) $\sqrt{5}/2$

(b) $2/5$

(c) $2/\sqrt{5}$

(d) $1/\sqrt{5}$

Q 30. The orthocentre of a triangle formed by the lines $x + y = 1$, $2x + 3y = 6$ and $4x - y + 4 = 0$ lies in the

[J&K CET]

(a) Ist quadrant

(b) IInd quadrant

(c) IIIrd quadrant

(d) IVth quadrant

Q 31. The equation of the base of an equilateral triangle is $x + y = 2$ and the vertex is $(2, -1)$, then the length of the side of the triangle is

[AMU]

(a) $\sqrt{3/2}/\sqrt{2/3}$

(b) $\sqrt{2}$

(c) $\sqrt{2/3}$

(d) $\sqrt{3/2}$

Q 32. Consider the fourteen lines in the plane given by $y = x + r$, $y = -x + r$, where $r \in \{0,1,2,3,4,5,6\}$. The number of squares formed by these lines, whose sides are of length $\sqrt{2}$, is

[EAMCET]

(a) 9

(b) 16

(c) 25

(d) 36

Q 33. The point on the axis of x, whose perpendicular distance from the straight line $\frac{x}{a}+\frac{y}{b}=1$ is a, are

[MP PET]

(a) $\frac{b}{a}(a \pm \sqrt{a^2 + b^2}, 0)$

(b) $\left(\frac{a}{b}(b \pm \sqrt{a^2 + b^2}), 0\right)$

(c) $\frac{b}{a}(a + b, 0)$

(d) $\frac{a}{b}(a \pm \sqrt{a^2 + b^2}, 0)$

Topic 3 Angle between Intersection Lines, Condition for Parallel and Perpendicular Lines

Q 1. The slopes of the lines which make an angle $45°$ with the line $3x - y = -5$ are

[Kerala CEE]

(a) $1, -1$

(b) $\frac{1}{2}, -1$

(c) $1, \frac{1}{2}$

(d) $2, -\frac{1}{2}$

(e) $-2, \frac{1}{2}$

Q 2. The locus of the orthocentre of the triangle formed by the lines $(1 + p)\, x - py + p\,(l + p) = 0$,

$(1 + q)\, x - qy + q\,(1 + q) = 0$ and $y = 0$, where $p \neq q$, is

[IIT JEE]

(a) a hyperbola

(b) a parabola

(c) an ellipse

(d) a straight line

Q 3. The lines $p(p^2 + 1)x - y + q = 0$ and $(p^2 + 1)^2x + (p^2 + 1)y + 2q = 0$ are perpendicular to a common line for

[AIEEE]

(a) exactly one value of p

(b) exactly two values of p

(c) more than two values of p

(d) no values of p

Q 4. All chords of the curve $3x^2 - y^2 - 2x + 4y = 0$ which subtend a right angle at the origin, pass through the fix point

[BITSAT]

(a) $(1, 2)$

(b) $(1, -2)$

(c) $(-1, 2)$

(d) $(-1, -2)$

Q 5. The pair of lines joining origin to the points of intersection of the two curves $ax^2 + 2hxy + by^2 + 2gx = 0$ and $a'x^2 + 2h'xy + b' y^2 + 2g'x = 0$ will be at right angles, if

[UP SEE]

(a) $(a' + b')g' = (a + b)g$

(b) $(a + b)g' = (a' + b')g$

(c) $h^2 - ab = h'^2 - a'b'$

(d) $a + b + h^2 = a' + b' + h'^2$

Q 6. If the lines $y = 3x + 1$ and $2y = x + 3$ are equally inclined to the line $y = mx + 4$, $\left(\frac{1}{2} < m < 3\right)$, then the values of m are

[Kerala CEE]

(a) $\frac{1}{7}(1 \pm 5\sqrt{3})$

(b) $\frac{1}{7}(1 \pm 5\sqrt{5})$

(c) $\frac{1}{7}(1 \pm 5\sqrt{2})$

(d) $\frac{1}{7}(1 \pm 2\sqrt{5})$

(e) $\frac{1}{7}(1 \pm 3\sqrt{2})$

Q 7. The area (in square unit) of the triangle formed by $x + y + 1 = 0$ and the pair of straight lines $x^2 - 3xy + 2y^2 = 0$ is

[EAMCET]

(a) $\frac{7}{12}$

(b) $\frac{5}{12}$

(c) $\frac{1}{12}$

(d) $\frac{1}{6}$

Q 8. A line passes through point $(2, 2)$ and perpendicular to the line $3x + y = 3$. Then, y–intercept is

[OJEE]

(a) 2

(b) $\frac{5}{3}$

(c) $\frac{4}{3}$

(d) $\frac{3}{4}$

Q 9. If the lines $px^2 - qxy - y^2 = 0$ make the angles α and β with x–axis, then the value of tan $(\alpha + \beta)$ is

[VITEEE]

(a) $\frac{-q}{1+p}$

(b) $\frac{q}{1+p}$

(c) $\frac{p}{1+q}$

(d) $\frac{-p}{1+q}$

Q 10. If 'θ' is the angle between the lines $ax^2 + 2hxy + by^2 = 0$, then angle between $x^2 + 2xy \sec\theta + y^2 = 0$ is

[MHT CET]

(a) θ

(b) 2θ

(c) $\frac{\theta}{2}$

(d) 3θ

Q 11. The lines $(a+2b)x + (a-3b)y = a-b$ for different values of a and b pass through the fixed point whose coordinates are

[Kerala CEE]

(a) $\left(\frac{2}{5}, \frac{2}{5}\right)$

(b) $\left(\frac{3}{5}, \frac{3}{5}\right)$

(c) $\left(\frac{1}{5}, \frac{1}{5}\right)$

(d) $\left(\frac{3}{5}, \frac{2}{5}\right)$

(e) $\left(\frac{2}{5}, \frac{3}{5}\right)$

Q 12. A line passes through the point of intersection of the lines $100x + 50y - 1 = 0$ and

$75x+25y + 3 = 0$ and makes equal intercept on the axes. Its equation is

[Kerala CEE]

(a) $25X + 25Y - 1 = 0$

(b) $5x - 5Y + 3 = 0$

(c) $25x + 25y - 4 = 0$

(d) $25x - 25y + 6 = 0$

(e) $5x - 5y + 7 = 0$

Q 13. A pair of perpendicular straight lines passes through the origin and also through the point of intersection of the curve $x^2 + y^2 = 4$ with $x + y = a$. The set containing the value of 'a' is

[AMU]

(a) $\{-2, 2\}$

(b) $\{-3, 3\}$

(c) $\{-4, 4\}$

(d) $\{-5, 5\}$

Q 14. The lines represents by $ax^2 + 2hxy + by^2 = 0$ are perpendicular to each other, if

[Jamia Millia Manila]

(a) $h^2 = a + b$

(b) $a + b = 0$

(c) $h^2 = ab$

(d) $h = 0$

Q 15. The equation of the bisector of the obtuse angle between the lines $3x - 4y + 7 = 0$

and $-12x - 5y + 2 = 0$, is

[Manipal]

(a) $21x + 77y - 101 = 0$

(b) $99x - 27y + 81 = 0$

(c) $21x - 77y + 101 = 0$

(d) None of these

Q 16. The acute angle between the lines joining the origin to the points of intersection of the line $\sqrt{3}x + y = 2$ and the circle $x^2 + y^2 = 4$, is

[Kerala CEE]

(a) $\pi/2$

(b) $\pi/3$

(c) $\pi/4$

(d) $\pi/6$

(e) $\pi/12$

Q 17. The angle between the line joining the points $(1,-2)$, $(3,2)$ and the line $x+2y-7=0$ is

[EAMCET]

(a) π

(b) $\pi/2$

(c) $\pi/3$

(d) $\pi/6$

Q 18. The angle between the pair of straight lines formed by joining the points of intersection of

$x^2 + y^2 = 4$ and $y = 3x + c$ to the origin is a right angle. Then, c^2 is equal to

[EAMCET]

(a) 20

(b) 13

(c) 1/5

(d) 5

Q 19. The angle between lines joining origin and intersection points of line $2x + y = 1$ and curve $3x^2 + 4yx - 4x + 1 = 0$ is

[WB JEE]

(a) $\pi/2$

(b) $\pi/3$

(c) $\pi/4$

(d) $\pi/6$

Q 20. The angle between the straight lines

$x - y\sqrt{3} = 5$ and $\sqrt{3}x + y = 7$ is

[WBJEE]

(a) $90°$

(b) $60°$

(c) $75°$

(d) $30°$

Q 21. The angle between the pair of lines $(x^2 + y^2)\sin^2 \alpha = (x \cos \theta - y \sin \theta)^2$ is

[J&KCET]

(a) θ

(b) 2θ

(c) α

(d) 2α

Q 22. If (a, a^2) falls inside the angle made by the lines $y = \frac{x}{2}$, $x > 0$ and $y = 3x$, $x > 0$, then a belongs to

[AIEEE]

(a) $(3, \infty)$

(b) $\left(\frac{1}{2}, 3\right)$

(c) $\left(-3, -\frac{1}{2}\right)$

(d) $\left(0, \frac{1}{2}\right)$

Q 23. The angle between the lines $\sqrt{3}x - y - 2 = 0$ and $x - \sqrt{3}y + 1 = 0$ is

[Kerala CEE]

(a) $90°$

(b) $60°$

(c) $45°$

(d) $15°$

(e) $30°$

Q 24. The angle between the lines represented by the equation $2x^2 + 3xy - 5y^2 = 0$, is [MHT CET]

(a) $\frac{\pi}{3}$

(b) $\frac{\pi}{2}$

(c) $\tan^{-1}\left|\frac{12}{5}\right|$

(d) $\tan^{-1}\left|\frac{7}{3}\right|$

Q 25. A line passing through origin and is perpendicular to two given lines $2x + y + 6 = 0$ and $4x + 2y - 9 = 0$. The ratio in which the origin divides this line, is

[DCE]

(a) $1 : 2$

(b) $2 : 1$

(c) $4 : 2$

(d) $4 : 3$

Q 26. The angle between the pair of straight lines $y^2 \sin^2\theta - xy \sin^2\theta + x^2(\cos^2\theta - 1) = 0$ is

[Kerala CEE]

(a) $\pi/3$

(b) $\pi/4$

(c) $\pi/6$

(d) $\pi/2$

(e) π

Q 27. A straight line through the point $(2, 2)$ intersects the lines $\sqrt{3}x + y = 0$ and $\sqrt{3}x - y = 0$ at the points A and B. The equation to the line AB so that the ΔOAB is equilateral is

[Jamia Millia Manila]

(a) $x - 2 = 0$

(b) $y - 2 = 0$

(c) $x + y - 4 = 0$

(d) None of these

Q 28. The angle between the lines in $x^2 - xy - 6y^2 - 7x + 31y - 18 = 0$ is

[Guj.CET]

(a) $60°$

(b) $45°$

(c) $30°$

(d) $90°$

Q 29. $ax + by - a^2 = 0$, where a,b are non-zero, is the equation to the straight line perpendicular to a line l and passing through the point where l crosses the x-axis. Then, equation to the line l is

[Kerala CEE]

(a) $\frac{x}{b}-\frac{y}{a}=1$

(b) $\frac{x}{a}-\frac{y}{b}=1$

(c) $\frac{x}{b}+\frac{y}{a}=ab$

(d) $\frac{x}{a}-\frac{y}{b}=ab$

(e) $\frac{x}{a}+\frac{y}{b}=ab$

Q 30. If the pair of straight lines given by Ax^2 $+2Hxy + By^2 = 0$ $(H^2 > AB)$ forms an equilateral

triangle with line ax + by + c = 0, then (A+3B) (3 A + B) is equal to

[EAMCET]

(a) H^2

(b) $-H^2$

(c) $2H^2$

(d) $4H^2$

Q 31. The line passing through $\left(-1, \frac{\pi}{2}\right)$ and perpendicular to $\sqrt{3}\sin\theta + 2\cos\theta = \frac{4}{r}$ is

[EAMCET]

(a) $2 = \sqrt{3}r\cos\theta - 2r\sin\theta$

(b) $5 = -2\sqrt{3}r\sin\theta + 4r\cos\theta$

(c) $2 = \sqrt{3}r\cos\theta + 2r\sin\theta$

(d) $5 = 2\sqrt{3}r\sin\theta + 4r\cos\theta$

Q 32. The lines $(lx + my)^2 - 3(mx - ly)^2 = 0$ and $lx + my + n = 0$ form

[MP PET]

(a) an isosceles triangle

(b) a right angled triangle

(c) an equilateral triangle (d) None of these

Topic 4 Position of a Point, Bisectors of the Lines

Q 1. The vertices of a triangle are A $(3, 7)$, B $(3, 4)$ and C $(5, 4)$. The equation of the bisector of the angle ABC is

[Kerala CEE]

(a) $y = x + 1$

(b) $y = x - 1$

(c) $y = 3x - 5$

(d) $y = x$

(e) $y = -x$

Q 2. If P is a point (x, y) on the line $y = -3x$ such that P and the point $(3, 4)$ are on the opposite sides of the line $3x - 4y - 8 = 0$, then

[AMU]

(a) $x > \frac{8}{15}, y < -\frac{8}{5}$

(b) $x > \frac{8}{5}, y < -\frac{8}{15}$

(c) $x = \frac{8}{15}, y = -\frac{8}{5}$

(d) None of these

Q 3. The bisector of the acute angle formed between the lines $4x - 3y + 7 = 0$ and $3x - 4y + 14 = 0$ has the equation

[DCE]

(a) $x + y + 3 = 0$

(b) $x - y - 3 = 0$

(c) $x - y + 3 = 0$

(d) $3x + y - 7 = 0$

Q 4. If the bisectors of angles represented by $ax^2 + 2hxy+by^2 = 0$ and $a'x^2 +2h'xy + b' y^2 = 0$ are same, then

[OJEE]

(a) $(a - b)h' = (a' - b')h$

(b) $(a - b)h = (a' - b')h'$

(c) $(a + b)h' = (a' - b')h$

(d) $(a - b)h' = (a' + b')h$

Q 5. If pairs straight lines $x^2 - 2pxy - y^2 = 0$ and $x^2 - 2qxy - y^2 = 0$ be such that each pair bisects the angle between the other pair, then

[MP PET]

(a) $pq = 1$

(b) $pq = -1$

(c) $pq = 2$

(d) $pq = -2$

Q 6. Lines $L_1 : y - x = 0$ and $L_2 : 2x + y = 0$ intersect the line $L_3 : y + 2 = 0$ at P and Q, respectively. The bisector of the acute angle between L_1 and L_2 intersects L_3 at R.

[IIT JEE]

Statement I The ratio PR : PQ equals $2\sqrt{2} : \sqrt{5}$.

Because Statement II In any triangle, bisector of an angle divides the triangle into two similar triangles.

(a) Statement I is true, Statement II is true;

Statement II is a correct explanation for Statement I

(b) Statement 1 is true, Statement II is true;

Statement II is not a correct explanation for

Statement I

(c) Statement I is true, Statement II is false

(d) Statement I is false, Statement II is true

Q 7. If one of the lines of $my^2 + (1 - m^2)xy - mx^2 = 0$ is a bisector of the angle between the lines $xy = 0$, then m is/are

[AIEEE]

(a) $-\frac{1}{2}$

(b) -2

(c) ± 1

(d) 2

Q 8. Let P = $(-1, 0)$, Q = $(0, 0)$ and R= $(3, 3\sqrt{3})$ be three points. The equation of the bisector of the angle PQR is

[AIEEE]

(a) $\sqrt{3}x + y = 0$

(b) $x + \frac{\sqrt{3}}{2} y = 0$

(c) $\frac{\sqrt{3}}{2} x + y = 0$

(d) $x + \sqrt{3} y = 0$

Q 9. The equation of bisectors of the angles between the lines | x | − | y | are

[WB JEE]

(a) $y = \pm x$ and $x = 0$

(b) $x = \frac{1}{2}$ and $y = \frac{1}{2}$

(c) $y = 0$ and $x - 0$

(d) None of these

Q 10. The equation of the bisector of the acute angle between the lines $3x - 4y + 7 = 0$ and

$12x + 5y - 2 = 0$ is

[WBJEE]

(a) $99x - 27y - 81 = 0$

(b) $11x - 3y + 9 = 0$

(c) $21x + 77y - 101 = 0$

(d) $21x + 77y + 101 = 0$

Q 11. The line $2x - y = 1$ bisects angle between two lines. If equation of one line is $y = x$, then the equation of the other line is

[Kerala CEE]

(a) $7x - y - 6 = 0$

(b) $x - 2y + 1 = 0$

(c) $3x - 2y - 1 = 0$

(d) $x - 7y + 6 = 0$

(e) $2x - 3y + 1 = 0$

Q 12. The position of reflection of the point $(4, 1)$ about the line $y = x - 1$ is

[Kerala CEE]

(a) $(1, 2)$

(b) $(3, 4)$

(c) $(-1, 0)$

(d) $(-2, -1)$

(e) $(2, 3)$

Q 13. Three straight lines $2x + 11y - 5 = 0$, $24x + 7y - 20 = 0$ and $4x - 3y - 2 = 0$

[OJEE]

(a) from a triangle

(b) are only concurrent

(c) are concurrent with one line bisecting the angle between the other two

(d) None of the above

Q 14. The image of the origin with reference to the line $4x + 3y - 25 = 0$, is

[MHT CET]

(a) $(-8,6)$

(b) $(8,6)$

(c) $(-3,4)$

(d) $(8,-6)$

Q 15. The bisector of the acute angle formed between the lines $4x - 3y + 7 = 0$ and $3x - 4y + 14 = 0$ has the equation

[Guj. CET]

(a) $x + y + 3 = 0$

(b) $x - y - 3 = 0$

(c) $x - y + 3 = 0$

(d) $3x + y - 7 = 0$

Q 16. The equation of the line bisecting perpendicularly the segment joining the points $(-4,6)$ and $(8, 8)$ is

[KCET]

(a) $y = 7$

(b) $6x + y - 19 = 0$

(c) $x + 2y - 7 = 0$

(d) $6x + 2y - 19 = 0$

Answers

Topic 1 : Equation of Line and Pair of Lines

1. (d) 2. (d) 3. (b) 4. (a) 5. (b) 6. (b) 7. (d) 8. (c)

9. (a) 10. (a) 11. (b) 12. (a) 13. (b) 14. (b) 15. (a)

16. (b) 17. (d) 18. (c) 19. (e) 20. (c) 21. (b) 22.

(c) 23. (c) 24. (b) 25. (a) 26. (a) 27. (c) 28. (a)

29. (d) 30. (a) 31. (a) 32. (b) 33. (a) 34. (c) 35.

(b) 36. (c) 37. (d) 38. (c) 39. (a) 40. (d) 41. (d)

42. (b) 43. (c) 44. (c) 45. (c) 46. (b) 47. (d) 48.

(b) 49. (c) 50. (d) 51. (b) 52. (c) 53. (b) 54. (a)

55. (b) 56. (b) 57. (d) 58. (a) 59. (a) 60. (c)

61. (a) 62. (a) 63. (c) 64. (d) 65. (d) 66. (d) 67.

(a) 68. (b) 69. (b) 70. (a) 71. (c) 72. (c) 73. (e)

74. (a) 75. (a) 76. (c) 77. (d) 78. (c) 79. (b) 80.

(d) 81. (d) 82. (b) 83. (c) 84. (a) 85. (c) 86. (b)

87. (c) 88. (c) 89. (c) 90. (b) 91. (d)

Topic 2 : Distance between Lines and Foot of Perpendicular

1. (d) 2. (d) 3. (b) 4. (c) 5. (c) 6. (a) 7. (c) 8. (d)

9. (a) 10. (d) 11. (a) 12. (b) 13. (a) 14. (b) 15. (a)

16. (d) 17. (d) 18. (c) 19. (a) 20. (d) 21. (a) 22.

(b) 23. (a) 24. (b) 25. (a) 26. (c) 27. (a) 28. (b)

29. (c) 30. (a) 31. (c) 32. (c) 33. (b)

Topic 3 : Angle between Intersection Lines, Condition for Parallel and Perpendicular Lines

1. (e) 2. (d) 3. (a) 4. (b) 5. (b) 6. (c). 7. (c) 8. (c)

9. (a) 10. (a) 11. (e) 12. (c) 13. (a) 14. (b) 15. (a)

16. (b) 17. (b) 18. (a) 19. (a) 20. (a) 21. (d) 22.

(b) 23. (e) 24. (d) 25. (d) 26. (d) 27. (b) 28. (b)

29. (b) 30. (d) 31. (a) 32. (c)

Topic 4 : Position of a Point, Bisectors of the Lines

1. (a) 2. (a) 3. (c) 4. (a) 5. (b) 6. (b) 7. (c) 8. (a)

9. (c) 10. (c) 11. (b) 12. (e) 13. (c) 14. (b) 15. (c)

16. (b)

Statistics

Useful Results & Formulae

1. (i) Arithmetic mean $\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$

$\overline{x} = A + \frac{\sum d}{n}$, where d = x – A

$$\overline{x} = \frac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i}$$

(ii) Weighted arithmetic mean

$$= \frac{w_1 x_1 + w_2 x_2 + ... + w_n x_n}{w_1 + w_2 + ... + w_n}$$

(iii) Combined mean, $\bar{x}_{12} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$, where $\bar{x}_1$ and n_1 are mean and size of first data and $\bar{x}_2$ and n_2 are mean and size of second data.

2. Geometric mean, GM = $(x_1 \cdot x_2 \cdot x_3 \ldots \cdot x_n)^{1/n}$

$$GM = (x_1^{f_1} . x_2^{f_2} . x_3^{f_3} ..... x_n^{f_n})^{1/N},$$

where $N = f_1 + f_2 + \ldots + f_n$

3. Harmonic mean, $HM = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + ... + \frac{1}{x_n}} = \frac{n}{\sum_{i=1}^{n} \frac{1}{x_i}}$

$HM = \frac{N}{\sum_{i=1}^{n} \left(\frac{f_i}{x_i} \right)}$, where $N = f_1 + f_2 + \ldots + f_n$

4. AM $\geq$ GM $\geq$ HM

5. (i) Median of individual series

(a) If n is odd, then median = value of $\left(\frac{n+1}{2}\right)$th

observation

(b) If n is even, then median $= \frac{1}{2}\left[\text{value of}\left\{\frac{n}{2}\text{th} + \left(\frac{n}{2}+1\right)\text{th}\right\}\right]$

observations

(ii) Median of continuous series

Median = $I + \left(\frac{\left(\frac{n}{2}\right) - C}{f}\right) \times h$

where l = lower limit, N = total frequency h = width of median class C = cumulative frequency of the class preceding the median class.

6. (i) Mode of individual series :

The maximum number of times, the observation repeated is a mode

(ii) Mode of continuous series:

Mode = $I_1 + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right] \times h$

where l_1 = lower limit and

h = width of modal class f_1 = frequency of the

modal class f_0 = frequency of the class preceding

the modal class f_2 = frequency of the class

succeeding the modal class

7. Mode = 3 Median − 2 Mean

8. (i) Range = $X_{max} - X_{min}$

(ii) Coefficient of range $= \frac{X_{max} - X_{min}}{X_{max} + X_{min}}$

(iii) Quartile deviation, QD $= \frac{1}{2}(Q_3 - Q_1)$

(iv) Coefficient of QD = $\frac{Q_3 - Q_1}{Q_3 + Q_1}$

9. (i) Mean deviation for ungrouped data (or individual series)

$$= \frac{\sum |x - M|}{n}$$

where M = mean (mean, median or mode)

n = number of terms

(ii) Mean deviation for continuous series

$$= \frac{\sum f |x - M|}{\sum f}$$

10. (i) Standard deviation

$$\sigma = \sqrt{\frac{\sum(x - \bar{x})^2}{n}}$$

Standard deviation $= \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2}$

where d = x − A (A = assumed mean)

(ii) Standard deviation is not depend on change of origin but it depends on change of scale.

11. (i) Variance = σ^2

(ii) Coefficient of variance $= \frac{\sigma}{\bar{x}} \times 100$

12. (i) In a symmetrical distribution,

Mean = Median = Mode

(ii) When the distribution is not symmetrical, it is called asymmetrical or skewed distribution

Mean $\neq$ Median $\neq$ Mode

(iii) Absolute measures of skewness (S_k)

S_k = Mean − Median = Mean − Mode

(iv) Karl Pearson's coefficient of skewness

$$S_k = \frac{\text{Mean} - \text{Mode}}{\text{Standard deviation}} = \frac{3(\text{Mean} - \text{Median})}{\text{Standard deviation}}$$

13. Covariance, cov (x, y) $= \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{n}$

$$= \left(\frac{\sum x_i y_i}{n} - \bar{x}\ \bar{y} \right)$$

14. (i) Karl Pearson's correlation coefficient

$$r = \frac{\text{cov}(x, y)}{\sqrt{\text{var}(x).\text{var}(y)}} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 . \sum(y_i - \bar{y})^2}}$$

$$= \frac{n\sum xy - \sum x \sum y}{\sqrt{\{n\sum x^2 - (\sum x)^2\}}\sqrt{\{n\sum y^2 - (\sum y)^2\}}}$$

(ii) Coefficient of rank correlation for

$$R = 1 - \frac{6\sum d_i^2}{n(n^2 - 1)}$$

where d_i, is rank difference.

(iii) $-1 \leq r \leq 1$

15. (i) Standard Error, $SE = \frac{1-r^2}{\sqrt{n}}$

(ii) Probable Error, PE = $0.6745 \left(\frac{1-r^2}{\sqrt{n}}\right)$

16. (i) Regression line of y on x

$$y - \bar{y} = \frac{cov(x, y)}{\sigma_x^2}(x - \bar{x})$$

or $y - \bar{y} = b_{yx}(x - \bar{x})$,

where $b_{yx} = r\frac{\sigma_y}{\sigma_x}$ = regression coefficient

(ii) Regression line of x on y

$$x - \bar{x} = \frac{\text{cov}(x, y)}{\sigma_y^2}(y - \bar{y})$$

or $x - \bar{x} = b_{xy}(y - \bar{y})$,

where $b_{xy} = r\frac{\sigma_x}{\sigma_y}$ = regression coefficient

17. Angle between two lines of regression is given by

$$\tan\theta = \left(\frac{1 - r^2}{r}\right)\frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2}$$

18. $r = \pm\sqrt{b_{yx}.b_{xy}}$, if both b_{yx} and b_{xy} are positive, then r will be positive and if both are negative, then r will be negative.

19. The interesection of two regression lines is on mean ie, $(\bar{x}.\bar{y})$

Topic 1 Mean, Median and Mode

Q 1. The mean of the values 0, 1, 2, 3, ..., n with the corresponding weights ${}^{n}C_{o}$, ${}^{n}C_{1}$,...., ${}^{n}C_{n}$ respectively, is

[Kerala CEE]

(a) $\frac{n+1}{2}$

(b) $\frac{n-1}{2}$

(c) $\frac{2^{n}-1}{2}$

(d) $\frac{2^{n}+1}{2}$

(e) $\frac{n}{2}$

Q 2. The mean age of a combined group of men and women is 25 yr. If the mean age of the group of men is 26 and that of the group of women is 21, then the percentage of men and women in the group is

[AMU]

(a) 46, 60

(b) 80, 20

(c) 20, 80

(d) 60, 40

Q 3. Median of ${}^{2n}C_0$, ${}^{2n}C_1$, ${}^{2n}C_2$, ${}^{2n}C_3$,..., ${}^{2n}C_n$

(where n is even) is

[UP SEE]

(a) ${}^{2n}C_{\frac{n}{2}}$

(b) ${}^{2n}C_{\frac{n+1}{2}}$

(c) ${}^{2n}C_{\frac{n-1}{2}}$

(d) None of these

Q 4. The average of the four-digit numbers that can be formed using each of the digits $3,5,7$ and 9 exactly once in each number, is

[Kerala CEE]

(a) 4444

(b) 5555

(c) 6666

(d) 7777

(e) 8888

Q 5. The values of mean, median and mode coincide, then the distribution is

[Jamia Millia Islamia]

(a) positive skewness

(b) symmetric distribution

(c) negative skewness

(d) All of the above

Q 6. The average marks of boys in a class is 52 and that of girls is 42. The average marks of boys and girls combined is 50. The percentage of boys in the class is

[AIEEE]

(a) 40%

(b) 20%

(c) 80%

(d) 60%

Q 7. In a class of 100 students there are 70 boys whose average marks in a subject are 75. If the average marks of the complete class is 72, then what is the average of the girls ?

[OJEE]

(a) 73

(b) 65

(c) 68

(d) 74

Q 8. If in a frequency distribution, the mean and median are 21 and 22 respectively, then its mode is approximately

[AIEEE]

(a) 24.0

(b) 25.5

(c) 20.5

(d) 22.0

Q 9. Mean marks scored by the students of a class is 53. The mean marks of the girls is 55 and the mean marks of the boys is 50. What is the percentage of girls in the class ?

[Kerala CEE]

(a) 60%

(b) 40%

(c) 50%

(d) 45%

(e) 55%

Q 10. The weight (in kilogram) of 15 students are as follows

31, 35, 27, 29, 32, 43, 37, 41, 34, 28, 36, 44, 45, 42, 30. If the weight 44 kg is replaced by 46 kg and 27 kg is by 25 kg, then new median is

[J&K CET]

(a) 32

(b) 33

(c) 34

(d) 35

Q 11. The median of a set of 9 distinct observations is 20.5. If each of the largest 4 observations of the set is increased by 2, then the median of the new set

[AIEEE]

(a) is increased by 2

(b) is decreased by 2

(c) is two times the original median

(d) remains the same as that of the original set

Q 12. The median from the table

Value	7	8	10	9	11	12	13
Frequency	9	1	4	5	6	1	3

is

[AMU]

(a) 100

(b) 10

(c) 110

(d) 1110

Q 13. The mode of the series 3,4,2,6,1,7,6,7,6, 8,9,5 is

[AMU]

(a) 5

(b) 6

(c) 7

(d) 8

Topic 2 Measure of Dispersion

Q 1. For two data sets, each of size 5, the variances are given to be 4 and 5 and the corresponding means are given to be 2 and 4, respectively. The variance of the combined data set is

[AIEEE]

(a) $\frac{5}{2}$

(b) $\frac{11}{2}$

(c) 6

(d) $\frac{13}{2}$

Q 2. The means and variance of n observations x_1, x_2 ,x_3,..., x_n are 5 and 0 respectively. If $\sum_{i=1}^{n} x_i^2 = 400$, then the value of n is equal to

[AIEEE]

(a) 80

(b) 25

(c) 20

(d) 16

(e) 4

Q 3. If the mean deviation of number $1, 1+ d, 1 + 2d, \ldots, 1 + 100c$? from their mean is 255, then the d is equal to

[AIEEE]

(a) 10.0

(b) 20.0

(c) 10.1

(d) 20.2

Q 4. **Statement I** The variance of first n even natural numbers is $\frac{n^2-1}{4}$.

[AIEEE]

Statement II The sum of first n natural numbers is $\frac{n(n+1)}{2}$ and the sum of squares of first n natural numbers is $\frac{n(n+1)(2n+1)}{6}$.

(a) Statement I is true, Statement II is true;

Statement II is a correct explanation for Statement I.

(b) Statement I is true, Statement II is true;

Statement II is not a correct explanation for Statement I.

(c) Statement I is true, Statement II is false.

(d) Statement I is false Statement II is true.

Q 5. If the variable takes the values $0, 1, 2,$ …, n with frequencies proportional to the binomial coefficients C(n, 0), C(n, 1), C(n, 2),…, C(n, n) respectively, then the variance of the distribution is

[DCE]

(a) n

(b) $\frac{\sqrt{n}}{2}$

(c) $\frac{n}{2}$

(d) $\frac{n}{4}$

Q 6. If the variance of $1,2,3, 4,5,\ldots, 10$ is $\frac{99}{12}$, then the standard deviation of $3, 6, 9, 12,\ldots, 30$ is

[Kerala CEE]

(a) $\frac{297}{4}$

(b) $\frac{3}{2}\sqrt{33}$

(c) $\frac{3}{2}\sqrt{99}$

(d) $\sqrt{\frac{99}{12}}$

(e) $3\sqrt{\frac{3}{2}}$

Q 7. If the coefficient of variation is 45% and the mean is 12 then its standard deviation is

[AMU]

(a) 5.2

(b) 5.3

(c) 5.4

(d) None of these

Q 8. The mean deviation from the mean of the set of observations, $-1, 0, 4$ is

[OJEE]

(a) 3

(b) 1

(c) -2

(d) 2

Q 9. The variance of first n numbers is

[BCECE]

(a) $\frac{n^2+1}{12}$

(b) $\frac{n^2-1}{12}$

(c) $\frac{(n+1)(2n+1)}{6}$

(d) $\left[\frac{n(n+1)}{2}\right]^2$

Q 10. If $x_1, x_2, \ldots x_{18}$ are observation such that $\sum_{j=1}^{18}(x_j - 8) = 9$ and $\sum_{j=1}^{18}(x_j - 8)^2 = 45$, then these standard deviation of these observations is

[J&K CET]

(a) $\sqrt{\frac{81}{34}}$

(b) 5

(c) $\sqrt{5}$

(d) $\frac{3}{2}$

Q 11. The mean of the numbers a, b, 8, 5, 10 is 6 and the variance is 6.80. Then, which one of the following gives possible values of a and b ?

[AIEEE]

(a) a = 3, b = 4

(b) a = 0, b = 7

(c) a = 5, b = 2

(d) a =1, b = 6

Q 12. If the standard deviation of the observations $-5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5$ is $\sqrt{10}$. The standard deviation of observations $15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25$ will be

[DCS]

(a) $\sqrt{10}+20$

(b) $\sqrt{10}+10$

(c) $\sqrt{10}$

(d) None of these

Q 13. The standard deviation for the scores 1, 2, 3, 4, 5, 6 and 7 is 2. Then, the standard deviation of

12, 23, 34, 45, 56, 67 and 78 is

[Kerala CEE]

(a) 2

(b) 4

(c) 22

(d) 11

(e) 44

Q 14. Mean and standard deviation from the following observations of marks of 5 students of a tutorial group (marks out of 25) 8, 12, 13, 15, 22 are respectively

[VITEEE]

(a) 14, 4.604

(b) 15, 4.604

(c) 14, 5.604

(d) None of these

Q 15. The standard deviation of a variable x is 10. Then, the standard deviation of $50 + 5x$ is

[OJEE]

(a) 50

(b) 550

(c) 10

(d) 500

Q 16. The standard deviation of the numbers

31, 32, 33, ..., 46,47 is

[Guj. CET]

(a) $\sqrt{\frac{17}{12}}$

(b) $\sqrt{\frac{47^2-1}{12}}$

(c) $2\sqrt{6}$

(d) $4\sqrt{3}$

Q 17. The quartile deviation for the following data is

X	2	3	4	5	6
F	3	4	8	4	1

[Kerala CEE]

(a) 0

(b) $\frac{1}{4}$

(c) $\frac{1}{2}$

(d) 1

(e) $\frac{1}{5}$

Q 18. If the mean of n observations $1^2, 2^2, 3^2,$ …, n^2 is $\frac{46n}{11}$, then n is equal to

[AMU]

(a) 11

(b) 12

(c) 23

(d) 22

Q 19. What is the standard deviation of the following series

Measurements	0−10	10−20	20−30	30−40
Frequency	1	3	4	2

[EAMCET]

(a) 81

(b) 7.6

(c) 9

(d) 2.26

Q 20. Suppose a population A has 100 observations 101, 102, ..., 200 and another population B has 100 observations 151, 152, ...,

250. If V_A and V_B represent the variances of the two populations respectively, then $\frac{V_A}{V_B}$ is

[AIEEE]

(a) $\frac{9}{4}$

(b) $\frac{4}{9}$

(c) $\frac{2}{3}$

(d) 1

Q 21. The standard deviation of n observations $x_1, x_2, \ldots, x_n$ is 2. If $\sum_{i=1}^{n} x_i = 20$ and $\sum_{i=1}^{n} x_i^2 = 100$, then n is

[Kerala CEE]

(a) 10 or 20

(b) 5 or 10

(c) 5 or 20

(d) 5 or 15

(e) 25

Q 22. For the arithmetic progression a, (a + d), (a + 2d), (a + 3d), ..., (a + 2nd), the mean deviation from mean is

[Kerala CEE]

(a) $\frac{n(n+1)d}{2n-1}$

(b) $\frac{n(n+1)d}{2n+1}$

(c) $\frac{n(n-1)d}{2n+1}$

(d) $\frac{(n+1)d}{2}$

(e) $\frac{n(n-1)d}{2n-1}$

Q 23. If σ is the standard deviation of a random variable x, then the standard deviation of the random variable ax + b, where a, b $\in$ R is

[J&K CET]

(a) $a\sigma + b$

(b) $|a|\ \sigma$

(c) $|a|\ \sigma + b$

(d) $a^2\ \sigma$

Q 24. Let $x_1, x_2, \ldots, x_x$, be n observations such that $\sum x_i^2 = 400$ and $\sum x_i = 80$. Then, a possible value of n among the following is

[AIEEE]

(a) 12

(b) 9

(c) 18

(d) 15

Q 25. Consider the following statements

(1) Mode can be computed from histogram

(2) Median is not independent of change of scale

(3) Variance is independent of change of origin

and scale Which of these is/are correct ?

[AIEEE]

(a) Only (1)

(b) Only (2)

(c) Only (1) and (2)

(d) Only (1), (2) and (3)

Q 26. In a series of 2n observations, half of them equal a and remaining half equal − a. If the standard deviation of the observations is 2, then | a | equals

[AIEEE]

(a) $\frac{1}{n}$

(b) $\sqrt{2}$

(c) 2

(d) $\frac{\sqrt{2}}{n}$

Q 27. In an experiment with 15 observations on x, the following results were available $\sum x^2 = 2830$, $\sum x = 170$. One observation that was 20, was found to be wrong and was replaced by the correct value 30. Then, the corrected variance is

[AIEEE]

(a) 78.0

(b) 188.66

(c) 177.33

(d) 8.33

Q 28. Standard deviation of the first $2n+1$ natural numbers is equal to

[Kerala CEE]

(a) $\sqrt{\frac{n(n+1)}{2}}$

(b) $\sqrt{\frac{n(n+1)(2n+1)}{3}}$

(c) $\sqrt{\frac{n(n+1)}{3}}$

(d) $\sqrt{\frac{n(n-1)}{2}}$

(e) $2n+1$

Q 29. If the standard deviation of a variable x is σ, then the standard deviation of another variable $\frac{ax+b}{c}$ is

[AMU]

(a) $\frac{\sigma a + b}{c}$

(b) $\frac{\sigma a}{c}$

(c) σ

(d) None of these

Q 30. The, coefficient of SD and coefficient of variance from the given data

Class interval	0–10	10–20	20–30	30–40	40–50

Frequency	2	10	8	4	6

is

[AMU]

(a) 50,48.1

(b) 51.9,48.1

(c) 0.481,48.1

(d) 0.481,51.8

Q 31. The mean-deviation and coefficient of mean deviation from the data. Weight (in kg) 54, 50,40, 42, 51,45,47, 55, 57 is

[AMU]

(a) 0.0900

(b) 0.0956

(c) 0.0056

(d) 0.0946

Topic 3 Correlation and Rank Correlation

Correlation

Q 1. If Z = aX + bY and r be the correlation coefficient between X and Y, then σ_Z^2 is equal to

[Jamia Millia Islamia]

(a) $a^2\sigma_X^2 + b^2\sigma_Y^2 + 2abr\,\sigma_X\sigma_Y$

(b) $a^2\sigma_X^2 + b^2\sigma_Y^2 - 2abr\,\sigma_X\sigma_Y$

(c) $2abr\,\sigma_X\sigma_Y$

(d) None of the above

Q 2. For the given data, the calculation corresponding to all values of variates (x, y) is following $\sum(x-\bar{x})^2 = 36,\ \sum(y-\bar{y})^2 = 25,\ \sum(x-\bar{x})(y-\bar{y}) = 20$

[Jamia Millia Islamia]

The Karl Pearson's correlation coefficient is

(a) 0.2

(b) 0.5

(c) 0.66

(d) 0.33

Q 3. The correlation coefficient between x and y from the following data $\Sigma x = 40$, $\Sigma y = 50$, $\Sigma xy = 220$, $\Sigma x^2 = 200$, $\Sigma y^2 = 262$, $n = 10$ is

[MP PET]

(a) 0.89

(b) 0.76

(c) 0.91

(d) 0.98

Q 4. Two numbers within the brackets denote the ranks of 10 students of a class in two subjects $(1, 10)$, $(2, 9)$, $(3, 8)$, $(4,7)$, $(5,6)$, $(6, 5)$, $(7,4)$, $(8,3)$, $(9,2)$, $(10,1)$, then rank correlation coefficient is

[MP PET]

(a) 0

(b) -1

(c) 1

(d) 0.5

Q 5. If $\bar{x} = \bar{y} = 0$, $\Sigma x_i y_i = 12$, $\sigma_x = 2$, $\sigma_y = 3$ and $n = 10$, then the coefficient of correlation is

[Jamia Millia Islamia]

(a) 0.1

(b) 0.3

(c) 0.2

(d) 0.1

Q 6. If var (x) = 8.25, var (y) = 33.96 and cov (x, y) = 102 then the correlation coefficient is

[AMU]

(a) 0.89

(b) -0.98

(c) 0.61

(d) −0.16

Q 7. If $\Sigma x = 15$, $\Sigma y = 36$, $\Sigma xy = 110$, $n = 5$,

then cov (x, y) equals

[AMU]

(a) 1/5

(b) −1/5

(c) 2/5

(d) $-2/5$

Q 8. If r is Karl Pearson's coefficient of correlation between two sets of variates, then

[J&K CET]

(a) $r < 1$

(b) $r > 1$

(c) $r < -1$

(d) $|r| \leq 1$

Topic 4 Regression

Q 1. If the variance of x = 9 and regression equations are $4x - 5y + 33 = 0$ and $20 - 9y - 10 = 0$, then the coefficient of correlation between x and y and the variance of y respectively are

[Jamia Millia Islamia]

(a) 0.6; 16

(b) 0.16; 16

(c) $0.3;4$

(d) $0.6;\ 4$

Q 2. The correlation coefficient of two variables x and y is 0.8. The regression coefficient of y on x is 0.2, then the regression coefficient of x on y is

[MP PET]

(a) 3.2

(b) -3.2

(c) 4

(d) 0.16

Q 3. Angle between two lines of regression is given by

[DCE]

(a) $\tan^{-1}\left(\frac{b_{xy}-\frac{1}{b_{yx}}}{1-\frac{b_{xy}}{b_{yx}}}\right)$

(b) $\tan^{-1}\left(\frac{b_{yx}.b_{xy}-1}{b_{yx}+b_{xy}}\right)$

(c) $\tan^{-1}\left(\dfrac{b_{xy}-\dfrac{1}{b_{yx}}}{1+\dfrac{b_{xy}}{b_{yx}}}\right)$

(d) $\tan^{-1}\left(\dfrac{b_{yx}-b_{xy}}{1+b_{yx}.b_{xy}}\right)$

Q 4. If θ is the angle between two regression lines with correlation coefficient γ, then

[UP SEE]

(a) $\sin\theta \geq 1 - \gamma^2$

(b) $\sin\theta \leq 1 - \gamma^2$

(c) $\sin\theta \leq \gamma^2 + 1$

(d) $\sin \theta \leq \gamma^2 - 1$

Q 5. In a bivariate data $\Sigma x = 30$, $\Sigma y = 400$, $\Sigma x^2 = 196$, $\Sigma xy = 850$ and $n = 10$. The regression coefficient of y on x is

[MPPET]

(a) -3.1

(b) -3.2

(c) -3.3

(d) -3.4

Q 6. If the lines of regression are $3x + 12y = 19$ and $3y + 9x = 46$, then r_x will be

[MP PET]

(a) 0.289

(b) -0.289

(c) 0.209

(d) None of these

Q 7. If two lines of regression are $3\bar{x} - 2\bar{y} + 1 = 0$ and $2\bar{x} - \bar{y} - 2 = 0$, then $(\bar{x}, \bar{y})$ is

[DCE]

(a) (8, 5)

(b) (5, 8)

(c) (5, 5)

(d) (8, 8)

Q 8. If there exists a linear statistical relationship between two variables x and y, then the regression coefficient of y on x is

[BCECE]

(a) $\frac{\text{cov}(x, y)}{\sigma_x \sigma_y}$

(b) $\frac{\text{cov}(x, y)}{\sigma_y^2}$

(c) $\frac{\text{cov}(x, y)}{\sigma_x^2}$

(d) None of these

Q 9. When the origin is changed, then the coefficient of correlation

[BCECE]

(a) becomes zero

(b) varies

(c) remains fixed

(d) None of these

Q 10. If the two lines of regression are $4x + 3y + 7 = 0$ and $3x + 4y + 8 = 0$, then the means of x and yare

[BCECE]

(a) $\frac{-4}{7}, \frac{-11}{7}$

(b) $\frac{-4}{7}, \frac{11}{7}$

(c) $\frac{4}{7}, \frac{-11}{7}$

(d) 4, 7

Q 11. If the values of regression coefficients are -0.33 and -1.33, then the value of coefficients of correlation between the two variables, is

[MP PET]

(a) 0.2

(b) -0.66

(c) 0.4

(d) -0.4

Q 12. The intersecting point of two regression lines is

[Kerala CEE]

(a) $(\bar{x}, 0)$

(b) $(0, \bar{y})$

(c) (b_{xy}, b_{yx})

(d) $(0,0)$

(e) $(\bar{x}, \bar{y})$

Q 13. The regression coefficient of y on x is $2/3$ and that of x on y is $4/3$. The acute angle between the two regression lines is $\tan^{-1} k$, where k is equal to

[Kerala CEE]

(a) $1/9$

(b) $2/9$

(c) $1/18$

(d) $1/3$

(e) $8/9$

Q 14. The two regression lines are $2x - 7y + 6 = 0$ and $7x - 2y + 1 = 0$. The correlation coefficient between x and y is

[BCECE]

(a) $-\frac{2}{3}$

(b) $\frac{2}{7}$

(c) $\frac{4}{9}$

(d) None of these

Q 15. If the two lines of regression are $x + 4y = 3$ and $3x + y = 15$, then value of x for $y = 3$ is

[BCECE]

(a) 4

(b) −9

(c) −4

(d) None of these

Q 16. If both the regression lines intersect perpendicularly, then

[Jamia Millia Islamia]

(a) $r < -1$

(b) $r = -1$

(c) $r = 0$

(d) $r = \frac{1}{2}$

Q 17. If the regression coefficients are 0.8 and 0.2, then the value of coefficient of correlation is

[DCE]

(a) 0.16

(b) 0.4

(c) 0.04

(d) 0.164

Q 18. The coefficient of correlation (r) and the two regression coefficients b_{yx}, b_{xy} are related as

(a) $r = \frac{b_{xy}}{b_{yx}}$

(b) $r = b_{xy} \times b_{yx}$

(c) $r = b_{xy} + b_{yx}$

(d) $r = (\text{sign } b_{yx}) \sqrt{b_{xy} b_{yx}}$

Q 19. The two lines of regression are given by $3x + 2y = 26$ and $6x + y = 31$. The coefficient of correlation between x and y is

[AMU]

(a) $-1/3$

(b) $1/3$

(c) $-1/2$

(d) $1/2$

Answers

Topic 1: Mean, Median and Mode

1. (e) 2. (b) 3. (a) 4. (c) 5. (b) 6. (c) 7. (b) 8. (a)

9. (a) 10. (d) 11. (d) 12. (b) 13. (b)

Topic 2 : Measure of Dispersion

1. (b) 2. (d) 3. (c) 4. (d) 5. (d) 6. (b) 7. (c) 8. (d)

9. (b) 10. (d) 11. (a) 12. (e) 13. (c) 14. (a) 15. (a)

16. (c) 17. (d) 18. (a) 19. (c) 20. (d) 21. (c) 22.

(b) 23. (b) 24. (c) 25. (c) 26. (c) 27. (a) 28. (c)

29. (b) 30. (c) 31. (b)

Topic 3 : Correlation and Rank Correlation

1. (a) 2. (c) 3. (c) 4. (b) 5. (c) 6. (c) 7. (c) 8. (d)

Topic 4 : Regression

1. (a) 2. (a) 3. (b) 4. (b) 5. (c) 6. (b) 7. (b) 8. (c)

9. (c) 10. (a) 11. (b) 12. (e) 13. (c) 14. (b) 15. (a)

16. (c) 17. (b) 18. (d) 19. (c)

Rectangular Cartesian Coordinates

Topic – 1

System of Coordinates and Distance between Points

Q 1. If the three points $(0, 1)$, $(0, -1)$ and $(x, 0)$ are vertices of an equilateral triangle, then the values of x are

[Kerala CEE]

(a) $\sqrt{3}, \sqrt{2}$

(b) $\sqrt{3}, -\sqrt{3}$

(c) $-\sqrt{5}, \sqrt{3}$

(d) $\sqrt{2}, -\sqrt{2}$

(e) $\sqrt{5}, -\sqrt{5}$

Q 2. If the distance between $(2, 3)$ and $(-5, 2)$ is equal to the distance between $(x, 2)$ and $(1, 3)$, then the values of x are

[BITSAT]

(a) $-6, 8$

(b) $6,8$

(c) $-8,6$

(d) $-7, 7$

(e) $-8, -6$

Q 3. Point Q is symmetric to P $(4, -1)$ with respect to the bisector of the first quadrant. The length of PQ is

[DCE]

(a) $3\sqrt{2}$

(b) $5\sqrt{2}$

(c) $7\sqrt{2}$

(d) $9\sqrt{2}$

Q 4. The point on the line $3x + 4y = 5$, which is equidistant from $(1,2)$ and $(3,4)$ is

[EAMCET]

(a) $(7,-4)$

(b) $(15,-10)$

(c) $\left(\frac{1}{7},\frac{8}{7}\right)$

(d) $\left(0,\frac{5}{4}\right)$

Q 5. If C is a point on the line segment joining A $(-3, 4)$ and B $(2,1)$ such that AC = 2 BC, then the coordinate of C is

[WB JEE]

(a) $\left(\frac{1}{3},2\right)$

(b) $\left(2, \frac{1}{3}\right)$

(c) $(2, 7)$

(d) $(7, 2)$

Q 6. If C is the reflection of A $(2, 4)$ in x−axis and B is the reflection of C in y−axis, then | AB | is

[WB JEE]

(a) 20

(b) $2\sqrt{5}$

(c) $4\sqrt{5}$

(d) 4

Q 7. Let A (k, 2) and B ($3,5$) are points. The point

(t, t) divide $\overline{AB}$ from A's side in the ratio of k, the: k

= ....k $\in$ R $-\{0,-1\}$

[Guj. CET]

(a) -4

(b) -2

(c) 4

(d) 2

Q 8. Consider three points

$P = (-\sin(\beta - \alpha), -\cos\beta)$,

$Q = (\cos(\beta - \alpha), \sin\beta)$

and $R = (\cos(\beta - \alpha + \theta), \sin(\beta - \theta))$,

where $0 < \alpha, \beta, \theta < \frac{\pi}{4}$. Then,

[IITJEE]

(a) P lies on the line segment RQ

(b) Q lies on the line segment PR

(c) R lies on the line segment QP

(d) P, Q, R are non-collinear

Q 9. If the sum of the distance of a point P from two perpendicular lines in a plane is 1, then the locus of P is a

[EAMCET]

(a) rhombus

(b) circle

(c) straight line

(d) pair of straight lines

Q 10. One possible condition for the three points (a, b), (b, a) and (a^2, $-b^2$) to be collinear, is

[WB JEE]

(a) $a - b = 2$

(b) $a + b = 2$

(c) a= 1 + b

(d) a = 1 − b

Q 11. The mid point of the line joining the points $(-10,8)$ and $(-6,12)$ divides the line joining the points $(4,-2)$ and $(-2,4)$ in the ratio

[Kerala CEE]

(a) $1:2$ internally

(b) $1:2$ externally

(c) $2:1$ internally

(d) $2:1$ externally

(e) $2:3$ externally

Q $12.$ The image of the centre of the circle $x^2 + y^2 = a^2$ with respect to the mirror image $x + y = 1$, is

[OJEE]

(a) $\left(\frac{1}{\sqrt{2}}, \sqrt{2}\right)$

(b) $(\sqrt{2},\sqrt{2})$

(c) $(\sqrt{2},2\sqrt{2})$

(d) None of these

Q 13. If P(l, 2), Q(4,6), R(5,7) and S (a, b) are the vertices of a parallelogram PQRS, then

[AMU]

(a) a = 2,b = 4

(b) a = b, b = 4

(c) a = 2, b = 3

(d) a = 3, b = 5

Q 14. The point P is equidistant from A$(1,3)$, B$(-3,5)$ and C$(5, -1)$, then PA is equal to

[RPET]

(a) 5

(b) $5\sqrt{5}$

(c) 25

(d) $5\sqrt{10}$

Q 15. The mid points of the sides of a triangle are D $(6,1)$, E $(3,5)$ and F $(-1, -2)$, then the vertex opposite to D is

[J&K CET]

(a) $(-4,2)$

(b) $(-4,5)$

(c) $(2,5)$

(d) $(10,8)$

Q 16. $(0, -1)$ and $(0, 3)$ are two opposite vertices of a square. The other two vertices are

[BITSAT]

(a) $(0, 1), (0, -3)$

(b) $(3,-1), (0,0)$

(c) $(2, 1), (-2, 1)$

(d) $(2, 2), (1, 1)$

Q 17. Point $\left(\frac{1}{2}, -\frac{13}{4}\right)$ divides the line joining the points $(3, -5)$ and $(-7, 2)$ in the ratio of

[Kerala CEE]

(a) $1:3$ internally

(b) $3:1$ internally

(c) $1:3$ externally

(d) $3:1$ externally

(e) None of these

Q 18. The ratio in which the line x + y = 4 divides the line joining the points $(1, -1)$ and $(5,7)$ is

[Kerala CEE]

(a) 1:2

(b) 2 : 1

(c) 1 : 3

(d) 3 : 1

(e) 3 : 2

Q 19. If $A(3, 5)$, $B(-5, -4)$, $C(7, 10)$ are the vertices of a parallelogram, taken in the order, then the coordinates of the fourth vertex are

[MHTCET]

(a) $(10, 19)$

(b) $(15, 19)$

(c) $(19, 10)$

(d) $(19, 15)$

Q 20. The intercepts on the straight line y = mx by the line y = 2 and y = 6 is less than 5, then m belongs to

[DCE]

(a) $\left]-\frac{4}{3},\frac{4}{3}\right[$

(b) $\left]\frac{4}{3},\frac{3}{8}\right[$

(c) $\left]-\infty,-\frac{4}{3}\right[\cup\left]\frac{4}{2},\infty\right[$

(d) $\left]\frac{4}{3},\infty\right[$

Q 21. If the distance between the points $(a \cos \theta, a \sin \theta)$ and $(a \cos \phi, a \sin \phi)$ is $2a$, then B is equal to

[EAMCET]

(a) $2n\pi \pm \pi + \phi$, $n \in Z$

(b) $n\pi + \frac{\pi}{2} + \phi$, $n \in Z$

(c) $n\pi - \phi$, $n \in Z$

(d) $2n\pi + \phi$, $n \in Z$

Q 22. The points $(1,3)$ and $(5,1)$ are the opposite vertices of a rectangle. The other two vertices lie on the line y = 2x + c, then the value of c will be

[Jamia Millia Islamia]

(a) 4

(b) -4

(c) 2

(d) -2

Q 23. The centre of circle inscribed in square formed by the lines $x^2 - 8x + 12 = 0$ and $y^2 - 14y + 45 = 0$, is

[IITJEE]

(a) (4,7)

(b) (7,4)

(c) (9, 4)

(d) (4, 9)

Q 24. The vertex of an equilateral triangle is $(2,-1)$ and the equation of its base is x + 2y = 1, the length of its sides is

[UP SEE]

(a) $\frac{2}{\sqrt{15}}$

(b) $\frac{4}{3\sqrt{3}}$

(c) $\frac{1}{\sqrt{5}}$

(d) $\frac{4}{\sqrt{5}}$

Topic 2 Orthocentre, Circumcentre and Incentre

Q 1. Three distinct points A, B and C given in the two dimensional coordinate plane such that the ratio of the distance of any one of them from the point $(1, 0)$ to the distance from the point $(-1,0)$ is equal to $\frac{1}{3}$. Then, the circumcentre of the triangle ABC is at the point

[AIEEE]

(a) $\left(\frac{5}{4}, 0\right)$

(b) $\left(\frac{5}{2}, 0\right)$

(c) $\left(\frac{5}{3}, 0\right)$

(d) $(0, 0)$

Q 2. The vertices of a triangle are $(6,0)$, $(0,6)$ and $(6,6)$. The distance between its circumcentre and centroid is

[KCET]

(a) 2

(b) $\sqrt{2}$

(c) 1

(d) $2\sqrt{2}$

Q 3. The coordinates of the incentre of the triangle having sides $3x - 4y = 0$, $5x + 12y = 0$ and $y - 15 = 0$ are

[OJEE]

(a) $-1,8$

(b) $1, -8$

(c) $2,6$

(d) None of these

Q 4. The circumcentre of a triangle formed by the lines $xy + 2x+2y + 4 = 0$ and $x + y + 2 = 0$ is

[DCE]

(a) $(-1,-1)$

(b) $(0,-1)$

(c) $(1, 1)$

(d) $(-1,0)$

Q 5. The circumcentre of the triangle formed by the lines y = x, y = 2x and y = 3x + 4 is

[UP SEE]

(a) $(6, 8)$

(b) $(6,-8)$

(c) $(3,4)$

(d) $(-3,-4)$

Q 6. Circumcentre of triangle whose vertices are

$(0,0)$, $(3,0)$ and $(0,4)$ is

[UP SEE]

(a) $\left(\frac{3}{2},2\right)$

(b) $\left(2,\frac{3}{2}\right)$

(c) $(0,0)$

(d) None of these

Q 7. The centroid of the triangle ABC, where A $\equiv$ $(2, 3)$, B $\equiv$ $(8, 10)$ and C $\equiv$ $(5,5)$ is

[KCET]

(a) $(5, 6)$

(b) $(6, 5)$

(c) $(6,6)$

(d) $(15,18)$

Q 8. The circumcentre of the triangle with vertices $(0, 30)$, $(4,0)$ and $(30,0)$ is

[Kerala CEE]

(a) $(10, 10)$

(b) $(10, 12)$

(c) $(12, 12)$

(d) $(15, 15)$

(e) $(17, 17)$

Q 9. The coordinates of the orthocentre of the triangle formed by $(0,0)$, $(8,0)$, $(4,6)$ is

[OJEE]

(a) $(4, 0)$

(b) $(6, 3)$

(c) $(6,0)$

(d) None of these

Q 10. The orthocentre of the triangle with vertices O$(0, 0)$, A$\left(0, \frac{3}{2}\right)$, B$(-5, 0)$ is

[Manipal]

(a) $\left(\frac{5}{2}, \frac{3}{4}\right)$

(b) $\left(\frac{-5}{2}, \frac{3}{4}\right)$

(c) $\left(-5, \frac{3}{2}\right)$

(d) $(0, 0)$

Q 11. The vertices P,Q,R of a triangle are $(2, 1)$, $(5, 2)$ and $(3, 4)$ respectively. Then, the circumcentre is

[Guj. CET]

(a) $\left(\frac{13}{4}, -\frac{9}{4}\right)$

(b) $\left(-\frac{13}{4}, \frac{9}{4}\right)$

(c) $\left(-\frac{13}{4}, -\frac{9}{4}\right)$

(d) $\left(\frac{13}{4}, \frac{9}{4}\right)$

Q 12. Let O$(0,0)$, P$(3,4)$, Q$(6,0)$ be the vertices of the triangle OPQ. The point R inside the triangle OPQ is such that the triangles OPR,PQR,OQR are of equal area. The coordinates of R are

[IITJEE]

(a) $\left(\frac{4}{3},3\right)$

(b) $\left(3,\frac{2}{3}\right)$

(c) $\left(3, \frac{4}{3}\right)$

(d) $\left(\frac{4}{3}, \frac{2}{3}\right)$

Q 13. If a vertex of a triangle is $(1, 1)$ and the mid points of two sides through the vertex are $(-1, 2)$ and $(3, 2)$, then the centroid of the triangle is

[BITSAT]

(a) $\left(1, \frac{7}{3}\right)$

(b) $\left(\frac{1}{3}, \frac{7}{3}\right)$

(c) $\left(-\frac{1}{3},\frac{7}{3}\right)$

(d) $\left(-1,\frac{7}{3}\right)$

Q 14. In Δ ABC, G is the centroid, D is the mid point of BC. If A = $(2,3)$ and G = $(7,5)$, then the point D is

[KCET]

(a) $\left(\frac{9}{2},4\right)$

(b) $\left(\frac{19}{2},6\right)$

(c) $\left(\frac{11}{2}, \frac{11}{2}\right)$

(d) $\left(8, \frac{13}{2}\right)$

Q 15. In the triangle with vertices at A$(6,3)$,B$(-6,3)$ and C$(-6, -3)$, the median through A meets BC at P, the line AC meets the x−axis at Q, while R and S respectively denote the orthocentre and centroid of the triangle. Then the correct matching of the coordinates of points in List−I to List−II is

[EAMCET]

List–I	**List–II**
(i) P	(A) $(0,0)$
(ii) Q	(B) $(6,0)$
(iii) R	(C) $(-2,l)$
(iv) S	(D) $(-6,0)$
(E) $(-6,-3)$	(F) $(-6,3)$

(i)	(ii)	(iii)	(iv)
(a) D	A	E	C

(b) D B E C

(c) D A F C

(d) B A F C

Q 16. Let ABC be a triangle, two of whose vertices are $(15, 0)$ and $(0, 10)$. If the orthocentre is $(6, 9)$, then the third vertex is

[OJEE]

(a) $(15, 10)$

(b) $(10, -15)$

(c) $(0, 0)$

(d) None of these

Q 17. If orthocentre and circumcentre of a triangle are respectively $(1,1)$ and $(3,2)$, then the coordinates of its centroid are

[OJEE]

(a) $\left(\frac{7}{3},\frac{5}{3}\right)$

(b) $\left(\frac{5}{3},\frac{7}{3}\right)$

(c) $(7, 5)$

(d) None of these

Q 18. If the centroid of the triangle formed by the points $(0,0)$, $(\cos\theta, \sin\theta)$ and $(\sin\theta, -\cos\theta)$ lies on the line $y = 2x$, then θ is equal to

[J&K CET]

(a) $\tan^{-1} 2$

(b) $\tan^{-1} 3$

(c) $\tan^{-1} (-3)$

(d) $\tan^{-1}(-2)$

Q 19. The incentre of the triangle with vertices $(1,\sqrt{3})$, $(0,0)$ and $(2,0)$ is

[OJEE]

(a) $\left(1,\frac{\sqrt{3}}{2}\right)$

(b) $\left(\frac{2}{3},\frac{1}{\sqrt{3}}\right)$

(c) $\left(\frac{2}{3},\frac{\sqrt{3}}{2}\right)$

(d) $\left(1, \frac{1}{\sqrt{3}}\right)$

Q 20. The orthocentre of the ΔOAB, where O is the origin, A $(6,0)$ and $B(3, 3\sqrt{3})$ is

[J&K CET]

(a) $(9/2, \sqrt{3}/2)$

(b) $(3, \sqrt{3})$

(c) $(\sqrt{3}, 3)$

(d) $(3, -\sqrt{3})$

Q 21. ABC is a triangle with vertices A$(-1, 4)$, B$(6,-2)$ and C$(-2, 4)$. D,E and F are the points which divide each AB, BC and CA respectively in the ratio $3 : 1$ internally. Then, the centroid of the triangle DEF is

[Kerala CEE]

(a) $(3, 6)$

(b) $(1, 2)$

(c) $(4,8)$

(d) $(-3,6)$

(e) $(-1,2)$

Q 22. The equations of the three sides of a triangle are $x = 2$, $y + 1 = 0$ and $x + 2y = 4$. The coordinates of the-circumcentre of the triangle are

[AMU]

(a) $(4,0)$

(b) $(2,-1)$

(c) $(0,4)$

(d) $(-1,2)$

Q 23. The x−coordinate of the incentre of the triangle where the mid points of the sides are $(0,$ $1)$, $(1,\ 1)$ and $(1,0)$, is

[MP PET]

(a) $2+\sqrt{2}$

(b) $1+\sqrt{2}$

(c) $2-\sqrt{2}$

(d) $1-\sqrt{2}$

Q 24. The orthocentre of the triangle with vertices $(-2,-6)$, $(-2,4)$ and $(l,3)$ is

[J&K CET]

(a) $(3, 1)$

(b) $(1, 1/3)$

(c) $(1,3)$

(d) None of these

Q 25. The incentre of a triangle with vertices (7, l),(−l,5)and $(3+2\sqrt{3},3+4\sqrt{3})$ is

[J&K CET]

(a) $\left(3+\frac{2}{\sqrt{3}},3+\frac{4}{\sqrt{3}}\right)$

(b) $\left(1+\frac{2}{3\sqrt{3}},1+\frac{4}{3\sqrt{3}}\right)$

(c) (7, 1)

(d) None of the above

Q 26. Orthocentre of the triangle formed by the lines $x + y = 1$ and $xy = 0$ is

[OJEE]

(a) $(0, 0)$

(b) $(0, 1)$

(c) $(1, 0)$

(d) $(-1, 1)$

Q 27. Orthocentre of triangle with vertices $(0, 0)$, $(3, 4)$ and $(\mathbf{4,0})$ is

[IITJEE]

(a) $\left(3, \frac{5}{4}\right)$

(b) $(3, 12)$

(c) $\left(3, \frac{3}{4}\right)$

(d) $(3, 9)$

Q 28. The incentre of the triangle formed by $(0,$ $0)$, $(5, 12)$, $(16,12)$ is

[AMU]

(a) $(7, 9)$

(b) $(9, 7)$

(c) $(-9,7)$

(d) $(-7,9)$

Topic 3 Area of Some Geometrical Figures

2010

Q 1. If the three points $(3q, 0)$, $(0, 3p)$ and $(1,1)$ are collinear then which one is true?

[WB JEE]

(a) $\frac{1}{p}+\frac{1}{q}=0$

(b) $\frac{1}{p}+\frac{1}{q}=1$

(c) $\frac{1}{p}+\frac{1}{q}=3$

(d) $\frac{1}{p}+\frac{3}{q}=1$

Q 2. The area of the triangle formed by the points

$(2,2)$, $(5, 5)$, $(6, 7)$ is equal to (in square unit)

[Kerala CEE]

(a) $\frac{9}{2}$

(b) 5

(c) 10

(d) $\frac{3}{2}$

(e) 14

Q 3. The vertices of a family of triangles have integer coordinates. If two of the vertices of all the triangles are $(0, 0)$ and $(6, 8)$, then the least value of areas of the triangles is

[Kerala CEE]

(a) 1

(b) $\frac{3}{2}$

(c) 2

(d) $\frac{5}{2}$

(e) 3

Q 4. The area between the curve y = 1 − j x j and the x−axis is equal to

[Kerala CEE]

(a) 1 sq unit

(b) $\frac{1}{2}$ sq unit

(c) $\frac{1}{3}$ sq unit

(d) 2 sq units

(e) 3 sq units

Q 5. Area of the triangle formed by the lines v =

2x, y = 3x and y = 5 is equal to (in square unit)

[Kerala CEE]

(a) $\frac{25}{6}$

(b) $\frac{25}{12}$

(c) $\frac{5}{6}$

(d) $\frac{17}{12}$

(e) 6

Q 6. Triangle ABC has vertices $(0, 0)$, $(11, 60)$ and $(91, 0)$. If the line y = kx cuts the triangle into two triangles of equal area, then k is equal to

[Kerala CEE]

(a) $\frac{30}{51}$

(b) $\frac{4}{7}$

(c) $\frac{7}{4}$

(d) $\frac{30}{91}$

(e) $\frac{27}{37}$

Q 7. One side of length 3 a of a triangle of area a square unit lies on the line x = a. Then, one of the lines on which the third vertex lies, is

[Kerala CEE]

(a) $x = -a^2$

(b) $x = a^2$

(c) $x = -a$

(d) $x = \frac{a}{3}$

(e) $x = -\frac{a}{3}$

Q 8. If t_1,t_2 and t_3 are distinct points $(t_1, 2at_1 + at_1^3)$.

$(t_2, 2at_2 + at_2^3)$ and $(t_3, 2at_3 + at_3^3)$ are collinear, if

[Jamia Millia Islamia]

(a) $t_1t_2t_3 = 1$

(b) $t_1 + t_2 + t_3 = t_1t_2t_3$

(c) $t_1 + t_2 + t_3 = 0$

(d) $t_1 + t_2 + t_3 = -1$

Q 9. If A$(0, 0)$, B$(12,0)$, C$(12, 2)$, D$(6, 7)$ and E$(0, 5)$ are the vertices of the pentagon ABCDE, then its area in square units, is

[Kerala CEE]

(a) 58

(b) 60

(c) 61

(d) 62

(e) 63

Q 10. If a > 0, b > 0 the maximum area of the triangle formed by the points O(0,0), A(a cos θ, b sin θ) and B (a cos 6, − bsin 0) is (in sq unit)

[Manipal]

(a) $\frac{ab}{2}$ when $\theta = \frac{\pi}{4}$

(b) $\frac{3ab}{4}$ when $\theta = \frac{\pi}{4}$

(c) $\frac{ab}{2}$ when $\theta = -\frac{\pi}{2}$

(d) a^2b^2

Q 11. The x−axis, p−axis and a line passing through the point A $(6,0)$ form a triangle ABC, If

$\angle A = 30°$, then the area of the triangle, in sq units is

[Guj. CET]

(a) $6\sqrt{3}$

(b) $12\sqrt{3}$

(c) $4\sqrt{3}$

(d) $8\sqrt{3}$

Q 12. Let A(h, k), B(1, 1) and C(2,l) be the vertices of a angled triangle with AC as its hypotenuse. If the area of the triangle is 1, then the set of values which 'k' can take is given by

[AIEEE]

(a) $\{1,3\}$

(b) $\{0,2\}$

(c) $\{-1,3\}$

(d) $\{-3,-2\}$

Q 13. If the points (a, b), (a', b') and (a − a', b − b') are collinear, then

[BITSAT]

(a) ab' = ab

(b) ab = a'b'

(c) aa' = bb'

(d) $a^2 + b^2 = 1$

Q 14. ABC is a triangle with $\angle A = 30°$, BC = 10 cm. The area of the circumcircle of the triangle is

[KCET]

(a) $100\ \pi$ sq cm

(b) 5 sq cm

(c) 25 sq cm

(d) $\frac{100\pi}{3}$ sq m

Q 15. ABC is a right angled triangle with $\angle$B = $90°$, a = 6 cm. If the radius of the circumcircle is 5 cm. Then the area of ΔABC is

[Kerala CEE]

(a) 25 cm^2

(b) 30 cm^2

(c) 36 cm^2

(d) 24 cm^2

(e) 48 cm^2

Q 16. The area of an equilateral triangle that can be inscribed in the circle $x^2 + y^2 - 4x - 6y - 12 = 0$, is

[Kerala CEE]

(a) $\frac{25\sqrt{3}}{4}$ sq units

(b) $\frac{35\sqrt{3}}{4}$ sq units

(c) $\frac{55\sqrt{3}}{4}$ sq units

(d) $\frac{75\sqrt{3}}{4}$ sq units

(e) $\frac{25}{4}$ sq units

Q 7. The area (in square unit) of the triangle formed by the points with polar coordinates $(1, 0)$, $\left(2, \frac{\pi}{3}\right)$ and $\left(3, \frac{2\pi}{3}\right)$ is

[EAMCET]

(a) $\frac{11\sqrt{3}}{4}$

(b) $\frac{5\sqrt{3}}{4}$

(c) $\frac{5}{4}$

(d) $\frac{11}{4}$

Q 18. If A$(-5,0)$ and B$(3, 0)$ are two vertices of a triangle ABC. Its area is 20 sq cm. The vertex C lies on the line x − y = 2. The coordinates of C are

[Guj.CET]

(a) $(-7,-5)$or $(3, 5)$

(b) $(-3,-5)$ or $(-5,7)$

(c) $(7, 5)$ or $(3, 5)$

(d) $(-3,-5)$ or $(7, 5)$

Q 19. The area of the segment of a circle of radius a subtending an angle of 2a at the centre is

[UP SEE, WB JEE]

(a) $a^2\left(\alpha+\frac{1}{2}\sin 2\alpha\right)$

(b) $\frac{1}{2}a^2\sin 2\alpha$

(c) $a^2\left(\alpha-\frac{1}{2}\sin 2\alpha\right)$

(d) $a^2\alpha$

Q 20. Area (in sq unit) enclosed by $y = 1$, $2x + y = 2$ and $x + y = 2$ is

[Kerala CEE]

(a) $\frac{1}{2}$ sq unit

(b) $\frac{1}{4}$ sq unit

(c) 1 sq unit

(d) 2 sq units

(e) 4 sq units

Q 21. If the area of the triangle with vertices (x, 0), (1, 1) and (.0. 2) is 4 sq unit, then the value of x is

[AMU]

(a) −2

(b) −4

(c) −6

(d) 8

Q 22. The area enclosed within the curve | x | + | y | = 1 is

[Jamia Millia Islamia]

(a) 1 sq unit

(b) $2\sqrt{2}$ sq units

(c) $\sqrt{2}$ sq units

(d) 2 sq units

Q 23. The straight lines $x + y = 0$, $3x + y - 4 = 0$ and $x + 3y - 4 = 0$ form a triangle which is

[UP SEE]

(a) right angled

(b) equilateral

(c) isosceles

(d) None of these

Q 24. The area (in square unit) of the triangle formed by the lines $x = 0$, $y = 0$ and $3x + 4y = 12$, is

[EAMCET]

(a) 3

(b) 4

(c) 6

(d) 12

Q 25. Three points are A$(6, 3)$, B$(-3,5)$, C$(4,-2)$ and P(x, y) is a point, then the ratio of area of ΔPBC and ΔABC is

[BCECE]

(a) $\left|\frac{x+y-2}{7}\right|$

(b) $\left|\frac{x-y+2}{2}\right|$

(c) $\left|\frac{x-y-2}{7}\right|$

(d) None of these

Q 26. A triangle with vertices $(4, 0)$, $(-1,-1)$, $(3, 5)$ is

[Jamia Millia Islamia]

(a) isosceles and right angled

(b) isosceles but not right angled

(c) right angled but not isosceles

(d) neither right angled nor isosceles

Q 27. Area of triangle formed by the lines x + y = 3 and angle bisectors of the pair of straight lines $x^2 - y^2 + 2y = 1$ is

[ITT JEE]

(a) 2 sq units

(b) 4 sq units

(c) 6 sq units

(d) 8 sq units

Q 28. Area of the triangle formed by the lines

$3x^2 - 4xy + y^2 = 0$, $2x - y = 6$ is

[EAMCET]

(a) 16 sq units

(b) 25 sq units

(c) 36 sq units

(d) 49 sq units

Q 29. The triangle formed by $x^2 - 3y^2 = 0$ and

$x = 4$ is

[OJEE]

(a) isosceles

(b) equilateral

(c) right angled

(d) None of these

Q 30. The area of triangle formed by the points

(a, b + c), (b, c + a), (c, a + b) is equal to

[Jamia Millia Islamia]

(a) abc

(b) $a^2 + b^2 + c^2$

(c) ab + bc + ca

(d) 0

Q 31. If the points $(1, 1)$, $(-1,-1)$, $(-\sqrt{3},\sqrt{3})$ are the vertices of a triangle, then this triangle is

[MPPET]

(a) right angled

(b) isosceles

(c) equilateral

(d) None of these

Q 32. Area of quadrilateral whose vertices are

$(2, 3)$, $(3, 4)$, $(4,5)$ and $(5,6)$ is equal to

[DCE]

(a) 0

(b) 4

(c) 6

(d) None of the above

Q 33. The points $(1, 1)$, $(-5, 5)$ and $(13, \lambda)$ lie on the same straight line, if λ is equal to

[Kerala CEE]

(a) 7

(b) −7

(c) ± 7

(d) 0

(e) 14

Topic 4 Transformation of Axes and Locus

Q 1. If the sum of the distances from two perpendicular lines in a plane is 1, then its locus is

[DCE]

(a) a square

(b) a circle

(c) a straight line

(d) two intersecting lines

Q 2. A variable line through the point $\left(\frac{1}{5},\frac{1}{5}\right)$ cuts the coordinate axes in the points A and B. If the point P divides AB internally in the ratio $3:1$, then the locus of P is

[AMU]

(a) $3y + x = 20xy$

(b) $y + 3x = 20xy$

(c) $x + y = 20xy$

(d) $3x + 3y = 20xy$

Q 3. The locus of a points which moves such that the sum of the squares of its distance from three vertices of the triangle is constant is a/an

[Jamia Millia Islamia]

(a) circle

(b) straight line

(c) ellipse

(d) None of the above

Q 4. The transformed equation of $3x^2 + 3y^2 + 2xy = 2$, when the coordinate axes are rotated through an angle of $45°$, is

[BITSAT, AMU]

(a) $x^2 + 2y^2 = 1$

(b) $2x^2 + y^2 = 1$

(c) $x^2 + y^2 = 1$

(d) $x^2 + 3y^2 = 1$

Q 5. A variable line $\frac{x}{a}+\frac{y}{b}=1$ is such that a + b = 4.

The locus of the mid point of the portion of the line

intercepted between the axes is

[KCET]

(a) x + y = 4

(b) x + y = 8

(c) x + y = l

(d) $x + y = 2$

Q 6. If a point $P(4, 3)$ is shifted by a distance $\sqrt{2}$ unit parallel to the line $y = x$, then coordinates of P in new position are

[AMU]

(a) $(5, 4)$

(b) $(5+\sqrt{2}, 4+\sqrt{2})$

(c) $(5-\sqrt{2}, 4-\sqrt{2})$

(d) None of these

Q 7. The pair of lines $\sqrt{3}x^2 - 4xy + \sqrt{3}y^2 = 0$ are rotated about the origin by $n/6$ in the anti-clockwise sense. The equation of the pair in the new position is

[OJEE]

(a) $\sqrt{3}y^2 - xy = 0$

(b) $\sqrt{3}x^2 - xy = 0$

(c) $x^2 - y^2 = 0$

(d) $\sqrt{3}x^2 + xy = 0$

Q 8. The locus of a point P which moves such thr

$2PA = 3PB$, where $A(0, 0)$ and $B(4, -3)$ are

points, is

(a) $5x^2 - 5y^2 - 72x + 54y + 225 = 0$

(b) $5x^2+5y^2 - 72x + 54y + 225=0$

[WB JEE]

(c) $5x^2 +5y^2 +72x - 54y+ 225=0$

(d) $5x^2 +5y^2 -72x - 54y - 225 =0$

Q 9. Let A$(2, -3)$ and B$(-2, 1)$ be vertices of a triangle ABC If the centroid of this triangle moves on the line $2x + 3y = l$, then the locus of the vertex C is the line

[Jamia Millia Islamia]

(a) $2x + 3y = 9$

(b) $2x - 3y = 7$

(c) $3x + 2y = 5$

(d) $3x - 2y = 3$

Q 10. What is the equation of the locus of a point which moves such that 4 times its distance from the x−axis is the square of its distance from the origin ?

[AMU]

(a) $x^2 - y^2 - 4y = 0$

(b) $x^2 + y^2 - 4|y| = 0$

(c) $x^2 + y^2 - 4x = 0$

(d) $x^2 + y^2 - 4|x| = 0$

Q 11. The transformed equation of $x^2 + 6xy + 8y^2 = 10$ when the axes are rotated through an angle $\frac{\pi}{4}$ is

(a) $15x^2 - 14xy + 3y^2 = 20$

(b) $15x^2 + 14xy - 3y^2 = 20$

[EAMCET]

(c) $15x^2 + 14xy + 3y^2 = 20$

(d) $15x^2 - 14xy - 3y^2 = 20$

Q 12. Let P be the point $(1,0)$ and Q be the point on $y^2 = 8x$. The locus of mid point of PQ is

[AIEEE]

(a) $x^2 - 4y + 2 = 0$

(b) $x^2 + 4y + 2 = 0$

(c) $y^2+4x+2 = 0$

(d) $y^2 - 4x+2 = 0$

Q 13. If the distance of any point P from the points $A(a + b, a - b)$ and $B(a - b, a + b)$ are equal, then the locus of P is

[GujCET]

(a) $ax + by = 0$

(b) $x - y = 0$

(c) $x + y = 0$

(d) $bx - ay = 0$

Q 14. The locus of the point (x, y) which is equidistant from the points (a + b,b − a) and (a − b,a + b) is

[Kerala CEE]

(a) ax = by

(b) ax + by = 0

(c) bx + ay = 0

(d) bx−ay = 0

(e) $\frac{x}{a}+\frac{y}{b}=1$

Q 15. If A and B are two fixed points, then the locus of a point which moves in such a way that the angle, APB is a right angle is

[RPET]

(a) a circle

(b) an ellipse

(c) a parabola

(d) None of these

Q 16. The equation of the locus of a point equidistant from the points (a_1, b_1) and (a_2, b_2) is $(a_1 - a_2)x + (b_1 - b_2)y + c = 0$, then the value of c is

[J&K CET]

(a) $\sqrt{(a_1^2 + b_1^2 + c_1^2)}$

(b) $a_1^2 - b_1^2 - c_1^2$

(c) $\frac{1}{2}(a_2^2 + b_2^2 - a_1^2 - b_1^2)$

(d) None of the above

Q 17. Locus of centroid of the triangle whose vertices are $(a \cos t, a \sin t)$, $(b \sin t, -b \cos t)$ and $(1,0)$, where t is a parameter, is

[AIEEE]

(a) $(3x - 1)^2+(3y)^2 = a^2-b^2$

(b) $(3x - 1)^2 +(3y)^2 =a^2 +b^2$

(c) $(3x + l)^2+(3y)^2 = a^2 + b^2$

(d) $(3x + l)^2+(3y)^2 = a^2 - b^2$

Q 18. If P = $(1,0)$, Q = $(-1,0)$ and R = $(2,0)$ are three given points, then the locus of a point S satisfying the relation $SQ^2 + SR^2 = 2SP^2$ is

[AMU]

(a) a straight line parallel to x−axis

(b) a circle through origin

(c) a circle with centre at the origin

(d) a straight line parallel to y−axis

Q 19. Given the points A $(0,4)$ and B $(0, -4)$, then the equation of the locus of the point P(x, y) such that, $|AP - BP| = 6$, is

[AMU]

(a) $\frac{x^2}{7}+\frac{y^2}{9}=1$

(b) $\frac{x^2}{9}+\frac{y^2}{7}=1$

(c) $\frac{x^2}{7}-\frac{y^2}{9}=1$

(d) $\frac{y^2}{9}-\frac{x^2}{7}=1$

Q 20. The coordinate axes are rotated through an angle $135°$. If the coordinates of a point P in the new system are known to be $(4,-3)$, then the coordinates of P in the original system are

[EAMCET]

(a) $\left(\frac{1}{\sqrt{2}},\frac{7}{\sqrt{2}}\right)$

(b) $\left(\frac{1}{\sqrt{2}},-\frac{7}{\sqrt{2}}\right)$

(c) $\left(-\frac{1}{\sqrt{2}},-\frac{7}{\sqrt{2}}\right)$

(d) $\left(-\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right)$

Q 21. The locus of the point of intersection of the lines x cot θ + y cosec θ = 2 and x cosec θ + y cot θ = 6 is

[J&K GET]

(a) a straight line

(b) circle

(c) a hyperbola

(d) an ellipse

Answers

Topic 1 : System of Coordinates and Distance between Points

1. (b) 2. (a) 3, (b) 4. (b) 5. (a) 6. (c) 7. (b) 8. (d)

9. (a) 10. (c) 11. (d) 12. (d) 13. (c) 14. (d) 15. (a)

16. (c) 17. (a) 18. (a) 19. (b) 20. (c) 21. (a) 22.

(b) 23. (a) 24. (a)

Topic 2 : Orthocentre, Circumcentre and Incentre

1. (a) 2. (b) 3. (a) 4. (a) 5. (b) 6. (a) 7. (a) 8. (e)

9. (d) 10. (d) 11. (d) 12. (c) 13. (a) 14. (b) 15. (c)

16. (c) 17. (a) 18. (c) 19. (d) 20. (b) 21. (b) 22.

(a) 23. (c) 24. (c) 25. (a) 2$. (a) 27. (c) 28. (a)

Topic 3 : Area of Some Geometrical Figures

1. (c) 2. (d) 3. (a) 4. (a) 5. (b) 6. (a) 7. (d) 8. (c)

9. (e) 10. (a) 11. (a) 12. (c) 13. (a) 14. (a) 15. (d)

16. (d) 17. (b) 18. (d) 19. (d) 20. (b) 21. (c) 22.

(d) 23. (c) 24. (c) 25. (a) 26. (a) 27. (a) 28.,(c)

29. (b) 30. (d) 31. (c) 32. (a) 33. (b)

Topic 4 : Transformation of Axes and Locus

1. (a) 2. (b) 3. (a) 4. (b) 5. (d) 6. (a) 7. (a) 8. (b)

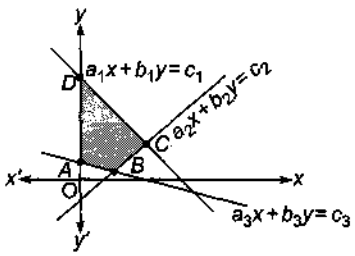
9. (a) 10. (b) 11. (c) 12. (d) 13. (b) 14. (d) 15. (a)

16. (c) 17. (b) 18. (d) 19. (d) 20. (d) 21. (c)

Linear Programming

Useful Results & Formulae

1. Let general form of inequations are $a_1x + b_1y \leq c_1$, $a_2x + b_2y \geq c_2$, $a_3x + b_3y \geq c_3$. The graph of these inequations is given below. Here



The bounded region ABCD is a feasible region. A, B, C and D are corner points.

2. The set of values of the variables which satisfies the set of constraints of LPP is a feasible solution.

3. A feasible solution for which the objective function is minimum or maximum is optimal solution.

4. If the line joining any two points of a set lies inside the set, then that set is a convex set. eg,

(i)  are convex set.

(ii)  are concave set.

5. In a unbounded region the minimum value of objective function lies at one vertex but there is no existence of maximum value.

Q 1. The linear programming problem Maximize z

$= x_1 + x_2$

Subject to constraints

$x_1 + 2x_2 \leq 2000$ $x_1 + x_2 \leq 1500$ $x_2 \leq 600$ $x_1 \geq 0$ has

[DCE]

(a) No feasible solution

(b) Unique optimal solution

(c) A finite number of optimal solutions

(d) Infinite number of optimal solutions

Q 2. The area of the feasible region for the following constraints $3y + x \geq 3$, $x \geq 0$, $y \geq 0$ will be

[BITSAT]

(a) bounded

(b) unbounded

(c) convex

(d) concave

Q 3. The optimal value of the objective function is

attained at the point

[BITSAT]

(a) given by intersection of inequations with axes

only

(b) given by intersection of inequations with x−axis

only

(c) given by corner points of the feasible region

(d) None of the above

Q 4. The maximum value of $z = 10x + 6y$ subject to constraints $x \geq 0$, $y \geq 0$, $x + y \leq 12$, $2x + y \leq 20$ is

[AMU]

(a) 72

(b) 80

(c) 104

(d) 110

Q 5. The maximum value of $z = 9x + 13y$ subject to constraints $2x + 3y \leq 18$, $2x + y \leq 10$, $x \geq 0$, $y \geq 0$ is

[MHT CET]

(a) 130

(b) 81

(c) 79

(d) 99

Q 6. Maximum value of $12x + 3y$ subject to constraints $x > 0$, $y \geq 0$, $x + y \leq 5$ and $3x + y \leq 9$ is

[MPPET]

(a) 15

(b) 36

(c) 60

(d) 40

Q 7. A vertex of a feasible region by the linear constraints $3x + 4y \leq 18$, $2x + 3y \geq 3$ and $x, y \geq 0$ is

[DCE]

(a) $(0,2)$

(b) $(4.8,0)$

(c) $(0,3)$

(d) None of these

Q 8. For the LPP Min z = 2x + y subject to

constraints $5x + 10y \leq 50$, $x + y \geq 1$, $y \leq 4$ and x,

$y \geq 0$, then z is

[DCE]

(a) 0

(b) 1

(c) 2

(d) 12

Q 9. For the LPP Min $z = x_1 + x_2$ such that inequalities

[MHT CET]

$5x_1 + 10x_2 \geq 0$, $x_1 + x_2 \leq 1$, $x_2 \leq 4$ and $x_1, x_2 \geq 0$

(a) There is a bounded solution

(b) There is no solution

(c) There are infinite solutions

(d) None of the above

Q 10. The region represented by the inequation system x, $y \geq 0$, $y \leq 6$, $x + y \leq 3$, is

[MHTCET]

(a) unbounded in first quadrant

(b) unbounded in first and second quadrants

(c) bounded in first quadrant

(d) None of the above

Q 11. A wholesale merchant wants to start the business of cereal with Rs 24000. Wheat is Rs 400

per quintal and rice is Rs 600 per quintal. He has capacity to store 200 quintal cereal. He earns the profit Rs 25 per quintal on wheat and Rs 40 per quintal on rice. If he stores x quintal rice and y quintal wheat, then for maximum profit the objective function is

[MHT CET]

(a) $25x + 40y$

(b) $40x+25y$

(c) $400x + 600y$

(d) $\frac{400}{40}x + \frac{600}{25}y$

Q 12. The maximum value of $z = 5x + 3y$,

subjected to the conditions $3x + 5y \leq 15, 5x + 2y \leq 10; x, y \geq 0$ is

[MP PET]

(a) $\frac{235}{19}$

(b) $\frac{325}{19}$

(c) $\frac{523}{19}$

(d) $\frac{532}{19}$

Q 13. The constraints $-x_1 + x_2 \leq 1, -x_1 + 3x_2 \leq 9$; $x_1, x_2 \geq 0$ defines on

[MHTCET]

(a) bounded feasible space

(b) unbounded feasible space

(c) both bounded and unbounded feasible space

(d) None of the above

Q 14. Which of the term is not used in a linear programming problem?

[MHTCET]

(a) Optimal solution

(b) Feasible solution

(c) Concave region

(d) Objective function

Q 15. The maximum value of $z = 4x+2y$

subjected to the constraints $2x + 3y \leq 18, x + y \geq 10; x, y \geq 0$ is

[MPPET]

(a) 20

(b) 36

(c) 40

(d) None of these

Q 16. $z = 4x + 2y, 4x + 2y \geq 46, x + 3y \leq 24$

and x and y are greater than or equal to zero, then

the maximum value of z is

[DCE]

(a) 46

(b) 96

(c) 52

(d) None of these

Q 17. If given constraints are $5x + 4y \geq 2$, $x \leq 6$, $y \leq 7$, then the maximum value of the function $z = x + 2y$ is

[MHTCET]

(a) 13

(b) 14

(c) 15

(d) 20

Q 18. The minimum value of the objective function $z = 2x + 10y$ for linear constraints

$x \geq 0, y \geq 0, x - y \geq 0, x - 5y \leq -5$ is

[MPPET]

(a) 10

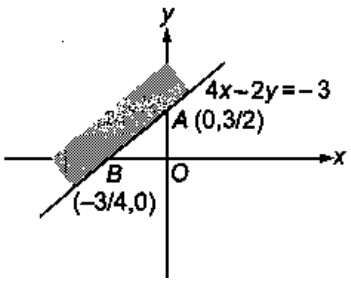
(b) 15

(c) 12

(d) 8

Q 19. Shaded region is represented by

[MPPET]



(a) $4x-2y \leq 3$

(b) $4x-2y \leq -3$

(c) $4x-2y \geq 3$

(d) $4x - 2y \geq -3$

Q 20. The coordinate of the point at which minimum value of $z = 7x - 8y$ subject to the conditions $x + y - 20 \leq 0$, $y \geq 5$, $x \geq 0$, $y \geq 0$ is attained, is

[DCE]

(a) $(20,0)$

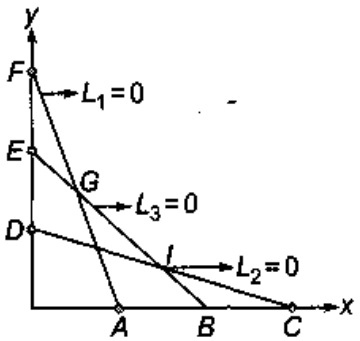
(b) $(15,5)$

(c) $(0, 5)$

(d) $(0, 20)$

Q 21. The feasible region for the following constraints $L_1 \leq 0$, $L_2 \geq 0$, $L_3 = 0$, $x \geq 0$, $y \geq 0$ in the diagram shown is

[Kerala CEE]



(a) area DHF

(b) area AHC

(c) line segment EG

(d) line segment GI

(e) line segment IC

Q 22. $z = 30x + 20y$, $x + y \leq 8$, $x + 2y \geq 4$, $6x + 4y \geq 12$, $x \geq 0$, $y \geq 0$ has

[MHTCET]

(a) unique solution

(b) infinitely many solution

(c) minimum at $(4, 0)$

(d) minimum 60 at point $(0, 3)$

Q 23. Maximum value of $z = 3x + 4y$ subject to

$x - y \geq -1$, $-x + y \leq 0$; $x, y \geq 0$, is given by

[MP PET]

(a) 1

(b) 4

(c) 6

(d) no feasible region

Q 24. The minimum value of $z = 5x-4y$ subject to constraints $x + y \leq 10$, $y \leq 4$; $x, y > 0$ will be at the point

[DCE]

(a) $(10,4)$

(b) $(-10,4)$

(c) $(6, 4)$

(d) $(0, 4)$

Q 25. Minimize $z = 3x + y$, subject to

constraints $2x + 3y \leq 6$, $x + y \geq 1$, $x \geq 0$, $y \geq 0$.

Then

[MHTCET]

(a) $x = 1, y = 1$

(b) $x = 0, y = 1$

(c) $x = 1, y = 0$

(d) $x = -1, y = -1$

Q 26. The shaded region for the inequality $x + 5y < 6$ is

[MHTCET]

(a) to the non−origin side of $x + 5y = 6$

(b) to the either side of $x + 5y = 6$

(c) to the origin side of $x + 5y = 6$

(d) to the neither side of $x + 5y = 6$

Q 27. ALPP means

[MHTCET]

(a) only objective function is linear

(b) only constraints are linear

(c) either objective function or constraints are linear

(d) all objective function and constraints are linear

Q 28. The maximum value of, $\mu = 3x + 4y$,

subjected to the conditions $x + y \leq 40$, $x + 2y \leq 60$;

$x, y \geq 0$, is

[MP PET]

(a) 130

(b) 140

(c) 40

(d) 120

Q 29. Consider the inequalities $x_1 + x_2 \leq 3$, $2x_1 + 5x_2 \geq 10$; $x_1, x_2 \geq 0$. Which of the point lies in the feasible region ?

[MPPET]

(a) $(2,2)$

(b) $(1,2)$

(c) $(2, 1)$

(d) $(4, 2)$

Q 30. Consider the inequalities $5x_1 + 4x_2 \geq 9$, $x_1 + x_2 \leq 3$, $x_1 \geq 0$, $x_2 \geq 0$. Which of the following point lies inside the solution set ?

[MP PET]

(a) (1,3)

(b) (l,2)

(c) (1, 4)

(d) (2, 2)

Q 31. The minimum value of $z = 2x_1 + 3x_2$

subjected to the constraints $2x_1 + 7x_2 \geq 22$, $x_1 + x_2 \geq 6$, $5x_1 + x_2 \geq 10$ and $x_1, x_2 \geq 0$ is

[MPPET]

(a) 14

(b) 20

(c) 10

(d) 16

Q 32. In case of a linear programming problem, feasible region is always

[J&.K CET]

(a) a convex set

(b) a concave set

(c) a bounded convex set

(d) a bounded concave set

Answers

1. (d) 2. (b) 3. (c) 4. (c) 5. (c) 6. (b) 7. (d) 8. (b)

9. (a) 10. (c) 11. (b) 12. (a) 13. (b) 14. (c) 15. (d)

16. (b) 17. (d) 18. (b) 19. (b) 20. (d) 21. (c) 22.

(d) 23. (d) 24. (d) 25. (b) 26. (c) 27. (d) 28. (b)

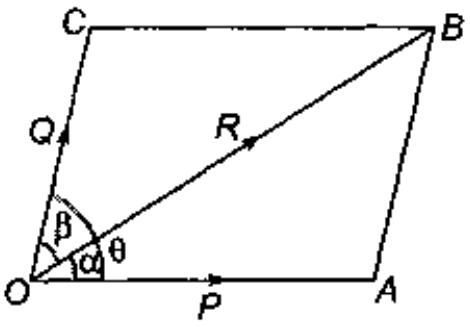
29. (b) 30. (b) 31. (a) 32. (a)

Statics and Dynamics

Useful Results & Formulae

1. (i) $R^2 = P^2 + Q^2 + 2PQ\cos\theta$

(ii) $\tan\alpha = \dfrac{Q\sin\theta}{P + Q\cos\theta}$



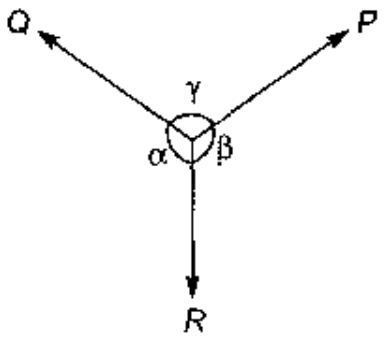
(iii) Component of a force B in two directions are

$$P = \frac{R \sin \beta}{\sin(\alpha + \beta)}$$

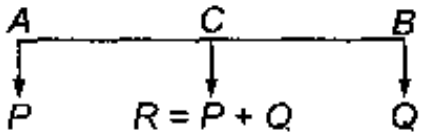
and $Q = \frac{R \sin \alpha}{\sin(\alpha + \beta)}$

(iv) Resultant of given coplanar forces acting at a point is $R = \sqrt{X^2 + Y^2}$ where $X = P_1 \cos \theta_1 + P_2 \cos \theta_2 + \ldots + P_n \cos \theta_n$ $Y = P_1 \sin \theta_1 + P_2 \sin \theta_2 + \ldots + P_n \sin \theta_n$

2. Lami's theorem is $\frac{P}{\sin \alpha} = \frac{Q}{\sin B} = \frac{R}{\sin \gamma}$



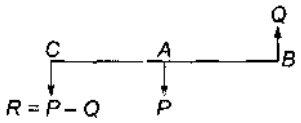
3. (i) For like parallel forces



$$AC = \left(\frac{AB}{P+Q}\right)Q$$

and $$BC = \left(\frac{AB}{P+Q}\right)P$$

(ii) For unlike parallel forces



$$AC = \left(\frac{AB}{P-Q}\right)Q$$

and $$BC = \left(\frac{AB}{P-Q}\right)P$$

4. Moment of force F about O =F . P

where P is the perpendicular distance from O to the line of action of the force

5. Moment of a couple = force x arm of the couple

6. Coefficient of friction,

$\mu = \tan \lambda = \frac{\text{Maximum force of friction}}{\text{Normal reaction}} = \frac{F}{R}$

7. (i) Centre of gravity $G(\bar{x}, \bar{y})$ is given by

$$\bar{x} = \frac{\sum w_1 x_1}{\sum w_1}, \quad \bar{y} = \frac{\sum w_1 y_1}{\sum w_1}$$

(ii) Centre of gravity of a compound body is given by

$$OG = \frac{w_1 OG_1 + w_2 OG_2}{w_1 + w_2}$$

8. (i) Average speed $= \frac{2v_1 v_2}{v_1 + v_2}$

(ii) Average velocity $= \frac{v_1 t_1 + v_2 t_2}{t_1 + t_2}$

(iii) Resultant velocity w $= \sqrt{u^2 + v^2 + 2uv\cos\alpha}$

and $\theta = \tan^{-1}\left(\frac{v\sin\alpha}{u + v\cos\alpha}\right)$

(iv) If the velocities of P and Q are $\vec{V}_P$ and $\vec{V}_Q$, then

relative velocity $\vec{V}_{QP} = \vec{V}_Q - \vec{V}_P$

9. (i) Rectilinear motion v = u + ft

(ii) $s = ut + \frac{1}{2}ft^2$

(iii) $v^2 = u^2 + 2fs$

(iv) Distance travelled in t th second,

$s_t = u + \frac{f}{2}(2t-1)$

10. (i) Force =mf

(ii) Momentum = mv

11. (i) $m_1u_1 + m_2 u_2 = m_1v_1 + m_2v_2$ (law of conservation of momentum)

(ii) $v_1 - v_2 = -e(u_1 - u_2)$ (Newton's Experimental law) where e is coefficient of restitution

(iii) Loss of KE = $\frac{m_1m_2}{2(m_1+m_2)}(u_1 - u_2)^2(1 - e^2)$

12. (i) Equation of the path of a projectile is parabola

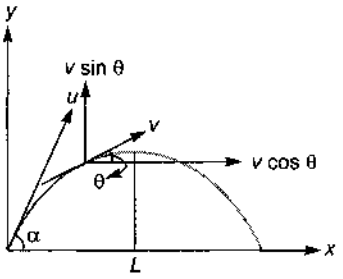
$$y = x\tan\alpha - \frac{gx^2}{2u^2\cos^2\alpha}$$

(ii) Vertex = $\left(\frac{u^2\sin\alpha\cos\alpha}{g}, \frac{u^2\sin^2\alpha}{2g}\right)$.

(iii) Directrix = $\frac{u^2}{2g}$

(iv) Focus = $\left(\frac{u^2\sin 2\alpha}{2g}, \frac{-u^2\cos 2\alpha}{2g}\right)$.

(v) Latusrectum $= \frac{2u^2\cos^2\alpha}{g}$



(vi) Time of flight $= \frac{2u \sin \alpha}{g}$

(vii) Horizontal range, $R = \frac{u^2 \sin 2\alpha}{g}$

(viii) Maximum horizontal range $= \frac{u^2}{g}$, at $\alpha = \frac{\pi}{4}$

(ix) Greatest height = $\frac{u^2 \sin^2 \alpha}{2g}$ and time taken to reach the greatest height = $\frac{u \sin \alpha}{g}$

(x) Greatest height gained by body, $H = \frac{u^2}{2g}$. Time taken to reach greatest height $= \frac{u}{g}$

(xi) Time of flight of body = $\frac{2u}{g}$

(xii) Velocity of projectile at the height $h = \sqrt{u^2 - 2gh}$ and its direction is $\theta = \tan^{-1}\left(\frac{u \sin \alpha - gt}{u \cos \alpha}\right)$

(xiii) Resultant velocity at time t is $\sqrt{u^2 - 2ugt \sin \alpha + g^2 t^2}$.

13. Maximum range on smooth inclined plane, when.

(i) Particle is projected upwards on the inclined

plane, $R_{max} = \frac{u^2}{g(1+\sin\beta)}$

(ii) Particle is projected downwards on the inclined

plane, $R_{max} = \frac{u^2}{g(1-\sin\beta)}$

14. (i) Work done = Force x Displacement

(ii) Kinetic energy, KE = $\frac{1}{2}mv^2$

(iii) Potential energy, PE = mgh

Topic 1 Velocity and Acceleration, Relative Velocities

Q 1. The displacement x of a particle at time t is given by $x = At^2 + Bt + C$ where A, B, C are constants and v is velocity of a particle, then the value of $4Ax - v^2$ is

[WBJEE]

(a) $4AC + B^2$

(b) $4AC-B^2$

(c) $2AC - B^2$

(d) $2AC + B^2$

Q 2. If the displacement, velocity and acceleration of a particle at time t be x, v and f respectively, then which one is true ?

[WB JEE, BITSAT]

(a) $f = v^3 \frac{d^2t}{dx^2}$

(b) $f = -v^3 \frac{d^2t}{dx^2}$

(c) $f = v^2 \frac{d^2t}{dx^2}$

(d) $f = -v^2 \frac{d^2t}{dx^2}$

Q 3. The displacement of a particle at time t is x, where $x = t^4 - kt^3$. If the velocity of the particle at time t = 2 is minimum, then

[WB JEE, BITSAT]

(a) k = 4

(b) k = − 4

(c) k = 8

(d) k = −8

Q 4. If the resultant of two forces of magnitude P and $P\sqrt{3}$ acting on a particle is of magnitude P, then the angle between them is

[UP SEE]

(a) $60°$

(b) $120°$

(c) $90°$

(d) $150°$

Q 5. The greatest and the least magnitudes of the resultant of two forces of constant magnitudes are F and G. When the forces act at an angle 2 a, the magnitude of the resultant is equal to

[J&K CET]

(a) $\sqrt{F^2 \cos^2 \alpha + G^2 \sin^2 \alpha}$

(b) $\sqrt{F^2 \sin^2 \alpha + G^2 \cos^2 \alpha}$

(c) $\sqrt{F^2 + G^2}$

(d) $\sqrt{F^2 - G^2}$

Q 6. A train of length 200 m, travelling at 30 m/s overtakes another train of length 300 m, travelling at 20 m/s in the same direction. The time taken by the first train to pass the second is

[UP SEE]

(a) 30 s

(b) 50 s

(c) 10 s

(d) 40 s

Q 7. The velocity of a particle at time t is given by the relation $v = 6t - \frac{t^2}{6}$. The distance travelled in 3 s is, if $s = 0$ at $t = 0$

[MHTCET]

(a) $\frac{39}{2}$

(b) $\frac{57}{2}$

(c) $\frac{51}{2}$

(d) $\frac{33}{2}$

Q 8. A particle moves in a straight line so that it covered a distance $(at^3+bt + 5)$ metre in t seconds. If its acceleration after 4 s is 48 m/s^2, then a is equal to

[MPPET]

(a) 1

(b) 2

(c) 3

(d) 4

Q 9. A particle is moving in a straight line with uniform velocity. Its acceleration is

[UP SEE]

(a) positive

(b) negative

(c) zero

(d) None of these

Q 10. Resultant velocity of two velocities 30 km/h and 60 km/h making an angle 60° with each other is

[AMU]

(a) 90 km/h

(b) 30 km/h

(c) $30\sqrt{7}$ km/h

(d) None of these

Q 11. A particle has two velocities of equal magnitude inclined to each other at an angle θ. If one of them is halved, the angle between the other

and the original resultant velocity is bisected by the new resultant. Then θ is

[AIEEE]

(a) 120°

(b) 45°

(c) 60°

(d) 90°

Q 12. A particle having simultaneous velocities 3 m/s, 5 m/s and 7 m/s is at rest. The angle between the first two velocities is

[AMU]

(a) 30°

(b) 45°

(c) 60°

(d) 90°

Q 13. A particle moves towards east from a point A to a point B at the rate of 4 km/h and then towards north from B to C at rate of 5 km/h. If AB = 12 km and BC = 5 km, then its average speed for its journey from A to C and resultant average velocity direct from A to C are respectively

[AIEEE]

(a) $\frac{17}{4}\text{km/h}$ and $\frac{13}{4}\text{km/h}$

(b) $\frac{13}{4}\text{km/h}$ and $\frac{17}{4}\text{km/h}$

(c) $\frac{17}{9}\text{km/h}$ and $\frac{13}{9}\text{km/h}$

(d) $\frac{13}{9}\text{km/h}$ and $\frac{17}{9}\text{km/h}$

Q 14. A velocity $1/4$ m/s is resolved into two components along OA and OB making angles $30°$ and $45°$ respectively with the given velocity. Then the component along OB is

[AIEEE]

(a) $\frac{1}{8}\text{m/s}$

(b) $\frac{1}{4}(\sqrt{3}-1)\mathrm{m/s}$

(c) $\frac{1}{4}\mathrm{m/s}$

(d) $\frac{1}{8}(\sqrt{6}-\sqrt{2})\mathrm{m/s}$

Q 15. A man in swimming with the uniform velocity of 6 km/h straight across a river which is flowing at the rate of 2 km/h. If the breadth of the river is 300 m, the distance between the point and the man is initially directed to and the point it will reach on the opposite bank of the river is equal to

(a) 100 m

(b) 200 m

(c) 300 m

(d) 400 m

Q 16. A particle possess two velocities simultaneously at an angle of $\tan^{-1}\frac{12}{5}$ to each other. Their resultant is 15 m/s. If one velocity is 13 m/s, then the other will be

(a) 5 m/s

(b) 4 m/s

(c) 12 m/s

(d) 13 m/s

Q 17. A train is running at 5 m/s and a man jumps out of it with a velocity 10 m/s in a direction making an angle of $60°$ with the direction of the

train. The velocity of the man relative to the ground is equal to

[UP SEE]

(a) 12.24 m/s

(b) 11.25 m/s

(c) 14.23 m/s

(d) 13.23 m/s

Q 18. Two trains are 2 km apart. Their lengths are 200 m and 300 m. They are approaching towards each other with speed of 20 m/s and 30 m/s respectively. They will cross each other after

[UP SEE]

(a) 150 s

(b) 100 s

(c) 50 s

(d) $\frac{25}{3}$s

Q 19. The distance s (in cm) travelled by a particle in t seconds is given by, $s = t^3 + 2t^2 + t$. The speed of the particle after 1 s will be

[UP SEE]

(a) 2 cm/s

(b) 8 cm/s

(c) 6 cm/s

(d) None of these

Topic 2

Rectilinear Motion with Acceleration, Motion Under Gravity and Laws of Motion

Q 1. A particle is dropped from a height 12 g m and 4s after another particle is projected from the ground towards it with a velocity $4 gms^{-1}$. The time after which the second particle meets first is

[UP SEE]

(a) 4s

(b) 2s

(c) $\frac{1}{2}$s

(d) 1 s

Q 2. A house has multi-storeys. The lowest storey is 20 ft high. A stone which is dropped from the top of the house passes the lowest storey in $\frac{1}{4}$s. The height of the house is

[DCE]

(a) 100 ft

(b) 110 ft

(c) 110.25 ft

(d) None of these

Q 3. If a falling body covers 75 m in its last second, then the height from which it is falling is (take g = 10 m/s^2) (in metres)

[UP SEE]

(a) 320

(b) 300

(c) 350

(d) 360

Q 4. A particle is projected vertically upwards and is at a height h after t_1 seconds and again after t_2 seconds, then

[WB JEE]

(a) $h = gt_1t_2$

(b) $h = \frac{1}{2} g t_1 t_2$

(c) $h = \frac{2}{g} t_1 t_2$

(d) $h = \sqrt{g t_1 t_2}$

Q 5. Rectilinear motion is performed in accordance with the formula $s = \frac{2}{9} \sin \frac{\pi t}{2} + s_0$. Then the acceleration at the end of the 1st second (in cm/s^2) is

[J&K CET]

(a) $\frac{\pi^2}{18}$

(b) $\frac{\pi^2}{7}$

(c) $-\frac{\pi^2}{9}$

(d) $-\frac{\pi^2}{18}$

Q 6. A point is moving with uniform acceleration in the eleventh and fifteenth seconds from the commencement it moves through 720 and 960 cm respectively. Its initial velocity and the acceleration with which it moves are respectively

[J&K CET]

(a) 60 m/s, 40 m/s^2

(b) 70 m/s, 30 m/s^2

(c) 90 m/s, 60 m/s^2

(d) None of these

Q 7. A body falls freely from a point A and passes through the points B and C. Given that AB $=2$BC. The ratio of the time taken by the body to cover the distances AB and BC is

[DCE]

(a) $\frac{2+\sqrt{6}}{1}$

(b) $\frac{2-\sqrt{6}}{1}$

(c) $\frac{1-\sqrt{6}}{2}$

(d) $\frac{1+\sqrt{6}}{2}$

Q 8. A ball falls from rest from top of a tower. If the ball reaches the foot of the tower in 3 s, then height of tower is(take g = $10 m/s^2$)

[UP SEE, AMU]

(a) 45 m

(b) 50 m

(c) 40 m

(d) None of these

Q 9. A particle is thrown vertically upwards with a velocity of 490 cm/s. It will return to this position after

[UP SEE]

(a) 1 s

(b) 0.5 s

(c) 2 s

(d) None of these

Q 10. Two trains A and B 100 km apart are travelling towards each other with starting speeds of 50 km/h. The train A is accelerating at 18 km/h^2 and B deaccelerating at 18 km/h^2. The distance

where the engines cross each other from the initial position of A is

[AMU]

(a) 50 km

(b) 68 km

(c) 32 km

(d) 59 km

Q 11. A body falls freely from the top of a tower and during the last second of its flight it falls $\frac{5}{9}$th of

the whole distance. The height of the tower and time of motion are respectively

[BCECE]

(a) 44.1 m and 3s

(b) 44.1 m and 5 s

(c) 4.41 m and 3 s

(d) None of these

Q 12. A body starts from rest and moves with a uniform acceleration. The ratio of the distance

covered in nth second to the distance covered in n seconds is

[Guj. CET]

(a) $\frac{2}{n}-\frac{1}{n^2}$

(b) $\frac{1}{n^2}-\frac{1}{n}$

(c) $\frac{2}{n^2}-\frac{1}{n}$

(d) $\frac{2}{n}+\frac{1}{n^2}$

Q 13. A body falling from rest under gravity passes a certain point P. It was at a distance of 400 m from P, 4 s prior to passing through P. If g = 10 m/s^2, then the height above the point P from where the body began to fall is

[AIEEE]

(a) 900 m

(b) 320 m

(c) 680 m

(d) 720 m

Q 14. Two points move in the same straight line starting at the same moment from the same point in the same direction. The first moves with constant velocity u and the second starts from rest with constant acceleration f, the distance between the two points will be maximum at time

[AMU]

(a) $t = \frac{2u}{f}$

(b) $t = \frac{u}{f}$

(c) $t = \frac{u}{2f}$

(d) $t = \frac{u^2}{f}$

Q 15. A particle is dropped under gravity from rest from a height h($g - 9.8$ m/s) and it travels a distance $\frac{9h}{25}$ in the last second. The height h is

[BCECE]

(a) 100 m

(b) 122.5 m

(c) 145 m

(d) 167.5 m

Q 16. A cyclist is beginning to move with an acceleration of 1 m/s^2 and a boy, who is $40\frac{1}{2}$m behind the cyclist, starts running at 9 m/s to meet him. The boy will be able to meet the cyclist after

[AMU]

(a) 6 s

(b) 8 s

(c) 9 s

(d) 10 s

Q 17. A bullet of 0.005 kg moving at 120 m/s penetrates deeply into a fixed target and is brought to rest in 0.01 s. The distance through which it penetrates, is equal to **[BCECE]**

(a) 3 cm

(b) 60 cm

(c) 30 cm

(d) 0

Q 18. A body travels a distance s in t seconds. It starts from rest and ends at rest. In the first part of the journey, it moves with constant acceleration f

and in the second part with constant retardation r.

The value of t is given by

[AIEEE]

(a) $2s\left(\frac{1}{f}+\frac{1}{r}\right)$

(b) $\frac{2s}{\frac{1}{f}+\frac{1}{r}}$

(c) $\sqrt{2s(f+r)}$

(d) $\sqrt{2s\left(\frac{1}{f}+\frac{1}{r}\right)}$

Q 19. The height of a tower is 7848 cm. A particle is thrown from the top of the tower with the horizontal velocity of 1784 cm/s. The time taken by the particle to reach the ground is ($g = 981$ cm/s^2)

[UP SEE]

(a) $\sqrt{8}$s

(b) 2 s

(c) 4 s

(d) 8 s

Q 20. Two bodies of different masses m_1 and m_2 are dropped from different heights h_1 and h_2. The ratio of the times taken by the two bodies to fall through these distance is

[BCECE]

(a) $h_1 : h_2$

(b) $\sqrt{h_1} : \sqrt{h_2}$

(c) $h_1^2 : h_2^2$

(d) $h_2 : h_1$

Topic 3

Projectiles, Momentum, Energy and Direct Impact of Smooth Bodies

Q 1. Two parabolic paths of angles α and β of projection aimed at a target on the horizontal plane through O, fall p units short and the other p units far from the target. If θ is the correct angle of

projection so as to hit the target, then sin 2θ is

equal to

[DCE]

(a) $\frac{1}{2}(\tan \alpha + \tan \beta)$

(b) $\frac{1}{2}(\tan \alpha - \tan \beta)$

(c) $\frac{1}{2}(\sin 2\alpha + \sin 2\beta)$

(d) $\frac{1}{2}(\sin 2\alpha - \sin 2\beta)$

Q 2. The angle of projection of a particle when its range on a horizontal plane is $4\sqrt{3}$ times the greatest height attained by it is

[BITSAT]

(a) $15°$

(b) $30°$

(c) $45°$

(d) $60°$

Q 3. A particle is thrown with the velocity v with the angle α from the horizontal plane and its range on the horizontal plane is twice to the maximum height gained. Then, tan α is equal to

[UP SEE]

(a) 9

(b) 5

(c) 2

(d) 1

Q 4. The product of two times of flight from a point P to another point Q with a given velocity of projection is

[UP SEE]

(a) $\frac{PQ}{g}$

(b) $\frac{2PQ}{g}$

(c) $\frac{PQ}{2g}$

(d) $\frac{3PQ}{g}$

Q 5. A stone of mass m is thrown vertically upwards with a velocity of $9.8\ ms^{-1}$. The height of the point where KE = PE is ($g = 9.8\ ms^{-2}$)

[UP SEE]

(a) 9.8 m

(b) 4.9 m

(c) 2.45 m

(d) 2 m

Q 6. A particle is projected from a point on the horizontal plane so as to just clear two walls each of height 20 m at distances 30 m and 170 m respectively from the point of projection. If a is the angle of projection, then

[J & K CET]

(a) $40 \tan \alpha = 51$

(b) $40 \cot \alpha = 51$

(c) $30 \tan \alpha = 23$

(d) $30 \cot \alpha = 23$

Q 7. The amount of force that is needed to accelerate a truck of mass 36000 kg from rest to a velocity of 60 km/h in 20 s is

[J&K CET]

(a) 6 kN

(b) 30 kN

(c) 60 kN

(d) 30000 kN

Q 8. A sphere S_1 impings directly on an equal sphere S_2 at rest. If the coefficient of restitution is e, then the velocities of S_1 and S_2 are in the ratio

[UP SEE]

(a) $\frac{1+e}{1-e}$

(b) $\frac{1-e}{1+e}$

(c) $\frac{e-1}{e+1}$

(d) $\frac{e+1}{e-1}$

Q 9. Two bodies of masses m and 4m are moving with equal momentum. The ratio of their KE is

[UP SEE]

(a) 1 : 4

(b) 4 : 1

(c) 1 : 1

(d) 1 : 2

Q 10. A ball is dropped from a height h above a horizontal plane. If the coefficient of restitution is e, through what height will the ball rebound?

[J&K CET]

(a) eh

(b) $2eh$

(c) e^2h

(d) $2e^2h$

Q 11. Given that the force acting on a material point is inversely proportional to the velocity of the moving point. Then the kinetic energy of the point is a ……function of time.

[J&K CET]

(a) exponential

(b) linear

(c) second degree

(d) non–linear

Q 12. A particle of mass m is projected from a fixed point O into the air with velocity win a direction making an angle a with the horizontal. Then the motion of the particle describes a parabola with the latusrectum

[J&K CET]

(a) $\frac{2}{g}$(horizontal velocity)^2

(b) $\frac{2}{g}$(vertical velocity)

(c) $\frac{2}{g^2}$(horizontal velocity)^2

(d) $\frac{2}{g^2}$(vertical velocity)^2

Q 13. If in a projectile motion range R is maximum, then relation between H and R is

[DCE]

(a) $H = \frac{R}{2}$

(b) $H = \frac{R}{4}$

(c) $H = 2R$

(d) $H = \frac{R}{8}$

Q 14. If a particle is projected with a velocity 49 m/s making an angle $60°$ with the horizontal, its time of flight is

[UP SEE]

(a) $10\sqrt{3}\,s$

(b) $5\sqrt{3}\,s$

(c) $\sqrt{3}\,s$

(d) None of these

Q 15. A ball of mass 3 kg moving with a velocity of 3 m/s collides with another ball of mass 1 kg moving with velocity u in the opposite direction. If the first ball comes to rest after the impact and $e = \frac{2}{7}$, then u in m/s, is

[UP SEE]

(a) $\frac{13}{3}$

(b) $\frac{17}{3}$

(c) $\frac{19}{3}$

(d) $\frac{23}{3}$

Q 16. A particle is projected from a point O with velocity u at an angle of $60°$ with the horizontal. When it is moving in a direction at right angle to its direction at O, then its velocity is given by

[AIEEE]

(a) $\frac{u}{\sqrt{3}}$

(b) $\frac{2u}{3}$

(c) $\frac{u}{2}$

(d) $\frac{u}{3}$

Q 17. A ball is projected at an angle α. Its range will be maximum at an angle

[DCE]

(a) 0

(b) $\frac{\pi}{4}$

(c) $\frac{\pi}{2}$

(d) π

Q 18. Two masses are projected with equal velocity u at angle $30°$ and $60°$ respectively. If the ranges covered by the masses be R_1 and R_2, then

[UP SEE]

(a) $R_1 > R_2$

(b) $R_1 = R_2$

(c) $R_1 = 4R_2$

(d) $R_2 > R_1$

Q 19. From a point on the ground at a distance 70 ft from the foot of a vertical wall, a ball is thrown at an angle of $45°$ which just clears the top of the wall and afterwards strikes the ground at a distance 30 ft on the other side of the wall. The height of the wall is

[AMU]

(a) 20 ft

(b) 21 ft

(c) 10 ft

(d) 105 ft

Q 20. If t_1 and t_2 are the times of flight of two particles having the same initial velocity u and range R on the horizontal, then $T_1^2 + T_2^2$ is equal to

[AIEEE]

(a) $\frac{u^2}{g}$

(b) $\frac{4u^2}{g}$

(c) $\frac{u^2}{2g}$

(d) 1

Q 21. The horizontal range of a projectile is $\frac{4}{\sqrt{3}}$ times its maximum height. The angle of projection is

[DCE]

(a) $30°$

(b) $45°$

(c) $60°$

(d) None of these

Q 22. If a body of mass m is thrown with initial velocity u with an angle α with the horizontal, then the maximum height attained by the body is

[DCE]

(a) $\frac{u^2 \sin^2 \alpha}{2g}$

(b) $\frac{u^2 \sin 2\alpha}{2g}$

(c) $\frac{u^2}{2g}$

(d) $\frac{u^2}{g}$

Q 23. The maximum horizontal range of a ball projected with a velocity of 39.2 m/s is (take g = $9.8 m/s^2$)

[UP SEE]

(a) 100 m

(b) 127 m

(c) 157 m

(d) 177 m

Q 24. A ball is projected vertically upward with a velocity 112 m/s. The time taken by it to return to the point of projection is ($g = 10 m/s^2$)

[UP SEE]

(a) 11s

(b) 33 s

(c) 5.5 s

(d) 22 s

Q 25. A man can throw a stone 90 m. The maximum height to which it will rise in metres, is

[AMU]

(a) 30 m

(b) 40 m

(c) 45 m

(d) 50 m

Q 26. Let R_1 and R_2 respectively be the maximum ranges up and down an inclined plane and R be the maximum range on the horizontal plane. Then R_1, R, R_2 are in

[AIEEE]

(a) arithmetico–geometric progression (AGP) (b) AP

(c) GP

(d) HP

Q 27. Two stones are projected from the top of a cliff h metres high, with the same speed u so as to hit the ground at the same spot. If one of the stones is projected horizontally and the other is

projected at an angle θ to the horizontal, then tan θ equals

[AIEEE]

(a) $\sqrt{\frac{2u}{gh}}$

(b) $2g\sqrt{\frac{u}{h}}$

(c) $2h\sqrt{\frac{u}{g}}$

(d) $u\sqrt{\frac{2}{gh}}$

Topic 4

Basic Laws of Mechanics, Composition and Equilibrium of Forces

Q 1. If forces $\vec{P}, \vec{Q}, \vec{R}$ acting at a point can be represented by the sides of a triangle taken in order, then

(a) $\vec{P}+\vec{Q}+\vec{R}=\vec{0}$

(b) $\vec{P}-\vec{Q}+\vec{R}=\vec{0}$

(c) $\vec{P}+\vec{Q}-\vec{R}=\vec{0}$

(d) $\vec{P}-\vec{Q}-\vec{R}=\vec{0}$

Q 2. Forces of magnitudes 5 N, 10N, 15 N and 20 N act on a particle in the direction of North, South, East and West respectively. The magnitude of their resultant is

[BCECE]

(a) $15\sqrt{2}$ N

(b) 10 N

(c) $25\sqrt{2}$ N

(d) $5\sqrt{2}$ N

Q 3. Three forces of magnitudes 30, 60 and P acting at a point are in equilibrium. If the angle between the first two is $60°$, the value of P is

[UP SEE]

(a) $30\sqrt{7}$

(b) $30\sqrt{3}$

(c) $20\sqrt{6}$

(d) $25\sqrt{2}$

Q 4. The resultant of two forces P and 2P is $P\sqrt{3}$. If 1st force is doubled and reversed, the resultant of forces is

[UP SEE]

(a) $2P\sqrt{3}$

(b) $P\sqrt{3}$

(c) AP

(d) None of the above

Q 5. If two equal components (P) of a force (F) are acting at an angle $120°$, then F is

[UP SEE]

(a) P

(b) $P\sqrt{2}$

(c) $2P$

(d) $\frac{P}{2}$

Q 6. If $3P$ and $4P$ are resultant of a force $5P$, then angle between $3P$ and $5P$ is

[AMU]

(a) $\sin^{-1}\left(\frac{3}{5}\right)$

(b) $\sin^{-1}\left(\frac{4}{5}\right)$

(c) $90°$

(d) None of the above

Q 7. If forces of magnitudes 12 kg-wt, 5 kg-wt and 13 kg-wt act at a point are in equilibrium, then the angle between the first two forces is

[BCECE]

(a) $30°$

(b) $90°$

(c) $60°$

(d) $45°$

Q 8. With two forces acting at a point, the maximum effect is obtained when their resultant is 4 N. If they act at right angles, then their resultant is 3N. Then the forces are

[Jamia Millia Islamia]

(a) $(2+\sqrt{2})\text{N}$ and $(2-\sqrt{2})\text{N}$

(b) $(2+\sqrt{3})\text{N}$ and $(2-\sqrt{3})\text{N}$

(c) $\left(2+\frac{1}{2}\sqrt{2}\right)\text{N}$ and $\left(2-\frac{1}{2}\sqrt{2}\right)\text{N}$

(d) $\left(2+\frac{1}{2}\sqrt{3}\right)\text{N}$ and $\left(2-\frac{1}{2}\sqrt{3}\right)\text{N}$

Q 9. The resultant R of two forces P and Q act at right angles to P. Then the angle between the forces is

[AMU]

(a) $\cos^{-1}\left(\frac{\text{P}}{\text{Q}}\right)$

(b) $\cos^{-1}\left(-\frac{\text{P}}{\text{Q}}\right)$

(c) $\sin^{-1}\left(\frac{P}{Q}\right)$

(d) $\sin^{-1}\left(-\frac{P}{Q}\right)$

Q 10. The resultant R of two forces acting on a particle is at right angles to one of them and its magnitude is one third of the other force. The ratio of larger force to smaller one is

[J&K CET]

(a) $3:2\sqrt{2}$

(b) $3 : 2$

(c) $3 : \sqrt{2}$

(d) $2 : 1$

Q 11. The angle between two forces each equal to P when their resultant is also equal to P is

[UP SEE]

(a) $\frac{2\pi}{3}$

(b) $\frac{\pi}{3}$

(c) π

(d) $\frac{\pi}{2}$

Q 12. The resultant of forces $\vec{P}$ and $\vec{Q}$ is $\vec{R}$. If $\vec{Q}$ is doubled, then $\vec{R}$ is doubled. If the direction of $\vec{Q}$ is reversed, then $\vec{R}$ is again doubled, then $P^2 : Q^2 : R^2$ is

[AIEEE]

(a) $3 : 1 : 1$

(b) $2 : 3 : 2$

(c) $1 : 2 : 3$

(d) $2 : 3 : 1$

Q 13. Two particles start simultaneously from the same point and move along two straight lines, one with uniform velocity $\vec{u}$ and the other from rest with uniform acceleration $\vec{f}$. Let α be the angle between their directions of motion. The relative

velocity of the second particle w.r.t. the first is least after a time

[AIEEE]

(a) $\frac{u \sin \alpha}{f}$

(b) $\frac{f \cos \alpha}{u}$

(c) $u \sin \alpha$

(d) $\frac{u \cos \alpha}{f}$

Q 14. The resultant of two forces P and Q is R. If the direction of P is reversed keeping the direction Q same, the resultant remains unaltered. The angle between P and Q is

[UP SEE]

(a) $90°$

(b) $60°$

(c) $45°$

(d) $30°$

Topic 5

Composition of Coplanar, Concurrent and Parallel Forces, Lami's Theorem

Q 1. Let ABC be an equilateral triangle formed by weightless inextensible strings, the side AB is horizontal. A and B are tied to fixed points D and E by equal weightless inextensible strings AD, BE, a weight of W gram is attached at C. The angle DAB,

ABE are each $150°$. Then, which of the following statements is true ?

[DCE]

(a) The tensions in the strings AB, BC and CA are equal

(b) The tensions in the strings BE, AD are inversely proportional

(c) The tension in BE is less than the tension in AB

(d) The tension in AD is twice the tension in BC

Q 2. P and Q are two like parallel forces. If Q is moved parallel to itself through a distance x, then the resultant of P and Q moves through a distance

[J&K CET]

(a) Q(P + Q)x

(b) $\frac{Qx}{(P+Q)}$

(c) $\frac{Px}{(P+Q)}$

(d) $\frac{(P+Q)x}{(P-Q)}$

Q 3. Three forces $\vec{P}, \vec{Q}, \vec{R}$ are acting at a point in a plane. The angle between $\vec{P}$ and $\vec{Q}$, $\vec{Q}$ and $\vec{R}$ are $150°$ and $120°$ respectively, then for equilibrium, forces P, Q, R are in the ratio

[DCE]

(a) $1 : 2 : 3$

(b) $1 : 2 : \sqrt{3}$

(c) $3 : 2 : 1$

(d) $\sqrt{3} : 2 : 1$

Q 4. If two like parallel forces $\frac{P}{Q}$ N and $\frac{Q}{P}$ N have a resultant 2 N, then

[UP SEE]

(a) P = Q

(b) 2P = Q

(c) P^2 = Q

(d) P = 2Q

Q 5. Three coplanar forces acting on a particle are in equilibrium. The angle between the first and the second is $60°$ and that between the second and the third is $150°$. The ratio of the magnitude of the forces are

[AMU]

(a) $1 : 1 : \sqrt{3}$

(b) $1 : \sqrt{3} : 1$

(c) $\sqrt{3} : 1 : 1$

(d) $\sqrt{3} : \sqrt{3} : 1$

Q 6. Three forces $\vec{P}, \vec{Q}$ and $\vec{R}$ acting along IA, IB and IC, where I is the incentre of a ΔABC, are in equilibrium. Then $\vec{P} : \vec{Q} : \vec{R}$ is

[AIEEE]

(a) $\cos\frac{A}{2} : \cos\frac{B}{2} : \cos\frac{C}{2}$

(b) $\sin\frac{A}{2}:\sin\frac{B}{2}:\sin\frac{C}{2}$

(c) $\sec\frac{A}{2}:\sec\frac{B}{2}:\sec\frac{C}{2}$

(d) $\operatorname{cosec}\frac{A}{2}:\operatorname{cosec}\frac{B}{2}:\operatorname{cosec}\frac{C}{2}$

Q 7. Three forces P, Q and R acting on a particle are in equilibrium. If the angle between P and Q is double the angle between P and R, then P is equal to

[BCECE]

(a) $\frac{Q^2 + R^2}{R}$

(b) $\frac{Q^2 - R^2}{Q}$

(c) $\frac{Q^2 - R^2}{R}$

(d) $\frac{Q^2 + R^2}{Q}$

Topic 6 Moment, Couple and Friction

Q 1. The magnitude of the two forces forming a couple is 36 N each and the arm of the couple is 4 m. The magnitude of each force of an equivalent couple whose arm is 9m, is(in Newtons)

[UP SEE]

(a) 18

(b) 26

(c) 16

(d) 15

Q 2. In a right angle ΔABC, $\angle$A = $90°$ and sides a, b, c are respectively, 5 cm, 4 cm and 3 cm. If a force $\vec{F}$ has moments 0, 9 and 16 in N cm unit respectively about vertices A, B and C, the magnitude of $\vec{F}$ is

[Jamia Millia Manila]

(a) 3

(b) 4

(c) 5

(d) 9

Q 3. Three forces P, Q, R act along the sides BC, CA, AB of a triangle ABC taken in order. The condition that the resultant passes through the incentre is

[AMU]

(a) P + O + R = 0

(b) P cos A + Q cos B + R cos C = 0

(c) P sec A + Q sec B + R sec C = 0

(d) $\frac{P}{\sin A}+\frac{Q}{\sin B}+\frac{R}{\sin C}=0$

Q 4. A uniform ladder rest in limiting equilibrium with its lower end on a rough horizontal plane and its upper end against a smooth vertical wall, if θ is an angle of inclination of the ladder to the vertical

wall and μ is the coefficient of friction, then tan 6 is equal to

[AMU]

(a) μ

(b) 2μ

(c) $\frac{3\mu}{2}$

(d) $\mu + 1$

Q 5. A couple is of moment $\vec{G}$ and the force forming the couple is $\vec{P}$. If $\vec{P}$ is turned through a right angle, the moment of the couple thus formed is $\vec{H}$. If instead, the forces $\vec{P}$ is turned through an angle a, then the moment of couple becomes

[AIEEE]

(a) $\vec{G} \sin \alpha - \vec{H} \cos \alpha$

(b) $\vec{H} \cos \alpha + \vec{G} \sin \alpha$

(c) $\vec{G} \cos \alpha + \vec{H} \sin \alpha$

(d) $\vec{H} \sin \alpha - \vec{G} \cos \alpha$

Q 6. A block weighing w, is supported on an inclined surface with the help of a horizontal force P. The same block can be supported with the help of another force Q acting parallel to the inclined surface, then the value of $\frac{1}{P^2} + \frac{1}{w^2}$ is

[UP SEE]

(a) w sin α

(b) 1

(c) $\frac{1}{Q}$

(d) $\frac{1}{Q^2}$

Q 7. Forces forming a couple are of magnitude 12N each and the arm of the couple is 8 m. The force of an equivalent couple whose arm is 6 m is of magnitude

[BCECE]

(a) 8N

(b) 16 N

(c) 12 N

(d) 4N

Answers

Topic 1 : Velocity and Acceleration, Relative Velocities

1. (b) 2. (b) 3. (a) 4. (d) 5. (a) 6. (b) 7. (c) 8. (b)

9. (c) 10. (c) 11. (a) 12. (c) 13. (a) 14. (d) 15. (a)

16. (b) 17. (d) 18. (c) 19. (b)

Topic 2 : Rectilinear Motion with Acceleration, Motion Under (Gravity and Laws of Motion

1. (?) 2. (c) 3. (a) 4. (b) 5. (d) 6. (c) 7. (a) 8. (a)

9. (a) 10. (d)11. (a) 12. (a) 13. (d) 14. (b) 15. (b)

16. (c) 17. (b) 18. (d) 19. (c) 20. (b)

Topic 3 : Projectiles, Momentum, Energy and Direct Impact of Smooth Bodies

1. (c) 2. (b) 3. (c) 4. (b) 5. (c) 6. (b) 7. (b) 8. (b)

9. (b) 10. (c) 11. (b) 12. (a) 13. (b) 14. (b) 15. (c)

16. (a) 17. (b) 18. (b) 19. (b) 20. (b) 21. (c) 22.

(a) 23. (c) 24. (d) 25. (c) 26. (d) 27. (d)

Topic 4 : Basic Laws of Mechanics, Composition and Equilibrium of Forces

1. (a) 2. (d) 3. (a) 4. (a) 5. (a) 6. (b) 7. (b) 8. (c)

9. (b) 10. (a) 11. (a) 12. (b) 13. (d) 14. (a)

Topic 5 : Composition of Coplanar, Concurrent and Parallel Forces, Lami's Theorem

1. (a) 2. (b) 3. (d) 4. (a) 5. (a) 6. (a) 7. (b)

Topic 6 : Moment, Couple and Friction

1. (c) 2. (c) 3. (a) 4. (b) 5. (c) 6. (d) 7. (b)

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